



RELATIONAL PSYCHOANALYSIS

the
Emergence
of a
Tradition

EDITED BY

STEPHEN A. MITCHELL • LEWIS ARON



FORENSIC USES OF CLINICAL ASSESSMENT INSTRUMENTS

This book provides, in a single volume, an extensive, research-based evaluation of the most popular clinical assessment tools as applied in forensic settings. These widely used instruments often require important modifications in their administration and interpretation when used for forensic purposes, and it is vital that the clinician is intimately familiar with their correct application, as well as their limitations. The test instruments included are analyzed by senior figures in the field of psychological assessment who are uniquely qualified to discuss them because they have had a crucial role in the development of the tests or they have dedicated their careers to advancing our understanding of these clinical assessment measures. Each chapter begins with a summary of the development of the assessment instrument in its more traditional applications in clinical settings, and then considers its utilization in forensic settings. The types of forensic issues which have been addressed with these instruments are reviewed, and an illustrative case example is given which reflects the uses and limitations of the assessment technique when applied in a forensic context. New in this edition are a chapter on the MMPI-2-RF and separate chapters for the adult and youth versions of the Hare Psychopathy Checklist. Psychologists using assessment instruments in applied forensic settings will find this to be a valuable and practical source of information, as will attorneys wishing to gain an understanding of the application of these psychological assessment approaches in the courtroom.

Robert P. Archer, PhD, ABPP, is the Frank Harrell Redwood Distinguished Professor, Department of Psychiatry and Behavioral Sciences, Eastern Virginia Medical School, Norfolk, Virginia.

Elizabeth M. A. Wheeler, PhD, is affiliated with Forensic Psychology Associates, a forensic psychology practice in Richmond, Virginia, which provides a broad range of criminal and civil forensic services.

FORENSIC USES OF CLINICAL ASSESSMENT INSTRUMENTS

Second edition

*Edited by Robert P. Archer and
Elizabeth M. A. Wheeler*

This edition published 2013
by Routledge
711 Third Avenue, New York, NY 10017

Simultaneously published in the UK
by Routledge
27 Church Road, Hove, East Sussex BN3 2FA

Routledge is an imprint of the Taylor & Francis Group, an informa business

© 2013 Taylor & Francis

The right of the editors to be identified as the authors of the editorial material, and of the authors for their individual chapters, has been asserted in accordance with sections 77 and 78 of the Copyright, Designs and Patents Act 1988.

All rights reserved. No part of this book may be reprinted or reproduced or utilized in any form or by any electronic, mechanical, or other means, now known or hereafter invented, including photocopying and recording, or in any information storage or retrieval system, without permission in writing from the publishers.

Trademark notice: Product or corporate names may be trademarks or registered trademarks, and are used only for identification and explanation without intent to infringe.

First edition published by Routledge 2006

Library of Congress Cataloging-in-Publication Data

Forensic uses of clinical assessment instruments / [edited by] Robert P. Archer
and Elizabeth M.A. Wheeler. — 2nd ed.

p. cm.

Includes bibliographical references and index.

1. Forensic psychology. 2. Psychological tests. I. Archer, Robert P. II. Wheeler,
Elizabeth M. A.

RA1148.F559 2013

614'.15—dc23

2012034930

ISBN: 978-0-415-52785-9 (hbk)

ISBN: 978-0-415-81522-2 (pbk)

ISBN: 978-0-203-11874-0 (ebk)

Typeset in Bembo

by Keystroke, Station Road, Codsall, Wolverhampton

CONTENTS

<i>List of Figures</i>	vii
<i>List of Tables</i>	viii
<i>About the Editors</i>	x
<i>About the Contributors</i>	xi
<i>Preface</i>	xiv
1 Introduction to Forensic Uses of Clinical Assessment Instruments <i>Robert P. Archer, Rebecca Stredny, and Elizabeth M. A. Wheeler</i>	1
2 The Minnesota Multiphasic Personality Inventory–2 <i>Martin Sellbom and Jaime L. Anderson</i>	21
3 Forensic Applications of the Minnesota Multiphasic Personality Inventory–2–Restructured Form <i>Yossef S. Ben-Porath</i>	63
4 The Minnesota Multiphasic Personality Inventory–Adolescent <i>Robert P. Archer and Richard W. Handel</i>	108
5 Forensic Assessment with the Personality Assessment Inventory <i>Leslie C. Morey and Justin K. Meyer</i>	140
6 The Millon Clinical Multiaxial Inventory–III: Use in Forensic Settings <i>Robert J. Craig</i>	175
7 The Rorschach Inkblot Method <i>Irving B. Weiner</i>	202

vi Contents

8	The Psychopathy Checklist–Revised: Forensic Applications and Limitations <i>Robert D. Hare, Pamela J. Black, and Zach Walsh</i>	230
9	The Hare Psychopathy Checklist–Youth Version <i>Angela S. Book, Adelle E. Forth, and Heather J. Clark</i>	266
10	Forensic Neuropsychological Assessment <i>Eric A. Zillmer and Christopher M. King</i>	291
11	The Achenbach System of Empirically Based Assessment (ASEBA): Applications in Forensic Contexts <i>Thomas M. Achenbach and Leslie A. Rescorla</i>	311
12	The Forensic Uses and Limitations of the Parenting Stress Index <i>Richard R. Abidin, William G. Austin, and James R. Flens</i>	346
	<i>Author Index</i>	380
	<i>Subject Index</i>	403

FIGURES

2.1	MMPI-2 Extended Score Report	44
3.1	MMPI-2-RF Score Report	94
4.1	MMPI-A Structural Summary Form	118
4.2	MMPI-A Profile for Basic Scales for Brian	131
4.3	MMPI-A Profile for Content and Supplementary Scales for Brian	133
5.1	PAI Full Scale Profile for Case Example: Lionel	165
5.2	PAI Subscale Profile for Case Example: Lionel	166
8.1	Four-Factor Structure of the PCL-R	236
8.2	Two-Factor Structure of the PCL-R	237
11.1	Pages 1 and 2 of the Child Behavior Checklist completed for Wayne Webster by his mother	312
11.2	Page 3 of the Child Behavior Checklist completed for Wayne Webster	314
11.3	Hand-scored Competence Profile for Wayne Webster's Child Behavior Checklist	317
11.4	Computer-scored Syndrome Profile for the Youth Self-Report completed by Wayne Webster	318
11.5	Hand-scored <i>DSM</i> -Oriented Profile scored from the Teacher's Report Form completed by one of Wayne Webster's teachers	320
11.6	Cross-informant comparison of syndrome scores obtained from instruments completed by Wayne Webster, his mother, father, and three teachers	322
11.7	Narrative Report summarizing Youth Self-Report results for Wayne Webster	323
12.1	Theoretical Model for the PSI's Development	349

TABLES

1.1	Advantages and Disadvantages of Specialized Forensic Assessment Instruments	6
1.2	Instruments Used in 20% or More of Forensic Neuropsychological Evaluations	8
1.3	Tests Rated by Diplomates in Forensic Psychology as Recommended, Acceptable, or Unacceptable for Forensic Evaluations	11
1.4	Percentage of Practitioners Reporting Use of Popular Clinical Assessment Instruments in Forensic Evaluations	14
1.5	Uses and Limitations of Clinical Assessment Instruments in Forensic Applications	15
3.1	The MMPI–2–RF Scales	65
4.1	Overview of the MMPI–A Scales and Subscales	112
4.2	Overview of MMPI–A Scales Added in the MMPI–A Manual Supplement	115
5.1	PAI Scales and Subscales	141
5.2	Supplementary PAI Indices	143
5.3	PAI Supplemental Indices for Case Example: Lionel	167
6.1	Millon's 15 Different Personality Spectra of Normal and Subclinical Types (Moderate) and Clinical Disorders	177
6.2	Millon's Bioevolutionary Characterization of Personality Styles and Personality Disorders	178
6.3	MCMI–III Scales and the Grossman Facet Scales	180
6.4	Stability and Reliability of Scale 6A	182
6.5	Convergent Validity of the MCMI–III Antisocial PD Scale with Similar Measures	184
6.6	Correspondence of the MCMI Major Depression Scale with Similar Measures	185
6.7	Diagnostic Power of the Borderline PD Scale	186
6.8	Antisocial Personality Personologic Domains	192
6.9	MCMI–III Scores for the Case Example Examinee	193
7.1	Sequence of Scores for Mr. Clark	221
7.2a	Structural Summary for Mr. Clark	222
7.2b	Ratios, Percentages and Derivatives	223
8.1	Items and Factors in the Hare PCL Scales	234
9.1	Hare Psychopathy Checklist: Youth Version (PCL:YV) Scores across Male and Female Adolescents	268
9.2	Items and Factors in the PCL:YV	269

10.1 Application of Principles of Forensic Mental Health Assessment to Forensic Neuropsychological Assessment	292
10.2 Commonly Employed Neuropsychological Instruments and References Discussing Their Utility for Detecting Neurocognitive Feigning	298
11.1 ASEBA Self-Report and Collateral Report Forms	315
11.2 Scales Scored from ASEBA Self-Report and Collateral Report Instruments	321
11.3 National Normative Samples for ASEBA Self-Report and Collateral Report Instruments	328
11.4 Psychometric Characteristics of ASEBA Scales	329
11.5 Validity Data for ASEBA Self-Report and Collateral Report Instruments	331
11.6 <i>T</i> -Scores for Lisa Based on Ratings by Four Informants	339
12.1 Child Characteristic Domain Subscales	347
12.2 Parent Characteristics Domain Subscales	347
12.3 PSI and PSI-SF Publishers and Approved Translations	353
12.4 Availability of PSI and PSI-SF Translations	353
12.5 Preliminary Descriptive PSI-3 Data for Use in the Child Custody Context	368

ABOUT THE EDITORS

Robert P. Archer, PhD, ABPP, is the Frank Harrell Redwood Distinguished Professor, and Director of Psychology, in the Department of Psychiatry and Behavioral Sciences at the Eastern Virginia Medical School in Norfolk, Virginia. Dr. Archer is the author of numerous books, book chapters, and articles related to psychological assessment and forensic evaluation. He is the author of *MMPI-A: Assessing Adolescent Psychopathology* (3rd edition) and the co-editor of *Personality Assessment*. Dr. Archer is a co-author of the *MMPI-A* and serves on the Advisory Committee for the development of the *MMPI-A-RF*.

Elizabeth M. A. Wheeler, PhD, is a graduate of the Clinical Psychology Program at Virginia Commonwealth University and completed her internship training at the Charleston Consortium Clinical Psychology Internship Program. Dr. Wheeler is the senior author of a recent publication concerning the *MMPI-2-RF* characteristics of child custody litigants published in *Assessment*. Dr. Wheeler is affiliated with Forensic Psychology Associates, a forensic psychology practice in the Richmond, Virginia area, that provides a broad range of criminal and civil forensic services.

ABOUT THE CONTRIBUTORS

Richard R. Abidin, EdD, ABPP-Clinical, is Professor Emeritus and founder of the Clinical and School Psychology PhD program at the University of Virginia. He is the Founding Editor of *Early Education and Development*, Past President of the Society for the Study of School Psychology, and Past President of Division 12 (Clinical Child Psychology) of the American Psychological Association.

Thomas M. Achenbach, PhD, is Professor of Psychiatry and Psychology and Director of the Center for Children, Youth, and Families at the University of Vermont. His honors include the APA's Distinguished Contribution Award for Clinical Child Psychology and the Institute for Scientific Information's Most Highly Cited in the World Psychiatry and Psychology Literature. He developed the Achenbach System of Empirically Based Assessment (ASEBA).

Jaime L. Anderson, MS, is a doctoral student in clinical psychology, with a concentration in psychology and law, at the University of Alabama. Her research interests focus on the assessment of personality disorders and malingering, with an emphasis on the MMPI-2/MMPI-2-RF.

Robert P. Archer, PhD, ABPP, is the Frank Harrell Redwood Distinguished Professor, Department of Psychiatry and Behavioral Sciences, Eastern Virginia Medical School, Norfolk, Virginia.

William G. Austin, PhD, has a private practice in clinical and forensic psychology with a specialty in child custody evaluation and trial consultation. He has developed forensic evaluation models for custody evaluators and family courts for the complex problems of relocation, intimate partner violence, and parental gatekeeping. He was the co-chair of the AFCC task force that created the Model Standards of Practice for Child Custody Evaluation and is currently on the AFCC Task Force for developing guidelines for conducting custody evaluations where there are allegations of intimate partner violence.

Yossef S. Ben-Porath, PhD, is Professor of Psychology at Kent State University. He is Associate Editor of the journal *Psychological Assessment* and former Editor of *Assessment*. He is Co-director of the MMPI Workshops and Symposia series, co-developer and co-author of the MMPI-2-RF, and

xii About the Contributors

co-author of the MMPI–2 and MMPI–A manuals. He also practices forensic psychological assessment.

Pamela J. Black is a PhD student in the Department of Psychology at the University of British Columbia, Okanagan.

Angela S. Book, PhD, is Associate Professor in the Psychology Department at Brock University in St. Catharines, Ontario, Canada. Her research examines the relationship between psychopathy and victim selection.

Heather J. Clark, MA, is pursuing her PhD in Forensic Psychology at Carleton University, Ottawa. Her research area of interest is enhancing treatment gain and effectiveness for young offenders with psychopathic traits.

Robert J. Craig, PhD, ABPP, is a Professor at Roosevelt University, Chicago, and has authored three books and published seminar papers on the MCMI in peer-reviewed journals. He was a fellow in APA and in the Society for Personality Assessment and is now in semi-retirement.

James R. Flens, PsyD, ABPP, is board-certified in clinical and forensic psychology. He has published several peer-reviewed articles and book chapters with colleagues, and regularly lectures at national and regional conferences regarding various topics in family law issues. His practice is devoted to family law-related issues, including evaluations and trial consultation.

Adelle E. Forth, PhD, is Associate Professor of Forensic Psychology at Carleton University, Ottawa. She is the senior author of the *Hare Psychopathy Checklist: Youth Version*, co-author of the *Structured Assessment of Violence Risk in Youth*, and co-author of the textbooks *Forensic Psychology* and *Criminal Behaviour: A Canadian Perspective*. Her research focuses on assessment of psychopathic traits, the relationship between psychopathy and emotional/cognitive processes, and victims of psychopaths.

Richard W. Handel, PhD, is a Professor in the Department of Psychiatry and Behavioral Sciences at Eastern Virginia Medical School. His primary research interest is the assessment of personality and psychopathology using the MMPI–2–RF and MMPI–A instruments.

Robert D. Hare, PhD, is Emeritus Professor of Psychology, University of British Columbia, Okanagan, and President of Darkstone Research Group, Ltd., a forensic research and consulting firm. He has published extensively on psychopathy: its nature, assessment, and implications for mental health and criminal justice. He is the author of the *Psychopathy Checklist–Revised* and a co-author of its derivatives. He also is the author of *Without Conscience* and a co-author of *Snakes in Suits*.

Christopher M. King is a JD–PhD student at Drexel University. His research interests include forensic mental health assessment, offender risk assessment and reduction, and mental health and corrections law and public policy.

Justin K. Meyer, BS, is a current PhD student at Texas A&M University working with Dr. Leslie C. Morey, author of the Personality Assessment Inventory.

Leslie C. Morey, PhD, is Professor of Psychology at Texas A&M University. He is past Associate Editor of the *Journal of Personality Assessment* and past Associate Editor of *Assessment*. He is the author of the Personality Assessment Inventory and related works, and has published over 200 articles, books, and chapters on the assessment and diagnosis of mental disorders. He is a member of the *DSM-5* Work Group on Personality and Personality Disorders.

Leslie A. Rescorla, PhD, is Professor in the Psychology Department and Director of the Child Study Institute at Bryn Mawr College. She is a licensed clinical psychologist and a certified school psychologist with a practice serving children and adults. She is a co-author of Achenbach System of Empirically Based Assessment (ASEBA) forms and profiles for children and adults, and also of numerous publications in the area of empirically based assessment of children, adolescents, and adults.

Martin Sellbom, PhD, is Assistant Professor of Psychology at the University of Alabama. He has contributed extensively to the MMPI-2 literature over the past 10 years. He is an Associate Editor for the journal *Assessment*, and Consulting Editor for the other major assessment journals. His clinical practice emphasizes forensic psychological evaluations and pre-employment screening.

Rebecca Stredny, PsyD, ABPP, is Clinical Director at Eastern State Hospital in Williamsburg, Virginia, and an adjunct faculty member in the Psychology Department at Virginia Commonwealth University, Richmond.

Zach Walsh, PhD, R.Psych, is Co-director of the Centre for the Advancement of Psychological Science and Law and Assistant Professor of Psychology at the University of British Columbia, Okanagan.

Irving B. Weiner, PhD, ABPP, is Clinical Professor of Psychiatry and Neuroscience at the University of South Florida. He is a former President of the APA Division of Clinical Psychology; the APA Division of Evaluation, Measurement, and Statistics; the Society for Personality Assessment; and the International Rorschach Society. He is also a former Editor of the *Journal of Personality Assessment*, and co-author with Roger Greene of the *Handbook of Personality Assessment*.

Elizabeth M.A. Wheeler, PhD, is affiliated with Forensic Psychology Associates, a forensic psychology practice in Richmond, Virginia, which provides a broad range of criminal and civil forensic services.

Eric A. Zillmer, PsyD, is Carl R. Pacifico Professor of Neuropsychology and the Athletics Director at Drexel University. He is the co-author of *Military Psychology* and *Principles of Neuropsychology*. He is a Fellow of American Psychological Association, the Society for Personality Assessment, and the National Academy of Neuropsychology, for which he also served as President.

PREFACE

The *Specialty Guidelines for Forensic Psychology* (2011) note that forensic psychologists have a duty to use traditional assessment procedures in a manner consistent with the research evidence available on their proper application in forensic settings. Forensic psychologists are cautioned to consider carefully the strengths and limitations of traditional assessment procedures when used in forensic evaluations, and to consider the validity of such techniques as specifically applied in forensic settings. The purpose of the first edition, and the current revision, of this text is to assist forensic psychologists in meeting those guidelines. *Forensic Uses of Clinical Assessment Instruments* provides an in-depth research-based evaluation of the most commonly used psychological assessment instruments in clinical practice in terms of their uses and limitations when applied in forensic settings.

The second edition was developed to summarize, in a single volume, the available research on the use of popular clinical assessment tools as applied in forensic settings, including the substantial research that has occurred in this field since the publication of the first edition in 2006. This research literature has significantly expanded for all of the popular clinical assessment tools covered in the first edition, and has substantively informed our understanding of the types of modifications in administration and interpretation most appropriate when these tests are used for forensic purposes.

The second edition is able to provide an authoritative review of the forensic uses of clinical assessment instruments because chapter authors have unique expertise in their fields. The authors for the second edition include the senior figures in psychological assessment uniquely qualified to discuss the specific tests covered in their chapters because of their crucial role in the development of this test instrument, or their unique contributions in terms of expanding the critical research base for the instrument. We are extremely pleased that the core group of authors who participated in the first edition have been willing to update and expand their contributions for the second edition. The format of the chapters that appear in the second edition remains the same as in the original text. Each chapter provides a summary of the development of the assessment instrument in its more traditional applications in clinical settings. The chapters then provide a summary of the literature on the utilization of the instrument in forensic settings, as well as a review of the types of forensic issues that have been addressed with that instrument. Finally, the authors of each chapter were asked to provide an illustrative case example that reflects the types of uses and limitations of the assessment technique when applied to forensic issues.

An important change that has occurred in the second edition involves the inclusion of Elizabeth Wheeler as co-editor of this edition, as well as co-author of the introductory chapter. Dr. Wheeler

completed her graduate training at Virginia Commonwealth University, and her clinical psychology internship training at the Charleston Consortium Psychology Internship Training Program. Dr. Wheeler has recently begun her forensic psychology practice in Richmond, Virginia, in a setting exclusively focused on civil and criminal forensic evaluations. Dr. Wheeler assumed a major role in organizing and integrating the second edition of this work, and it was a joy for me to have this opportunity to work with her.

Dr. Wheeler and I would like to specifically acknowledge several individuals who played a very important role in the creation of the second edition. First and foremost, we would like to express our gratitude to Dr. George Zimmar, Publisher at Routledge, for his initial encouragement and support for the development of a second edition of this book. In addition, we would like to express our gratitude to our editors at Routledge, Marta Moldvai and Christopher Tominich, for their work in the development and production of this book. Further, we would like to recognize the important contributions of Dr. Eric Zillmer, the Carl R. Pacifico Professor of Neuropsychology at Drexel University; Dr. Richard Handel, Associate Professor in the Department of Psychiatry and Behavioral Sciences at Eastern Virginia Medical School; and Dr. Elisabeth Gibbings, Director of Admissions and Practicum Training, Institute for Graduate Clinical Psychology, Widener University, for their thoughtful critiques of our first editions and their very crucial constructive recommendations and suggestions for issues that should be addressed in the revision of this work. Finally, I am quite confident that I also speak for Dr. Wheeler when I express my deepest appreciation and gratitude to my wife, Dr. Linda R. Archer, for her enthusiastic support for this project.

In the original edition, I concluded the Preface with the observation that traditional clinical assessment instruments will continue to play a vital role in forensic psychology in the foreseeable future. In the ensuing years since the publication of the original work, there is certainly no evidence in any test utilization surveys that the preeminent role of traditional clinical assessments instruments in forensic psychology has been, or will be, supplanted or minimized. On a pragmatic basis, the issue is not whether traditional assessment instruments will be used in forensic settings, but whether those instruments will be used poorly, or thoughtfully and constructively. The objective of the second edition is to provide the most recent information on these important tests for clinical and forensic psychologists who use the assessment instruments in forensic evaluations, and for attorneys who seek a better understanding of these instruments as employed in courtroom settings.

Robert P. Archer
Elisabeth M.A. Wheeler

1

INTRODUCTION TO FORENSIC USES OF CLINICAL ASSESSMENT INSTRUMENTS

Robert P. Archer, Rebecca Stredny, and Elizabeth M. A. Wheeler

What Is Forensic Psychology?

As defined in the 2011 edition of the American Psychological Association's (APA) *Specialty Guidelines for Forensic Psychologists*, the term "forensic psychology" refers to "professional practice by any psychologist working within any sub-discipline of psychology (e.g., clinical, developmental, social, cognitive) when applying the scientific, technical, or specialized knowledge of psychology to the law to assist in addressing legal, contractual, and administrative matters" (p. 1).¹ Additionally, the APA Council of Representatives notes that the application of the term and the specialty guidelines provided by APA do not rely on the individual's typical area of practice, but instead rely on the services that the professional is providing. However, the APA also states that simply being involved in a court-related matter does not mean that one is practicing forensic psychology. For instance, when a psychologist testifies solely on the basis of the "provision of psychotherapy," this is not considered forensic practice (p. 1). A psychologist, in terms of the APA's definition, is engaging in the practice of forensic psychology when they are providing a psycholegal opinion in a court-related matter, regardless of whether their typical practice consists of forensic work.

The Rapidly Developing Field of Forensic Psychology

Although still relatively young, the field of forensic psychology has developed rapidly since the landmark 1962 ruling in *Jenkins v. United States* that facilitated the ability of psychologists to testify as expert witnesses in the courtroom. The origins of forensic psychology, however, can be dated back at least to the early 20th century, when psychological science began contributing to an understanding of the limitations of eyewitness testimony. The scope of psychology's role in the courtroom has expanded significantly since that time, and a number of court rulings have helped define and clarify what is considered admissible testimony from a psychological expert. Psychologists now provide expert testimony to the courts on a wide range of issues, including competency issues in both civil and criminal matters, issues concerning personal injury, and child custody issues. Furthermore, the growth of forensic psychology as a field has fueled the formation of professional societies, specialized journals, and advanced certifications. On the basis of these developments, forensic psychology has clearly emerged as a defined and well-established specialty practice area for psychologists.

Most of the development of forensic psychology as a field of practice occurred during the latter half of the 20th century. However, Hugo Münsterberg, a German immigrant to the United States

in the early 20th century, was the first person to promulgate research into forensic psychology in the United States, and to call for an increased role for psychologists within the legal system (Vaccaro & Hogan, 2004). Münsterberg was an applied psychologist with wide-ranging interests in the areas of memory, fatigue, social influence, and the effects of advertising. He also provided clinical treatment to individuals with mental illness, primarily through the mechanism of autosuggestion. Münsterberg's primary interest in forensic psychology was in the area of the reliability of eyewitness testimony, and his research demonstrated the malleability of memory, as well as individual differences in the perception of events. Münsterberg believed that individual experiences and biases played a role in recollection of events, and in 1908 he authored a book, *On the Witness Stand*, that called for increased research into the reliability of eyewitness testimony. The book also explored the phenomenon of false confessions, suggesting that people with strong needs for punishment or compliance were especially likely to admit to crimes they had not committed. Münsterberg also called for an increased use of applied psychological knowledge in the courtroom, and decried judges and lawyers for relying solely on "common sense." However, his book was of minimal immediate impact (Vaccaro & Hogan, 2004). Research on psychological aspects of the legal process continued to be limited and tended to have little effect on the justice system. Moreover, psychologists did not begin to serve widely as expert witnesses until the 1960s. Prior to that time, physicians, predominantly psychiatrists, typically provided expert testimony regarding mental disorders.

Psychologists began to be qualified as expert witnesses as the direct result of a United States Court of Appeals decision in the Washington, DC, Circuit Court in 1962. In *Jenkins v. United States*, the defense of an individual on trial for housebreaking, assault, and intent to commit rape had originally involved the testimony of three psychologists in support of an insanity defense. At the trial's end, the judge instructed the jury to disregard the psychologists' opinions, ruling that psychologists were not qualified to provide expert opinions regarding mental functioning and disease. When the case was appealed, the Circuit Court reversed this instruction, stating specifically that individuals with training and expertise in mental disorders, including psychologists, were qualified to provide expert testimony on questions including a defendant's mental condition. This decision provided an important basis for psychologists as expert witnesses, and since 1962 there has been an exponential rate of growth in the role of psychological science within the United States' justice system, resulting in psychologists providing expert opinions across a widely varying spectrum of legal matters.

At the time of the *Jenkins v. United States* decision, the operable federal standard for the admission of scientific knowledge as evidence during legal proceedings (as well as the standard in many states) was the *Frye* standard, determined in the case of *Frye v. United States* in 1923. The standard specified in that case was that testimony was admissible only if "the principle it is based on is sufficiently established to have a general acceptance to the field in which it belongs." This was a fairly restrictive standard that tended to limit the admission into evidence of new or emerging knowledge in a field of study, and efforts to revise or expand this standard occurred with increasing frequency in the 1980s and early 1990s. In the case of *Daubert v. Merrell Dow Pharmaceuticals* (1993), the U.S. Supreme Court stated that the Federal Rules of Evidence were applicable to scientific testimony and laid out several factors that could be used by the courts in making determinations about the admissibility of such testimony. These factors involved the extent to which:

- the theory or technique has been scientifically tested;
- the theory or technique has been subjected to peer review and publication;
- the technique has a known or quantified error rate;
- the theory or technique has acceptance in the relevant scientific community.

The identification of these factors was intended to create a more flexible test than the *Frye* standard afforded while still allowing the court to exclude testimony in those cases where the methodology

underlying the opinion did not meet basic criteria for sound science. The *Daubert* test is now the standard in federal courts, although individual states are not required to apply it. However, many states have begun to employ this standard, although some continue to rely upon either the *Frye* standard or other standards in making decisions about the admissibility of expert testimony. The *Daubert* criteria have been further defined by a number of subsequent decisions. These include *General Electric Company v. Joiner* (1997), which permits the court to consider whether an expert has extrapolated inappropriately from an accepted premise to an unfounded conclusion. In *Kumho Tire Co. v. Carmichael* (1999), the U.S. Supreme Court further expanded the *Daubert* factors beyond “scientific” expert testimony to include expert testimony based on “technical” or “other specialized” knowledge, and it also stated that the criteria specified in *Daubert* are not required or exclusive in qualifying experts for testimony. Although the *Daubert* decision expanded the criteria so as to allow expert opinions to rely on new or evolving science, the specific factors identified, in conjunction with the *Kumho Tire* and *Joiner* decisions, directly imply that expert opinions must be based on sound science, thereby requiring forensic experts to be very familiar with the scientific underpinnings of their methods and testimony (Medoff, 2003).

For forensic psychologists, the potential admissibility and usefulness of their conclusions are crucially related to the scientific basis of their testimony. Furthermore, the *Ethical Principles of Psychologists and Code of Conduct*, published by the American Psychological Association (APA, 2010), underscores the central role of empirical knowledge in formulating professional judgments, and the importance of interpreting psychological assessment results within the context of research on the instrument’s psychometric properties. The *Specialty Guidelines for Forensic Psychologists* (APA, 2011) similarly emphasize scientific knowledge in the selection and utilization of methods and procedures in forensic evaluations. In fact, the most recent revision of the specialty guidelines provides more detail regarding what is expected in the use of assessment measures in forensic contexts. For instance, the guidelines state that practitioners should strive to use assessment measures with established validity and reliability in the population being assessed (forensic). When this is not possible, it is advised that the examiner describe the strengths and limitations of their findings and seek to make clear what differences may be seen between a more traditional assessment context and a forensic context (APA, 2011). Given these ethical standards, and in order to maximize the likelihood that data and opinions will be both accurate and admissible in court, it is especially important that forensic psychologists choose and utilize test instruments in a manner that is consistent with available scientific evidence and the instruments’ intended use, and with an understanding of the strengths and limitations of the use of these assessment measures in a forensic context.

Currently, forensic psychologists provide expert knowledge to the courts on a wide variety of criminal issues, such as competency to stand trial, and criminal law constitutes a large area of forensic psychology practice. A number of other competencies are relevant to criminal questions as well, such as competency to waive Miranda rights, competency to waive counsel and act as one’s own attorney, and competency to be executed. Beyond these competency issues, forensic psychologists working in the criminal arena may also conduct evaluations to assess potential mitigating factors in capital sentencing, offer recommendations for disposition in juvenile delinquency cases, determine mental status or diminished capacity at the time of the offense, or provide risk assessments for sex offenders and other potentially violent offenders. The assessment of malingering is an additional significant concern in criminal forensic psychology, as well as in most other areas of forensic practice (e.g., personal injury litigation and workers’ compensation).

There are a number of civil questions that forensic psychologists may also address. These include competencies to release information, consent to medication treatment, refusal of hospitalization, and managing one’s financial affairs. Other civil matters may involve personal injury or workers’ compensation cases, in which a psychologist may address such questions as the extent and proximate

cause of psychological or neuropsychological deficits, including conditions like post-traumatic stress disorder. As in criminal forensic practice, personal injury and workers' compensation cases usually involve individuals who are submitting claims that may involve opportunities for external gain, and thus the assessment of the accuracy of an individual's self-report is often of critical importance. Instruments such as the Structured Interview of Reported Symptoms, Second Edition (SIRS-2; Rogers, Sewell, & Gillard, 2010) and the Test of Memory Malingering (TOMM; Tombaugh, 1996) are used frequently to discern the presence of over-reported or malingered symptoms, and the validity scales of the MMPI-2 are relevant to the detection of symptom under-reporting and over-reporting, as well as response patterns that are random or inconsistent.

Another major area of forensic practice is that of child custody and protection. The court may often turn to psychologists to help determine what is in the best interests of the child in cases where the parents' separation or divorce leads to custody disputes, or where there are parenting capacity issues resulting from allegations of child abuse or neglect. Grounds for termination of parental rights can include abuse, neglect, abandonment, or the presence of parental conditions that cause incapacity to care for a child (such as mental illness or substance abuse). Because of the enormous difficulty and consequences of decisions related to parental custody, courts often tend to weight expert opinion in child protection proceedings to a greater extent than in many other areas of law (Goodman, Emery, & Haugaard, 1998).

The areas listed above represent merely a sampling of the kinds of issues addressed by forensic psychologists. These and the many other issues (e.g., violence and sexual offense risk assessment, sexual harassment, jury selection) represent a very broad range of forensic applications of psychological science. The expansion of the field of forensic psychology has directly contributed to the formation of Division 41 of the American Psychological Association (the American Psychology-Law Society) in 1985 as well as the development of specialty guidelines for the practice of forensic psychology (APA, 2011). There are currently several academic journals focusing on research and practice issues relevant to forensic psychology; including, among others, the *American Journal of Forensic Psychology*, *Behavioral Sciences and the Law*, *Criminal Justice and Behavior*, the *Journal of the American Academy of Psychiatry and Law*, the *Journal of Forensic Psychology Practice*, *Law and Human Behavior*, and *Psychology, Public Policy, and Law*. The American Board of Forensic Psychology began awarding a diploma in forensic psychology in 1979, providing a way of obtaining advanced credentials in the field. In August 2001, the American Psychological Association's Council of Representatives voted to recognize forensic psychology as an officially designated specialty area (Otto & Heilbrun, 2002). All of these developments have occurred within the past 30 years, indicating the rapid rate of growth for forensic psychology.

The use of traditional clinical assessment instruments in forensic settings holds tremendous promise for assisting the psychologist in reaching useful conclusions and recommendations, as well as substantial potential for the misuse or misapplication of these tests. Melton, Petrila, Poythress, and Slobogin (2007) highlight the important differences between therapeutic assessment and forensic assessment, including:

- The scope of the evaluation: evaluations for the court may be centered on whether impairment of a specific type is present, but do not usually address treatment concerns.
- The client's perspective: the client's point of view is the primary concern in psychotherapy, while in forensic evaluations extensive collateral information is sought and the examinee's perspective tends to be secondary.
- The voluntariness of the subject's participation: most therapeutic clients embark on assessment voluntarily, while forensic examinees are often reluctant, extremely guarded, or overtly uncooperative.

- Threats to validity: forensic examinees (and the collateral sources required) may have a wide variety of reasons to be dishonest during the evaluation, including avoidance of criminal consequences or motivations related to the possibility of material gain. Deliberate or intentional dishonesty in clinical evaluations is rarer, and threats to validity are more likely to come from low self-awareness.
- Relationship dynamics: treating clinicians generally focus on demonstrating empathy and achieving an interpersonal bond with their clients, whereas forensic evaluators, whose primary responsibility is typically to the court and not the client, are usually more detached and at times even confrontational. The presentation of oneself as an empathic and supportive examiner is, in fact, considered misleading and unethical by some forensic evaluators.
- The pace and setting of the evaluation: in forensic evaluations, access to the subject may be limited, time may be very short because of the constraints of court orders and schedules, and the physical realities of conducting evaluations in prisons and hospitals may be quite challenging. Additionally, forensic work products are time-limited and not generally subject to change after being submitted to the court.

Forensic evaluations do not have a therapeutic objective but are typically performed solely to assist the trier of fact, and confidentiality is significantly different from assessments done in therapeutic contexts. There is an obvious implication, therefore, that the chosen assessment tools are selected to provide maximum benefit to the court and not necessarily to benefit the examinee. This forensic perspective represents a quite significant conceptual shift for clinicians new to this work.

Specialized Forensic Instruments versus Traditional Clinical Tests

Standardized assessment measures employed in forensic evaluations include instruments expressly designed to be used in a forensic setting, often for a specific population or purpose, and traditional clinical assessment tools applied to answer a forensic question or issue. One example of an instrument designed for use specifically in a forensic setting is the Massachusetts Youth Screening Instrument–Second Version (MAYSI–2; Grisso & Barnum, 2003). This test is a brief, easy-to-read symptom self-report measure that can be administered to youths entering detention or correctional programs. It is used to identify significant levels of a variety of psychiatric symptoms, and in particular to identify youths in psychological crisis who might require immediate intervention. While many of the MAYSI–2 items relate to a broad array of psychiatric symptoms, the test was designed specifically for use in detention and correctional settings, and norms are provided only for delinquent youths in these settings.

A series of tests developed for the assessment of competency to waive Miranda rights also provide an excellent illustration of instruments designed solely to address a specific forensic issue. These tests include the Comprehension of Miranda Rights (CMR), and related instruments, including the Comprehension of Miranda Rights–Recognition (CMR–R), the Comprehension of Miranda Rights–Vocabulary (CMR–V), and the Function of Rights in Interrogation (FRI) (Grisso, 1981, 1986). These four instruments are designed to assess the competencies of both adult and juvenile defendants to meaningfully waive their Miranda rights. The CMR uses a semi-structured interview format in which each of the four parts of the standard Miranda warning are presented to a subject with a request that he or she describe in their own words the meaning of each sentence. The CMR is also available in a true–false format that assesses the individual’s ability to recognize accurately (rather than verbally express) the meaning of the content of Miranda warnings (CMR–R), and a version that focuses on the vocabulary of the Miranda language (CMR–V) to help determine whether the individual has sufficient vocabulary knowledge to understand the Miranda warnings.

The FRI consists of pictures and vignettes that serve to assess a subject’s understanding of the functions of the subject’s right to remain silent and right to an attorney during interrogation. For example, one picture shows an individual seated at the defense table before a judge, and the subject is asked specific questions about critical aspects of the right to remain silent during court proceedings.

Otto and Heilbrun (2002) have discussed what they described as an “explosion” in the development and publication of forensic assessment instruments and forensically relevant instruments. Forensically relevant instruments can be distinguished from forensic assessment instruments in that the former do not focus on specific legal standards (or specific functional capacities associated with the standard), but rather address clinical constructs that are often directly pertinent to evaluations conducted within the legal system (e.g., evaluations of malingering, impulsivity, and psychopathy). Otto and Heilbrun noted that more forensic assessment instruments and forensically relevant instruments had been published in the immediately preceding decade than in the 40 years prior to that, and placed the current minimum number of such instruments at between 40 and 50 in 2002. This number has undoubtedly continued to increase since then.

Instruments designed specifically for forensic use have both advantages and disadvantages (see Table 1.1 for a summary). One obvious advantage is that they allow for a more precise focus on a legal issue or standard. For example, in the case of the CMR, the test quickly (within 10–15 minutes) provides a circumscribed evaluation of whether an individual is capable of understanding his or her Miranda rights, a salient legal concern. Relatively less extrapolation or inference beyond the subject’s direct responses may be necessary, because the test itself employs the exact language used in the Miranda warnings. Such tests often have the additional advantage of providing normative data based on forensic populations, allowing individual results comparable with overall results for the specific population of interest.

Many specialized forensic tests, however, possess the disadvantage of having a very limited research literature, and thus of having reliability and validity characteristics that are not fully understood. Indeed, Otto and Heilbrun (2002) observed that a number of forensic instruments have been published and distributed following only preliminary research, sometimes in the absence of a comprehensive test manual. Some instruments designed to assess competency to stand trial, for example, have generally demonstrated inconsistent research results regarding their factor structure and, more broadly, construct validity (Rogers, Jackson, Sewell, Tillbrook, & Martin,

TABLE 1.1 Advantages and Disadvantages of Specialized Forensic Assessment Instruments

<i>Advantages</i>	<i>Disadvantages</i>
Specialized tests may offer a specific focus on the legal issues at hand.	Specialized tests often have a limited research literature and have been the subject of minimal independent research.
Specialized tests typically require less reliance on extrapolation and inference to address a given legal issue.	The applicability of specialized tests may be limited by their specificity (e.g., state-specific language for Miranda warnings).
Specialized tests often provide normative data based on forensic populations, allowing for direct comparison of individual findings to the population of interest.	There are numerous specific legal issues for which there are no specialized forensic assessment measures.
	Specialized tests generally do not provide a broad understanding of overall psychological functioning.

2003). In addition, there is often little independent research on forensic assessment instruments beyond the group of researchers directly involved in their development. Another potential disadvantage is that tests focused on specific forensic issues offer, by design, a very restricted assessment range, providing information on only one area of interest and thus possibly requiring the use of one or more additional instruments to address other important issues. The major benefit of the specialized tests, therefore, simultaneously acts as a significant limitation. To illustrate this issue, the specific wording of Miranda rights varies across states, thus potentially limiting the generalizability of the CMR to only those states that employ the specific language and vocabulary used in the test. There are also many specific legal issues, such as competency to manage one's financial affairs and competency to consent to treatment, for which no specialized assessment instruments have been developed. Finally, because specialized forensic assessment instruments typically focus strictly on legal standards, they do not usually provide sufficient data to permit a broader and contextual understanding of the individual's overall psychological functioning. Many forensic psychologists frequently use traditional clinical assessment instruments to address this latter assessment issue.

The Role of Traditional Clinical Tests in Forensic Psychology

Despite the existence of a rapidly growing body of specialized forensic instruments, traditional clinical assessment tools continue to play a crucial role in forensic evaluations. Research surveys report that instruments such as the Wechsler Intelligence Scales (e.g. Wechsler, 2004), the Minnesota Multiphasic Personality Inventory–2 (MMPI–2; Butcher, Dahlstrom, Graham, Tellegen, & Kaemmer, 2001), and the Millon Clinical Multiaxial Inventory–Third Edition (MCMI–III; Millon, Davis, & Millon, 1997) are commonly used in forensic assessment, and a growing body of research literature is providing guidance and support for using clinical instruments in a forensic context. Although some cautions have been raised regarding the use of traditional clinical measures in forensic assessment (e.g., Melton et al., 2007), such tests have become widely accepted for use in forensic evaluations (e.g., Archer, Buffington-Vollum, Stredny, & Handel, 2006; Borum & Grisso, 1995; Gould & Stahl, 2000; Greenberg, Otto, & Long, 2003; Otto, Edens, & Barcus, 2000). There are some distinct challenges and difficulties, however, associated with “retrofitting” clinical instruments for addressing a forensic issue or standard. Clinicians who routinely perform forensic evaluations must be aware of these limitations and the appropriate uses of tests as identified by the research literature.

In reviewing the literature on the use of clinical assessment instruments in forensic settings, we will organize studies into two topic areas, with individual studies presented chronologically within these broad categories. We will first review studies of test usage based on surveys of neuropsychologists, including surveys regarding tests of malingering (an area often associated with neuropsychological assessment). We will then turn to a more general category of studies that have surveyed clinicians in terms of their test use in specific or general forensic settings.

Surveys of Neuropsychological Tests (Including Studies of Malingering)

In a survey of forensic neuropsychological evaluations (Lees-Haley, Smith, Williams, & Dunn, 1996), the WAIS–R and MMPI or MMPI–2 have been found to be the two most popular tests, followed by the Wechsler Memory Scale (Wechsler, 1987), Trail Making Tests A and B (Army Individual Test Battery, 1944), the Finger Oscillation Test (Halstead, 1947), the Bender-Gestalt Visual Motor Test (Bender, 1946), the Category Test (Halstead, 1947), and the Wisconsin Card Sorting Test (Berg, 1948). Table 1.2 lists the most popular neuropsychological instruments in forensic evaluations as reported by Lees-Haley and his colleagues.

TABLE 1.2 Instruments Used in 20% or More of Forensic Neuropsychological Evaluations

Rank	Test
1	WAIS and WAIS-R (76%)
2	MMPI and MMPI-2 (68%)
3	WMS and WMS-R (51%)
4	Trails A (48%)
5	Trails B (47%)
6	Finger Oscillation Test, Finger Tapping Test (38%)
7	Bender Gestalt (34%)
8	Category Test—all forms
9	Wisconsin Card Sorting Test (29%)
10	Sentence Completion Test—various forms
11	Tactual Performance Test (26%)
12	WRAT and WRAT-R (25%)
13	Beck Depression Inventory (25%)
14	Reitan-Klove Sensory Perceptual Examination (23%)
15	Grooved Pegboard (23%)
16	Seashore Rhythm (23%)
17	Speech Sounds Perception Test (21%)
18	Boston Naming Test, Boston Naming Test—Revised (21%)
19	Hooper Visual Organization Test (20%)

Source: From “Forensic Neuropsychological Test Usage: An Empirical Survey,” by P. R. Lees-Haley, H. H. Smith, C. W. Williams, and J. T. Dunn, 1996, *Archives of Clinical Neuropsychology*, 11, p. 48. Copyright 1996 by Elsevier. Adapted with permission from Elsevier.

Regarding the assessment of malingering, Slick, Tan, Strauss, and Hultsch (2004) surveyed forensic neuropsychologists and found that 79% of respondents used specialized forensic tests to assess malingering, with 50% reporting that they routinely gave such tests at the outset of each evaluation. The most commonly reported tests in this survey were the Rey 15-Item Test (Rey, 1964) and the Test of Memory Malingering (TOMM; Tombaugh, 1996).

In 2006, Boccaccini, Boothby, and Overduin conducted a survey of 350 members of the American Pain Society who identified their discipline as psychology, and 302 members of both APA Division 2 and Division 41. A total of 116 professionals responded. The participants were provided with a short vignette about a hypothetical case and then asked which measures they would use to assess malingering or exaggerating in a forensic assessment. In the study, the most commonly reported measure was the MMPI-2, followed by the SIRS (Structured Interview of Reported Symptoms), the VIP (Validity Indicator Profile; Frederick, 1997) and the TOMM (Test of Memory Malingering). The results reported by Boccaccini et al. (2006) regarding the dominance of the SIRS, TOMM, and VIP as measures of malingering are similar to the findings reported by Archer et al. (2006) in the section of their survey dealing with tests used by forensic psychologists to assess malingering. Results regarding the prevalence of use of the last three tests differed depending on whether the respondent self-identified as a pain specialist, a forensic specialist, or a forensic pain specialist.

Surveys in Specific or General Forensic Settings

A study by Borum and Grisso (1995) demonstrates that traditional psychological testing is widely used among forensic psychologists and psychiatrists in addressing a broad variety of forensic issues.

These authors conducted a survey of highly experienced forensic psychologists and psychiatrists, predominantly board-certified for forensic practice, regarding their usage of psychological tests in evaluations for competency to stand trial and criminal responsibility. Among forensic psychologists, the use of psychological testing was rated as either essential or recommended by 68% of evaluators in criminal responsibility evaluations, and by 51% in their evaluations of competency to stand trial. In criminal responsibility evaluations, the most commonly used type of test was an objective personality instrument, with the MMPI or MMPI-2 reported as being commonly used by 94% of these survey respondents. Next most commonly used were measures of intellectual functioning (mentioned by 80% of respondents), neuropsychological tests (50%), and projective tests (42%). In contrast, forensic instruments were reported as being used less frequently than clinical instruments, with 46% of forensic psychologists reporting that they never used such tests in criminal responsibility evaluations, and 20% reporting that they rarely used them. Regarding test usage in assessments of competency to stand trial, similar results were found overall. Psychological testing was rated as essential or recommended by 51% of respondents, and objective personality inventories or intellectual testing were mentioned by 90% of the evaluators in this type of assessment. Neuropsychological tests were also frequently mentioned (42%), and projective tasks somewhat less frequently (33%). Forensic psychiatrists generally rated psychological testing as essential or recommended for assessing criminal responsibility and competency to stand trial at rates similar to forensic psychologists, although psychiatrists reported less frequent actual utilization of such testing.

When completing child custody evaluations, intelligence tests, projective tests, and broad personality measures are most frequently used by evaluators. Ackerman and Ackerman (1997), for example, found that most custody evaluators (58%) reported using some type of intelligence test when assessing children. A projective test, typically either the Children's Apperception Test (Bellak & Bellak, 1949) or the Thematic Apperception Test (Murray, 1943), was identified as being the next most popular (37%). Results from the same survey also showed that when assessing adults in custody evaluations, 92% of survey respondents used either the MMPI or MMPI-2, 48% used the Rorschach (the scoring system used was not specified), and 43% used the Wechsler Adult Intelligence Scale-Revised (WAIS-R; Wechsler, 1981). A similar pattern of test usage in child custody evaluations has been noted by Hagen and Castagna (2001), who found that the MMPI-2 and Rorschach remained the most popular tools for assessing adults (used in 84% and 31% of evaluations, respectively), while either intelligence tests (26%) or the Bricklin Perceptual Scales (Bricklin, 1984) (26%) were most popular in assessing children.

In a survey specifically focused on personal injury evaluations related to emotional damages (Boccacini & Brodsky, 1999), forensic psychologists reported most commonly using the MMPI or MMPI-2 (94%), the WAIS-R or WAIS-III (54%), and the MCMI-II or MCMI-III (50%), with the Rorschach and the Beck Depression Inventory (Beck, Ward, Mendelson, Mock, & Erbaugh, 1961) being used somewhat less often (41% and 33%, respectively).

In evaluations of juveniles' competency to stand trial, Ryba, Cooper, and Zapf (2003) reported that 82% of clinicians report using the Wechsler Intelligence Scale for Children-Third Edition (WISC-III; Wechsler, 1991), and 56% report using either the MMPI-2 (presumably used with 18-year-olds) or the Minnesota Multiphasic Personality Inventory-Adolescent (MMPI-A; Butcher et al., 1992). Although evaluators in this survey reported occasional use of specific forensic assessment instruments, general clinical instruments were used much more frequently.

Surveys of clinicians who perform forensic evaluations on a regular basis indicate that a number of traditional clinical assessment tools are commonly employed. In a survey of psychologists who had been awarded the diploma in forensic psychology by the American Board of Forensic Psychology (Table 1.3), Lally (2003) found that the Wechsler Adult Intelligence Scale-Third Edition (WAIS-III; Wechsler, 1997) and the MMPI-2 were rated as "Recommended" for sanity

evaluations by 60% and 54% of experts, respectively. Additional instruments considered “Acceptable” for such evaluations included the Halstead–Reitan Neuropsychological Battery (Reitan, 1979) (71%), the Personality Assessment Inventory (PAI) (Morey, 1991) (69%), the Luria–Nebraska Neuropsychological Battery (Golden, Hammeke, & Purisch, 1978) (58%), the MCMI–III (54%), and the Stanford–Binet–Revised (Thorndike, Hagen, & Sattler, 1986) (52%). In conducting risk for violence assessments, 63% of the clinicians in the survey rated the Psychopathy Checklist–Revised (PCL–R; Hare, 1991) as “Recommended.” Instruments additionally named as “Acceptable” for such evaluations included the MMPI–2 (82%), the Psychopathy Checklist–Screening Version (PCL:SV; Hart, Cox, & Hare, 1995) (73%), the Violence Risk Appraisal Guide (VRAG; Harris, Rice, & Quinsey, 1993) (73%), the WAIS–III (67%), and the PAI (61%). The latter survey covered several other types of evaluations, including risk for sexual violence, competency to stand trial, competency to waive Miranda rights, and malingering.

In 2006, Bow, Gould, Flens and Greenhut surveyed 89 psychologists regarding test selection, usage, and whether they felt the tests met the *Daubert* admissibility standard for use in child custody evaluations. They found that the most commonly used assessment measure was the MMPI–2, with 90.6% of respondents saying that they typically used the instrument. The second most commonly used instrument was the MCMI, with 58% reporting that they typically used the measure. Additionally, 38.8% reported typical use of the MMPI–A, 32% the Conners’ Parent Rating Scale (Conners, 2008), 27% the Parenting Stress Index (Abidin, 1995), and 23% the Child Behavior Checklist (Achenbach, 1991).

Archer et al. (2006) surveyed 152 doctoral-level psychologists who were members of Division 41 (the American Psychology–Law Society) of the American Psychological Association and/or diplomates of the American Academy of Forensic Psychology. A large majority of the respondents identified themselves as forensic psychologists who typically spent over 50% of their time in forensic practice. Survey respondents were asked to report their test usage in relationship to specific categories such as custody evaluations, competency to stand trial, and sex offender risk assessment. Survey results were also gathered for specific types of testing such as multiscale inventories, cognitive/intellectual tests, neuropsychological tests, and single scale tests. Overall, results indicated that multiscale inventories, particularly the MMPI–2, were utilized by the largest proportion of respondents. The Wechsler Intelligence Scales were dominant in the assessment of cognitive, intellectual ability, and the MMPI–A emerged as the most widely used self-report instrument in evaluating children’s psychological functioning in forensic evaluations. The Beck Depression Inventory (Beck et al., 1961) and the Beck Anxiety Inventory (Beck & Steer, 1993) were the most widely used single scales in the forensic evaluation of adults, and the Rorschach (Exner, 1995) was the most widely used unstructured test. In general, the survey demonstrated the continued and marked popularity of traditional assessment instruments in forensic evaluations.

Viljoen, McLachlan, and Vincent (2010) conducted a survey of psychologists involved in juvenile and adult violence risk and psychopathy evaluations. Their results indicated that 75.3% of the 215 psychologists (85 who completed the survey related to child assessments and 130 who completed the survey related to adult assessments) reported using the WISC–IV (Wechsler, 2004), WAIS–III, or WASI (Wechsler, 1999) in their practice. Further, 28% reported that they always or almost always used one of these cognitive tests. Sixty-six percent reported use of the MMPI–2 or MMPI–A (Butcher et al., 1992), 35.1% used the SAVRY (Borum, Bartel, & Forth, 2003), 31.2% used the MACI (Millon, Millon, & Davis, 1993) or the MCMI–III (Millon, 1994), 26% the PCL:YV (Forth, Kosson, & Hare, 2003) or the PCL–R (Hare, 1991, 2003), 24.7% used the WRAT–3 or WRAT–4, 19.5% used the Rorschach, and 14.3 the Child Behavior Checklist (CBCL; Achenbach, 1991) or Youth Self-Report (YSR; Achenbach, 1991; Achenbach & Rescorla, 2001). Remaining measures were reported to be used in less than 15% of cases. This

TABLE 1.3 Tests Rated by Diplomates in Forensic Psychology as Recommended, Acceptable, or Unacceptable for Forensic Evaluations

<i>Type of Evaluation and Tests</i>	<i>Recommended</i>	<i>Acceptable</i>	<i>Unacceptable</i>
Mental status at the time of offense			
WAIS-III	60%	96%	—
MMPI-2	54%	94%	—
R-CRAS	—	94%	—
Halstead-Reitan	—	71%	—
PAI	—	69%	—
Luria-Nebraska	—	58%	—
MCMI-III	—	54%	—
S-B-R	—	52%	—
Projective drawings	—	—	81%
TAT	—	—	65%
Sentence completion	—	—	60%
Risk for violence			
PCL-R	63%	88%	—
MMPI-2	—	88%	—
PCL-SV	—	73%	—
VRAG	—	73%	—
WAIS-III	—	67%	—
PAI	—	61%	—
Projective drawings	—	—	90%
TAT	—	—	82%
Sentence completion	—	—	71%
Rorschach	—	—	53%
16PF	—	—	53%
Risk for sexual violence			
PCL-R	62%	91%	—
MMPI-2	—	81%	—
PCL-SV	—	71%	—
WAIS-III	—	71%	—
VRAG	—	67%	—
SORAG	—	62%	—
Plethysmograph	—	60%	—
SVR-20	—	57%	—
PAI	—	55%	—
Projective drawings	—	—	95%
Sentence completion	—	—	76%
TAT	—	—	76%
16PF	—	—	60%
Rorschach	—	—	52%
MCMI-III	—	—	50%
Competency to stand trial			
WAIS-III	62%	90%	—
MacCAT-CA	56%	90%	—
CAI	—	85%	—
CST	—	77%	—
MMPI-2	—	73%	—
GCCT	—	65%	—
Halstead-Reitan	—	64%	—
IFI-R	—	62%	—

TABLE 1.3 Continued

<i>Type of Evaluation and Tests</i>	<i>Recommended</i>	<i>Acceptable</i>	<i>Unacceptable</i>
S-B-R	—	54%	—
PAI	—	52%	—
Luria-Nebraska	—	50%	—
Projective drawing	—	—	87%
TAT	—	—	77%
Sentence completion	—	—	60%
Rorschach	—	—	60%
16PF	—	—	58%
MCMI-III	—	—	50%
Competency to waive Miranda rights			
WAIS-III	83%	100%	—
Grisso instruments	55%	88%	—
MMPI-2	—	70%	—
Halstead-Reitan	—	68%	—
S-B-R	—	63%	—
Luria-Nebraska	—	55%	—
Projective drawings	—	—	95%
TAT	—	—	78%
Sentence completion	—	—	70%
Rorschach	—	—	63%
16PF	—	—	63%
MCMI-III	—	—	55%
Malingering			
MMPI-2	64%	92%	—
SIRS	58%	89%	—
WAIS-III	—	75%	—
Rey	—	68%	—
PAI	—	53%	—
VIP	—	53%	—
TOMM	—	64%	—
Halstead-Reitan	—	51%	—
Projective drawings	—	—	89%
Sentence completion	—	—	72%
TAT	—	—	72%
16PF	—	—	66%
Rorschach	—	—	55%

Source: Adapted from "What Tests Are Acceptable for Use in Forensic Evaluations? A Survey of Experts," by S. J. Lally, 2003, *Professional Psychology: Research and Practice*, 34, pp. 491–498. Copyright 2003 by the American Psychological Association. Adapted with permission of the American Psychological Association.

Note:

Percentages across columns may total in excess of 100% because respondents could rate tests as "Recommended" and "Acceptable." Results for tests reported by Lally as producing only "No Opinion" ratings are omitted from this table. WAIS-III = Wechsler Adult Intelligence Scale-Third Edition; MMPI-2 = Minnesota Multiphasic Personality Inventory Scale-2; R-CRAS = Rogers Criminal Responsibility Assessment Scales; PAI = Personality Assessment Inventory; MCMI-III = Millon Clinical Multiaxial Inventory-Third Edition; S-B-R = Stanford-Binet-Revised; TAT = Thematic Apperception Test; PCL-R = Psychopathy Checklist Revised; PCL-SV = Psychopathy Checklist Short Version; VRAG = Violence Risk Appraisal Guide; 16PF = 16 Personality Factors; SORAG = Sex Offender Risk Appraisal Guide; SVR-20 = Sexual Violence Risk-20; MacCAT-CA = MacArthur Competency Assessment Tool-Criminal Adjudication; CAI = Competency to Stand Trial Assessment Instrument; CST = Competency Screening Test; GCCT = Georgia Court Competency Test; IFI-R = Interdisciplinary Fitness Interview-Revised; SIRS = Structured Interview of Reported Symptoms; Rey = Rey 15-Item Visual Memory Test; VIP = Validity Indicator Profile; TOMM = Test of Memory Malingering.

survey, consistent with other survey findings, indicates that traditional clinical assessment tools are prominently named as the instruments most frequently recommended for use in nearly all forms of forensic evaluations, while more specialized forensic instruments were identified more rarely and for much more restrictive uses.

Wygant and Austin (in preparation) utilized a web-based survey of clinicians across a variety of specialties, including forensic psychology, to assess clinicians' recent use (prior two years) of adult personality and psychopathology measures. The survey was distributed online to several groups that included large numbers of psychologists whose practice includes forensic cases, including members of the American Psychology-Law Society (APLS), the Society for Personality Assessment (SPA), and an email listserv (Psychology-Law Listserv). Four hundred and seventy-nine respondents completed the survey, of whom 284 reported practicing in one (or more) of the following six forensic contexts: Competency/Criminal Responsibility, Criminal Risk Assessment, Medico-Legal, Medico-Legal Neuropsychological, Parental Fitness, and Child Custody. The format of the survey was similar to that used by Archer et al. (2006) and allowed for calculation of a weighted score as an indication of the test's popularity in clinical practice. Across all forensic settings, the MMPI-2 emerged as the most popular psychological test and was utilized sometimes (at least 25% of the time in practice) or more often by 59% of the respondents. Rounding out the most popular psychological tests across all forensic settings (in order) along with the percentage of respondents who used the test at least sometimes or more often in clinical practice were:

2. the Personality Assessment Inventory (38%);
3. the MMPI-2-Restructured Form (29%);
4. the Millon Clinical Multiaxial Inventory-III (26%);
5. the Rorschach Ink Blot Test-Comprehensive System (22%);
6. the Psychopathy Checklist-Revised (20%);
7. the Beck Depression Inventory-II (20%);
8. the Beck Anxiety Inventory (13%);
9. the Trauma Symptom Inventory (10%).

Table 1.4 provides a summary of testing reported in surveys of evaluations conducted for child custody, competency to stand trial, determinations of criminal responsibility, and personal injury.

In addition to the consistent evidence of the marked popularity of clinical instruments in forensic evaluations, there is a growing body of research literature that supports the forensic use of these instruments. Some of the most popular instruments, as indicated by survey research, have also garnered the most extensive research regarding their use in forensic evaluations. For example, as summarized by Pope, Butcher, and Seelen (2006), there is considerable research on the forensic applications of the MMPI-2 and MMPI-A. Reviews by Baum, Archer, Forbey and Handel (2009) and by Pennuto and Archer (2008) provide more data on the applications and validity of the MMPI-A as used in forensic settings, the former reporting external validity data from a variety of studies. Neuropsychological tests commonly used in forensic evaluations have also gained substantial research attention (e.g., Horton & Hartlage, 2003; Sweet, 1999; Sweet & King, 2002). Further, McCann and Dyer (1996) have summarized the research literature on forensic uses of the MCMI-III, Blood (2008) provided a summary of the uses of the MCMI-III in parenting capacity assessments, and there are several reviews available on the use of the Rorschach in forensic evaluation (e.g., McCann, 1998; Weiner, 2003). Other tests featured in this book have generated less substantive forensic research bases, but appear quite likely to do so in the coming years. Therefore, an ample research base is rapidly developing to provide the psychologist with critical information concerning the strengths and limitations of several clinical assessment instruments for various forensic applications.

TABLE 1.4 Percentage of Practitioners Reporting Use of Popular Clinical Assessment Instruments in Forensic Evaluations

Study	Type of Evaluation	MMPI, MMPI-2 or MMPI-A		Wechsler Intellectual Assessment Instruments		Millon Assessment Instruments		Rorschach		Beck Depression Inventory		Non-Wechsler Intellectual Assessment Instruments		TAT, Projective Drawings, and Other Projective Tasks	
		MMPI, MMPI-2 or MMPI-A		Wechsler Intellectual Assessment Instruments		Millon Assessment Instruments		Rorschach		Beck Depression Inventory		Non-Wechsler Intellectual Assessment Instruments		TAT, Projective Drawings, and Other Projective Tasks	
		MMPI, MMPI-2 or MMPI-A		Wechsler Intellectual Assessment Instruments		Millon Assessment Instruments		Rorschach		Beck Depression Inventory		Non-Wechsler Intellectual Assessment Instruments		TAT, Projective Drawings, and Other Projective Tasks	
Ackerman and Ackerman (1997)	CC	92%		43%		—		48%		—		58%		37%	
Hagen and Castagna (2001)	CC	84%		—		—		31%		—		26%		—	
Ryba, Cooper, and Zapf (2003)	CST (juvenile)	56%		82%		—		—		—		—		—	
Borum and Grisso (1995)	CST	94%		78%		32%		32%		2%		—		—	
Borum and Grisso (1995)	CR	86%		83%		18%		30%		4%		—		—	
Boccaccini and Brodsky (1999)	PI	94%		54%		50%		41%		33%		—		—	
Lees-Haley, Smith, Williams, and Dunn (1996)	PI	68%		76%		9%		14%		25%		—		6%	
Viljoen, McLachlan, and Vincent (2010)	RA (juvenile)	66.2%		75.3%		31.2%		19.5%		—		—		—	
Bow, Gould, Flens, and Greenhut (2006)	CC (Adult)	90.6%		5.0%		58.0%		—		—		—		—	
Bow, Gould, Flens, and Greenhut (2006)	CC (juvenile)	38.8%		11.0%		19.7%		—		—		—		—	

Note:

CC = child custody; CR = criminal responsibility; CST = competence to stand trial, PI = personal injury, RA = risk assessment. Data from Borum and Grisso (1995) are presented separately for two types of evaluations, but these data are from the same study. Data are presented by Bow et al. (2006) for projective, the BDI, and non-Wechsler assessments but cannot be broken into frequencies for adults versus juveniles; results are presented separately for adults and juveniles but data are from the same study.

However, there are also inherent difficulties associated with using tests developed for clinical purposes to address forensic issues that involve clinical constructs. One concern is that the original development of these instruments did not usually focus on forensic applications, but on diagnostic and treatment planning issues. The MMPI–2 basic scales, for example, are named for diagnostic categories, not for legal issues or standards (e.g., there is no “competency to stand trial” scale on the MMPI–2). Podboy and Kastl (1993) have pointed out that the potential forensic applications of traditional psychological tests were generally never considered during their development, and such tests have been subjected to notable misuse in the courtroom setting. To illustrate this latter point, Podboy and Kastl observe that attorneys have employed such tactics as presenting highly selective quotations from computerized interpretations, and drawing overly broad conclusions based on a single, narrowly focused instrument designed to assess traits such as anxiety or depression. While there have been some attempts to devise new scales for traditional clinical instruments that would be relevant to forensic evaluation, such as the Malingered Depression scale for the MMPI–2 (Steffan, Clopton, & Morgan, 2003), these efforts have generally been limited, and the effectiveness of these measures is not yet established by systematic research findings.

Another concern regarding the use of clinical instruments for forensic purposes is the observation that the research base for forensic applications varies quite widely depending on the individual instrument. As has already been pointed out, many of the most popular clinical tests used in forensic evaluations have extensive research to support their use in this context. However, other tests that evaluators may be accustomed to using in their clinical practices may have more limited research support for their use in forensic evaluations. Some clinical tests that are employed in forensic settings may lack research-based correlates for forensic use, and few clinical instruments have appropriate forensic norms. One rare exception to the former limitation is the Megargee classification system for the MMPI and MMPI–2, which does provide forensic correlates for these instruments, as well as a profile classification system designed specifically for adult offender populations (e.g., Megargee, 1994; Megargee & Carbonell, 1985). See Table 1.5 for a summary of the uses and limitations of clinical assessment instruments in forensic applications.

Clinical assessment tools may be employed for forensic purposes when they are appropriate and relevant to the specific legal question, but evaluators must maintain an awareness of how their methods will be received by the trier of fact (Medoff, 2003). Similarly, clinical assessment tools should not be utilized to attempt to address forensic questions beyond the scope of the instrument.

TABLE 1.5 Uses and Limitations of Clinical Assessment Instruments in Forensic Applications

<i>Uses</i>	<i>Limitations</i>
Traditional clinical tests can provide a broad understanding of an individual’s intellectual, emotional, and personality functioning	The original development of most traditional clinical tests focused on diagnoses and treatment planning, not on forensic applications
Instruments such as the MMPI–2 contain validity indicators that are an important source for evaluating defensiveness and/or malingering	Few clinical assessment instruments have developed appropriate forensic norms
The broader scope of these instruments helps to generate hypotheses for further evaluation	The research base for forensic applications varies widely by individual instrument
Many traditional tests have extensive research bases and well-established psychometric properties, which can enhance the conclusions in court testimony	Results from traditional tests often cannot be used to directly address specific legal issues

For example, forensic evaluators are sometimes asked to reconstruct an individual's mental state at a prior point in time; however, traditional clinical tests are typically of very limited value for such reconstructions (Melton et al., 2007). Moreover, Greenberg et al. (2003) suggest that because standard clinical assessment instruments may be limited in their ability to directly answer legal questions, such tests should instead be used only to generate hypotheses to be addressed in further evaluation. Rogers (2003) also cautions that when using multiscale inventories such as the MMPI-2 in forensic evaluations, clinicians should be especially wary of exercising a "confirmatory bias" through highly selective interpretation of test results. To illustrate this latter point, the MMPI-2 contains 567 items and in excess of 100 scales and subscales; it is quite likely that some isolated result can be found to support almost any point or contention regarding an individual's psychological functioning. Fortunately, knowledgeable interpreters understand, and can explain to the trier of fact, that MMPI-2 results are interpreted not in isolation but within the context of overall test and evaluation findings.

Formal psychological assessment may provide an important additional source of information, in combination with interviews and collateral data, which sets psychological evaluations apart from psychiatric evaluations (Heilbrun, Marczyk, & DeMatteo, 2002). Instruments designed to assess specific legal issues, such as various competencies, serve a valuable function in many types of evaluations. Nonetheless, many forensic evaluations call for a broad understanding of personality, emotional functioning, and cognitive functioning that may best be addressed through the use of existing, well-validated measures that illuminate multiple aspects of individual functioning.

Clinical tests must be used wisely and appropriately in the forensic setting, and it is critically important that forensic psychologists recognize the appropriate and inappropriate application of clinical instruments in forensic evaluations. Applications such as child custody, personal injury litigation, and sentencing recommendations in capital punishment cases, for example, represent highly sensitive areas in which the opinion of a psychological expert can have enormous, and life-altering, consequences. Psychologists evaluating competencies to stand trial or waive Miranda rights, for example, are charged with tasks that contribute to upholding the integrity of judicial proceedings, and assuring that individual rights are protected. Providing an expert opinion in legal proceedings comes with a weighty responsibility to serve the court, and the public good, to the greatest degree possible. Therefore, it is incumbent upon clinicians who perform forensic evaluations to use psychological tests appropriately, and for purposes for which their use is empirically supported. If psychological testing were to be regularly misapplied in the forensic arena, this utilization would have the potential to undermine the integrity of the justice system, harm the individuals who rely upon it, and diminish psychological science in the eyes of the public. Furthermore, the work of forensic psychologists is often subject to intense scrutiny and cross-examination in the courtroom, and evaluators must frequently justify their choice of methods (as well as test findings and conclusions) in much greater detail than is generally the case in a clinical evaluation (Heilbrun et al., 2003). In general, the most persuasive basis for the successful defense of one's assessment procedures is to be familiar with the research on the measures that were used in the evaluation (Hilsenroth & Stricker, 2004). Given the potential benefit of using psychological tests in forensic evaluations, as well as the potential pitfalls of their improper use, knowledge of the appropriate and inappropriate forensic uses of traditional clinical assessment instruments is vital for forensic psychologists.

The purpose of the current edited text is to provide a firm basis for psychologists to understand the appropriate uses and limitations of popular clinical assessment measures when they are used to assess clinical constructs relevant to forensic issues. The instruments were selected because of their wide utilization and importance in both clinical and forensic settings. The first six instruments in the book are broad-spectrum, multidimensional personality instruments that are used in a variety

of forensic evaluations and settings. The book includes chapters on the MMPI-2 (authored by Martin Sellbom and Jaime L. Anderson); the MMPI-2-RF (authored by Yossef S. Ben-Porath; new to this edition of this book); the MMPI-A (by Robert P. Archer and Richard W. Handel); the PAI (by Leslie C. Morey and Justin K. Meyer); the MCMI-III (by Robert J. Craig); and the Rorschach (authored by Irving B. Weiner). The later instrument is the most frequently used unstructured and/or projective instrument in forensic evaluations.

In the next grouping of two chapters, measures of psychopathy are presented. These included the PCL-R, reviewed by Robert D. Hare, Pamela J. Black, and Zach Walsh, and the PCL-YV, presented by Angela S. Book, Adelle E. Forth, and Heather J. Clark. These measures of psychopathy in adults and adolescents are particularly relevant to assessments that are typically conducted in forensic settings. The separation between presentations of the adult and youth versions of the PCL into two distinct chapters represents a change from the first edition of the book, a difference deemed necessary because additional literature has been published on both measures. The following chapter, by Eric A. Zillmer and Christopher M. King, is on neuropsychological assessment instruments—a diverse group of clinical tests that are frequently associated with personal injury evaluation but are also used in criminal forensic evaluations.

The final two chapters of the revised text address tests often used in child custody, child protection, and juvenile forensic evaluations. These include the Achenbach System of Empirically Based Assessment by Thomas M. Achenbach and Leslie A. Rescorla, and the Parenting Stress Index by Richard R. Abidin, William G. Austin, and James R. Flens.

The chapter authors were selected because of their unique expertise and authority regarding the chosen instruments, resulting from their role in the development of the test and/or their crucial contributions to its research base. The authors were asked to present and discuss the strengths and weaknesses of the clinical assessment instrument in the forensic domain and to discuss appropriate and inappropriate uses of the instrument in this context. Additionally, they were asked to update the chapters with any research literature published on the use of the instruments in forensic contexts since the publication of the first edition of this book in 2006. The overarching goal of each chapter remains the same: to objectively describe the application of the instrument in the most responsible and appropriate manner for forensic settings, creating a comprehensive overview, within a single text, of the prominent clinical assessment instruments widely used in forensic populations. Like the first edition of this text, the second edition is not intended to provide sufficient training to allow a clinical psychologist to develop the expertise necessary to present themselves as a forensic psychologist; rather, it is designed to facilitate the optimal use of clinical assessment instruments among psychologists who have undertaken the training necessary to understand and apply psychological principles and test findings to salient legal standards or issues.

Note

- 1 Currently the most recent version of the APA *Specialty Guidelines for Forensic Psychologists* is available in final form online. The PDF is available at <http://www.apa.org/practice/guidelines/forensic-psychology.pdf>. The Guidelines will appear in a future edition of *American Psychologist*.

References

- Abidin, R. R. (1995). *Manual for the Parenting Stress Index*. Charlottesville, VA: Pediatric Psychology Press.
- Achenbach, T. M. (1991). *Manual for the Child Behavior Checklist/4-18 and 1991 Profile*. Burlington: University of Vermont.
- Achenbach, T. M., & Rescorla, L. A. (2001). *Manual for ASEBA school-age forms and profiles*. Burlington: University of Vermont, Research Center for Children, Youth, and Families.

- Ackerman, M. J., & Ackerman, M. C. (1997). Custody evaluations practices: A survey of experienced professionals (revisited). *Professional Psychology: Research and Practice*, 28, 137–145.
- American Psychological Association. (2010). Ethical principles of psychologists and code of conduct. Retrieved from <http://www.apa.org/ethics/code/index.aspx#> (downloaded February 13, 2012).
- American Psychological Association. (2011). Specialty Guidelines for forensic psychologists. Retrieved from <http://www.apa-ls.org/aboutpsychlaw/SpecialtyGuidelines.php> (downloaded October 8, 2012).
- Archer, R. P., Buffington-Vollum, J. K., Stredny, R. V., & Handel, R. W. (2006). A survey of psychological test use patterns among forensic psychologists. *Journal of Personality Assessment*, 87, 84–94.
- Army Individual Test Battery. (1944). *Manual of directions and scoring*. Washington, DC: War Department, Adjutant General's Office.
- Baum, L. J., Archer, R. P., Forbey, J. D., & Handel, R. W. (2009). A review of the Minnesota Multiphasic Personality Inventory–Adolescent (MMPI–A) and the Millon Adolescent Clinical Inventory (MACI) with an emphasis on juvenile justice settings. *Assessment*, 16, 384–400.
- Beck, A. T., & Steer, R. A. (1993). *Beck Anxiety Inventory Manual*. San Antonio, TX: Harcourt Brace.
- Beck, A. T., Ward, C. H., Mendelson, M., Mock, J. E., & Erbaugh, J. K. (1961). An inventory for measuring depression. *Archives of General Psychiatry*, 4, 53–63.
- Bellak, L., & Bellak, S. S. (1949). *Children's Apperception Test (CAT)*. Larchmont, NY: CPS.
- Bender, L. (1946). *Instructions for the use of the Visual Motor Gestalt Test*. New York: American Orthopsychiatric Association.
- Berg, E. A. (1948). A simple objective treatment for measuring flexibility in thinking. *Journal of General Psychology*, 39, 15–22.
- Blood, L. (2008). The use of the MCMI–III in completing parenting capacity assessments. *Journal of Forensic Psychology Practice*, 8, 24–38.
- Boccaccini, M. T., Boothby, J. L., & Overduin, L. Y. (2006). Evaluating the validity of pain complaints in personal injury cases: Assessment approaches of forensic and pain specialists. *Journal of Forensic Psychology Practice*, 6, 51–62.
- Boccaccini, M. T., & Brodsky, S. L. (1999). Diagnostic test use by forensic psychologists in emotional injury cases. *Professional Psychology, Research and Practice*, 30, 253–259.
- Borum, R., Bartel, P., & Forth, A. (2003). *Manual for the Structured Assessment of Violence Risk in Youth, version 1.1*. Tampa: University of South Florida.
- Borum, R., & Grisso, T. (1995). Psychological test use in criminal forensic evaluations. *Professional Psychology: Research and Practice*, 26, 465–473.
- Bow, J. N., Gould, J. W., Flens, J. R., & Greenhut, D. (2006). Testing in child custody evaluations: Selections, usage, and Daubert admissibility: A survey of Psychologists. *Journal of Forensic Psychology Practice*, 6, 17–38.
- Bricklin, B. (1984). *Bricklin Perceptual Scales*. Furlong, PA: Village.
- Butcher, J. N., Dahlstrom, W. G., Graham, J. R., Tellegen, A., & Kaemmer, B. (2001). *Minnesota Multiphasic Personality Inventory–2 (MMPI–2): Manual for administration and scoring*. Minneapolis: University of Minnesota.
- Butcher, J. N., Williams, C. L., Graham, J. R., Archer, R. P., Tellegen, A., Ben-Porath, Y. S., & Kaemmer, B. (1992). *MMPI–A (Minnesota Multiphasic Personality Inventory–Adolescent): Manual for administration, scoring, and interpretation*. Minneapolis: University of Minnesota Press.
- Conners, C. K. (2008). *Conners' 3rd edition manual*. Toronto: Multi-Health Systems.
- Daubert v. Merrell Dow Pharmaceuticals, Inc.*, 113 S.Ct. 2786 (1993).
- Exner, J. E. (1995). *The Rorschach: A comprehensive system*. Vol. 1: *Basic Foundations*. New York: John Wiley.
- Forth, A. E., Kosson, D., & Hare, R. D. (2003). *The Hare PCL: Youth Version*. Toronto: Multi-Health Systems.
- Frederick, R. I. (1997). *Validity Indicator Profile manual*. Minnetonka, MN: NCS Assessments.
- Frye v. United States*, 408 A.2d 364 (1923).
- General Electric Company v. Joiner*, 118 S.Ct. 512 (1997).
- Golden, C. J., Hammeke, T. A., & Purisch, A. O. (1978). Diagnostic validity of a standardized neuropsychological battery derived from Luria's neuropsychological tests. *Journal of Consulting and Clinical Psychology*, 46, 1258–1265.
- Goodman, G. S., Emery, R. E., & Haugaard, J. J. (1998). Developmental psychology and law: Divorce, child maltreatment, foster care, and adoption. In W. Damon (Ed.), *Handbook of child psychology* (5th ed., Vol. 4, pp. 775–874). New York: Wiley.

- Gould, J. W., & Stahl, P. M. (2000). The art and science of child custody evaluations: Integrating clinical and forensic mental health models. *Family and Conciliation Courts Review*, 38, 392–414.
- Greenberg, S. A., Otto, R. L., & Long, A. C. (2003). The utility of psychological testing in assessing emotional damages in personal injury litigation. *Assessment*, 10, 411–419.
- Grisso, T. (1981). *Juveniles' waiver of rights: Legal and psychological competence*. New York: Plenum.
- Grisso, T. (1986). *Evaluating competencies: Forensic assessments and instruments*. New York: Plenum.
- Grisso, T., & Barnum, R. (2003). *Massachusetts Youth Screening Instrument—Version 2: User's manual and technical report*. Sarasota, FL: Professional Resource Press.
- Hagen, M. A., & Castagna, N. (2001). The real numbers: Psychological testing in custody evaluations. *Professional Psychology: Research and Practice*, 32, 269–271.
- Halstead, W. C. (1947). *Brain and intelligence*. Chicago: University of Chicago Press.
- Hare, R. D. (1991). *The Hare Psychopathy Checklist—Revised*. Toronto: Multi-Health Systems.
- Hare, R. D. (2003). *Manual for the Revised Psychopathy Checklist* (2nd ed.). Toronto: Multi-Health Systems.
- Harris, G. T., Rice, M. E., & Quinsey, V. L. (1993). Violent recidivism of mentally disordered offenders: The development of a statistical prediction instrument. *Criminal Justice and Behavior*, 20, 315–335.
- Hart, S. D., Cox, D. N., & Hare, R. D. (1995). *The Hare Psychopathy Checklist—Screening Version (PCL:SV)*. Toronto: Multi-Health Systems.
- Heilbrun, K., Marczyk, G. R., & DeMatteo, D. (2002). *Forensic mental health assessment: A casebook*. New York: Oxford University Press.
- Heilbrun, K., Marczyk, G. R., DeMatteo, D., Zillmer, E. A., Harris, J., & Jennings, T. (2003). Principles of forensic mental health assessment: Implications for neuropsychological assessment in forensic contexts. *Assessment*, 10, 329–343.
- Hilsenroth, M. J., & Stricker, G. (2004). A consideration of challenges to psychological assessment instruments used in forensic settings: Rorschach as exemplar. *Journal of Personality Assessment*, 83, 141–152.
- Horton, A. M. Jr., & Hartlage, L. C. (Eds.). (2003). *Handbook of forensic neuropsychology*. New York: Springer.
- Jenkins v. United States*, 307 F. 2d 637 (1962).
- Kumho Tire Co. v. Carmichael*, 526 U.S. 137 (1999).
- Lally, S. J. (2003). What tests are acceptable for use in forensic evaluations? A survey of experts. *Professional Psychology: Research and Practice*, 34, 491–498.
- Lees-Haley, P. R., Smith, H. H., Williams, C. W., & Dunn, J. T. (1996). Forensic neuropsychological test usage: An empirical survey. *Archives of Clinical Neuropsychology*, 11, 45–51.
- McCann, J. T. (1998). Defending the Rorschach in court: An analysis of admissibility using legal and professional standards. *Journal of Personality Assessment*, 70, 125–144.
- McCann, J. T., & Dyer, F. J. (1996). *Forensic assessment with the Millon inventories*. New York: Guilford Press.
- Medoff, D. (2003). The scientific basis of psychological testing: Considerations following *Daubert*, *Kumho*, and *Joiner*. *Family Court Review*, 41, 199–213.
- Megargee, E. I. (1994). Using the Megargee MMPI-based classification system with MMPI–2s of male prison inmates. *Psychological Assessment*, 6, 337–344.
- Megargee, E. I., & Carbonell, J. L. (1985). Predicting prison adjustment with MMPI correctional scales. *Journal of Consulting and Clinical Psychology*, 53, 874–883.
- Melton, G. B., Petrila, J., Poythress, N. G., & Slobogin, C. (2007). *Psychological evaluations for the courts: A handbook for mental health professionals and lawyers* (3rd ed.). New York: Guilford Press.
- Millon, T. (1994). *Millon Clinical Multiaxial Inventory—III manual*. Minneapolis, MN: National Computer Systems.
- Millon, T., Davis, R. D., & Millon, C. (1997). *Manual for the Millon Clinical Multiaxial Inventory—III (MCMI—III)*. Minneapolis, MN: National Computer Systems.
- Millon, T., Millon, C., & Davis, R. (1993). *Millon Adolescent Clinical Inventory*. Minneapolis, MN: National Computer Systems.
- Morey, L. C. (1991). *Personality Assessment Inventory: Professional manual*. Odessa, FL: Psychological Assessment Resources.
- Murray, H. A. (1943). *Thematic Apperception Test*. New York: Oxford University Press.
- Otto, R. K., Edens, J. F., & Barcus, E. H. (2000). The use of psychological testing in child custody evaluations. *Family and Conciliation Courts Review*, 38, 312–340.
- Otto, R. K., & Heilbrun, K. (2002). The practice of forensic psychology: A look toward the future in the light of the past. *American Psychologist*, 57, 5–18.

- Pennuto, T. O., & Archer, R. P. (2008). MMPI-A Forensic Case Studies: Uses in documented court decisions. *Journal of Personality Assessment*, 90, 215–226.
- Podboy, J. W., & Kastl, A. J. (1993). The intentional misuse of standard psychological tests in complex trials. *American Journal of Forensic Psychology*, 11, 47–54.
- Pope, K. S., Butcher, J. N., & Seelen, J. (2006). *The MMPI, MMPI-2 and MMPI-A in court: A practical guide for expert witnesses and attorneys* (2nd ed.). Washington, DC: American Psychological Association.
- Reitan, R. M. (1979). *Manual for administration of neuropsychological test batteries for adults and children*. Tucson, AZ: Author.
- Rey, A. (1964). *L'examen clinique en psychologie*. Paris: Presses Universitaires de France.
- Rogers, R. (2003). Forensic use and abuse of psychological tests: Multiscale inventories. *Journal of Psychiatric Practice*, 9, 316–320.
- Rogers, R., Jackson, R. L., Sewell, K. W., Tillbrook, C. E., & Martin, M. A. (2003). Assessing the dimensions of competency to stand trial: Construct validation of the ECST-R. *Assessment*, 10, 344–351.
- Rogers, R., Sewell, K. W., & Gillard, N. D. (2010). *Structured Interview of Reported Symptoms: Professional manual* (2nd ed.). Lutz, FL: Psychological Assessment Resources.
- Ryba, N. L., Cooper, V. G., & Zapf, P. A. (2003). Juvenile competence to stand trial evaluations: A survey of current practices and test usage among psychologists. *Professional Psychology: Research and Practice*, 34, 499–507.
- Slick, D. J., Tan, J. E., Strauss, E. H., & Hultsch, D. F. (2004). Detecting malingering: A survey of experts' practices. *Archives of Clinical Neuropsychology*, 19, 465–473.
- Steffan, J. S., Clopton, J. R., & Morgan, R. D. (2003). An MMPI-2 scale to detect malingered depression (Md scale). *Assessment*, 10, 382–392.
- Sweet, J. J. (1999). *Forensic neuropsychology: Fundamentals and practice*. Lisse, Netherlands: Swets & Zeitlinger.
- Sweet, J. J., & King, J. H. (2002). Category test validity indicators: Overview and practice recommendations. *Journal of Forensic Neuropsychology*, 3, 241–274.
- Thorndike, R. L., Hagen, E. P., & Sattler, J. M. (1986). *Stanford-Binet Intelligence Scale: Guide for administering and scoring the fourth edition*. Chicago: Riverside Publishing.
- Tombaugh, T. N. (1996). *Test of Memory Malingering (TOMM)*. New York: Multi-Health Symptoms.
- Vaccaro, T. P., & Hogan, J. D. (2004). The origins of forensic psychology in America: Hugo Münsterberg on the witness stand. *New York State Psychologist*, 16, 14–17.
- Viljoen, J. L., McLachlan, K., & Vincent, G. M. (2010). Assessing violence risk and psychopathy in juvenile and adult offenders: A survey of clinical practices. *Assessment*, 17, 377–395.
- Wechsler, D. (1981). *Wechsler Adult Intelligence Scale-Revised manual*. New York: Psychological Corporation.
- Wechsler, D. (1987). *Wechsler Memory Scale-Revised*. New York: Psychological Corporation.
- Wechsler, D. (1991). *Wechsler Intelligence Scale for Children-third edition*. San Antonio, TX: Psychological Corporation.
- Wechsler, D. (1997). *Wechsler Adult Intelligence Scale-third edition*. San Antonio, TX: Psychological Corporation.
- Wechsler, D. (1999). *Wechsler Abbreviated Scale of Intelligence*. New York: Psychological Corporation.
- Wechsler, D. (2004). *Wechsler Adult Intelligence Scale-fourth edition*. London: Pearson Assessment.
- Weiner, I. B. (2003). *Principles of Rorschach interpretation* (2nd ed.). Mahwah, NJ: Lawrence Erlbaum.
- Wygant, D. B., & Austin, K. D. (in preparation). A survey of psychological test use patterns: Adult personality and psychopathology measures.

2

THE MINNESOTA MULTIPHASIC PERSONALITY INVENTORY–2

Martin Sellbom¹ and Jaime L. Anderson

Basic Description of the MMPI–2

The original version of the Minnesota Multiphasic Personality Inventory (MMPI; Hathaway & McKinley, 1943) was the most frequently used self-report inventory measuring personality and psychopathology (Lubin, Larsen, & Matarazzo, 1984; Piotrowski & Keller, 1989) as well as the most researched (Reynolds & Sundberg, 1976). This pattern has continued with the MMPI–2 (Butcher et al., 2001), which is used in a variety of settings (Camara, Nathan, & Puente, 2000; Archer, Buffington-Vollum, Stredny, & Handel, 2006) and remains the most frequently investigated psychological test (Butcher & Rouse, 1996).

Administration of the MMPI–2

The MMPI–2 should be administered only to adults who are 18 years of age or older. Younger individuals should be administered the adolescent version of the test, the MMPI–A (Butcher et al., 1992). There are certain test conditions that may preclude an individual from taking the MMPI–2. The authors of the test manual (Butcher et al., 2001) recommend that individuals who have less than a sixth-grade reading level not be administered the MMPI–2 in the standard format; however, some persons with limited reading ability can complete the test if it is presented using a standard audio version of the test, available on cassette or CD. Other conditions that might preclude an MMPI–2 administration include altered cognitive states or confusion stemming from brain impairment, or severe psychopathology.

The MMPI–2 should be administered in a quiet and comfortable place for the test-taker. It takes about 60–90 minutes to administer the test in standard booklet and answer sheet form for individuals of normal intellectual functioning (Graham, 2011). Complicating factors such as disabling psychopathology, low reading level, or lower intellectual functioning may result in a longer time, such as two hours or more. Administration by computer using standard software available through Pearson Assessments reduces the amount of time needed to complete the inventory.

MMPI-2 Scales

The standard scales of the MMPI-2 currently include 9 Validity Scales, 10 Clinical Scales (and 31 subscales), 9 Restructured Clinical (RC) Scales, 15 Content Scales (and 28 Content Component Scales), 15 Supplementary Scales, and 5 Personality Psychopathology Five (PSY-5) Scales. Many other scales have been developed or proposed for the test; however, only the standard scales are recommended for use by the test publisher.

The MMPI-2 *Validity Scales* are an essential component of the test. Validity Scales were first introduced when the original MMPI was published to address a challenge inherent in self-report inventories: their susceptibility to misleading responding and scoring error. The scales are discussed in the order in which they appear on the MMPI-2 profile, which is also the order recommended for interpretation (see Butcher et al., 2001).

The Cannot Say count measures non-responding. This index should always be consulted first, as item omissions affect all remaining scales, including the other Validity Scales. A protocol where 30 or more items cannot be scored may be of questionable validity. Automated scoring systems indicate the percentage of items answered on each MMPI-2 scale, allowing for a more refined interpretation of the affects of item omissions. Dragon, Ben-Porath and Handel (2012) recently showed that the validity of MMPI-2 RC Scales was not significantly affected when up to 30–50% of items were randomly omitted from the scale. Although they showed that omissions were rare in most testing circumstances, they found that even with omissions, clinically significant elevations were still identifiable.

The Variable Response Inconsistency (*VRIN*) and True Response Inconsistency (*TRIN*) scales measure inconsistent responding. *VRIN* assesses random responding, whereas *TRIN* measures fixed responding in either the true or false directions. The scales are scored by comparing the test-taker's responses to pairs of items. A *T*-score greater than 79 on either of these scales indicates that the MMPI-2 protocol is invalid and uninterpretable (Butcher et al., 2001).

The Infrequency (*F*) scale was originally developed to detect random responding and clerical scoring error, but is currently used primarily to identify over-reporting of psychopathology. The test manual states that *T*-scores greater than 100 *may* indicate an invalid protocol; however, this possibility needs to be evaluated by considering scores on the *VRIN*, *TRIN*, and *F_p* (discussed later) scales (Butcher et al., 2001; see also Graham, 2011). Because all the *F* scale items appear within the first 361 items on the test, a scale to measure infrequent responding to items in the second half of the test booklet was developed. The Back-Infrequency (*F_b*) scale has been found to successfully detect shifts in response sets across the inventory (Berry, Baer, & Harris, 1991; Clark, Gironda, & Young, 2003).

Arbisi and Ben-Porath (1995) developed the Infrequency-Psychopathology (*F_p*) scale to address a problem posed by the confounding of severe psychopathology and distress with over-reporting as sources of elevation on the *F* scale. The *F_p* scale consists of items that are infrequently endorsed in both normal and abnormal populations; thus, individuals scoring high on this scale report symptoms that even individuals experiencing severe symptoms of psychopathology are unlikely to endorse. The manual authors state that *T*-scores of 100 on *F_p* indicate that the protocol is likely invalid. We recommend that a *T*-score of 99 rather than 100 be used, because a raw score of 8 corresponds to a 99*T* in the male normative sample. In addition, results from a meta-analysis have supported a raw score of 8 as the optimal cut score for *F_p* (Rogers et al., 2003).

The Symptom Validity (formerly Fake Bad; *FBS*) scale was originally developed to provide a validity measure specific to individuals attempting to feign symptoms within the context of a personal injury evaluation (Lees-Haley, English, & Glenn, 1991). Research has later shown that the scale is also effective in the detection of exaggerated somatic symptoms (Larrabee, 2003b). Although the *FBS* scale was initially faced with criticism for its narrow focus and possible high

false-positive rate (e.g., Rogers, Sewell, Martin, & Vitacco, 2003; Butcher, Arbisi, Atlas, & McNulty, 2003), results from a meta-analysis by Nelson, Hoelzle, Sweet, Arbisi, and Demakis (2010) indicated that *FBS* performed as well as, if not better than, other scales used in detecting the over-reporting of symptoms in forensic neuropsychological evaluations. Ben-Porath, Graham, and Tellegen (2009) authored a monograph in which they provide sufficient research foundation to justify the test publisher including *FBS* as a standard scale on the MMPI-2.

The Lie (*L*) scale was developed to detect under-reporting. High scorers present in an unusually virtuous manner and deny personal flaws that most people would be willing to admit to. The manual authors indicate that a *T*-score of 80 and higher on *L* identifies a protocol as being likely invalid. The Correction (*K*) scale is a more subtle indicator of under-reporting. *T*-scores of 75 or higher may indicate an invalid profile (Butcher et al., 2001) in non-clinical settings.

Finally, the Superlative (*S*) scale is also designed to assess under-reporting. *T*-scores of 70 and higher on this scale raise the possibility that the profile may be invalid.

The *Clinical Scales* were developed using an empirical keying method that identified items that discriminated between a specific diagnostic group from a normal group. The MMPI-2 Clinical Scales include Scales 1 (Hypochondriasis [*Hi*]), 2 (Depression [*Di*]), 3 (Hysteria [*Hy*]), 4 (Psychopathic Deviate [*Pd*]), 5 (Masculinity [*Mf*]), 6 (Paranoia [*Pa*]), 7 (Psychasthenia [*Pt*]), 8 (Schizophrenia [*Sc*]), 9 (Hypomania [*Ma*]), and 0 (Social Introversion [*Si*]). Because of the methodology used in their development, the Clinical Scales are very heterogeneous, making their interpretation difficult. As a result, Harris and Lingoes (1955) developed subscales for the Clinical Scales that focused on grouping items of specific content domains together to facilitate interpretation. Unfortunately, these subscales have poor psychometric properties and have been only marginally successful (cf. Ben-Porath, 2012; Graham, 2011). Another method to tease out the overlapping variance among the Clinical Scales involves classifying protocols into “code types,” based on the two or three most elevated scales in the profile. The test manual recommends interpreting only well-defined code types (i.e. the lowest scale in the code type is at least 5 *T*-score points greater than the highest scale outside the code type).

The MMPI-2 *Restructured Clinical (RC) Scales* (Tellegen et al., 2003) are the most recent major addition to the MMPI-2. The RC Scales were developed to remove a common Demoralization factor that saturates the original Clinical Scales and improve their discriminant and convergent validity (Tellegen et al., 2003). Demoralization was conceptualized through Tellegen’s (1985) framework of positive and negative emotionality as corresponding to the pleasantness–unpleasantness vector between these two orthogonal affective dimensions. Tellegen’s (1985) model links depression to low positive emotionality and anxiety to high negative emotionality. Therefore, as a first step in RC Scale construction, Demoralization markers were identified based on factor analyses of Clinical Scales 2 and 7. The next step in developing these scales involved identifying the unique core components of each Clinical Scale after Demoralization items had been removed. This was followed by analyses designed to maximize the distinctiveness of each core component and identify items throughout the MMPI-2 pool that are related uniquely to this core. The final set of RC Scales includes 192 non-overlapping items scored in scales labeled Demoralization (*RCd*), Somatic Complaints (*RC1*), Low Positive Emotional (*RC2*), Cynicism (*RC3*), Antisocial Behavior (*RC4*), Ideas of Persecution (*RC6*), Dysfunctional Negative Emotions (*RC7*), Aberrant Experiences (*RC8*), and Hypomanic Activation (*RC9*). In comparison with the Clinical Scales, the RC Scales are more homogeneous and less intercorrelated, resulting in improved convergent and discriminant validity (see, for example, Ben-Porath & Tellegen, 2008; Tellegen, Ben-Porath, & Sellbom, 2009, for reviews). The RC Scales have also become the core set of scales on the recently released MMPI-2 Restructured Form (MMPI-2-RF; Ben-Porath & Tellegen, 2008; see Ben-Porath, this volume, Chapter 3).

The MMPI-2 *Content Scales* (Butcher, Graham, Williams, & Ben-Porath, 1990) were developed through a series of rational-conceptual and empirical analyses modeled after Wiggins's (1969) original set of Content Scales for the MMPI. They are designed to facilitate test interpretation by providing a reliable indication of the individual's self-presentation and expanding the content domains represented by the original Clinical Scales. The 15 Content Scales include Anxiety (*ANX*), Fears (*FRS*), Obsessiveness (*OBS*), Depression (*DEP*), Health Concerns (*HEA*), Bizarre Mentation (*BIZ*), Anger (*ANG*), Cynicism (*CYN*), Antisocial Practices (*ASP*), Type A Behavior (*TPA*), Low Self-Esteem (*LSE*), Social Discomfort (*SOD*), Family Problems (*FAM*), Work Interference (*WRK*), and Negative Treatment Indicators (*TRT*) scales. Twelve of the scales also have Content Component Scales (Ben-Porath & Sherwood, 1993) that were developed through principal components analyses and designed to further focus Content Scale interpretation.

The *Supplementary Scales* represent an ad hoc collection of MMPI-2 measures developed over the test's history. They include Welsh's (1956) Anxiety (*A*) and Repression (*R*) scales, which measure the two factors representing internalizing and reverse externalizing (see Krueger & Markon, 2006) that are commonly derived when the Clinical Scales are factor-analyzed. The *A* scale is a measure of general emotional distress. Other Supplementary Scales include broad personality measures (Ego Strength [*Es*], Dominance [*Do*], and Social Responsibility [*Re*]), general maladjustment scales, including the College Maladjustment and Marital Distress Scale (*MDS*; Hjemboe, Almagor, & Butcher, 1992), and behavioral dyscontrol scales (Hostility [*Ho*] and Over-controlled Hostility [*O-H*]). The substance abuse scales include the MacAndrew Alcoholism Scale-Revised (*MAC-R*; MacAndrew, 1965), the Addiction Admission Scale (Weed, Butcher, Ben-Porath, & McKenna, 1992) and the Addiction Potential Scale (*APS*; Weed et al., 1992).

Finally, the *Personality Psychopathology Five* (*PSY-5*; Harkness, McNulty, & Ben-Porath, 1995) scales provide a measurement model for five dimensional personality constructs that describe normal to abnormal range personality traits. Harkness and McNulty (1994) identified these major dimensions of personality pathology in a series of latent root analyses using a mixture of pathological descriptors (per *DSM-III-R* personality disorder criteria) which were embedded within a pool of normal-range descriptors. The *PSY-5* Scales were developed to assist in assessing personality pathology from a dimensional perspective. The scales include Aggressiveness (*AGGR*), Psychoticism (*PSYC*), Disconstraint (*DISC*), Neuroticism/Negative Emotionality (*NEGE*), and Introversion/Low Positive Emotionality (*INTR*). Furthermore, it should be highlighted that the five construct domains mirror those five domain traits currently being proposed for the *DSM-5* (see, for example, Skodol et al., 2011): Antagonism, Psychoticism, Disinhibition, Negative Affectivity, and Detachment. Indeed, the *DSM-5* Personality and Personality Disorders workgroup explicitly states that the proposed *DSM-5* personality trait model "at the domain-level, bears a strong resemblance to Dr. Allan Harkness' Personality Psychopathology Five (*PSY-5*) model of clinically relevant personality variants" (APA, 2011).

History of the MMPI and MMPI-2's Development

The MMPI was developed by Hathaway and McKinley (1943) to facilitate psychodiagnosis. Although the MMPI has often been described as being constructed in an atheoretical vacuum, this was actually not the case. The then contemporary descriptive nosology and a combination of psychodynamic and behavioral theory all played a role in generating the test's items and scale construction (Ben-Porath, 2012). In developing the instrument, Hathaway and McKinley gathered an item pool by examining the literature concerning the major psychiatric diagnoses of the time and complementing it with their own extensive clinical experience. This procedure generated close

to 1,000 test items. The authors employed an empirical keying method to assign items to a particular scale without consideration of their content. By ignoring item content in the construction process, Hathaway and McKinley believed that the resulting scales would be less susceptible to distortion because items were statistically, but not necessarily manifestly, related to the target constructs. Moreover, as expanded by Meehl (1945), because the scales were constructed and to be interpreted empirically, there was no need to assume that the items had similar meaning to different individuals.

Hathaway and McKinley administered the candidate items to various groups of psychiatric patients as well as a normative group consisting of approximately 750 individuals, primarily skilled laborers and farmers, many of whom were visitors to the University of Minnesota Hospital. Items were selected for a particular diagnostic scale if they discriminated sufficiently between members of a specific diagnostic group (e.g., depression) and the normal group. This procedure yielded the eight original MMPI Clinical Scales.

Although initially insufficiently effective in predicting membership in specific diagnostic categories (Hathaway, 1960), the MMPI underwent a transformation from an instrument designed to predict diagnostic taxonomies to one that relies on empirically derived correlates in assessing symptomatology and personality patterns. The call for establishing empirical correlates for the MMPI was made by Meehl (1945, 1954), who suggested that the test would be optimally used if such correlates were established. These recommendations were subsequently followed by Meehl's (1956) call for a "good cookbook" that could be used to inform clinicians about the empirical correlates of the MMPI scales.

Several large-scale studies establishing the empirical correlates of MMPI code types were conducted after Meehl issued his call. Marks and Seeman (1963) published an empirical codebook that provided correlates of code types for psychiatric inpatients. Gilberstadt and Duker (1965) published empirical correlates of code types for 360 male VA inpatients. These and other source books serve as the foundation for contemporary interpretation texts such as those written by Graham (2011) and Greene (2010).

In the 1970s, it was becoming increasingly apparent that some changes were in order for the MMPI. One of the greatest needs was to update the test's norms. As mentioned earlier, the original normative data were collected in the late 1930s and consisted primarily of Caucasian working-class rural Minnesotans possessing an average of eight years of education. Although this sample was appropriate during the MMPI's initial conception, it was no longer adequate given its widespread use across the United States and the world. The second need was to revise some MMPI items that were outdated or awkwardly phrased. Some items had content that was no longer clear, relevant, or appropriate in light of modern language use, cultural practices, and social norms. Other items contained sexist language or controversial religious statements. A final issue concerned the need to include a broader range of content domains that were not necessarily salient in the 1930s but had grown very relevant in contemporary assessment (e.g., suicidal ideation, Type A behavior, substance abuse, low self-esteem, and treatment indicators).

In order to deal with the problems just noted, the University of Minnesota Press, the test's publisher, formed a restandardization committee to coordinate the revision of the MMPI. This committee had two major goals: to improve the test while maintaining continuity with the original version to the extent possible. The second goal was necessary to ensure that the large amount of research generated on the original MMPI could still be applied with the revision.

The restandardization project produced the MMPI-2 (Butcher, Dahlstrom, Graham, Tellegen, & Kaemmer, 1989), including a new normative sample (described in greater detail next) consisting of 1,138 men and 1,462 women of diverse ethnic backgrounds and from different regions of the United States. The MMPI-2 consists of 567 items. Of the 383 items on the basic Validity and Clinical Scales, 372 were retained for the MMPI-2. Eleven items were deleted and a total of 64

items were slightly revised. Ben-Porath and Butcher (1989a; 1989b) found that these changes did not alter the psychometric functioning of the Validity or Clinical Scales.

Psychometric Characteristics of the MMPI-2

Standardization and Norms

The MMPI-2 normative sample consists of 1,138 male and 1,462 female community volunteers who were tested in seven states (California, Minnesota, North Carolina, Ohio, Pennsylvania, Virginia, and Washington), and their demographic make-up was designed to match the 1980 U.S. census. A comparison with the 1997 annual census information revealed even closer similarities (Butcher et al., 2001), except that Hispanics were under-represented in the normative sample.

The racial distribution of the MMPI-2 normative sample includes 82% Caucasians, 11% African Americans, 3% Hispanics, 3% Native Americans, and less than 1% Asian Americans for the men, whereas the women include 81% Caucasians, 13% African Americans, 3% Hispanics, 3% Native Americans, and 1% Asian Americans. Individuals in the normative sample have an average of 14.72 years of education, which is slightly higher than for the general adult U.S. population (according to 1997 census information). Normative sample members range in age from 18 to 85. Analyses were conducted for various age subgroupings and it was determined that separate norms were not needed.

Several studies have examined the associations between demographic variables and scores on MMPI-2 scales. Schinka, LaLone, and Greene (1998) concluded that demographics contributed little to the majority of MMPI-2 scales' ability to discriminate between psychiatric inpatients and the normative sample, with the exception of gender, which added significantly to the discrimination for *ASP*, *FRS*, and *MAC-R*, but also gender-based scales such as Scale 5 and the two Gender Roles Supplementary Scales. Moreover, Long, Graham, and Timbrook (1994) concluded that education and family income levels likely have little association with MMPI-2 scores. Some researchers have criticized the MMPI-2 normative sample for its high educational level. However, Schinka and LaLone (1997) examined this issue and concluded that the higher-than-average education level of the MMPI-2 normative sample has no meaningful impact on test scores.

Because of concerns that there may be confounding gender differences in test-taking approaches to the MMPI and MMPI-2, the instrument has traditionally had separate norms for men and women. This approach reflected concerns that women might be more willing to endorse symptoms of psychopathology than men (Graham, 2011). Ben-Porath and Forbey (2003) developed non-gender-based norms for MMPI-2 scores as this had become necessary to comply with the congressional Civil Rights Act of 1991, which prohibits gender-based norming in pre-employment evaluations. Ben-Porath and Forbey (2003) reported that moving from gendered to non-gendered norms had negligible impact on MMPI-2 *T*-scores, indicating that concerns about a possible gender-based confound are unwarranted.

MMPI-2 raw scores are converted to *T*-scores to facilitate standardized interpretation. For several MMPI-2 scale sets, uniform *T*-scores are used to ensure percentile comparability across scales (Tellegen & Ben-Porath, 1992). Uniform *T*-scores are used for the Clinical, RC, Content, and PSY-5 scales.

Reliability

The MMPI-2 manual (Butcher et al., 2001), the PSY-5 test report (Harkness, McNulty, Ben-Porath, & Graham, 2002), and the RC Scale monograph (Tellegen et al., 2003) provide detailed

information concerning the reliability of the various MMPI-2 scales. Therefore, only a concise summary is provided here.

In terms of internal consistency, the Clinical Scales are the least internally consistent, which would be expected, as they were not designed to be homogeneous scales. The Content, PSY-5, and RC Scales were all constructed with an emphasis on internal consistency. In the normative sample, internal consistencies for the Clinical Scales range from .34 to .85 for men and from .37 to .87 for women, the RC Scales from .63 to .87 for men and .62 to .89 for women, and the Content Scales from .72 to .86 for men and from .68 to .86 for women. The Supplementary Scales' internal consistencies range from .34 to .89 for men and .24 to .90 for women, whereas those for the PSY-5 Scales range from .65 to .84 for both men and women.

In the normative sample, test-retest correlations for the Clinical Scales range from .67 to .93 for men and from .54 to .92 for women, those for the RC Scales from .62 to .88 for the combined sample, and those for the Content Scales from .77 to .91 for men and .78 to .91 for women. The Supplementary Scales had test-retest correlations that range from .63 to .91 for men and from .69 to .91 for women. Harkness et al. (2002) reported PSY-5 Scale test-retest coefficients for the overall sample that range from .78 to .88.

Validity

The MMPI-2 is unparalleled in terms of the number of validity studies that it has generated (e.g., Butcher & Rouse, 1996). Dahlstrom, Welsh, and Dahlstrom (1975) referenced over 6,000 research studies conducted with the original MMPI. Many of these studies followed Meehl's (1956) call for a "good cookbook," where he urged researchers to identify empirical correlates for the test's scales and code types. Numerous studies were conducted with psychiatric inpatients (e.g., Marks & Seeman, 1963; Gilberstadt & Duker, 1956), medical patients (Guthrie, 1949), adolescents (Archer, Gordon, Giannetti, & Singles, 1988; Hathaway & Monachesi, 1963), and normal college students (e.g., J. D. Black, 1953).

The research with the MMPI-2 has continued in the same vein. Graham (2006) indicated that since the MMPI-2 was published, 2,800 journal articles, book chapters, and textbooks about the test have been published. Graham (2011) did not provide a specific update five years later, but it is safe to say that this number now well exceeds 3,000.² Although it is well beyond the scope of this chapter to summarize these studies, we provide some overall conclusions regarding the validity of the various MMPI-2 scales.

Many research studies have supported the use of the Validity Scales (e.g., Graham, Watts, & Timbrook, 1991). In regard to the *VRIN* scale, previous research has supported the cutoff suggested in the manual (Berry, Wetter et al. 1991; Gallen & Berry, 1996; Paolo & Ryan, 1992) to identify random responding. Handel, Arnau, Archer, and Dandy (2006) found that the *TRIN* scale showed utility in identifying protocols with randomly inserted true and false responses. However, they did find a cutoff of 75 (rather than the manual-recommended 80) to be optimal. Rogers, Sewell, Martin, and Vitacco (2003) conducted a meta-analysis on the MMPI-2 over-reporting scales and found that the infrequency scales (F , F_B , and F_p) were effective in detecting malingering. They also noted that F_p consistently had the largest effect size in differentiating individuals asked to mangle from those who took the test under standard instructions. Arbisi, Ben-Porath, & Yossef (1998) found that F_p added incrementally to F in differentiating psychiatric inpatients asked to over-report from those who took the test honestly. In addition, meta-analyses by Nelson et al. (2010) and Nelson, Sweet, & Demakis (2006) have found that *FBS* has utility in detecting non-credible somatic and cognitive responding, particularly in cases of reported traumatic brain injury. Another

meta-analysis by Baer and Miller (2002) indicated that the *L* scale was consistently the best predictor of under-reporting, but noted effectiveness for *K* as well.

The convergent validity of the Clinical, Content, Supplementary, and PSY-5 Scales and Code Types has been established in outpatient (Graham, Ben-Porath, & McNulty, 1999; Harkness et al., 2002), inpatient (Arbisi, Ben-Porath, & McNulty, 2003; Archer, Aiduk, Griffin, & Elkins, 1996; Archer, Griffin, & Aiduk, 1995), forensic (Petroskey, Ben-Porath, & Stafford, 2003), college student (Ben-Porath, McCully, & Almagor, 1993), and private practice samples (Sellbom, Graham, & Schenk, 2005), to mention a few. These correlates have been remarkably similar across studies and also congruent with those of the original MMPI (Graham, 2011), indicating that the correlates of the MMPI-2 scale scores generalize well across settings.

The discriminant validity for several MMPI-2 scale scores, however, has been problematic. Many of the correlate studies just mentioned demonstrate that MMPI-2 scales, and most substantially so the Clinical Scales, are significantly correlated with affective-laden (i.e., depression, anxiety) criteria, including scales for which this would not be expected (e.g., Clinical Scales 4, 6, and 8—measures of antisocial personality traits, persecutory ideation, and thought dysfunction, respectively). Indeed, in some settings these scales were more strongly correlated with emotional distress criteria than those indexing core components of their respective target constructs (see, for example, Graham et al., 1999). Welsh (1956) indicated early through factor analysis that a common higher-order factor labeled Anxiety saturates the Clinical Scales. Tellegen et al. (2003) more recently labeled this component Demoralization. In restructuring the Clinical Scales, they sought to minimize its impact to the extent theoretically indicated and empirically feasible, which empirical evidence has clearly shown to be a success (e.g., Sellbom, Ben-Porath, & Graham, 2006; Simms, Casillas, Clark, Watson, & Doebbeling, 2005; Tellegen et al., 2003; Tellegen et al., 2006).

Psychometric Strengths and Weaknesses

The strengths of the MMPI-2 include the vast research literature on the scales supporting their convergent validity as well as effectiveness in detecting a variety of clinically relevant problems, such as substance abuse difficulties and malingering. As a result, the MMPI-2 has also been one of the most frequently used tests in psychological practice for many decades and it continues to be a centerpiece graduate and internship training. However, limitations of the test include the just-discussed excessive intercorrelations among the Clinical Scales, which are without question the scale set with which clinicians are most familiar. This challenge requires test interpreters to rely on other MMPI-2 sources (e.g., code types, RC Scales, Content Scales). Moreover, the inclusion of “subtle items” and use of the *K* correction may attenuate the convergent validity of the Clinical Scales (Barthlow, Graham, Ben-Porath, Tellegen, & McNulty, 2002; Detrick, Chibnall, & Rosso, 2001; Sellbom, Ben-Porath, McNulty, Arbisi, & Graham, 2006; Sellbom, Fischler, & Ben-Porath, 2007; Weed, Ben-Porath, & Butcher, 1990). Most of these challenges have been remedied through the development of the RC Scales and to a lesser degree the PSY-5 scales, but these scale sets do not cover the full range of clinical and personality symptomatology available through test items.

Clinical Uses and Limitations

The original purpose of the MMPI was to assist in psycho-diagnosis, but it was quickly determined that this could not be accomplished with sufficient accuracy. Instead, MMPI-2 information now focuses on assessing the test-taker’s level of adjustment, specific symptoms of psychopathology, personality characteristics, and interpersonal functioning. This information can, of course, be

considered in generating diagnostic impressions, as well as treatment planning, and to form opinions for various decisions including psychological functioning.

Mental Health Settings

Not surprisingly, the most common settings in which the MMPI-2 is used are mental health assessment and treatment centers (Graham, 2011). Several researchers have provided detailed information on the correlates of the MMPI and MMPI-2 in these settings as well as recommendations for interpretation (Archer et al., 1995; Gilberstadt & Duker, 1965; Graham et al., 1999; Harkness et al., 2002; Marks & Seeman, 1963; Tellegen et al., 2003). Relatedly, Sellbom, Graham, and Schenk (2005) reported the correlates of MMPI-2 scales and code types for clients in private practice and found that the correlational patterns were virtually identical to those obtained in other outpatient studies (e.g., Graham et al., 1999).

The MMPI-2 is generally used to identify clinical conditions in these settings. The test can generate inferences regarding general maladjustment and psychopathology symptoms (e.g., depression, anxiety, hallucinations) to aid in diagnosis and case conceptualization. The MMPI-2 can also be used to screen for substance abuse problems, as indicated by scores on *MAC-R*, *AAS*, and *APS* (Graham & Strenger, 1988; Weed et al., 1992). Finally, the MMPI-2 can produce information about personality traits that should be considered in case conceptualization and treatment planning (cf. Harkness & McNulty, 2002, 2006).

Medical Settings

The MMPI-2 is frequently used in medical settings. Arbisi and Butcher (2004) provide a detailed review of the instrument's use in this setting. One important use of the MMPI-2 with medical patients has been to screen for serious psychopathology that may not have been detected (Graham, 2011). The MMPI-2 can further be used to understand how persons with medical problems are affected by them psychologically by consulting the variety of scales that give indications of general emotional distress (e.g., *RC Demoralization*, *Mean 8 Clinical Scales*). Conversely, the role of psychological factors in physical health complaints can also be examined with the MMPI-2, particularly via the 1-3/3-1 code type, *Somatic Complaints (RC1)*, and the *Health Concerns Content and Content Component Scales* (cf. Arbisi & Butcher, 2004).

Substance abuse problems are quite common among persons being treated primarily for medical conditions. The MMPI-2 can be useful in alerting clinicians to the possibility of substance abuse. Fordyce (1979) suggested that the MMPI could be helpful in identifying individuals who misuse prescribed medication. He argued that high scores on Clinical Scales 2 and 9 were particularly helpful. The Supplementary Scales *MAC-R*, *AAS*, and *APS* are helpful in indicating increased risk for or actual substance abuse.

The MMPI-2 can also provide important information concerning how medical patients are likely to respond psychologically to medical interventions, including the prediction of pre- and post-operative adjustment (Henrichs & Waters, 1972), psychosocial adjustment of cancer patients (Sobel & Worden, 1979), gastric bypass surgery (Tsushima, Bridenstine, & Balfour, 2004; Wygant et al., 2007), and response to chronic-pain treatment programs (Akerlind, Hornquist, & Bjurulf, 1992; Vendrig, Derksen, & de Mey, 1999). The MMPI/MMPI-2 has also demonstrated some success in identifying individuals at higher risk for coronary heart disease (Williams et al., 1980; Barefoot, Dahlstrom, & Williams, 1983; Kawachi et al., 1998).

Pre-employment Screening

The MMP or MMPI-2 is frequently used in selecting employees for sensitive occupations (e.g., law enforcement officers, airline pilots, nuclear power plant operators) (Butcher, 1979, 1985). These screenings are particularly justified when the presence of psychological dysfunction may jeopardize the safety of the community. Consider, for example, an airline pilot with a substance abuse problem or a law enforcement officer with an impulsive disposition. Several research studies have demonstrated that the MMPI can be useful in screening for psychopathology in non-clinical settings (e.g., Lachar, 1974). These studies cover a wide range of occupations, including police applicants (e.g., Bernstein, 1980; Beutler, Storm, Kirkish, Scogin, & Gaines, 1985; Costello, Schoenfeld, & Kobos, 1982; Hartman, 1987; Inwald, 1988), physicians' assistants (Crovitz, Huse, & Lewis, 1973), medical assistants (Stone, Bassett, Brousseau, Demers, & Stiening, 1972), psychiatric residents (Garetz & Anderson, 1973), clinical psychology graduate students (Butcher, 1979), nurses (Kelly, 1974), firefighters (Avery, Mussio, & Payne, 1972; Liao, Arvey, Butler, & Nutting, 2000), probation officers (Solway, Hays, & Zieben, 1976), and nuclear power plant personnel (Dunnette, Bownas, & Bosshardt, 1981). Most of these studies have focused primarily on the ability of the Clinical Scales to identify potential problems in applicants.

Detrick et al. (2001) found that MMPI-2 scores correlated moderately with scores on the Inwald Personality Inventory (IPI), which was designed to identify problem areas specific to law enforcement candidates. Other studies found that Clinical Scales 4 and 9 were particularly, but modestly, related to worse job outcomes (Bartol, 1991; Costello, Schneider, & Schoenfeld, 1996). More recently, Sellbom, Fischler, and Ben-Porath (2007) reported that the MMPI-2 Clinical, RC, and specific addiction-relevant scales were useful in predicting future negative outcomes in a large sample ($n = 426$) of police officers. More specifically, the findings suggested that the RC Scales (in particular, *RC3*, *RC4*, *RC6*, and *RC8*) were the most potent predictors of sustained internal affairs complaints and negative supervisor ratings subsequent to being hired. Caillouet, Boccaccini, Varela, David, and Rostow (2010) examined the PSY-5 scales in a sample of 600 police officers and found that these scales significantly predicted future job termination and officer misconduct.

Strengths and Limitations

The major strength of the MMPI-2 is the vast amount of research that has been conducted with the instrument, which has established its utility for many purposes. This research has focused on symptom and personality correlates, classification accuracy of various conditions, and predictive validity, to mention a few areas. Moreover, the MMPI-2 Validity Scales are unparalleled in terms of their research support in effectively identifying misleading response styles. This makes the instrument very appealing for most types of evaluations, including those in which individuals might have a strong incentive to distort their self-report.

As discussed earlier, in spite of the broad literature available to guide their interpretation, there exist significant challenges to the Clinical Scales' interpretation stemming from the methods used to construct them. These have been addressed previously through the development of a variety of strategies to augment interpretation of the Clinical Scales, including reliance on code types, subscales, and a variety of supplementary sources of information. As a result, the MMPI-2 interpretation can be rather complex. The RC Scales were designed to provide a more parsimonious solution to these challenges and by most accounts appear to have been quite successful (e.g., Archer, 2006; Finn & Kamphuis, 2006; Simms, 2006; Tellegen et al., 2006; Tellegen et al., 2009).

Another limitation concerns generating diagnostic impressions. Although the MMPI-2 provides information regarding symptomatology associated with various forms of psychopathology, the MMPI-2 in itself cannot be relied upon to generate diagnostic impressions. Efforts to generate

MMPI-2 profiles that are consistent with specific types of psychopathology have failed. Examples include so-called floating profiles for borderline personality disorder (e.g., Gartner, Hurt, & Gartner, 1989) or post-traumatic stress disorder (see Keane, Malloy, & Fairbank, 1984). Instead, clinicians have to rely on the MMPI-2 to generate information about clinical symptoms and dysfunctional personality traits to integrate with other sources of information (e.g., interviews, records, third party sources) for optimal diagnostic decision making and treatment planning in clinical psychology settings.

Forensic Uses and Limitations

Early use of the MMPI focused primarily on the characterization and explanation of criminal behavior (see Dahlstrom et al., 1975). Studies by Capwell (1945a; 1945b) and Hathaway and Monachesi (1953; 1957) examined personality patterns of juvenile delinquents. Hathaway and Monachesi found that certain Clinical Scales were of an “excitatory” nature (4, 8, and 9), meaning that they predicted greater likelihood for acting out, whereas other scales were of an “inhibitory” nature (2, 5, and 0). Delinquents were found to score higher on the excitatory scales compared to the non-delinquents. A number of early researchers focused on the etiology of criminal behavior, using the MMPI as a proxy for underlying personological explanations for various crimes, including heroin dependency (Sutker & Allain, 1973), and aggressive individuals (e.g., Megargee & Mendelsohn, 1962). For instance, Pothast (1956) conducted an early study in which he examined personological differences between instrumental murderers and affective/reactive murderers. He found that the psychoticism scales were higher among the instrumental murderers, whereas the neuroticism scales were higher among the affective murderers.

Present forensic applications of the MMPI-2 are extensive. Several surveys have indicated that the MMPI-2 is more frequently used in forensic settings than any other test (Archer et al., 2006; Lees-Haley, 1992) and considered appropriate for most forensic questions (Lally, 2003). The MMPI-2 is second only to the Wechsler Intelligence Scales in its use for forensic neuropsychological evaluations (Lees-Haley, Smith, Williams, & Dunn, 1996). The test is also used quite frequently in countries other than the United States, such as Australia (Martin, Allan, & Allan, 2001) for forensic purposes. Given the instrument’s widespread use in forensic assessments, Otto (2002) indicates that the question is not whether it is appropriate to use the MMPI-2 in forensic settings, but rather for what purposes. We concur, and address this question later. We first discuss some general issues concerning using the MMPI-2 in forensic assessment.

Issues Concerning MMPI-2 Use in Forensic Assessment

Admissibility of MMPI-2 Testimony

If an expert relies on the MMPI-2 to form an opinion to be offered in testimony, the basis for this opinion may be scrutinized and admissibility denied if it fails to meet certain standards. In federal court cases and in most state courts, these standards are outlined in the U.S. Supreme Court decision in *Daubert v. Merrell Dow Pharmaceuticals, Inc.* (1993). In its ruling on *Daubert*, the Supreme Court established that trial judges must determine the validity of inferences based on scientific techniques by considering whether (1) the technique can be and has been tested empirically, (2) the technique has been subjected to peer review, (3) the error rates of the technique are known, (4) there are standards for applying the technique, and (5) the technique is generally accepted in its scientific discipline. The last criterion refers back to the *Frye v. U.S.* (1923) ruling regarding expert testimony, which still applies in several states.

At a broad, general level, the MMPI-2 meets these criteria. It was developed through an empirical method and much of its interpretation is based on empirical correlates derived for the scales. Moreover, the applications of the test are inherently testable. The MMPI or MMPI-2 has been studied in over 9,000 publications that have appeared in peer-reviewed journals. Psychometric techniques lend themselves well to analyses of error rates (by considering test-score reliability), and numerous research studies have examined the false positive and negative rates of the test's scales in predicting a variety of phenomena. The MMPI-2 test manual provides direct guidelines for standardized administration and scoring, and, as mentioned earlier, the MMPI-2 is one of the most-used psychological tests in clinical (Camara et al., 2000) and forensic (Archer et al., 2006) practice. Finally, Bow, Gould, Flens, and Greenhut (2006) presented results from a survey of 89 forensic psychologists, which indicated that 95% believed that the MMPI-2 met *Daubert* criteria for admissibility for most purposes.

However, these general statements alone do not justify using the MMPI-2 in forensic evaluations. In any given case, the question is whether a specific MMPI-2-based conclusion reached by an expert can meet the standards for admissibility. Thus, when faced with a *Daubert* challenge the expert must be prepared to cite original peer-reviewed research to support their conclusions as well as to answer any other *Daubert*-relevant questions. For example, in *U.S. v. Huberty* (2000), Lt. Col. Huberty was facing criminal charges of indecent exposure. An expert witness testifying for the defense argued that Huberty could not be guilty of the crime because his MMPI profile was not indicative of exhibitionism. The testimony was not admitted, however, because no empirical research has demonstrated that exhibitionists produce specific MMPI-2 profiles.

Malingering and Defensiveness

The external incentive to distort one's responses on the MMPI-2 is high in both criminal and civil evaluations. The *Diagnostic and Statistical Manual of Mental Disorders (DSM-IV-TR*; American Psychiatric Association, 2000) defines malingering as the exaggeration or fabrication of physical and psychological symptoms for external incentives. For instance, an individual faced with criminal charges might want to appear more psychologically disturbed than they really are in order to be found not guilty by reason of insanity. Several research studies have suggested that malingering is in fact common in forensic evaluations. Grossman and Wasylw (1988) estimated that almost 41% of all insanity defendants malingering during psychological evaluations, whereas Ardolf, Denney, and Houston (2007) estimated that around 54% of individuals evaluated for neuropsychological deficits in criminal populations fall within at least a probable malingering range. In personal injury settings, it has been estimated that 18–60% of clients intentionally distort their responses (Binder, 1993; Chafetz, 2008; Mittenburg, Patton, Canyock, & Condit, 2002).

As mentioned earlier, several MMPI-2 scales have been designed to measure over-reporting of psychopathology symptoms. The *F* scales (*F*, *F_B*, *F_p*) are quite useful for this purpose. The *F_p* scale is the most effective indicator when the base rate for psychopathology is high (Arbisi et al., 1998), and a meta-analysis (Rogers et al., 2003) indicated that it has the highest effect size of all MMPI-2 over-reporting indicators in identifying protocols evidencing malingered psychopathology. Although relatively few research studies have examined the *F* scales' utility specifically in forensic settings, some studies have shown that the scales can differentiate individuals who have been asked to malingering on the MMPI-2 from forensic inpatients and criminal defendants (Bagby, Buis, & Nicholson, 1995; Bagby, Rogers, & Buis, 1994; Gallagher, 1997; Hawk & Cornell, 1989; Roman, Tuley, Villanueva, & Mitchell, 1990). More recent research employing "known groups" designs using the SIRS as the criterion has shown significant utility for the *F* and *F_p* scales in identifying

over-reported psychopathology (Kucharski & Duncan, 2007; Toomey, Kucharski, & Duncan, 2009).

The *FBS* is a frequently studied scale in forensic neuropsychology contexts (see Ben-Porath et al., 2009). It was proposed as an alternative to the *F* scales as the latter set are sensitive to “acting crazy” exaggeration, which presumably does not frequently occur in personal injury settings in which the individual “acts hurt” (Larrabee, 2003a). Thus, the *FBS* focuses primarily on somatic and cognitive embellishment. Although considerable debate initially took place with regard to the validity and utility of this scale (see, for example, Arbisi & Butcher, 2004; Butcher et al., 2003; Greve & Bianchini, 2004; Lees-Haley & Fox, 2004), more recent research has offered solutions and new empirical data to discount many of the serious problems initially proposed, including high rates of false positive prediction (see Ben-Porath et al., 2009; Greiffenstein, Fox, & Lees-Haley, 2006) and gender bias (Lee, Graham, Sellbom, & Gervais, 2012). We recommend using this scale for the assessment of non-credible somatic and cognitive responding, particularly in forensic neuropsychological settings.

Some individuals undergoing forensic assessments may minimize or deny psychological problems. Examples include parents undergoing a child custody evaluation or patients seeking early release from a forensic hospital. Several MMPI-2 scales mentioned earlier (*L* and *K*) can be useful in identifying under-reporting. Many studies have supported their clinical use in general (see Baer & Miller, 2002, for a meta-analysis), but few studies have examined their ability to detect defensiveness in forensic settings. Although there is no a priori reason to assume that findings from other settings would not generalize to forensic test-takers, future research should nonetheless explore specifically the utility of the *L* and *K* scales in forensic evaluations.

Another issue concerning the assessment of malingering and defensiveness involves coaching. Wetter and Corrigan (1995) found that almost 50% of attorneys and 33% of law students felt that it was their responsibility to inform their clients about the test’s Validity Scales and their operation. Such information certainly presents potential limitations for the MMPI-2’s ability to detect response distortion. Mere knowledge and professional understanding of clinical syndromes does not appear to aid in avoiding detection by the Validity Scales (Bagby et al., 1997; Rogers, Bagby, & Chakraborty, 1993; Wetter, Baer, Berry, & Reynolds, 1994); however, instruction about how the Validity Scales function does have an effect (Rogers et al., 1993; Rogers et al., 2003; Storm & Graham, 2000). Despite this effect, a large portion of coached malingerers were still correctly identified in these studies.

The Utility of the K Correction

Although it has a long history with the MMPI-2, the *K* correction, a procedure that involves adding a certain proportion of the *K* raw score to five of the Clinical Scales, has long been of questionable utility (Graham, 2011; Greene, 2010; see also Dahlstrom & Welsh, 1960). The MMPI-2’s publisher recommends that in cases where the test-takers produce deviant scores on *K* (i.e., those cases when the *K* correction is likely to significantly alter Clinical Scale scores), the non-*K*-corrected profile be reviewed in order to determine the impact of the *K* correction. Moreover, none of the supplementary sources used to refine Clinical Scale interpretation (i.e., Harris-Lingoes subscales, RC, Content, PSY-5 Scales, etc.) is *K*-corrected, confounding differences between the Clinical Scales and these supplementary sources.

Norms for the non-*K*-corrected Clinical Scale scores have been available since the MMPI-2 was first published in 1989. The Extended Score Report, an automated scoring system for the MMPI-2 distributed by Pearson Assessments, provides scores on both *K*- and non-*K*-corrected profiles. As early as 1960, Dahlstrom and Welsh (1960) raised questions about the applicability of

the K correction in a correctional setting, but it has nonetheless been used routinely because of concerns that the correlate literature is based on K -corrected scales (Dahlstrom et al., 1972). Research indicates that in clinical settings the K -corrected scores are no more valid than the non- K -corrected scores (e.g., Barthlow et al., 2002), and in non-clinical settings characterized by a significant incentive to under-report, the K correction may substantially attenuate if not altogether eliminate the validity of the Clinical Scale scores (Detrick et al., 2001; Sellbom, Fischler, & Ben-Porath, 2007). In a comprehensive analysis, Ben-Porath and Forbey (2004) examined the impact of the K correction in three clinical and three non-clinical archival samples. They concluded that the K correction did not improve Clinical Scale score validity in any setting and significantly attenuated its validity in some instances. They also reported that correlates of code types classified based on non- K -corrected scores of the Clinical Scales are similar in content but tend to be stronger in magnitude when compared to those based on K -corrected scores. Thus, MMPI-2 code types based on non- K -corrected Clinical Scales can be interpreted on the basis of the existing research literature, but with greater confidence. Therefore, it is our recommendation that forensic examiners consider the non- K -corrected profile in their MMPI-2 interpretation. If it points to a different inference than with the K -corrected profile, interpreters should rely on other MMPI-2 sources and the research findings just described in determining which inference is more likely to be correct.

Ethnic Minority Considerations

The original MMPI was developed and normed primarily with Caucasians. As a result, it was questioned whether the test would be appropriate for use with members of ethnic minorities, which sparked a plethora of studies examining group differences on MMPI scores. A review by Greene (1987) concluded that there were no systematic differences between Caucasians and any of the ethnic minorities studied on MMPI scores, and the few differences that were found were not clinically meaningful.

The MMPI-2 normative sample included a more racially diverse group, more closely matching the most recent U.S. census. However, studies have continued to examine potential differences between Caucasians and ethnic minorities on MMPI-2 scores. Hall, Bansal, and Lopez (1999) conducted a meta-analysis that revealed no systematic mean group differences between Caucasians and African Americans, Latinos, and Asian Americans. However, to study test bias it is not sufficient to examine group differences, as these differences may actually reflect real differences and not bias. Conversely, test bias may still exist in the absence of group differences. Therefore, researchers have begun to examine differential predictive validity for Caucasians and ethnic minority groups on the MMPI-2. Timbrook and Graham (1994) found that MMPI-2 scale scores predicted extra test criteria equally well for Caucasian and African American individuals in the MMPI-2 normative sample. This finding has been replicated in two community mental health centers (Castro, Gordon, Brown, Anestis, & Joiner, 2008; McNulty, Graham, Ben-Porath, & Stein, 1997). Arbisi, Ben-Porath, and McNulty (2002) examined differential predictive validity of MMPI-2 scores for Caucasian and African American inpatients using hierarchical regression analyses. They found very few differences across the Clinical and Content Scales, and where there were differences, the MMPI-2 scores appeared, unexpectedly, to underpredict psychopathology for Caucasians rather than African Americans. These differences, however, were not clinically meaningful.

A study comparing Native Americans with the MMPI-2 normative sample revealed few systematic group differences (Robin, Greene, Albaugh, Caldwell, & Goldman, 2003). The differences that did emerge were expected and likely indicated real differences in psychopathology (e.g., Scales 4, 8, 9, CYN, MAC-R, and AAS). Greene, Robin, Albaugh, Caldwell, & Goldman (2003) conducted a follow-up study in which they examined the correlations for MMPI-2 scores

with extra-test criteria for the Native American samples. They generally found expected correlates with a few exceptions (e.g., *MAC-R*). Most recently, Pace et al. (2006) found that Native American groups had significantly higher scores than the normative sample on the majority of the Validity and Clinical Scales. For this reason, they suggest using the MMPI-2 with caution in this population. Unfortunately, these researchers did not directly assess for test bias, and without external criteria, group differences are of ambiguous utility.

Recent studies with Asian Americans have focused on the effect of acculturation (the degree to which an individual has assimilated the values and behaviors of the majority culture) on MMPI-2 scores (Tsai & Pike, 2000). Tsai and Pike found that the low-acculturated students had higher scores than Caucasian students on most MMPI-2 scales, with the greatest differences on the *F* Scale and Scale 8. The high-acculturated students did not differ meaningfully from the Caucasian students. Okazaki and Sue (1995) have stressed the importance of considering the acculturation levels of Asian Americans when conducting psychological evaluations. No studies have examined differential predictive validity of MMPI-2 scores for Caucasian and Asian American individuals. This research is needed, as the acculturation-related differences just noted may also reflect adjustment difficulties that are accurately captured by the MMPI-2 scale elevations found in less acculturated individuals.

Although several research studies have focused on mean group differences between Caucasians and Hispanics (see Greene, 1987; Hall et al., 1999), no studies have examined differential predictive validity of MMPI-2 scores for Hispanic individuals. This research is also very much needed, especially given the large growth of the Hispanic population in the United States.

A few research studies have examined MMPI and MMPI-2 differences between Caucasians and ethnic minorities in forensic settings. These studies have primarily focused on mean group differences. African Americans tend to score higher on Scale 9, lower on Scale 0 (Holcomb & Adams, 1982), and higher on Scale 4 (Elion & Megargee, 1975). Ben-Porath, Shondrick, and Stafford (1995) compared MMPI-2 scores of African American and Caucasian men who underwent court-ordered psychological evaluations. The African American group scored higher on the Antisocial Practices and Cynicism Content scales. Gironde (1999) has found in the same court-ordered sample that MMPI-2 scale scores predicted external criteria equally well for Caucasian and African American men.

Finally, it is worth noting that the MMPI-2 has been translated into dozens of languages and adopted successfully in many diverse countries and cultures (e.g., Butcher, 1996). This indicates that the instrument is quite robust to cultural differences and can be applied appropriately with a broad range of individuals undergoing forensic examinations. In conclusion, additional research is needed to examine the differential predictive validity for MMPI-2 scales across ethnic minority groups, especially in forensic settings. However, the empirical findings to date suggest that MMPI-2 scales will be equally valid for Caucasian and African American test-takers, including those undergoing forensic evaluations.

Administrative and Interpretative Issues

Graham (2011) offers several recommendations concerning the administration and interpretation of the MMPI-2 in forensic settings. It is very important that individuals taking the MMPI-2 in such assessments be monitored by trained professionals or technicians. This issue becomes especially salient when the forensic examiner is faced with the task of testifying as to whether the defendant took the MMPI-2 under supervised conditions to ensure that no external influences contaminated the responses.

The standard administration requires a test booklet or computer, but if the defendant cannot read at a sixth-grade level there are audio versions of the MMPI-2 available on cassette or CD. It

is not appropriate to read MMPI–2 items to the test-taker as a variety of non-verbal behaviors might influence the answers.

Short forms of the MMPI–2 should be avoided. Research has generally indicated that most short forms of the MMPI and MMPI–2 are not as valid as the full-length version (Graham, 2011). One potential short form would be to administer only the first 370 items of the test, which include the basic Validity (*L*, *F*, *K*) and Clinical Scales. We advise against this practice, because a number of standard components of the test (e.g., the *VRIN*, *TRIN*, *F_p*, Content, PSY–5, and RC Scales) cannot be scored on the basis of the abbreviated administration.

Scoring the MMPI–2 in forensic assessments can be accomplished using any of the standard techniques described earlier. These include hand-scoring with templates, or automatic scoring with a number of available systems. Because of the possibility of error and the tendency to score fewer scales when hand-scoring, this approach, though possible, is not recommended for forensic practice. Automated scoring is more time-efficient and less error-prone, provided that the forensic examiner use only officially licensed and approved software. In addition to raising questions about the knowledge and credibility of the examiner, reliance on unlicensed (i.e., bootleg) software to score an MMPI–2 often results in scoring errors that could significantly skew the results.

As indicated, the MMPI–2 interpretation can be informed by an extensive empirical literature. Implications of this literature are considered effectively in standard interpretative texts such as Graham (2011) and Greene (2010). However, in the adversarial forensic environment it behooves forensic examiners to be able to cite original empirical research, not just secondary sources, to support their opinions and conclusions.

Specific Forensic Applications

Competency to Stand Trial

Psychologists are often asked to provide opinions regarding a defendant's competency to stand trial. Current criteria for competency were first outlined in *Dusky v. United States* (1960), which established that to be considered competent to stand trial, defendants must (1) reasonably understand the legal proceedings they face including the roles of various participants in this process, and (2) have the ability to meaningfully assist in their own defense. These criteria are of a functional nature and Ogloff (1995) has stated that the MMPI–2 may have little to offer in this regard. We disagree with this statement for two main reasons (see also Stafford & Sellbom, 2012, for a detailed discussion of psychological testing in competency evaluations more broadly). First, a threshold question in conducting competency to stand trial evaluations is whether a defendant has a mental condition that could impair these functions. Second, the possibility of malingering must always be considered in conducting these evaluations. Thus, although the MMPI–2 cannot directly answer the psycholegal question of competency (other information is needed for that purpose), there are several ways in which MMPI–2 data can add to the information upon which a competency opinion is based. The Validity Scales can provide relevant information regarding the possibility of malingering, or the lack thereof, and the MMPI–2 substantive scales (Clinical, RC, Content, etc.) can provide information regarding the likelihood that the individual has a mental condition that could impair their competency to stand trial. For example, Stafford and Wygant (2005) reported that psychotic symptoms are frequently found in defendants who are incompetent to stand trial. The MMPI–2 has a number of valid scales to predict psychotic symptomatology.

The MMPI–2 has been specifically examined with individuals adjudicated competent or incompetent to stand trial, though the literature has been somewhat sparse in this regard. An early study by Cooke (1969) found that 93 individuals adjudicated incompetent scored significantly

higher on the *F* Validity Scale, as well as Clinical Scales 2, 3, 4, 6, 7, and 8 relative to 122 patients who had been found competent to stand trial. Maxson and Neuringer (1970) replicated these findings in a larger sample ($n = 594$), but found only differences on the *F* scale and Clinical Scale 6—a measure of paranoid delusions. In a similar design at a Canadian inpatient unit, Rogers, Gillis, McMain, & Dickens (1988) found only negligible differences on Clinical Scale 9—a measure of manic symptomatology. Most recently, Otto et al. (1998) reported moderate negative correlations between the MMPI-2 Psychoticism scale and the Reasoning and Understanding domains of the MacCAT-CA in a large sample of individuals evaluated for competency to stand trial. These results have been summarized in a recent meta-analysis. Pirelli, Zapf, and Gottdiener (2011) reported that the effect sizes for MMPI scales *F*, 6, and 8 were in the small to moderate range (.33, .39 and .33, respectively), as incompetent defendants produced higher than average scores on these scales that assess symptoms of psychopathology. Thus, confirming Stafford and Wygant's (2005) findings, elevations on MMPI-2 scales reflective of severe psychopathology—in particular, psychotic symptomatology—appear to best differentiate competent and incompetent defendants.

Criminal Responsibility

In order to be found guilty of a crime, a defendant must have the capacity to form criminal intent. Otherwise, in the presence of a major mental illness at the time of the offense, they are typically determined to be not guilty by reason of insanity (NGRI). Over the past two decades, several definitions have been employed concerning the determination of sanity. These generally require a mental incapacity that affects behavior. Most standards also include cognitive incapacity that influences the defendant's ability to appreciate the nature or wrongfulness of their actions, and/or volitional capacity that includes the inability to control behavior (Melton, Petrila, Poythress, & Slobogin, 1997).

The assessment of criminal responsibility is retrospective in nature. The forensic examiner will evaluate the defendant weeks, months, or (on rare occasions) years after the alleged criminal act. Because the determination of legal sanity requires the answering of specific questions regarding an individual's mental state at the time of the offense, the MMPI-2 and similar tests have more limited utility in NGRI evaluations than in the case of competency-to-stand-trial evaluations. Moreover, there is little research to support MMPI-2 use in answering legal questions directly related to insanity. For example, Rogers and McKee (1995) found that there is no research to suggest that insane defendants score differently from sane defendants on the MMPI-2 with the exception of the antisocial behavior scales (e.g., Scale 4, *ASP*), on which insane defendants score lower than those adjudicated sane. Barendregt, Muller, Nijman, and de Beuers (2008) replicated this finding in a Dutch study on criminal responsibility, indicating that experts who use MMPI-2 results are less likely to opine insanity in cases where individuals exhibit personality disorders, particularly of an antisocial or psychopathic nature.

Despite the questionable utility of MMPI-2 scores in these evaluations, Lally (2003) reported that the majority of the diplomate forensic examiners in his study recommended using the MMPI-2 for criminal responsibility evaluations. This finding is likely because the MMPI-2 generates information that is relevant to the evaluation even though it does not directly address the psycholegal questions. The MMPI-2 Validity Scales can assist in the identification of *current* attempts to malingering psychopathology. This is noteworthy because if a defendant attempts to malingering psychopathology in the course of an NGRI evaluation, it occurs at the time of the evaluation, not that of the alleged crime. In addition, individuals who were found NGRI are likely to have chronic and severe disorders that would likely be reflected in a current MMPI-2 protocol. Finally, such defendants have sometimes had prior contact with the mental health system, and pre-offense

MMPI–2 protocols may exist that can be compared with a current protocol to further establish the presence or absence of a severe disorder prior to an alleged offense.

Risk Assessment

Risk assessment is generally defined as the prediction of future dangerousness towards oneself or others. Researchers (e.g., Heilbrun & Heilbrun, 1995; Monahan, 2003) have indicated that a variety of factors should be considered when making a dangerousness prediction. These can broadly be grouped into static and dynamic factors. The *static* factors refers to those historical and demographic factors that are unchangeable, including gender, age, arrest history, juvenile delinquency, number of past violent offenses, age at first violent offense, psychopathic personality, to mention a few. The *dynamic* risk factors center on clinical and risk management factors that are amenable to treatment and thus changeable in each individual, and arguably more interesting from a clinical perspective (Douglas & Skeem, 2005). After all, clinicians are interested in the prevention of violence (Hart, 1998). These include clinical risk factors, exemplified by active symptoms of mental illness, substance abuse, negative affectivity (especially anger), interpersonal difficulties, antisocial attitudes, and negative treatment attitudes (Douglas & Skeem, 2005), as well as future risk management (including contextual) factors, such as availability of weapons and victims, social support, and treatment monitoring (Monahan, 2003; Webster & Hucker, 2007). Given that most conceptual and measurement models of violence risk assessment now incorporate both static and dynamic risk factor domains (e.g., Monahan, 2003; Webster & Hucker, 2007), it is important for clinicians to consider how the MMPI–2 can add useful information from this perspective as well.

Although MMPI–2 scale scores are likely of limited utility in predicting who will be violent in the future, they can provide some information that can aid in such prediction with regard to both static and dynamic risk factors. With regard to static risk factors, the MMPI–2 can likely be of most use in the assessment of personality psychopathology. The MMPI–2 PSY–5 scales reflect a dimensional model of personality disorder that has undergone extensive validity research (Harkness, Finn, McNulty, & Shields, 2012). The underlying constructs indexed by the PSY–5 scales are conceptual and empirical cognates of those proposed for the *DSM–5* (Anderson et al., in press; Harkness et al., 2012). Furthermore, several other scales, such as Clinical Scales 4 and 9, *RC4* and *RC9*, and *ASP*, can provide assessment of antisocial personality disorder traits (Lilienfeld, 1996; Sellbom, Ben-Porath, & Stafford, 2007).

Along the theme of static risk factor assessment, Heilbrun and Heilbrun (1995) have made some recommendations as to how the MMPI–2 can aid in risk assessments. One of these is in the assessment of psychopathy, which is one of the single best predictors of violence risk (see, for example, Monahan, 2006). Although Heilbrun (1979) recommended high scores on Scale 4 coupled with low intelligence as a marker for high risk, we would caution against this approach. Clinical Scale 4 is too heterogeneous and does not correlate strongly with other measures of psychopathy (Hare, 1985; Harpur, Hare, & Hakstian, 1989). On the other hand, the MMPI–2 *RC* Scales do provide valid information for psychopathy assessment, especially considering high scores on *RC4* and *RC9* coupled with low scores on *RC2* and *RC7* (Sellbom, Ben-Porath, Lilienfeld, Patrick, & Graham, 2005; Sellbom, Ben-Porath, & Stafford, 2007).

The MMPI–2 has several scales that can identify many of the dynamic risk factors discussed earlier. In terms of active symptoms of major mental illness, Clinical Scales 6, 8, and 9, and, with more precision, *RC6*, *RC8*, and *RC9*, along with the *BIZ* Content Scale, can provide valid assessment of psychotic and manic illness symptoms. Several scales provide broad-based measurement of negative affectivity, including Clinical Scale 7, *RC7*, and *NEGE*, which constitutes an important dynamic risk factor (Douglas & Skeem, 2005). More specifically, the MMPI–2

Content Scales can provide more fine-tuned assessment of specific negative emotions, including anger-proneness, which is an important dynamic risk factor (Douglas & Skeem, 2005; Monahan, 2003). Furthermore, the MMPI-2 Supplementary Scales *MAC-R*, *AAS*, and *APS* provide assessment of alcohol and substance abuse. The PSY-5 scale *DISC* is a particularly good predictor of impulse control problems (Harkness et al., 2012). Other MMPI-2 scales, such as the Content Component Scale Antisocial Attitudes (*ASP1*), can also assist in such measurement.

Only limited research on the predictive validity of MMPI-2 scale scores concerning violence risk assessment exist. In correctional settings, Megargee, Bohn, Meyer, and Sink (1979) have developed a configural classification system with the MMPI, and more recently the MMPI-2, Clinical Scales. They found that certain groups were more likely to reoffend after release (Group Foxtrot, 64%) and act more aggressively in prison (Group Jupiter). Sellbom, Ben-Porath, Baum, Erez, and Gregory (2008) published a recent study that used the MMPI-2 RC scales in the prediction of violent recidivism among offenders undergoing treatment in a Batterers Intervention Program. They found that *RC4* and *RC9* were able to predict recidivism. For example, offenders who produced elevated scores on *RC9* were 2.5 times more likely to reoffend. The RC Scales predicted recidivism even after both historical (e.g., arrest history) and demographic (e.g., age, income) variables had been accounted for. Davis and Archer (2010) reviewed the literature on objective personality inventories in the assessment of sexual offenders. All but one of the studies focused on the original MMPI. Although they did not focus on violence risk per se, their analysis of the literature revealed that Clinical Scale 4 differentiated certain types of sexual offenders. Unfortunately, this information is of limited practical utility, as Davis and Archer (2010) point out, in that this Clinical Scale is sensitive to externalizing proclivities in general and likely does not have much specificity with regard to sexual offenders.

Finally, much more research on predicting violence risk with the MMPI-2, especially in conjunction with other important variables, is needed. The strongest studies will include a clear definition of dangerousness, and data on violent reoffending, and will examine the MMPI-2 scales' predictive abilities in conjunction with other static and dynamic risk factors. Moreover, informative studies would also address how the MMPI-2 can provide information to inform ratings of structured professional guides of violence risk assessment, such as the Historical/Clinical/Risk Management-20 (HCR-20; Webster, Douglas, Eaves, & Hart, 1997).

Child Custody

The Uniform Marriage and Divorce Act (UMDA) 1979 states that child custody arrangements should be made in the "best interest of the child." Consideration of a child's best interest typically includes the mental health of all parties involved. As is the case with all other forensic assessments, psychological testing is not sufficient for making such decisions. Interviews with parents and children, review of relevant records, and interviews with less subjective third parties (e.g., neighbors, teachers) provide important information as well. In terms of testing, surveys have clearly established that the MMPI-2 is the most frequently used psychological test in child custody evaluations (e.g., Bow et al., 2006; Hagen & Castagna, 2001). Bow et al.'s (2006) survey results indicated that child custody evaluators believe that the MMPI-2 would also be the most likely to withstand a *Daubert* challenge among all psychological tests considered.

As might be expected, most individuals undergoing child custody evaluations approach the examination defensively, and attempt to make a favorable impression (Bagby, Nicholson, Buis, Radovanovic, & Fidler, 1999; Bathurst, Gottfried, & Gottfried, 1997; Cooke, 2010). Cooke (2010) examined the effects of socioeconomic status and education on MMPI-2 *L*, *K*, and *S* scores and concluded that high scores (65–70*T*) on these scales should not invalidate the MMPI-2 profiles in

high-educated individuals, as they still fall within one standard deviation of the mean. We concur, and argue that psychologists should consider the degree to which defensive responding is generally expected in child custody settings, and compare a test-taker's MMPI-2 protocol to child custody groups in addition to using standard norms. If an individual's Validity Scale score (*L* or *K*) is greater than one standard deviation higher compared to what is typically expected in these settings, then that score should warrant particular suspicion of under-reporting.

Four general types of MMPI-2 protocols are often obtained in child custody evaluations: (1) a defensive protocol with no elevations on the substantive scales, (2) a non-defensive protocol with no elevations on the substantive scales, (3) a defensive profile with substantive scale elevations, and (4) a non-defensive protocol with substantive scale elevations. In cases 3 and 4, the substantive elevations may be relied upon in describing characteristics associated with the elevated indicators. As discussed earlier, caution should be exercised in relying on elevations on *K*-corrected Clinical Scale scores when the non-*K*-corrected scores are not elevated. Of course, in case 3 the possibility remains that the test-taker is only admitting part of their maladaptive functioning. In case 2, cautious inferences about the absence of significant maladjustment may be reached. Caution is indicated because, generally speaking, MMPI-2 scale scores are more informative when they are elevated rather than non-elevated. Finally, in case 1 no information can be reached about the test-taker's adjustment because it is impossible to determine whether the absence of elevation is a product of defensiveness, good adjustment, or some combination of both.

Only limited research has been conducted with the MMPI-2 in child custody settings. Ezzo, Pinsoneault, and Evans (2008) recently showed that parents documented to engage in child maltreatment had substantially higher scores on MMPI-2 Clinical Scales relative to those who did not engage in such maltreatment. Although it is unlikely that MMPI-2 research will yield any particular "profiles" indicative of a bad parent (Otto & Collins, 1995), it would nonetheless be important to examine whether individual scales might be more effective in highlighting characteristics associated with poor outcomes after custody decisions have been made.

Personal Injury

Psychologists are often asked to conduct psychological evaluations with individuals who have been exposed to a traumatic event and claim to have suffered psychological or emotional difficulties as a result. Pope, Butcher, and Seelen (2006) raise three important questions that need to be addressed in personal injury evaluations: (1) the actual presence of psychological or emotional problems, (2) whether the trauma caused the psychological problems as opposed to a preexisting condition or subsequent events, and (3) the likely course of recovery.

The MMPI-2 can aid to some degree in answering these questions. The Validity Scales can help determine whether the individual is malingering. A substantial amount of research has supported the use of the Validity Scales in this regard, with a particular focus in forensic neuropsychological settings. As mentioned earlier, Nelson and colleagues (2006, 2010) have published meta-analyses that reveal that the MMPI-2 Validity Scales show substantial utility in differentiating those deemed to be malingering from those who are not malingering. The *FBS* scale performs the best in these meta-analyses. Gervais, Ben-Porath, Wygant, and Green (2007) recently developed the Response Bias Scale (*RBS*) by identifying items that specifically differentiated those who failed symptom validity tests from those who did not in a very large forensic disability sample. This scale was too new to be included in Nelson et al.'s meta-analyses, and subsequent empirical evaluations of the *RBS* indicate that it typically performs among the best in identifying malingering of cognitive complaints in these settings (Gervais, Ben-Porath, Wygant, & Green, 2008; Tolin, Steenkamp, Marx, & Litz, 2010; Tsushima, Geling, & Fabregas, 2011; Wygant et al., 2010). Finally,

it should be noted that when malingering of PTSD is suspected in these settings, the over-reporting scales focused on psychopathology (F , F_B , and, in particular, F_p) tend to perform better than FBS (e.g., Bury & Bagby, 2002; Efendov, Sellbom, & Bagby, 2008; Elhai, Gold, Sellers, & Dorfman, 2001).

The MMPI-2 can provide useful information about psychological and emotional difficulties once the profile has been determined to be interpretable. Many personal injury claimants specifically indicate that they may suffer from post-traumatic stress disorder (PTSD). Although individuals with PTSD often produce multiple Clinical Scale elevations (especially 2, 4, 7 and 8) (Graham, 2011), these profiles do not discriminate PTSD individuals from others with extensive psychopathology (Keane et al., 1984). Other researchers have found that there is no prototypical PTSD profile for personal injury litigants (Platt & Husband, 1987). Keane et al. (1984) developed a PTSD scale (PK), but its utility presently is limited to combat veterans. Thus, as is the case with other disorders, the MMPI-2 is best relied upon to generate information about possible symptoms associated with PTSD rather than to predict presence or absence of the disorder. Indeed, recent research has shown that the RC Scales, and in particular RC Demoralization, is sensitive to PTSD symptoms in clinical and disability settings (Sellbom, Ben-Porath, & Bagby, 2008; Sellbom, Lee, Ben-Porath, Arbisi, & Gervais, 2012).

MMPI-2 data provide very limited information on the association between traumatic events and any psychopathology reflected in the test protocol. As is the case with NGRI evaluations, occasionally a pre-event MMPI-2 profile may exist that can shed some light on the individual's psychological functioning prior to the trauma. In such cases, limited inferences about the relation between a traumatic event and current psychopathology may be indicated, but would need to be corroborated by collateral information.

Correctional Settings

The rate of incarceration in the United States continues to increase exponentially. The total prison population now exceeds 2 million inmates, and is associated with an annual cost of \$70 billion (Bureau of Justice Statistics from 2006). A large proportion of prison inmates suffer from mental illnesses or severe personality disorders (Department of Justice statistics from 2006); therefore, assessment of personality and psychopathology characteristics in offenders is an extremely worthwhile endeavor for both efficiently allocating mental health treatment resources and security risk classification (M. S. Black et al., 2004; Wilkinson, 2000).

In a landmark prospective study, Hathaway and Monachesi (1953, 1957) found that Scale 4 was particularly valid for predicting juvenile delinquency in both boys and girls. More recent research concerning the use of MMPI-2 in correctional settings has been directed by Megargee and Carbonell (1995), who have emphasized the importance of efficient classification within the correctional system, primarily because of lack of resources. This research focused initially on aggression and conduct problems among prison inmates, and eventually led to the classification system mentioned earlier. Megargee and colleagues (1979) developed a configural system through cluster analysis of the Clinical Scales, yielding 10 profile types. They were able to classify 85–95% of MMPI profiles into one of 10 groups. This classification system has also been adapted for the MMPI-2 (see Megargee, 1994). The classification system had originally been developed for male prisoners; Megargee (1997) adapted the classification rules for women as well. Almost 98% of women offenders could be classified. Extra-test correlates, including institutional infractions, psychological adjustment, educational, work and health histories, prison adjustment, and recidivism were generated for each of the ten types. Megargee, Carbonell, Bohn, and Sliger (2001) summarize these for the MMPI-2.

Most recently, Forbey, Ben-Porath, and Gartland (2009) examined a computer-adaptive version of the MMPI-2 (MMPI-2-CA) in a large male correctional sample. These researchers found a considerable reduction in the number of items administered, as well as corresponding time savings for the MMPI-2-CA relative to standard paper-and-pencil administration of the test. They also demonstrated comparability with regard to test-retest reliability and correlations with extra-test criterion measures for scale scores derived from the MMPI-2-CA relative to standard format.

CASE EXAMPLE

The following case study is designed to illustrate a number of the points made in this chapter about forensic applications of the MMPI-2. In order to mask the identity of the forensic examinee whose MMPI-2 protocol is analyzed, the background information presented is based on a composite of several actual forensic evaluations conducted by the first author.

Background

M.J. is a 52-year-old African American man referred by the court for a second opinion evaluation of the hospital recommendation for conditional release into the community following a not guilty by reason of insanity (NGRI) adjudication for aggravated murder approximately 20 years earlier. He has been committed to a state-operated forensic psychiatric hospital since that time. With regard to the index offense, Mr. J had shot his father dead because he believed that his father had been hypnotized by communist spies and was in imminent danger of murdering Mr. J's mother. At the time of the offense, he also exhibited significant persecutory delusions: that his place of employment was hiding assets from him, that local law enforcement agencies were specifically targeting him, and that Russians and communists had infiltrated local universities and government agencies.

A review of Mr. J's history revealed that he had been hospitalized for Schizophrenia, Paranoid Type, on at least two occasions prior to the index offense. He did not use alcohol on a regular basis, nor had he ever abused any illicit drugs. He did not have a criminal arrest history prior to the index offense.

At the time of the evaluation, it was clear that he continued to exhibit substantial persecutory delusions about imposters, communist spies, his former workplace, and local sheriff's and police departments. His delusions had also evolved over the past 20 years to incorporate various governors and FBI task forces who were secretly working to free him from the hospital. Mr. J believes that his hospitalization is a cover-up to prevent him from exposing the many secrets that comprise his delusional network. A review of hospital treatment records indicated that Mr. J has failed to respond to any psychotropic, psychoeducational, or psychotherapeutic treatments available. However, because he has not been violent during his hospitalization, and the staff believe they have been able to help him gain some insight into his illness, it is their opinion that he is ready for conditional release into the community under court jurisdiction. It is noteworthy that two prior requests for conditional release have been denied. Mr. J reports that he believes he is ready to be released back into the community, and if this were to happen, he would attempt to return to his old place of employment, infiltrate, and eventually obtain the assets they are hiding from him.

MMPI-2 Results and Interpretation

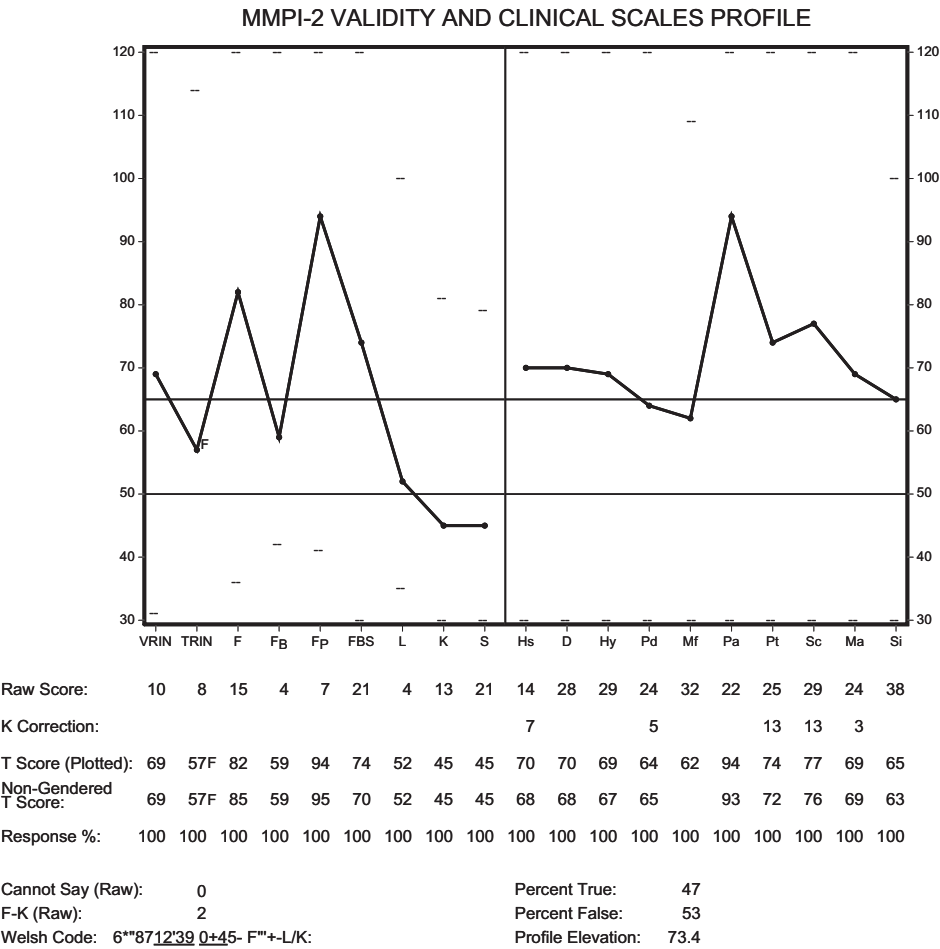
As part of the court-ordered evaluation, M.J. was administered the MMPI-2, and produced the protocol that appears in Figure 2.1. The protocol was scored using the MMPI-2 Extended Score Report (ESR) distributed by Pearson Assessments. Page 2 of the ESR (the first page reproduced in Figure 2.1) provides scores on the standard MMPI-2 Validity Scales and the *K*-corrected Clinical Scale profile. Examination of the Cannot Say score indicates that the test-taker responded to all test items and, as a result, the last row of numbers under the profile (labeled "Response %") shows that 100% of the items were answered on all of the scales. Both gendered and non-gendered *T*-scores (discussed earlier) are provided for all of the MMPI-2 scales and, as reported by Ben-Porath and Forbey (2004), differences between the two sets of scores are negligible. Therefore, the focus of this interpretation will be on the gender-based *T*-scores, which are also the ones plotted throughout the protocol.

Examination of scores on the consistency scales (*VRIN* and *TRIN*) indicates that Mr. J responded with some inconsistency to the MMPI-2 items but not to a degree that would invalidate the profile. Of particular importance here is the very high score (94*T*) on the *F_p* scale, which is very close to where the profile could be invalidated due to over-reporting. However, in this instance it behooves the examiner to consider, first, the context in which he took the MMPI-2, and second, Mr. J's level of psychopathology. Most individuals undergoing second opinion evaluations would have an incentive to under-report rather than over-report their psychopathology. The fact that Mr. J, despite his desire to leave the hospital, did not approach the test in the expected manner is very informative of his level of insight into his mental illness. Indeed, Mr. J's mental illness is so severe that he remains symptomatic despite over 20 years of treatment. It was therefore concluded that Mr. J's profile was valid for interpretation with caution.

Mr. J's scores on the *K*-corrected (page 2 of the ESR) and non-*K*-corrected (page 3 of the ESR) Clinical Scale profile revealed a fairly similar pattern of elevations, with the exception of Clinical Scale 4, which is elevated in the latter profile. In addition to the extreme elevation on Clinical Scale 6, which is likely reflective of his delusional disorder, a number of additional hypotheses must also be considered in light of elevations on Clinical Scales 1, 2, 3, 4, 7, 8, 9, and 0. Indeed, these scores would suggest in isolation that Mr. J likely exhibits symptoms associated with most major domains of psychopathology. Without additional sources from the MMPI-2 and no well-defined code type, these hypotheses are difficult to rule out.

The RC Scales (page 4 of the ESR) provide a much more defined picture of Mr. J's personality and psychopathology functioning. Indeed, his only elevation is on *RC6* (Ideas of Persecution; 82*T*), with subclinical scores on *RCd* (Demoralization; 64*T*) and *RC8* (Aberrant Experiences; 63*T*). It also provides a likely explanation for many of the elevations on the Clinical Scales. Sellbom, Ben-Porath, McNulty et al. (2006) showed that demoralization accounts for many of the cases in which Clinical and RC Scale counterparts are discrepant with regard to elevation status. In other words, most of those Clinical Scales elevations can likely be attributed to demoralization rather than the core features associated with the constructs underlying those scales.

Tellegen et al. (2003) indicate that individuals who produce elevated scores on *RC6* "feel mistreated or picked on, and may have difficulties forming trusting relationships. Those with particularly high scores . . . are likely to be characterized by paranoid thinking that may be symptomatic of a schizophrenic or delusional disorder" (p. 56).

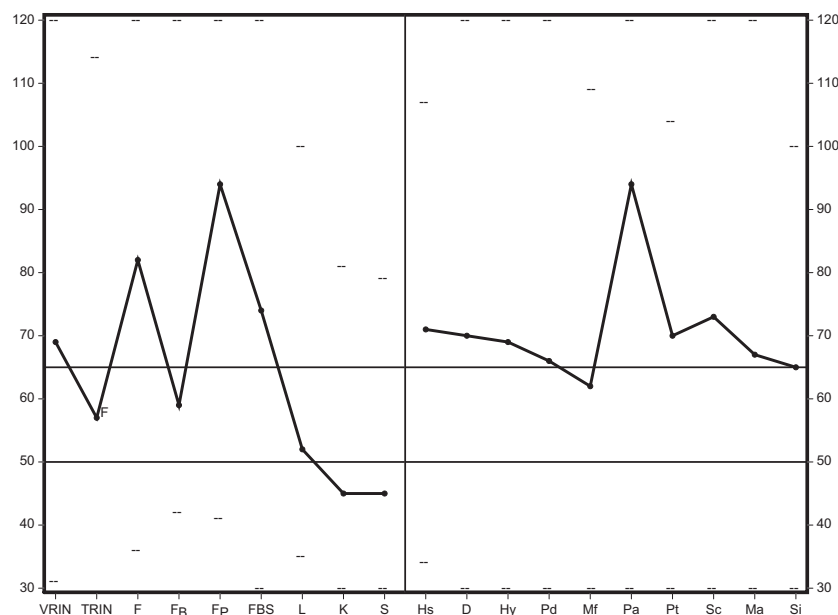


The highest and lowest T scores possible on each scale are indicated by a "--".
For information on FBS, see Ben-Porath, Y. S., & Tellegen, A. (2006). The FBS: Current Status, a report on the Pearson web site (www.pearsonassessments.com/tests/mmpi_2.htm).

FIGURE 2.1 MMPI-2 Extended Score Report.

Copyright © 1989, 1994, 2000, 2003, 2006 by the Regents of the University of Minnesota. All rights reserved. Portions reproduced from the MMPI-2 test booklet. Copyright © 1942, 1943 (renewed 1970), 1989 by the Regents of the University of Minnesota. All rights reserved. Portions excerpted from the MMPI-2 Manual for Administration, Scoring, and Interpretation, Revised Edition. Copyright © 2001 by the Regents of the University of Minnesota. All rights reserved. Used by permission of the University of Minnesota Press. "MMPI" and "Minnesota Multiphasic Personality Inventory" are trademarks owned by the Regents of the University of Minnesota.

MMPI-2 NON-K-CORRECTED VALIDITY/CLINICAL SCALES PROFILE



Raw Score:	10	8	15	4	7	21	4	13	21	14	28	29	24	32	22	25	29	24	38
T Score (Plotted):	69	57F	82	59	94	74	52	45	45	71	70	69	66	62	94	70	73	67	65
Non-Gendered T Score:	69	57F	85	59	95	70	52	45	45	68	68	67	67		93	68	72	68	63
Response %:	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100

Cannot Say (Raw): 0 Percent True: 47
 Profile Elevation: 72.5 Percent False: 53

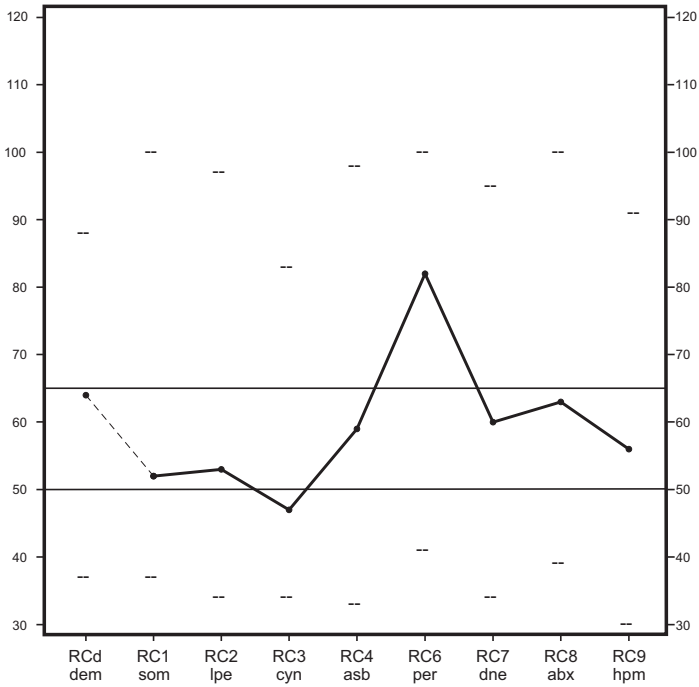
The highest and lowest T scores possible on each scale are indicated by a "--".

Non-K-corrected T scores allow interpreters to examine the relative contributions of the Clinical Scale raw score and the K correction to K-corrected Clinical Scale T scores. Because all other MMPI-2 scores that aid in the interpretation of the Clinical Scales (the Harris-Lingoes subscales, Restructured Clinical Scales, Content and Content Component Scales, PSY-5 Scales, and Supplementary Scales) are not K-corrected, they can be compared most directly with non-K-corrected T scores.

For information on FBS, see Ben-Porath, Y. S., & Tellegen, A. (2006). The FBS: Current Status, a report on the Pearson web site (www.pearsonassessments.com/tests/mmpi_2.htm).

The MMPI-2 Content Scales (page 5 of the ESR) show several elevations that further contribute to the clinical picture. Mr. J produced an elevated score on Bizarre Mentation (77T), and the Content Component scales indicate that *BIZ1* (Psychotic Symptomatology) should be emphasized in this interpretation. In addition, he also produced elevations on Anxiety and Fears, indicating some general apprehensiveness and generalized fearfulness (based on *FRS1*) which is consistent with his delusional network. He also produced an elevated score on the Type A Behavior scale, which reflects irritability, hostility, grudging attitudes, competitive drive, and high energy, but is also consistent with paranoid ideation in clinical settings (Graham,

MMPI-2 RESTRUCTURED CLINICAL SCALES PROFILE



Raw Score:	11	3	5	5	9	9	10	5	16
T Score (plotted):	64	52	53	47	59	82	60	63	56
Non-Gendered T Score:	62	51	54	47	62	94	57	63	58
Response %:	100	100	100	100	100	100	100	100	100

The highest and lowest T scores possible on each scale are indicated by a "--".

LEGEND

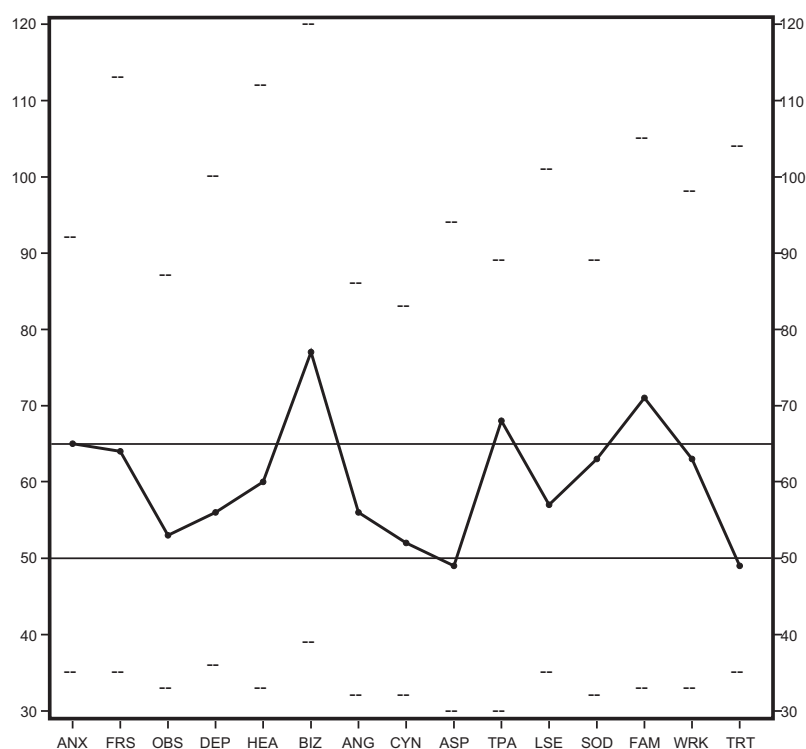
dem = Demoralization	cyn = Cynicism	dne = Dysfunctional Negative Emotions
som = Somatic Complaints	asb = Antisocial Behavior	abx = Aberrant Experiences
lpe = Low Positive Emotions	per = Ideas of Persecution	hpm = Hypomanic Activation

For information on the RC scales, see Tellegen, A., Ben-Porath, Y.S., McNulty, J.L., Arbisi, P.A., Graham, J.R., & Kaemmer, B. 2003. The MMPI-2 Restructured Clinical (RC) Scales: Development, Validation, and Interpretation. Minneapolis: University of Minnesota Press.

2011). Finally, Mr. J also elevated the Family Problems scale, which should be focused on familial alienation rather than discord per the Content Component Scales.

The MMPI-2 Supplementary Scales (page 6 of the ESR) show elevations on *A*, *Mt*, *Pk*, and *MDS*, which are all consistent with general emotional distress and demoralization (Graham, 2011). Importantly, Mr. J also produced an elevated score on Hostility (66*T*), which reflects “higher levels of experienced anger and overtly hostile behavior” (p. 206), which is important

MMPI-2 CONTENT SCALES PROFILE



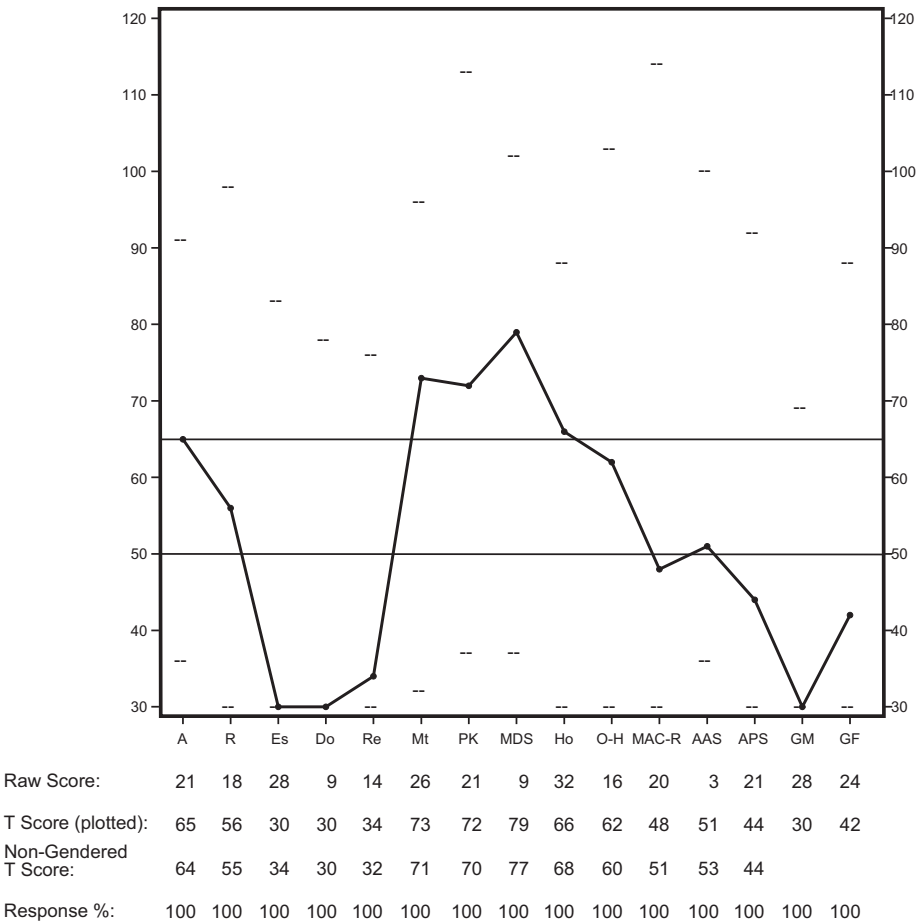
Raw Score:	12	8	6	7	9	10	8	12	8	14	7	14	13	14	4
T Score (plotted):	65	64	53	56	60	77	56	52	49	68	57	63	71	63	49
Non-Gendered T Score:	63	57	52	55	58	77	56	53	51	70	55	63	69	61	49
Response %:	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100

The highest and lowest T scores possible on each scale are indicated by a "--".

given the possibility that he might perceive himself to be threatened once released into the community.

Finally, the PSY-5 Scales (page 7 of the ESR) show an elevation on Psychoticism (PSYC). This scale reflects a general disconnect from reality which is grounded in a theoretical model of individual differences as an underlying dispositional proclivity rather than any particular diagnostic category (e.g., Harkness, 2012). This elevation is consistent with Mr. J's chronic presentation of psychosis over the past 20 years. No other PSY-5 scale elevations were observed.

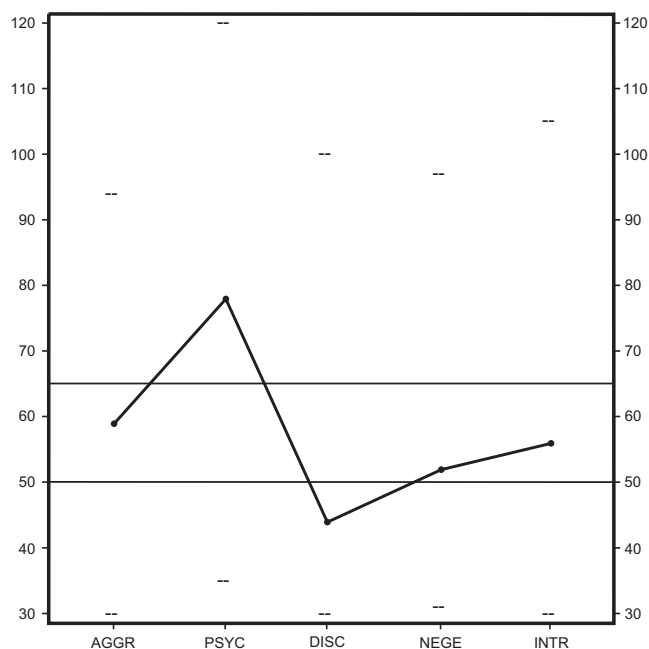
MMPI-2 SUPPLEMENTARY SCALES PROFILE



The highest and lowest T scores possible on each scale are indicated by a "--".

Considered in their totality, the MMPI-2 substantive scale results are highly consistent with a marked presence of current delusional symptomatology, particularly of a persecutory nature. Anxious apprehension and generalized fearfulness are likely reflective of paranoid hypervigilance, and expressive anger and potential for overt hostile behavior further add to the clinical picture. Diagnostically, they suggest the possibility of Schizophrenia, Paranoid Type. These diagnoses should be considered by determining whether Mr. J meets *DSM-IV* diagnostic criteria for these or other disorders.

MMPI-2 PSY-5 SCALES PROFILE



Raw Score:	11	12	12	11	14
T Score (plotted):	59	78	44	52	56
Non-Gendered T Score:	62	78	49	51	57
Response %:	100	100	100	100	100

The highest and lowest T scores possible on each scale are indicated by a "--".

The MMPI-2 results represent a useful addition to this risk assessment. They confirm the presence of significant persecutory delusions, but also introduce the possibility that any perceived threat might result in overt hostile behavior. Indeed, previous behavior has indicated the capacity for such behavior during a threatening situation. The current case also illustrates the importance of examining dynamic risk factors in risk evaluations, for which MMPI-2 results are particularly helpful. Mr. J would have scored low on most actuarial risk assessment instruments that emphasize static factors, but his active symptoms of major mental illness, negative affectivity, hypervigilance to threats, lack of insight, poor treatment response, and poor social support constitute major risk factors in this particular case. Furthermore, as mentioned earlier, MMPI-2 data cannot be relied upon to generate directly an ultimate

MMPI®-2 Extended Score Report
09/27/2007, Page 8

ID: 2007306

ADDITIONAL SCALES

(to be used as an aid in interpreting the parent scales)

	Raw Score	T Score	Non-Gendered T Score	Resp %
Harris-Lingoes Subscales				
Depression Subscales				
Subjective Depression (D ₁)	13	66	64	100
Psychomotor Retardation (D ₂)	7	59	58	100
Physical Malfunctioning (D ₃)	5	67	65	100
Mental Dullness (D ₄)	7	72	71	100
Brooding (D ₅)	4	62	60	100
Hysteria Subscales				
Denial of Social Anxiety (Hy ₁)	2	40	40	100
Need for Affection (Hy ₂)	8	55	55	100
Lassitude-Malaise (Hy ₃)	11	88	86	100
Somatic Complaints (Hy ₄)	4	57	55	100
Inhibition of Aggression (Hy ₅)	2	40	39	100
Psychopathic Deviate Subscales				
Familial Discord (Pd ₁)	7	84	82	100
Authority Problems (Pd ₂)	3	47	50	100
Social Imperturbability (Pd ₃)	1	33	34	100
Social Alienation (Pd ₄)	8	71	71	100
Self-Alienation (Pd ₅)	6	63	63	100
Paranoia Subscales				
Persecutory Ideas (Pa ₁)	10	100	100	100
Poignancy (Pa ₂)	5	68	67	100
Naivete (Pa ₃)	6	56	55	100
Schizophrenia Subscales				
Social Alienation (Sc ₁)	10	80	78	100
Emotional Alienation (Sc ₂)	2	59	59	100
Lack of Ego Mastery, Cognitive (Sc ₃)	5	72	73	100
Lack of Ego Mastery, Conative (Sc ₄)	7	76	76	100
Lack of Ego Mastery, Defective Inhibition (Sc ₅)	3	61	60	100
Bizarre Sensory Experiences (Sc ₆)	2	51	50	100
Hypomania Subscales				
Amorality (Ma ₁)	2	50	52	100
Psychomotor Acceleration (Ma ₂)	8	63	64	100
Imperturbability (Ma ₃)	1	35	36	100
Ego Inflation (Ma ₄)	7	76	75	100
Social Introversion Subscales				
Shyness/Self-Consciousness (Si ₁)	10	65	64	100
Social Avoidance (Si ₂)	4	54	55	100
Alienation--Self and Others (Si ₃)	10	65	64	100

psycholegal opinion; however, these results are consistent with a rating of high risk for future violence in Mr. J's case.

This case illustrates several MMPI-2 interpretive issues addressed in this chapter. First, Validity Scale interpretation needs to consider other sources before opinions about the presence

MMPI-2 Extended Score Report
09/27/2007, Page 9

ID: 2007306

Content Component Scales		Non-Gendered			
		Raw Score	T Score	T Score	Resp %
Fears Subscales					
Generalized Fearfulness (FRS ₁)		3	71	65	100
Multiple Fears (FRS ₂)		5	59	54	100
Depression Subscales					
Lack of Drive (DEP ₁)		2	51	51	100
Dysphoria (DEP ₂)		2	58	55	100
Self-Depreciation (DEP ₃)		2	55	55	100
Suicidal Ideation (DEP ₄)		0	45	46	100
Health Concerns Subscales					
Gastrointestinal Symptoms (HEA ₁)		1	57	55	100
Neurological Symptoms (HEA ₂)		2	54	52	100
General Health Concerns (HEA ₃)		3	64	64	100
Bizarre Mentation Subscales					
Psychotic Symptomatology (BIZ ₁)		5	94	95	100
Schizotypal Characteristics (BIZ ₂)		3	60	61	100
Anger Subscales					
Explosive Behavior (ANG ₁)		3	58	60	100
Irritability (ANG ₂)		4	56	56	100
Cynicism Subscales					
Misanthropic Beliefs (CYN ₁)		5	47	48	100
Interpersonal Suspiciousness (CYN ₂)		7	66	68	100
Antisocial Practices Subscales					
Antisocial Attitudes (ASP ₁)		6	49	51	100
Antisocial Behavior (ASP ₂)		2	52	56	100
Type A Subscales					
Impatience (TPA ₁)		4	57	58	100
Competitive Drive (TPA ₂)		6	66	68	100
Low Self-Esteem Subscales					
Self-Doubt (LSE ₁)		3	54	54	100
Submissiveness (LSE ₂)		3	62	60	100
Social Discomfort Subscales					
Introversion (SOD ₁)		8	59	61	100
Shyness (SOD ₂)		5	63	62	100
Family Problems Subscales					
Family Discord (FAM ₁)		5	60	59	100
Familial Alienation (FAM ₂)		4	76	77	100

of response bias is determined. Second, the Clinical Scale profile should be interpreted only after considering other interpretative sources, including the RC Scales. The RC Scales identified a significant proclivity toward delusional thinking and the likely possibility that most of the Clinical Scale elevations were due to general emotional distress. Further, the Content and Supplementary Scales assisted with identifying specific characteristics of hypervigilance to threat and hostile behavior, which completed the clinical picture.

MMPI®-2 Extended Score Report
09/27/2007, Page 10

ID: 2007306

	Raw Score	T Score	Non-Gendered T Score	Resp %
Negative Treatment Indicators Subscales				
Low Motivation (TRT ₁)	0	42	42	100
Inability to Disclose (TRT ₂)	3	60	61	100

Linear T scores are used for Hs D Hy Pd Pa Pt Sc Ma the content scales the content component scales and the PSY-5 scales The remaining scales and subscales use linear T scores

Notes

- 1 Corresponding author: Martin Sellbom, Ph.D., Assistant Professor, Department of Psychology, 348 Gordon Palmer Hall, The University of Alabama, Tuscaloosa, AL 35487-0348, USA. Phone: +1-205-348-9936; Fax: +1-205-348-8648; E-mail: msellbom@ua.edu.
- 2 A PsycINFO search between 2006 and 2012 (the years that have passed since the writing of Graham, 2006) with the term MMPI-2 suggests that almost 600 articles and chapters were published in this span; almost 400 had "MMPI-2" in the title.

References

- Akerlind, I., Hornquist, J. O., & Bjurulf, P. (1992). Psychological factors in the long-term prognosis of chronic low back pain patients. *Journal of Clinical Psychology, 48*, 596–606.
- American Psychiatric Association. (2000). *Diagnostic and statistical manual of mental disorders* (4th ed., rev.). Washington, DC: Author.
- American Psychiatric Association. (2011). *American Psychiatric Association DSM-5 development: Personality disorders*. Retrieved from <http://www.dsm5.org/ProposedRevision/Pages/PersonalityDisorders.aspx>
- Anderson, J. L., Sellbom, M., Bagby, M., Quilty, L. C., Veltri, C. O. C., Markon, K. E., & Krueger, R. F. (in press). Examining the MMPI-2-RF PSY-5 scales for the assessment of DSM-5 personality trait dimensions. *Assessment*.
- Arbisi, P. A., & Ben-Porath, Y. S. (1995). An MMP-2 infrequent response scale for use with psychopathological populations: The Infrequency Psychopathology Scale, F(p). *Psychological Assessment, 7*, 424–431.
- Arbisi, P. A., & Ben-Porath, Y. S., & Yossef, S. (1998). The ability of Minnesota Multiphasic Personality Inventory-2 validity scales to detect fake-bad responses in psychiatric inpatients. *Psychological Assessment, 10*, 221–228.
- Arbisi, P. A., Ben-Porath, Y. S., & McNulty, J. (2002). A comparison of MMPI-2 validity in African American and Caucasian psychiatric inpatients. *Psychological Assessment, 14*, 3–15.
- Arbisi, P. A., Ben-Porath, Y. S., & McNulty, J. (2003). Empirical correlates of common MMPI-2 two-point codes in male psychiatric inpatients. *Assessment, 10*, 237–247.
- Arbisi, P. A., & Butcher, J. N. (2004). Relationship between personality and health symptoms: Use of the MMPI-2 in medical assessments. *International Journal of Clinical and Health Psychology, 4*, 571–595.
- Archer, R. P. (2006). A Perspective on the Restructured Clinical (RC) Scale Project. *Journal of Personality Assessment, 87*(2), 179–185.
- Archer, R. P., Aiduk, R., Griffin, R., & Elkins, D.E. (1996). Incremental validity of the MMPI-2 content scales in a psychiatric sample. *Assessment, 3*, 79–90.
- Archer, R. P., Buffington-Vollum, J. K., Stredny, R. V., & Handel, R. W. (2006). A survey of psychological test use patterns among forensic psychologists. *Journal of Personality Assessment, 87*, 84–94.
- Archer, R. P., Gordon, R. A., Giannetti, R., & Singles, J. M. (1988). MMPI scale clinical correlates for adolescent inpatients. *Journal of Personality Assessment, 52*, 707–721.

- Archer, R. P., Griffin, R., & Aiduk, R. (1995). MMPI-2 clinical correlates for the common codes. *Journal of Personality Assessment*, 65, 391-407.
- Ardolf, B. R., Denney, R. L., & Houston, C. M. (2007). Base rates of negative response bias and malingered neurocognitive dysfunction among criminal defendants referred for neuropsychological evaluation. *Clinical Neuropsychologist*, 21(6), 899-916.
- Avery, R. D., Mussio, S. J., & Payne, G. (1972). Relationships between MMPI scores and job performance measures of fire fighters. *Psychological Reports*, 31, 199-202.
- Baer, R. A., & Miller, J. (2002). Underreporting of psychopathology on the MMPI-2: A meta-analytic review. *Psychological Assessment*, 14, 16-26. doi:10.1037/1040-3590.14.1.16
- Bagby, R. M., Buis, T., & Nicholson, R. A. (1995). Relative effectiveness of the standard validity scales in detecting fake-bad and fake-good responding: Replication and extension. *Psychological Assessment*, 7, 84-92.
- Bagby, R. M., Nicholson, R. A., Buis, T., Radovanovic, H., & Fidler, B. J. (1999). Defensive responding on the MMPI-2 in family custody and access evaluations. *Psychological Assessment*, 11, 24-28.
- Bagby, R. M., Rogers, R., & Buis, T. (1994). Detecting malingered and defensive responding on the MMPI-2 in a forensic inpatient sample. *Journal of Personality Assessment*, 62, 191-203.
- Bagby, R. M., Rogers, R., Nicholson, R. A., Buis, T., Seeman, M. V., & Rector, N. A. (1997). Effectiveness of the MMPI-2 validity indicators in the detection of defensive responding in clinical and nonclinical samples. *Journal of Personality Assessment*, 68, 650-664.
- Barefoot, J. C., Dahlstrom, W. G., & Williams, R. B. (1983). Hostility, CHD incidence, and total mortality: A 25-year follow-up study of 255 physicians. *Psychosomatic Medicine*, 45, 59-63.
- Barendregt, M., Muller, E., Nijman, H., & de Beuers, E. (2008). Factors associated with experts' opinions regarding criminal responsibility in the Netherlands. *Behavioral Sciences and the Law*, 26, 619-631.
- Barthlow, D. L., Graham, J. R., Ben-Porath, Y. S., Tellegen, A., & McNulty, J. L. (2002). The appropriateness of the MMPI-2 K correction. *Assessment*, 9, 219-229.
- Bartol, C. R. (1991). Predictive validation of the MMPI for small-town police officers who fail. *Professional Psychology: Research and Practice*, 22, 127-132.
- Bathurst, K., Gottfried, A. W., & Gottfried, A. E. (1997). Normative data for the MMPI-2 in child custody litigation. *Psychological Assessment*, 9(3), 205-211.
- Ben-Porath, Y. S. (2012). *Interpreting the MMPI-2-RF*. Minneapolis: University of Minnesota Press.
- Ben-Porath, Y. S., & Butcher, J. N. (1989a). Psychometric stability of rewritten MMPI items. *Journal of Personality Assessment*, 53, 645-653.
- Ben-Porath, Y. S., & Butcher, J. N. (1989b). The comparability of MMPI and MMPI-2 scales and profiles. *Psychological Assessment*, 1, 345-347.
- Ben-Porath, Y. S., & Forbey, J. D. (2003). *Non-gendered norms for the MMPI-2*. Minneapolis: University of Minnesota Press.
- Ben-Porath, Y. S., & Forbey, J. D. (2004, May). Detrimental effects of the K correction on Clinical Scale validity. Paper presented at the 39th Annual Symposium on Recent Developments of the MMPI-2/MMPI-A, Minneapolis, MN.
- Ben-Porath, Y. S., Graham, J. R., & Tellegen, A. (2009). *The MMPI-2 Symptom Validity (FBS) scale development, research findings, and interpretive recommendations*. Minneapolis: University of Minnesota Press.
- Ben-Porath, Y. S., McCully, E., & Almagor, M. (1993). Incremental validity of the MMPI-2 content scales in the assessment of personality and psychopathology by self-report. *Journal of Personality Assessment*, 61, 557-575.
- Ben-Porath, Y. S., & Sherwood, N. E. (1993). *The MMPI-2 content component scales*. Minneapolis: University of Minnesota Press.
- Ben-Porath, Y. S., Shondrick, D. D., & Stafford, K. P. (1995). MMPI-2 and race in a forensic diagnostic sample. *Criminal Justice and Behavior*, 22, 19-32.
- Ben-Porath, Y. S., & Tellegen, A. (2008). *Minnesota Multiphasic Personality Inventory-2 Restructured Form: Manual for administration, scoring, and interpretation*. Minneapolis: University of Minnesota Press.
- Bernstein, I. H. (1980). Security guards' MMPI profiles: Some normative data. *Journal of Personality Assessment*, 44, 377-380.
- Berry, D. T. R., Baer, R. A., & Harris, M. J. (1991). Detection of malingering on the MMPI: A meta-analysis. *Clinical Psychology Review*, 11, 585-598.

- Berry, D. R., Wetter, M. W., Baer, R. A., Widiger, T. A., Sumpter, J. C., Reynolds, S. K., & Hallam, R. A. (1991). Detection of random responding on the MMPI-2: Utility of F, back F, and VRIN scales. *Psychological Assessment*, 3(3), 418–423. doi:10.1037/1040-3590.3.3.418
- Beutler, L. E., Storm, A., Kirkish, P., Scogin, F., & Gaines, J. A. (1985). Parameters in the prediction of police officer performance. *Professional Psychology: Research and Practice*, 16, 324–335.
- Binder, L. M. (1993). "Assessment of malingering after mild head trauma with the Portland Digit Recognition Test": Erratum. *Journal of Clinical and Experimental Neuropsychology*, 15(6), 852.
- Black, J. D. (1953). The interpretation of MMPI profiles of college women. Unpublished doctoral dissertation, University of Minnesota, Minneapolis.
- Black, M. S., Forbey, J. D., Ben-Porath, Y. S., Graham, J. R., McNulty, J. L., Anderson, S. V., & Burlew, A. (2004). Using the Minnesota Multiphasic Personality Inventory-2 (MMPI-2) to detect psychological distress and dysfunction in a state correctional setting. *Criminal Justice and Behavior*, 31(6), 734–751.
- Bow, J. N., Gould, J. W., Flens, J. R., & Greenhut, D. (2006). Testing in child custody evaluations: Selection, usage, and Daubert admissibility: A survey of psychologists. *Journal of Forensic Psychology Practice*, 6(2), 17–38.
- Bury, A. S., & Bagby, R. (2002). The detection of feigned uncoached and coached posttraumatic stress disorder with the MMPI-2 in a sample of workplace accident victims. *Psychological Assessment*, 14, 472–484.
- Butcher, J. N. (1979). Use of the MMPI in personnel selection. In J. N. Butcher (Ed.), *New developments in the use of the MMPI* (pp. 165–201). Minneapolis: University of Minnesota Press.
- Butcher, J. N. (1985). Personality assessment in industry: Theoretical issues and illustrations. In H. J. Bernardin (Ed.), *Personality assessment in organizations* (pp. 277–310). New York: Praeger.
- Butcher, J. N. (1996). Translation and adaptation of the MMPI-2 for international use. In J. N. Butcher (Ed.), *International adaptations of the MMPI-2: Research and clinical applications* (pp. 26–43). Minneapolis: University of Minnesota Press.
- Butcher, J. N., Arbisi, P. A., Atlis, M. M., & McNulty, J. L. (2003). The construct validity of the Lees-Haley Fake Bad Scale: Does this measure somatic malingering and feigned emotional distress? *Archives of Clinical Neuropsychology*, 18, 473–485.
- Butcher, J. N., Dahlstrom, W. G., Graham, J. R., Tellegen, A., & Kaemmer, B. (1989). *Minnesota Multiphasic Personality Inventory-2 (MMPI-2): Manual for administration and scoring*. Minneapolis: University of Minnesota Press.
- Butcher, J. N., Graham, J. R., Ben-Porath, Y. S., Tellegen, A., Dahlstrom, W. G., & Kaemmer, B. (2001). *MMPI-2 (Minnesota Multiphasic Personality Inventory-2): Manual for administration, scoring, and interpretation, revised edition*. Minneapolis: University of Minnesota Press.
- Butcher, J. N., Graham, J. R., Williams, C. L., & Ben-Porath, Y. S. (1990). *Development and use of the MMPI-2 content scales*. Minneapolis: University of Minnesota Press.
- Butcher, J. N., & Rouse, S. V. (1996). Personality: Individual differences and clinical assessment. *Annual Review of Psychology*, 47, 87–111.
- Butcher, J. N., Williams, C. L., Graham, J. R., Archer, R. P., Tellegen, A., Ben-Porath, Y. S., & Kaemmer, B. (1992). *Minnesota Multiphasic Personality Inventory-Adolescent (MMPI-A): Manual for administration, scoring, and interpretation*. Minneapolis: University of Minnesota Press.
- Caillouet, B. A., Boccaccini, M. T., Varela, J. G., Davis, R. D., & Rostow, C. D. (2010). Predictive validity of the MMPI-2 PSY-5 scales and facets for law enforcement officer employment outcomes. *Criminal Justice and Behavior*, 37(2), 217–238. doi:10.1177/0093854809351948.
- Camara, W. J., Nathan, J. S., & Puente, A. E. (2000). Psychological test usage: Implications in professional psychology. *Professional Psychology: Research and Practice*, 31, 141–154.
- Capwell, D. F. (1945a). Personality patterns of adolescent girls: I. Girls who show improvement in IQ. *Journal of Applied Psychology*, 29, 212–228.
- Capwell, D. F. (1945b). Personality patterns of adolescent girls: II. Delinquents and non-delinquents. *Journal of Applied Psychology*, 29, 289–297.
- Castro, Y., Gordon, K. H., Brown, J. S., Anestis, J. C., & Joiner, T. R. (2008). Examination of racial differences on the MMPI-2 clinical and restructured clinical scales in an outpatient sample. *Assessment*, 15(3), 277–286.
- Chafetz, M. D. (2008). Malingering on the social security disability consultative exam: Predictors and base rates. *Clinical Neuropsychologist*, 22(3), 529–546.
- Clark, M. E., Gironde, R. J., & Young, R. W. (2003). Detection of back random responding: Effectiveness of MMPI-2 and Personality Assessment validity indices. *Psychological Assessment*, 15, 223–234.

- Cooke, G. (1969). The court study unit: Patient characteristics and differences between patients judged competent and incompetent. *Journal of Clinical Psychology*, 25, 140–143.
- Cooke, G. (2010). MMPI-2 defensiveness in child custody evaluations: The role of education and socioeconomic level. *American Journal of Forensic Psychology*, 28(2), 5–16.
- Costello, R. M., Schneider, S. L., & Schoenfeld, L. S. (1996). Validation of a preemployment MMPI index correlated with interdisciplinary suspension days of police officers. *Psychology: Crime and Law*, 2, 299–306.
- Costello, R. M., Schoenfeld, L. S., & Kobos, J. (1982). Police applicant screening: An analogue study. *Journal of Clinical Psychology*, 38, 216–221.
- Crovitz, E., Huse, M. N., & Lewis, D. E. (1973). Selection of physicians' assistants. *Journal of Medical Education*, 48, 551–555.
- Dahlstrom, W. G., & Welsh, G. S. (1960). *An MMPI handbook: A guide to use in clinical practice and research*. Minneapolis: University of Minnesota Press.
- Dahlstrom, W. G., Welsh, G. S., & Dahlstrom, L. E. (1975). *An MMPI handbook*. Vol. 2: *Research applications*. Minneapolis: University of Minnesota Press.
- Daubert v. Merrell Dow Pharmaceuticals*, 727 F. Supp. 570 (S.D.Cal. 1989), aff'd, 951F.2d 1128 (9th Cir. 1990), vacated, 1123 S.Ct. 2786 (1993).
- Davis, K. M., & Archer, R. P. (2010). A critical review of objective personality inventories with sex offenders. *Journal of Clinical Psychology*, 66(12), 1254–1280.
- Detrick, P., Chibnall, J. T., & Rosso, M. (2001). Minnesota Multiphasic Personality Inventory-2 in police officer selection: Normative data and relation to the Inwald Personality Inventory. *Professional Psychology: Research and Practice*, 32, 484–490.
- Douglas, K. S., & Skeem, J. L. (2005). Violence risk assessment: Getting specific about being dynamic. *Psychology, Public Policy, and Law*, 11(3), 347–383.
- Dragon, W. R., Ben-Porath, Y. S., & Handel, R. W. (2012). Examining the impact of unscorable item responses on the validity and interpretability of MMPI-2/MMPI-2-RF Restructured Clinical (RC) scale scores. *Assessment*, 19 (1), 101–113. doi:10.1177/1073191111415362.
- Dunnette, M. D., Bownas, D. A., & Bosshardt, M. J. (1981). *Electric power plant study: Prediction of inappropriate, unreliable or aberrant job behavior in nuclear power plant settings*. Minneapolis: Personnel Decisions Research Institute.
- Dusky v. United States*, 362 U.S. 402 (1960).
- Efendov, A. A., Sellbom, M., & Bagby, R. (2008). The utility and comparative incremental validity of the MMPI-2 and Trauma Symptom Inventory validity scales in the detection of feigned PTSD. *Psychological Assessment*, 20(4), 317–326.
- Elhai, J. D., Gold, S. N., Sellers, A. H., & Dorfman, W. J. (2001). The detection of malingered posttraumatic stress disorder with MMPI-2 Fake Bad indices. *Assessment*, 8(2), 221–236. doi:10.1177/107319110100800210
- Elion, V. H., & Megargee, E. I. (1975). Validity of the MMPI Pd scale among black males. *Journal of Consulting and Clinical Psychology*, 43, 166–172.
- Ezzo, F. R., Pinsonneault, T. B., & Evans, T. M. (2008). A comparison of MMPI-2 profiles between child maltreatment cases and two types of custody cases. *Forensic Psychology Practice*, 7, 29–43.
- Finn, S. E., & Kamphuis, J. H. (2006). The MMPI-2 Restructured Clinical (RC) Scales and restraints to innovation, or "What have they done to my song?" *Journal of Personality Assessment*, 87(2), 202–210.
- Fischler, G. L. (2005, April). MMPI-2 Predictors of police officer integrity problems. Paper presented at the 40th Annual Symposium on Recent Research with the MMPI-2/MMPI-A, Fort Lauderdale, FL.
- Forbey, J. D., Ben-Porath, Y. S., & Gartland, D. (2009). Validation of the MMPI-2 Computerized Adaptive Version (MMPI-2-CA) in a correctional intake facility. *Psychological Services*, 6(4), 279–292. doi:10.1037/a0016195.
- Fordyce, W.E. (1979). Use of the MMPI in the assessment of chronic pain (Clinical Notes on the MMPI No. 3). Minneapolis, MN: National Computer Systems.
- Frye v. United States*, 293 F.1012 (D.C.Cir. 1923).
- Gallagher, R. W. (1997). Detection of malingering at the time of intake in a correctional setting with the MMPI-2 validity scales. Unpublished doctoral dissertation, Kent State University, Kent, OH.
- Gallen, R. T., & Berry, D. R. (1996). Detection of random responding on MMPI-2 protocols. *Assessment*, 3(2), 171–178.

- Garetz, F. K., & Anderson, R. W. (1973). Patterns of professional activities of psychiatrists: A follow-up of 100 psychiatric residents. *American Journal of Psychiatry*, 130, 981–984.
- Gartner, J., Hurt, S. W., & Gartner, A. (1989). Psychological test signs of borderline personality disorder: A review of the empirical literature. *Journal of Personality Assessment*, 53(3), 423–441.
- Gervais, R. O., Ben-Porath, Y. S., Wygant, D. B., & Green, P. (2007). Development and validation of a Response Bias Scale (RBS) for the MMPI-2. *Assessment*, 14(2), 196–208.
- Gervais, R. O., Ben-Porath, Y. S., Wygant, D. B., & Green, P. (2008). Differential sensitivity of the Response Bias Scale (RBS) and MMPI-2 validity scales to memory complaints. *Clinical Neuropsychologist*, 22(6), 1061–1079.
- Gilberstadt, H., & Duker, J. (1965). *A handbook for clinical and actuarial MMPI interpretation*. Philadelphia: W. B. Saunders.
- Gironde, R. J. (1999). Comparative validity of MMPI-2 scores of African-Americans and Caucasians in a forensic diagnostic sample. Unpublished doctoral dissertation, Kent State University, Kent, OH.
- Graham, J. R. (2006). *The MMPI-2: Assessing personality and psychopathology* (4th ed.). New York: Oxford University Press.
- Graham, J. R. (2011). *The MMPI-2: Assessing personality and psychopathology* (5th ed.). New York: Oxford University Press.
- Graham, J. R., Ben-Porath, Y. S., & McNulty, J. L. (1999). *MMPI-2 correlates for outpatient mental health settings*. Minneapolis: University of Minnesota Press.
- Graham, J. R., & Strenger, V. E. (1988). MMPI characteristics of alcoholics: A review. *Journal of Consulting and Clinical Psychology*, 56, 197–205.
- Graham, J. R., Watts, D., & Timbrook, R. E. (1991). Detecting fake-good and fake-bad MMPI-2 profiles. *Journal of Personality Assessment*, 57, 264–277.
- Greene, R. L. (1987). Ethnicity and MMPI performance: A review. *Journal of Consulting and Clinical Psychology*, 55, 497–512.
- Greene, R. L. (2010). *The MMPI-2/MMPI-2-RF: An interpretive manual* (3rd ed.). Upper Saddle River, NJ: Pearson.
- Greene, R. L., Robin, R. W., Albaugh, B., Caldwell, A., & Goldman, D. (2003). Use of the MMPI-2 in American Indians: II. Empirical correlates. *Psychological Assessment*, 15, 360–369.
- Greiffenstein, M. F., Fox, D., & Lees-Haley, P. R. (2006). The MMPI-2 Fake Bad Scale in detection of noncredible brain injury claims. In K. Boone (Ed.), *Detection of noncredible cognitive performance*. New York: Guilford Press.
- Greve, K. W., & Bianchini, K. J. (2004). Response to Butcher et al., “The construct validity of the Lees-Haley Fake-Bad Scale.” *Archives of Clinical Neuropsychology*, 19(3), 337–339.
- Grossman, L. S., & Wasylw, O. E. (1988). A psychometric study of stereotypes: Assessment of malingering in a criminal forensic group. *Journal of Personality Assessment*, 52, 549–563.
- Guthrie, G. M. (1949). A study of the personality characteristics associated with the disorders encountered by an internist. Unpublished doctoral dissertation, University of Minnesota, Minneapolis.
- Hagen, M. A., & Castagna, N. (2001). The real numbers: Psychological testing in custody evaluations. *Professional Psychology: Research and Practice*, 32, 269–271.
- Hall, G. N. C., Bansal, A., & Lopez, I. R. (1999). Ethnicity and psychopathology: A meta-analytic review of 31 years of comparative MMPI/MMPI-2 research. *Psychological Assessment*, 11, 186–197.
- Handel, R. W., Arnau, R. C., Archer, R. P., & Dandy, K. L. (2006). An evaluation of the MMPI-2 and MMPI-A True Response Inconsistency (TRIN) scales. *Assessment*, 13(1), 98–106. doi:10.1177/1073191105284453
- Hare, R. D. (1985). Comparison of the procedures for the assessment of psychopathy. *Journal of Clinical and Consulting Psychology*, 53, 111–119.
- Harkness, A. R., Finn, J. A., McNulty, J. L., & Shields, S. M. (2012). The Personality Psychopathology-Five (PSY-5): Recent constructive replication and assessment literature review. *Psychological Assessment*, 24, 432–443.
- Harkness, A. R., & McNulty, J. L. (1994). The Personality Psychopathology Five (PSY-5): Issues from the pages of a diagnostic manual instead of a dictionary. In S. Strack & M. Lorr (Eds.) *Differentiating normal and abnormal personality* (pp. 291–315). New York: Springer.
- Harkness, A. R., & McNulty, J. L. (2002). Implications of personality individual differences science for clinical work on personality disorders. In P. T. Costa Jr. & T. A. Widiger (Eds.), *Personality disorders and the five-factor model of personality* (2nd ed., pp. 391–403). Washington, DC: American Psychological Association.

- Harkness, A. R., & McNulty, J. L. (2006). An overview of personality: The MMPI-2 Personality Psychopathology Five (PSY-5) Scales. In J. N. Butcher (Ed.), *MMPI-2: A practitioner's guide* (pp. 73–97). Washington, DC: American Psychological Association.
- Harkness, A. R., McNulty, J. L., & Ben-Porath, Y. S. (1995). The Personality Psychopathology Five (PSY-5): Constructs and MMPI-2 scales. *Psychological Assessment*, 7, 104–114.
- Harkness, A. R., McNulty, J. L., Ben-Porath, Y. S., & Graham, J. R. (2002). *MMPI-2 Personality Psychopathology Five (PSY-5) Scales: Gaining an overview for case conceptualization and treatment planning*. Minneapolis: University of Minnesota Press.
- Harpur, T. J., Hare, R. D., & Hakstian, A. R. (1989). Two-factor conceptualization of psychopathy: Construct validity and assessment implications. *Psychological Assessment*, 1, 6–17.
- Harris, R., & Lingoes, J. (1955). Subscales for the Minnesota Multiphasic Personality Inventory. Mimeographed materials, Langley Porter Clinic.
- Hart, S. (1998). The role of psychopathy in assessing risk for violence: Conceptual and methodological issues. *Legal and Criminological Psychology*, 3, 121–137.
- Hartman, B. J. (1987). Psychological screening of law enforcement candidates. *American Journal of Forensic Psychology*, 1, 5–10.
- Hathaway, S. R. (1960). Foreword to the first edition. In W. G. Dahlstrom & G. S. Welsch (Eds.), *An MMPI handbook: A guide to use in clinical practice and research* (pp. vii–xi). Minneapolis: University of Minnesota Press.
- Hathaway, S. R., & McKinley, J. C. (1943). *The Minnesota Multiphasic Personality Inventory manual*. New York: Psychological Corporation.
- Hathaway, S. R., & Monachesi, E. D. (1953). *Analyzing and predicting juvenile delinquency with the MMPI*. Minneapolis: University of Minnesota Press.
- Hathaway, S. R., & Monachesi, E. D. (1957). The personalities of predelinquent boys. *Journal of Criminal Law, Criminology, and Political Science*, 48, 149–163.
- Hathaway, S. R., & Monachesi, E. D. (1963). *Adolescent personality and behavior: MMPI patterns of normal, delinquent, dropout, and other outcomes*. Minneapolis: University of Minnesota Press.
- Hawk, G. L., & Cornell, D. G. (1989). MMPI profiles of malingers diagnosed in pretrial forensic evaluations. *Journal of Clinical Psychology*, 45, 673–678.
- Heilbrun, A. B. (1979). Psychopathy and violent crime. *Journal of Consulting and Clinical Psychology*, 47, 509–516.
- Heilbrun, K., & Heilbrun, A. B. (1995). Risk assessment with the MMPI-2 in forensic evaluations. In Y. S. Ben-Porath, J. R. Graham, G. C. N. Hall, R. D. Hirshman, & M. S. Zaragoza (Eds.), *Forensic applications of the MMPI-2* (pp. 160–178). Thousand Oaks, CA: Sage.
- Henrichs, T. F., & Waters, W. F. (1972). Psychological adjustment and responses to open-heart surgery: Some methodological considerations. *British Journal of Psychiatry*, 120, 491–496.
- Hjemboe, S., Almagor, M., & Butcher, J. N. (1992). Empirical assessment of marital distress: The Marital Distress Scale (MDS) for the MMPI-2. In J. N. Butcher & C. D. Spielberger (Eds.), *Advances in personality assessment* (Vol. 9, pp. 141–152). Hillsdale, NJ: Lawrence Erlbaum.
- Holcomb, W. R., & Adams, N. (1982). Racial influences on intelligence and personality measures of people who commit murder. *Journal of Clinical Psychology*, 38, 793–796.
- Inwald, R. E. (1988). Five-year follow-up study of departmental terminations as predicted by 16 preemployment psychological indicators. *Journal of Applied Psychology*, 4, 703–710.
- Kawachi, I., Sparrow, D., Kubazansky, L. D., Spiro, A., Vokonas, P. S., & Weiss, S. T. (1998). Prospective study of a self-report Type A scale and risk of coronary heart disease: Test of the MMPI-2 Type A scale. *Circulation*, 98, 405–412.
- Keane, T. M., Malloy, P. F., & Fairbank, J. A. (1984). Empirical development of an MMPI subscale for the assessment of combat-related posttraumatic stress disorder. *Journal of Consulting and Clinical Psychology*, 52, 888–891.
- Kelly, W. L. (1974). Psychological prediction of leadership in nursing. *Nursing Research*, 23, 38–42.
- Krueger, R. F., & Markon, K. E. (2006). Understanding psychopathology: Melding behavior genetics, personality, and quantitative psychology to develop and empirically based model. *Current Directions in Psychological Science*, 15, 113–117.
- Kucharski, L., & Duncan, S. (2007). Differentiation of mentally ill criminal defendants from malingers on the MMPI-2 and PAI. *American Journal of Forensic Psychology*, 25(3), 21–42.

- Lachar, D. (1974). Prediction of early U.S. Air Force cadet adaptation with the MMPI. *Journal of Counseling Psychology, 21*, 404–408.
- Lally, S. J. (2003). What tests are acceptable for use in forensic evaluations? A survey of experts. *Professional Psychology: Research and Practice, 34*, 491–498.
- Larrabee, G. J. (2003a). Detection of symptom exaggeration with the MMPI–2 in litigants with malingered neurocognitive dysfunction. *Clinical Neuropsychologist, 17*, 54–68.
- Larrabee, G. J. (2003b). Exaggerated MMPI–2 symptoms report in personal injury litigants with malingered neurocognitive deficit. *Archives of Clinical Neuropsychology, 18*, 673–686.
- Lee, T. C., Graham, J. R., Sellbom, M., & Gervais, R. O. (2012). Examining the potential for gender bias in the prediction of symptom validity test failure by MMPI–2 symptom validity scale scores. *Psychological Assessment, 24*, 618–627.
- Lees-Haley, P. R. (1992). Psychodiagnostic test usage by forensic psychologists. *American Journal of Forensic Psychology, 10*, 25–30.
- Lees-Haley, P. R., English, L. T., & Glenn, W. J. (1991). A fake-bad scale on the MMPI–2 for personal injury claimants. *Psychological Reports, 68*, 203–210.
- Lees-Haley, P. R., & Fox, D. D. (2004). Commentary on Butcher, Arbisi, Atlis, and McNulty (2003) on the Fake Bad Scale. *Archives of Clinical Neuropsychology, 19*(3), 333–336.
- Lees-Haley, P. R., Smith, H. H., Williams, C. W., & Dunn, J. T. (1996). Forensic neuropsychological test usage: An empirical survey. *Archives of Clinical Neuropsychology, 11*, 45–51.
- Liao, H., Arvey, R. D., Butler, R. J., & Nutting, S. M. (2000). Correlates of work injury frequency and duration among firefighters. *Journal of Occupational Health Psychology, 6*, 229–242.
- Lilienfeld, S. O. (1996). The MMPI–2 Antisocial Practices Content Scale: Construct validity and comparison with the Psychopathic Deviate Scale. *Psychological Assessment, 8*(3), 281–293.
- Long, K. A., Graham, J. R., & Timbrook, R. E. (1994). Socioeconomic status and MMPI–2 interpretation. *Measurement and Evaluation in Counseling and Development, 27*, 158–177.
- Lubin, B., Larsen, R. M., & Matarazzo, J. D. (1984). Patterns of psychological test usage in the United States: 1935–1982. *American Psychologist, 39*, 451–454.
- MacAndrew, C. (1965). The differentiation of male alcoholic outpatients from nonalcoholic psychiatric outpatients by means of the MMPI. *Quarterly Journal of Studies on Alcohol, 26*, 238–246.
- Marks, P. A., & Seeman, W. (1963). *Actuarial description of abnormal personality*. Baltimore, MD: Williams & Wilkins.
- Martin, M. A., Allan, A., & Allan, M. M. (2001). The use of psychological tests by Australian psychologists who do assessments for the courts. *Australian Journal of Psychology, 53*, 77–82.
- Maxson, L. S., & Neuringer, C. (1970). Evaluating legal competency. *Journal of Genetic Psychology, 117*, 267–273.
- McNulty, J. L., Graham, J. R., Ben-Porath, Y. S., & Stein, L. A. R. (1997). Comparative validity of MMPI–2 scores of African American and Caucasian mental health center clients. *Psychological Assessment, 9*(4), 464–470.
- Meehl, P. E. (1945). The dynamics of “structured” personality tests. *Journal of Clinical Psychology, 1*, 296–303.
- Meehl, P. E. (1954). *Clinical versus statistical prediction: A theoretical analysis and a review of the evidence*. Minneapolis: University of Minnesota Press.
- Meehl, P. E. (1956). Wanted—a good cookbook. *American Psychologist, 11*, 263–272.
- Megargee, E. I. (1994). Using the Megargee MMPI-based classification system with MMPI–2s of male prison inmates. *Psychological Assessment, 6*, 337–344.
- Megargee, E. I. (1997). Using the Megargee MMPI-based classification system with the MMPI–2’s of female prison inmates. *Psychological Assessment, 9*, 75–82.
- Megargee, E. I., Bohn, M. J., Meyer, J. E. Jr., & Sink, F. (1979). *Classifying criminal offenders: A new system based on the MMPI*. Beverly Hills, CA: Sage.
- Megargee, E. I., & Carbonell, J. L. (1995). Use of the MMPI–2 in correctional settings. In Y. S. Ben-Porath, J. R. Graham, G. C. N. Hall, R. D. Hirshman, & M. S. Zaragoza (Eds.), *Forensic applications of the MMPI–2* (pp. 127–159). Thousand Oaks, CA: Sage.
- Megargee, E. I., Carbonell, J. L., Bohn, M. J., & Sliger, G. L. (2001). *Classifying criminal offenders with the MMPI–2: The Megargee System*. Minneapolis: University of Minnesota Press.
- Megargee, E. I., & Mendelsohn, A. (1962). A cross-validation of twelve MMPI indices of hostility and control. *Journal of Abnormal and Social Psychology, 65*, 431–438.

- Melton, G. B., Petrila, J., Poythress, N., & Slobogin, C. (1997). *Psychological evaluations for the courts: A handbook for mental health professionals and lawyers*. New York: Guilford Press.
- Mittenburg, W., Patton, C., Canyock, E. M., & Condit, D. C. (2002). Base rates of malingering and symptom exaggeration. *Journal of Clinical and Experimental Neuropsychology*, 24, 1094–1102.
- Monahan, J. (2003). Violence risk assessment. In A. M. Goldstein (Ed.), *Handbook of forensic psychology* (Vol. 11, pp. 527–540). New York: Wiley.
- Nelson, N. W., Hoelzle, J. B., Sweet, J. J., Arbisi, P. A., & Demakis, G. J. (2010). Updated meta-analysis of the MMPI-2 Symptom Validity Scale (FBS): Verified utility in forensic practice. *Clinical Neuropsychologist*, 24, 701–724.
- Nelson, N. W., Sweet, J. J., & Demakis, G. J. (2006). Meta-analysis of the MMPI-2 Fake Bad Scale: Utility in forensic practice. *Clinical Neuropsychologist*, 20, 39–58.
- Ogloff, J. R. P. (1995). The legal basis of forensic applications of the MMPI-2. In Y. S. Ben-Porath, J. R. Graham, G. C. N. Hall, R. D. Hirschman, & M. S. Zaragoza (Eds.), *Forensic Applications of the MMPI-2* (pp. 18–47). Thousand Oaks, CA: Sage.
- Okazaki, S., & Sue, S. (1995). Cultural considerations in psychological assessment of Asian-Americans. In J. N. Butcher (Ed.), *Clinical personality assessment: Practical approaches* (pp. 107–119). New York: Oxford University Press.
- Otto, R. K. (2002). Use of the MMPI-2 in forensic settings. *Journal of Forensic Psychology Practice*, 2, 71–92.
- Otto, R. K., & Collins, R. P. (1995). Use of the MMPI-2/MMPI-A in child custody evaluations. In Y. S. Ben-Porath, J. R. Graham, G. C. N. Hall, R. D. Hirschman, & M. S. Zaragoza (Eds.), *Forensic applications of the MMPI-2* (pp. 222–252). Thousand Oaks, CA: Sage.
- Otto, R. K., Poythress, N. G., Nicholson, R. A., Edens, J. F., Monahan, J., Bonnie, R. J., . . . Eisenber, M. (1998). Psychometric properties of the MacArthur Competence Assessment Tool–Criminal Adjudication. *Psychological Assessment*, 10, 435–443.
- Pace, T. M., Robbins, R. R., Choney, S. K., Hill, J. S., Lacey, K., & Blair, G. (2006). A cultural-contextual perspective on the validity of the MMPI-2 with American Indians. *Cultural Diversity and Ethnic Minority Psychology*, 12(2), 320–333.
- Paolo, A. M., & Ryan, J. J. (1992). Detection of random response sets on the MMPI-2. *Psychotherapy in Private Practice*, 11(4), 1–8.
- Petroskey, L. J., Ben-Porath, Y. S., & Stafford, K. P. (2003). Correlates of the Minnesota Multiphasic Personality Inventory-2 (MMPI-2) Personality Psychopathology Five (PSY-5) scales in a forensic assessment setting. *Assessment*, 10, 393–399.
- Piotrowski, C., & Keller, J. W. (1989). Psychological testing in outpatient mental health facilities: A national survey. *Professional Psychology: Research and Practice*, 20, 423–425.
- Pirelli, G., Zapf, P. A., & Gottdiener, W. H. (2011). Competency to stand trial research: Guidelines and future directions. *Journal of Forensic Psychiatry and Psychology*, 22, 340–370.
- Platt, J. R., & Husband, S. D. (1987). Posttraumatic stress disorder and motor vehicle accident victims. *American Journal of Forensic Psychology*, 5, 35–42.
- Pope, K. S., Butcher, J. N., & Seelen, J. (2000). *The MMPI, MMPI-2, and MMPI-A in court: A practical guide for expert witnesses and attorneys* (2nd ed.). Washington, DC: American Psychological Association.
- Pothast, M. D. (1956). A personality study of two types of murderers. Unpublished doctoral dissertation, University of Minnesota.
- Reynolds, W. M., & Sundberg, D. (1976). Recent research trends in testing. *Journal of Personality Assessment*, 40, 228–233.
- Robin, R. W., Greene, R. L., Albaugh, B., Caldwell, A., & Goldman, D. (2003). Use of the MMPI-2 in American Indians: I. Comparability of the MMPI-2 between two tribes and with the MMPI-2 normative group. *Psychological Assessment*, 15, 351–359.
- Rogers, R., Bagby, R. M., & Chakraborty, D. (1993). Feigning schizophrenic disorders on the MMPI-2: Detection of coached simulators. *Journal of Personality Assessment*, 60, 215–226.
- Rogers, R., Gillis, J. R., McMain, S., & Dickens, S. E. (1988). Fitness evaluations: A retrospective study of clinical, criminal, and sociodemographic characteristics. *Canadian Journal of Behavioural Science*, 20, 192–200.
- Rogers, R., & McKee, G. R. (1995). Use of the MMPI-2 in the assessment of criminal responsibility. In Y. S. Ben-Porath, J. R. Graham, G. C. N. Hall, R. D. Hirschman, & M. S. Zaragoza (Eds.), *Forensic applications of the MMPI-2* (pp. 103–126). Thousand Oaks, CA: Sage.

- Rogers, R., Sewell, K. W., Martin, M. A., & Vitacco, M. J. (2003). Detection of feigned mental disorders: A meta-analysis of the MMPI-2 and malingering. *Assessment*, 10, 160-177.
- Roman, D. D., Tuley, M. R., Villanueva, M. R., & Mitchell, W. E. (1990). Evaluating MMPI validity in a forensic psychiatric population: Distinguishing between malingering and genuine psychopathology. *Criminal Justice and Behavior*, 17, 186-198.
- Schinka, J. A., & LaLone, L. (1997). MMPI-2: Comparisons with a census-matched subsample. *Psychological Assessment*, 9, 307-311.
- Schinka, J. A., LaLone, L., & Greene, G. L. (1998). Effects of psychopathology and demographic characteristics on MMPI-2 scale scores. *Journal of Personality Assessment*, 70, 197-211.
- Sellbom, M., Ben-Porath, Y. S., & Bagby, R. (2008). On the hierarchical structure of mood and anxiety disorders: Confirmatory evidence and elaboration of a model of temperament markers. *Journal of Abnormal Psychology*, 117(3), 576-590.
- Sellbom, M., Ben-Porath, Y. S., Baum, L. J., Erez, E., & Gregory, C. (2005). Empirical correlates of the MMPI-2 Restructured Clinical (RC) scales in a forensic setting. Paper presented at the 2005 Annual Meeting of the Society for Personality Assessment, Chicago.
- Sellbom, M., Ben-Porath, Y. S., Baum, L. J., Erez, E., & Gregory, C. (2008). Predictive validity of the MMPI-2 Restructured Clinical (RC) scales in a batterers' intervention program. *Journal of Personality Assessment*, 90(2), 129-135. doi:10.1080/00223890701845153.
- Sellbom, M., Ben-Porath, Y. S., & Graham, J. R. (2006). Correlates of the MMPI-2 Restructured Clinical (RC) scales in a college counseling setting. *Journal of Personality Assessment*, 86, 89-99.
- Sellbom, M., Ben-Porath, Y. S., Lilienfeld, S. O., Patrick, C. J., & Graham, J. R. (in press). Assessing psychopathic personality traits with the MMPI-2. *Journal of Personality Assessment*.
- Sellbom, M., Ben-Porath, Y. S., McNulty, J. L., Arbisi, P. A., & Graham, J. R. (2006). Elevation differences between MMPI-2 Clinical and Restructured Clinical (RC) scales: Frequency, origins, and interpretative implications. *Assessment*, 13, 430-441.
- Sellbom, M., Ben-Porath, Y. S., & Stafford, K. P. (2007). A comparison of MMPI-2 measures of psychopathic deviance in a forensic setting. *Psychological Assessment*, 19(4), 430-436. doi:10.1037/1040-3590.19.4.430.
- Sellbom, M., Fischler, G. L., & Ben-Porath, Y. S. (2007). Identifying MMPI-2 predictors of police officer integrity and misconduct. *Criminal Justice and Behavior*, 34, 985-1004.
- Sellbom, M., Graham, J. R., & Schenk, P. W. (2005). Symptom correlates of MMPI-2 scales and code types in a private practice setting. *Journal of Personality Assessment*, 84, 163-171.
- Sellbom, M., Lee, T. T., Ben-Porath, Y. S., Arbisi, P. A., & Gervais, R. O. (2012). Differentiating PTSD symptomatology with the MMPI-2-RF (Restructured Form) in a forensic disability sample. *Psychiatry Research*, 197, 172-179.
- Simms, L. J. (2006). Bridging the divide: Comments on the Restructured Clinical Scales of the MMPI-2. *Journal of Personality Assessment*, 87(2), 211-216.
- Simms, L. J., Casillas, A., Clark, L. A., Watson, D., & Doebbeling, B. N. (2005). Psychometric evaluation of the restructured clinical scales of the MMPI-2. *Psychological Assessment*, 17, 345-358.
- Skodol, A. E., Clark, L. A., Bender, D. S., Krueger, R. F., Morey, L. C., Verheul, R., . . . Oldham, J. M. (2011). Proposed changes in personality and personality disorder assessment and diagnosis for DSM-5 Part I: Description and rationale. *Personality Disorders: Theory, Research, and Treatment*, 2(1), 4-22. doi:10.1037/a0021891.
- Slick, D. J., Hopp, G. A., Strauss, E. H., & Spellacy, F. J. (1996). Victoria Symptom Validity Test: Efficiency for detecting feigned memory impairment and relationship to neuropsychological tests and MMPI-2 validity scales. *Journal of Clinical and Experimental Neuropsychology*, 18, 911-922.
- Sobel, H. J., & Worden, W. (1979). The MMPI as a predictor of psychosocial adaptation to cancer. *Journal of Consulting and Clinical Psychology*, 47, 716-724.
- Solway, K. S., Hays, J. R., & Zieben, M. (1976). Personality characteristics of juvenile probation officers. *Journal of Community Psychology*, 4, 152-156.
- Stafford, K. P., & Sellbom, M. (2012). Competency to stand trial. In R. K. Otto (Ed.), *Handbook of psychology*. Vol. 11: *Forensic psychology* (2nd ed., pp. 412-439). New York: Wiley.
- Stafford, K. P., & Wygant, D. B. (2005). The role of competency to stand trial in mental health courts. *Behavioral Sciences and the Law*, 23, 245-258.
- Stone, L. A., Bassett, G. R., Brousseau, J. D., Demers, J., & Stiening, J. A. (1972). Psychological test scores for a group of MEDEX trainees. *Psychological Reports*, 31, 827-831.

- Storm, J., & Graham, J. R. (2000). Detection of coached general malingering on the MMPI-2. *Psychological Assessment, 12*, 158–165.
- Sutker, P. B., & Allain, N. (1973). Incarcerated and street heroin addicts: A personality comparison. *Psychological Reports, 32*, 243–246.
- Tellegen, A. (1985). Structures of mood and personality and their relevance to assessing anxiety, with an emphasis on self-report. In A. H. Tuma & J. D. Maser (Eds.), *Anxiety and the anxiety disorders* (pp. 681–706). Hillsdale, NJ: Lawrence Erlbaum.
- Tellegen, A., & Ben-Porath, Y. S. (1992). The new uniform T scores for the MMPI-2: Rationale, derivation, and appraisal. *Psychological Assessment, 4*, 145–155.
- Tellegen, A., Ben-Porath, Y. S., McNulty, J. L., Arbisi, P. A., Graham, J. R., & Kaemmer, B. (2003). *MMPI-2 Restructured Clinical (RC) scales: Development, validation, and interpretation*. Minneapolis: University of Minnesota Press.
- Tellegen, A., Ben-Porath, Y. S., & Sellbom, M. (2009). Construct validity of the MMPI-2 Restructured Clinical (RC) scales: Reply to Rouse, Greene, Butcher, Nichols, and Williams. *Journal of Personality Assessment, 91*, 211–221.
- Tellegen, A., Ben-Porath, Y. S., Sellbom, M., Arbisi, P. A., McNulty, J. L., & Graham, J. R. (2006). Further evidence on the validity of the MMPI-2 Restructured Clinical (RC) scales: Addressing questions raised by Rogers, Sewell, Harrison, and Jordan and Nichols. *Journal of Personality Assessment, 87*, 148–171.
- Timbrook, R. E., & Graham, J. R. (1994). Ethnic differences on the MMPI-2? *Psychological Assessment, 6*, 212–217.
- Tolin, D. F., Steenkamp, M. M., Marx, B. P., & Litz, B. T. (2010). Detecting symptom exaggeration in combat veterans using the MMPI-2 symptom validity scales: A mixed group validation. *Psychological Assessment, 22*, 729–736.
- Toomey, J. A., Kucharski, L., & Duncan, S. (2009). The utility of the MMPI-2 Malingering Discriminant Function Index in the detection of malingering: A study of criminal defendants. *Assessment, 16*(1), 115–121. doi:10.1177/1073191108319713.
- Tsai, D. C., & Pike, P. L. (2000). Effects of acculturation on the MMPI-2 scores of Asian American students. *Journal of Personality Assessment, 74*, 216–230.
- Tsushima, W. T., Bridenstine, M. P., & Balfour, J. F. (2004). MMPI-2 scores in the outcome prediction of gastric bypass surgery. *Obesity Surgery, 14*, 528–532.
- Tsushima, W. T., Geling, O., & Fabrigas, J. (2011). Comparison of MMPI-2 validity scale scores of personal injury litigants and disability claimants. *Clinical Neuropsychologist, 25*(8), 1403–1414.
- United States v. Huberty*, 53 MJ 369 (2000).
- Vendrig, A. A., Derksen, J. J. L., & de Mey, H. R. (1999). Utility of selected MMPI-2 scales in the outcome prediction for patients with chronic back pain. *Psychological Assessment, 11*, 381–385.
- Webster, C. D., Douglas, K. S., Eaves, D., & Hart, S. D. (1997). Assessing risk of violence to others. In C. D. Webster, M. A. Jackson, C. D. Webster, & M. A. Jackson (Eds.), *Impulsivity: Theory, assessment, and treatment* (pp. 251–277). New York: Guilford Press.
- Webster, C. D., & Hucker, S. J. (2007). *Violence risk: Assessment and management*. New York: John Wiley.
- Weed, N. C., Ben-Porath, Y. S., & Butcher, J. N. (1990). Failure of Wiener and Harmon Minnesota Multiphasic Personality Inventory (MMPI) subtle scales as personality descriptors and as validity indicators. *Psychological Assessment, 2*, 281–285.
- Weed, N. C., Butcher, J. N., McKenna, T., & Ben-Porath, Y. S. (1992). New measures for assessing alcohol and drug abuse with the MMPI-2: The APS and AAS. *Journal of Personality Assessment, 58*, 389–404.
- Welsh, G. S. (1956). Factor dimensions A and R. In G. S. Welsh & W. G. Dahlstrom (Eds.), *Basic readings on the MMPI in psychology and medicine* (pp. 264–281). Minneapolis: University of Minnesota Press.
- Wetter, M. W., Baer, R. A., Berry, D. T. R., & Reynolds, S. K. (1994). The effect of symptom information on faking on the MMPI-2. *Assessment, 1*, 199–207.
- Wetter, M. W., & Corrigan, S. K. (1995). Providing information to clients about psychological tests: A survey of attorneys' and law students' attitudes. *Professional Psychology: Research and Practice, 26*, 1–4.
- Wiggins, J. S. (1969). Content dimensions in the MMPI. In J. N. Butcher (Ed.), *MMPI: Research developments and clinical applications* (pp. 127–180). New York: McGraw-Hill.
- Wilkinson, R. A. (2000, January/February). Mental health care for Ohio state prisoners: The view from the director's office. *Correctional mental health report*. Retrieved from <http://www.drc.ohio.gov/web/Articles/Mental%20Health%20Care%20for%20Ohio%20Sta.pdf>.

- Williams, R. B., Haney, T. L., Lee, K. L., Kong, Y. H., Blumenthal, J. A., & Whalen, R. E. (1980). Type A behavior, hostility, and coronary atherosclerosis. *Psychosomatic Medicine*, 42, 539–549.
- Wygant, D. B., Boutacoff, L. I., Arbisi, P. A., Ben-Porath, Y. S., Kelly, P. H., & Rupp, W. M. (2007). Examination of the MMPI–2 Restructured Clinical (RC) scales in a sample of bariatric surgery candidates. *Journal of Clinical Psychology in Medical Settings*, 14(3), 197–205.
- Wygant, D. B., Sellbom, M., Gervais, R. O., Ben-Porath, Y. S., Stafford, K. P., Freeman, D. B., & Heilbronner, R. L. (2010). Further validation of the MMPI–2 and MMPI–2–RF Response Bias Scale: Findings from disability and criminal forensic settings. *Psychological Assessment*, 22, 745–756.

3

FORENSIC APPLICATIONS OF THE MINNESOTA MULTIPHASIC PERSONALITY INVENTORY–2–RESTRUCTURED FORM

Yossef S. Ben-Porath

Basic Description of the MMPI–2–RF

The Minnesota Multiphasic Personality Inventory–2 Restructured Form (MMPI–2–RF; Ben-Porath & Tellegen, 2008/2011; Tellegen & Ben-Porath, 2008/2011) consists of 338 items that are presently scored on 51 scales. Ben-Porath (2012) notes that the authors' goal for the MMPI–2–RF was to represent the clinically significant substance of the MMPI–2 item pool with a comprehensive set of psychometrically adequate measures. Shortly after the MMPI–2 was published in 1989, one of the co-authors of the revised inventory, Auke Tellegen, launched an effort to address long-recognized concerns about the original Clinical Scales, which were essentially unchanged on the revised test. The product of this effort, a set of Restructured Clinical (RC) Scales (Tellegen et al., 2003), was added to the MMPI–2 in 2003 and forms the core of the MMPI–2–RF. The nine RC Scales were designed to assess major distinctive core components of the original MMPI Clinical Scales. They are augmented on the MMPI–2–RF by 33 substantive measures and 9 validity indicators that were developed to canvass the full range of constructs that can be reliably and validly assessed with the MMPI–2 item pool. A central aspect of our¹ approach to developing the MMPI–2–RF was to link the test to contemporary concepts and models of personality and psychopathology.

Administration of the MMPI–2–RF

Standard procedures for administering and scoring the MMPI–2–RF are described in detail by Ben-Porath and Tellegen (2008/2011). Deviations from these procedures should be avoided, but if they occur, this should be noted and considered in the forensic report. Briefly, the examiner should inquire about prior testing, and if the individual has previously completed the test, her or his experiences, impressions, and expectations for the current evaluation should be discussed to help identify and correct any misconceptions. However, particularly in the context of a forensic assessment, the examiner should refrain from providing any information beyond that which is included in the standard administration instructions for the inventory. The individual administering the test should conduct a preliminary assessment of the testability of the test-taker by determining whether he or she has any limitations or conditions that might impair his or her ability to respond meaningfully to the test items and, if so, whether corrective steps can properly be taken without

deviating substantially from the standard administration procedures. Individuals experiencing acute psychotic episodes or delirium and some potential test-takers with severe cognitive impairments may be unable to produce a valid and interpretable MMPI-2-RF.

After confirming that the test-taker can see the test items or providing appropriate accommodations, the individual's reading ability should be considered. The reading level required for test-takers to respond meaningfully to the MMPI-2 (and, by extension, the MMPI-2-RF) has been the subject of some discussion and confusion. The authors of the second edition of the MMPI-2 manual indicate that the test requires a sixth-grade reading level (Butcher et al., 2001). In this regard, the inventory is frequently compared with the Personality Assessment Inventory (PAI; Morey, 2007), which is reported in its manual to require a fourth-grade reading level. Some authors have inferred that this makes the PAI a more appropriate instrument in settings where test-takers may have lower reading levels. For example, Morey, Warner, and Hopwood (2007) commented that the lower reading level required for the PAI is "an important issue in forensic settings where reading ability is commonly lower than average" (p. 108). However, comparisons of the MMPI-2 and PAI reading levels, estimated at the sixth- and fourth-grade levels, respectively, are inaccurate because the two estimates were derived with different methodologies. Butcher et al. (2001) relied on analyses using Lexile values (Stenner, Horabin, Smith, & Smith, 1988), whereas Morey (2007) derived the PAI estimate on the basis of analysis of the test's entire item pool using the Flesch-Kincaid reading level index available in the Microsoft Word spell-checker, which yields a score of 4.0. Applying the same procedure to the MMPI-2-RF item pool produces a Flesch-Kincaid index of 4.5. Establishing and comparing reading levels for self-report inventories is complicated further when different response formats are used. The true-false response format of the MMPI-2-RF is less complex than Likert-scale response formats and, all things being equal, should be less challenging for test-takers with limited cognitive abilities.

If a test-taker's ability to read the MMPI-2-RF items is in doubt (e.g., because of possible reading limitations or because the test-taker is not a native English speaker), administration of a standard test of reading ability is recommended. If the test-taker's reading level is less than sixth grade, or if it is not feasible to administer a reading test when reading comprehension is in doubt, audio administration of the MMPI-2-RF using the standard compact disc or computer software is recommended. Test-takers who have the visual ability to respond to a conventional administration of the MMPI-2-RF and the requisite reading skills (or who receive an audio administration of the test) may still be unable to respond meaningfully to the test items because of a lack of comprehension. This may occur with non-native English speakers as well as test-takers whose cognitive functioning is compromised by dementia, delirium, intoxication, or acute primary psychopathology. The test-taker's English language proficiency and mental status should be examined prior to administration of the MMPI-2-RF.

Standard MMPI-2-RF administration modalities include a test booklet, computer administration, and audio recordings of the test items. Audio administration can be accomplished with a standard compact disc or as an optional accompaniment to computerized test administration. Reading the test items aloud to the test-taker is not an acceptable alternative, particularly in forensic assessments, because it deviates from standard administration procedures and introduces an interpersonal element to the assessment. When the booklet or audio recording is used to administer the test, an answer sheet is the recording modality. Several studies have established that booklet and computer administrations of the MMPI yield comparable results (see Forbey, Ben-Porath, & Gartland, 2009). Computer administration offers two primary advantages: First, it is appreciably faster than using a booklet and answer sheet. Test-takers with normal-range cognitive functioning and reading skills can typically complete a computer administration of the MMPI-2-RF in 25–35 minutes. Booklet and answer sheet administration typically requires 35–50 minutes. Computerized

administration is faster because the test-taker does not need to go back and forth between the booklet and answer sheet and fill in 338 bubbles. In addition, with computer administration the test-taker inputs his or her responses directly into the scoring software, saving the time needed to input the responses and avoiding potential error associated with manual entry.

MMPI-2-RF Scales

The MMPI-2-RF is made up of 51 scales described in detail by Ben-Porath (2012). Listed and described briefly in Table 3.1, the MMPI-2-RF scales are divided into six sets. The *Validity Scales* include nine MMPI-2-RF measures designed to alert the interpreter to various threats to the validity of an individual test protocol. They include measures of inconsistent responding, over-reporting, and under-reporting, which constitute the major threats to the validity of self-report measures of personality and psychopathology. The remaining 42 “substantive” scales include three *Higher-Order Scales*, which indicate whether and to what extent a test-taker is likely experiencing problems in the domains of mood and affect, thought processes, and/or behavior; nine *Restructured Clinical Scales* that provide an indication of the individual’s standing on the nine psychological constructs identified by Tellegen et al. (2003) as major distinctive core components of the original MMPI Clinical Scales; 23 *Specific Problems Scales*, the most narrowly focused MMPI-2-RF measures, which are subdivided into indicators of somatic and cognitive complaints, internalizing difficulties, externalizing behaviors, and interpersonal functioning; two *Interest Scales*, derived from the original MMPI Masculinity/Femininity scale; and five *Personality Psychopathology Five (PSY-5) Scales*, which are revised versions of similarly labeled MMPI-2 scales, designed to provide a dimensional perspective on features of personality disorder-related psychopathology.

TABLE 3.1 The MMPI-2-RF Scales

<i>Abbreviation</i>	<i>Meaning</i>	<i>Description</i>
Validity Scales		
<i>VRIN-r</i>	Variable Response Inconsistency	Random responding
<i>TRIN-r</i>	True Response Inconsistency	Fixed responding
<i>F-r</i>	Infrequent Responses	Responses infrequent in the general population
<i>Fp-r</i>	Infrequent Psychopathology Responses	Responses infrequent in psychiatric populations
<i>Fs</i>	Infrequent Somatic Responses	Somatic complaints infrequent in medical patient populations
<i>FBS-r</i>	Symptom Validity	Somatic and cognitive complaints associated at high levels with over-reporting
<i>RBS</i>	Response Bias Scale	Exaggerated memory complaints
<i>L-r</i>	Uncommon Virtues	Rarely claimed moral attributes or activities
<i>K-r</i>	Adjustment Validity	Avowals of good psychological adjustment associated at high levels with under-reporting
Higher-Order (H-O) Scales		
<i>EID</i>	Emotional/Internalizing Dysfunction	Problems associated with mood and affect
<i>THD</i>	Thought Dysfunction	Problems associated with disordered thinking

TABLE 3.1 Continued

<i>Abbreviation</i>	<i>Meaning</i>	<i>Description</i>
<i>BXD</i>	Behavioral/Externalizing Dysfunction	Problems associated with under-controlled behavior
Restructured Clinical (RC) Scales		
<i>RCd</i>	Demoralization	General unhappiness and dissatisfaction
<i>RC1</i>	Somatic Complaints	Diffuse physical health complaints
<i>RC2</i>	Low Positive Emotions	Lack of positive emotional responsiveness
<i>RC3</i>	Cynicism	Non-self-referential beliefs expressing distrust and a generally low opinion of others
<i>RC4</i>	Antisocial Behavior	Rule breaking and irresponsible behavior
<i>RC6</i>	Ideas of Persecution	Self-referential beliefs that others pose a threat
<i>RC7</i>	Dysfunctional Negative Emotions	Maladaptive anxiety, anger, irritability
<i>RC8</i>	Aberrant Experiences	Unusual perceptions or thoughts
<i>RC9</i>	Hypomanic Activation	Over-activation, aggression, impulsivity, and grandiosity
Specific Problems (SP) Scales		
<i>Somatic/Cognitive Scales</i>		
<i>MLS</i>	Malaise	Overall sense of physical debilitation, poor health
<i>GIC</i>	Gastrointestinal Complaints	Nausea, recurring upset stomach, and poor appetite
<i>HPC</i>	Head Pain Complaints	Head and neck pain
<i>NUC</i>	Neurological Complaints	Dizziness, weakness, paralysis, loss of balance, etc.
<i>COG</i>	Cognitive Complaints	Memory problems, difficulties concentrating
<i>Internalizing Scales</i>		
<i>SUI</i>	Suicidal/Death Ideation	Direct reports of suicidal ideation and recent suicide attempts
<i>HLP</i>	Helplessness/Hopelessness	Belief that goals cannot be reached or problems solved
<i>SFD</i>	Self-Doubt	Lack of confidence, feelings of uselessness
<i>NFC</i>	Inefficacy	Belief that one is indecisive and inefficient
<i>STW</i>	Stress/Worry	Preoccupation with disappointments, difficulty with time pressure
<i>AXY</i>	Anxiety	Pervasive anxiety, frights, frequent nightmares
<i>ANP</i>	Anger Proneness	Becoming easily angered, impatient with others
<i>BRF</i>	Behavior-Restricting Fears	Fears that significantly inhibit normal activities
<i>MSF</i>	Multiple Specific Fears	Fears of blood, fire, thunder, etc.
<i>Externalizing Scales</i>		
<i>JCP</i>	Juvenile Conduct Problems	Difficulties at school and at home, stealing
<i>SUB</i>	Substance Abuse	Current and past misuse of alcohol and drugs

TABLE 3.1 Continued

<i>Abbreviation</i>	<i>Meaning</i>	<i>Description</i>
<i>AGG</i>	Aggression	Physically aggressive, violent behavior
<i>ACT</i>	Activation	Heightened excitation and energy level
<i>Interpersonal Scales</i>		
<i>FML</i>	Family Problems	Conflictual family relationships
<i>IPP</i>	Interpersonal Passivity	Being unassertive and submissive
<i>SAV</i>	Social Avoidance	Avoiding or not enjoying social events
<i>SHY</i>	Shyness	Bashful, prone to feel inhibited and anxious around others
<i>DSF</i>	Disaffiliativeness	Disliking people and being around them
<i>Interest Scales</i>		
<i>AES</i>	Aesthetic-Literary Interests	Literature, music, the theater
<i>MEC</i>	Mechanical-Physical Interests	Fixing and building things, the outdoors, sports
<i>Personality Psychopathology Five (PSY-5) Scales</i>		
<i>AGGR-r</i>	Aggressiveness-Revised	Instrumental, goal-directed aggression
<i>PSYC-r</i>	Psychoticism-Revised	Disconnection from reality
<i>DISC-r</i>	Disconstraint-Revised	Under-controlled behavior
<i>NEGE-r</i>	Negative Emotionality/Neuroticism-Revised	Anxiety, insecurity, worry, and fear
<i>INTR-r</i>	Introversion/Low Positive Emotionality-Revised	Social disengagement and anhedonia

History of the MMPI-2-RF's development

The Original MMPI

A detailed description of the evolution of the MMPI from its initial publication through the MMPI-2-RF is provided by Ben-Porath (2012). Briefly, Hathaway and McKinley (1943) developed the MMPI as a differential diagnostic measure designed to identify eight commonly occurring (at the time) psychiatric disorders. The then widely used Krapelinian descriptive nosology and psychodynamic and behavioral theory informed the authors' process of generating items and developing the MMPI scales (Ben-Porath, 2012). Hathaway and McKinley assembled their item pool guided by the content of existing psychiatric interviews, the research literature, and their own extensive clinical experience. The authors relied on empirical keying to select items without considering their content, believing that doing so would produce measures less susceptible to distortion because responses to these items were associated with the diagnoses statistically, but not necessarily based on item content. Meehl (1945) noted that because the scales were constructed, and were to be interpreted empirically, they need not be assumed to have similar meaning for different individuals.

Hathaway and McKinley administered the candidate items to different psychiatric patient groups as well as to a normative group consisting of approximately 750 individuals, primarily skilled laborers and farmers. Many of these individuals were visitors to the University of Minnesota Hospital who were approached in the hallways and invited to participate in a psychology study. Items that discriminated sufficiently between members of a specific diagnostic group (e.g., schizophrenia) and the normal group were assigned to the relevant scale. This procedure was followed when developing the eight original MMPI Clinical Scales.

The Clinical Scales turned out to be insufficiently effective in predicting membership in specific diagnostic categories (Hathaway, 1960). As a consequence, the MMPI evolved from an instrument designed to predict diagnoses to one that relied on empirical correlates in assessing symptomatology and personality patterns. Meehl (1956) called for a “good cookbook” that could be used to inform clinicians about clinically relevant features associated empirically with MMPI scales and in particular with code types that were based on patterns of Clinical Scale scores. Comprehensive studies designed to establish correlates of MMPI code types were conducted by Marks and Seeman (1963), who published a codebook that identified code types correlates for psychiatric inpatients, and Gilberstadt and Duker (1965) who reported empirical correlates for 360 male VA inpatients. These and other source books still serve as the foundation for contemporary MMPI–2 interpretation texts such as those written by Graham (2012) and Greene (2011).

The paradigm shift just described succeeded in providing an alternative framework for using the MMPI when the instrument was found not to function well as a direct differential diagnostic indicator. However, at a conference convened in 1970 in honor of the test’s author, Starke Hathaway, several personality assessment experts (Loevinger, 1972; Meehl, 1972; Norman, 1972) discussed some significant psychometric shortcomings of the original Clinical Scales and offered suggestions for updating the then 30-year-old instrument. Another decade was to pass before a research project that generated a revised inventory was launched, and almost two decades passed before the MMPI–2 was published. However, the problems identified at the 1970 MMPI symposium were carried over to the MMPI–2 as a consequence of the development team’s decision to keep the Clinical Scales intact and focus on other pressing needs in the revision.

The MMPI–2

One of the greatest needs was to update the test’s norms. The original normative data were collected in the late 1930s and early 1940s and consisted primarily of Caucasian working-class rural Minnesotans possessing an average of eight years of education. Although this sample was appropriate during the MMPI’s initial conception, it was no longer adequate, given its eventual widespread use across the United States. A second need was to revise some MMPI items that were outdated or phrased awkwardly. Some items had content that was no longer clear, relevant, or appropriate in light of modern language use, cultural practices, and social norms. Other items contained sexist language or controversial religious statements. A final issue concerned the need to include a broader range of content domains that were not necessarily salient in the 1930s but had become very relevant in contemporary assessment (e.g., suicidal ideation, Type A behavior, substance abuse, low self-esteem, and treatment indicators).

To update the inventory, the test’s publisher, the University of Minnesota Press, formed a committee eventually dubbed the *Restandardization Committee*, and charged it with the task of coordinating a revision of the MMPI. The committee had two major, but potentially conflicting goals: improve the test while maintaining continuity with the original version. Continuity was deemed necessary so that the large amount of research generated on the original MMPI Clinical Scales could continue to be relied upon in MMPI–2 interpretation. As discussed later, this emphasis on continuity did, however, come at a significant price.

The restandardization project culminated with the publication of the MMPI–2 (Butcher, Dahlstrom, Graham, Tellegen, & Kaemmer, 1989), for which a new normative sample was assembled consisting of 1,138 men and 1,462 women of various ethnic backgrounds and from different regions of the United States. Continuity was accomplished by leaving the 13 basic Validity and Clinical Scales of the MMPI largely intact. Scale level improvement was accomplished by introduction of 21 new measures, including three new Validity Scales, the MMPI–2 Content Scales

(Butcher, Graham, Williams, & Ben-Porath, 1990), and three Supplementary Scales, two designed to measure gender roles and a third intended to serve as a posttraumatic stress disorder (PTSD) indicator.

During the decade following its publication, the MMPI-2 was the subject of several hundred journal articles, book chapters, books, and doctoral dissertations. Some research focused initially on comparing Clinical Scale scores based on the MMPI and MMPI-2 norms. Concerns about possible incongruence between the two sets of norms were resolved after it was determined that if code-type interpretation was limited to well-defined cases, where there was sufficient separation of the scales defining the code type from the remaining Clinical Scales, the two sets of norms yielded comparable Clinical Scale profiles (Graham, Timbrook, Ben-Porath & Butcher, 1991; McNulty, Ben-Porath & Graham, 1998; Tellegen & Ben-Porath, 1993). The focus then shifted to validating the new scales and exploring further scale development based (in part) on the new items in the inventory. A revised edition of the MMPI-2 manual (Butcher et al., 2001) was designed to update interpretive guidelines for some scales, formalize the discontinuation of other scales, and provide guidelines for interpreting several new measures developed during the decade following the revision. The revised manual did not introduce any changes to the norms or item composition of the MMPI-2 scales included in the 1989 manual.

Structural problems with the MMPI/MMPI-2 Clinical Scales were discussed at length by Jackson (1972), Loevinger (1972), Meehl (1972), and Norman (1972), and revisited by Helmes and Reddon (1993) led Auke Tellegen, a member of the Restandardization Committee, to begin exploring ways they might be addressed. This work culminated in the development of the RC Scales (Tellegen et al., 2003), which was the first step toward the construction of the MMPI-2-RF.

The MMPI-2-RF

MMPI-2-RF development proceeded in two steps: construction of the nine RC Scales followed by augmentation with 42 measures designed to produce a comprehensive assessment of clinically relevant constructs that can be assessed with the MMPI-2 item pool. Discussed in detail by Tellegen et al. (2003) and Ben-Porath (2012), derivation of the RC Scales did not conform to one particular test construction method, or recipe. Methodologically, exploratory factor analyses guided dimensional choices and delineations, and additional correlational analyses served to optimize the internal and intercorrelational structures and external validities of the evolving scales. The conceptual and empirical interplay that produced the RC Scales can be broken down into the following four-step sequence:

1. *Capturing Demoralization.* In a series of analyses, the Demoralization construct, hypothesized to parallel the general bipolar “Pleasantness-versus-Unpleasantness” dimension of self-reported mood (Tellegen et al., 1999a; 1999b; Watson & Tellegen, 1985), and related conceptually (and subsequently empirically) to Frank’s (1974) similarly labeled construct, was empirically tested and corroborated. On the basis of these analyses, a set of Demoralization marker items was identified.
2. *Identifying major distinctive components of the Clinical Scales.* The items of each Clinical Scale were combined with the Demoralization items, and each of these item sets was factor-analyzed separately. These analyses were performed on the eight designated Clinical Scales, as well as on the subsequently developed Scales 5 and 0, each augmented with Demoralization items. A Demoralization factor was recovered from each item set, defined, as expected, by not only the original Demoralization markers but significant numbers of Clinical Scale items as well. In addition, two- or three-factor rotations allowed identification of a meaningful core

non-Demoralization factor in 9 of the 10 item sets, and a four-factor rotation uncovered 2 such factors in the Scale 5 set, producing a total of 11 identified mutually distinctive core factors.

3. *Constructing seed scales.* Carefully delineated *seed scales*, designed to be optimally distinctive, were developed to represent Demoralization and the 11 identified core factors. This entailed a series of correlational analyses designed to ensure that the resulting restructured scales would be optimally distinctive from each other (as well as from Demoralization).
4. *Deriving final RC Scales.* Correlational analyses were conducted, in which the 12 seed scales were correlated with all 567 MMPI-2 items (with due corrections for item overlap), to identify a set of adequately converging measures representing Demoralization and core factors of the eight original Clinical Scales.² These analyses, combined with evaluations of item content, item-scale correlations, and available external item validity data, resulted in the final RC Scale set.

The resulting nine RC Scales are described in Table 3.1.

At the conclusion of the monograph that introduced the RC Scales, its authors noted:

The introduction of the RC Scales may stimulate additional MMPI-2 scale development. It may prove worthwhile to search for and measure distinctive core features of important MMPI-2 scales other than the MMPI-2 Clinical Scales, some of which may also be confounded with a strong Demoralization component. Investigations along these lines may lead to additional measures that are incrementally informative beyond the RC Scales. Through such efforts it may be possible eventually to capture the full range of attributes represented by the large body of MMPI-2 constructs with a set of new scales more transparent and effective than those currently available.

(Tellegen et al., 2003, pp. 85–86)

Tellegen and Ben-Porath (2008/2011) and Ben-Porath (2012) describe the work conducted along these lines to develop the remaining scales of the MMPI-2-RF. This effort produced 9 measures of protocol validity and 33 additional substantive scales.

The MMPI-2-RF Validity Scales (described in Table 3.1) were designed to produce a comprehensive assessment of threats to the validity and interpretability of a test protocol in an efficient manner. Comprehensiveness was accomplished by ensuring that the MMPI-2-RF Validity Scales adequately assess the threats to protocol validity measured by the MMPI-2 Validity Scales. Efficiency was achieved by developing scales that, for the most part, are shorter than their MMPI-2 counterparts while adequately measuring the validity problems they were designed to assess.

Factor analyses of the RC Scales conducted with three clinical samples consistently identified a coherent three-factor structure representing emotional, thought, and behavioral dysfunction. The three MMPI-2-RF Higher-Order (H-O) Scales (described in Table 3.1) are item composites derived to produce measures of these three dimensions in the following manner. First, the combined items of those RC Scales that had been found to be the primary markers of the three higher-order were factor-analyzed in each derivation sample, and from each of these item-level analyses the rotated three-factor solution and corresponding factor scores were obtained. Next, the three factor scores were correlated with each of the 567 MMPI-2 items in each sample. Finally, a set of item-composite measures was obtained by selecting from the MMPI-2 item pool, on the basis of the combined results, a set of diverse and distinctive item markers for each higher-order factor.

As was noted earlier, our goal in constructing the MMPI-2-RF was to represent the clinically significant substance of the MMPI-2 item pool with a comprehensive set of psychometrically

adequate measures. Needed to accomplish this task, in addition to the RC and H-O Scales, were scales measuring distinctive Clinical Scale components *not* represented by the RC Scales (e.g., shyness, a reversed component of Scale 3) and RC Scale facets warranting separate assessment (e.g., substance abuse, a facet of *RC4*), as well as clinically significant attributes not, or not directly, assessed by the RC Scales (e.g., suicidal ideation, which is related to but not specifically assessed by the Demoralization Scale). In developing additional scales to assess the identified target areas, we generally followed the procedures used to construct the RC Scales. Using the same developmental samples, we factor-analyzed item sets representing each construct along with the Demoralization markers used in the second step of constructing the RC Scales. Items that loaded on the Demoralization factor were dropped, except for those that were considered important facets of Demoralization and were used to develop separate measures (e.g., suicidal ideation). Next, seed scales were constructed by dropping candidate items that were too highly correlated with other seed scales. In a final step, correlations were calculated between a set of seed scales and the 567 MMPI-2 items, and items were added to a scale if they were sufficiently correlated with its seed scale, and more highly so than with the other seeds. The resulting 23 Specific Problem (SP) Scales and 2 Interest Scales are described in Table 3.1.

A major addition to the MMPI-2 was the Personality Psychopathology Five (PSY-5) Scales, documented in the revised edition of the MMPI-2 manual (Butcher et al., 2001). The scales were introduced by Harkness, McNulty, and Ben-Porath (1995) as measures of a personality model developed by Harkness and McNulty (1994), who discussed how the five constructs compare with other five-factor models of personality. They reported that the origin for the PSY-5 Scales was research done by Harkness (1992) on the *DSM-III-R* criteria for diagnosing personality disorders (APA, 1987). In contrast to other five-factor models that have been explored *ex post facto* as dimensional models of personality disorder symptomatology, the PSY-5 constructs originated from clinical criteria for diagnosing personality disorders. Since their introduction, a considerable body of research has established that the PSY-5 Scales provide MMPI-2 users an important link to a well-studied model of personality (Harkness, Finn, McNulty, & Shields, 2012). Allan Harkness and John McNulty revised the scales based on the 338-item MMPI-2-RF pool following an iterative process involving internal (item-scale) and external (item-criterion) analyses. The resulting revised PSY-5 Scales are described in Table 3.1.

Psychometric Characteristics of the MMPI-2-RF

Standardization and Norms

Details about the MMPI-2-RF norms and procedures used to derive standard scores for the 51 scales are provided by Ben-Porath and Tellegen (2008/2011). Briefly, with the exception of the transition from gender-specific to non-gendered norms, the MMPI-2-RF normative sample is essentially the same one as used in standardizing the MMPI-2. The practice of reporting and interpreting gendered norms for MMPI scales began with the original test and was maintained with the MMPI-2 (Butcher et al., 1989). Gendered norms were also reported for all the scales included in the revised edition of the MMPI-2 manual (Butcher et al., 2001). However, the use of tests in certain areas, particularly in personnel screening, is governed by laws that prohibit the use of group-specific norms. The federal Civil Rights Act of 1991 explicitly prohibits consideration of race, color, religion, sex, or national origin in employment practices and has been interpreted to prohibit using gendered norms in personnel screening. To address the resulting need for gender-neutral norms, a set of non-gendered norms for all MMPI-2 scales included in the test manual (Butcher et al., 2001) was developed by Ben-Porath and Forbey (2003), who reported finding little to no

interpretable differences between gender-normed and non-gendered MMPI-2 scale scores. Consequently, the MMPI-2-RF standard scores were developed based on the non-gendered MMPI-2 normative sample.

Butcher (2010) is critical of use of non-gendered norms with the MMPI-2-RF, stating:

Unlike the original MMPI and MMPI-2, in which separate gender norms were provided, the MMPI-2-RF authors combined genders into one comparison sample. This situation may result in different standards being applied for men and women in assessment and prediction. Further study of this potential bias needs to be conducted. However, the MMPI-2-RF manuals do not provide the information necessary for exploring this question because raw score data by gender are not reported.

(p. 14)

This criticism reflects a fundamental misunderstanding of group-specific norms. Contrary to Butcher's assertion, gender-based norms *create* different standards for men and women, which can mask meaningful gender differences. Non-gendered norms apply the same standard to men and women's test scores and reflect rather than mask actual gender differences. Butcher's (2010) assertion that the MMPI-2-RF manuals do not provide information necessary to explore this question is also incorrect. Means and standard deviations of scores on the 51 MMPI-2-RF scales are reported in the Technical Manual (Tellegen & Ben-Porath, 2008/2011) by gender for a wide range of samples, including the normative sample. Examination of these data indicates that in the normative sample, mean *T*-scores for the two genders are within 2 points of each other for 37 of the 51 MMPI-2-RF scales. They are within 3–4 points of each other for 9 additional scales. For the remaining 5 scales, men score 6 *T*-score points higher than women on Behavioral Externalizing Dysfunction (*BXD*) and 8 *T*-score points higher on Mechanical Physical Interests (*MEC*) and Disconstraint (*DISC-r*). Women score 6 *T*-score points higher than men on Aesthetic-Literary Interests (*AES*) and 8 *T*-score points higher on Multiple Specific Fears (*MSF*). Gender-based norms would have masked these differences by setting the mean *T*-score for each gender at 50. Moreover, inclusion of extensive, gender-specific descriptive data in the Technical Manual allows MMPI-2-RF users to compare a test-taker's results with those of samples of men and women tested in a wide range of mental health, medical, forensic, personnel screening, and non-clinical settings.

The non-gendered MMPI-2-RF normative sample was created by combining data for the 1,138 men of the MMPI-2 normative sample with those of 1,138 women selected randomly from the 1,462 normative women. Ethnic origin data reported by Ben-Porath and Tellegen (2008/2011) indicate that the MMPI-2-RF normative sample slightly over-represents Caucasians and under-represents Asians and Hispanics. Given the increased rate of growth of the Hispanic population in the United States, the under-representation of this group in the normative sample is even greater. This stems in part from the fact that many Hispanics (and Asians) are immigrants who are not native English speakers and may not have the requisite language skills to complete the MMPI-2 instruments. The MMPI-2 nevertheless has been translated and is in use in several Spanish-speaking and Asian countries. As of the writing of this chapter, a Spanish translation of the MMPI-2-RF, developed and normed in Spain, and a Korean version, developed and normed in South Korea, are already in use, while several additional adaptation projects are ongoing.

Standard *T*-scores for the original MMPI Clinical Scales were derived through a linear transformation of raw scores to standard scores with a mean of 50 and a standard deviation of 10 based on the responses of members of the original normative sample (Hathaway & McKinley, 1942). Because the distribution of raw scores varied across scales, the linear transformation

procedure yielded differential amounts of skewness in the T -score distributions, resulting in the same T -score value (e.g., 70), possibly reflecting different degrees of deviation from the norm in percentile terms. Examination of the raw-score distributions of the MMPI-2 Clinical Scales revealed that the positive skew characterized not only the normative distributions of most of the scales in the 1940s but also the 1989 normative data. A positive skew is to be expected for measures of psychopathology and can be viewed as typical for such measures and also as more appropriate than, for example, a normal distribution (Tellegen & Ben-Porath, 1992). However, for both sets of norms the degree of skewness of the raw-score distributions, and of the corresponding linear T -score distributions, varied from scale to scale. Consequently, the same linear T -score value (e.g., a T -score of 70) could have different percentile values for different scales.

For the MMPI-2, this undesirable feature of linear T -scores was corrected with a minimum of change in the original linear T -score distributions, preserving the typical positive skewness of the MMPI Clinical Scales. This was accomplished by the derivation of uniform T (UT) scores (Tellegen & Ben-Porath, 1992), which were designed to have distributions approximating the “prototypical” linear T -score distribution of the MMPI-2 Clinical Scales. Such a distribution was derived empirically for that purpose and was adopted as a standard. This prototypical distribution is essentially a composite of 16 individual distributions, namely, the non- K -corrected linear T -score distributions of the eight original Clinical Scales in the normative samples for both men and women (omitting Scales 5 and 0 because of their distinctive distributions and methods of derivation). As expected, this distribution is positively skewed. Regression methods were then developed to transform the raw scores of each scale directly into UT-scores that would conform as closely as possible to the composite standard. The UT-score transformation largely eliminated the original linear T -score distributional differences between the Clinical Scales, since the obtained UT-score distributions do conform well to the adopted standard and are consequently quite similar (Tellegen & Ben-Porath, 1992). UT scores were developed for the (eight original) MMPI-2 Clinical Scales as well as for the Content Scales (Butcher et al., 1990), the PSY-5 Scales (Harkness et al., 1995), the RC Scales (Tellegen et al., 2003), and other selected Supplementary Scales. UT scores were also adopted for the MMPI-2-RF, with the exception of the Validity and Interest Scales (for which linear T -scores were adopted because these scales have distinctive distributions dissimilar to the composite uniform distribution). The composite distribution that was targeted to derive UT scores for the MMPI-2 was also used in calculating UT scores for the MMPI-2-RF, so that the standard scores are directly comparable across the two versions of the instrument.

Reliability

Reliability estimates and associated standard error of measurement (SEM) statistics are reported by Tellegen and Ben-Porath (2008/2011) for the 51 MMPI-2-RF scales based on analyses of the test normative data, and several clinical samples. In discussing this information, Tellegen and Ben-Porath (2008/2011) note the need to focus on the SEM estimates, which incorporate both the reliability estimates and consideration of scale score variance in the relevant samples. Along the same lines, it is the SEMs, not the reliability estimates, that directly address the issue of known and potential rate of error associated with a technique (i.e., scale scores) that is of particular relevance in forensic assessments. The importance of focusing on SEMs can be illustrated with some of the data reported in the Technical Manual. For example, reliability estimates for the inconsistent responding measures—Variable Response Inconsistency ($VRIN-r$) and True Response Inconsistency ($TRIN-r$)—are, as expectable, considerably lower (ranging from .16 to .52) than those obtained for the remaining validity scales. However, because scale score variances are also consistently smaller for these scales, the resulting standard errors of measurement (ranging from 7

to 9 *T*-score points) are not appreciably higher than those reported for the remaining validity indicators. As another example, the reliability estimate for the Symptom Validity scale (*FBS-r*) based on Cronbach's alpha for normative sample men is .50. In the same sample, Cronbach's alpha for the Infrequent Responding (*F-r*) scale is .69. However, the SEM for both scales in this sample is 6 *T*-score points, indicating comparable measurement error.

Higher SEM values, such as those found generally for scores on the MMPI-2-RF Validity Scales when compared with the substantive scales, indicate a need to use higher cutoffs to identify significant deviations from the norm. Ben-Porath (2012) notes that this is consistent with the interpretive recommendations for the MMPI-2-RF Validity Scales, which do in fact require greater deviation from the norm (than is required for identification of clinically significant elevation on the substantive scales) to raise substantial concerns about the validity of a test protocol. Exceptions to the requirement for greater deviations from the mean are the Uncommon Virtues (*L-r*) and Adjustment Validity (*K-r*) scales, for which SEM findings are comparable to those reported for the substantive scales. Tellegen and Ben-Porath (2008/2011) similarly indicate that more extreme *T*-scores are needed to justify clinically significant inferences for the substantive scales reported to have greater SEMs.

Validity

Correlations between scores on the MMPI-2-RF substantive scales and other information collected with large samples representing settings in which the test is intended to be used are reported in Appendix A of the Technical Manual (Tellegen & Ben-Porath, 2008/2011). For samples of men and women tested in an outpatient community mental health center, correlations are reported with ratings and information provided by the intake workers who conducted the initial assessments of these individuals and with ratings provided by their therapists after they had been in treatment at the facility for approximately 1 month. For three large psychiatric inpatient samples, correlations are reported with indicators (e.g., psychosocial history, presenting symptoms and complaints, mental status on admission, initial diagnostic impressions, medication at intake and discharge) recorded by research assistants who systematically reviewed their medical charts. Correlations with indicators based on chart data (including psychosocial history, legal history, mental status, and diagnostic impressions) are also reported for two large samples of men and women undergoing forensic assessments related to competency to stand trial evaluations and insanity pleas. In addition, correlations with a wide range of standard self-report measures are reported for samples of mental health and medical patients, individuals assessed at intake to a substance abuse treatment program, large samples of men and women assessed in connection with disability claims and personal injury litigation, and non-clinical samples of male and female college students.

All told, the thousands of external correlates (53,970) reported in Appendix A are based on data provided by 4,336 men and 2,337 women using 605 different criteria. These validity findings served as the primary source for identifying the empirical correlates of substantive scale scores listed in the interpretive guidelines provided in the MMPI-2-RF Manual for Administration, Scoring, and Interpretation (Ben-Porath & Tellegen, 2008/2011). Correlates were included on these lists if they replicated across setting, gender, and criterion source.

As of December 2012, the MMPI-2-RF literature includes over 160 articles published in peer-reviewed journals. Most of these publications report empirical findings on MMPI-2-RF scale sets using data collected in various settings where the test is in use. This substantial peer-reviewed literature has accumulated in a relatively short period of time because existing MMPI-2 data can be used to conduct MMPI-2-RF research, and also because one set of measures, the Restructured Clinical (RC) Scales, was first introduced in early 2003, five years prior to publication of the

MMPI-2-RF. An up-to-date list of MMPI-2-RF references, organized by topic, is maintained at the test publisher's website.³

Approximately a quarter of the MMPI-2-RF peer-reviewed literature focuses on the Validity Scales. Two studies provide empirical findings on measures used to detect content non-responsiveness (Dragon, Ben-Porath, & Handel, 2012; Handel, Ben-Porath, Tellegen, & Archer, 2010) and one investigation explores the utility of the under-reporting measures. Most of the investigations focus on assessment and identification of over-reporting with the MMPI-2-RF. Sixteen of these studies examine the utility of various over-reporting indicators with forensic samples (Dionysus, Denney, & Halfaker, 2011; Gervais, Ben-Porath, Wygant, & Green, 2007; 2008; Gervais, Ben-Porath, & Wygant, 2009; Gervais, Ben-Porath, Wygant, & Sellbom, 2010; Gervais, Wygant, Sellbom, & Ben-Porath, 2011; Nelson, Sweet, & Heilbronner, 2007; Rogers, Gillard, Berry, & Granacher, 2011; Sellbom, Toomey, Wygant, Kucharski, & Duncan, 2010; Smart et al., 2008; Tsushima, Geling, & Fabriga, 2011; Wygant et al., 2011; Wygant et al., 2009; Wygant et al., 2010; Youngjohn, Wershba, Stevenson, Sturgeon, & Thomas, 2011).

Other forensically focused peer-reviewed MMPI-2-RF publications include studies with civil litigants (Gervais et al., 2009; Thomas & Youngjohn, 2009), and research related to the assessment of psychopathy (e.g., Rock, Sellbom, Ben-Porath, & Salekin, in press; Sellbom, 2011; Sellbom, Ben-Porath, & Stafford, 2007; Wygant & Sellbom, in press). Additional investigations focus on criminal defendants (e.g., Mattson, Powers, Halfaker, Akenson, & Ben-Porath, 2012; Sellbom, Ben-Porath, Baum, Erez, & Gregory, 2008; Wygant et al., 2007), an examination of scores of individuals undergoing parental competency evaluations (Stredny, Archer, & Mason, 2006), and a descriptive study of scores generated by a sample of child custody litigants (Archer, Hagan, Mason, Handel, & Archer, 2012).

Most of the peer-reviewed MMPI-2-RF publications did not focus specifically on forensic settings or samples, but many of these investigations can nonetheless inform forensic users of the inventory. Studies conducted in medical settings, for example an investigation on use of the MMPI-2-RF to differentiate between patients experiencing epileptic or non-epileptic seizures (Locke et al., 2010), or a study on use of the RC Scales to identify depression in patients with chronic pain (McCord & Drerup, 2011), provide findings that can be of particular relevance in civil litigation.

MMPI-2-RF research conducted in mental health settings can similarly inform use of the measure in forensic assessments when psychological functioning is at issue. For example, possible thought dysfunction, as indicated by various MMPI-2-RF measures, would not be expected to appear differently on the inventory in individuals undergoing forensic evaluations than it does in test-takers assessed in traditional mental health settings. Examples of such research include the work of Arbisi, Polusny, Erbes, Thuras, and Reddy (2011), who report findings on screening for post-traumatic stress disorder (PTSD) and mild traumatic brain injury (mTBI) in a sample of National Guard troops following a 15-month deployment in Iraq; Arbisi, Sellbom, & Ben-Porath (2008); Handel and Archer (2008), who identify psychopathology-focused empirical correlates of the RC Scales in psychiatric inpatients; Ozonoff, Garcia, Clark, and Lainhart's (2005) report of RC Scale findings in high-functioning adults with autism spectrum disorders; Purdon, Purser, and Goddard's (2011) report of MMPI-2-RF over-reporting scale scores of individuals with first-episode psychosis; Sellbom, Bagby, Kushner, Quilty, and Ayeast's (in press) examination of the diagnostic construct validity of the MMPI-2-RF; and Watson, Quilty, and Bagby's (2011) study of use of the MMPI-2-RF to identify individuals with bipolar disorder.

Psychometric Strengths and Weaknesses

In addition to the external correlate data just discussed, analyses reported in the Technical Manual include 39,855 correlations between various MMPI-2-RF and MMPI-2 scale scores. These analyses, conducted with a broad range of samples representing settings in which the MMPI-2-RF is used, provide detailed information about the internal structure of the inventory. In addition, descriptive data (means and standard deviations on the 51 MMPI-2-RF scales) are reported for 68,377 individuals assessed in various mental health, medical, forensic, personnel screening, and non-clinical settings where the test is used.

The research just described indicates that the psychometric strengths of the MMPI-2-RF include substantially improved discriminant validity that was achieved without sacrificing (and actually often enhancing) convergent validity or reliability. Data reported in the Technical Manual provide an unparalleled (both in quantity and quality) amount of psychometric data that can be used to evaluate and guide interpretation of the MMPI-2-RF. Complemented by the growing peer reviewed literature just described, these data document the psychometric strengths of the test.

A related strength involves the substantially enhanced construct validity of the instrument. As discussed in detail by Ben-Porath (2012), beginning in the 1960s, MMPI research became increasingly focused on the cookbook approach to test interpretation. As investigators compiled sizable data sets and identified clinically relevant empirical correlates of the code types, MMPI interpretation became almost exclusively focused on these correlates—that is, on criterion validity rather than more broadly on construct validity. Basic researchers, on the other hand, found limited value in the practically convenient but theoretically undeveloped code types and turned instead to alternative measures for studying personality and psychopathology. This schism had unfortunate consequences for both camps. MMPI research became increasingly divorced from developments in the fields of personality and psychopathology, and basic researchers lost access to rich sources of clinical data available on literally hundreds of thousands of individuals tested with the instrument in a broad array of mental health, medical, forensic, and various non-clinical settings.

A critical aspect of restructuring the MMPI involved identifying links between the various restructured scales and contemporary models and concepts in personality and psychopathology. These associations, discussed and documented in detail by Ben-Porath (2012), provide a solid foundation for relying on construct validity (i.e., an empirically grounded understanding of the underlying constructs assessed by its scales) in MMPI-2-RF interpretation.

Substantial as it is, the empirical literature available to guide MMPI-2-RF interpretation is not as broad as the literature accumulated with the original MMPI or MMPI-2.⁴ The restructured inventory has not, at this point, been studied in as wide a range of settings and applications as were its predecessors, and documentation of its psychometric functioning in additional settings will further strengthen its foundations.

Clinical Uses and Limitations

The MMPI-2-RF is primarily a clinical assessment instrument, although it is also used in a variety of non-clinical settings, including in forensically focused assessments. Clinically, the test is used as a measure of personality and psychopathology in a broad range of mental health and medical settings. To facilitate its use in these settings, Appendix D of the Technical Manual provides *Comparison Group* data, descriptive findings (mean *T*-scores and their associated standard deviations) on the 51 MMPI-2-RF scales for several large samples representing mental health and medical populations. These samples include 370 men and 582 women tested at an outpatient community mental health center, 246 men and 432 women who completed the test at an outpatient private

practice, 659 men and 498 women tested at a psychiatric inpatient unit of a community hospital, 1,059 men tested in an inpatient psychiatric unit of a VA hospital, 1,151 men assessed at intake to a substance abuse treatment unit of a VA hospital, 367 men and 894 women assessed at intake to a college counseling clinic, 228 men and 435 women evaluated as part of a pre-surgical evaluation for bariatric surgery, and 263 men and 265 women evaluated as part of a pre-surgical screening of candidates for spine surgery or spinal cord stimulator implants.

Interpretation guidelines provided by Ben-Porath and Tellegen (2008/2011) identify a broad range of clinically relevant features associated with scores on the 42 substantive MMPI-2-RF scales. These include psychopathology symptoms, personality characteristics, and behavioral tendencies linked empirically and conceptually to the substantive scales as well as diagnostic and treatment considerations. The empirical correlates listed in the test manual are based primarily on the validation data reported in the MMPI-2-RF Technical Manual (described earlier). These findings are augmented by a growing, robust literature that reports clinically relevant MMPI-2-RF findings. The following selective review⁵ of this literature is designed to illustrate the types of clinically relevant information that can be obtained with the instrument.

Psychodiagnostic Assessments

The MMPI was originally designed to serve as a direct differential diagnostic instrument. And though it did not succeed in this narrow capacity, the test has been used to assess constructs of relevance to psychodiagnostic assessments throughout its history. Consistent with this focus, several studies have been published that can inform use of the MMPI-2-RF as a psychodiagnostic instrument. The first set of studies reviewed here examined the construct validity of MMPI-2-RF scales by testing a series of conceptually based hypotheses. The second set of investigations includes findings related to the assessment of post-traumatic stress disorder (PTSD) and mild traumatic brain injury (mTBI) in combat veterans. Another topic covered in this area involves a study pertinent to assessment of psychopathy, a topic of particular relevance to forensic psychologists.

Psychodiagnostic Construct Validity

Sellbom et al. (in press) examined the diagnostic construct validity of the MMPI-2-RF in a sample of patients diagnosed with Bipolar Disorder, Major Depressive Disorder, or Schizophrenia on the basis of the Axis I module of the Structured Clinical Interview for *DSM-IV* (SCID). The authors tested a series of conceptually based expectations, and their findings were generally congruent with their predictions, lending support to construct validity of the instrument. In univariate analyses, the Specific Problems (SP) scale Activation (*ACT*) provided the strongest differentiation between patients diagnosed with Bipolar Disorder and Major Depression. The Higher-Order (H-O) scale Emotional/Internalizing Dysfunction (*EID*) and Restructured Clinical (RC) scales Demoralization (*RCd*) and Ideas of persecution (*RC6*) best differentiated between individuals diagnosed with Bipolar Disorder and Schizophrenia. *EID*, *RCd*, the RC scale Low Positive Emotions (*RC2*) and the SP scale Malaise (*MLS*) differentiated best between patients diagnosed with Major Depression and Schizophrenia. Using hierarchical logistic regression analyses, the authors found that the RC Scales provided incrementally improved group membership prediction beyond the H-O Scales and that the SP Scales added incrementally to predictions based on a best-weighted combination of H-O and RC scale scores.

In a related study, Watson et al. (2011) examined the ability of the MMPI-2-RF to differentiate patients with lifetime Bipolar Disorder from patients with lifetime Major Depressive Disorder and to differentiate patients with Bipolar Disorder who were depressed at the time they were assessed

from individuals diagnosed with Major Depressive Disorder. The ACT scale performed best in both analyses. The authors followed up on their multivariate analyses with a series of Receiver Operator Curve (ROC) analyses, seeking to identify for each scale the cutoff score that produced the largest combined sensitivity and specificity values. An ACT raw score of 4 was identified as optimal for both differential diagnostic tasks. This translates into a *T*-score value of 53, which falls well below the traditional cutoff of 65 *T*. The mean ACT *T*-score (and standard deviation) for the Bipolar Disorder group was 58.79 (13.07), compared with 47.15 (10.23) for patients diagnosed with Major Depression. The finding that a *T*-score of 53 best differentiated between the groups reflects the tendency of Bipolar patients to score somewhat above average and Major Depression patients to score somewhat below average on ACT. It also illustrates that, though a single point of demarcation for clinically significant elevation at 65 *T* on the substantive scales serves a useful heuristic purpose, there is no reason to expect the same cutoff to be optimal across differential diagnostic tasks and scales.

PTSD and Mild Traumatic Brain Injury

In a study of particular relevance to psychologists working in the military or VA systems, Arbisi and colleagues (2011) reported MMPI-2-RF findings for a sample of National Guard soldiers tested following a 15-month deployment in Iraq during the height of one of the most difficult periods of operations in that country. The sample consisted of soldiers who, prior to being deployed, had agreed to be contacted after their return and completed the instrument within 3–11 months post-deployment. These individuals also completed the PTSD Checklist (PCL)–Military Version and a mild traumatic brain injury (mTBI) screen. The sample was classified into four groups: 166 soldiers who did not screen positive for PTSD or mTBI, 9 who screened positive for both conditions, and two groups made up of soldiers who screened positive for one condition (PTSD, $n = 21$ or mTBI, $n = 33$) but not the other.

Descriptive findings indicated that participants who did not screen positive for either condition produced mean *T*-scores that fell well within normal limits, with quite a few falling below the general population mean. The soldiers who screened positive only for mTBI also produced mean scores that fell largely within half a standard deviation of the general population mean, with the exception of the RC scale Somatic Complaints (*RC1*; $M = 56.8$, $SD = 13.4$), the SP scales Head Pain Complaints (*HPC*; $M = 56.5$, $SD = 12.4$), and Neurological Complaints (*NUC*; $M = 58.2$, $SD = 13.5$). Returning soldiers who screened positive only for PTSD produced clinically elevated mean scores on the SP scales Anxiety (*AXY*; $M = 65.6$, $SD = 21.1$) and Gastrointestinal Complaints (*GIC*; $M = 65.4$, $SD = 16.2$), and mean elevations at least 1 standard deviation above the general population mean on *RC1* ($M = 61.0$, $SD = 13.6$), and the SP scale Cognitive Complaints (*COG*; $M = 62.6$, $SD = 15.2$). The small group of individuals who screened positive for both PTSD and mTBI produced the most deviant mean MMPI-2-RF scores, with a clinically significant mean elevation on *GIC* ($M = 65.4$, $SD = 13.8$), and mean elevations at least 1 standard deviation above the general population norms on *RC1* ($M = 63.3$, $SD = 12.6$), *MLS* ($M = 61.8$, $SD = 11.8$), *HPC* ($M = 61.2$, $SD = 15.5$), *NUC* ($M = 63.5$, $SD = 18.1$), *COG* ($M = 63.7$, $SD = 12.8$), *AXY* ($M = 63.6$, $SD = 15.8$), Social Avoidance (*SAV*; $M = 60.4$, $SD = 6.2$), and the Personality Psychopathology Five (PSY-5) scale Introversion/Low Positive Emotionality (*INTR-r*; $M = 61.0$, $SD = 12.3$).

A noteworthy aspect of these findings is that the vast majority of this sample of soldiers returning from a difficult deployment showed no evidence of psychological dysfunction when tested in a research context within several months of their return. As noted, those who did not screen positive for either condition (approximately two-thirds of the sample) scored below average on several

MMPI-2-RF scales, perhaps reflecting a psychological resiliency that served as a protective factor. Consistent with the literature in this area, the group that screened positive only for mTBI showed no indications of residual psychopathology, whereas the groups that screened positive for PTSD (either with or without mTBI) did evidence considerable psychological dysfunction in the emotional, somatoform, and interpersonal domains.

In a subsequent study that included some participants from the investigation just discussed, Nelson et al. (2011) reported MMPI-2-RF findings for a sample of 128 Operation Enduring Freedom/Operation Iraqi Freedom (OEF/OIF) veterans and active duty personnel. All participants reported a history of at least one combat-related brain injury that was no greater than mild in severity according to the American Congress of Rehabilitation Medicine (ACRM; Kay et al., 1993) criteria. Individuals with histories of mild “complicated” concussions, as indicated by demonstrable evidence of structural brain injury on neuroimaging, were not included in this investigation. The sample as a whole reported a mean of 3.4 combat-related concussions ($SD = 4.4$). On average, these individuals completed the MMPI-2 (transformed for this investigation to the MMPI-2-RF) 35.8 months after their most recent injury ($SD = 18.8$). A total of 45 participants reported a history of definite brief loss of consciousness as a result of a combat-related concussion; 49 reported a history of alteration without complete loss of consciousness; and 34 veterans reported experiencing some neurologic signs resulting from a concussion, but denied any loss or alteration of consciousness.

Participants were divided first into three groups. A forensic group was made up of 42 individuals who were consecutively assessed as part of a VA compensation and pension (C&P) evaluation. A clinical group consisted of 43 OEF/OIF veterans who consecutively underwent neuropsychological assessments within a physical medicine and rehabilitation setting related to a self-reported history of concussion. A research group included 43 OEF/OIF veterans who were selected from the Arbisi et al. (2011) study just reviewed. The authors compared scores generated by the three groups on the MMPI-2 and MMPI-2-RF over-reporting indicators, the RC Scales, and select SP scales, finding many substantial differences between both the forensic and the clinical groups when compared with the research group, and only relatively small differences in comparisons of the forensic and clinical groups.

The last comparison was confounded by follow-up findings indicating that 55.8% of the clinical group had indicated that they were involved in a pending C&P claim. The follow-up analysis indicated that 20.9% of the research participants also had an active C&P claim; however, they had been informed that their MMPI-2-RF data would have no bearing on any current or subsequent service-connected claims. The authors therefore regrouped the study participants into two subsamples consisting of individuals with ($n = 66$) and without ($n = 53$) a pending active C&P claim. Comparisons of these two groups yielded very large differences (Cohen’s $d > .70$) on *FBS*, *FBS-r*, and *RC1* and large differences (Cohen’s $d = .50-.69$) on *RBS*, *RCd*, *MSL*, *GIC*, *HPC*, *NUC*, and *COG*, with the active claim group scoring higher on all scales.

Each of the two groups was subdivided further into three subgroups consisting of those diagnosed with PTSD, those diagnosed with another Axis I disorder, and those with no diagnosis. Because a comparison of the six groups did not produce a significant multivariate effect (most likely owing to low power), the authors did not report any univariate comparisons. However, examination of the descriptive data reported for the groups points to some instructive patterns. In each of the three diagnostic groups (PTSD, other diagnosis, no diagnosis), those with active claims scored consistently higher than those with no active claims on both the over-reporting indicators and the substantive MMPI-2-RF scales. For example, the mean *FBS-r* *T*-score for military personnel and veterans diagnosed with PTSD with no active claim was 65.5 ($SD = 12.0$). In contrast, the mean *FBS-r* *T*-score for those diagnosed with PTSD who did have an active claim was 80.3 ($SD = 13.4$). The mean *RC1* *T*-score for the non-claimant group of individuals diagnosed

with PTSD was 66.1 (SD = 13.0), whereas the claimants with the same diagnosis produced a mean *T*-score of 78.7 (SD = 12.6).

Overall, Nelson et al.'s (2011) findings indicated that individuals with self-reported concussions but no co-morbid PTSD or other Axis I disorders showed no evidence of over-reporting or psychopathology on the MMPI-2-RF. Those diagnosed with an Axis I disorder who did not have active claims also showed no evidence of over-reporting (with the exception of RBS, which was higher than expected for this group), along with clinically significant elevations on measures of anxiety, anger, and somatic complaints (including neurological complaints). Corresponding groups with active claims produced considerably higher scores on measures of over-reporting, anxiety, and somatic complaints.

Psychopathy

In a study focusing on the assessment of Psychopathy, Sellbom, Ben-Porath et al. (2012) reported correlations between scores on select MMPI-2-RF substantive scales and measures of global psychopathy and its two primary facets Fearless-Dominance (FD) and Impulsive Antisociality (IA) as assessed by the Psychopathic Personality Inventory (PPI, Lilienfeld & Andrews, 1996). The MMPI-2-RF scales included in their analyses were selected to test specific hypotheses on the basis of their construct validity-based links to the psychopathy literature. Overall, the findings were quite consistent with a priori hypotheses, lending support to the construct validity of the MMPI-2-RF in the assessment of psychological facets of psychopathy. The single best MMPI-2-RF predictor of global psychopathy was the RC scale Antisocial Behavior (*RC4*), followed closely by Hypomanic Activation (*RC9*). The strongest predictor of FD was a lower score on the SP scale Shyness (*SHY*), followed closely by a lower score on Social Avoidance (*SAV*). *RC4* was also the best predictor of IA. The authors also derived regression equation-based MMPI-2-RF measures of the three psychopathy-related constructs, which they found to be differentially correlated as expected with scores on the Screening Version of the Psychopathy Checklist (PCL-SV; Hart, Cox, & Hare, 1995) in a forensic sample of pre-trial criminal defendants, as well as with a broad range of other measures used commonly in the assessment and study of psychopathy. The newly derived MMPI-2-RF psychopathy measures thus showed considerable promise and will hopefully be the subject of further empirical investigation.

Psychological Assessment of Medical Patients

Several studies have examined the utility of MMPI-2-RF scales in assessments of patients with seizure disorders. These studies are reviewed first, followed by research with chronic pain patients and spine surgery candidates.

Seizure Disorder

The MMPI has been used in assessments of non-psychiatric medical patients since it was first developed. Two of the original Clinical Scales, 1 and 3, were designed to assist with the long-challenging task of identifying functional (i.e., psychogenic) somatic complaints. Hathaway and McKinley went through multiple iterations in their efforts to make the two scales maximally independent of each other while sufficiently sensitive to their targeted diagnoses (Hypochondriasis and Conversion Disorder respectively). Ben-Porath (2012) notes that when Tellegen was developing the RC Scales, somatic complaints emerged as the major non-demoralization component of both scales, leading to the construction of a single RC scale Somatic Complaints (*RC1*) representing this

common variance of Clinical Scales 1 and 3. When the remaining substantive scales of the MMPI-2-RF were developed, additional measures—Malaise (*MLS*), Gastrointestinal Complaints (*GIC*), Head Pain Complaints (*HPC*), and Neurological Complaints (*NUC*)—were derived to provide more specific information related to possible somatoform psychopathology.

Thomas and Locke (2010) conducted a comprehensive study of the MMPI-2-RF Somatic Scales, using data generated in the assessment of 399 medical patients diagnosed with either epilepsy or psychogenic non-epileptic seizures. Using a series of sophisticated analyses, the authors examined the internal structure of *RC1* and the additional somatic scales, applying taxometric, confirmatory factor analysis, and item response theory-based procedures. They concluded that the unidimensional model adopted by its developers provided an adequate fit for the structure of *RC1* and that the additional Somatic Scales and that *RC1* scores are most precise in the *T*-score range of 55–90. Their overall conclusion was that *RC1* appears to be well suited for the assessment of somatization in medical patients.

Using the same sample, Locke et al. (2010) examined the utility of the MMPI-2-RF in differentiating between patients diagnosed with epilepsy (ES) and psychogenic non-epileptic seizures (NES) in a hospital-based epileptic monitoring unit. A comparison of mean *T*-scores for the two groups, indicated that the NES patients scored substantially higher than those with ES on *FBS-r*, *RC1*, *MLS*, *HPC*, and *NUC*. Among these scales, the only one on which the ES group produced a mean at or above 65*T* was *NUC* ($M = 71.4$, $SD = 13.0$), with mean scores on *RC1* and *MLS* approaching, but falling below, this cutoff. In contrast, the NES group produced clinically elevated scores on *RC1* ($M = 73.7$, $SD = 12.4$), *MLS* ($M = 71.1$, $SD = 11.8$), and *NUC* ($M = 82.1$, $SD = 12.9$). These findings illustrate that the presence of a significant medical condition, ES, is generally not sufficient to produce clinically significant elevation on *RC1* or *MLS*, and in the case of *NUC*, the scale most directly linked to the presenting symptoms in both groups, the NES sample produced a mean score more than 3 standard deviations above the general population norm.

Locke et al. (2010) also reported classification accuracy findings for *FBS-r*, *RC1*, and *NUC*. For over-reporting indicators, using cutoffs that produce high specificity levels is critical to avoid an inaccurate conclusion that a patient presenting with seizures is over-reporting. The lowest cutoff designated in the MMPI-2-RF *Manual for Administration, Scoring, and Interpretation* (Ben-Porath & Tellegen, 2008) for identifying possible over-reporting based on *FBS-r* is 80*T*, which in this study produced a .85 specificity estimate. At the highest *T*-score cutoff reported (90), specificity was .98, indicating that virtually none of the patients with confirmed ES diagnoses scored this high on *FBS-r*. On the substantive scales, Ben-Porath and Tellegen's (2008/2011) interpretive recommendations for *RC1* designate two levels of clinically significant elevation on *RC1*, namely 65*T* and 80*T*, with the latter indicating a particularly strong likelihood of somatoform psychopathology. A 65*T* cutoff on *RC1* produced a sensitivity estimate of .76 with specificity estimated at .60 in Locke et al.'s sample. At 80*T*, the respective sensitivity and specificity estimates were .31 and .88, indicating that about a third of the NES individuals and only 12% of the ES sample would be identified as very likely presenting with somatoform symptoms. The corresponding cutoffs for *NUC* are 65*T* (sensitivity = .91, specificity = .27) and 90*T* (sensitivity = .37, specificity = .90).

In another study, Locke and Thomas (2011) used the same sample to develop MMPI-2-RF scales designed to optimally identify NES patients. After first identifying MMPI-2-RF items that best differentiated between the ES and NES groups, the authors conducted a series of statistical and conceptual analyses that led them to develop two scales, labeled Psychogenic Nonepileptic Seizures Physical Complaints (*PNES-c*) and Psychogenic Nonepileptic Seizures Attitudes (*PNES-a*). Using the same sample to examine the classification accuracy of these scales, the authors reported a modest improvement over single or combined MMPI-2-RF scales, and recommended that further work be conducted to cross-validate their findings with independent samples.

In a follow-up study, Locke et al. (2011) reported the results of a randomized comparison of the PAI and MMPI-2/MMPI-2-RF in differentiating between a new set of patients diagnosed with ES and NES. A sample of patients enrolled in an epilepsy monitoring unit study was randomly assigned to be tested with either the PAI or the MMPI-2 (which was also converted to the MMPI-2-RF). Application of standard exclusion criteria produced 73 valid PAI protocols, 62 valid MMPI-2 protocols, and 63 valid MMPI-2-RF protocols. The reported analyses were limited to classification accuracy statistics for two PAI scales, Somatic Complaints (SOM) and Somatic Complaints-Conversion (SOM-C), two MMPI-2-based indexes that had been developed previously for this differential diagnostic task, and RC1, which can be scored on both versions of the MMPI and a combination of scores on Locke and Thomas's (2011) MMPI-2-RF *PNES-a* and *PNES-c* scales. Examining first the generally recommended cutoffs, 70T for the PAI scales, 65T for RC1, and optimized combinations of the specially developed MMPI-2 and MMPI-2-RF measures, Locke et al. (2011) concluded that the PAI scales performed best, with estimated hit rates of .79, followed by RC1, with an estimated hit rate of .73. None of the specialized MMPI-2 or MMPI-2-RF indicators outperformed RC1. Following up with examination of alternative cutoffs on the PAI and MMPI-2-RF scales, Locke et al. (2011) found that the RC1 cutoff used in their initial analyses, 65T, provided the best MMPI-2-RF-based differentiation of ES- and NES-diagnosed patients. The authors concluded that the PAI may be the better personality assessment measure for use in distinguishing patients with epilepsy from those with psychogenic seizures, though they noted that they did not compare the instruments' ability to assess other constructs of interest in these assessments, such as anxiety and mood disorders.

Several aspects of this investigation warrant further consideration. First, the MMPI-2-RF-based classification accuracy estimates reported by Locke et al. (2011) differed substantially from the ones reported by Locke et al. (2010). For example, for the RC1 cutoff 65T, Locke et al. (2010) reported estimated sensitivity and specificity rates of .76 and .60 respectively, whereas in the later investigation (Locke et al., 2011) the respective estimates were .97 and .50. A critical difference that may account for these discrepant findings is that the Locke et al. (2010) analyses were based on a sample of 399 patients (196 ES and 203 NES), whereas the Locke et al. (2011) analyses were based on a considerably smaller sample of 63 patients (32 ES and 31), which is less likely to yield stable classification accuracy estimates.

Second, the sensitivity rate for RC1 at 65T (.97) substantially outperformed the best PAI scale (SOM at 70T, with sensitivity estimated at .83). Personality tests are not actually used to diagnose an NES disorder. They are best viewed as screening devices for this condition. As such, the measure with the highest sensitivity level may be best suited to assist in identifying patients whose symptoms are suspect and should therefore be referred for a costly hospital study. Based on Locke et al.'s (2011) findings, using RC1 would result in the identification of nearly 97% of NES patients at the expense of also having to test and verify the condition of 50% of individuals with ES.

Third, Locke et al. (2010) identified several other (than RC1) MMPI-2-RF scales (*FBS-r*, *MLS*, *GIC*, *HPC*, and *NUC*) that were able to differentiate between ES and NES patients. These scales were not included in the analyses reported by Locke et al. (2011). Finally, the MMPI-2-RF scales in Locke et al.'s (2011) comparative study were scored on the basis of an administration of the full MMPI-2. It is possible that using the shorter MMPI-2-RF booklet (comparable in length to the PAI) may enhance the quality of data produced by patients undergoing a comprehensive neuropsychological assessment. Further investigation using larger samples, all of the potentially relevant MMPI-2-RF scales, and the shorter MMPI-2-RF booklet is needed before any definitive conclusions may be drawn about the relative utility of the MMPI-2-RF in these assessments.

Chronic Pain and Depression

McCord and Drerup (2011) compared the ability of the RC Scales and the original Clinical Scales to differentiate between depressed and non-depressed individuals with chronic pain undergoing a neuropsychological assessment. Relying on clinical diagnoses derived from a number of sources, including the Clinical Scales but not the RC Scales, the authors classified a sample of 316 chronic pain evaluatees into those diagnosed with any form of depression and those who were not. In univariate comparisons, the depressed group scored substantially higher (5 *T*-scores points or more) than the non-depressed group on 8 of the 10 Clinical Scales (the exception being Scales 5 and 9), with mean scores around 80 *T* on Clinical Scales 1, 2, and 3, and clinically elevated means (65 *T* or higher) on scales 4, 6, 7, and 8. The non-depressed group produced clinically significant mean elevations on Clinical Scales 1, 2, 3, and 8. By contrast, the depressed group scored substantially higher than the non-depressed group on four of the RC Scales (*RCd*, *RC1*, *RC2*, and *RC7*) and substantially lower on *RC9*. The depressed group produced clinically significant elevations on *RCd*, *RC1*, and *RC2*, whereas the non-depressed sample of chronic pain patients produced clinically significant elevations only on *RC1*. Thus, consistent with one of the main goals of the restructuring, the RC Scales provided a much more discriminating pattern of results.

Assessing Spine Surgery Candidates

Block, Ben-Porath, and Marek (2012) present empirical findings with spine surgery candidates who completed the MMPI-2-RF as part of a pre-surgical psychological screening. They report descriptive analyses with a sample of 1,054 individuals and correlations between MMPI-2-RF scale scores and measures of the primary risk factors for negative surgical outcomes in a smaller sample of 197 pre-surgical candidates. The descriptive findings showed that, by and large, spine surgery and spinal cord implant candidates of both genders produce comparable MMPI-2-RF scores, indicating against the need to rely on separate comparison groups when contrasting a back surgery candidate's test results with a relevant cohort. Similar findings with a subset of these data led Tellegen and Ben-Porath (2008/2011) to combine spine surgery and spinal cord implant candidates into a single comparison group in Appendix D of the MMPI-2-RF Technical Manual.

With several noteworthy exceptions, back surgery candidates produced mean MMPI-2-RF substantive scale scores that were well within normal limits. Scores on *RC1*, *MLS*, *HPC*, and *NUC*, all measures of somatic complaints, were considerably above average. Ben-Porath (2012) notes that a moderate *RC1* elevation may reflect contributions of actual physical health problems; however, as the *RC1* score increases, so does the likelihood that psychological factors play a significant role in the somatic symptom reporting of the test-taker. Block et al. (in press) indicate that *RC1* *T* scores above 75 are uncommon in back surgery candidates, and point to likely somatization. Consistent with prior findings, the *MLS* results indicated that back pain surgery candidates are likely to present as being debilitated by physical health problems and having low energy levels. *T*-scores of 80 and above on *MLS* and 75 and higher on *HPC* and *NUC* are uncommon for this population and can be interpreted as indicating a substantially increased likelihood of somatoform psychopathology contributing to the pain experienced and reported by spine surgery and spinal cord stimulator patients. The empirical correlates reported by these authors link MMPI-2-RF substantive scale scores to a number of risk factors for failed back surgery.

Limitations

As noted earlier, although a robust empirical literature is already available to help guide MMPI-2-RF interpretation, additional research focused on various clinical applications of the test can build further on this foundation. Additional studies reporting correlations between MMPI-2-RF scale scores and findings of structured diagnostic interviews will assist in providing empirical support (or refutation) for the largely inferential⁶ diagnostic considerations listed in the MMPI-2-RF interpretive guidelines. Treatment outcome studies will similarly be needed to elaborate on the largely inferential treatment considerations listed in the test manual. Ongoing studies of the ability to predict negative outcomes based on MMPI-2-RF scale scores of spine surgery and bariatric surgery candidates will assist in using the test as a screening measure in medical settings.

Forensic Uses and Limitations

As a clinical assessment device, the MMPI-2-RF does not provide direct answers to psycholegal questions such as whether a criminal defendant is competent to stand trial, or whether a personal injury litigant sustained psychological damages for which another party is liable, or what is the best custody arrangement consistent with the needs of a child. On the other hand, as an empirically validated assessment instrument the test provides relevant information that can be used to assist the examiner in addressing these and other forensic referral questions. To the extent that an individual's psychological functioning (or dysfunction), personality characteristics, and behavioral tendencies are at issue in a forensic evaluation, the research cited in this chapter, the broader peer-reviewed MMPI-2-RF literature, and the extensive empirical data reported in the Technical Manual for the instrument provide a solid foundation for using the test in forensic assessments.

An additional area where the MMPI-2-RF may be of particular value in forensic assessments is in the assessment of how the individual involved approached the evaluation. Incentives to over-report or under-report psychological problems are inherent in virtually all forensic assessments. Availability of well-validated measures of over-reporting and under-reporting is likely one of the reasons why the MMPI-2 became one of the psychological tests most widely used in forensic evaluations—a pattern that is likely to continue with the MMPI-2-RF, for which a set of well-validated measures of test-taking approach is available.

Forensic practitioners contemplating using the MMPI-2-RF (or any other clinical instrument for that matter) in their assessments need also to consider potential challenges and drawbacks. This section begins with a discussion of this topic and how such challenges can be addressed, which is followed by a review of the literature on the MMPI-2-RF Validity Scales and additional research of particular relevance to forensic users of the test.

Potential Challenges to MMPI-2-RF Use in Forensic Assessments

Forensic practitioners face unique challenges when a new version of a psychological test is released. An expert who uses the newer version of the test may be challenged for relying on a “new, unproven device.” On the other hand, a psychologist who uses the older version may be challenged for using an “old, antiquated instrument.” Thus, at least for a period of time, forensic users of an updated measure encounter a “damned if you do and damned if you don’t” situation that may be accentuated by the adversarial nature of the legal system.

One way to avoid this dilemma is to never update our instruments. Had this approach been adopted in the case of the MMPI (Hathaway & McKinley, 1943), forensic psychologists would

still be relying on scales developed and norms collected during the late 1930s and early 1940s. Most experts and, for that matter, most legal decision makers are unlikely to find this solution satisfactory over the long term. Alternatively, when a psychological test is revised, it is incumbent upon forensic users of the measure to (1) become familiar with the updated instrument, including the rationale for, methods used in, and outcome of the revision; (2) make an informed decision about whether to use the revised test in their forensic assessments; and (3) be prepared to defend their decision.

Ben-Porath (2012) identifies challenges forensic psychologists may encounter with the introduction of the MMPI-2-RF and provides answers to a series of questions that are likely to come up if MMPI-2-RF-based testimony is contested. In relatively rare cases, this may involve a pre-trial effort to suppress testimony under the governing admissibility standards. More commonly, this will occur during pre-trial depositions, or at trial, under cross-examination vis-à-vis weight-of-the-evidence considerations. Although their goals are different, the same issues are likely to be raised by attorneys seeking to suppress MMPI-2-RF-based testimony and those seeking to mitigate its impact through cross-examination. The questions and answers discussed by Ben-Porath (2012) apply to both types of challenges.

Using the factors identified by the U.S. Supreme Court in its landmark *Daubert v. Merrell Dow Pharmaceuticals, Inc.* (1993) decision, Ben-Porath (2012) notes that the MMPI-2-RF has been tested extensively, and the resulting empirical findings on the psychometric properties of the inventory are reported in detail the Technical Manual for the inventory. A similar compilation of data has never been available in a single source for any version of the MMPI or for other psychological tests. The studies cited in this chapter are examples (but by no means an exhaustive list) of a broad and growing peer-reviewed literature on the MMPI-2-RF. Which and how many of these investigations will apply in a given forensic case will of course depend on the specific issues at hand. However, in general the MMPI-2-RF has been subjected to extensive peer review. Reliability estimates and their associated standard errors of measurement, classification accuracy estimates reported in the Technical Manual, and other published validation data provide information about the known or potential rate of error associated with MMPI-2-RF scale scores. Proper use of these statistics requires that their inherent limitations be considered. Standards controlling all aspects of using the MMPI-2-RF—administration, scoring, and interpretation—are spelled out in detail in the test manual (Ben-Porath & Tellegen, 2008/2011). Inclusion of detailed interpretive guidelines in this document facilitates cross-interpreter reliability and the ability of judicial fact-finders to assess whether an expert has adhered to these guidelines.

Finally, Ben-Porath (2012) notes that in jurisdictions where case law indicates the need to consider *Daubert*-like criteria in determining general acceptance, the issues just discussed are also relevant to this factor. In addition, citation of MMPI-2-RF findings in legal decisions (e.g., *Michigan v. Espinoza*, 2011; *Wood v. Thaler*, 2011), detailed coverage of and guidance on how to use the MMPI-2-RF in the two leading MMPI textbooks (Graham, 2012; Greene, 2011), coverage of the test in recent assessment reference sources (e.g., Groth-Marnat, 2012; Hebben & Milberg, 2009; Kitaëff, 2011; Larrabee, 2012; Neukrug & Fawcett, 2009; Plante, 2010), incorporation of the instrument in graduate assessment classes, opportunities for MMPI-2-RF training offered by many national and state-level organizations, and international adaptation of the instrument all point to acceptance of the test.

Use of the MMPI-2-RF Validity Scales in Forensic Assessments

Recognizing the inherent vulnerability of self-report measures to misleading responding, Hathaway and McKinley (1943) incorporated validity scales in the original version of the MMPI. Ben-Porath (2012) describes a conceptual framework that has guided MMPI-2 (Butcher et al., 2001) and

MMPI-2-RF (Ben-Porath & Tellegen, 2008/2011) validity scale interpretation for some time. This approach identifies two general classes of threats to the validity of an MMPI protocol: *non-content-based invalid responding*, consisting of non-responding, random responding, and fixed responding (in the “True” or False” direction); and *content-based invalid responding*, consisting of over-reporting or under-reporting. Studies of MMPI-2-RF measures designed to assess the two types of threats are reviewed next. The final part of this section addresses research related to the utility of validity scales in general.

Measures of Non-Content-Based Threats to Protocol Validity

Two simulation studies have examined the MMPI-2-RF measures of non-content-based invalid responding. The first focuses on non-responding and the second investigation involves the identification of random and fixed responding.

Non-responding

Non-responding to the MMPI items has traditionally been measured with the “Cannot Say” index, a count of the number of items to which the test-taker failed to respond, or responded both “true” and “false.” Non-responding artifactually deflates scale scores. In an extreme case, if a test-taker does not respond to any of the items on a scale, the resulting raw score is zero, and cannot be distinguished from one where the individual responded to all of the items on that scale in the non-keyed direction. Ben-Porath and Tellegen (2008/2011) recommend that non-responding be addressed by considering the percentage of scorable responses to the items of each MMPI-2-RF scale. As long as at least 90% of the items on a scale are scorable, the result can be interpreted in the standard manner. As the percentage of scorable responses falls below 90, the absence of elevation on a scale becomes increasingly uninterpretable.

In support of this recommendation, Ben-Porath and Tellegen (2008/2011) cite an investigation that was unpublished at the time the MMPI-2-RF manual was written. The study by Dragon et al. (2012) has since been published. The authors examined the impact of unscorable item responses on the validity and interpretability of scores on the MMPI-2/MMPI-2-RF Restructured Clinical (RC) Scales. Dragon and colleagues first reported that in five archival samples they examined, unscorable responses at the level of 10% or more of a scale’s items were relatively uncommon, occurring most often in forensics samples. The authors then simulated non-responding by inserting varying proportions of non-responses (10–90% in 10 percent increments) in place of the actual responses of subjects in two of their archival samples. They then examined two distinct attributes of the affected scales, their validity, as reflected in correlations between RC Scale scores and external criteria, and interpretability, indicated by examination of the percentage of cases with clinically elevated scores as a function of different levels of simulated non-responding.

Dragon and colleagues found that scale score validity was relatively robust up to the level of 50% non-responding. For instance, in an outpatient mental health sample, the correlation between low positive emotions (RC2) and therapist-rated depression was .58 with no simulated non-responding, .57 with 10% simulated non-responding to the RC2 items, .56 at 20%, .54 at 40%, .52 at 50%, and .45 at 60% simulated non-responding. Interpretability, on the other hand, was affected much more rapidly by non-responding. For example, 33% of the men and 45% of the women in this sample produced clinically significant elevations (*T*-scores greater than or equal to 65) on RC2. With just 10% simulated non-responding, these percentages declined to 26 and 37 respectively, and at 20% simulated non-responding the percentage of individuals with elevated scores on RC2 went down to 17 and 26 (i.e., nearly half the cases with elevated scores on RC2

no longer showed clinically significant elevation). Thus, consistent with Ben-Porath and Tellegen's (2008/2011) recommendations, scale score interpretability is significantly compromised if the percentage of scorable responses to the items of a scale falls below 90. If a scale remains elevated in spite of non-responding, that is an interpretable finding. However, the absence of elevation on a scale that has less than 90% scorable responses is uninterpretable.

Random and Fixed Responding

Random and fixed responding to the MMPI-2-RF items are assessed with the Variable Response Inconsistency (*VRIN-r*) and True Response Inconsistency (*TRIN-r*) scales respectively. In the Technical Manual for the inventory, Tellegen and Ben-Porath (2008/2011) cite an unpublished study in support of the validity of these scales. The study has since been published by Handel et al. (2010), who investigated the identification of random and fixed responding with *VRIN-r* and *TRIN-r*, as well as the impact of these test-taking approaches on the RC Scales. The authors evaluated the effect of increasing levels of simulated random and fixed responding. Their results supported the interpretation of scores on *VRIN-r* and *TRIN-r* as measures of random and fixed responding, respectively, and are consistent with the interpretive guidelines in the MMPI-2-RF manual, indicating that caution should be exercised if *VRIN-r* or *TRIN-r* *T*-scores reach 70, and that protocols should be deemed invalid if *T*-scores on either of these scales reach or exceed 80.

As was the case with non-responding, substantive scale score validity was relatively robust to simulated inconsistent responding. For example, with simulated 30% fixed "true" responding in a psychiatric inpatient sample, the average validity coefficient for Demoralization (*RCd*) declined from .38 only to .34. However, in the MMPI-2-RF normative sample, with 30% fixed "true" response simulation the mean *T*-score on *RCd* increased from 50 to 60 and the standard deviation declined from 10 to 6.6, indicating that interpretability was significantly compromised at this level of fixed responding. The mean *T*-score on *TRIN-r* climbed from 50 to 80*T*, and nearly 60% of the sample scored at or above 80 in this condition, supporting the utility of this scale in the detection of fixed "true" responding. Similar results were obtained with simulated fixed "false" responding and *TRIN-r*, and simulated random responding and *VRIN-r*.

Measures of Content-Based Threats to Protocol Validity

Detecting Over-reported Psychopathology

Differentiating between genuine psychopathology and symptom over-reporting has been one of the key functions of the MMPI validity scales since the early days of the instrument (Ben-Porath, 2012). A significant confound between severe psychopathology (e.g., psychotic disorders) and scores on the MMPI-2 *F* scale led Arbisi and Ben-Porath (1995) to develop a scale, *Fp*, designed to perform this task optimally. Revised versions of both of these scales are included on the MMPI-2-RF.

Utilizing an analog simulation study design, Marion, Sellbom, and Bagby (2011) examined the ability of the Infrequent Responses (*F-r*), Infrequent Psychopathology Responses (*Fp-r*), Infrequent Somatic Responses (*Fs*), and Symptom Validity (*FBS-r*) scales to detect participants who were simulating one of three mental disorders: Major Depressive Disorder, Schizophrenia, or Post-traumatic Stress Disorder. The authors used data provided by two groups of simulators: naïve undergraduates with no training in mental disorders, and sophisticated participants with advanced training in psychopathology or personal experience with the disorder. Both groups' scores on the MMPI-2-RF validity indicators were compared with those of psychiatric patients diagnosed with

one of the three disorders. Overall, when manual-recommended cutoffs were used, the MMPI-2-RF over-reporting indicators were able to differentiate between the simulators and patients quite effectively, with specificities always exceeding .95. Classification accuracy was good for both types of simulators, but, as expected, better for the naïve ones.

In a similar simulation study, Sellbom and Bagby (2010) examined the ability of the MMPI-2-RF over-reporting indicators to differentiate between college students simulating psychopathology and genuine psychiatric patients. The simulators were divided into two groups: one received coaching about the existence of the validity scales, how they work, and how best to avoid detection; the other did not. The *Fp-r* scale again best differentiated both groups of simulators from the patients, with stronger effect sizes found for the non-coached participants. Classification accuracy analyses provided support for the interpretive guidelines in the test manual. Specificities were generally well in excess of .90.

Utilizing a known-groups research design, Sellbom et al. (2010) examined the ability of the MMPI-2-RF over-reporting indicators to identify criminal defendants classified as malingering on the basis of the Structured Interview of Reported Symptoms (SIRS; Rogers, Bagby, & Dickens, 1992). The two MMPI-2-RF validity scales designed to detect over-reporting of psychopathology symptoms, *F-r* and *Fp-r*, best differentiated between those classified as malingering on the basis of the SIRS and those not so classified. Each scale added incrementally to the other in this differentiation task. Both scales had specificities reaching or exceeding .90 when cutoffs recommended in the test manual were applied.

Purdon et al. (2011) examined the MMPI-2-RF over-reporting scale scores of patients hospitalized with a first-episode psychotic disorder. Although characterized as acutely ill upon presentation (and testing), consistent and responsive profiles were generated by 79% of these individuals when they were tested upon admission to an inpatient early psychosis intervention unit of a neuropsychology service. Patients were diagnosed with a first-episode psychotic disorder using the Structured Clinical Interview of *DSM-IV* (SCID; First, Spitzer, Gibbon, & Williams, 2005). For those who produced content-consistent protocols, mean *T*-scores on *Fp-r* were found to be higher than those reported for psychiatric patients in the MMPI-2-RF Technical Manual (Tellegen & Ben-Porath, 2008/2011). This is not surprising, because the inpatient samples included in the Technical Manual were diagnostically more diverse, including large numbers of individuals with internalizing disorders, who do not typically score as high on this scale as those diagnosed with thought disorders. Indeed, the overall mean *T*-score on *Fp-r* for this group (70.61; *SD* = 23.99) is nearly identical to the mean *Fp* *T*-score (70.20; *SD* = 21.65) reported by Arbisi and Ben-Porath (1997) for patients diagnosed with Schizophrenia. Purdon et al. (2011) noted that 13% of the patients included in their sample scored at or above the 100*T* score cutoff identified in the MMPI-2-RF manual as very likely indicating over-reporting (for a .87 specificity rate). They note, however, that absent an independent measure of embellishment, it is not possible to determine whether all of these cases are indeed false positives. By contrast, in the study by Marion et al. (2011) just reviewed, the sensitivity estimate was .93 when the same cutoff was applied to patients diagnosed with Schizophrenia.

Overall, the findings just reviewed indicate that *Fp-r* is the most effective MMPI-2-RF validity indicator when the differential diagnosis involves genuine psychopathology versus over-reporting. Specificity estimates were generally well in excess of .90, indicating a false positive rate generally well below 10% when a cutoff of 100*T* is applied.

Detecting Over-reported Somatic and Cognitive Complaints

As noted earlier, the original MMPI validity scales focused on detection of over-reported psychopathology. However, as the test became used widely in neuropsychological evaluations, and in particular in evaluations related to personal injury litigation and disability claims, assessment of over-reported somatic and cognitive complaints became the focus of some neuropsychological assessments and the original *F* scale proved generally ineffective for this task (Lees-Haley, 1989). Lees-Haley, English, and Glenn (1991) developed the scale originally called “Fake Bad,” now labeled Symptom Validity (but still abbreviated *FBS* to identify its origin) to address this problem. A revised version of *FBS*, *FBS-r* (also labeled Symptom Validity), is complemented by *Fs* and the Response Bias Scale (*RBS*; Gervais, Ben-Porath, Wygant, & Green, 2008) as measures of over-reported somatic and cognitive complaints on the MMPI-2-RF.

Studying a sample of individuals undergoing compensation-seeking evaluations, Wygant et al. (2011) examined the ability of the MMPI-2-RF over-reporting indicators to detect malingered neurocognitive dysfunction and malingered pain dysfunction on the basis of the respective criteria developed by Slick, Sherman, and Iverson (1999) and Bianchini, Greve, and Glynn (2005). The MMPI-2-RF scales produced large effect sizes in contrasting those classified as likely malingering versus those not so classified. The largest effect sizes were found for *F-r* and *RBS*. Cutoffs of 110 and 105, respectively, yielded a specificity estimate of .93 for both scales.

Using a sample collected at the same setting, Rogers et al. (2011) sought to examine the effectiveness of the MMPI-2-RF Validity Scales in detecting feigned mental disorder (FMD) and feigned cognitive impairment (FCI). FMD was defined on the basis of scores on the Structured Interview of Reported Symptoms (SIRS-2; Rogers, Sewell, & Gillard, 2010) and FCI was determined on the basis of below chance performance on either the Victoria Symptom Validity Test (Slick, Hopp, Strauss, & Thompson, 1997) or the Test of Memory Malingering (Tombaugh, 1996). The authors concluded that *F-r* and *Fp-r* performed well in detecting FMD and that the MMPI-2-RF over-reporting indicators were less effective at identifying FCI, though they noted that *FBS-r* and *RBS* may be useful for this purpose if interpreted in conjunction with other clinical data.

The analyses by Rogers et al. (2011) regarding FCI are of limited value because the authors deviated from standard research practice by requiring below chance rather than below established cutoff performance in symptom validity tests to classify test-takers as engaging in feigned cognitive impairment. Their definition produced an unrealistic base rate of 10 percent for the FCI group. Rogers et al. (2011) also excluded from their analyses any individual who met their criteria for both FMD and FCI, mistakenly assuming that the two forms of feigning are mutually exclusive in compensation-seeking individuals.

Schroeder et al. (2012) investigated the utility of the MMPI-2-RF validity indicators in detecting malingering based on the Slick criteria in a sample of individuals involved in litigation related to a traumatic brain injury (TBI). The authors found that *RBS* was associated with the strongest effect size in differentiating TBI litigants who passed versus failed the Slick criteria, followed by *FBS-r*. The authors also compared the scores of individuals diagnosed with epileptic disorders, mixed neurological disorders, and mixed psychiatric disorders with those of the individuals who failed the Slick criteria, finding similar results. *T*-score frequency tables indicated that cutoffs of 90 T –100 T on *F-r*, *Fs*, *FBS-r*, and *RBS* were associated with specificities in excess of .90, indicating acceptable false positive rates for the MMPI-2-RF over-reporting scales based on the MMPI-2-RF manual guidelines. Youngjohn et al. (2011) similarly investigated the utility of the MMPI-2-RF Validity Scales (not including *RBS*) in a sample of individuals evaluated in connection with TBI claims. Scores on *FBS-r* were associated significantly with effort test failure,

whereas *F-r* and *Fs* scores showed similar trends. The authors did not report classification accuracy analyses.

Gervais et al. (2010) investigated the incremental validity of the MMPI-2-RF over-reporting scales and their MMPI-2 counterparts in assessing the veracity of memory complaints. They reported finding that scores on all of the MMPI-2-RF measures were more strongly correlated with exaggerated memory complaints than their MMPI-2 counterparts, with RBS demonstrating the best prediction of this variable. Wygant et al. (2010) investigated the ability of RBS scores to predict Symptom Validity Test (SVT) failure in civil and criminal forensic samples, demonstrating specificity estimates in excess of .90 and good sensitivity at recommended cutoffs for the scale.

Overall, the literature reviewed here shows that four of the MMPI-2-RF over-reporting measures—*F-r*, *Fs*, *FBS-r*, and *RBS*—provide empirically verified indications of over-reported somatic and cognitive complaints. *FBS-r* and *Fs* have been found to be particularly effective in mTBI-related assessments, and *RBS* provides a strong incremental contribution in the assessment of exaggerated memory complaints. *F-r* is a more general over-reporting indicator, sensitive to all three varieties of symptom over-reporting (psychological, somatic, and cognitive).

Detecting Under-reporting

As was the case with the MMPI-2, research on detection of under-reporting with the MMPI-2-RF has been relatively limited, but supportive of use of the under-reporting indicators of the inventory, Uncommon Virtues (*L-r*) and Adjustment Validity (*K-r*). Sellbom and Bagby (2008) examined the utility of these scales using three samples. In their first study, the authors found that both *L-r* and *K-r* significantly differentiated two groups of individuals (patients with schizophrenia and university students) who had been instructed to under-report from participants who took the test under standard instructions. *L-r* and *K-r* also added incremental predictive variance to one another in differentiating these groups. In a second study, a similar set of outcomes emerged when a differential prevalence design was used. *L-r* and *K-r* significantly differentiated a group of child custody litigants from university students taking the test under standard instructions.

Do the MMPI-2-RF Validity Scales Identify Invalid Responding?

With the availability of hundreds of publications that document the utility of the MMPI, MMPI-2, and MMPI-2-RF Validity Scales, this may appear to be a peculiar question. However, from a review of the validity scale literature, McGrath, Mitchell, Kim, and Hough (2010) concluded that “[d]espite many years of research, a sufficient justification for the use of bias indicators [i.e., validity scales] in applied settings remains elusive” (p. 450). McGrath et al.’s sweeping conclusion, published in the APA’s flagship review journal, likely surprised many readers. It appears, for example, to be at odds with the literature on detecting over-reported psychological, somatic, and cognitive symptoms just reviewed. To reach their conclusion, McGrath et al. (2010) employed a methodology that would have excluded all of these studies, and did indeed exclude all but 40 of approximately 4,000 relevant investigations. The authors limited their review to ones they believed allowed for the testing of their *response bias hypothesis*: “[C]riterion-related validity of a response bias indicator [i.e., validity scale] is not reflected in its correlation with a criterion but in the degree to which it use enhances the criterion-related validity of a second substantive indicator” (p. 452). In other words, according to McGrath et al. (2010), a study showing that scores on a validity scale are associated with invalid responding (for example using simulation or known group designs) is not sufficient to indicate that the scale is valid. Rather, according to these authors, what is required is a demonstration that scores on a validity scale predict the reduction of the validity of substantive

scale scores. In MMPI-2-RF terms, it is not sufficient, for example, to show that scores on *Fp-r* are higher in individuals who simulate over-reporting when compared with those who do not. To establish the validity of the *Fp-r* as an over-reporting measure, McGrath et al. require a demonstration that its scores moderate or suppress the validity of substantive scale scores.

Two MMPI-2-RF studies illustrate the flaws in McGrath et al.'s logic and conclusion. Gervais et al. (2011) examined associations between Symptom Validity Test (SVT) failure and scores on the MMPI-2-RF Validity Scales and substantive scales. Using a known groups design, the authors divided a sample of disability claimants who completed the MMPI-2-RF as part of their evaluation into four subgroups defined by rates of failure (0, 1, 2, or 3 SVTs failed respectively) and compared scores on the Validity and substantive scales across the four groups. As has been demonstrated repeatedly, scores on several Validity indicators, particularly the over-reporting indicators, were associated substantially with SVT failure. Significant associations with SVT failure were also found for numerous MMPI-2-RF substantive scales. Not surprisingly, Cognitive Complaints (*COG*) showed the strongest association with SVT failure in men, and among the strongest such associations in women. For example, the mean *COG* scores (and associated standard deviation) in men for the four groups (ranging from 0 to 3 failed SVTs) were 66.8 (14.7), 72.6 (14.8), 79.9 (12.9), and 82.7 (8.8) respectively. By McGrath et al.'s logic, these findings are insufficient to support the use of the MMPI-2-RF over-reporting indicators as measures of response bias because no evidence is provided that (A) the higher *COG* score in those who fail 3 SVTs (82.7) is less valid than the lower one in those who fail none (66.8), and (B) an MMPI-2-RF validity scale can account statistically for reduced validity that would satisfy A.

In a unique simulation study, Burchett and Ben-Porath (2010) provided findings that would satisfy McGrath et al.'s criterion A. Using an undergraduate sample, the authors studied the impact of over-reporting on MMPI-2-RF substantive scale score validity by first administering a set of criterion measures under standard instructions to all their participants before randomly dividing them into three groups. The first group also completed the MMPI-2-RF following standard instructions. A second group completed the test under instructions to feign psychopathology. The third group completed the inventory under instructions to feign somatic complaints. The unique aspect of this study involved the administration of criterion measures to members of all groups under standard instructions, which allowed for a direct, experimentally controlled test of the impact of simulated over-reporting.

As expected, the impact of over-reporting on substantive scale score validity was substantial. For example, in the group that completed both the criterion measures and the MMPI-2-RF under standard instructions, the correlation between the Higher-Order Emotional Internalizing Dysfunction (*EID*) scale and a criterion measure of depression was .78. For the group that completed the depression measure under standard instructions and the MMPI-2-RF under instructions to feign psychopathology, the correlation between these variables was -.02. For the group that completed the test under instructions to feign somatic complaints, the correlation was .26. Consistent with many prior investigations, scores on the MMPI-2-RF over-reporting scales varied substantially as a function of group membership, with *F-r* and *Fp-r* providing the best separation between the standard instructions and feign psychopathology groups, and *Fs* best differentiating between standard instructions and feign somatic complaints groups.

However, even this study would not satisfy McGrath et al.'s requirement for evidence of Validity Scale validity because although it satisfies their criterion A—the validity of substantive scale scores was shown to decline as a function of over-reporting—it would not satisfy their criterion B—that MMPI-2-RF validity scale scores be shown to account statistically for the reduction in substantive scale validity. For such a demonstration, McGrath et al. would require that MMPI-2-RF Validity Scale scores account for the reduction in substantive scale score validity in regression analyses as

suppressor or moderator variables. By McGrath et al.'s logic, in the absence of such evidence the fact that the same experimental manipulation reduced the validity of substantive scale scores while increasing scores on the MMPI-2-RF over-reporting indicators may simply be a coincidence, and until this is ruled out, they contend that a sufficient justification for use of Validity Scales in applied settings remains elusive.

McGrath et al.'s critique may best be characterized as requiring that Validity Scale researchers demonstrate what most might consider the obvious: that it is more than a coincidence when Validity and substantive scale scores are simultaneously affected by invalid responding (simulated or deduced). Because of the prominent outlet for McGrath et al.'s critique, and the potential it creates to question standard assessment practices in a host of applied settings, such demonstrations are likely to appear soon in the literature.

Specific Forensic Applications

Availability of forensic comparison group data in the MMPI-2-RF Technical Manual can facilitate forensic uses of the test. As of the writing of this chapter, data for five forensically relevant comparison groups are reported. They include 243 men and 238 women who completed the test in connection with child custody litigation, 523 men and 480 women assessed as part of a disability claim or personal injury litigation, 463 men and 578 women tested as part of a forensic independent neuropsychological examination, 551 men and 223 women assessed in connection with pre-trial evaluations of criminal defendants, and 34,933 men and 7,353 women tested at intake to a state correctional system. Although they are not included presently in the Technical Manual, data for a sixth comparison group made up of 226 men and 336 women assessed in the context of parental fitness evaluations are available in the scoring software for the instrument.

A growing body of published studies can further facilitate use of the MMPI-2-RF in forensic assessments. These include studies of child custody litigants and parental fitness evaluatees (Archer et al., 2012; Pinsoeneault & Ezzo, 2012; Resendes & Lecci, in press; Stredny et al., 2006), an investigation of predicting drug court completion with the MMPI-2-RF (Matson et al., in press), studies of the predictive validity of MMPI-2-RF scales in a batterers' intervention program (Sellbom et al., 2008; Rock et al., in press), research on assessing PTSD symptoms in workplace disability claimants (Sellbom & Bagby, 2009; Sellbom, Lee, Ben-Porath, Arbisi, & Gervais, 2012), and the psychopathy research cited earlier in this chapter.

CASE EXAMPLE

Background

Mr. C is a 28-year-old unmarried male referred for an evaluation of his competency to stand trial in connection with drug-trafficking charges. Mr. C was arrested by members of a local drug task force after selling illegal drugs to an undercover agent. He informed his attorney that he was set up by the local authorities, who were engaged in a long-standing dispute with his family. When the attorney attempted to reason with his client, explaining that the prosecutor had recordings of the defendant at various stages of the transaction, Mr. C. insisted that the recordings were fabricated and that he expected to be able to persuade a jury that he was innocent. After the attorney continued to pressure Mr. C to enter plea negotiations, the

defendant informed the trial judge that he could not work with his court-appointed attorney, and at his request another attorney was appointed to represent him. After a similar pattern of exchanges ensued with the second attorney, the judge ordered a competency-to-stand-trial evaluation.

Mr. C was raised in a rural community in an intact family with three older brothers and no sisters. Consistent with his report to his attorneys, members of the defendant's extended family had indeed been embroiled in long-standing conflicts with local county and state authorities. These problems, instigated by years of unpaid property taxes, began during Mr. C's early childhood and continued through the time of his arrest for the alleged offense. The defendant's father and uncle had been incarcerated on several occasions during the course of this conflict. Mr. C himself had no prior involvement with the criminal justice system. He was home-schooled, and thus no academic records were available for the evaluation. He did not have any prior involvement with the mental health system. A drug screen conducted following his arrest against the defendant's wishes was negative. Mr. C resided on his parents' property and worked for his uncle's business.

At the time of his evaluation, Mr. C was out on bond in the community. He arrived early for his evaluation and was observed joking with the clinic's administrative staff while performing magic tricks. He was fully oriented to person, place, time, and his situation. His appearance was unremarkable. Mr. C's mood and affect were normal. He appeared unperturbed by the serious charges against him. He spoke in a normal tone at a normal rate and the content of his speech was unremarkable, with the exception of his claim that he and his family members were victims of a decades-long conspiracy. There were no indications that he was experiencing hallucinations or non-persecutory delusions. Mr. C's attention and concentration appeared to be intact and he was judged to possess an average level of intelligence (corroborated later with a measure of cognitive functioning). The defendant denied any suicidal or homicidal ideation.

MMPI-2-RF Results and Interpretation

Mr. C's MMPI-2-RF results are reproduced in Figure 3.1. The male pre-trial criminal defendant comparison group was used in this case. Mr. C's MMPI-2-RF scores are indicated by the small filled-in circles and connected by solid lines. Open diamonds connected by dotted lines mark the mean scores for the comparison group. Shaded bars reflect a range of plus/minus one standard deviation from the mean score of the comparison group on each scale. The mean *T*-score for the comparison group on each scale and its associated standard deviation are also conveyed numerically in the test report. The final row of numbers indicates the percentage of individuals in the comparison group who scored at or below Mr. C on each scale.

Examination of page 2 of the report indicates that Mr. C responded to all the test items, and there are no signs (based on *VRIN-r* or *TRIN-r*) of inconsistent responding. His scores on *F-r*, *Fp-r*, *Fs*, *FBS-r*, and *RBS* all fall below the recommended cutoff for identifying over-reporting. Mr. C just exceeds the cutoff for identifying under-reporting based on *L-r* and scored just below the cutoff recommended in the test manual for *K-r*. With the exception of *K-r*, his Validity Scale scores fall within a standard deviation from the mean for pre-trial criminal defendants. As seen in the last row of numbers under *K-r*, 88% of pre-trial criminal defendants scored at or below the defendant's level on this scale. Overall, Mr. C's Validity Scale results indicate a greater level of under-reporting than would be expected of someone evaluated under these circumstances.



Minnesota Multiphasic
Personality Inventory-2
Restructured Form®

Score Report

MMPI-2-RF®

Minnesota Multiphasic Personality Inventory-2-Restructured Form®

Yossef S. Ben-Porath, PhD, & Auke Tellegen, PhD

ID Number:	Mr. C
Age:	28
Gender:	Male
Marital Status:	Never Married
Years of Education:	12
Date Assessed:	10/09/2011



Copyright © 2008, 2011, 2012 by the Regents of the University of Minnesota. All rights reserved.
Distributed exclusively under license from the University of Minnesota by NCS Pearson, Inc. Portions reproduced from the MMPI-2-RF test booklet. Copyright © 2008 by the Regents of the University of Minnesota. All rights reserved. Portions excerpted from the *MMPI-2-RF Manual for Administration, Scoring, and Interpretation*. Copyright © 2008, 2011 by the Regents of the University of Minnesota. All rights reserved.
Used by permission of the University of Minnesota Press.

MMPI-2-RF, the MMPI-2-RF logo, and Minnesota Multiphasic Personality Inventory-2-Restructured Form are registered trademarks of the University of Minnesota. Pearson, the PSI logo, and PsychCorp are trademarks in the U.S. and/or other countries of Pearson Education, Inc., or its affiliate(s).

TRADE SECRET INFORMATION

Not for release under HIPAA or other data disclosure laws that exempt trade secrets from disclosure.

[2.1 / 1 / 0.0.0]

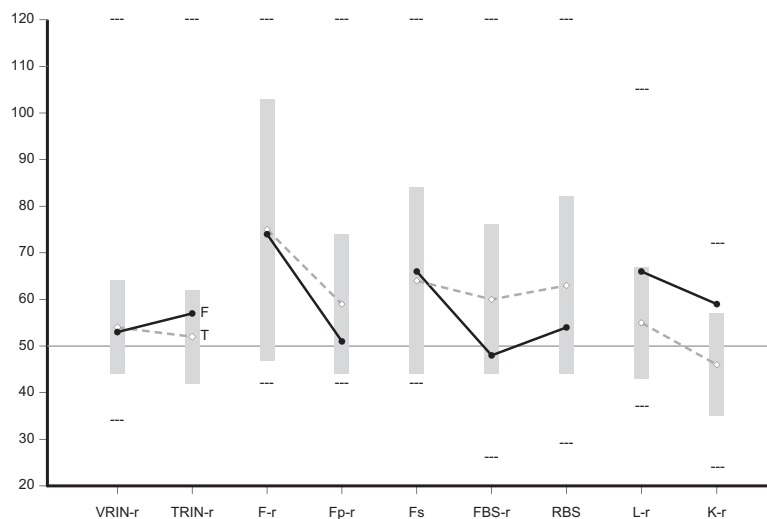
ALWAYS LEARNING

PEARSON

FIGURE 3.1 MMPI-2-RF Score Report.

Reproduced by permission of the University of Minnesota Press. All rights reserved. "MMPI-2-RF®" and Minnesota Multiphasic Personality Inventory-2-Restructured Form® are registered trademarks owned by the University of Minnesota.

MMPI-2-RF Validity Scales



Raw Score: 4 10 7 1 3 7 6 6 10
 T Score: 53 57 F 74 51 66 48 54 66 59
 Response %: 100 100 100 100 100 100 100 100 100
 Cannot Say (Raw): 0 Percent True (of items answered): 36%

Comparison Group Data: Forensic, Pre-trial Criminal (Men), N = 551

Mean Score (---): 54 52 T 75 59 64 60 63 55 46
 Standard Dev (±1SD): 10 10 28 15 20 16 19 12 11
 Percent scoring at or below test taker: 59 74 61 50 70 30 44 85 88

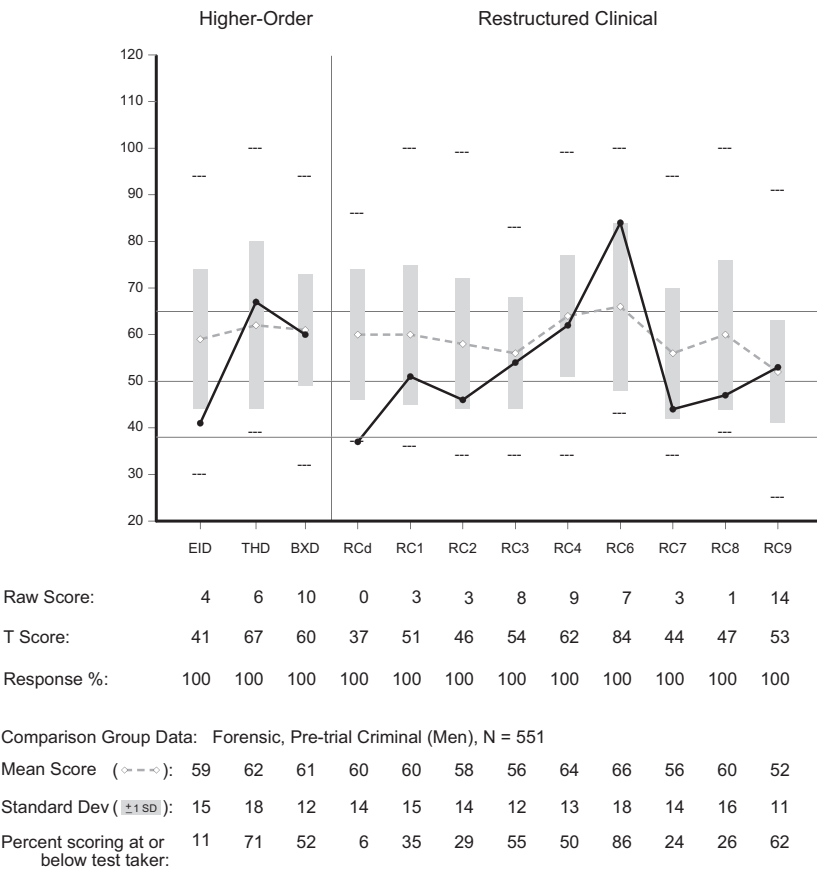
The highest and lowest T scores possible on each scale are indicated by a "---"; MMPI-2-RF T scores are non-gendered.

VRIN-r	Variable Response Inconsistency	Fs	Infrequent Somatic Responses	L-r	Uncommon Virtues
TRIN-r	True Response Inconsistency	FBS-r	Symptom Validity	K-r	Adjustment Validity
F-r	Infrequent Responses	RBS	Response Bias Scale		
Fp-r	Infrequent Psychopathology Responses				

His substantive scale scores are nonetheless likely to provide an accurate indication of his psychological functioning, although they may underestimate the magnitude of any identified dysfunction.

Page 3 of Mr. C's MMPI-2-RF Score Report includes his results on the Higher-Order and RC Scales. His elevated score on the Thought Dysfunction (*THD*) scale indicates the presence of some disordered thinking. His score on *RC6* indicates that Mr. C reports prominent persecutory ideation that likely rises to the level of paranoid delusions, including a strong belief

MMPI-2-RF Higher-Order (H-O) and Restructured Clinical (RC) Scales

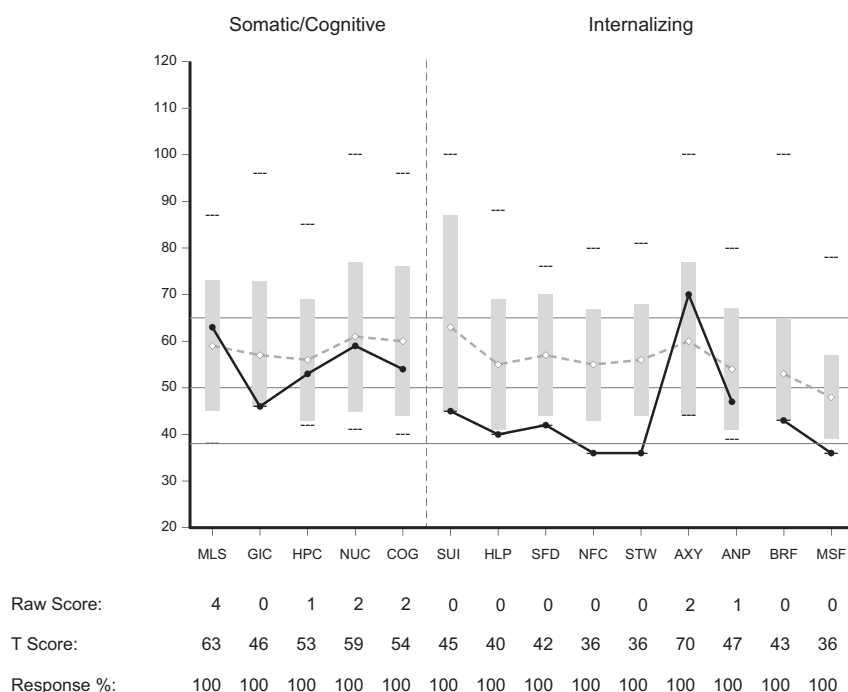


The highest and lowest T scores possible on each scale are indicated by a "----"; MMPI-2-RF T scores are non-gendered.

EID	Emotional/Internalizing Dysfunction	RCd	Demoralization	RC6	Ideas of Persecution
THD	Thought Dysfunction	RC1	Somatic Complaints	RC7	Dysfunctional Negative Emotions
BXD	Behavioral/Externalizing Dysfunction	RC2	Low Positive Emotions	RC8	Aberrant Experiences
		RC3	Cynicism	RC9	Hypomanic Activation
		RC4	Antisocial Behavior		

that others seek to harm him. He is very likely to be suspicious and distrustful, to experience serious interpersonal difficulties as a result of pervasive interpersonal suspiciousness, and to lack insight. Mr. C's other noteworthy score on page 3 is on the Demoralization scale (*RCd*), which indicates a higher than average level of morale and life satisfaction, an unexpected finding in someone assessed under these circumstances. The data reported on page 3 indicate that 94% of criminal defendants score higher than Mr. C on *RCd*. Although unexpected, this finding is

MMPI-2-RF Somatic/Cognitive and Internalizing Scales



Comparison Group Data: Forensic, Pre-trial Criminal (Men), N = 551

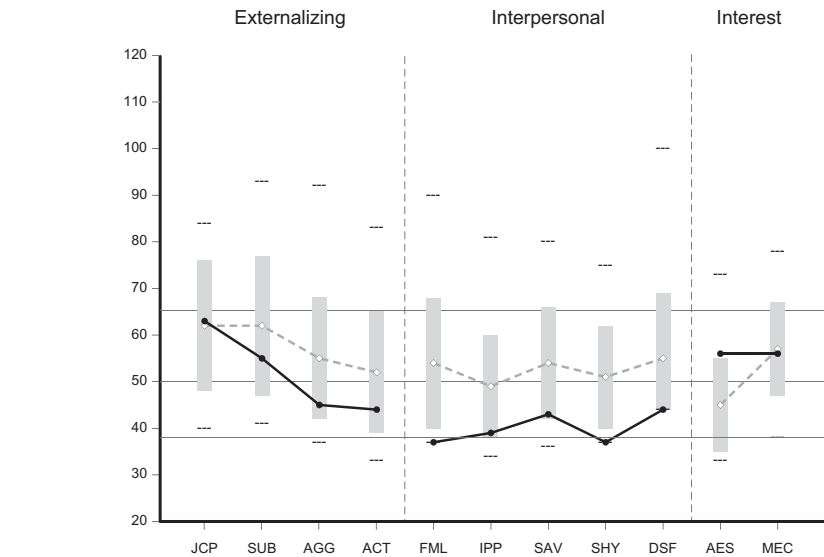
Mean Score ($\bar{x}$):	59	57	56	61	60	63	55	57	55	56	60	54	53	48
Standard Dev (± 1 SD):	14	16	13	16	16	24	14	13	12	12	17	13	12	9
Percent scoring at or below test taker:	69	61	53	55	50	58	34	33	8	6	79	40	47	15

The highest and lowest T scores possible on each scale are indicated by a "----"; MMPI-2-RF T scores are non-gendered.

MLS	Malaise	SUI	Suicidal/Death Ideation	AXY	Anxiety
GIC	Gastrointestinal Complaints	HLP	Helplessness/Hopelessness	ANP	Anger Proneness
HPC	Head Pain Complaints	SFD	Self-Doubt	BRF	Behavior-Restricting Fears
NUC	Neurological Complaints	NFC	Inefficacy	MSF	Multiple Specific Fears
COG	Cognitive Complaints	STW	Stress/Worry		

congruent with Mr. C's demeanor during the evaluation. Finally, although his scores on the higher-order Behavioral/Externalizing (*BXD*) and Antisocial Behavior (*RC4*) scales do not quite exceed the cutoff for clinically significant elevation, just-discussed findings of under-reporting indicate that problems associated with these externalizing behavior scales cannot be ruled out. Indeed, problems in this domain are indicated by an elevated score on the PSY-r Disconstraint (*DISC-r*) scale, which is discussed later.

MMPI-2-RF Externalizing, Interpersonal, and Interest Scales



Raw Score:	3	2	1	2	0	1	1	0	0	4	4
T Score:	63	55	45	44	37	39	43	37	44	56	56
Response %:	100	100	100	100	100	100	100	100	100	100	100

Comparison Group Data: Forensic, Pre-trial Criminal (Men), N = 551

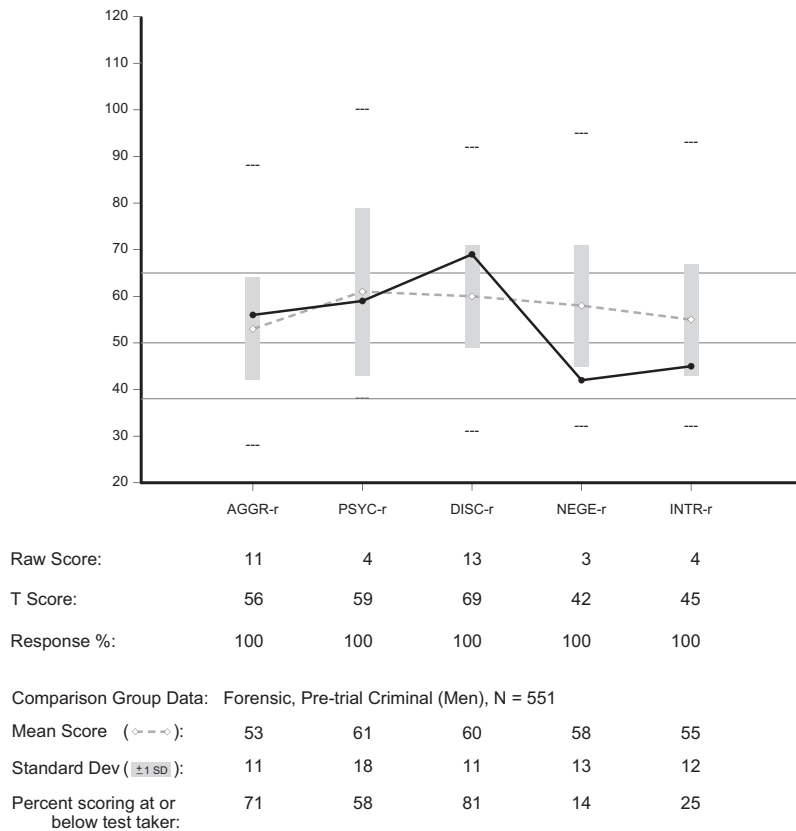
Mean Score (> - <):	62	62	55	52	54	49	54	51	55	45	57
Standard Dev (± 1 SD):	14	15	13	13	14	11	12	11	14	10	10
Percent scoring at or below test taker:	60	45	34	36	16	22	25	17	52	90	56

The highest and lowest T scores possible on each scale are indicated by a "----"; MMPI-2-RF T scores are non-gendered.

JCP	Juvenile Conduct Problems	FML	Family Problems	AES	Aesthetic-Literary Interests
SUB	Substance Abuse	IPP	Interpersonal Passivity	MEC	Mechanical-Physical Interests
AGG	Aggression	SAV	Social Avoidance		
ACT	Activation	SHY	Shyness		
		DSF	Disaffiliativeness		

Page 4 includes Mr. C's scores on the Somatic/Cognitive and Internalizing Specific Problems Scales. His scores on the former are unremarkable. On the internalizing scales, Mr. C produced a clinically significant elevation on Anxiety (AXY), indicating that he reports being anxious, and is likely to experience significant anxiety and anxiety-related problems, intrusive ideation, and nightmares. His scores on the remaining internalizing scales are generally within normal limits. His low scores on Inefficacy (NFC), Stress and Worry (STW), and Multiple Specific Fears (MSF)

MMPI-2-RF PSY-5 Scales



The highest and lowest T scores possible on each scale are indicated by a "----"; MMPI-2-RF T scores are non-gendered.

AGGR-r Aggressiveness-Revised
 PSYC-r Psychoticism-Revised
 DISC-r Disconstraint-Revised
 NEGE-r Negative Emotionality/Neuroticism-Revised
 INTR-r Introversion/Low Positive Emotionality-Revised

indicate that Mr. C does not endorse items indicative of feeling inefficacious or indecisiveness, he reports a below-average level of stress and worries, and he reports a lower than average number of specific fears. However, these and other low scores need to be interpreted cautiously in light of the indications of under-reporting just discussed. Mr. C's scores on the Externalizing, Interpersonal, and Interest Scales (reported on page 5) are all within normal limits, with the exception of low scores on Family Problems (*FML*) and Shyness (*SHY*), which respectively

MMPI-2-RF T SCORES (BY DOMAIN)**PROTOCOL VALIDITY**

Content Non-Responsiveness	0	53	57 F			
	CNS	VRIN-r	TRIN-r			
Over-Reporting	74	51		66	48	54
	F-r	Fp-r		Fs	FBS-r	RBS
Under-Reporting	66	59				
	L-r	K-r				

SUBSTANTIVE SCALES

Somatic/Cognitive Dysfunction	51	63	46	53	59	54
	RC1	MLS	GIC	HPC	NUC	COG
Emotional Dysfunction	41	37	45	40	42	36
	EID	RCd	SUI	HLP	SFD	NFC
		46	45			
		RC2	INTR-r			
		44	36	70	47	43
		RC7	STW	AXY	ANP	BRF
						36
						MSF
						42
						NEGE-r
Thought Dysfunction	67	84				
	THD	RC6				
		47				
		RC8				
		59				
		PSYC-r				
Behavioral Dysfunction	60	62	63	55		
	BXD	RC4	JCP	SUB		
		53	45	44	56	69
		RC9	AGG	ACT	AGGR-r	DISC-r
Interpersonal Functioning	37	54	39	43	37	44
	FML	RC3	IPP	SAV	SHY	DSF
Interests	56	56				
	AES	MEC				

Note. This information is provided to facilitate interpretation following the recommended structure for MMPI-2-RF interpretation in Chapter 5 of the *MMPI-2-RF Manual for Administration, Scoring, and Interpretation*, which provides details in the text and an outline in Table 5-1.

indicate that Mr. C reports a past and current family environment that is relatively conflict-free, and little or no social anxiety. Mr. C's PSY-5 Scale scores are reported on page 6. His elevated score on Disconstraint (*DISC-r*) indicates various manifestations of under-controlled behavior and that he is likely to act out impulsively.

Page 7 of the report provides a summary of Mr. C's MMPI-2-RF results. The test scores are reported following the recommended structure for MMPI-2-RF interpretation described in the

ITEM-LEVEL INFORMATION

Unscorable Responses

The test taker produced scorable responses to all the MMPI-2-RF items.

Critical Responses

Seven MMPI-2-RF scales--Suicidal/Death Ideation (SUI), Helplessness/Hopelessness (HLP), Anxiety (AXY), Ideas of Persecution (RC6), Aberrant Experiences (RC8), Substance Abuse (SUB), and Aggression (AGG)--have been designated by the test authors as having critical item content that may require immediate attention and follow-up. Items answered by the individual in the keyed direction (True or False) on a critical scale are listed below if his T score on that scale is 65 or higher. The percentage of the MMPI-2-RF normative sample (NS) and of the Forensic, Pre-trial Criminal (Men) comparison group (CG) that answered each item in the keyed direction are provided in parentheses following the item content.

Anxiety (AXY, T Score = 70)

- 79. Item content removed. (True; NS 6.2%, CG 30.5%)
- 289. Item content removed. (True; NS 12.7%, CG 26.1%)

Ideas of Persecution (RC6, T Score = 84)

- 71. Item content removed. (True; NS 2.0%, CG 23.4%)
- 110. Item content removed. (True; NS 9.9%, CG 36.3%)
- 194. Item content removed. (True; NS 17.1%, CG 42.5%)
- 212. Item content removed. (False; NS 9.1%, CG 38.1%)
- 264. Item content removed. (True; NS 5.3%, CG 28.5%)
- 287. Item content removed. (True; NS 3.1%, CG 16.7%)
- 332. Item content removed. (True; NS 3.2%, CG 21.1%)

End of Report

This and previous pages of this report contain trade secrets and are not to be released in response to requests under HIPAA (or any other data disclosure law that exempts trade secret information from release). Further, release in response to litigation discovery demands should be made only in accordance with your profession's ethical guidelines and under an appropriate protective order.

Manual for Administration, Scoring, and Interpretation (Ben-Porath & Tellegen, 2008/2011) and by Ben-Porath (2012). Page 8 provides item-level information, which in Mr. C's case includes his responses⁷ that generated elevated scores on the *AXY* and *RC6* scales. The percentages associated with each item represent the percent of individuals in the normative sample (NS) and comparison groups (CG) who responded to each of these items responded in the keyed direction.

Case Discussion

Mr. C's MMPI-2-RF results indicate that he produced a valid and interpretable (although somewhat guarded) protocol that raises the possibility that he has a delusional disorder. Other diagnostic possibilities include an anxiety-related disorder and an antisocial personality

disorder. The former was ruled out because of the absence of other indications of an anxiety disorder. A review of all available records and information indicated that Mr. C likely did indeed have features of an antisocial personality disorder.

The possibility that Mr. C has a delusional disorder is particularly salient in a competency to stand trial evaluation. Because there is no evidence of over-reporting in his protocol, greater confidence can be placed in an inference that his test findings do in fact reflect a thought disorder. Moreover, although for test security reasons the actual content of the items he answered in the keyed direction on RC6 is not included in Figure 3.1, responses indicated that Mr. C believed he is being plotted against and that he has been punished unjustly, that he is talked about and has enemies who wish to harm him, and that others are trying to influence his thoughts. None of these responses reflects bizarre beliefs that would counterindicate a diagnosis of delusional disorder. In addition to requiring the absence of bizarre beliefs, the diagnostic criteria for this disorder indicate the need to rule out cultural factors as explanations for Mr. C's persecutory beliefs. In this case, the possibility that a family environment marked by constant conflict with authorities within which Mr. C was raised was considered as a potential cultural factor. However, it was deemed to be insufficient to account for the self-referential nature of Mr. C's persecutory beliefs.

Having established that Mr. C did indeed have a delusional disorder, the examiner next had to consider whether this condition played a role in the defendant's lack of cooperation with his attorneys. The MMPI-2-RF does not provide information that can be used to address this question directly. The examiner in this case ultimately concluded that Mr. C's failure to cooperate with his attorneys in planning and carrying out a defense was not a product of his delusional disorder. There was no reason to conclude that Mr. C was unable to recall his documented sale of illegal substances to an undercover agent, and his unrealistic insistence that all the evidence against him was fabricated appeared to be self-serving and more likely a product of antisocial behavior than of a delusional disorder.

The examiner informed the court of his opinion that Mr. C's uncooperative relationship with his attorneys was not the product of a mental disorder, and the defendant was found competent to stand trial. He was eventually convicted and sentenced to a fairly lengthy period of incarceration. He subsequently engaged in disruptive, unruly behavior at the state correctional facility, leading to an even longer period of incarceration.

Notes

- 1 MMPI-2-RF co-authors Auke Tellegen and myself.
- 2 RC Scales were not developed for the original Clinical Scales 5 and 0 because they are not measures of psychopathology. The three distinctive core components of the scales identified in Step 2 were targeted in subsequent development of substantive MMPI-2-RF measures.
- 3 <http://www.upress.umn.edu/test-division/MMPI-2-RF/mmpi-2-rf-references>.
- 4 It is worth noting in this context that many articles included in counts of MMPI or MMPI-2 investigations did not report studies of the test itself. Rather, these were investigations in which the MMPI or MMPI-2 was used as dependent measures of specific psychological constructs.
- 5 The following is not intended as an exhaustive review of the clinically relevant MMPI-2-RF literature, something that is beyond the scope of this chapter.
- 6 These inferences were made by the test authors on the basis of their understanding of the constructs assessed by the MMPI-2-RF substantive scales.
- 7 To protect test security, the content of the items Mr. C answered in the keyed direction is not included here, but would be included in an actual report.

References

- American Psychiatric Association (APA) (1987). *Diagnostic and statistical manual of mental disorders: DSM-III-R* (3rd ed. rev.). Washington, DC: Author.
- Arbisi, P. A., & Ben-Porath, Y. S. (1995). An MMPI-2 infrequent response scale for use with psychopathological populations: The Infrequency-Psychopathology Scale, F(p). *Psychological Assessment*, 7, 424–431.
- Arbisi, P. A., & Ben-Porath, Y. S. (1997). Characteristics of the F(p) scale as a function of diagnosis in an inpatient sample of veterans. *Psychological Assessment*, 9, 102–105.
- Arbisi, P. A., Polusny, M. A., Erbes, C. R., Thuras, P., & Reddy, M. K. (2011). The Minnesota Multiphasic Personality Inventory-2 Restructured Form in National Guard soldiers screening positive for posttraumatic stress disorder and mild traumatic brain injury. *Psychological Assessment*, 23(1), 203–214.
- Arbisi, P. A., Sellbom, M., & Ben-Porath, Y. S. (2008). Empirical correlates of the MMPI-2 Restructured Clinical (RC) scales in psychiatric inpatients. *Journal of Personality Assessment*, 90, 122–128.
- Archer, E. M., Hagan, L. D., Mason, J., Handel, R. W., & Archer, R. P. (2012). MMPI-2-RF characteristics of custody evaluation litigants. *Assessment*, 19, 14–20.
- Ben-Porath, Y. S. (2012). Addressing challenges to MMPI-2-RF-based testimony: Questions and answers. *Archives of Clinical Neuropsychology*, 27, 691–705.
- Ben-Porath, Y. S. (2012). *Interpreting the MMPI-2-RF*. Minneapolis: University of Minnesota Press.
- Ben-Porath, Y. S., & Forbey, J. D. (2003). *Non-gendered norms for the MMPI-2*. Minneapolis: University of Minnesota Press.
- Ben-Porath, Y. S., & Tellegen, A. (2008). *MMPI-2-RF: Manual for administration, scoring, and interpretation*. Minneapolis: University of Minnesota Press.
- Ben-Porath, Y. S., & Tellegen, A. (2011). *MMPI-2-RF: User's guide for reports* (2nd ed.). Minneapolis: University of Minnesota Press.
- Bianchini, K. J., Greve, K. W., & Glynn, G. (2005). On the diagnosis of malingered pain-related disability: Lessons from cognitive malingering research. *Spine Journal*, 5, 404–417.
- Block, A., Ben-Porath, Y. S., & Marek, R. J. (2012). Psychological risk factors for poor outcome of spine surgery and spinal cord stimulator implant: A review of the literature and their assessment with the MMPI-2-RF. *Clinical Neuropsychologist*.
- Burchett, D. L., & Ben-Porath, Y. S. (2010). The impact of overreporting on MMPI-2-RF Substantive Scale score validity. *Assessment*, 17, 497–516.
- Butcher, J. N. (2010). Personality assessment from the 19th to the early 21st century: Past achievements and contemporary challenges. *Annual Review of Clinical Psychology*, 6, 1–20.
- Butcher, J. N., Dahlstrom, W. G., Graham, J. R., Tellegen, A., & Kaemmer, B. (1989). *Manual for the restandardized Minnesota Multiphasic Personality Inventory: MMPI-2*. Minneapolis: University of Minnesota Press.
- Butcher, J. N., Graham, J. R., Ben-Porath, Y. S., Tellegen, A., Dahlstrom, W. G., & Kaemmer, B. (2001). *MMPI-2: Manual for administration and scoring* (rev. ed.). Minneapolis: University of Minnesota Press.
- Butcher, J. N., Graham, J. R., Williams, C. L., & Ben-Porath, Y. S. (1990). *Development and use of the MMPI-2 content scales*. Minneapolis: University of Minnesota Press.
- Daubert v. Merrell Dow Pharmaceuticals, Inc.* (1993) 509 U.S. 579, 589.
- Dionysius, K. E., Denney, R. L., & Halfaker, D. A. (2011). Detecting negative response bias with the Fake Bad Scale, Response Bias Scale, Henry-Heilbronner Index of the Minnesota Multiphasic Personality Inventory-2. *Archives of Clinical Neuropsychology*, 26, 81–88.
- Dragon, W. R., Ben-Porath, Y. S., & Handel, R. W. (2012). Examining the impact of unscorable item responses on the validity and interpretability of MMPI-2/MMPI-2-RF Restructured Clinical (RC) scale scores. *Assessment*, 19(1), 101–113.
- First, M. B., Spitzer, R. L., Gibbon, M., & Williams, J. B. W. (2001). *Structured clinical interview for DSM-IV axis I disorders*. New York: Biometrics Research Department, New York State Psychiatric Institute.
- Forbey, J. D., Ben-Porath, Y. S., & Gartland, D. (2009). Validation of the MMPI-2 Computerized Adaptive version (MMPI-2-CA) in a correctional intake facility. *Psychological Services*, 6(4), 279–292.
- Frank, J. D. (1974). Psychotherapy: The restoration of morale. *American Journal of Psychiatry*, 131, 271–274.
- Gervais, R. O., Ben-Porath, Y. S., & Wygant, D. B. (2009). Empirical correlates and interpretation of the MMPI-2-RF Cognitive Complaints (COG) scale. *Clinical Neuropsychologist*, 23, 996–1015.

- Gervais, R. O., Ben-Porath, Y. S., Wygant, D. B., & Green, P. (2007). Development and validation of a Response Bias Scale (RBS) for the MMPI-2. *Assessment, 14*(2), 196–208.
- Gervais, R. O., Ben-Porath, Y. S., Wygant, D. B., & Green, P. (2008). Differential sensitivity of the Response Bias Scale and MMPI-2 Validity Scales to memory complaints. *Clinical Neuropsychologist, 22*, 1061–1079.
- Gervais, R. O., Ben-Porath, Y. S., Wygant, D. B., & Sellbom, M. (2010). Incremental validity of the MMPI-2-RF over-reporting scales and RBS in assessing the veracity of memory complaints. *Archives of Clinical Neuropsychology, 25*, 274–284.
- Gervais, R. O., Wygant, D. B., Sellbom, M., & Ben-Porath, Y. S. (2011). Associations between Symptom Validity Test failure and scores on the MMPI-2-RF validity and substantive scales. *Journal of Personality Assessment, 93*, 508–517.
- Gilberstadt, H., & Duker, J. (1965). *A handbook for clinical and actuarial MMPI Interpretation*. Philadelphia: W. B. Saunders.
- Graham, J. R. (2012). *MMPI-2: Assessing personality and psychopathology* (3rd ed.). New York: Oxford University Press.
- Graham, J. R., Timbrook, R. E., Ben-Porath, Y. S., & Butcher, J. N. (1991). Congruence between MMPI and MMPI-2: Separating fact from artifact. *Journal of Personality Assessment, 57*, 205–215.
- Greene, R. L. (2011). *MMPI-2/MMPI-2-RF: An interpretive manual* (3rd ed.). Boston: Allyn & Bacon.
- Groth-Marnat, G. (2012). *Integrated psychological assessment reports: Theories, guidelines, and strategies*. Hoboken, NJ: John Wiley.
- Handel, R. W., & Archer, R. P. (2008). An investigation of the psychometric properties of the MMPI-2 Restructured Clinical (RC) Scales with mental health inpatients. *Journal of Personality Assessment, 90*, 239–249.
- Handel, R. W., Ben-Porath, Y. S., Tellegen, A., & Archer, R. P. (2010). Psychometric functioning of the MMPI-2-RF VRIN-r and TRIN-r scales with varying degrees of randomness, acquiescence, and counter-acquiescence. *Psychological Assessment, 22*, 87–95.
- Harkness, A. R. (1992). Fundamental topics in the personality disorders: Candidate trait dimensions from lower regions of the hierarchy. *Psychological Assessment, 4*, 251–259.
- Harkness, A. R., Finn, J. A., McNulty, J. L., & Shields, S. M. (2011). The Personality Psychopathology-Five (PSY-5): Recent constructive replication and assessment literature review. *Psychological Assessment, 24*, 432–443.
- Harkness, A. R., & McNulty, J. L. (1994). The Personality Psychopathology Five (PSY-5): Issues from the pages of a diagnostic manual instead of a dictionary. In S. Strack and M. Lorr (Eds.), *Differentiating normal and abnormal personality* (pp. 291–315). New York.
- Harkness, A. R., McNulty, J. L., & Ben Porath, Y. S. (1995). The Personality Psychopathology Five (PSY-5): Constructs and MMPI-2 scales. *Psychological Assessment, 7*, 104–114.
- Hart, S., Cox, D., & Hare, R. (1995). *The Hare PCL: SV Psychopathy Checklist: Screening Version*. New York: Multi-Health Systems.
- Hathaway, S. R. (1960). Foreword. In W. G. Dahlstrom & G. S. Welsh, *An MMPI handbook: A guide to use in clinical practice and research*. Minneapolis: University of Minnesota Press.
- Hathaway, S. R., & McKinley, J. C. (1942). A multiphasic personality schedule (Minnesota): III. The measurement of symptomatic depression. *Journal of Psychology, 14*, 73–84.
- Hathaway, S. R., & McKinley, J. C. (1943). *The Minnesota Multiphasic Personality Inventory*. Minneapolis: University of Minnesota Press.
- Hebben, N., & Milberg, W. (2009). *Essentials of neuropsychological assessment* (2nd ed.). New York: John Wiley.
- Helmes, E., & Reddon, J. R. (1993). A perspective on developments in assessing psychopathology: A critical review of the MMPI and MMPI-2. *Psychological Bulletin, 113*, 453–471.
- Jackson, D. N. (1971). The dynamics of structured personality tests. *Psychological Review, 78*(3), 229–248.
- Kay, T., Harrington, D. E., Adams, R. E., Anderson, T. W., Berrol, S., Cicerone, K., . . . Malec, J. (1993). Definition of mild traumatic brain injury: Report from the Mild Traumatic Brain Injury Committee of the Head Injury Interdisciplinary Special Interest Group of the American Congress of Rehabilitation Medicine. *Journal of Head Trauma Rehabilitation, 8*(3), 86–87.
- Kitaef, J. (2011). *Handbook of police psychology*. New York: Routledge.
- Larrabee, G. J. (2012). *Forensic neuropsychology: A scientific approach* (2nd ed.). New York: Oxford University Press.

- Lees-Haley, P. R. (1989). Malingering post-traumatic stress disorder on the MMPI. *Forensic Reports*, 2, 89–91.
- Lees-Haley, P. R., English, L. T., & Glenn, W. J. (1991). A Fake Bad Scale on the MMPI-2 for personal injury claimants. *Psychological Reports*, 68, 203–210.
- Lilienfeld, S. O., & Andrews, P. (1996). Development and preliminary validation of a self-report measure of psychopathic personality traits in noncriminal populations. *Journal of Personality Assessment*, 66(3), 488–524.
- Locke, D. E. C., Kirlin, K. A., Thomas, M. L., Osborne, D., Hurst, D. F., Drazkowski, J. F., . . . Noe, K. H. (2010). The Minnesota Multiphasic Personality Inventory–Restructured Form in the epilepsy monitoring unit. *Epilepsy and Behavior*, 17, 252–258.
- Locke, D. E. C., Kirlin, K. A., Wershaba, R., Osborne, D., Drazkowski, J. F., Sirven, J. I., & Noe, K. H. (2011). Randomized comparison of the Personality Assessment Inventory and the Minnesota Multiphasic Personality Inventory-2 in the epilepsy monitoring unit. *Epilepsy and Behavior*, 21, 397–401.
- Locke, D. E., & Thomas, M. L. (2011). Initial development of Minnesota Multiphasic Personality Inventory-2–Restructured Form (MMPI-2–RF) scales to identify patients with psychogenic nonepileptic seizures. *Journal of Clinical and Experimental Neuropsychology*, 33(3), 335–343.
- Loevinger, J. (1972). Some limitations of objective personality tests. In J. N. Butcher (Ed.), *Objective personality assessment: Changing perspectives* (pp. 45–58). Oxford: Academic Press.
- Marion, B. E., Sellbom, M., & Bagby, R. M. (2011). The detection of feigned psychiatric disorders using the MMPI-2–RF Overreporting Validity Scales: An analog investigation. *Psychological Injury and Law*, 4, 1–12.
- Marks, P. A., & Seeman, W. (1963). *The actuarial description of abnormal personality: An atlas for use with the MMPI*. Baltimore, MD: Williams & Wilkins.
- Mattson, C. A., Powers, B. K., Halfaker, D., Akenson, S. T., & Ben-Porath, Y. S. (in press). Predicting drug court completion with the MMPI-2–RF. *Psychological Assessment*.
- McCord, D. M., & Drerup, L. C. (2011). Relative practical utility of the MMPI-2 RC Scales versus the Clinical Scales in a chronic pain patient sample. *Journal of Clinical and Experimental Neuropsychology*, 33(1), 140–146.
- McGrath, R. E., Mitchell, M., Kim, B. H., & Hough, L. (2010). Evidence for response bias as a source of error variance in applied assessment. *Psychological Bulletin*, 136, 450–470.
- McNulty, J. L., Ben Porath, Y. S., & Graham, J. R. (1998). An empirical examination of the correlates of well-defined and not defined MMPI-2 code types. *Journal of Personality Assessment*, 71, 393–410.
- Meehl, P. E. (1945). The dynamics of “structured” personality tests. *Journal of Clinical Psychology*, 1, 296–303.
- Meehl, P. E. (1956). Wanted—a good cookbook. *American Psychologist*, 11, 263–272.
- Meehl, P. E. (1972). Reactions, reflections, projections. In J. N. Butcher (Ed.), *Objective personality assessment: Changing perspectives* (pp. 131–189). Oxford: Academic Press.
- Michigan v. Espinoza* (Mic. Ct. App. 2011).
- Morey, L. C. (2007). *Personality Assessment Inventory Professional Manual* (2nd ed.). Odessa, FL: Psychological Assessment Resources.
- Morey, L. C., Warner, M. B., & Hopwood, C. J. (2007). The Personality Assessment Inventory. In Goldstein, A. (Ed.), *Forensic psychology: Advanced topics for forensic mental experts and attorneys* (pp. 97–126). Hoboken, NJ: John Wiley.
- Nelson, N. W., Hoelzle, J. B., McGuire, K. A., Sim, H. A., Goldman, D. J., Ferrier-Auerbach, A. G., . . . Sponheim, H. R. (2011). Self-report of psychological function among OEF/OIF personnel who also report combat-related concussion. *Clinical Neuropsychology*, 25, 716–740.
- Nelson, N. W., Sweet, J. J., & Heilbronner, R. L. (2007). Examination of the new MMPI-2 Response Bias Scale (Gervais): Relationship with MMPI-2 validity scales. *Journal of Clinical and Experimental Neuropsychology*, 29, 67–72.
- Neukrug, E. S., & Fawcett, R. C. (2009). *Essentials of testing and assessment: A practical guide for counselors, social workers, and psychologists* (2nd ed.). Belmont, CA: Thompson Brooks/Cole.
- Norman, W. (1972). Psychometric considerations for a revision of the MMPI. In J. N. Butcher (Ed.), *Objective personality assessment: Changing perspectives* (pp. 59–83). Oxford: Academic Press.
- Ozonoff, S., Garcia, N., Clark, E., & Lainhart, J. E. (2005). MMPI-2 personality profiles of high-functioning adults with autism spectrum disorders. *Assessment*, 12, 86–95.
- Pinoneault, T. B., & Ezzo, F. R. (2012). A comparison of MMPI-2–RF profiles between child maltreatment and non-maltreatment custody cases. *Journal of Forensic Psychology Practice*, 12, 227–237.
- Plante, T. G. (2010). *Contemporary clinical psychology* (3rd ed.). New York: John Wiley.

- Purdon, S. E., Purser, S. M., & Goddard, K. M. (2011). MMPI-2 Restructured Form over-reporting scales in first-episode psychosis. *Clinical Neuropsychologist*, 25, 829–842.
- Resendes, J., & Lecci, L. (in press). Comparing the MMPI-2 scale scores of parents involved in parental competency and child custody assessments. *Psychological Assessment*. Advance online publication.
- Rock, R. C., Sellbom, M., Ben-Porath, Y. S., & Salekin, R. T. (in press). Concurrent and predictive validity of psychopathy in a batterers intervention sample. *Law and Human Behavior*.
- Rogers, R., Bagby, R. M., & Dickens, S. E. (1992). *Structured Interview of Reported Symptoms professional manual*. Odessa, FL: Psychological Assessment Resources.
- Rogers, R., Gillard, N. D., Berry, D. T. R., & Granacher, R. P. (2011). Effectiveness of the MMPI-2-RF Validity Scales for feigned mental disorders and cognitive impairment: A known-groups study. *Journal of Psychopathology and Behavioral Assessment*, 33, 355–367.
- Rogers, R., Sewell, K. W., & Gillard, N. D. (2010). *Structured Interview of Reported Symptoms (SIRS) and professional manual*. Lutz, FL: Psychological Assessment Resources.
- Schroeder, R. W., Baade, L. E., Peck, C. P., VonDran, E. J., Brockman, C. J., Webster, B. K., & Heinrichs, R. J. (2012). Validation of the MMPI-2-RF Validity Scales in criterion group neuropsychological samples. *Clinical Neuropsychologist*, 26, 129–146.
- Sellbom, M. (2011). Elaborating on the construct validity of the Levenson Self-Report Psychopathy Scale in incarcerated and non-incarcerated samples. *Law and Human Behavior*, 35, 440–451.
- Sellbom, M., & Bagby, R. M. (2008). Validity of the MMPI-2-RF (Restructured Form) L-r and K-r scales in detecting underreporting in clinical and nonclinical samples. *Psychological Assessment*, 20, 370–376.
- Sellbom, M., & Bagby, R. M. (2010). Detection of overreported psychopathology with the MMPI-2 RF validity scales. *Psychological Assessment*, 22(4), 757–767.
- Sellbom, M., Bagby, R. M., Kushner, S., Quilty, L. C., & Ayeart, L. E. (2012). Diagnostic construct validity of the MMPI-2 Restructured Form (MMPI-2-RF) scale scores. *Assessment*, 19, 176–186.
- Sellbom, M., Ben-Porath, Y. S., Baum, L. J., Erez, E., & Gregory, C. (2008). Predictive validity of the MMPI-2 Restructured Clinical (RC) scales in a batterers' intervention program. *Journal of Personality Assessment*, 90, 129–135.
- Sellbom, M., Ben-Porath, Y. S., Patrick, C. J., Wygant, D. B., Gartland, D. M., & Stafford, K. P. (2012). Development and construct validation of MMPI-2-RF measures assessing global psychopathy, fearless-dominance, and impulsive-antisociality. *Personality Disorders: Theory, Research, and Treatment*, 3, 17–38.
- Sellbom, M., Ben-Porath, Y. S., & Stafford, K. P. (2007). A comparison of MMPI-2 measures of psychopathic deviance in a forensic setting. *Psychological Assessment*, 19, 430–436.
- Sellbom, M., Lee, T. T. C., Ben-Porath, Y. S., Arbisi, P. A., & Gervais, R. O. (2012). Differentiating PTSD symptomatology with the MMPI-2-RF (Restructured Form) in a forensic disability sample. *Psychiatry Research*, 197, 172–179.
- Sellbom, M., Toomey, J. A., Wygant, D. B., Kucharski, L. T., & Duncan, S. (2010). Utility of the MMPI-2-RF (Restructured Form) validity scales in detecting malingering in a criminal forensic setting: A known-groups design. *Psychological Assessment*, 22, 22–31.
- Slick, D. J., Hopp, G., Strauss, E., & Thompson, G. B. (1997). *Victoria Symptom Validity Test: Professional manual*. Odessa, FL: Psychological Assessment Resources.
- Slick, D. J., Sherman, E. M. S., & Iverson, G. L. (1999). Diagnostic criteria for malingered neurocognitive dysfunction: Proposed standards for clinical practice and research. *Clinical Neuropsychologist*, 13, 545–561.
- Smart, C. M., Nelson, N. W., Sweet, J. J., Bryant, F. B., Berry, D. T. R., Granacher, R. P., & Heilbronner, R. L. (2008). Use of MMPI-2 to identify cognitive effort: A hierarchically optimal classification tree analysis. *Journal of the International Neuropsychological Society*, 14, 842–852.
- Stenner, A. J., Horabin, I., Smith, D. R., & Smith, M. (1988). Most comprehension tests do measure reading comprehension: A response to McLean and Goldstein. *Phi Delta Kappan*, 69, 765–769.
- Stredny, R. V., Archer, R. P., & Mason, J. A. (2006). MMPI-2 and MCMI-III characteristics of parental competency examinees. *Journal of Personality Assessment*, 87, 113–115.
- Tellegen, A., & Ben-Porath, Y. S. (1992). The new uniform T-scores for the MMPI-2: Rationale, derivation, and appraisal. *Psychological Assessment*, 4, 145–155.
- Tellegen, A., & Ben-Porath, Y. S. (1993). Code-type comparability of the MMPI and MMPI-2: Analysis of recent findings and criticisms. *Journal of Personality Assessment*, 61, 489–500.
- Tellegen, A., & Ben-Porath, Y. S. (2008). *MMPI-2-RF technical manual*. Minneapolis: University of Minnesota Press.

- Tellegen, A., Ben-Porath, Y. S., McNulty, J. L., Arbisi, P. A., Graham, J. R., & Kaemmer, B. (2003). *The MMPI-2 Restructured Clinical Scales: Development, validation, and interpretation*. Minneapolis: University of Minnesota Press.
- Tellegen, A., Watson, D., & Clark, L. A. (1999a). Further support for a hierarchical model of affect: Reply to Green and Salovey. *Psychological Science*, 10(4), 307–309.
- Tellegen, A., Watson, D., & Clark, L. A. (1999b). On the dimensional and hierarchical structure of affect. *Psychological Science*, 10, 297–303.
- Thomas, M. L., & Locke, D. E. C. (2010). Psychometric properties of the MMPI-2-RF Somatic Complaints (RC1) Scale. *Psychological Assessment*, 22, 492–503.
- Thomas, M. L., & Youngjohn, J. R. (2009). Let's not get hysterical: Comparing the MMPI-2 Validity, Clinical, and RC Scales in TBI litigants tested for effort. *Clinical Neuropsychologist*, 23, 1067–1084.
- Tombaugh, T. N. (1997). The Test of Memory Malingering (TOMM): Normative data from cognitively intact and cognitively impaired individuals. *Psychological Assessment*, 9(3), 260–268.
- Tsushima, W. T., Geling, O., & Fabriga, L. (2011). Comparison of MMPI-2 Validity Scale scores of personal injury litigants and disability claimants. *Clinical Neuropsychologist*, 25, 1403–1414.
- Watson, C., Quilty, L. C., & Bagby, R. M. (2011). Differentiating bipolar disorder from major depressive disorder using the MMPI-2-RF: A receiver operating characteristics (ROC) analysis. *Journal of Psychopathology and Behavioral Assessment*, 33, 368–374.
- Watson, D., & Tellegen, A. (1985). Toward a consensual structure of mood. *Psychological Bulletin*, 98, 219–235.
- Wood v. Thaler*, 131 S.Ct. 2451 (2011).
- Wygant, D. B., Anderson, J. L., Sellbom, M., Rapiet, J. L., Allgeier, L. M., & Granacher, R. P. (2011). Association of MMPI-2 Restructured Form (MMPI-2-RF) validity scales with structured malingering criteria. *Psychological Injury and Law*, 4, 13–23.
- Wygant, D. B., Ben-Porath, Y. S., Arbisi, P. A., Berry, D. T. R., Freeman, D. B., & Heilbronner, R. L. (2009). Examination of the MMPI-2 Restructured Form (MMPI-2-RF) validity scales in civil forensic settings: Findings from simulation and known group samples. *Archives of Clinical Neuropsychology*, 27, 671–680.
- Wygant, D. B., & Sellbom, M. (in press). Viewing psychopathy from the perspective of the Personality Psychopathology Five model: Implications for DSM-5. *Journal of Personality Disorders*.
- Wygant, D. B., Sellbom, M., Ben-Porath, Y. S., Stafford, K. P., Freeman, D. B., & Heilbronner, R. I. (2007). The relation between symptom validity testing and MMPI-2 scores as a function of forensic evaluation context. *Archives of Clinical Neuropsychology*, 22, 488–499.
- Wygant, D. B., Sellbom, M., Gervais, R. O., Ben-Porath, Y. S., Stafford, K. P., Freeman, D. B., & Heilbronner, R. L. (2010). Further validation of the MMPI-2 and MMPI-2-RF Response Bias Scale: Findings from disability and criminal forensic settings. *Psychological Assessment*, 22(4), 745–756.
- Youngjohn, J. R., Wershba, R., Stevenson, M., Sturgeon, J., & Thomas, M. L. (2011). Independent validation of the MMPI-2-RF Somatic/Cognitive and Validity Scales in TBI litigants tested for effort. *Clinical Neuropsychologist*, 25, 463–476.

4

THE MINNESOTA MULTIPHASIC PERSONALITY INVENTORY–ADOLESCENT

Robert P. Archer and Richard W. Handel

History and Development

The Minnesota Multiphasic Personality Inventory–Adolescent (MMPI–A) is a 478-item objective personality assessment instrument designed for use with adolescents. It provides an array of Validity and Clinical Scales, and interpretation is facilitated by a substantial research literature. The MMPI–A is an adaptation of the Minnesota Multiphasic Personality Inventory (MMPI), and therefore an understanding of its development, psychometric characteristics, and research base necessarily begins with a history and description of the MMPI.

In 1937, Starke R. Hathaway and J. C. McKinley began construction on what eventually became the Minnesota Multiphasic Personality Inventory. Their work was stimulated, in part, by their observation that many patients in medical settings displayed symptoms that were psychological or “psychoneurotic in nature” (McKinley & Hathaway, 1943, p. 161). Furthermore, they believed that an objective self-report instrument could identify and describe the psychological features of patients more accurately than the traditional psychiatric interview used as the primary diagnostic procedure in the 1930s. The MMPI was also designed to provide a standard diagnostic tool for researchers evaluating the effectiveness of newly developed interventions. An additional goal was the creation of a personality measure that would be sensitive to psychological changes over time, allowing for a longitudinal view of psychological functioning through repeat administrations. Although Hathaway and McKinley originally sought to develop their instrument for use at the University of Minnesota Hospitals, the MMPI was published in 1942 and rapidly became established as the most commonly utilized objective personality assessment instrument in the United States across a variety of clinical settings (e.g., Lubin, Larsen, & Matarazzo, 1984; Lubin, Larsen, Matarazzo, & Seever, 1985; Lubin, Wallis, & Paine, 1971; Piotrowski & Keller, 1989).

Hathaway and McKinley developed the scales of the MMPI using a method known as empirical or criterion keying, and its application in the development of a personality assessment instrument was revolutionary. Previous personality test development methods had often relied on the rational selection of items using face validity as the primary criterion; that is, items were selected on the basis of the degree to which they appeared to the test developer to be related to a given construct. In contrast, the empirical keying approach identifies test items solely on the basis of the results of statistical procedures. Potential test items were administered to a normal sample and a variety of criterion samples, the latter containing individuals who appeared to display clear and unambiguous

symptoms related to a specific diagnosis. Items were then selected for scale membership according to their ability to differentiate effectively between normal and criterion groups.

Prior to its release in 1942, preliminary versions of the MMPI were termed the “Medical and Psychiatric Inventory” and the “Minnesota Personality Schedule” (Colligan & Offord, 1992). The initial item pool was created by reviewing prominent psychiatric textbooks as well as previously established scales of social attitude, temperament, and psychological functioning. Indeed, the Humm–Wadsworth Temperament Scale (1935) appears to have served as a source for approximately 150 items eventually selected for the MMPI (Colligan & Offord, 1992). The early item pool comprised over 1,000 items and then was reduced to 504 by eliminating redundant items as well as items that appeared to hold little utility for a personality inventory. The potential items were then administered to 724 visitors of patients at the University of Minnesota Hospital, and to this core group Hathaway and McKinley eventually added smaller normative samples consisting of students receiving pre-college counseling, individuals employed through the Federal Works Progress Administration (WPA), and a limited number of medical patients who did not display psychiatric disorders. The criterion groups used for scale development were relatively small subgroups of clinical patients divided into discrete diagnostic categories, including Hypochondriasis (*Hi*), Depression (*D*), Hysteria (*Hy*), Psychopathic Deviancy (*Pd*), Paranoia (*Pa*), Psychasthenia (*Pt*), Schizophrenia (*Sc*), and Mania (*Ma*). Eventually the Masculinity–Femininity (*Mf*) and Social Introversion (*Si*) scales were also added, although these were considered non-clinical scales and were often omitted by early test-users.

Hathaway and McKinley recognized that test-takers did not always respond in an open and candid manner to psychological tests, and therefore they created several Validity Scales in addition to the MMPI basic Clinical Scales. These scales were developed to provide a standardized assessment of the individual’s test-taking attitude or response style. The Cannot Say or (?) scale is the total of the items omitted by the test taker or endorsed in both the true and false direction. The Lie (*L*) scale comprises 15 items that were rationally derived to evaluate a defensive test-taking attitude involving the denial of common human failings or faults. A 64-item Infrequency (*F*) scale was also developed containing statements that were endorsed in the deviant direction by 10% or fewer of the normative sample. The content of the *F* scale varies widely, but elevations on this scale may be the result of carelessness, problems in literacy, and conscious or unconscious efforts to over-report or exaggerate symptomatology, as well as more severe forms of genuine psychopathology. Finally, the Defensiveness (*K*) scale was added to identify more subtle efforts to deny symptomatology or psychopathology as part of a defensive response style. In addition to serving as a measure of psychological defensiveness, the *K* scale was proposed by McKinley, Hathaway, and Meehl (1948) as a correction factor for five of the MMPI–A basic scales. Thus, the *K* scale also serves as a correction adjustment to basic scales *Hi*, *Pd*, *Pt*, *Sc*, and *Ma* to compensate for suppression of scores due to psychological defensiveness.

History of the Original MMPI with Adolescents

Dora Capwell undertook the first research investigation of the MMPI with adolescents in 1941, prior to the actual publication of the test. Capwell (1945a) demonstrated the ability of the MMPI to accurately discriminate between groups of delinquent and non-delinquent girls on the basis of *Pd* scale elevations, and a further investigation showed that these differences were maintained in follow-up studies conducted 4–15 months following the initial administration (Capwell, 1945b). In the largest MMPI data set ever collected with adolescents, Hathaway and Monachesi (1953; 1963) conducted a large-scale longitudinal study of the relationship between the MMPI findings from 15,000 high school students in the late 1940s and early 1950s and subsequent delinquent

behaviors as reflected in school, police, and court records. Hathaway and Monachesi (1963) and Monachesi and Hathaway (1969) summarized their findings by proposing that elevations on MMPI scales *Pd*, *Sc*, and *Ma* (the excitatory scales) were associated with higher delinquency rates, whereas elevations on scales *D*, *Mf*, and *Si* (the inhibitory scales) were related to reduced risk of delinquent or antisocial behaviors.

Marks and Briggs (1972) generated the most frequently used set of adolescent norms for the original MMPI based on 1,766 normal adolescents. Their norms were grouped separately for girls and boys and were provided for ages 17, 16, 15, and a category of 14 and below. Marks, Seeman, and Haller (1974) reported the first actuarially based personality descriptors for a series of 29 MMPI codetypes. Their codetype study was based on the responses of approximately 1,250 adolescents aged 12–18 who had undergone at least 10 hours of psychotherapy between 1965 and 1973. This study also employed 172 therapists from 30 states, who provided descriptive ratings for these adolescents from a correlate pool of 1,265 descriptors that appeared relevant to both male and female adolescents. The Marks et al. (1974) study was critical in providing clinicians with the first correlate information necessary to interpret adolescent codetype patterns. In 1987, Archer produced a comprehensive guide to using the MMPI with adolescents, and in 1991 a survey conducted by Archer, Maruish, Imhof, and Piotrowski found that the original form of the MMPI was the most frequently utilized objective personality assessment instrument in evaluations of adolescents in the United States.

History of the Development of the MMPI–A

Several areas of concern in regard to the use of the MMPI with adolescents emerged in the 1991 survey conducted by Archer and his colleagues. Disadvantages cited included the length of the test, the lack of contemporary norms, the demanding reading level associated with the instrument, and the presence of inappropriate or outdated items (e.g., “I used to play drop-the-handkerchief”). In addition, because the MMPI was primarily developed as an adult assessment instrument, questions designed to assess important aspects of adolescent experience were not included in the original inventory. These content areas included adolescent substance abuse, eating disorders, and school problems.

Regarding the lack of contemporary norms cited as a problem by respondents in the 1991 survey, the Marks and Briggs (1972) norms commonly employed with adolescents had been collected from the mid-1940s to the mid-1960s in a sample that contained very few non-Caucasian adolescents (Archer, 2005). Pancoast and Archer (1988) examined the adequacy of the Marks and Briggs norms in an analysis of MMPI results for normal adolescent samples collected between 1965 and 1974, contrasted with samples of more contemporary adolescents collected following 1975. Their findings indicated that the Marks and Briggs adolescent norms were not providing an adequate normative baseline for evaluations of more contemporary samples of adolescents.

Description of the MMPI–A

In July 1989, an advisory committee was appointed by the University of Minnesota Press to develop an adolescent form of the MMPI. An overarching goal was to maintain substantive continuity with the original MMPI, including the preservation of the basic validity and clinical scales. Therefore, an effort was made to minimize the changes to most MMPI basic scales while recognizing that more extensive changes would be needed for scales *F*, *Mf*, and *Si*. Although it understood the need for continuity with the original MMPI, the committee also perceived the revision as an opportunity to improve the item pool for adolescents and to provide several new scales directly relevant to

adolescent development and psychopathology. A final goal of the project was the collection of a new normative sample representative of a diverse and contemporary population of adolescents.

Data were initially collected from about 2,500 adolescents in Minnesota, Ohio, California, Virginia, Pennsylvania, New York, North Carolina, and Washington. Participants were excluded if they produced incomplete data, their *F* raw score value exceeded 25, their Cannot Say scale value exceeded 35, or their age was above 18 or below 14. The resulting normative sample of 1,620 adolescents included 805 boys and 815 girls. The mean age was 15.5 ($SD = 1.17$) for boys and 15.6 ($SD = 1.18$) for girls. The ethnic makeup of the sample was 76% Caucasian, 12% African American, and 12% from other ethnic groups. Although the MMPI-A normative data set was a clear improvement over the Marks and Briggs (1972) normative sample in terms of geographic and ethnic diversity, the contemporary norms also have some less desirable characteristics. As was noted by Archer (2005) and Black (1994), many adolescents included in the MMPI-A normative set were children of highly educated parents. For example, about 50% of the adolescents' fathers and about 40% of their mothers reported obtaining a bachelor's degree or higher; in contrast, according to U.S. census data from 1980, only about 20% of men and 13% of women held college degrees. Butcher et al. (1992) also noted that the MMPI-A norms were less representative of adolescents who had a history of truancy or delinquency or who had dropped out of school, particularly important issues for adolescents evaluated in juvenile justice detention or correctional facilities.

Despite efforts to minimize the length of the MMPI-A, the final form continues to be time-consuming in comparison with other, similar tests. The adolescent version of the Personality Assessment Inventory (PAI-A) contains 264 items, or a little over half of the number of items on the MMPI-A (Morey, 2000). The Millon Adolescent Clinical Inventory (MACI) is a 160-item test that typically requires only about 30 minutes to administer (Millon, 1993). In contrast, the 478 items of the MMPI-A typically require 60–90 minutes for completion (Archer & Krishnamurthy, 2002). Although it is possible to administer an abbreviated version of the MMPI-A limited to the first 350 items, this format should be considered only as a last resort, because the Validity Scales *VRIN*, *TRIN*, *F*, and *F2*, and the Content and Supplementary Scales, cannot be scored.

The MMPI-A is designed to be used with adolescents aged 14–18 and should never be given to an individual older than 18 (Butcher et al., 1992). At the other end of the age continuum, the MMPI-A can be used selectively with 12- and 13-year-old examinees if they are developmentally advanced and have sufficient cognitive and reading skills (Butcher et al., 1992). However, many younger adolescents assessed in juvenile justice settings will lack the sixth- to seventh-grade reading ability necessary to respond successfully to the MMPI-A (Archer & Krishnamurthy, 2002). An audiotaped version of the MMPI-A is available for reading-disabled adolescents who otherwise have the cognitive skills necessary to understand the MMPI-A item pool, and this administration form typically requires about 1 hour 40 minutes.

MMPI-A Scales

The MMPI-A consists of 478 items, and an abbreviated administration may be conducted using the first 350 items, which permits scoring of Validity Scales *L*, *F1*, and *K*, as well as the 10 basic Clinical Scales. However, all 478 items of the MMPI-A must be administered in order to obtain the Content and Supplementary Scale results. The MMPI-A was adapted from the original MMPI form with the following changes: 58 standard scale items were deleted from the basic scales, with 88% of these items coming from the *F*, *Mf*, and *Si* scales. Items eliminated from the original form of the MMPI in the development of the MMPI-A typically pertained to religious attitudes and practices, sexual preferences, bowel and bladder functioning, or topics deemed inappropriate in evaluating adolescents (e.g., voting in elections). The resulting MMPI-A included the original

10 basic Clinical Scales and 3 basic Validity Scales, 4 newly developed Validity Scales, 15 Content Scales, 6 Supplementary Scales, and 28 Harris–Lingoes and 3 *Si* subscales. The new Validity Scales included in the MMPI–A are the *F1* and *F2* subscales, the True Response Inconsistency (*TRIN*) scale, and the Variable Response Inconsistency (*VRIN*) scale. Table 4.1 provides an overview of the MMPI–A scales presented in the 1992 manual.

TABLE 4.1 Overview of the MMPI–A Scales and Subscales

<i>Basic Profile Scales (17 scales)</i>	
<i>Standard Scales (13)</i>	
<i>L</i> through <i>Si</i>	
<i>Additional Validity Scales (4)</i>	
<i>F1/F2</i> (Subscales of <i>F</i> Scale)	
<i>VRIN</i> (Variable Response Inconsistency)	
<i>TRIN</i> (True Response Inconsistency)	
<i>Content and Supplementary Scales (21 scales)</i>	
<i>Content Scales (15)</i>	
<i>A-anx</i> (Anxiety)	
<i>A-obs</i> (Obsessiveness)	
<i>A-dep</i> (Depression)	
<i>A-hea</i> (Health Concerns)	
<i>A-aln</i> (Alienation)	
<i>A-biz</i> (Bizarre Mentation)	
<i>A-ang</i> (Anger)	
<i>A-cyn</i> (Cynicism)	
<i>A-con</i> (Conduct Problems)	
<i>A-lse</i> (Low Self-esteem)	
<i>A-las</i> (Low Aspirations)	
<i>A-sod</i> (Social Discomfort)	
<i>A-fam</i> (Family Problems)	
<i>A-sch</i> (School Problems)	
<i>A-trt</i> (Negative Treatment Indicators)	
<i>Supplementary Scales (6)</i>	
<i>MAC-R</i> (MacAndrew Alcoholism–Revised)	
<i>ACK</i> (Alcohol/Drug Problem Acknowledgment)	
<i>PRO</i> (Alcohol/Drug Problem Proneness)	
<i>IMM</i> (Immaturity)	
<i>A</i> (Anxiety)	
<i>R</i> (Repression)	
<i>Harris–Lingoes and Si Subscales (31 scales)</i>	
<i>Harris–Lingoes Subscales (28)</i>	
<i>D₁</i> (Subjective Depression)	
<i>D₂</i> (Psychomotor Retardation)	
<i>D₃</i> (Physical Malfunctioning)	
<i>D₄</i> (Mental Dullness)	
<i>D₅</i> (Brooding)	
<i>Hy₁</i> (Denial of Social Anxiety)	
<i>Hy₂</i> (Need for Affection)	
<i>Hy₃</i> (Lassitude–Malaise)	
<i>Hy₄</i> (Somatic Complaints)	
<i>Hy₅</i> (Inhibition of Aggression)	
<i>Pd₁</i> (Familial Discord)	

TABLE 4.1 Continued

<i>Pd</i> ₂ (Authority Problems)
<i>Pd</i> ₃ (Social Imperturbability)
<i>Pd</i> ₄ (Social Alienation)
<i>Pd</i> ₅ (Self-Alienation)
<i>Pa</i> ₁ (Persecutory Ideas)
<i>Pa</i> ₂ (Poignancy)
<i>Pa</i> ₃ (Naiveté)
<i>Sc</i> ₁ (Social Alienation)
<i>Sc</i> ₂ (Emotional Alienation)
<i>Sc</i> ₃ (Lack of Ego Mastery, Cognitive)
<i>Sc</i> ₄ (Lack of Ego Mastery, Conative)
<i>Sc</i> ₅ (Lack of Ego Mastery, Defective Inhibition)
<i>Sc</i> ₆ (Bizarre Sensory Experiences)
<i>Ma</i> ₁ (Amorality)
<i>Ma</i> ₂ (Psychomotor Acceleration)
<i>Ma</i> ₃ (Imperturbability)
<i>Ma</i> ₄ (Ego Inflation)
<i>Si</i> Subscales (3)
<i>Si</i> ₁ (Shyness/Self-Consciousness)
<i>Si</i> ₂ (Social Avoidance)
<i>Si</i> ₃ (Alienation—Self and Others)

Source: From Archer, R. P. *MMPI-A: Assessing Adolescent Psychopathology*, Third Edition (Routledge, 2005).

There is a considerable degree of overlap between the 15 Content Scales on the MMPI-2 and the MMPI-A. A thorough discussion of the development of the MMPI-A content scales is provided by Williams, Butcher, Ben-Porath, and Graham (1992). After the identification of items in the MMPI-2 Content Scales deemed appropriate for the assessment of adolescents, the MMPI-A Content Scales were refined by adding or deleting items depending on their relative contributions to the overall reliability of the scale. A rational review of scale-item content was then completed to ensure that items appeared appropriate for measuring underlying scale constructs. Finally, items correlating more strongly with scales other than the content scale to which they were originally assigned were deleted. The development process for the MMPI-A Content Scales rendered the scales face-valid and thereby easily influenced by response style in terms of over-reporting or under-reporting symptoms. When interpreting the MMPI-A Content Scales, it is important to carefully consider the validity indicators, particularly scores on the Defensiveness (*K*) scale. Furthermore, although the MMPI-A Content Scales have relatively high alpha coefficient values, most of the scales also possess one or more subcomponents. Sherwood, Ben-Porath, and Williams (1997) developed a set of Content Component Scales for 13 of the 15 MMPI-A Content Scales to facilitate the evaluation of specific areas of content endorsement. The Content Component Scales were officially added to the MMPI-A in the MMPI-A manual supplement (Ben-Porath, Graham, Archer, Tellegen, & Kaemmer, 2006).

The Supplementary Scales of the MMPI-A include three scales developed for the original MMPI: Anxiety (*A*), Repression (*R*), and the MacAndrew Alcoholism Revised (*MAC-R*). Additional Supplementary Scales include the Immaturity (*IMM*) scale developed by Archer, Pancoast, and Gordon (1994), the Alcohol/Drug Problem Acknowledgment (*ACK*) scale, and the Alcohol/Drug Problem Proneness (*PRO*) scale developed by Weed, Butcher, and Williams (1994). The relatively low number of item deletions made to the MMPI-A clinical scales made it possible to retain the Harris-Lingoes Content subscales and extend these subscales to the MMPI-A content

scales. Additionally, the *Si* subscales developed for the MMPI-2 by Ben-Porath, Hostetler, Butcher, and Graham (1989) were also included on the MMPI-A Subscale Profile Sheet. Thus, a review of MMPI-A scale and subscale features reveals numerous similarities between the MMPI-A and both the original MMPI and the MMPI-2.

Finally, the MMPI-A includes measures of the Personality Psychopathology Five (PSY-5) constructs (Harkness & McNulty, 1994). The PSY-5 is a dimensional model of personality developed from empirical studies of both personality disorder and personality descriptors (Harkness & McNulty, 1994). Subsequently, McNulty, Harkness, Ben-Porath, and Williams (1997) developed MMPI-A PSY-5 scales. These scales were officially added to the MMPI-A in the MMPI-A manual supplement (Ben-Porath et al., 2006). The scales added to the MMPI-A in the 2006 manual supplement are presented in Table 4.2.

A large body of research is available to support the interpretation of scores on various MMPI-A scales. Forbey (2003) estimated that approximately 120 studies were completed on the MMPI-A from the time of its initial publication in 1992 through 2002. In addition, Archer and Krishnamurthy (2002) noted that the MMPI-A has been the subject of numerous master's thesis studies and doctoral dissertations, further reflecting the strong research interest in the MMPI-A. More recently, Baum, Archer, Forbey, and Handel (2009) identified 277 articles, books, book chapters, monographs, and dissertation abstracts on the MMPI-A between 1992 and 2007.

Psychometric Characteristics

Information on the test-retest reliability, internal consistency, and factor structure of the MMPI-A, as well as correlate information for normative and clinical samples, is provided in the MMPI-A manual (Butcher et al., 1992) and was updated by Archer (2005). The MMPI-A is a psychometrically sound instrument that exhibits test-retest correlates for basic scales ranging from $r = .19$ for the F_i subscale to $r = .84$ for the *Si* scale (Butcher et al., 1992). Stein, McClinton, and Graham (1998) evaluated the 1-year test-retest reliability of the MMPI-A clinical scales and reported values ranging from $r = .51$ for the *Pa* scale to $r = .75$ for the *Si* scale. Similarly, test-retest correlations for the content scales ranged from $r = .40$ for the *A-trt* to $r = .73$ for the *A-sch* scale. The standard error of measurement for the clinical scales on the MMPI-A is typically estimated to fall between 2 and 3 raw score points (Butcher et al., 1992). In the MMPI-A normative sample, the internal consistency (alpha coefficient) values for the MMPI-A scales range from lower values on scales *Mf* ($\alpha = .43$) and *Pa* ($\alpha = .57$) to relatively higher values ($\alpha = .80$) on many of the content scales and the *IMM* scale. Handel, Archer, Elkins, Mason, and Simonds-Bisbee (2011) recently reported alpha coefficients for the Clinical, Content, and Supplementary Scales in a forensic sample of boys and girls. For the clinical scales, Handel et al. reported alpha coefficients ranging from .17 for girls and .29 for boys on *Mf* to .90 on *Sc* for both genders. For the Content Scales, alpha coefficients ranged from .51 for girls and .52 for boys on *A-las*, to .84 on *A-sod* for girls and .81 on *A-hea* for boys. Finally, for the Supplementary Scales, alpha coefficients ranged from low values on *MAC-R* (.38 for girls and .42 for boys) to higher values ($\alpha \geq .80$) on *IMM* and *A*.

Because the MMPI-A retains many features of the MMPI, particularly regarding the basic validity and clinical scales, the research done with adolescents on the original MMPI is largely generalizable to the MMPI-A (Archer, 2005). Furthermore, research has found that correlates for the clinical scales and for several supplementary scales appear similar for adolescents and adults (Archer, Gordon, Anderson, & Giannetti, 1989; Archer, Gordon, Giannetti, & Singles, 1988; Williams & Butcher, 1989). Furthermore, two recent studies on the MMPI-A added support to the validity of scores on the Clinical Scales. Veltri et al. (2009) identified correlates of the Clinical Scales using criterion variables from psychiatrists' and parents' reports as well as psychosocial

TABLE 4.2 Overview of MMPI-A Scales Added in the MMPI-A Manual Supplement (Ben-Porath et al., 2006)

<i>Content Component Scales (31)</i>
<i>A-dep₁</i> (Dysphoria)
<i>A-dep₂</i> (Self-Depreciation)
<i>A-dep₃</i> (Lack of Drive)
<i>A-dep₄</i> (Suicidal Ideation)
<i>A-hea₁</i> (Gastrointestinal Complaints)
<i>A-hea₂</i> (Neurological Symptoms)
<i>A-hea₃</i> (General Health Concerns)
<i>A-ahn₁</i> (Misunderstood)
<i>A-ahn₂</i> (Social Isolation)
<i>A-ahn₃</i> (Interpersonal Skepticism)
<i>A-biz₁</i> (Psychotic Symptomatology)
<i>A-biz₂</i> (Paranoid Ideation)
<i>A-ang₁</i> (Explosive Behavior)
<i>A-ang₂</i> (Irritability)
<i>A-cyn₁</i> (Misanthropic Beliefs)
<i>A-cyn₂</i> (Interpersonal Suspiciousness)
<i>A-con₁</i> (Acting-Out Behaviors)
<i>A-con₂</i> (Antisocial Attitudes)
<i>A-con₃</i> (Negative Peer Group Influences)
<i>A-lse₁</i> (Self-Doubt)
<i>A-lse₂</i> (Interpersonal Submissiveness)
<i>A-las₁</i> (Low Achievement Orientation)
<i>A-las₂</i> (Lack of Initiative)
<i>A-sod₁</i> (Introversion)
<i>A-sod₂</i> (Shyness)
<i>A-fam₁</i> (Familial Discord)
<i>A-fam₂</i> (Familial Alienation)
<i>A-sch₁</i> (School Conduct Problems)
<i>A-sch₂</i> (Negative Attitudes)
<i>A-trt₁</i> (Low Motivation)
<i>A-trt₂</i> (Inability to Disclose)
<i>Personality Psychopathology Five (PSY-5) scales</i>
<i>AGGR</i> (Aggressiveness)
<i>PSYC</i> (Psychoticism)
<i>DISC</i> (Disconstraint)
<i>NEGE</i> (Negative Emotionality/Neuroticism)
<i>INTR</i> (Introversion/Low Positive Emotionality)

histories. Their sample included 157 boys from a forensic setting and 197 girls from an acute psychiatric inpatient setting. Handel et al. (2011) also identified Clinical Scale correlates for both boys and girls in a large forensic sample ($n = 496$) using both caregiver ratings and other self-report data as criterion measures. In both of these studies, Clinical Scale correlate patterns were largely consistent with expectations.

Research pertaining to the MMPI-A content and supplementary scales, including scales such as *ACK*, *PRO*, and *IMM*, has also begun to accumulate. For example, Archer and Krishnamurthy (1997) reported a hit rate of .68 using either the MMPI-A scale 2 or the *A-dep* scale for identifying adolescents given depression diagnoses. For conduct disorder, scales *A-con* and *A-cyn* yielded hit rates of .61 and .66, respectively, using the *T*-score cutoff of $T = 65$. Further, Imhof and Archer (1997) evaluated the concurrent validity of the *IMM* scale for 66 adolescents aged 13–18 in a

residential treatment setting. They reported that elevated *IMM* scales were related to greater impulsivity and lower levels of intelligence. Similar *IMM* scale correlates have been reported by Milne and Greenway (1999) and by Zinn, McCumber, and Dahlstrom (1999). Both Veltri et al. (2009) and Handel et al. (2011) provided a wide range of correlate data for the content and supplementary scales. Here too, correlate patterns were largely consistent with expectations for these scale sets.

Another area of MMPI-A research interest has been the issue of critical items. Archer and Jacobson (1993) examined the endorsement frequency of the Koss-Butcher (1973) and the Lachar-Wrobel (1979) critical items for normative and clinical samples for the MMPI-A and the MMPI-2. They found that adolescents in both the normative and the clinical samples endorsed items at a higher frequency than did normal adults. Further, results indicated that adolescents in clinical settings did not generally endorse critical items with a higher frequency than normal adolescents. The authors observed that these findings underscored the difficulties inherent in creating a critical item list for adolescents by selecting items with differences in endorsement frequency between normal and clinical settings. Forbey and Ben-Porath (1998) developed a set of MMPI-A critical items by comparing the item-level responses of the MMPI-A normative sample with 419 adolescents receiving treatment in a Midwestern residential treatment facility. The result of this effort was a critical item set composed of 82 items grouped into 15 content areas, including aggression, conduct problems, and depression-suicidal ideation. The critical item set developed by Forbey and Ben-Porath was officially added to the MMPI-A in 2006.

The *K*-correction procedure used with the original MMPI was not carried over to the MMPI-A. The original MMPI *K* scale included 30 items that were empirically selected by contrasting the item endorsements of clearly disturbed psychiatric patients who produced normal-range scores on the clinical scales with the item responses of the normative group. The main function of the *K* scale was to improve the ability of the clinical scales to detect psychopathology by adding varying proportions of *K*-scale raw score values to the scales *Hs*, *Pd*, *Pt*, *Sc*, and *Ma*. Although *K* correction was carried over to the MMPI-2 (Butcher et al., 2001), the procedure was not retained for the MMPI-A. Alperin, Archer, and Coates (1996) derived experimental *K*-weights for the MMPI-A to determine whether they could improve test accuracy in adolescent samples. Empirically determined *K*-weights were systematically added to the raw score values from the eight basic scales (excluding *Mf* and *Si*) and used to predict membership in the MMPI-A normative sample versus membership in a clinical sample of 122 adolescents receiving inpatient treatment. The results indicated that the use of the *K*-correction procedure did not lead to any systematic improvement in test accuracy in this classification task. Alperin and her colleagues concluded that their findings failed to provide support for the use of the *K*-correction procedure with the MMPI-A.

In addition to studies of the *K*-correction procedure, a substantial amount of research has been devoted to the effectiveness of the MMPI-A validity scales. A major advantage of using the MMPI-A in forensic settings is certainly its array of Validity Scales capable of accurately identifying problems in test-taking attitudes or response patterns. Profile validity is a particularly important issue for MMPI-A evaluations of adolescents in a forensic setting. In this setting, response biases involving both under-reporting and over-reporting frequently occur, depending on the perceived advantages associated with these approaches. In addition, adolescents from lower socioeconomic backgrounds or with various forms of conduct disorder may produce invalid results, owing to reading deficits and/or carelessness. It is therefore critical for psychologists to be able to use MMPI-A validity scales in order to detect response distortions or biases. Preliminary evaluations of the *VRIN* and *TRIN* validity scales, reported in the test manual (Butcher et al., 1992), indicated that these scales were useful in detecting inconsistent responding (*VRIN*) as well as an acquiescent or nay-saying response style (*TRIN*). The test manual also noted that the *T*-score difference between *F1* and *F2* could prove useful in identifying changes in an adolescent's test-taking approach between the first and

second half of the test. Archer (2005) presented a random response pattern on the MMPI-A, with the resulting profile showing elevations on scales *L*, *VRIN*, *F1*, and *F2*, an invalid profile easily detected by most interpreters. Baer, Ballenger, Berry, and Wetter (1997) evaluated varying degrees of random responding on the MMPI-A. Their results demonstrated a pattern of increasing scores on *F1*, *F2*, *F*, and *VRIN* as profile randomness increased. Archer and Elkins (1999) also found that scores on the Validity Scales *F*, *F1*, and *F2* were effective in differentiating profiles of a clinical sample of 354 adolescents from randomly generated protocols. To examine the *TRIN* scale, Handel, Arnau, Archer, and Dandy (2006) randomly inserted varying degrees of true responses into existing MMPI-A profiles and then randomly inserted false responses. Handel et al. demonstrated that *TRIN* *T*-scores increase with increasing degrees of fixed responding.

Baer, Ballenger, and Kroll (1998) compared community and clinical adolescent samples instructed to portray themselves as having superior psychological adjustment (i.e., the instructions were designed to produce under-reporting of psychopathology). The results included significantly higher scores on the MMPI-A defensiveness scales *L* and *K*, and these scales provided accurate discrimination between under-reported and standard profiles. Baer, Kroll, Rinaldo, and Ballenger (1999) further demonstrated that over-reporting on the MMPI-A was effectively detected using an *F* scale *T*-score of 80 as the optimal cutoff value. In addition, the *F* scale was sensitive to both random responding and over-reporting of symptoms, whereas *VRIN*-scale results were effective only for the identification of random responding. Finally, an Infrequency Psychopathology Scale (*Fp-A*) for the MMPI-A was constructed by McGrath et al. (2000) that was very similar in purpose to the *Fp* scale developed for the MMPI-2 by Arbisi and Ben-Porath (1995) by selecting items rarely endorsed by normative adults or by VA psychiatric patients. Using a similar approach, McGrath and his colleagues developed the *Fp-A* scale by selecting items that were rarely endorsed by normal adolescents or by two inpatient adolescent clinical samples. The initial results obtained by McGrath et al., however, showed relatively little gain from using the *Fp-A* scale rather than relying on the MMPI-A *F* scale.

Fabry, Bertinetti, and Guzman-Cavazos (2011) reported frequencies of MMPI-A Validity Scale elevations in a sample of 1,054 adolescent offenders. Using the Validity Scale criteria outlined in Archer and Krishnamurthy (2002), Fabry et al. found that 37% of adolescents elevated at least one Validity Scale. *L* and *K* were the most commonly elevated Validity Scales in their sample.

Finally, a substantive group of research studies have centered on the factor structure of the MMPI-A. Archer, Belevich, and Elkins (1994) conducted a factor analysis using the MMPI-A normative sample. The results indicated that eight factors accounted for a majority of raw score variance produced by the 69 scales and subscales of the MMPI-A. The factor structure identified by these authors included the following dimensions: General Maladjustment, Immaturity, Disinhibition-Excitatory Potential, Social Discomfort, Health Concerns, Naivete, Familial Alienation, and Psychoticism. These factor analysis results were used by Archer and Krishnamurthy (1994a; 1994b) to construct the MMPI-A Structural Summary. The Structural Summary is a means of effectively focusing a clinician's attention on the dimensions of greatest importance for understanding an adolescent's performance on the MMPI-A. Archer and Krishnamurthy (1994b) recommended that more than half of the scales and subscales associated with each factor should reach critical values before an interpreter emphasizes a given dimension in describing an adolescent's functioning. Additionally, these authors recommended that when more than one factor structure was suggested, the factor with the highest percentage of scales or subscales producing clinical values should receive greater weight in the interpretation. Archer, Krishnamurthy, and Jacobson (1994) provided factor descriptions for all of the external correlates that have been identified for the Structural Summary factors, and these factor correlates are also found in Archer (2005). Figure 4.1 presents the MMPI-A Structural Summary form developed by Archer and Krishnamurthy (1994).

MMPI-A Structural Summary

Robert P. Archer and Radhika Krishnamurthy

Name: _____ Date: _____
 Age: _____ Grade: _____
 Gender: _____ School: _____

Test-Taking Attitudes

1. Omissions (raw score total)

_____ ? (Cannot Say scale)

2. Consistency (T-score values)

_____ VRIN

_____ TRIN

_____ F₁ vs. _____ F₂

3. Accuracy (check if condition present)

Overreport

_____ F scale T score ≥ 90

_____ All clinical scales except 5 and 0 ≥ 60

Underreport

_____ High L (T ≥ 65)

_____ High K (T ≥ 65)

_____ All clinical scales except 5 and 0 < 60

Factor Groupings (enter T-score data)

1. General Maladjustment

- _____ Welsh's A
- _____ Scale 7
- _____ Scale 8
- _____ Scale 2
- _____ Scale 4
- _____ D₁ (Subjective Depression)
- _____ D₄ (Mental Dullness)
- _____ D₅ (Brooding)
- _____ Hy₃ (Lassitude-Malaise)
- _____ Sc₁ (Social Alienation)
- _____ Sc₂ (Emotional Alienation)
- _____ Sc₃ (Lack of Ego Mastery - Cognitive)
- _____ Sc₄ (Lack of Ego Mastery - Conative)
- _____ Si₃ (Alienation)
- _____ Pd₄ (Social Alienation)
- _____ Pd₅ (Self-Alienation)
- _____ Pa₂ (Polgnancy)
- _____ A-dep
- _____ A-anx
- _____ A-lse
- _____ A-aln
- _____ A-obs
- _____ A-trt

_____/23 Number of scales with T ≥ 60

2. Immaturity

- _____ IMM
- _____ Scale F
- _____ Scale 8
- _____ Scale 6
- _____ ACK
- _____ MAC-R
- _____ Pa₁ (Persecutory Ideas)
- _____ Sc₂ (Emotional Alienation)
- _____ Sc₆ (Bizarre Sensory Experiences)
- _____ A-sch
- _____ A-biz
- _____ A-aln
- _____ A-con
- _____ A-fam
- _____ A-trt

_____/15 Number of scales with T ≥ 60

FIGURE 4.1 MMPI-A Structural Summary Form.

Reproduced by special permission of the Publisher, Psychological Assessment Resources, Inc., 16204 North Florida Avenue, Lutz, FL 33549. From the *MMPI-A Structural Summary* by Robert P. Archer and Radhika Krishnamurthy. Copyright © 1994 by Psychological Assessment Resources, Inc. Further reproduction is prohibited from PAR, Inc.

3. Disinhibition/Excitatory Potential

- ☐ Scale 9
☐ Ma₂ (Psychomotor Acceleration)
☐ Ma₄ (Ego Inflation)
☐ Sc₅ (Lack of Ego Mastery, Defective Inhibition)
☐ D₂ (Psychomotor Retardation) (low score)*
☐ Welsh's R (low score)*
☐ Scale K (low score)*
☐ Scale L (low score)*
☐ A-ang
☐ A-cyn
☐ A-con
☐ MAC-R

☐ /12 Number of scales with $T \geq 60$ or ≤ 40 for scales with asterisk

5. Health Concerns

- ☐ Scale 1
☐ Scale 3
☐ A-hea
☐ Hy₄ (Somatic Complaints)
☐ Hy₃ (Lassitude-Malaise)
☐ D₃ (Physical Malfunctioning)

☐ /6 Number of scales with $T \geq 60$

7. Familial Alienation

- ☐ Pd₁ (Familial Discord)
☐ A-fam
☐ Scale 4
☐ PRO

☐ /4 Number of scales with $T \geq 60$

4. Social Discomfort

- ☐ Scale 0
☐ Si₁ (Shyness/Self-Consciousness)
☐ Hy₁ (Denial of Social Anxiety) (low score)*
☐ Pd₃ (Social Imperturbability) (low score)*
☐ Ma₃ (Imperturbability) (low scores)*
☐ A-sod
☐ A-lse
☐ Scale 7

☐ /8 Number of scales with $T \geq 60$ or $T \leq 40$ for scales with asterisk

6. Naivete

- ☐ A-cyn (low score)*
☐ Pa₃ (Naivete)
☐ Hy₂ (Need for Affection)
☐ Si₃ (Alienation-Self and Others) (low score)
☐ Scale K

☐ /5 Number of scales with $T \geq 60$ or $T \leq 40$ for scales with asterisk

8. Psychoticism

- ☐ Pa₁ (Persecutory Ideas)
☐ Scale 6
☐ A-biz
☐ Sc₆ (Bizarre Sensory Experiences)

☐ /4 Number of scales with $T \geq 60$

Note. The presentation of scales under each factor label is generally organized in a descending order from the best to the least effective marker. Within this overall approach, scales are grouped logically in terms of basic clinical scales, Harris-Lingoes and Si subscales, and content scale. The majority of scales included in this summary sheet were correlated $\geq .60$ or $\leq -.60$ with the relevant factor for the MMPI-A normative sample.

PAR Psychological Assessment Resources, Inc.
 P.O. Box 998/Odessa, Florida 33556/Toll-Free 1-800-331-TEST

Copyright © 1994 by Psychological Assessment Resources, Inc. All rights reserved. May not be reproduced in whole or in part in any form or by any means without written permission of Psychological Assessment Resources, Inc. "MMPI-A" and "Minnesota Multiphasic Personality Inventory-Adolescent" are trademarks owned by the University of Minnesota.

9 8 7 6 5 4 3 2 1

Reorder #RO-2698

Printed in the U.S.A.

This form is printed in blue ink on white paper. Any other version is unauthorized.

Archer and Krishnamurthy (1997) replicated the structural summary in a clinical sample of adolescents, and the same authors provided an extensive array of empirical correlates for each factor dimension (1994b). This work was extended to juvenile forensic populations by Archer, Bolinsky, Morton, and Farris (2003). Krishnamurthy, Bolinsky, and Archer (2005) showed the viability of the eight-factor Structural Summary model when the 69 standard MMPI-A scales and subscales are factored in combination with new MMPI-A scales, including the Content Component scales and the Personality Psychopathology Five (PSY-5) scales developed by McNulty et al. (1997).

Clinical Uses and Limitations

The MMPI-A was developed to evaluate adolescent psychopathology in a variety of settings, particularly outpatient and inpatient psychiatric and alcohol and drug treatment settings, schools, and medical clinics and hospitals. The MMPI-A is typically administered at the beginning of treatment in order to assess an adolescent's ability to actively participate and engage in the therapeutic process and to provide a baseline of psychological functioning. The information gathered from the MMPI-A is routinely integrated with data from other tests, such as reports and/or ratings provided by teachers, parents, and treatment team members, as well as the adolescent's treatment history and any interview findings, to allow a comprehensive overview of the adolescent's psychological functioning.

The MMPI-A serves as an important indicator of therapeutic progress when administered on a follow-up basis to evaluate treatment effectiveness. It is particularly useful in this regard because the test is capable of reflecting changes in psychological functioning over relatively short periods of time. Archer (2005) observed that most adolescents are not resistant to the readministration of the MMPI-A to evaluate treatment progress if they receive feedback on the test and thus have an investment in the testing outcome. Archer and Krishnamurthy (2002) noted that adolescents are capable of receiving and utilizing a great deal of feedback information from the MMPI-A if the examiner avoids technical jargon and articulates the feedback in a manner that provides information to them concerning issues they perceive as important.

Like all psychometric instruments, the MMPI-A also has several limitations. It requires a substantial amount of cognitive maturation and reading ability for successful administration, and these factors are likely to be particularly important in the assessment of adolescents in the juvenile justice system, where problems in reading comprehension are frequently encountered. Further, adolescents completing the MMPI-A must have the capacity and motivation to complete a relatively long and demanding test instrument. Although short forms have been developed for the MMPI-A, these are not recommended for the evaluation of adolescents in forensic settings. Archer, Tirrell, and Elkins (2001), for example, developed an MMPI-A short form based on the administration of the first 150 items of the test, and it illustrates the limitations of all short-form approaches. It is useful for detecting the presence of psychopathology, but the profiles it produces often differ significantly from those produced by full-length test administrations.

Another area of potential limitation is the tendency of the MMPI-A to produce relatively low basic-scale elevations, even for adolescents who display significant psychiatric symptoms. This problem was originally noted by Archer (1987) in reference to the Marks and Briggs (1972) norms for the original MMPI, and it continues to occur with adolescents who complete the MMPI-A (e.g., Archer, 2005; Archer & Krishnamurthy, 2002). Alperin and her colleagues (1996) evaluated the relative efficacy of applying an MMPI-A *T*-score value of 60 or above (as opposed to the standard value of 65) to define "clinical levels" of symptomatology. Lowering the clinical criterion *T*-score in this study reduced rather than increased the overall classification accuracy. These results

are consistent with guidance provided in the MMPI-A manual (Butcher et al., 1992), specifically the statement that “a clinically significant elevation is defined as an MMPI-A T -score = 65” (p. 43). Fontaine, Archer, Elkins, and Johansen (2001) replicated and expanded the work of Alperin et al. by comparing the accuracy of the MMPI-A in identifying normal and clinical adolescents utilizing the $T = 60$ and $T = 65$ criteria at differing base rates for the occurrence of psychopathology. Once again, the $T = 65$ criterion resulted in a higher level of accuracy, minimizing misclassification of both normal and clinical cases. Most recently, Archer, Handel, and Lynch (2001) showed that the MMPI-A item endorsement frequencies for normal and clinical adolescent samples often exhibit no significant differences, in contrast to the MMPI-2, which typically demonstrates a high rate of effectiveness in separating normal from clinical samples. Archer et al. concluded that a markedly high rate of MMPI-A item endorsement in the normative sample (rather than unusually high rates of item endorsement associated with adolescents in clinical samples) was most likely responsible for this phenomenon. Overall, these research studies indicate that there is a high rate of within-normal-limits (W-N-L) MMPI-A profiles among adolescents in treatment settings, and findings by Archer et al. (2003) indicate that the high frequency of W-N-L profiles is also found among adolescents evaluated in juvenile justice or forensic settings.

The MMPI-A, like the MMPI-2, is best viewed as an instrument that provides a description of an individual's psychological functioning at a “moment in time” (Archer & Baker, 2005). MMPI-A profiles have a very limited capacity for making long-term behavioral or diagnostic predictions regarding adult functioning (Gottesman & Hanson, 1990; Lowman, Galinsky, & Gray-Little, 1980). This limitation is not unique to the MMPI-A but is possessed by all psychological instruments because of the behavioral and affective instability characteristic of adolescent development (Archer, 2005). This fluidity in personality structure is a major reason why the *Diagnostic and Statistic Manual of Mental Disorders, Fourth Edition, Test Revision (DSM-IV-TR*; American Psychiatric Association, 2000) strongly encourages clinicians to avoid the application of personality disorder diagnoses to adolescents. Further, the manual explicitly prohibits the use of the Antisocial Personality Disorder diagnosis with individuals under the age of 18.

MMPI-A-RF

At the time of the revision of this chapter, collaborative work was being done under the sponsorship of the University of Minnesota Press to develop the Minnesota Multiphasic Personality Inventory-Adolescent-Restructured Form (MMPI-A-RF). The MMPI-A-RF was undertaken to address some of the disadvantages associated with the MMPI-A, including its 478-item length, the extensive degree of content heterogeneity found within the clinical scales, and the extensive degree of item overlap that existed between scales (Archer, Handel, Tellegen, and Ben-Porath, 2011). The development committee consists of the authors of this chapter from the Eastern Virginia Medical School, Dr. Auke Tellegen from the University of Minnesota, and Dr. Yossef Ben-Porath from Kent State University. This committee used the MMPI-2-RF as a general template for the development of the MMPI-A-RF; however, emphasis was also placed on modifying and adapting the MMPI-A-RF scales to deal specifically with issues of adolescent development and psychopathology. All of the items selected for the MMPI-A-RF were taken from the item pool of the MMPI-A, and the MMPI-A normative sample will serve as the general basis for the MMPI-A-RF normative sample. At this point, there is also mutual agreement that the optimal length of the MMPI-A-RF will be targeted at fewer than 250 items in the overall test booklet, including MMPI-A-RF Restructured Clinical (RC) and PSY-5 Scales, Higher-Order and Specific Problem Scales, and Validity Scales. It is anticipated that the MMPI-A-RF will be validated in clinical samples that are separate from the samples used for scale development, and the

MMPI–A–RF will be released with an accompanying manual for administration, scoring, and interpretation, as well as software services supporting each of these functions.

Forensic Uses and Limitations

Common Uses

Hathaway and McKinley (1943) did not develop the MMPI with the intent of utilizing this instrument in forensic evaluations, and the MMPI–A manual (Butcher et al., 1992) provides little guidance on the application of this instrument in forensic settings. Therefore, the use of the MMPI–A in forensic settings represents an extension of this instrument in which test findings must be “translated” to address questions beyond the original scope of the instrument. There are a variety of referral questions that the MMPI–A can be useful for answering, as the clinical constructs assessed by this instrument are often relevant to forensic issues or to individuals being treated in forensic settings. O’Connor Pennuto and Archer (2008), for example, noted eight documented legal cases that included the use of the MMPI–A in issues related to competency to stand trial and insanity evaluations, sexual abuse and sexual predator assessments, and assessment of psychological functioning for purposes of sentencing placement, mitigation, and transfers of adolescents to adult status for trial and sentencing. Archer (2005) also noted that the MMPI–A has been used in custody evaluations with adolescents and in personal injury and disability assessments. However, caution should always be used in making the transition from clinical conclusions to inferences regarding forensic issues. As the area of forensic psychology has rapidly expanded, the opportunities for the use and misuse of traditional clinical assessment instruments in forensic settings have also dramatically increased.

Grisso (1998) noted that juvenile courts, prosecutors, and defense attorneys may request a clinical evaluation of an adolescent at a number of points in the juvenile justice system. The most frequent requests for an evaluation are to assist the court in dispositional decisions prior to adjudication and/or to assist in identifying appropriate sentencing after adjudication. Psychologists may also be asked to evaluate the emergency medical needs of an adolescent who is being held in detention and exhibiting severely disorganized or depressed (and potentially suicidal) behaviors. Forensic evaluations may also be requested at the pre-trial stage to determine whether the adolescent should be retained in a secure facility or could be released back to the community to await trial.

Psychological evaluations may also be requested in preparation for hearings on the waiver of adolescents from juvenile courts to the adult criminal justice system. Competency to stand trial is another type of pretrial issue for which evaluations may be requested by the court. These evaluations typically focus on the juvenile’s awareness and understanding of the charges, knowledge of the legal system, and ability to meaningfully assist an attorney with defense preparations. Psychologists may also be asked to perform evaluations of a juvenile’s competence to waive Miranda rights “knowingly, intelligently, and voluntarily.” Finally, psychologists may be asked to evaluate an adolescent’s mental state at the time of the offense, although this latter issue rarely arises. Grisso (1998), for example, noted that many clinicians working in the juvenile justice system have never been asked to evaluate an adolescent’s sanity at the time of the offense.

While the most frequent forensic use of the MMPI–A is within the juvenile justice system, another important application is the assessment, in personal injury cases, of adolescents who may have suffered emotional damage. The legal framework for personal injury litigation differs substantially from the criminal justice system, in that the former is largely defined by the accumulation of relevant case law whereas criminal cases are typically tried on state or federal statutes. The basis

of personal injury litigation is tort, defined by Greenberg, Otto, and Long (2003) as “a civil wrong that gives rise to a remedy in the form of a claim for compensation that is commenced with the filing of a complaint or petition” (p. 412). The usefulness of the MMPI-A for evaluating adolescents in civil cases rests on its ability to provide an overview of the plaintiff’s psychological or emotional functioning in reference to well-established adolescent norms. Additionally, the MMPI-A provides an important opportunity to evaluate an adolescent’s response style, such as a tendency to over-report symptomatology, a dimension particularly relevant to examinations in personal injury cases.

Other uses of the MMPI-A in forensic settings include the evaluation of adolescents who have been sexually abused. The literature on the use of the MMPI and MMPI-A with this population was briefly reviewed by Archer and Krishnamurthy (2002). As noted by these authors, the MMPI-A profiles of sexually abused adolescents typically contain multiple basic scale elevations reflective of general emotional distress. In addition, these adolescents are likely to produce elevations on scales reflecting alienation, interpersonal sensitivity, and social withdrawal. Hillary and Schare (1993), for instance, found that sexually abused adolescents typically produced elevations on scales measuring depression (*D*) and anxiety (*Pt*), with elevations on scales *Sc* and *Ma* occurring frequently as well. Scott and Stone (1986) compared the MMPI profiles of adolescents and adults who had been molested by their fathers and/or stepfathers and found that both groups produced elevations on *Sc*, with adolescents also producing *Ma*-scale elevations. Forbey and Ben-Porath (2000) examined the MMPI-A profiles of adolescents receiving residential treatment and reported that sexual abuse histories were related to higher scores on basic scales *D*, *Pd*, *Si*, and *Pa* as well as Content Scales *A-dep*, *A-lse*, *A-sod*, *A-fam*, and *A-ang*. Williams et al. (1992) found a similar Content Scale pattern for adolescents with histories of sexual abuse, with sexually abused boys particularly likely to produce elevations on scales *A-dep*, *A-ang*, *A-lse*, and *A-sch*. Williams and her colleagues also found that sexually abused boys most commonly had higher scores on basic Clinical Scales *Pt* and *Sc* and that sexually abused girls were more likely to produce elevations on scales *Pd* and *Sc*. Archer and Krishnamurthy (2002) concluded that sexually abused adolescents are likely to produce multiple elevations on the MMPI-A scales, and no single profile can be used to identify sexually abused adolescents effectively.

The MMPI-A is also frequently used in child custody evaluations. Most domestic or family relations courts render decisions concerning child custody based on a generally accepted standard involving “the best interests of the child.” The MMPI-A can be used to evaluate the emotional and psychological needs of adolescents involved in child custody proceedings and to provide information or offer recommendations to the court regarding optimal custodial placements. Otto and Collins (1995) noted that the MMPI-A provides a well-researched basis for describing the emotional needs of adolescents, although they caution that little is currently known about how to use the MMPI-2 characteristics of parents and the MMPI-A characteristics of adolescents in combination to predict specific outcomes of child placement decisions.

Archer, Buffington-Vollum, Stredny, and Handel (2006) surveyed 152 forensic psychologists and reported that the MMPI-A was the most widely used measure in child-related forensic evaluations. In comparison, parental rating measures of child behavior or functioning ranked second and third, and the Millon Adolescent Clinical Inventory (Millon, 1993) was ranked fourth.

Research on Forensic Uses of the MMPI and MMPI-A

The history of the research on the MMPI and MMPI-A in juvenile offender populations spans six decades. As previously noted, Dora Capwell (1945a; 1945b) examined the effectiveness of scale *Pd* in accurately identifying delinquent and non-delinquent girls. Hathaway and Monachesi (1952;

1953; 1961; 1963) launched a landmark study of the ability of the MMPI to predict delinquent and antisocial behavior. They administered the MMPI to 3,971 ninth graders during the 1947–1948 school year and combined these data with the results of a statewide sample collected in 1954 on an additional 3,329 ninth graders in 86 Minnesota communities. Their researchers also collected data from school records, teacher ratings, and follow-up data from public agencies, including police and court reports. Hathaway and Monachesi undertook collection of this massive data set in order to implement a longitudinal study that would potentially identify personality variables related to the later onset of delinquent behaviors. Extensive follow-up research based on the data originally collected by Hathaway and Monachesi has provided mixed support for their proposal that elevations on scales *Pd*, *Sc*, and *Ma* tend to increase the probability of adolescent delinquency. Some early studies did appear to demonstrate the usefulness of the MMPI for identifying adolescent predisposal to delinquency (e.g., Briggs, Wirt, & Johnson, 1961; Huesmann, Lefkowitz, & Eron, 1978; Wirt & Briggs, 1959). More recent studies, however, have produced mixed results on the usefulness of the excitatory scales in predicting delinquent behavior (e.g., Archer et al., 2003).

The MMPI–A does appear to hold promise as a means of effectively identifying and describing juvenile delinquents, however. As noted earlier, Handel et al. (2011) published the results of a long-term project aimed at evaluating the utility of the MMPI–A in a forensic setting. These investigators presented extensive data (e.g., mean scale scores, intercorrelations, alpha coefficients, and external correlates) using a large sample ($n = 496$) of boys and girls who were court-ordered to receive psychological evaluations for the purpose of assisting in dispositional planning. In addition to use of the MMPI–A, adolescents were rated by caregivers on the Child Behavior Checklist (Achenbach & Rescorla, 2001) and the Disruptive Behavior Rating Scale–Parent Form (DBRS; Barkley & Murphy, 1998). Further, adolescents also completed the Youth Self-Report (Achenbach & Rescorla, 2001). Using a *T*-score cutoff of 65, *Pd* was the most frequently elevated scale for both boys and girls. Approximately one-third of the sample produced an elevated score on *Pd*. For the content scales, *A-sch* was the most frequently elevated scale for boys and *A-fam* and *A-sch* were the most frequently elevated scales for girls. *MAC-R* and *PRO* were the most frequently elevated Supplementary Scales for both boys and girls. Handel et al. provided extensive correlate data for the Clinical, Content, and Supplementary Scales that largely supports the validity of scores on MMPI–A scales in this population. Veltri et al. (2009) also presented MMPI–A correlate data using a sample of 157 boys court-ordered to receive a psychological evaluation. Here too, Veltri et al.’s results generally supported the convergent and discriminant validity of MMPI–A scale scores in assessing internalizing and externalizing psychopathology. Earlier, Cashel, Rogers, Sewell, and Holliman (1998) examined MMPI–A scale correlates for 99 male delinquents held in a Texas juvenile correction facility. These adolescents were also evaluated with the Schedule of Affective Disorders and Schizophrenia with School Age Children (Angold, 1989). Hicks, Rogers, and Cashel (2000) evaluated the usefulness of the screening version of the Psychopathology Checklist (PCL–SV) and the MMPI–A for predicting violent, non-violent, self-injurious, and total infractions in male adolescent offenders. Among the findings of this study, the MMPI–A proved to be more effective than the PCL–SV in predicting the total number of infractions during incarceration. The combination of scores from the MMPI–A *Pa* and *Ma* basic scales, for example, resulted in the accurate classification of 60% of residents with violent infractions and 66% of residents without violent infractions. Hicks and his colleagues suggested that clinicians routinely evaluate *Pa*-scale elevations in assessing the risk of violent infractions among adolescent offenders. Interestingly, *Pd*-scale elevations in this study were not correlated with scores from the PCL–SV, the number of infractions, or the presence of conduct disorder symptoms. Thus, Hicks et al. also provided limited support for the predictive utility of *Ma* as one of Hathaway and Monachesi’s excitatory scales but no support for the predictive utility of excitatory scales *Pd* and *Sc*.

A number of other researchers have also examined the MMPI-A characteristics of male juvenile offenders. Hume, Kennedy, Patrick, and Partyka (1996) evaluated the relationship between the MMPI-A and the Psychopathy Checklist-Revised (PCL-R) by Hare (1991) in a sample of 101 male juvenile delinquents in a state correctional facility. The authors reported that scores from these two instruments were essentially independent and unrelated. Pena, Megargee, and Brody (1996) compared the MMPI-A profiles of 162 incarcerated delinquent boys with several groups, including 805 non-delinquent male adolescents from the MMPI-A normative sample. The results indicated that the profiles of delinquents often showed elevations on *Pa*, *Pd*, and *Ma* and that the most frequent two-point codetype classification was the 4-9/9-4. Pena et al. concluded that their findings supported the construct validity of the MMPI-A in forensic settings and also noted it was possible to evaluate 94% of the boys in the sample by using an oral administration of the test instrument. Glaser, Calhoun, and Petrocelli (2002) also evaluated the ability of the MMPI-A to classify 72 male juvenile offenders according to whether they committed crimes against person, crimes against property, or alcohol- or drug-related offenses. The instrument correctly classified 79.2% of these adolescents. Adolescents who scored higher on scales *Hs* and *Si*, for example, were less likely to develop alcohol and drug problems and more likely to engage in property crimes. In contrast, adolescents who produced lower scores on the basic scale *Ma* and higher scores on the *A-sch* Content Scale were more likely to have committed drug-related offenses. Peterson and Robbins (2008) evaluated the ability of scores on the *Pd*, *A-con*, and *A-cyn* scales to predict recidivism in a sample of 107 adjudicated boys. These investigators concluded that the *A-con* scale was the strongest predictor of recidivism.

Morton, Farris, and Brenowitz (2002) examined the ability of the MMPI-A to discriminate accurately between 855 male delinquents in a South Carolina correctional facility and 805 males from the MMPI-A normative sample. The single most effective scale in accurately discriminating normal from delinquent adolescents was MMPI-A basic scale *Mf*, with more masculine scores characteristic of the delinquent sample. In addition, elevations on MMPI-A basic scales *Pd* and *Pa* were also found to be more frequent among male delinquents. These latter findings are consistent with the findings of Pena et al. (1996), but the relative importance of scale *Mf* found by Morton et al. has not yet been confirmed by other researchers. The results of these MMPI-A studies indicate that conduct disorder (*Pd* scale), identification with a stereotypically male gender role (*Mf* scale), and interpersonal suspiciousness, distrust, and sensitivity to injury or slight (*Pa* scale) are characteristics that distinguish male juvenile delinquents from their non-delinquent adolescent counterparts.

Archer et al. (2003) followed up the Morton et al. (2002) study to further evaluate the extent to which MMPI-A profiles from 196 male juveniles in detention facilities could be successfully discriminated from the MMPI-A protocols produced by 200 adolescent male psychiatric patients and 151 dually diagnosed adolescent males receiving inpatient treatment for co-morbid substance abuse and psychiatric disorders. The findings included significant differences in mean *T*-score elevations between these three groups of adolescents across a variety of MMPI-A scales and subscales. The profiles of delinquent boys were characterized by greater psychological immaturity, as seen on the *IMM* scale, and by superficial efforts to appear emotionally controlled and well adjusted, as seen on the *Hy* scale. Although the MMPI-A profiles of adolescents in detention, psychiatric, and dual diagnosis facilities showed some distinctive features, Archer and his colleagues also noted that these adolescents shared many common MMPI-A profile features across settings.

Katz and Marquette (1996) attempted to identify personality features exhibited on the MMPI-A for a group of 29 males who were convicted of first- or second-degree murder. This group was contrasted with a cohort of non-violent juvenile offenders as well as a group of normal high school students. Contrary to the researchers' expectations, the MMPI-A profiles for the murderers in this

study demonstrated little evidence of serious psychopathology. Further, the MMPI-A profiles of the murderers did not differ significantly from the profiles of their non-violent counterparts or from the normal high school student group. These authors concluded that adolescent males who commit violent crimes may do so for reasons that extend beyond traditionally defined psychopathology, including such factors as involvement with violent gangs. Kaser-Boyd (2002), in studying children who had committed murder, observed that no single personality style or dynamic appears to be common to juveniles who commit homicide. The histories of these children, however, often include exposure to family violence and abuse.

Strengths of the MMPI-A in Forensic Settings

Several factors contribute to the value of the MMPI-A as a potential component of a comprehensive forensic evaluation. It is the most widely researched and validated objective personality assessment measurement for adolescents (Archer, 2005; Forbey, 2003) and is more widely used with adolescents than any other objective test (Archer & Newsom, 2000; Archer et al., 2006). The MMPI-A contains a number of validity scales, including measures of consistency (*VRIN* and *TRIN*), defensiveness (*L* and *K*), and over-reporting (*F*, *F1*, and *F2*). These validity scales provide detailed and reliable information concerning the test-taking attitudes of the adolescent. The MMPI-A clinical scales and subscales have extensive reliability and validity data, and test-retest studies have demonstrated that the MMPI-A is sensitive to changes in psychological functioning over time (Archer & Baker, 2005). The MMPI-A provides important information concerning an adolescent's emotional distress, drug and alcohol abuse problems, perception of family relationships, and overall degree of impulse control. The MMPI-A also appears to have usefulness in predicting violent or aggressive behaviors in detention or correctional settings, a feature particularly important in the juvenile justice system. A valuable area of future research in forensic psychology is the usefulness of the MMPI-A in assessing the potential risk of development of conduct problems, including aggression, among adolescents receiving treatment in psychiatric settings.

Given the widespread use of the MMPI-2 and MMPI-A among psychologists and other mental health professionals, it is not surprising that these instruments are well accepted as the basis of expert opinions in the courtroom. Pope, Butcher, and Seelen (2006), for example, noted that the MMPI-2 and MMPI-A have become widely used for the assessment of personality features in forensic settings because the validity scales of these instruments allow for the evaluation of the credibility of an individual's responses. These authors also underscored the ability of these tests to provide important descriptive data through well-established empirically based correlates. Finally, Pope and his colleagues suggested that the MMPI-2 and MMPI-A are extensively used in forensic settings because the results produced by these instruments are relatively easy to communicate to non-psychologists in such settings. The MMPI-A scoring procedure, scale structure, and interpretation process can be presented to laypeople in a manner that requires little psychological training on the part of the audience. Pope et al. recommended that experts utilizing MMPI-A or MMPI-2 test results in forensic settings be prepared to present data relevant to the admissibility of these instruments. Decisions concerning admissibility tend to focus on the relevance of test results to the issue before the trier of fact and on evidence that the MMPI-A has adequate reliability to serve as a component of the assessment performed by the forensic psychologist to address a specific standard or legal question. Pope and his colleagues concluded that MMPI-2 results should be admissible under most interpretations of the *Daubert* standard established by the U.S. Supreme Court in 1993 and as subsequently refined by the Court in *Kumho* (1999) and other rulings.

Limitations of the MMPI-A in Forensic Settings

Even though the MMPI-A has an extensive number of advantages when used in forensic settings, there are also a number of limitations. First, the MMPI-A (as well as the original form of the MMPI) was developed as a measure of psychopathology rather than to address forensic issues. Therefore, the examiner must be aware that the forensic use of the MMPI-A represents a substantive extension of the instrument, and the applicability of MMPI-A findings to specific forensic issues will vary widely depending on the nature of the legal issue or standard. In addition, the correlates developed for the MMPI-A scales have predominantly been based on clinical populations rather than on the behaviors, attitudes, and beliefs of adolescents in the juvenile justice system or involved in civil litigation (e.g., a personal injury suit). Therefore, the MMPI-A is more valuable for describing an adolescent's psychological functioning, for example, than for predicting his or her behavior in detention or correctional facilities. Second, as previously noted, it is important to recognize that the MMPI-A is best viewed as an assessment instrument that can only provide a description of an adolescent's current psychological functioning. Although there is some research evidence that certain psychological features tend to be more stable than others in the description of both adolescents and adults, the MMPI-A remains quite limited in its ability to make long-term behavioral or diagnostic predictions. The turbulent and flexible nature of adolescent personality and the difficulty inherent in making long-term predictions based on adolescent characteristics served as an important basis of the U.S. Supreme Court decision in *Roper v. Simmons* (2005), in which the Court forbade the execution of juveniles.

A practical limitation of the MMPI-A involves the restrictions imposed by the reading requirement for this test instrument. The MMPI-A test manual (Butcher et al., 1992) reports the reading difficulty level for each item in the MMPI-A test booklet based on indices of reading ease, reading difficulty, and sentence structure. Results showed that most test items are at a fifth- to seventh-grade reading level. Archer and Krishnamurthy (2002), offering a conservative estimate, recommended a seventh-grade reading level for the successful administration of the MMPI-A, a standard which is not always met by adolescents in the criminal justice system. For some adolescents with problems in the area of reading disability, an audio-CD administration of the MMPI-A may render testing possible.

Forensic Utility of the MMPI-A with American Minorities

A controversial issue for almost every psychological assessment instrument concerns the appropriate use of the instrument with ethnic minorities. Fortunately, an extensive amount of research data regarding this issue is available for the MMPI, MMPI-2, and MMPI-A. Research on the effect of ethnic background on MMPI and MMPI-A profiles has produced mixed results (Archer, 2005). Some earlier investigations found that Black adults, for example, produced higher scores than White adults on scales *F*, *Sc*, and *Ma* (e.g., Gynther, 1972). Early research studies with adolescents yielded similar findings. For example, a study by Ball (1960) found that Black ninth graders scored higher than their White counterparts on scale *Hs* for boys and scales *F*, *Sc*, and *Si* for girls. Archer (1987) found minimal differences between the MMPI profiles of Black and White adolescents in a sample obtained from a middle-class public high school. Dahlstrom, Lachar, and Dahlstrom (1986) found that MMPI differences between ethnic groups were typically small and frequently non-significant when the groups were matched on demographic variables such as socioeconomic status and education. On the basis of an extensive review, Greene (2000) concluded that no significant or consistent patterns of ethnic differences were discernible on the MMPI when other demographic factors were controlled.

More recent MMPI-2 studies involving adult samples have generally failed to find meaningful differences in profile elevation between Blacks and Whites. Timbrook and Graham (1994) compared the responses of 75 Black men and 65 Black women with those of 725 White men and 724 White women from the MMPI-2 normative sample. They found that the mean differences between ethnic groups on the MMPI-2 basic scales were relatively small in magnitude for both genders, and these differences were further reduced when groups were matched for age, socioeconomic status, and income level. Similarly, Archer (2005) noted that there are few differences between Black and White adolescents in the MMPI-A normative sample for the basic scales. Archer and Krishnamurthy (2002) concluded that the current literature indicates that the standard norms for adult and adolescent forms of the MMPI may be used for assessing individuals from ethnic minority groups and that norms specifically developed for various ethnic groups are not required. Archer and Krishnamurthy offer several cautions, however, when the MMPI-A is used with ethnic minorities. First, because the bulk of MMPI research studies on the effects of ethnicity have been focused on the profiles of Blacks and Whites, relatively less is known about the response patterns of Hispanics, Asian Americans, Native Americans, and other ethnic minorities. One exception is a comparative study by Gómez, Johnson, Davis, and Velásquez (2000) that examined the MMPI-A scale elevations of Black and Mexican American youthful offenders. Gomez and his colleagues found few significant differences in MMPI-A response patterns related to ethnicity. Among individual MMPI-A scales, for example, they found only one scale, namely, the Repression (*R*) scale, on which Black adolescents scored significantly higher than Mexican American teenagers. In addition to normative differences between ethnic groups, we also know significantly less about the correlate patterns of some minority groups. Archer and Krishnamurthy (2002) reviewed evidence indicating that the MAC-R scale may produce higher rates of false classification with ethnic minorities than with White respondents (e.g., Gottesman & Prescott, 1989). Similarly, Wrobel and Lachar (1995) cautioned that MMPI Content Scale elevations produced by minorities, including Blacks, may be related to a different set of correlates than those typically found for Whites. Although the research by Wrobel and Lachar focused on the Wiggins Content Scales developed for the original MMPI, it is possible that these concerns may also be applicable to the MMPI-A Content Scales.

Administration and Interpretive Issues in Forensic Applications

Archer and Krishnamurthy (2002) suggested that examiners rule out the presence of detrimental factors such as intoxication, disorientation, or confusional states that might inhibit an adolescent's ability to respond to the MMPI-A reliably and accurately. It is also imperative that a quiet, comfortable test environment be provided (an admittedly difficult task in many detention and correctional settings) and that supervision is provided during the entirety of the testing session. It is permissible to provide breaks in the administration as needed to prevent excessive fatigue or distractibility, which may interfere with test validity. It is also important to review item omissions with the adolescent before leaving the testing setting and to have the examinee fill in missing information if he or she is capable of reading and understanding the items. Several available sources provide extensive information on appropriate administration and scoring procedures (e.g., Butcher et al., 1992; Archer, 2005).

Administrative guidelines and standards do not change as a function of administering the MMPI-A in a forensic setting, but it is reasonable to conclude that even relatively small variations from standard administrative practice can cause substantive problems in forensic uses. For example, the MMPI-A is essentially designed to evaluate adolescents 14–18 years old, but the MMPI manual (Butcher et al., 1992) also observes that the test may be administered to 12- and 13-year-olds who

are developmentally and cognitively capable of responding to the MMPI-A item pool. In most forensic applications, it would be reasonable to assume that a psychologist might be required to render some independent evidence that a 12- or 13-year-old respondent had the cognitive capacity and reading ability necessary to respond meaningfully to the MMPI-A. This issue extends to older adolescents (in the 14- to 18-year-old range) with receptive language disorders, reading disorders, or other learning disorders or cognitive deficits that might impair their ability to understand the items. It is also particularly critical that non-standard or short forms of the MMPI-A be avoided in the forensic setting. In addition, supervision standards in the testing environment take on additional importance in forensic applications, as “chain of evidence” requirements may apply. The MMPI-A should never be left with an adolescent to complete in an unsupervised environment or be sent home with the adolescent to be completed under the supervision of parents. Such concessions would severely compromise the usefulness of test results in a courtroom setting, and issues regarding supervision often result in the exclusion of test results.

Scoring accuracy is also crucial in forensic settings. Hand-scoring procedures exist for all MMPI-A scales, and these require the examiner to ensure that separate score templates are available for each scale and that gender-appropriate templates and profile sheets are utilized. Hand-scoring templates for the MMPI-A may be purchased from Pearson Assessments, with different answer keys required for the MMPI-A softcover and hardcover test booklets. In scoring *VRIN/TRIN*, the examiner must remember to first transfer the relevant item responses to the recording grid. Given the large number of MMPI-A scales and subscales, hand-scoring procedures tend to produce a significant amount of error, even when the MMPI-A is administered and scored with the anticipation that the data will serve as a foundation for expert testimony. For this reason, it is highly desirable to employ computerized scoring methods available through Pearson Assessments for MMPI-2 and MMPI-A profiles that will be used in court. Computerized scoring yields a basic scale Profile Report or an Extended Score Report, depending on the amount of information sought by the examiner. The basic scale Profile Report provides a single-page profile for the Validity and Clinical Scales, with raw scores and *T*-scores for each scale. The Extended Score Report offers additional content and supplementary scale scoring as well as the raw scores and *T*-scores for the Harris-Lingoes and *Si* subscales. Computerized scoring procedures are particularly appropriate for adolescents evaluated for forensic purposes because the frequency of errors generated by hand-scoring procedures is relatively high. Some forensic psychologists follow a process of entering the item-level responses into the computerized scoring package (i.e., the Q Local system) twice in order to rule out the possibility of data entry error through comparison of the resulting profile scores.

In addition to the administration and scoring issues just noted, there may be some differences in interpretive emphasis associated with using the MMPI-A in forensic settings. Although the MMPI-A validity scale findings are always important in terms of evaluating the reliability and validity of the adolescents' self-reports, they are likely to be particularly crucial in forensic settings. Individuals evaluated in a criminal justice setting or as part of a civil law case (e.g., a child custody or personal injury case) may have a substantial motivation to over-report or under-report symptomatology in order to influence the testing outcome. Adolescents charged with serious crimes, for example, may attempt to convince the court that the crime was committed when they were in a psychotic state; adolescents involved in personal injury litigation may exaggerate psychological dysfunction and emotional distress to influence jury awards; and adolescents involved in a child custody dispute may under-report or over-report symptoms depending on the manner in which they perceive these self-reports to influence the custody process. The MMPI-A defensiveness scales (*L* and *K*) are particularly useful in accurately identifying conscious and unconscious attempts to under-report symptoms, and the frequency scales (*F*, *F1*, and *F2*) are

reliable indicators of the tendency to over-report or exaggerate symptoms. Given that adolescents evaluated in forensic settings may be less cooperative than those assessed in clinical settings, the value of Validity Scale information is correspondingly increased in forensic evaluations.

Computerized test interpretation packages are available for the MMPI-A, including the MMPI-A Interpretive System (third edition) developed by Archer (2003) and distributed by Psychological Assessment Resources (PAR), and the Minnesota Report: Adolescent Interpretive System developed by Butcher and Williams (2007) and distributed by Pearson Assessments. As of this writing, the fourth edition of the MMPI-A Interpretive System by Archer is being prepared for release. Both the PAR and Pearson computer reports provide interpretive output specifically designed for adolescents evaluated in forensic settings. Automated computer interpretive reports for the MMPI-A are based on combinations of research findings and clinical experiences, resulting in what has been described as an actuarial-clinical approach to test interpretation (Graham, 2000). Specifically, a series of interpretive statements are generated to match particular sets of test scores based on the research literature combined with the authors' clinical hypotheses and experiences. Therefore, the accuracy of computer-based test interpretations depends heavily on the knowledge and skill of the individual generating the interpretive statements.

CLINICAL CASE EXAMPLE

Brian W., a 16-year-old Caucasian male, is the only child of an upper-middle-class Jewish family. Brian's parents are divorced, and he was living with his mother at the time of the evaluation. He was court-referred for a pre-adjudication evaluation following his arrest for possession of a gun on school property. The purpose of the assessment was to provide the court with information pertaining to Brian's psychological functioning and to make recommendations for interventions to meet possible mental health needs prior to and following adjudication. Brian had no prior history of criminal charges, but he did have a history of conduct problems that had resulted in his expulsion from a private school and recent transfer to a public school, where he committed the current offense. Following his arrest on school property, Brian was briefly held in a juvenile detention facility and subsequently released to his family to await adjudication. Brian had obtained the gun from his father's home three days prior to his arrest, during a scheduled weekend visitation. He used the unloaded gun to threaten another student, who then alerted school officials to Brian's possession of the weapon.

Brian was originally suspended from a private school, then expelled as a result of a series of confrontations and fights with his peers, and a pattern of bullying and intimidating behaviors. These problem behaviors included aggressive and threatening verbal and physical actions with his classmates. His behavioral problems in school were extensive and well documented in teacher evaluations; they included repeated outbursts of anger and hostility, combined with substantial interpersonal manipulateness. For example, Brian had frequently manipulated younger and more impressionable children into engaging in inappropriate behaviors, yet denied any responsibility for involvement in their actions. His expulsion resulted from involvement in influencing younger children to distribute marijuana to their classmates, drugs ultimately traced back to Brian by school authorities. Brian's academic performance had suffered over the past two years, and he dropped from a grade point average of 3.2 in the ninth grade to getting Cs and Ds during the current academic year.

Although Brian's problems in the academic environment were clearly evident, he was substantially more compliant and less openly hostile and rebellious within his home setting. Brian's parents were quite divergent in their perceptions of the seriousness of his problem behaviors. Brian's mother provided ratings for Brian on the Child Behavior Checklist (CBCL; Achenbach & Rescorla, 2001), and the resulting CBCL profile showed marked elevations on Delinquency Behavior, Aggressive Behavior, and Social Problems. In contrast, the CBCL ratings of Brian by his father reflected his perception that Brian was a moderately anxious and depressed child with few significant interpersonal conflicts or problem behaviors. On the Wechsler Intelligence Scale-3, Brian produced a full-scale IQ score of 113, a verbal IQ score of 115, and a performance IQ score of 110. Brian's scores on the Stanford Achievement Test, however, were below expected grade-level performance, with a spelling score at the 26th percentile and an arithmetic performance score at the 30th percentile.

Brian was administered the MMPI-A as part of a comprehensive evaluation that included an extensive clinical interview, collateral teacher and parent interviews, and school and arrest record reviews. His MMPI-A basic scale profile can be seen in Figure 4.2.

The first step in the profile interpretation is to assess the technical validity of this protocol using the validity assessment model proposed by Greene (2000). Brian produced no item omissions on the *Cannot Say* (?) scale, and his *T*-scores on the *VRIN* ($T = 51$) and *TRIN* ($T = 54$)



Profile for Basic Scales

James N. Butcher, Carolyn L. Williams, John R. Graham, Robert P. Archer,
Auke Tellegen, Yusef S. Ben-Porath, and Beverly Kanneker
S. R. Hathaway and J. C. McKinley

Copyright © 1992 REGENTS OF THE UNIVERSITY OF MINNESOTA. All rights reserved.
Copyright © 1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025, 2026, 2027, 2028, 2029, 2030, 2031, 2032, 2033, 2034, 2035, 2036, 2037, 2038, 2039, 2040, 2041, 2042, 2043, 2044, 2045, 2046, 2047, 2048, 2049, 2050, 2051, 2052, 2053, 2054, 2055, 2056, 2057, 2058, 2059, 2060, 2061, 2062, 2063, 2064, 2065, 2066, 2067, 2068, 2069, 2070, 2071, 2072, 2073, 2074, 2075, 2076, 2077, 2078, 2079, 2080, 2081, 2082, 2083, 2084, 2085, 2086, 2087, 2088, 2089, 2090, 2091, 2092, 2093, 2094, 2095, 2096, 2097, 2098, 2099, 2100, 2101, 2102, 2103, 2104, 2105, 2106, 2107, 2108, 2109, 2110, 2111, 2112, 2113, 2114, 2115, 2116, 2117, 2118, 2119, 2120, 2121, 2122, 2123, 2124, 2125, 2126, 2127, 2128, 2129, 2130, 2131, 2132, 2133, 2134, 2135, 2136, 2137, 2138, 2139, 2140, 2141, 2142, 2143, 2144, 2145, 2146, 2147, 2148, 2149, 2150, 2151, 2152, 2153, 2154, 2155, 2156, 2157, 2158, 2159, 2160, 2161, 2162, 2163, 2164, 2165, 2166, 2167, 2168, 2169, 2170, 2171, 2172, 2173, 2174, 2175, 2176, 2177, 2178, 2179, 2180, 2181, 2182, 2183, 2184, 2185, 2186, 2187, 2188, 2189, 2190, 2191, 2192, 2193, 2194, 2195, 2196, 2197, 2198, 2199, 2200, 2201, 2202, 2203, 2204, 2205, 2206, 2207, 2208, 2209, 2210, 2211, 2212, 2213, 2214, 2215, 2216, 2217, 2218, 2219, 2220, 2221, 2222, 2223, 2224, 2225, 2226, 2227, 2228, 2229, 2230, 2231, 2232, 2233, 2234, 2235, 2236, 2237, 2238, 2239, 2240, 2241, 2242, 2243, 2244, 2245, 2246, 2247, 2248, 2249, 2250, 2251, 2252, 2253, 2254, 2255, 2256, 2257, 2258, 2259, 2260, 2261, 2262, 2263, 2264, 2265, 2266, 2267, 2268, 2269, 2270, 2271, 2272, 2273, 2274, 2275, 2276, 2277, 2278, 2279, 2280, 2281, 2282, 2283, 2284, 2285, 2286, 2287, 2288, 2289, 2290, 2291, 2292, 2293, 2294, 2295, 2296, 2297, 2298, 2299, 2300, 2301, 2302, 2303, 2304, 2305, 2306, 2307, 2308, 2309, 2310, 2311, 2312, 2313, 2314, 2315, 2316, 2317, 2318, 2319, 2320, 2321, 2322, 2323, 2324, 2325, 2326, 2327, 2328, 2329, 2330, 2331, 2332, 2333, 2334, 2335, 2336, 2337, 2338, 2339, 2340, 2341, 2342, 2343, 2344, 2345, 2346, 2347, 2348, 2349, 2350, 2351, 2352, 2353, 2354, 2355, 2356, 2357, 2358, 2359, 2360, 2361, 2362, 2363, 2364, 2365, 2366, 2367, 2368, 2369, 2370, 2371, 2372, 2373, 2374, 2375, 2376, 2377, 2378, 2379, 2380, 2381, 2382, 2383, 2384, 2385, 2386, 2387, 2388, 2389, 2390, 2391, 2392, 2393, 2394, 2395, 2396, 2397, 2398, 2399, 2400, 2401, 2402, 2403, 2404, 2405, 2406, 2407, 2408, 2409, 2410, 2411, 2412, 2413, 2414, 2415, 2416, 2417, 2418, 2419, 2420, 2421, 2422, 2423, 2424, 2425, 2426, 2427, 2428, 2429, 2430, 2431, 2432, 2433, 2434, 2435, 2436, 2437, 2438, 2439, 2440, 2441, 2442, 2443, 2444, 2445, 2446, 2447, 2448, 2449, 2450, 2451, 2452, 2453, 2454, 2455, 2456, 2457, 2458, 2459, 2460, 2461, 2462, 2463, 2464, 2465, 2466, 2467, 2468, 2469, 2470, 2471, 2472, 2473, 2474, 2475, 2476, 2477, 2478, 2479, 2480, 2481, 2482, 2483, 2484, 2485, 2486, 2487, 2488, 2489, 2490, 2491, 2492, 2493, 2494, 2495, 2496, 2497, 2498, 2499, 2500, 2501, 2502, 2503, 2504, 2505, 2506, 2507, 2508, 2509, 2510, 2511, 2512, 2513, 2514, 2515, 2516, 2517, 2518, 2519, 2520, 2521, 2522, 2523, 2524, 2525, 2526, 2527, 2528, 2529, 2530, 2531, 2532, 2533, 2534, 2535, 2536, 2537, 2538, 2539, 2540, 2541, 2542, 2543, 2544, 2545, 2546, 2547, 2548, 2549, 2550, 2551, 2552, 2553, 2554, 2555, 2556, 2557, 2558, 2559, 2560, 2561, 2562, 2563, 2564, 2565, 2566, 2567, 2568, 2569, 2570, 2571, 2572, 2573, 2574, 2575, 2576, 2577, 2578, 2579, 2580, 2581, 2582, 2583, 2584, 2585, 2586, 2587, 2588, 2589, 2590, 2591, 2592, 2593, 2594, 2595, 2596, 2597, 2598, 2599, 2600, 2601, 2602, 2603, 2604, 2605, 2606, 2607, 2608, 2609, 2610, 2611, 2612, 2613, 2614, 2615, 2616, 2617, 2618, 2619, 2620, 2621, 2622, 2623, 2624, 2625, 2626, 2627, 2628, 2629, 2630, 2631, 2632, 2633, 2634, 2635, 2636, 2637, 2638, 2639, 2640, 2641, 2642, 2643, 2644, 2645, 2646, 2647, 2648, 2649, 2650, 2651, 2652, 2653, 2654, 2655, 2656, 2657, 2658, 2659, 2660, 2661, 2662, 2663, 2664, 2665, 2666, 2667, 2668, 2669, 2670, 2671, 2672, 2673, 2674, 2675, 2676, 2677, 2678, 2679, 2680, 2681, 2682, 2683, 2684, 2685, 2686, 2687, 2688, 2689, 2690, 2691, 2692, 2693, 2694, 2695, 2696, 2697, 2698, 2699, 2700, 2701, 2702, 2703, 2704, 2705, 2706, 2707, 2708, 2709, 2710, 2711, 2712, 2713, 2714, 2715, 2716, 2717, 2718, 2719, 2720, 2721, 2722, 2723, 2724, 2725, 2726, 2727, 2728, 2729, 2730, 2731, 2732, 2733, 2734, 2735, 2736, 2737, 2738, 2739, 2740, 2741, 2742, 2743, 2744, 2745, 2746, 2747, 2748, 2749, 2750, 2751, 2752, 2753, 2754, 2755, 2756, 2757, 2758, 2759, 2760, 2761, 2762, 2763, 2764, 2765, 2766, 2767, 2768, 2769, 2770, 2771, 2772, 2773, 2774, 2775, 2776, 2777, 2778, 2779, 2780, 2781, 2782, 2783, 2784, 2785, 2786, 2787, 2788, 2789, 2790, 2791, 2792, 2793, 2794, 2795, 2796, 2797, 2798, 2799, 2800, 2801, 2802, 2803, 2804, 2805, 2806, 2807, 2808, 2809, 2810, 2811, 2812, 2813, 2814, 2815, 2816, 2817, 2818, 2819, 2820, 2821, 2822, 2823, 2824, 2825, 2826, 2827, 2828, 2829, 2830, 2831, 2832, 2833, 2834, 2835, 2836, 2837, 2838, 2839, 2840, 2841, 2842, 2843, 2844, 2845, 2846, 2847, 2848, 2849, 2850, 2851, 2852, 2853, 2854, 2855, 2856, 2857, 2858, 2859, 2860, 2861, 2862, 2863, 2864, 2865, 2866, 2867, 2868, 2869, 2870, 2871, 2872, 2873, 2874, 2875, 2876, 2877, 2878, 2879, 2880, 2881, 2882, 2883, 2884, 2885, 2886, 2887, 2888, 2889, 2890, 2891, 2892, 2893, 2894, 2895, 2896, 2897, 2898, 2899, 2900, 2901, 2902, 2903, 2904, 2905, 2906, 2907, 2908, 2909, 2910, 2911, 2912, 2913, 2914, 2915, 2916, 2917, 2918, 2919, 2920, 2921, 2922, 2923, 2924, 2925, 2926, 2927, 2928, 2929, 2930, 2931, 2932, 2933, 2934, 2935, 2936, 2937, 2938, 2939, 2940, 2941, 2942, 2943, 2944, 2945, 2946, 2947, 2948, 2949, 2950, 2951, 2952, 2953, 2954, 2955, 2956, 2957, 2958, 2959, 2960, 2961, 2962, 2963, 2964, 2965, 2966, 2967, 2968, 2969, 2970, 2971, 2972, 2973, 2974, 2975, 2976, 2977, 2978, 2979, 2980, 2981, 2982, 2983, 2984, 2985, 2986, 2987, 2988, 2989, 2990, 2991, 2992, 2993, 2994, 2995, 2996, 2997, 2998, 2999, 3000, 3001, 3002, 3003, 3004, 3005, 3006, 3007, 3008, 3009, 3010, 3011, 3012, 3013, 3014, 3015, 3016, 3017, 3018, 3019, 3020, 3021, 3022, 3023, 3024, 3025, 3026, 3027, 3028, 3029, 3030, 3031, 3032, 3033, 3034, 3035, 3036, 3037, 3038, 3039, 3040, 3041, 3042, 3043, 3044, 3045, 3046, 3047, 3048, 3049, 3050, 3051, 3052, 3053, 3054, 3055, 3056, 3057, 3058, 3059, 3060, 3061, 3062, 3063, 3064, 3065, 3066, 3067, 3068, 3069, 3070, 3071, 3072, 3073, 3074, 3075, 3076, 3077, 3078, 3079, 3080, 3081, 3082, 3083, 3084, 3085, 3086, 3087, 3088, 3089, 3090, 3091, 3092, 3093, 3094, 3095, 3096, 3097, 3098, 3099, 3100, 3101, 3102, 3103, 3104, 3105, 3106, 3107, 3108, 3109, 3110, 3111, 3112, 3113, 3114, 3115, 3116, 3117, 3118, 3119, 3120, 3121, 3122, 3123, 3124, 3125, 3126, 3127, 3128, 3129, 3130, 3131, 3132, 3133, 3134, 3135, 3136, 3137, 3138, 3139, 3140, 3141, 3142, 3143, 3144, 3145, 3146, 3147, 3148, 3149, 3150, 3151, 3152, 3153, 3154, 3155, 3156, 3157, 3158, 3159, 3160, 3161, 3162, 3163, 3164, 3165, 3166, 3167, 3168, 3169, 3170, 3171, 3172, 3173, 3174, 3175, 3176, 3177, 3178, 3179, 3180, 3181, 3182, 3183, 3184, 3185, 3186, 3187, 3188, 3189, 3190, 3191, 3192, 3193, 3194, 3195, 3196, 3197, 3198, 3199, 3200, 3201, 3202, 3203, 3204, 3205, 3206, 3207, 3208, 3209, 3210, 3211, 3212, 3213, 3214, 3215, 3216, 3217, 3218, 3219, 3220, 3221, 3222, 3223, 3224, 3225, 3226, 3227, 3228, 3229, 3230, 3231, 3232, 3233, 3234, 3235, 3236, 3237, 3238, 3239, 3240, 3241, 3242, 3243, 3244, 3245, 3246, 3247, 3248, 3249, 3250, 3251, 3252, 3253, 3254, 3255, 3256, 3257, 3258, 3259, 3260, 3261, 3262, 3263, 3264, 3265, 3266, 3267, 3268, 3269, 3270, 3271, 3272, 3273, 3274, 3275, 3276, 3277, 3278, 3279, 3280, 3281, 3282, 3283, 3284, 3285, 3286, 3287, 3288, 3289, 3290, 3291, 3292, 3293, 3294, 3295, 3296, 3297, 3298, 3299, 3300, 3301, 3302, 3303, 3304, 3305, 3306, 3307, 3308, 3309, 3310, 3311, 3312, 3313, 3314, 3315, 3316, 3317, 3318, 3319, 3320, 3321, 3322, 3323, 3324, 3325, 3326, 3327, 3328, 3329, 3330, 3331, 3332, 3333, 3334, 3335, 3336, 3337, 3338, 3339, 3340, 3341, 3342, 3343, 3344, 3345, 3346, 3347, 3348, 3349, 3350, 3351, 3352, 3353, 3354, 3355, 3356, 3357, 3358, 3359, 3360, 3361, 3362, 3363, 3364, 3365, 3366, 3367, 3368, 3369, 3370, 3371, 3372, 3373, 3374, 3375, 3376, 3377, 3378, 3379, 3380, 3381, 3382, 3383, 3384, 3385, 3386, 3387, 3388, 3389, 3390, 3391, 3392, 3393, 3394, 3395, 3396, 3397, 3398, 3399, 3400, 3401, 3402, 3403, 3404, 3405, 3406, 3407, 3408, 3409, 3410, 3411, 3412, 3413, 3414, 3415, 3416, 3417, 3418, 3419, 3420, 3421, 3422, 3423, 3424, 3425, 3426, 3427, 3428, 3429, 3430, 3431, 3432, 3433, 3434, 3435, 3436, 3437, 3438, 3439, 3440, 3441, 3442, 3443, 3444, 3445, 3446, 3447, 3448, 3449, 3450, 3451, 3452, 3453, 3454, 3455, 3456, 3457, 3458, 3459, 3460, 3461, 3462, 3463, 3464, 3465, 3466, 3467, 3468, 3469, 3470, 3471, 3472, 3473, 3474, 3475, 3476, 3477, 3478, 3479, 3480, 3481, 3482, 3483, 3484, 3485, 3486, 3487, 3488, 3489, 3490, 3491, 3492, 3493, 3494, 3495, 3496, 3497, 3498, 3499, 3500, 3501, 3502, 3503, 3504, 3505, 3506, 3507, 3508, 3509, 3510, 3511, 3512, 3513, 3514, 3515, 3516, 3517, 3518, 3519, 3520, 3521, 3522, 3523, 3524, 3525, 3526, 3527, 3528, 3529, 3530, 3531, 3532, 3533, 3534, 3535, 3536, 3537, 3538, 3539, 3540, 3541, 3542, 3543, 3544, 3545, 3546, 3547, 3548, 3549, 3550, 3551, 3552, 3553, 3554, 3555, 3556, 3557, 3558, 3559, 3560, 3561, 3562, 3563, 3564, 3565, 3566, 3567, 3568, 3569, 3570, 3571, 3572, 3573, 3574, 3575, 3576, 3577, 3578, 3579, 3580, 3581, 3582, 3583, 3584, 3585, 3586, 3587, 3588, 3589, 3590, 3591, 3592, 3593, 3594, 3595, 3596, 3597, 3598, 3599, 3600, 3601, 3602, 3603, 3604, 3605, 3606, 3607, 3608, 3609, 3610, 3611, 3612, 3613, 3614, 3615, 3616, 3617, 3618, 3619, 3620, 3621, 3622, 3623, 3624, 3625, 3626, 3627, 3628, 3629, 3630, 3631, 3632, 3633, 3634, 3635, 3636, 3637, 3638, 3639, 3640, 3641, 3642, 3643, 3644, 3645, 3646, 3647, 3648, 3649, 3650, 3651, 3652, 3653, 3654, 3655, 3656, 3657, 3658, 3659, 3660, 3661, 3662, 3663, 3664, 3665, 3666, 3667, 3668, 3669, 3670, 3671, 3672, 3673, 3674, 3675, 3676, 3677, 3678, 3679, 3680, 3681, 3682, 3683, 3684, 3685, 3686, 3687, 3688, 3689, 3690, 3691, 3692, 3693, 3694, 3695, 3696, 3697, 3698, 3699, 3700, 3701, 3702, 3703, 3704, 3705, 3706, 3707, 3708, 3709, 3710, 3711, 3712, 3713, 3714, 3715, 3716, 3717, 3718, 3719, 3720, 3721, 3722, 3723, 3724, 3725, 3726, 3727, 3728, 3729, 3730, 3731, 3732, 3733, 3734, 3735, 3736, 3737, 3738, 3739, 3740, 3741, 3742, 3743, 3744, 3745, 3746, 3747, 3748, 3749, 3750, 3751, 3752, 3753, 3754, 3755, 3756, 3757, 3758, 3759, 3760, 3761, 3762, 3763, 3764, 3765, 3766, 3767, 3768, 3769, 3770, 3771, 3772, 3773, 3774, 3775, 3776, 3777, 3778, 3779, 3780, 3781, 3782, 3783, 3784, 3785, 3786, 3787, 3788, 3789, 3790, 3791, 3792, 3793, 3794, 3795, 3796, 3797, 3798, 3799, 3800, 3801, 3802, 3803, 3804, 3805, 3806, 3807, 3808, 3809, 3810, 3811, 3812, 3813, 3814, 3815, 3816, 3817, 3818, 3819, 3820, 3821, 3822, 3823, 3824, 3825, 3826, 3827, 3828, 3829, 3830, 3831, 3832, 3833, 3834, 3835, 3836, 3837, 3838, 3839, 3840, 3841, 3842, 3843, 3844, 3845, 3846, 3847, 3848, 3849, 3850, 3851, 3852, 3853, 3854, 3855, 3856, 3857, 3858, 3859, 3860, 3861, 3862, 3863, 3864, 3865, 3866, 3867, 3868, 3869, 3870, 3871, 3872, 3873, 3874, 3875, 3876, 3877, 3878, 3879, 3880, 3881, 3882, 3883, 3884, 3885, 3886, 3887, 3888, 3889, 3890, 3891, 3892, 3893, 3894, 3895, 3896, 3897, 3898, 3899, 3900, 3901, 3902, 3903, 3904, 3905, 3906, 3907, 3908, 3909, 3910, 3911, 3912, 3913, 3914, 3915, 3916, 3917, 3918, 3919, 3920, 3921, 3922, 3923, 3924, 3925, 3926, 3927, 3928,

scales fall well within acceptable ranges provided in the MMPI-A manual (Butcher et al., 1992). Brian's scores on the scales related to defensiveness, *L* ($T = 42$) and *K* ($T = 51$), indicate a relatively candid and accurate self-report. Brian's elevations on scales *F1* ($T = 47$) and *F2* ($T = 42$) also suggest an accurate self-report without evidence of exaggerated or over-reported symptomatology. Thus, Brian's Validity Scale configuration indicates that his responses are both consistent and accurate, and his profile is subject to valid and meaningful interpretation.

Brian's basic scale profile shows a single clinical range elevation on MMPI-A basic scale *Pd* ($T = 72$), with all other basic scales producing T -score values ≤ 55 . Elevations on the *Pd* scale are common in adolescents in criminal justice and psychiatric settings (Archer, 2005). Scores in this range are typically produced by juveniles characterized as rebellious, hostile toward authority figures, and defiant. These adolescents often have histories of poor school adjustment and conduct problems in school. The greater the elevation on scale *Pd*, the more likely it is that the adolescent exhibits a variety of overtly delinquent behaviors. Adolescents who produce *Pd* scale scores in ranges similar to Brian's typically display problems in delaying gratification and are described as impulsive and easily bored and frustrated (Archer, 2005). The most common diagnosis for individuals with MMPI-A profiles similar to Brian's is conduct disorder. Elevations on scale *Pd* are common among adolescents who engage in criminal and antisocial behaviors. Their primary defense mechanisms involve acting out, and their acting-out behavior is often not accompanied by feelings of guilt or remorse. These adolescents may create a good first impression and maintain a relatively extroverted interpersonal style, but they are eventually viewed by others as self-centered, egocentric, and selfish (Archer & Krishnamurthy, 2002). In detention or correctional facilities, these adolescents are more likely to commit institutional infractions and become involved in conflicts and altercations with other residents. In addition to his clinical range elevation on *Pd*, other basic scale test results produced by Brian indicate a marked lack of emotional distress (reflected by his normal-range scores on scales *D* and *Pt*). The low scale *Mf* score (in the masculine direction) seen in Brian's profile is often found among boys who externalize the blame or responsibility for problems and place great value on characteristics such as strength and dominance (Archer, 2005).

Although high scores on scale *Mf* have been related to a lower likelihood of acting out behaviors (e.g., Hathaway & Monachesi, 1963), low scores similar to Brian's have been associated with a greater propensity to engage in antisocial or delinquent behaviors. For example, research findings by Archer et al. (2003) show that boys in correctional facilities produce lower *Mf* scores than boys in psychiatric or substance abuse treatment settings. Further, Hathaway and Monachesi (1963) associated low scale *Mf* scores among boys with a substantially higher rate of delinquency and school dropout. However, recent MMPI-A correlate studies have called into question the importance of low *Mf* scores. Neither Handel et al. (2011) nor Veltri et al. (2009) found any relation between *Mf* scores and external measures of delinquent behaviors.

Brian's MMPI-A Content and Supplementary Scale profiles are shown in Figure 4.3. His Content and Supplementary Scale profile features are relatively consistent with his basic scale profile in showing little evidence of emotional or affective distress, including anxiety (*A-anx*, $T = 54$) or depression (*A-dep*, $T = 47$). In contrast, Brian does display a clinical range elevation on the *A-ang* scale ($T = 79$), characteristic of adolescents with hostile and misanthropic attitudes (Butcher et al., 1992). Adolescents producing similar *A-ang* scores feel misunderstood and believe that others are jealous of their accomplishments. They are likely to be perceived as

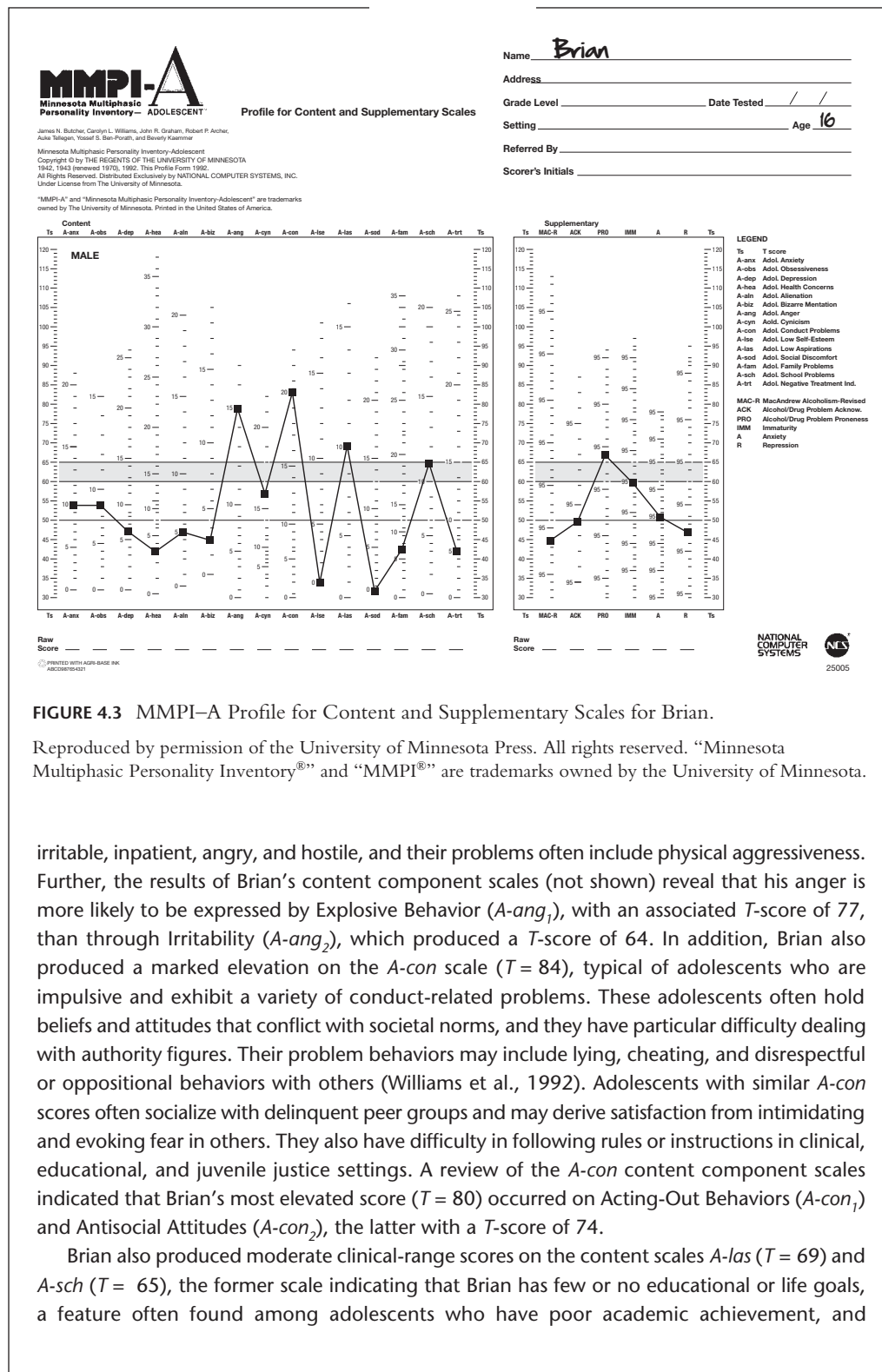


FIGURE 4.3 MMPI-A Profile for Content and Supplementary Scales for Brian.

Reproduced by permission of the University of Minnesota Press. All rights reserved. "Minnesota Multiphasic Personality Inventory" and "MMPI" are trademarks owned by the University of Minnesota.

irritable, inpatient, angry, and hostile, and their problems often include physical aggressiveness. Further, the results of Brian's content component scales (not shown) reveal that his anger is more likely to be expressed by Explosive Behavior ($A\text{-}ang_1$), with an associated T -score of 77, than through Irritability ($A\text{-}ang_2$), which produced a T -score of 64. In addition, Brian also produced a marked elevation on the $A\text{-}con$ scale ($T = 84$), typical of adolescents who are impulsive and exhibit a variety of conduct-related problems. These adolescents often hold beliefs and attitudes that conflict with societal norms, and they have particular difficulty dealing with authority figures. Their problem behaviors may include lying, cheating, and disrespectful or oppositional behaviors with others (Williams et al., 1992). Adolescents with similar $A\text{-}con$ scores often socialize with delinquent peer groups and may derive satisfaction from intimidating and evoking fear in others. They also have difficulty in following rules or instructions in clinical, educational, and juvenile justice settings. A review of the $A\text{-}con$ content component scales indicated that Brian's most elevated score ($T = 80$) occurred on Acting-Out Behaviors ($A\text{-}con_1$) and Antisocial Attitudes ($A\text{-}con_2$), the latter with a T -score of 74.

Brian also produced moderate clinical-range scores on the content scales $A\text{-}las$ ($T = 69$) and $A\text{-}sch$ ($T = 65$), the former scale indicating that Brian has few or no educational or life goals, a feature often found among adolescents who have poor academic achievement, and

supported by his elevated score ($T = 74$) on the Low Achievement Orientation ($A-las_2$) content component scale. These adolescents tend to become frustrated quickly and do not apply themselves in challenging situations. Brian's elevated score on $A-sch$ ($T = 65$) is typical of adolescents who experience significant behavioral and academic problems in the school setting. These adolescents maintain a negative attitude toward academic achievement and are commonly viewed as lazy or unmotivated by others (Archer, 2005). Brian's elevated ($T > 65$) scores on both the School Conduct Problems ($A-sch_1$) and Negative Attitudes ($A-sch_2$) content component scales indicate that his academic problems reflect both attitudinal and behavioral components. Finally, Brian's moderate clinical-range elevation on the Alcohol-Drug Problem Proneness (PRO) scale ($T = 67$) is characteristic of adolescents who are at increased risk for drug and alcohol problems. In contrast, Brian's scores on the $MAC-R$ scale and the Alcohol Drug Acknowledgement Scale (ACK) are well within normal limits. This mixed pattern of findings on the three supplementary scales related to drug or alcohol use problems clearly warrants further assessment of Brian's history of drug and alcohol use.

Finally, Brian's T -scores for the MMPI-A PSY-5 scales were as follows: Aggressiveness ($AGGR$) = 87; Psychoticism ($PSYC$) = 55; Disconstraint ($DISC$) = 86; Negative Emotionality/Neuroticism ($NEGE$) = 47; and Introversion/Low Positive Emotionality ($INTR$) = 54. These scores reflect Brian's use of aggression as an instrumental means of accomplishing goals through intimidation and threats ($AGGR$) and his problems with impulsivity, acting out, and irresponsible/delinquent behaviors ($DISC$). Brian's relative absence of affective distress is clearly indicated by his normal range scores on the $NEGE$ and $INTR$ scales.

Clinical Case Recommendations

The MMPI-A findings presented in this chapter, in combination with results from other components of this forensic evaluation, suggested that Brian was at low to moderate risk for recidivism. It was recommended that his rehabilitation needs could be best served by placement on probationary status, with the requirement that Brian undergo a comprehensive drug and alcohol abuse evaluation, that he engage in individual psychotherapy, and that he and his parents participate in family therapy sessions. It was further recommended that family therapy focus on increasing communication between Brian and his parents as well as providing assistance to his parents in placing firm limits and controls on Brian's behaviors. Individual psychotherapy was encouraged in part to explore the underlying aspects of Brian's frustration and anger but also to assist Brian in developing more appropriate anger management techniques. Finally, it was suggested that Brian's probationary period be closely supervised, and that any significant occurrence of antisocial or conduct disordered behaviors might serve as a basis for reevaluation of Brian's probationary status and placement in a juvenile correctional facility.

References

- Achenbach, T. M., & Rescorla, L. A. (2001). *Manual for the Achenbach System of Empirically Based Assessment school-age forms and profiles*. Burlington: University of Vermont, Department of Psychiatry.
- Alperin, J. J., Archer, R. P., & Coates, G. D. (1996). Development and effects of an MMPI-A K-correction procedure. *Journal of Personality Assessment*, 67, 155-168.
- American Psychiatric Association. (2000). *Diagnostic and statistic manual of mental disorders* (4th ed., test revision). Washington, DC: Author.

- Angold, A. (1989). Structured assessments of psychopathology in children and adolescents. In C. Thompson (Ed.), *The instruments of psychiatric research* (pp. 271–304). New York: John Wiley.
- Arbisi, P. A., & Ben-Porath, Y. S. (1995). An MMPI-2 infrequent response scale for use with psychopathological populations: The infrequency psychopathology scale F(p). *Psychological Assessment*, 7, 424–431.
- Archer, R. P. (1987). *Using the MMPI with adolescents*. Hillsdale, NJ: Lawrence Erlbaum.
- Archer, R. P. (2003). MMPI-A Interpretive System (Version 4) [Computer software]. Tampa, FL: Psychological Assessment Resources.
- Archer, R. P. (2005). *MMPI-A: Assessing adolescent psychopathology* (3rd ed.). Mahwah, NJ: Lawrence Erlbaum.
- Archer, R. P. (in preparation). MMPI-A Interpretive System (Version 4) [Computer software]. Tampa, FL: Psychological Assessment Resources.
- Archer, R. P., & Baker, E. M. (2005). Use of the Minnesota Multiphasic Personality Inventory-Adolescent (MMPI-A) in juvenile justice settings. In D. Seagraves & T. Grisso (Eds.), *Handbook of screening and assessment tools for juvenile justice*. New York: Guilford Press.
- Archer, R. P., Belevich, J. K. S., & Elkins, D. E. (1994). Item-level and scale-level factor structures of the MMPI-A. *Journal of Personality Assessment*, 62, 332–345.
- Archer, R. P., Bolinsky, P. K., Morton, T. L., & Farris, K. L. (2003). MMPI-A characteristics of male adolescents in juvenile justice and clinical treatment settings. *Assessment*, 10, 400–410.
- Archer, R. P., Buffington-Vollum, J. K., Stredny, R., & Handel, R. W. (2006). A survey of psychological test use patterns among forensic psychologists. *Journal of Personality Assessment*, 87, 84–94.
- Archer, R. P., & Elkins, D. E. (1999). Identification of random responding on the MMPI-A. *Journal of Personality Assessment*, 73, 407–421.
- Archer, R. P., Gordon, R. A., Anderson, G. L., & Giannetti, R. A. (1989). MMPI special scale clinical correlates for adolescent inpatients. *Journal of Personality Assessment*, 53, 654–664.
- Archer, R. P., Gordon, R. A., Giannetti, R. A., & Singles, J. M. (1988). MMPI scale clinical correlates for adolescent inpatients. *Journal of Personality Assessment*, 52, 707–721.
- Archer, R. P., Handel, R. W., & Lynch, K. D. (2001). The effectiveness of MMPI-A items in discriminating between normative and clinical samples. *Journal of Personality Assessment*, 77, 420–435.
- Archer, R. P., Handel, R. W., Tellegen, A., & Ben-Porath, Y. S. (2011, May). Status report on the MMPI-A-RF. Paper presented at the 46th Annual Symposium on MMPI research, Minneapolis, MN.
- Archer, R. P., & Jacobson, J. M. (1993). Are critical items “critical” for the MMPI-A? *Journal of Personality Assessment*, 61, 547–556.
- Archer, R. P., & Krishnamurthy, R. (1994a). *MMPI-A structural summary* [Summary form]. Odessa, FL: Psychological Assessment Resources.
- Archer, R. P., & Krishnamurthy, R. (1994b). A structural summary approach for the MMPI-A: Development and empirical correlates. *Journal of Personality Assessment*, 63, 554–573.
- Archer, R. P., & Krishnamurthy, R. (1997). MMPI-A scale-level factor structure: Replication in a clinical sample. *Assessment*, 4, 337–349.
- Archer, R. P., & Krishnamurthy, R. (2002). *Essentials of MMPI-A assessment*. New York: John Wiley.
- Archer, R. P., Krishnamurthy, R., & Jacobson, J. M. (1994). *MMPI-A casebook*. Tampa, FL: Psychological Assessment Resources.
- Archer, R. P., Maruish, M., Imhof, E. A., & Piotrowski, C. (1991). Psychological test usage with adolescent clients: 1990 survey findings. *Professional Psychology: Research and Practice*, 22, 247–252.
- Archer, R. P., & Newsom, C. R. (2000). Psychological test usage with adolescent clients: Survey update. *Assessment*, 7, 227–235.
- Archer, R. P., Pancoast, D. L., & Gordon, R. A. (1994). The development of the MMPI-A Immaturity (IMM) scale: Findings for normal and clinical samples. *Journal of Personality Assessment*, 62, 145–156.
- Archer, R. P., Tirrell, C. A., & Elkins, D. E. (2001). Evaluation of an MMPI-A short form: Implications for adaptive testing. *Journal of Personality Assessment*, 76, 76–89.
- Baer, R. A., Ballenger, J., Berry, D. T. R., & Wetter, M. W. (1997). Detection of random responding on the MMPI-A. *Journal of Personality Assessment*, 68, 139–151.
- Baer, R. A., Ballenger, J., & Kroll, L. S. (1998). Detection of underreporting on the MMPI-A in clinical and community samples. *Journal of Personality Assessment*, 71, 98–113.

- Baer, R. A., Kroll, L. S., Rinaldo, J., & Ballenger, J. (1999). Detecting and discriminating between random responding and overreporting on the MMPI-A. *Journal of Personality Assessment*, 72, 308–320.
- Ball, J. C. (1960). Comparison of MMPI profile differences among Negro–White adolescents. *Journal of Clinical Psychology*, 16, 304–307.
- Barkley, R. A., & Murphy, K. R. (1998). *Attention deficit hyperactivity disorder: A clinical workbook* (2nd ed.). New York: Guilford Press.
- Baum, L. J., Archer, R. P., Forbey, J. D., & Handel, R. W. (2009). A review of the Minnesota Multiphasic Personality Inventory–Adolescent (MMPI–A) and the Millon Clinical Inventory (MACI) with an emphasis on juvenile justice samples. *Assessment*, 16, 384–400.
- Ben-Porath, Y. S., Graham, J. R., Archer, R. P., Tellegen, A., & Kaemmer, B. (2006). *Supplement to the MMPI–A manual for administration, scoring, and interpretation: The Content Component Scales, the Personality Psychopathology Five (PSY-5) Scales, and the Critical Items*. Minneapolis: University of Minnesota Press.
- Ben-Porath, Y. S., Hostetler, K., Butcher, J. N., & Graham, J. R. (1989). New subscales for the MMPI–2 Social Introversion (Si) scale. *Psychological Assessment*, 1, 169–174.
- Black, K. (1994). A critical review of the MMPI–A. *Child Assessment News*, 4, 9–12.
- Briggs, P. F., Wirt, R. D., & Johnson, R. (1961). An application of prediction tables to the study of delinquency. *Journal of Consulting Psychology*, 25, 46–50.
- Butcher, J. N., Graham, J. R., Ben-Porath, Y. S., Tellegen, A., Dahlstrom, W. G., & Kaemmer, B. (2001). *Minnesota Multiphasic Personality Inventory–2 (MMPI–2): Manual for administration, scoring, and interpretation* (rev. ed.). Minneapolis: University of Minnesota Press.
- Butcher, J. N., & Williams, C. L. (2007). *Minnesota Multiphasic Personality Inventory–Adolescent (MMPI–A): User's guide for the Minnesota Report: Adolescent Interpretive System* (2nd ed.). Minneapolis: University of Minnesota Press.
- Butcher, J. N., Williams, C. L., Graham, J. R., Archer, R. P., Tellegen, A., Ben-Porath, Y. S., & Kaemmer, B. (1992). *Minnesota Multiphasic Personality Inventory–Adolescent (MMPI–A): Manual for administration, scoring, and interpretation*. Minneapolis: University of Minnesota Press.
- Capwell, D. F. (1945a). Personality patterns of adolescent girls: I. Girls who show improvement in IQ. *Journal of Applied Psychology*, 29, 212–228.
- Capwell, D. F. (1945b). Personality patterns of adolescent girls: II. Delinquents and non-delinquents. *Journal of Applied Psychology*, 29, 284–297.
- Cashel, M. L., Rogers, R., Sewell, K. W., & Holliman, N. G. (1998). Preliminary validation of the MMPI–A for a male delinquent sample: An investigation of clinical correlates and discriminant validity. *Journal of Personality Assessment*, 71, 46–69.
- Colligan, R. C., & Offord, K. P. (1992). Age, stage, and the MMPI: Changes in response patterns over an 85-year age span. *Journal of Clinical Psychology*, 48, 476–493.
- Dahlstrom, W. G., Lachar, D., & Dahlstrom, L. E. (1986). *MMPI patterns of American minorities*. Minneapolis: University of Minnesota Press.
- Fabry, J. J., Bertinetti, J. F., & Guzman-Cavazos, L. (2011). Characteristics of juvenile offenders with invalid and valid MMPI–A profiles. *Psychological Reports*, 108, 908–918.
- Fontaine, J. L., Archer, R. P., Elkins, D. E., & Johansen, J. (2001). The effects of MMPI–A T-score elevation on classification accuracy for normal and clinical adolescent samples. *Journal of Personality Assessment*, 76, 264–281.
- Forbey, J. D. (2003, June). A review of the MMPI–A research literature. Paper presented at the 38th Annual Symposium on Recent Developments in the Use of the MMPI–2 and MMPI–A, Minneapolis, MN.
- Forbey, J. D., & Ben-Porath, Y. S. (1998). *A critical item set for the MMPI–A*. MMPI–2/MMPI–A Test Rep. No. 4. Minneapolis: University of Minnesota Press.
- Forbey, J. D., & Ben-Porath, Y. S. (2000). A comparison of sexually abused and non-sexually abused adolescents in a clinical treatment facility using the MMPI–A. *Child Abuse and Neglect*, 24, 557–568.
- Glaser, B. A., Calhoun, G. B., & Petrocelli, J. V. (2002). Personality characteristics of male juvenile offenders by adjudicated offenses as indicated by the MMPI–A. *Criminal Justice and Behavior*, 29, 183–201.
- Gómez, F. C., Johnson, R., Davis, Q., & Velásquez, R. J. (2000). MMPI–A performance of African and Mexican American adolescent first-time offenders. *Psychological Reports*, 87, 309–314.
- Gottesman, I. I., & Hanson, D. R. (1990, August). Can the MMPI at age 15 predict schizophrenics-to-be? In R. C. Colligan (Chair), *The MMPI and adolescents: Historical perspective, current research, future*

- developments. Symposium conducted at the annual convention of the American Psychological Association, Boston.
- Gottesman, I. I., & Prescott, C. A. (1989). Abuses of the MacAndrew Alcoholism scale: A critical review. *Clinical Psychology Review, 9*, 223–242.
- Graham, J. R. (2000). *MMPI-2: Assessing personality and psychopathology* (3rd ed.). New York: Oxford University Press.
- Greenberg, S. A., Otto, R. K., & Long, A. C. (2003). The utility of psychological testing in assessing emotional damages in personal injury litigation. *Assessment, 10*, 411–419.
- Greene, R. L. (2000). *MMPI-2: An interpretive manual* (2nd ed.). Boston: Allyn & Bacon.
- Grisso, T. (1998). *Forensic evaluation of juveniles*. Sarasota, FL: Professional Resources Press.
- Gynther, M. D. (1972). White norms and Black MMPIs: A prescription for discrimination? *Psychological Bulletin, 78*, 386–402.
- Handel, R. W., Archer, R. P., Elkins, D. E., Mason, J. A., & Simonds-Bisbee, E. C. (2011). Psychometric properties of the Minnesota Multiphasic Personality Inventory–Adolescent (MMPI–A) clinical, content, and supplementary scales in a forensic sample. *Journal of Personality Assessment, 93*, 566–581.
- Handel, R. W., Arnau, R. C., Archer, R. P., & Dandy, K. L. (2006). An evaluation of the MMPI–2 and MMPI–A True Response Inconsistency (TRIN) scales. *Assessment, 13*, 98–106.
- Hare, R. D. (1991). *The Hare Psychopathy Checklist–Revised*. Toronto: Multi-Health Systems.
- Harkness, A. R., & McNulty, J. L. (1994). The Personality Psychopathology Five (PSY-5): Issue from the pages of a diagnostic manual instead of a dictionary. In S. Strak & M. Lorr (Eds.), *Differentiating normal and abnormal personality* (pp. 291–315). New York: Springer.
- Hathaway, S. R., & McKinley, J. C. (1943). *The Minnesota Multiphasic Personality Inventory* (rev. ed.). Minneapolis: University of Minnesota Press.
- Hathaway, S. R., & Monachesi, E. D. (1952). The Minnesota Multiphasic Personality Inventory in the study of juvenile delinquents. *American Sociological Review, 17*, 704–710.
- Hathaway, S. R., & Monachesi, E. D. (Eds.). (1953). *Analyzing and predicting juvenile delinquency with the MMPI*. Minneapolis: University of Minnesota Press.
- Hathaway, S. R., & Monachesi, E. D. (1961). *An atlas of juvenile MMPI profiles*. Minneapolis: University of Minnesota Press.
- Hathaway, S. R., & Monachesi, E. D. (1963). *Adolescent personality and behavior: MMPI patterns of normal, delinquent, dropout, and other outcomes*. Minneapolis: University of Minnesota Press.
- Hicks, M. M., Rogers, R., & Cashel, M. L. (2000). Prediction of violent and total infractions among institutionalized male juvenile offenders. *Journal of the American Academy of Psychiatry and Law, 28*, 183–190.
- Hillary, B. E., & Schare, M. L. (1993). Sexually and physically abused adolescents: An empirical search for PTSD. *Journal of Clinical Psychology, 49*, 161–165.
- Huesmann, L. R., Lefkowitz, M. M., & Eron, L. D. (1978). Sum of MMPI Scales F, 4, and 9 as a measure of aggression. *Journal of Consulting and Clinical Psychology, 46*, 1071–1078.
- Hume, M. P., Kennedy, W. A., Patrick, C. J., & Partyka, D. J. (1996). Examination of the MMPI–A for the assessment of psychopathy in incarcerated adolescent male offenders. *International Journal of Offender Therapy and Comparative Criminology, 40*, 224–233.
- Humm, D. G., & Wadsworth, G. W. (1935). The Humm–Wadsworth Temperament Scale. *American Journal of Psychiatry, 92*, 163–200.
- Imhof, E. A., & Archer, R. P. (1997). Correlates of the MMPI–A immaturity (IMM) scale in an adolescent psychiatric sample. *Assessment, 5*, 169–179.
- Kaser-Boyd, N. (2002). Children who kill. In N. G. Ribner (Ed.), *The California School of Professional Psychology handbook of juvenile forensic psychology* (pp. 195–229). San Francisco: Jossey-Bass.
- Katz, R. C., & Marquette, J. (1996). Psychosocial characteristics of young violent offenders: A comparative study. *Criminal Behaviour and Mental Health, 6*, 339–348.
- Koss, M. P., & Butcher, J. N. (1973). A comparison of psychiatric patients' self-report with other sources of clinical information. *Journal of Research in Personality, 7*, 225–236.
- Krishnamurthy, R., Bolinsky, P. K., & Archer, R. P. (2005, March). MMPI–A Structural Summary: Integrating new scales and subscales. Paper presented at the Annual Conference of the Society for Personality Assessment, Chicago.
- Kumho Tire Co. v. Carmichael*, 526 U.S. 138 (1999).

- Lachar, D., & Wrobel, T. A. (1979). Validating clinicians' hunches: Construction of a new MMPI critical item set. *Journal of Consulting and Clinical Psychology, 47*, 277–284.
- Lowman, J., Galinsky, M. D., & Gray-Little, B. (1980). *Predicting achievement: A ten-year follow-up of black and white adolescents*. Chapel Hill: University of North Carolina at Chapel Hill, Institute for Research in Social Science.
- Lubin, B., Larsen, R. M., & Matarazzo, J. D. (1984). Patterns of psychological test usage in the United States: 1935–1982. *American Psychologist, 39*, 451–454.
- Lubin, B., Larsen, R. M., Matarazzo, J. D., & Seever, M. F. (1985). Psychological test usage patterns in five professional settings. *American Psychologist, 40*, 857–861.
- Lubin, B., Wallis, R. R., & Paine, C. (1971). Patterns of psychological test usage in the United States: 1935–1969. *Professional Psychology, 2*, 70–74.
- Marks, P. A., & Briggs, P. F. (1972). *Adolescent norm tables for the MMPI*. In W. G. Dahlstrom, G. S. Welsh, & L. E. Dahlstrom (Eds.), *An MMPI handbook*. Vol. 1: *Clinical interpretation* (rev. ed., pp. 388–399). Minneapolis: University of Minnesota Press.
- Marks, P. A., Seeman, W., & Haller, D. L. (1974). *The actuarial use of the MMPI with adolescents and adults*. Baltimore: Williams & Wilkins.
- McGrath, R. E., Pogge, D. L., Stein, L. A. R., Graham, J. R., Zaccario, M., & Piacentini, T. (2000). Development of an Infrequency-Psychopathology scale for the MMPI-A: The *Fp-A* scale. *Journal of Personality Assessment, 74*, 282–295.
- McKinley, J. C., & Hathaway, S. R. (1943). The identification and measurement of the psychoneuroses in medical practice. *Journal of the American Medical Association, 122*, 161–167.
- McKinley, J. C., Hathaway, S. R., & Meehl, P. E. (1948). The Minnesota Multiphasic Personality Inventory: VI. The *K* scale. *Journal of Consulting Psychology, 12*, 20–31.
- McNulty, J. L., Harkness, A. R., Ben-Porath, Y. S., & Williams, C. L. (1997). Assessing the Personality Psychopathology Five (PSY-5) in adolescents: New MMPI-A scales. *Psychological Assessment, 9*, 250–259.
- Millon, T. (1993). *Millon Adolescent Clinical Inventory*. Minneapolis, MN: Pearson Assessments.
- Milne, L. C., & Greenway, P. (1999). Do high scores on the Adolescent-School Problems and Immaturity Scales of the MMPI-A have implications for cognitive performance as measured by the WISC-III? *Psychology in the Schools, 36*, 199–203.
- Monachesi, E. D., & Hathaway, S. R. (1969). The personality of delinquents. In J. N. Butcher (Ed.), *MMPI: Research developments and clinical applications* (pp. 207–219). Minneapolis: University of Minnesota Press.
- Morey, L. C. (2000). The challenge of construct validity in the assessment of psychopathology. In R. D. Goffin & E. Helmes (Eds.), *Problems and solutions in human assessment: Honoring Douglas N. Jackson at seventy* (pp. 141–171). New York: Kluwer Academic/Plenum Publishers.
- Morton, T. L., Farris, K. L., & Brenowitz, L. H. (2002). MMPI-A scores and high points of male juvenile delinquents: Scales 4, 5, and 6 as markers of juvenile delinquency. *Psychological Assessment, 14*, 311–319.
- O'Connor Pennuto, T., & Archer, R. P. (2008). MMPI-A forensic case studies: Uses in documented court decisions. *Journal of Personality Assessment, 90*, 215–226.
- Otto, R. K., & Collins, R. P. (1995). Use of the MMPI-2/MMPI-A in child custody evaluations. In Y. S. Ben-Porath, J. R. Graham, G. C. N. Hall, R. D. Hirschman, & M. S. Zaragoza (Eds.), *Forensic applications of the MMPI-2* (pp. 222–252). Thousand Oaks, CA: Sage.
- Pancoast, D. L., & Archer, R. P. (1988). MMPI adolescent norms: Patterns and trends across four decades. *Journal of Personality Assessment, 52*, 691–706.
- Pena, L. M., Megargee, E. I., & Brody, P. (1996). MMPI-A patterns of male juvenile delinquents. *Psychological Assessment, 8*, 388–397.
- Peterson, M., & Robbins, B. (2008). Using the MMPI-A to predict recidivism in adjudicated minors. *Applied Psychology in Criminal Justice, 4*, 172–181.
- Piotrowski, C., & Keller, J. W. (1989). Psychological testing in outpatient mental health facilities: A national study. *Professional Psychology: Research and Practice, 20*, 423–425.
- Pope, H. S., Butcher, J. N., & Seelen, J. (2006). *The MMPI, MMPI-2, MMPI-A in court: A practice guide for expert witnesses and attorneys* (3rd ed.). Washington, DC: American Psychological Association.
- Roper v. Simmons*, 543 U.S. 551 (2005).
- Scott, R., & Stone, D. (1986). MMPI measures of psychological disturbance in adolescent and adult victims of father-daughter incest. *Journal of Clinical Psychology, 42*, 251–259.

- Sherwood, N. E., Ben-Porath, Y. S., & Williams, C. L. (1997). *The MMPI-A content component scales: Development, psychometric characteristics and clinical application*. MMPI-2/MMPI-A Test Rep. No. 3. Minneapolis: University of Minnesota Press.
- Stein, L. A. R., McClinton, B. K., & Graham, J. R. (1998). Long-term stability of MMPI-A scales. *Journal of Personality Assessment*, 70, 103–108.
- Timbrook, R. E., & Graham, J. R. (1994). Ethnic differences on the MMPI-2? *Psychological Assessment*, 6, 212–217.
- Veltri, C. O. C., Graham, J. R., Sellbom, M., Ben-Porath, Y. S., Forbey, J. D., O'Connell, C. O., . . . White, R. S. (2009). Correlates of MMPI-A scales in acute psychiatric and forensic samples. *Journal of Personality Assessment*, 91, 288–300.
- Weed, N. C., Butcher, J. N., & Williams, C. L. (1994). Development of the MMPI-A alcohol/drug problem scales. *Journal of Studies on Alcohol*, 55, 296–302.
- Williams, C. L., & Butcher, J. N. (1989). An MMPI study of adolescents: II. Verification and limitations of code type classifications. *Psychological Assessment*, 1, 260–265.
- Williams, C. L., Butcher, J. N., Ben-Porath, Y. S., & Graham, J. R. (1992). *MMPI-A content scales: Assessing psychopathology in adolescents*. Minneapolis: University of Minnesota Press.
- Wirt, R. D., & Briggs, P. F. (1959). Personality and environmental factors in the development of delinquency. *Psychological Monographs: General and Applied (Whole No. 485)*, 1–47.
- Wrobel, N. H., & Lachar, D. (1995). Racial differences in adolescent self-report: A comparative validity study using homogeneous MMPI content measures. *Psychological Assessment*, 7, 140–147.
- Zinn, S., McCumber, S., & Dahlstrom, W. G. (1999). Cross-validation and extension of the MMPI-A IM scale. *Assessment*, 6, 1–6.

5

FORENSIC ASSESSMENT WITH THE PERSONALITY ASSESSMENT INVENTORY

Leslie C. Morey and Justin K. Meyer

The *Personality Assessment Inventory* (PAI; Morey, 1991) was designed to provide psychological assessment information in a variety of contexts in which psychopathology, personality, and psychosocial environment are a concern. The breadth of coverage of important psychological constructs allows for a smooth translation to forensic settings (Boccaccini & Brodsky, 1999; Edens, Cruise, & Buffington-Vollum, 2001). This chapter provides a brief overview of the theory and procedures employed in developing the PAI, a review of PAI psychometric and validity data with particular emphasis on forensic research, a discussion of the use of the PAI in forensic settings, and an illustrative case example. More detailed coverage than is possible in this chapter can be found in several primary sources (Morey, 1991, 1996, 2003), and an independent review of the PAI in forensic settings is also available (Edens et al., 2001), as well as a review of the clinical applications of the PAI in criminal justice settings (Ruiz & Ochshorn, 2010). To orient the reader to the structure of the PAI, scale and subscale descriptions are listed in Table 5.1, and validity and predictive indices developed following the construction of the PAI are listed in Table 5.2.

Theoretical Basis and Test Development

The development of the PAI was based on a construct validation framework that emphasized a theoretical and rational as well as a quantitative method of scale development. This framework places a strong emphasis on a theoretically informed approach to the development and selection of items, as well as on the assessment of their stability and correlates. As a first step, the theoretical and empirical literature for each of the constructs to be measured was closely examined, since this articulation had to serve as a guide to the content of information sampled and to the subsequent assessment of content validity. The development of the test then went through four iterations in a sequential construct validation strategy similar to that described by Loevinger (1957) and Jackson (1970), and including the consideration of a number of item parameters not described by those authors. Of paramount importance at each point of the development process was the assumption that no single quantitative item parameter should be used as the sole criterion for item selection. An over-reliance on a single parameter typically leads to a scale with one desirable psychometric property and numerous undesirable ones. Both the conceptual nature and empirical adequacy of the items played an important role in their inclusion in the final version of the PAI.

TABLE 5.1 PAI Scales and Subscales

<i>Scale</i>		<i>Interpretation of High Scores</i>
Validity Scales		
<i>ICN</i>	Inconsistency	Poor concentration or inattention
<i>INF</i>	Infrequency	Idiosyncratic or random response set
<i>NIM</i>	Negative Impression Management	Negative response set due to pessimistic world view and/or intentional dissimulation
<i>PIM</i>	Positive Impression Management	Positive response set due to naïveté or intentional dissimulation
Clinical Scales		
<i>SOM</i>	Somatic Complaints	
<i>SOM-C</i>	Conversion	Rare sensorimotor symptoms associated with conversion disorders or certain medical conditions
<i>SOM-S</i>	Somatization	The occurrence of common physical symptoms or vague complaints of ill health or fatigue
<i>SOM-H</i>	Health Concerns	Preoccupation with physical functioning and symptoms
<i>ANX</i>	Anxiety	
<i>ANX-C</i>	Cognitive	Ruminative worry and impaired concentration and attention
<i>ANX-A</i>	Affective	Experience of tension, difficulty relaxing, nervousness, and fatigue
<i>ANX-P</i>	Physiological	Overt signs of anxiety, including sweating, trembling, shortness of breath, and irregular heartbeat
<i>ARD</i>	Anxiety-Related Disorders	
<i>ARD-O</i>	Obsessive-Compulsive	Intrusive thoughts, compulsive behaviors, rigidity, indecision, perfectionism, and affective constriction
<i>ARD-P</i>	Phobias	Common fears, including social situations, heights, and public or enclosed places; low scores suggest fearlessness
<i>ARD-T</i>	Traumatic Stress	Experience of trauma that continue to cause distress
<i>DEP</i>	Depression	
<i>DEP-C</i>	Cognitive	Worthlessness, hopelessness, indecisiveness, and difficulty concentrating; low scores indicate personal confidence
<i>DEP-A</i>	Affective	Feelings of sadness, diminished interest, and anhedonia
<i>DEP-P</i>	Physiological	Level of physical functioning, activity, and sleep and diet patterns
<i>MAN</i>	Mania	
<i>MAN-A</i>	Activity Level	Disorganized over-involvement in activities, accelerated thought processes and behavior
<i>MAN-G</i>	Grandiosity	Inflated self-esteem and expansiveness; low scores indicate low self-esteem
<i>MAN-I</i>	Irritability	Frustration intolerance, impatience, and resulting strained relationships
<i>PAR</i>	Paranoia	
<i>PAR-H</i>	Hypervigilance	Suspiciousness and tendency to closely monitor environment; low scores suggest interpersonal trust
<i>PAR-P</i>	Persecution	Belief that others have intentionally constructed obstacles to one's achievement

TABLE 5.1 continued

<i>Scale</i>		<i>Interpretation of High Scores</i>
<i>PAR-R</i>	Resentment	Bitterness and cynicism in relationships, tendency to hold grudges, and externalization of blame
<i>SCZ</i>	Schizophrenia	
<i>SCZ-P</i>	Psychotic Experiences	Unusual perceptions and sensations, magical thinking, and unusual ideas
<i>SXZ-S</i>	Social Detachment	Social isolation, discomfort, and awkwardness
<i>SCZ-T</i>	Thought Disorder	Confusion, concentration difficulties, and disorganization
<i>BOR</i>	Borderline Features	
<i>BOR-A</i>	Affective Instability	Emotional responsiveness, rapid mood change, poor modulation
<i>BOR-I</i>	Identity Problems	Uncertainty about major life issues and feelings of emptiness or lack of fulfillment or purpose
<i>BOR-N</i>	Negative Relationships	History of intense, ambivalent relationships and feelings of exploitation or betrayal
<i>BOR-S</i>	Self-Harm	Impulsivity in areas likely to be dangerous
<i>ANT</i>	Antisocial Features	
<i>ANT-A</i>	Antisocial Behaviors	History of antisocial and illegal behavior
<i>ANT-E</i>	Egocentricity	Lack of empathy or remorse, exploitative approach to relationships
<i>ANT-S</i>	Stimulus-Seeking	Cravings for excitement, low boredom tolerance, recklessness
<i>ALC</i>	Alcohol Problems	Use of and problems with alcohol
<i>DRG</i>	Drug Problems	Use of and problems with drugs
Treatment Consideration Scales		
<i>AGG</i>	Aggression	
<i>AGG-A</i>	Aggressive Attitude	Hostility, poor control over anger, and belief instrumental utility of violence
<i>AGG-V</i>	Verbal Aggression	Assertiveness, abusiveness, and readiness to express anger to others
<i>AGG-P</i>	Physical Aggression	Tendency to be involved in physical aggression
<i>SUI</i>	Suicidal Ideation	Frequency and intensity of thoughts of self-harm or fantasies about suicide
<i>STR</i>	Stress	Perception of an uncertain or difficult environment
<i>NON</i>	Nonsupport	Perception that others are not available or willing to provide support
<i>RXR</i>	Treatment Rejection	Attitudes that represent obstacles or indicate low motivation for treatment
Interpersonal Scales		
<i>DOM</i>	Dominance	Desire and tendency for control in relationships; low scores suggest meekness and submissiveness
<i>WRM</i>	Warmth	Interest and comfort with close relationships; low scores suggest hostility, anger, and mistrust

TABLE 5.2 Supplementary PAI Indices

<i>Index</i>		<i>Development</i>	<i>Interpretation of High Scores</i>
Validity Indices			
<i>MI</i>	Malingering Index	Eight configural features observed with relatively high frequency in malingering samples	Negative response set, malingering
<i>RDF</i>	Rogers Discriminant Function	Function found to discriminate patients from naïve and coached malingerers	Malingering
<i>DI</i>	Defensiveness Index	Eight configural features observed with relatively high frequency in positive dissimulation samples	Self and/or other deception in the positive direction
<i>CDF</i>	Cashel Discriminant Function	Function found to discriminate real from fake good inmates and college students	Intentional concealment of specific problems
<i>ALCp</i>	<i>ALC</i> Predicted Score	<i>ALC</i> predicted by other elements of the profile	<i>ALCp</i> > <i>ALC</i> suggests deception regarding alcohol use
<i>DRGp</i>	<i>DRG</i> Predicted Score	<i>DRG</i> predicted by other elements of the profile	<i>DRGp</i> > <i>DRG</i> suggests deception regarding drug use
<i>BRR</i>	Back Random Responding	Differences >5 <i>T</i> on front and back halves of <i>ALC</i> and <i>SUI</i> scales	Random responding on back half of PAI
<i>ACS^a</i>	Addictive Characteristics Scale	Algorithm of <i>BOR-A</i> , <i>BOR-S</i> , and <i>ANT-E</i> used to predict addictive potential	Deception regarding substance use (with low <i>ALC</i> , <i>DRG</i>)
<i>INF-F^a</i>	Infrequency-Front	First four <i>INF</i> items	Random responding on first half of PAI
<i>INF-B^a</i>	Infrequency-Back	Last four <i>INF</i> items	Random responding on second half of PAI
<i>ICN-C^a</i>	Inconsistency-Corrections		Inconsistent responses to two similar items regarding illegal behavior Inattention
Predictive Indices			
<i>TPI</i>	Treatment Process Index	Twelve configural features of the PAI associated with treatment amenability	Difficult treatment process, high probability of reversals
<i>VPI</i>	Violence Potential Index	Twenty configural features of the PAI associated with dangerousness to others	Increased likelihood of violence to others
<i>SPI</i>	Suicide Potential Index	Twenty configural features of the PAI associated with suicide	Increased likelihood of suicide

Note:

^a Developed for use in correctional settings (Edens & Ruiz, 2005).

The construction of the PAI sought to develop scales that provided a balanced sampling of the most important elements of the constructs being measured. This content coverage was designed to include a consideration of the *breadth* as well as the *depth* of the construct. The *breadth* of content coverage refers to the diversity of elements subsumed within a construct. For example, in measuring depression it is important to inquire about physiological and cognitive symptoms as well as features of affect. Depression scales that focus exclusively on one of these elements have limited breadth of

coverage and compromised content validity. The PAI sought to insure breadth of content coverage through the use of subscales representing the major elements of the measured constructs, as indicated by the theoretical and empirical literature.

The *depth* of content coverage refers to the need to sample across the full range of construct severity. To assure adequate depth of coverage, the scales were designed to include items reflecting both its milder and most severe forms. The use of four-alternative scaling provides each item with the capacity to capture differences in the severity of the manifestation of a feature of a particular disorder, and is further justified psychometrically in that it allows a scale to capture more true variance per item, meaning that even scales of modest length can achieve satisfactory reliability. This item type may also be preferred by clinicians considering particular items (e.g., risk indicators) or clients themselves, who often express dissatisfaction with forced-choice alternatives because they feel that the truth is between the two extremes presented.

In addition to differences in depth of severity reflected in response options, the items themselves were constructed to tap different levels of severity. For example, cognitive elements of depression can vary from mild pessimism to severe feelings of hopelessness, helplessness, and despair. Item characteristic curves were used to select items that provide information across the full range of construct severity. The nature of the severity continuum varies across the constructs. For example, severity on the Suicidal Ideation (*SUI*) scale involves the imminence of the suicidal threat. Thus, items on this scale vary from vague and ill-articulated thoughts about suicide to immediate plans for self-harm.

One implication of a careful consideration of content validity in the construction of a test is that it is assumed that item content is critical in determining an item's ability to capture the phenomenology of various disorders and traits, and hence its relevance for the assessment of the construct. Empirically derived tests may include items on a construct scale that have no apparent relation to the construct in question. However, research (e.g., Holden, 1989; Holden & Fekken, 1990; Peterson, Clark, & Bennett, 1989) has consistently indicated that such items add little or no validity to self-report tests. The available empirical evidence is entirely consistent with the assumption that the content of a self-report item is critical in determining its utility in measurement. This assumption does not preclude the potential utility of items that are truly "subtle" in the sense that a lay audience cannot readily identify the relationship of the item to mental health status. However, the assumption does suggest that the implications of such items for mental health status should be apparent to the expert diagnosticians for the item to be useful.

Although discriminant validity has been long recognized as an important facet of construct validity, it traditionally has not played a major role in the construction of psychological tests, and it continues to represent one of the largest challenges in the assessment of psychological constructs. There are a variety of threats to validity where discriminability plays a vital role. One such area involves *test bias*. A test that is intended to measure a psychological construct should not be measuring a demographic variable, such as gender, age, or sex. This does not mean that psychological tests should never be correlated with demographic variables, but rather that the magnitude of any such correlations should not exceed the theoretical overlap of the demographic feature with the construct. For example, nearly every indicator of antisocial behavior suggests that it is more common in men than in women; thus, it would be expected that an assessment of antisocial behavior would yield average scores for men that are higher than those for women. However, the instrument should demonstrate a considerably greater correlation with other indicators of antisocial behavior than it does with gender, otherwise it may be measuring gender rather than measuring the construct it was designed to assess.

The issue of test bias is one that is particularly salient in light of past abuses of testing and current legislation designed to prevent such abuses. However, such bias is just one form of potential

problems with discriminant validity. It is particularly common in the field of clinical assessment to find that a measure that supposedly measures one construct (such as anxiety or schizophrenia) is in fact highly related to many constructs. It is this tendency that makes many instruments quite difficult to interpret. How does the clinician evaluate an elevated score on a scale measuring “schizophrenia” if that scale is also a measure of alienation, indecisiveness, family problems, and depression? At each stage of the development of the PAI, items were selected that had maximal associations with indicators of the pertinent construct and minimal associations with the other constructs measured by the test.

Normative Data

The PAI was developed and standardized for use in the assessment of individuals in the age range of 18 through adulthood, and has been translated into several languages. Items were written to be easily understood and applicable across cultures; the initial reading level analyses of the PAI test items indicated that reading ability at the fourth-grade level was necessary to complete the inventory. A comparative study of similar instruments by Schinka and Borum (1993) supported the conclusion that the PAI items are written at a grade equivalent lower than estimates for comparable instruments—an important issue in forensic settings, where reading ability is commonly lower than average.

PAI scale and subscale raw scores are transformed to *T*-scores (mean of 50, standard deviation of 10) in order to provide interpretation relative to a standardization sample of 1,000 community-dwelling adults. This sample was carefully selected to match 1995 U.S. census projections on the basis of gender, race and age; the educational level of the standardization sample (mean of 13.3 years) was representative of a community group with the required fourth-grade reading level. For each scale and subscale, the *T*-scores were linearly transformed from the means and standard deviations derived from the census-matched standardization sample. Unlike several similar instruments, the PAI does not calculate *T*-scores differently for men and women; instead, combined norms are used for both genders. Separate norms are necessary only when the scale contains some bias that alters the interpretation of a score on the basis of the respondent's gender. To use separate norms in the absence of such bias would only distort the natural epidemiological differences between genders. In the example discussed above, women are less likely than men to receive the diagnosis of antisocial personality disorder, and this is reflected in the lower mean scores for women on the Antisocial Features (*ANT*) scale. A separate normative procedure for men and women would result in similar numbers of each gender scoring in the clinically significant range, a result that does not reflect the established gender ratio for this disorder. The PAI included several procedures to eliminate items that might be biased as a result of demographic features, and items that displayed any signs of being interpreted differently as a function of these features were eliminated in the course of selecting final items for the test. As it turns out, with relatively few exceptions, differences as a function of demography were negligible in the community sample. The most noteworthy effects involve the tendency for younger adults to score higher on the Borderline Features (*BOR*) and *ANT* scales, and the tendency for men to score higher on the *ANT* and Alcohol Problems (*ALC*) relative to women.

Because *T*-scores are derived from a community sample, they provide a useful means for determining whether certain problems are clinically significant, because relatively few normal adults will obtain markedly elevated scores. However, other comparisons are often of equal importance in clinical decision making. For example, nearly all patients report depression at their initial evaluation; the question confronting the clinician considering a diagnosis of Major Depressive Disorder is one of relative severity of symptomatology. Knowing that the individual's score on the

PAI Depression scale is elevated in comparison to the standardization sample is of value, but a comparison of the elevation relative to a clinical sample may be more critical in forming diagnostic hypotheses.

To facilitate these comparisons, the PAI profile form also indicates the *T*-scores that correspond to marked elevations when referenced against a representative clinical sample. This profile “skyline” indicates the score for each scale and subscale that represents the raw score that is 2 standard deviations above the mean for a clinical sample of 1,246 patients selected from a wide variety of professional settings. Scores above this skyline represent a marked elevation of scores relative to those of patients in clinical settings. Thus, interpretation of the PAI profiles can be accomplished in comparison to both normal and clinical samples.

The PAI manual provides normative transformations for a number of different comparisons. Various appendices provide *T*-score transformations referenced against the clinical sample and a large sample of college students, as well as for various demographic subgroups of the community standardization sample. Although the differences between demographic groups were generally quite small, there are occasions where it may be useful to make comparisons with reference to particular groups. Thus, the raw score means and standard deviations needed to convert raw scores to *T*-scores with reference to normative data provided by particular groups are provided in the manual for this purpose. Personnel selection norms and scoring software are also available through the test publisher. Although it is appropriate to make normative comparisons to a sample most representative of the individual being assessed, for most clinical and research applications the use of *T*-scores derived from the full normative data is strongly recommended because of the latter’s representativeness and large sample size.

However, comparison to correctional norms may importantly supplement decisions made in correctional settings. Edens and Ruiz (2005) have developed a Correctional Software Report using normative data gathered from multiple correctional settings in order to enhance the forensic utility of the PAI. Their normative sample consisted of inmates in a pre-release treatment facility in New Jersey ($N = 542$), a treatment program for convicted sex offenders in Texas ($N = 98$), state prison inmates in Washington ($N = 515$), and forensic inpatients in New Hampshire ($N = 57$). Overall, the sample averaged 33.6 years of age ($SD = 8.9$) and 10.8 years of education. A large proportion of the sample were male (68.1%), and racial/ethnic representation mirrored that of corrections settings (45.2% African American, 37.7% Caucasian, 7.8% Hispanic). As would be predicted, the correctional norm group tended to score roughly a standard deviation higher than the community standardization sample on Antisocial Features ($d = 1.15$), Drug Problems ($d = 1.62$), Stress ($d = 0.99$) and Borderline Features ($d = 0.84$), whereas they scored roughly a standard deviation lower on Treatment Rejection ($d = -0.96$). Furthermore, the correctional group also scored approximately 56 *T* ($d = 0.62$) on the Infrequency validity scale, suggesting that the correctional context may necessitate some adjustment in determinations of profile validity using the Infrequency (*INF*) scale in particular.

Reliability

The reliability of the PAI scales and subscales has been examined in terms of internal consistency (Alterman et al., 1995; Boyle & Lennon, 1994; Morey, 1991; Rogers, Flores, Ustad, & Sewell, 1995; Schinka, 1995), test-retest reliability (Boyle & Lennon, 1994; Morey, 1991; Rogers et al., 1995) and configural stability (Morey, 1991). Internal consistency alphas for the full scales are generally found to be in the .80s, whereas the subscales yield alphas in the .70s. The reliability of the PAI in corrections populations appears similar to that in the standardization sample (Edens & Ruiz, 2005). Although these numbers are acceptable, internal consistency estimates are generally

not the ideal basis for deriving the standard error of measurement (SEM) in clinical measures, because temporal instability is often of greater concern than inter-item correlations.

For the standardization studies, the median test-retest reliability value over a 4-week interval for the 11 full clinical scales was .86 (Morey, 1991), leading to SEM for these scales of the order of 3–4 *T*-score points, with 95% confidence intervals of ± 6 –8 *T*-score points. Absolute *T*-score change values over time were quite small across scales, on the order of two to three *T*-score points for most of the full scales (Morey, 1991). Boyle and Lennon (1994) reported a median test-retest reliability of .73 in their normal sample over 28 days. Rogers et al. (1995) found an average stability of .71 for the Spanish version of the PAI, administered over a 2-week interval.

Because multiscale inventories are often interpreted configurally, additional questions should be asked concerning the stability of configurations on the 11 PAI clinical scales. One such analysis involved determining the inverse (or Q-type) correlation between each participant's profile at Time 1 and the profile at Time 2. Correlations were obtained for each of the 155 participants in the full retest sample, and a distribution of the within-subject profile correlations was obtained. Conducted in this manner, the median correlation of the clinical scale configuration was .83, indicating a substantial degree of stability in profile configurations over time (Morey, 1991).

Validity

In the examination of test validity presented in the manual (Morey, 1991), a number of the best available clinical indicators were administered concurrently to various samples to determine their convergence with corresponding PAI scales. Since that time, a number of studies have investigated the validity of the PAI in a variety of populations and for numerous purposes. A comprehensive presentation of available validity evidence for the various scales is beyond the scope of this chapter; the PAI manual alone contains information about correlations of individual scales with over 50 concurrent indices of psychopathology (Morey, 1991). The following paragraphs provide some of the more noteworthy findings from such studies, in particular those related to forensic assessment, divided into the four broad classes of PAI scales: Validity, Clinical, Treatment Consideration, and Interpersonal.

Validity Scales

The assessment of profile validity is a critical component of any method intended for use in an evaluative context; the issue is compounded in most forensic settings where incentives for dishonest responding often exist. The PAI validity scales were developed to provide an assessment of the potential influence of certain response tendencies on PAI test performance, including both random and systematic influences upon test responding. A comparison of profiles derived from normal subjects, clinical subjects, and random response simulations in the standardization sample demonstrated a clear separation of scores of actual respondents from the random simulations, and 99% of these random profiles were identified as such by either the Inconsistency (*ICN*) or the Infrequency (*INF*) scale (Morey, 1991). To address the potential problem of partially random response sets, Morey and Hopwood (2004) developed an indicator of back random responding (e.g., a respondent discontinues attending to item content midway through the examination) involving front to back scaled score discrepancies on two PAI scales (*ALC* and *SUI*) with satisfactory positive and negative predictive power across levels and base rates of back random responding. Using another approach, Edens and Ruiz (2005) divided the *INF* scale into a front (*INF-F*) and back (*INF-B*) in the correctional software to assess partial random responding. They also introduced an Inconsistency scale for use in correctional settings (*ICN-C*) and consisting of a

single pair of items regarding admitting illegal behavior in the past. Individuals who admit to illegal behavior on one item and deny it on the other are suspected of inattention, as this occurred in only 4% of the correctional standardization sample.

Systematic profile distortion can occur in the positive or negative directions (Caruso, Benedek, Auble, & Bernet, 2003; Holden, Book, Edwards, Wasylikiw, & Starzyk, 2003). It can also occur in both directions on the same profile. For example, an individual may wish to emphasize feelings of anxiety and depression (i.e., negatively dissimulate) and also disguise their use of alcohol or drugs (i.e., positively dissimulate). Systematic profile distortion can also occur in at least two ways. Individuals may consciously present themselves in a manner that is inconsistent with their experience or historical fact based on their perception of the consequences of their presentation (i.e., malingering or fake good). Conversely, individuals may present themselves in a way that is consistent with their subjective experience but which most other individuals would see as an exaggeration (if overly negative) or as lack of insight (if overly positive). This latter type of profile distortion has been referred to as self-deception, and is related to specific forms of psychopathology. For example, cognitive symptoms of depression involve an underestimation of one's own abilities and worth and an overestimation of obstacles to emotional health or achievement. Conversely, narcissistic or manic disorders often involve an expansive approach to life, and limited capacity for a critical appraisal of one's real abilities and prospects. PAI indices of systematic profile distortion were developed to assess self- and other-deception in both the positive and negative directions.

The three indices developed to directly assess positive dissimulation have been subjected to numerous empirical studies. In the initial validation studies described in the test manual, individuals scoring above the threshold on the Positive Impression Management (*PIM*) scale were 13.9 times more likely to be in the positive dissimulation sample than a community sample (Morey, 1991). Subsequent studies have generally supported the ability of these scales to distinguish simulators from actual protocols under a variety of response set conditions. For example, the studies described in the test manual found that the point of rarity on *PIM* between the distributions of the impression management sample (i.e., "fake good") and the community normative sample was at a raw score of 57*T*. Application of this cut score resulted in a sensitivity in the identification of defensiveness of 82%, and a specificity with respect to normal individuals of 70%. Although not reported in the manual, Positive and Negative Predictive Power (PPP, NPP) can be computed from these values, given a base-rate value for intentional positive distortion. For example, at 50% base rate, PPP = 73% and NPP = 79.5%; at 25% base rate, PPP = 57.7% and NPP = 88.6%. These findings have been replicated in several samples (Morey & Lanier, 1998); for example, a study by Cashel, Rogers, Sewell, and Martin-Cannici (1995) also identified 57*T* as their optimal cutting score. Their study, in which respondents were coached regarding believability of results, yielded sensitivity and specificity rates of 48% and 81%, respectively (PPP = 71.6% and NPP = 60.9% when base rate = 50%, PPP = 55.8% and NPP = 75.7% when base rate = 25%). Peebles and Moore (1998) also found a cutting score of 57*T* to be optimal for their sample, resulting in a hit rate of 85% in distinguishing forthright from fake-good responders. A study by Fals-Stewart (1996) found that the 57*T* cut score on *PIM* had a sensitivity of 88% in identifying "questionable responding" in substance abusers (e.g., forensic patients who denied substance use but had positive urine screens), with a specificity of 80% in honest responding groups (PPP = 81.5% and NPP = 87.0% when base rate = 50%, PPP = 68.8% and NPP = 93.0% when base rate = 25%). Finally, Edens and Ruiz (2006) found that the 57*T* cut score for *PIM* moderated the prediction of institutional misconduct from PAI clinical scales in correctional settings, underscoring the potential importance of validity scale elevations in forensic decision making.

The Defensiveness Index (*DEF*; Morey, 1996) is a composite of PAI scale configurations that tend to occur more frequently in profiles of individuals instructed to form a positive impression

than in honest respondents from clinical or community populations. Peebles and Moore (1998) reported a hit rate of 83% in distinguishing honest from fake-good respondents with the *DEF*. However, Cashel et al. (1995) reported that coached dissimulators obtained a lower mean score on the *DEF* than naïve dissimulators, though the coached mean was still roughly one standard deviation above the norm for community samples. Cashel et al. (1995) developed the Cashel Discriminant Function (*CDF*) to measure positive dissimulation unassociated with self-deception to indicate the possibility that an examinee is “faking good.” The *CDF* is a function composed of weighted scale scores that was found to optimally distinguish defensive and honest respondents in their study. Cashel et al. demonstrated *CDF* sensitivities ranging from 79% to 87% in identifying the falsified profiles of college students and jail inmates, and a specificity of 88%. Using the conservative value of 79% as the sensitivity estimate, $PPP = 86.8\%$ and $NPP = 80.7\%$ at 50% base rate, and $PPP = 76.9\%$ and $NPP = 8.9\%$ at 25% base rate. These results, which have been replicated with naïve dissimulators (Morey, 1996; Morey & Lanier, 1998) suggest the ability of the *CDF* to increment *PIM* and *DEF* in the detection of positive dissimulation.

A number of examinations of the utility of the Negative Impression Management (*NIM*) scale in the evaluation of malingering have also been reported in the literature. In the initial validation studies cited in the test manual, individuals scoring above the critical level of *NIM* were 14.7 times more likely to be a member of the malingering group than of the clinical sample. Rogers, Ornduff, and Sewell (1993) examined the effectiveness of *NIM* in identifying both naïve and sophisticated simulators (advanced graduate students in clinical and counseling psychology) who were given a financial incentive to avoid detection as malingerers while attempting to feign specific disorders. Rogers et al. found that the recommended *NIM* scale cutoff (84*T*) successfully identified 91% of participants attempting to feign schizophrenia, 56% of participants simulating depression, and 39% of participants simulating an anxiety disorder. In contrast, only 3% of control participants were identified as simulators. Rogers et al. concluded that *NIM* is most effective in identifying the malingering of more severe mental disorders. Interestingly, there was no effect of subject sophistication: the scale was equally effective in identifying naïve and sophisticated malingerers. Along similar lines, Bagby, Nicholson, Bacchocchi, Ryder, and Bury (2002) found *NIM* to yield only a moderate effect size ($d = .44$ for coached and $.53$ for uncoached malingerers vs. bona fide patients) in detecting a group that appeared to be malingering depression, with average *NIM* scores of 74*T* for uncoached and 72*T* for coached malingerers. Gaies (1993) conducted a similar study of malingering, focusing upon the feigning of clinical depression, and reported average scores on *NIM* of 92*T* for sophisticated malingerers and 81*T* for naïve malingerers. While the Gaies (1993) and Bagby et al. (2002) studies demonstrated that *NIM* was elevated relative to honest responding groups, the results are similar to those of Rogers et al. (1993) in suggesting that individuals attempting to simulate milder forms of mental disorder (in these studies, depression) will obtain more “moderate” elevations on *NIM*. Finally, Kucharski, Toomey, Fila, and Duncan (2007) found that *NIM* demonstrated convergence in the detection of malingering in criminal defendants as assessed by the Structured Interview of Reported Symptoms (SIRS; Rogers, Bagby, & Dickens, 1992), a widely used measure of malingering in forensic settings.

The utility of *NIM* in detecting simulated post-traumatic stress disorder (PTSD) has also received attention. Liljequist, Kinder, and Schinka (1998) reported significant *NIM* elevations in both clinical (alcohol mean = 62.93, $SD = 18.25$; PTSD mean = 67.93, $SD = 16.94$) and malingering (mean = 92.48, $SD = 28.40$) groups relative to community norms or their college student samples. Calhoun, Earnst, Tucker, Kirby, and Beckham (2000) compared PAI scores among PTSD-diagnosed veterans, college student PTSD simulators, and community controls. A *NIM* score of 80*T* detected 75% of feigned protocols and demonstrated a hit rate of 59%, whereas a score of 102*T* detected 44% of simulators and demonstrated a hit rate of 63%. Scragg, Bor, and Mendham

(2000) reported a sensitivity of 54% and a specificity of 100% for distinguishing malingered from true PTSD with the *NIM* scale (PPP = 100.0% and NPP = 68.5% when base rate = 50%, PPP = 100.0% and NPP = 81.3% when base rate = 25%). Finally, Thomas, Hopwood, Orlando, Weathers, and McDevitt-Murphy (2011) more recently examined *NIM*'s ability to detect feigned PTSD in comparison to genuine responders, and were able to correctly classify 75% of participants with a *NIM* score greater than 69*T*, and Rogers, Gillard, Wooley, and Ross (2012) were able to correctly identify 72% of genuine PTSD responders with a *NIM* cutoff score of 84*T*.

The Malingering Index (*MAL*; Morey, 1996), a composite of PAI configural indicators, was introduced to measure malingering more directly than *NIM*, which is often affected by response styles consequent to psychopathology in addition to overt attempts at negative dissimulation. *MAL* has been shown to differentiate individuals in a malingering sample from the community and clinical standardization samples (Morey, 1996). Wang et al. (1997) reported a statistically significant positive correlation between *MAL* and two scales of the SIRS: Improbable or Absurd Symptoms ($r = 0.39$) and Symptom Combinations ($r = 0.42$). Gaies (1993) reported *MAL* (cut score = 3) sensitivity estimates of 56.6% and 34.2% in identifying informed and naïve malingerers, respectively. Specificity with respect to depressed controls in that study was 89% and with respect to community controls was 100%. Calhoun et al. (2000) reported that a *MAL* cut score of 3 detected 44% of simulators in their sample and misclassified several individuals with PTSD as malingerers. Scragg et al. (2000) reported sensitivity of 45% and specificity of 94% in distinguishing malingered from true PTSD with *MAL* (PPP = 88.2% and NPP = 63.1% when base rate = 50%, PPP = 78.6% and NPP = 77.7% when base rate = 25%). These results suggest both an association with psychopathology, though less so than is the case with *NIM*, and a sensitivity which declines to the extent that less severe forms of psychopathology are malingered. However, Blanchard, McGrath, Pogge, and Khadivi (2003) reported that *MAL* was the most efficient PAI indicator of malingering in their data set, with a hit rate of 94% and PPP = 96%, NPP = 94% at their base rate of 11% malingerers. On a cautionary note, Bagby et al. (2002) found that *MAL* yielded a moderate effect size in distinguishing actual patients from uncoached malingerers, but their coached malingering group achieved similar *MAL* scores to their patient sample, suggesting that *MAL* may not be as effective in detecting sophisticated malingerers.

Rogers, Sewell, Morey, and Ustad (1996) developed the Rogers Discriminant Function (*RDF*), an index that appears able to indicate malingering but which is not associated with psychopathology (i.e., does not systematically correlate with the clinical scales). This research group demonstrated and cross-validated sensitivity and specificity estimates above 80% in distinguishing both coached and naïve malingerers from honest respondents. Morey (1996) replicated these results using naïve college student simulators, and found *RDF* to be the most accurate among the PAI negative validity indicators in identifying malingerers. In fact, *RDF* resulted in PPP estimates $>.80$ across malingering base rates of 20%, 50%, and 80%. The Bagby et al. (2002) study found that *RDF* was superior to *NIM* and *DEF* and incremented every Minnesota Multiphasic Personality Inventory–2 (MMPI–2; Butcher et al., 2001) malingering index, although the MMPI–2 indices *F* and *F(p)* were able to increment the predictive capability of *RDF*. In a similar study, Blanchard et al. (2003) reported that *NIM*, *MAL*, and *RDF* all demonstrated hit rates $>.93$ in identifying simulators, and that the PAI indicators significantly incremented MMPI–2 indicators of malingering, and vice versa. However, these authors also noted that eight MMPI–2 malingering indicators, considered as a group, yielded a larger *r*-square in regression prediction of malingering status than the grouping of the four PAI indicators (including the *CDF*) studied. Scragg et al. (2000) reported a sensitivity of 63% and specificity of 94% for the *RDF* in their study of malingered PTSD (PPP = 91.3% and NPP = 71.8% when base rate = 50%, PPP = 83.8% and NPP = 83.9% when base rate = 25%). However, Wang et al. (1997) failed to find a relationship between *RDF* and the SIRS malingering

classification in a small correctional sample. Similarly, Rogers, Sewell, Cruise, Wang, and Ustad (1998) tested *RDF* in a forensic setting against a criterion using a *SIRS* classification and observed a hit-rate shrinkage to 62%. On the basis of these results, Rogers et al. recommended use of *RDF* to screen for, but not directly indicate, malingering in forensic settings. However, Edens, Poythress, and Watkins-Clay (2007) found that *RDF* was more successful than the *SIRS* in distinguishing staff-identified malingering patients from bona fide patients, implying that the selection of a criterion standard for determining malingering may play at least as large a role in study results as the influence of the forensic context.

Clinical Scales

The clinical scales of the PAI were assembled to provide information about clinical constructs relevant in a variety of forensic and clinical contexts. A number of instruments have been used to provide information on the convergent and discriminant validity of the PAI clinical scales. Correlations tend to follow hypothesized patterns; for example, strong associations are found between neurotic-spectrum scales such as Anxiety (*ANX*), Anxiety Related Disorder (*ARD*), and Depression (*DEP*) and other psychometric measures of neuroticism (Costa & McCrae, 1992; Montag & Levin, 1994; Morey, 1991). The *ARD* scale has also been found to correlate with the probability of experiencing nightmares, with *ARD-T* in particular being associated with night terrors (Greenstein, 1995). The *ARD* scale (particularly *ARD-T*) has also been found to differentiate women psychiatric patients who were victims of childhood abuse from other women patients who did not experience such abuse (Cherepon & Prinzhorn, 1994). Edens and Ruiz (2005) reported that *ARD-T* was effective at detecting inmates released from a forensic inpatient unit with a diagnosis of PTSD ($AUC = 0.83$, $SE = .05$). The *DEP* scale demonstrates its largest correlations with various widely used indicators of depression, such as the Beck Depression Inventory (*BDI*; Beck & Steer, 1987), the Hamilton Rating Scale for Depression (Hamilton, 1960), and the Wiggins (1966) *MMPI* Depression content scale (Morey, 1991; Ban, Fjetland, Kutcher, & Morey, 1993). Several indicators have been found useful in discriminating PTSD from autism spectrum disorder (*ASD*) among individuals traumatized in motor-vehicle accidents (G. E. Holmes, Williams, & Haines, 2001), and the PAI has demonstrated effectiveness in distinguishing true from malingered PTSD in psychiatric samples (Liljeist et al., 1998) and samples of combat veterans (Calhoun et al., 2000). It has also been suggested that the PAI provides important diagnostic information about individuals with pain (George & Wagner, 1995; Karlin et al., 2005) and eating disorder (Tasca, Wood, Demidenko, & Bissada, 2002) problems. Interestingly, the Somatic Complaints (*SOM*) scale, particularly *SOM-C*, has proven useful in the distinction between epileptic and non-epileptic seizures (Wagner, Wymer, Topping, & Pritchard, 2005; Testa, Lesser, Krauss, & Brandt, 2011), suggesting that the PAI may be helpful in cases where functional versus organic origins of physical complaints are an issue.

The assessment of psychotic disorders is particularly relevant in forensic settings, where the base rate of psychosis is often higher (e.g., correctional settings) and where individuals' mental status is often a prominent issue. Within the psychotic spectrum, PAI scales such as Paranoia (*PAR*), Mania (*MAN*), and Schizophrenia (*SCZ*) have been correlated with a variety of other indicators of severe psychopathology (Morey, 1991). For example, *PAR* has been found to correlate particularly well with diagnostic assessments of paranoia made via structured clinical interview (Rogers, Ustad, & Salekin, 1998). Also, the *SCZ* scale has been found to distinguish schizophrenic patients from controls (Boyle & Lennon, 1994). In that study, the schizophrenic sample did not differ significantly from a sample of alcoholics on *SCZ* scores, although certain characteristics of their criterion group (patients on medication maintenance) and their alcoholic group (alcoholics undergoing

detoxification) might have in part accounted for their findings (Morey, 1995). *SCZ* has also been shown to significantly increment the Rorschach Comprehensive System Schizophrenia Index (*SCZI*; Exner, 1993) in differentiating inpatients with schizophrenic spectrum disorders from inpatients with other diagnoses, whereas *SCZI* was unable to increment *SCZ* (Klonsky, 2004). These data suggest that efforts during the construction of the PAI to enhance discriminant validity on the *SCZ* scale were justified, as the control group comprised psychiatric inpatients, many of whom met criteria for disorders with symptoms similar to schizophrenia (e.g., organic psychoses, cognitive deficits). Nonetheless, given the limited research and mixed findings (e.g., Edens & Ruiz, 2005) with the *SCZ* scale, some have suggested it be interpreted as a measure of general impairment, rather than as a specific marker of schizophrenia (Rogers, Ustad, & Salekin, 1998). Combining the PAI profile with information from other assessment sources may be particularly important for differential diagnosis of psychotic disorders (Edens et al., 2001).

Personality disorders are common in forensic examinees, and represent risk factors for a variety of problematic behaviors common in forensic settings, such as violence, substance use, and poor treatment response. Two scales on the PAI directly target personality disorders: the Borderline Features (*BOR*) scale and the Antisocial Features (*ANT*) scale. The choice to include these two constructs on the PAI was based on the fact that the majority of the literature on personality disorders relates to these constructs. Both *BOR* and *ANT* have been found to relate to other measures of these constructs as well as to predict relevant behavioral outcomes (e.g., Salekin, Rogers, Ustad, and Sewell, 1998; Trull, Useda, Conforti, & Doan, 1997). The *BOR* scale has been found to correlate with the MMPI Borderline scale (Morey, 1991), the Bell Object Relations Inventory (Bell, Billington, & Becker, 1985; Kurtz, Morey, & Tomarken, 1992), and the NEO–Personality Inventory (NEO–PI; Costa & McCrae, 1992) Neuroticism scale (Costa & McCrae, 1992). Edens and Ruiz (2005) reported an impressive ability for *BOR* to detect forensic inmates discharged with a diagnosis of borderline personality disorder ($AUC = 0.97$, $SE = .032$). Other studies have supported the validity and utility of this scale in borderline treatment samples (e.g., Evershed et al., 2003; Yeomans, Hull, & Clarkin, 1994) and in predicting general maladjustment in college students (Bagge et al., 2004). It has also been demonstrated that an assessment of *BOR* increments the information provided by a diagnosis of major depressive disorder (Kurtz & Morey, 2001). The *BOR* scale in isolation has been found to distinguish borderline patients from unscreened controls with an 80% hit rate, and successfully identified 91% of these participants as part of a discriminant function (Bell-Pringle, Pate, & Brown, 1997). Classifications based upon the *BOR* scale have been validated in a variety of domains related to borderline functioning, including depression, personality traits, coping, Axis I disorders, and interpersonal problems (Trull, 1995). These *BOR* scale classifications were also found to be predictive of 2-year outcome on academic indices in college students, even controlling for academic potential and substance abuse (Trull et al., 1997).

The *ANT* scale demonstrated its largest correlations in initial validation studies (Morey, 1991) with the MMPI Antisocial personality disorder (Morey, Waugh, & Blashfield, 1985) and the Self-Report Psychopathy test designed by Hare (1985) to assess his model of psychopathy. Subsequent studies have also supported the validity of *ANT*. Salekin, Rogers, and Sewell (1997) examined the relationship between *ANT* and psychopathic traits in a sample of female offenders and found that elevations on *ANT* among this population were primarily the result of endorsements on Antisocial Behaviors (*ANT-A*). Also, support was found for the convergent validity of *ANT* with other measures of psychopathy, including the Revised Psychopathy Checklist (PCL–R; Douglas, Guy, Edens, Boer, & Hamilton, 2007; Hare, 1991) total score and the Personality Disorder Examination (Loranger, 1988) Antisocial scale. Edens, Hart, Johnson, Johnson, and Olver (2000) examined the relationship of the *ANT* scale to the screening version of the Psychopathy Checklist (PCL:SV;

Hart, Cox, & Hare, 1995) and the PCL-R. Moderately strong correlations were found between *ANT* and the PCL:SV and the PCL-R total score, the highest correlations with these measures being found with *ANT-A*. Salekin et al. (1998) demonstrated the ability of the *ANT* and the Aggression (*AGG*) scales of the PAI to predict recidivism among female inmates over a 14-month follow-up interval. *ANT* has also demonstrated validity in predicting violence in a sample of incarcerated mentally ill individuals (Wang & Diamond, 1999), and in predicting treatment course for a sample of borderline females (Clarkin, Hull, Yeomans, Kakuma, & Cantor, 1994). In a number of comparisons, *ANT* has shown to both increment and outperform the PCL-R in predicting disciplinary infractions as well as recidivism at follow up (Buffington-Vollum, Edens, Johnson, & Johnson, 2002; Walters & Duncan, 2005). Finally, the PAI, and in particular the *BOR*, *ANT*, and Aggression (*AGG*) scales, has demonstrated the capacity to differentiate various types of antisocial individuals, such as generally violent men versus men who abuse family members but are not violent outside the home (DeSol, Margolin, & John, 2003) and individuals convicted of violent versus non-violent crimes (Edwards, Scott, Yarvis, Paizis, & Panizzon, 2003).

The PAI contains two scales, Alcohol Problems (*ALC*) and Drug Problems (*DRG*), that inquire directly about behaviors and consequences related to alcohol and drug use, abuse, and dependence. These scales demonstrate a similar pattern of correlates: strong correlations with corresponding measures of substance abuse and moderate associations with indicators of behavior problems and antisocial personality (Alterman et al., 1995; Parker, Daleiden, & Simpson, 1999). The *ALC* scale has been found to differentiate patients in an alcohol rehabilitation clinic from patients with schizophrenia (Boyle & Lennon, 1994) as well as normal controls (Ruiz, Dickinson, & Pincus, 2002). The *DRG* scale has been found to successfully discriminate drug abusers and methadone maintenance patients from general clinical and community samples (Alterman et al., 1995; Kellogg et al., 2002). *DRG* has also demonstrated strong correlations with the Addiction Severity Index (*ASI*; McLellan et al., 1992) and predicted both involvement (as measured by urine samples) with and negative consequences of substance use in a clinical sample (Kellogg et al., 2002). Another interesting finding from the Kellogg et al. study was that individuals who had been in methadone maintenance treatment for more than 1 year produced lower *DRG* scores than individuals who had been treated for less than 1 year. The authors attributed this effect to the biological and psychosocial stabilization that occur as individuals maintain successful methadone treatment. In a correctional sample, Edens & Ruiz (2008) found that the *DRG* scale demonstrated significant associations with independent substance abuse diagnosis in a correctional inpatient facility, further supporting the validity of this scale in the correctional context. However, because the items for *ALC* and *DRG* inquire directly about substance use, it should be recognized that these scales are susceptible to denial or under-reporting. Empirically derived procedures to assess the likelihood that a profile under-represents the extent of alcohol or drug problems (Edens & Ruiz, 2005; Fals-Stewart, 1996; Fals-Stewart & Lucente, 1997; Morey, 1996) will be described later in the chapter.

Treatment Consideration Scales

The treatment consideration scales of the PAI were assembled to provide indicators of potential complications in treatment that would not necessarily be apparent from diagnostic data, although they also have a variety of applications to forensic settings, where issues such as aggression, suicide, and stress are important and commonly encountered. The treatment consideration scales were developed to specifically assess the potential for harm to self or others, environmental circumstances, and motivation for psychological treatment.

Correlations between the PAI treatment consideration scales and an array of validation measures provide support for the construct validity of these scales (Costa & McCrae, 1992; Morey, 1991).

Substantial correlations are described between the Aggression (*AGG*) scale and the NEO-PI Hostility (.83) and the Spielberger State-Trait Anxiety Inventory (STAXI; Spielberger, 1988) Trait Anger scale (.75) and Anger Control scales (–.57) in the test manual. *AGG* has also been extensively researched in forensic settings. Wang et al. (1997) found significant positive relations between *AGG* and all three of its subscales to the Overt Aggression Scale (OES; Yudofsky, Silver, Jackson, Endicott, & Williams, 1986) in a sample of inmates receiving or requesting psychiatric services. Wang and Diamond (1999) identified the Physical Aggression (*AGG-P*) and Verbal Aggression (*AGG-V*) scales as related to institutional misbehavior among mentally ill offenders. Salekin et al. (1998) found that *AGG* predicted recidivism among female offenders over a 14-month follow-up. Walters, Duncan, and Geyer (2003) reported that *AGG* significantly incremented the PCL–R and demographic variables in predicting disciplinary adjustment in prison inmates. Magyar et al. (2012) found that *AGG* demonstrated incremental validity, above and beyond the predictive validity of other psychopathological constructs, in the identification of general non-compliance and aggressive behavior. The *AGG* scale can also be used in conjunction with *ANT*, *BOR*, and other scales in risk assessments (e.g., Delsol et al., 2003).

The Suicidal Ideation (*SUI*) scale was positively correlated with the Beck Hopelessness Inventory (Beck & Steer, 1988; .64), the BDI, and the Suicidal Ideation (.56) and Total Score (.40) of the Suicide Probability Scale (SPS; Cull & Gill, 1982). It was also found to be negatively correlated with the Perceived Social Support (PSS; Procidano & Heller, 1983) scales in the standardization studies. Wang et al. (1997) found that *SUI* had the largest correlation among PAI scales to the number of suicide risk assessments in a large sample of incarcerated men, and that the scale could reliably differentiate individuals who had or had not made suicidal gestures. Rogers, Ustad, and Salekin (1998) found that *SUI* demonstrated moderate to large correlations with other self-report measures of suicidal ideation in a sample of correctional emergency referrals.

As expected, the Nonsupport (*NON*) scale was found to be highly (and inversely) correlated with the social support measures: –.67 with PSS–Family, –.63 with PSS–Friends in the standardization studies. It was also moderately associated with numerous measures of distress and tension. The Stress (*STR*) scale displayed its largest correlations with the Schedule of Recent Events (SRE; .50), a unit-scoring adaptation of the widely used T. H. Holmes and Rahe (1967) checklist of recent stressors, and was also associated with various indices of depression and poor morale.

Finally, the Treatment Rejection (*RXR*) scale is found to be negatively associated with Wiggins Poor Morale (–.78) and the NEO–PI Vulnerability (–.54) scales, consistent with the idea that distress motivates treatment utilization and effort. *RXR* has been shown to be positively associated with indices of social support, implying that people are less likely to be motivated for treatment if they have an intact and available support system as an alternative. This scale has predicted treatment non-compliance in a sample of sex-offending inmates (Caperton, Edens, & Johnson, 2004).

Interpersonal Scales

The interpersonal scales of the PAI were designed to provide an assessment of the interpersonal style of subjects along two dimensions: (1) a warmly affiliative versus a cold, rejecting axis; and (2) a dominating, controlling versus a meekly submissive style. These axes provide a useful way of conceptualizing variation in normal personality as well as many different mental disorders, and people at the extremes of these dimensions may present with a variety of disorders. The PAI manual describes a number of studies indicating that diagnostic groups differ on these dimensions; for example, spouse-abusers usually score relatively high on the Dominance (*DOM*) scale, whereas schizophrenics tend to be low on the Warmth (*WRM*) scale (Morey, 1991). Correlations with related measures also provide support for the construct validity of these scales. For example, the

correlations with the Interpersonal Adjective Scales–Revised (IAS–R; Wiggins, 1979) vector scores are consistent with expectations, with *DOM* associated with the IAS–R dominance vector (.61) and *WRM* associated with the IAS–R love vector (.65). The NEO–PI Extroversion scale roughly bisects the high-*DOM*/high-*WRM* quadrant, as it is moderately positively correlated with both scales; this finding is consistent with previous research (Trapnell & Wiggins, 1990). The *WRM* scale was also correlated with the NEO–PI Gregariousness scale (.46), while *DOM* was associated with the NEO–PI Assertiveness facet (.71). Several studies have found that high *DOM* and low *WRM* scores are associated with non-compliance, aggressive acts, and overall poor treatment progress in offender populations (Doyle & Dolan, 2006; Edens, 2009; Magyar et al., 2012).

Forensic Assessment with the PAI

Forensic assessment may involve questions of competency, criminal responsibility, diagnosis, treatment planning, or risk prediction in correctional, child custody, emotional injury, disability, or other psycholegal contexts as well as questions regarding the valid use of the instrument in forensic decision making. A multiscale instrument used in forensic settings must be able to answer questions regarding response distortion, clinical diagnosis, substance abuse, risk of violence, suicide, and treatment planning. The PAI appears to be especially useful in a forensic context because of its ability to measure constructs related to that population (e.g., *BOR*, *ANT*, *DRG*, *ALC*) as well as its ability to detect invalid profiles (e.g., Douglas, Hart, & Kropp, 2001; Edens et al., 2001; Wang et al., 1997), and the PAI is currently being used in a wide variety of civil and criminal cases both within the United States and internationally (Mullen & Edens, 2008). The following is a brief review of the PAI with respect to the areas most relevant to forensic issues.

Response Distortion

The accuracy of information provided in forensic assessments is a persistent concern; in addition to the possibility of inattention or poor concentration leading to random responding, there is often motivation in such settings for individuals to appear more or less pathological than they actually are (Caruso et al., 2003). For example, inmates in correctional settings may malingering psychopathology in order to procure services (e.g., psychopharmacological agents), whereas individuals on probation may effortfully disguise substance use in order to eschew more stringent probationary requirements. The PAI offers a number of validity indices designed to assess the factors that could lead to distorted profiles.

As already discussed, *INF* and *ICN* are useful as indicators of random responding. Moderate elevations on *INF* may additionally suggest an idiosyncratic, as opposed to random, approach to the test, as might be expected from an individual with a cognitive deficit or psychotic disorder. Such elevations should be followed up by asking the individual to explain their responses to particular *INF* items if a non-random and non-normative response set is suspected. Moderate *ICN* elevations may suggest inattention, as might be found in someone who is agitated, intoxicated, or who has attentional problems. Finally, if it is suspected that the respondent began attending appropriately but finished the test randomly, the first half of the test can be used to estimate the full scale scores for the entire profile. Back random responding can be detected by comparing discrepancies on the first and last half of the *SUI* and *ALC* scales; differences greater than 5*T* on both scales suggest that the two halves of the test were not approached in the same way (Morey & Hopwood, 2004).

In general, research on the detection of more systematic forms of distortion has supported the utility of various PAI indicators for this purpose. For example, Hawes and Boccaccini (2009)

conducted a meta-analysis examining the ability of the PAI to detect general malingering, describing consistently large effect sizes across studies. Specifically, when comparing uncoached malingerers to genuine responders, *NIM* exhibited an overall effect size of $d = 1.48$, *MAL* exhibited an overall effect size of $d = 1.15$, and *RDF* exhibited an overall effect size of $d = 1.13$. In addition, it was determined that cut scores of $NIM \geq 81$ and $MAL \geq 3$ were the most optimal for overall correct identification of malingering. These results suggest a generally robust ability of the PAI to detect the over-reporting of psychopathology across a variety of criteria and settings.

However, in evaluating test data it is often important to distinguish effortful from non-effortful forms of distortion in the respondent's self-report. Toward this end, the pattern of correlations between the validity indices for both positive and negative dissimulation with clinical phenomena suggest that whereas *NIM* and *PIM* are heavily influenced by self-deception, *MAL* and *DEF* are less so, and *RDF* and *CDF* are not associated with either psychopathology or *NIM* or *PIM*, respectively. As such, the combination of three scales representative of positive and negative distortion provides the evaluator with a method to tease out the relative influences of psychopathology and intentional distortion in making decisions about invalidity (Morey, 2003), as will be discussed.

Although moderate elevations may be due to either conscious or unconscious response bias, more extreme *NIM* elevations make malingering the more likely hypothesis. *RDF* does not correlate with *NIM* or with psychopathology, and it displays moderate correlations with *MAL*. This suggests that *RDF* elevations represent intentional malingering and are free from the negative response set common among individuals with clinical disorders. This pattern of correlations also suggests that *MAL* is somewhere between *NIM* and *RDF*, in that it is prone to ego-syntonic negative coloration less than *NIM* but is not as clean a measure of malingering as is *RDF*. As such, a configuration in which *NIM* is moderately elevated, *MAL* is mildly elevated, and *RDF* is in the moderate range suggests an individual who is not intentionally misrepresenting their experience. Nevertheless, the profile ought to be viewed with their generally negative view of themselves and their situation in mind. Conversely, an elevation on *RDF*, which may also typically involve elevations on *NIM* and is likely to involve an elevation on *MAL*, suggests conscious malingering.

There are also three main indices of positive distortion. The *PIM* scale is affected by both self- and other-deception. Some individuals may obtain moderate elevations on *PIM* because they have a Pollyanna-like view of life and their situation, and may also under-report clinical phenomena for the same reason. Individuals naïve to psychopathology who are trying to present themselves in a favorable light may also obtain elevations on *PIM*. *DEF* is somewhat less prone to the influence of self-deception, as it is moderately correlated with *PIM* and has been shown to increment *PIM* in distinguishing individuals trying to fake good. *CDF* is uncorrelated with *PIM* and moderately correlated with *DEF*, and thus provides an index of intentional deception free from the effects of repression or naïveté. Thus, a similar strategy as was used with respect to negative dissimulation can be employed here, with elevations on *PIM* in the absence of *CDF* representing naïveté, and elevations on *CDF* more likely representing conscious distortion.

Clinical Diagnosis

Clinical diagnosis may be relevant in a variety of forensic contexts, including the assessment of risk factors, treatment selection, and in determining an individual's mental status, for example in a competency evaluation. No single method should be used in isolation to make psychiatric diagnoses in any setting. However, the PAI provides several scales that are useful in suggesting or confirming diagnoses in conjunction with other methods (Edens & Ruiz, 2008). In addition, several configural algorithms have been suggested as indicative of particular diagnoses (Morey, 1996).

Disorders associated with high degrees of negative affect, which have high base rates in clinical and forensic settings relative to the community, are assessed by four PAI full scales, each having three subscales: *SOM*, *ARD*, *ANX*, and *DEP*. The Somatic Complaints (*SOM*) scale was designed to address the phenomenology of individuals reporting pain or somatic deficits. It includes the Conversion Disorders (*SOM-C*), Somatic Complaints (*SOM-S*), and Health Concerns (*SOM-H*) subscales. The use of *SOM* in diagnostic formulations is most helpful when used in concert with medical information, reported symptoms, and an understanding of the assessment context. *SOM* subscales can be used in combination with the validity indices and external data to differentiate individuals who are malingering somatic problems, converting psychological problems to somatic symptoms as a characteristic strategy to minimize anxiety, or reporting genuine symptoms and severity (e.g., Wagner et al., 2005). The difference between these classes of individuals, all of whom are likely to obtain elevations on *SOM*, is a common and important forensic issue, particularly in contexts where there is motivation to exaggerate symptoms, such as disability evaluations. The administration of a self-report instrument such as the PAI is the industry standard in such cases (Bianchini, Etherton, & Greve, 2004; Slick, Sherman, & Iverson, 1999).

The Anxiety Related Disorders (*ARD*) scale assesses three anxiety phenomena: fears and phobias (*ARD-P*), compulsions and overall need for order (*ARD-O*) and trauma history and post-traumatic symptoms (*ARD-T*). *ARD* is one of the least internally consistent PAI scales (Morey, 1991) because the disorders represented by the *ARD* subscales represent somewhat distinct clinical phenomena, though all have in common a high degree of negative affect. Thus, it is common for the subscales to operate independently and thus reflect specific anxiety reactions rather than generalized anxiety. In concert with other scales, these subscales can also indicate personological factors important in clinical and forensic prediction, such as likely reaction to stress, interpersonal style, and organization of experience. For example, individuals with *ARD-O* elevations tend to be affectively constricted and interpersonally rigid, whereas individuals with *ARD-P* elevations tend to be cautious and fearful. *ANX* elevations indicate general tension or agitation and are useful in indicating individuals' capacity to cope with stressful environments (e.g., incarceration). The *ANX* subscales can be used to predict whether anxiety symptoms will be expressed as subjective feelings of distress and nervousness (*ANX-A*), rumination and obsessive or chaotic thinking (*ANX-C*), or physical symptoms (*ANX-P*). The Depression (*DEP*) scale measures depressive symptoms, and a diagnosis of major depressive disorder is most likely if all three subscales are elevated. The *DEP* subscales assess cognitive (*DEP-C*), affective (*DEP-A*), and physiological (*DEP-P*) aspects of depression. *DEP-C* can be thought of as a measure of self-efficacy, and higher scores suggest less confidence in a test subject's ability to master their environment. Elevations are also common among individuals ruminating about their situation or who experience cognitive deficits such as problems with attention or concentration. *DEP-A* is perhaps the most direct measure of overall life satisfaction on the PAI, and is thus often elevated in clinical and correctional settings. *DEP-P* elevations are also likely in an environment where exercise, sleep, and dietary schedules are abnormal or restricted, such as in correctional settings.

The PAI includes three full scales related to the psychotic spectrum of disorders: *MAN*, *PAR*, and *SCH*.

MAN was developed to measure symptoms of a manic disorder, although its subscales also have utility independent of that diagnosis. An elevation on *MAN-A* suggests someone who is overactive, and the higher the score, the less ability the individual is likely to demonstrate in managing their activity level effectively. Individuals with *MAN-A* elevations will tend to do poorly in restricted environments and are high risks for poorly planned behavior. High scores on *MAN-G* suggest grandiosity, as associated with bipolar and narcissistic disorders, and as also commonly observed in individuals with an incentive to represent themselves in a positive manner, such as those undergoing

personnel selection or child custody evaluations. *MAN-G* can also be conceptualized as a measure of self-esteem, so that both high and low scores on the scale are meaningful. *MAN-I* is a measure of irritability and is likely to be predictive of a poor response to incarceration, probation, employment, and treatment.

PAR and its subscales are useful in assessing the level of trust with which an individual is likely to approach most situations. The Resentment (*PAR-R*) and Hypervigilance (*PAR-H*) subscales address personological issues of hostility and bitterness, and predisposition to distrust others, respectively. Low scores on *PAR-H* are interpretable as high levels of interpersonal trust, and vice versa, an important factor in most assessment contexts. High scores on Persecution (*PAR-P*) indicate the behaviors and experiences of individuals diagnosed with delusional or other paranoid disorders, particularly if *SCZ-P* is also elevated.

An elevation on the Schizophrenia (*SCZ*) scale may suggest a psychotic disorder or high levels of psychological impairment associated with other clinical disorders. The *SCZ* subscales, designed to represent positive and negative symptoms of schizophrenia, are represented by the Psychotic Experiences (*SCZ-P*) and Social Detachment (*SCZ-S*) subscales, respectively. *SCZ-S* is also useful in indicating introversion, which may range from quiet and self-interested to schizoid or socially phobic. The Thought Disorder (*SCZ-T*) subscale measures organization of thought processes, and may suggest thought disorder related to non-psychotic disorders, such as severe depression or dementia.

Although most personality disorders have characteristic profile configurations (Morey, 2003), two full scales directly assess the most commonly researched personality disorders: Borderline Features (*BOR*) and Antisocial Features (*ANT*). Individuals with moderate elevations on all four *BOR* subscales are very likely to meet diagnostic criteria for borderline personality disorder, and probably several other personality disorders. *BOR* can be thought of as an index of overall psychological maturity, with low scorers being characteristically immature. The Identity Problems (*BOR-I*) and Negative Relationships (*BOR-N*) subscales reflect the common interpersonal pattern of need for interpersonal affection combined with the expectation and experience of disappointment and rejection characteristic of the borderline disorder. Elevations on the Affective Instability (*BOR-A*) and Self-Harm (*BOR-S*) subscales suggest emotional lability and impulsivity, respectively, traits likely to be common and problematic in forensic correctional settings. Antisocial personality is among the most common diagnoses in correctional settings, and the Antisocial Features (*ANT*) scale measures three key aspects of the disorder. *ANT-A* is a measure of historical and current antisocial behavior; it correlates most highly with the notoriously behavioral diagnosis of antisocial personality disorder. *ANT-E* reflects the character trait egocentricity, which is also common in individuals diagnosed as antisocial. High scorers are likely to be callous and unremorseful about past or future crimes. Elevations on Stimulus-Seeking (*ANT-S*) suggest an increased risk for substance use or other potentially self- or other-harming behaviors.

Substance Use

The Alcohol Problems (*ALC*) and Drug Problems (*DRG*) scales directly assess the degree to which alcohol or substance use is involved in the clinical picture. Each scale includes items addressing historical and current substance involvement and adverse consequences of substance use. Moderate elevations (e.g., 70T) suggest abuse, whereas marked elevations (e.g., 85T) suggest dependence. It is also possible to get a moderate elevation with individuals who report past substance use problems but do not admit to current use. Individuals entering substance abuse treatment often demonstrate elevations on NIM and other indicators of negative dissimulation (Alterman et al., 1995; Boyle & Lennon, 1994), a phenomenon that may be related to state affects associated with intoxication or

the detoxification process. Ruiz, Douglas, Edens, Nikolova, and Lilienfeld (2012) have demonstrated that *ALC* and *DRG* are often related to aggression and suicide risk in offender populations, with higher *ALC* and *DRG* scores being indicative of overall mental health problems.

As items on *ALC* and *DRG* are face-valid, individuals who wish to avoid disclosing the full extent of their substance use can obtain low scores somewhat easily. Therefore, several empirically derived procedures to assess the likelihood that a profile under-represents the extent of alcohol or drug problems (Fals-Stewart, 1996; Morey, 1996) have been described. As discussed on p. 153, Fals-Stewart and his research group have demonstrated some success using *PIM* and a combination of *PIM*, *ALC*, and *DRG* in identifying questionable substance use respondents. Additionally, Morey (1996, 2003) described a linear regression strategy that successfully distinguished the groups described in Fals-Stewart's study with greater accuracy than the composite of *PIM*, *ALC*, and *DRG*. In this strategy, a predicted score is derived for both *ALC* and *DRG* based on the scores of other scales in the regression model (*ALCest* and *DRGest*). The difference between the predicted and observed scores is then used to indicate the probability of under-reporting; this is thought to be a linear relation. Patry, Magaletta, Diamond, and Weinman (2010) examined the validity of the *DRG* and *ALC* scales in a correctional sample consisting of 1,120 non-clinical federal inmates, and found that not only did the *DRG* and *ALC* scales demonstrate strong convergent and discriminant validity in this sample, but *DRGest* and *ALCest* provided incremental validity in several areas, providing support for their validity in populations in which there is reason to doubt the accuracy of the respondent's self-report on the *DRG* and *ALC* scales. Edens and Ruiz (2005) developed the Addictive Characteristics Scale (*ACS*) to assess personality characteristics indicative of addictive behavior. Their scale consists of the summed raw scores on *BOR-A*, *BOR-S*, and *ANT-E*. As these authors note, this algorithm is very similar to that described by Morey (1996), and these indicators have yet to be compared empirically.

Risk of Violence

The Aggression scale (*AGG*) of the PAI was constructed to estimate aggressive potential, a common assessment issue in forensic settings. Moderate scores on *AGG* indicate chronic anger, whereas marked elevations suggest substantial risk for the expression of anger. As discussed above, *AGG* has been found useful in the prediction of violence, institutional misbehavior, and recidivism in forensic settings. Because of the scale's soft floors, low scores can be interpreted as meekness or aggression avoidance. *AGG* subscales comprise the three major elements of aggression identified in factor-analytic work on the construct (e.g., Riley & Treiber, 1989): Aggressive Attitude (*AGG-A*), Verbal Aggression (*AGG-V*), and Physical Aggression (*AGG-P*). The relative elevation of subscales suggests the typical mode through which anger is expressed. For example, a moderate elevation on *AGG-A* but not the other subscales would suggest an individual who experiences but does not express anger, whereas more explosive individuals might be likely to demonstrate elevations on *AGG-P*.

Elevations on several other scales (e.g., *ANT*, *BOR*, and *MAN*) also elevate the risks of behavioral acting out and recidivism. Research has found *ANT* and *ANT-E* in particular to be valid in the classification of recidivists (Salekin et al., 1998). Newberry and Shuker (2012) examined how institutional misconduct and risk of reconviction correlated to PAI profiles of criminal offenders, and found a moderate positive correlation between the *ANT* scale and general institutional misconduct. In addition, Newberry and Shuker (2012) found that *ANT*, *AGG*, and *DRG* were all positively correlated with risk of reconviction, and examined a variety of different cut scores in relation to these three scales across several levels of institutional misconduct, including violent infractions, with a general finding that a cut score of 60 *T* for *ANT-A* provided the highest predictive

accuracy. With regard to female inmates, the *ANT* scale was the most significant predictor of violence and general misconduct (Skopp, Edens, & Ruiz, 2007). In a sample of defensive prison inmates, Edens and Ruiz (2009) found that *NON* provided some insight into violence risk, with *NON* being negatively associated with physical aggression and aggressive defiance. In addition, Morey (1996) developed the Violence Potential Index (VPI) to increment the utility of the PAI in assessing risk. The VPI samples 20 indicators of violence potential based on PAI-scale elevations or configurations, such as impulsivity and sensation-seeking. The VPI has demonstrated an anticipated pattern of correlations (Edens et al., 2001; Morey, 1996; Wang et al. 1997). Scores of 9 (1 standard deviation above the mean of the clinical standardization sample) and 17 (2 SDs) were suggested by Morey (2003) as indicative of moderate and marked risk for violence, respectively.

Boccaccini, Murrie, Hawes, Simpler, and Johnson (2010) examined the ability of the PAI to predict recidivism in sexually violent criminals. Overall, the PAI demonstrated comparable effect sizes when predicting recidivism to measures specifically designed to predict recidivism in sex offenders, particularly with the *AGG* scale of the PAI. In addition, the PAI demonstrated incremental validity in the prediction of sex offender recidivism over background characteristics (such as offender age, prior arrests, etc.) as well as over the *STATIC-99* (Hanson & Thornton, 1999), an instrument specifically designed to predict sexual recidivism, though this effect was limited to the *DOM* and *AGG* scales of the PAI, and effect sizes were relatively small.

Suicide

Suicide risk is another common concern in forensic settings. The Suicidal Ideation (*SUI*) scale was designed to indicate suicide potential, and, as discussed on p. 154, encouraging validity data exist for *SUI* from several forensic samples. Several issues should be kept in mind when predicting suicide on the basis of an observed score on *SUI*. First, the base rate of suicide is generally quite low, and is variable across populations. The base rate in the sample from which the data are being collected must be kept carefully in mind, and inflexible cut scores are not recommended. However, scores above 85*T* typically suggest a need for close monitoring or other precautions. Second, *SUI* is a measure of ideation, not probability, and it has an extremely high ceiling. Individuals can achieve scores well above 70*T* without being an imminent risk for suicide. In fact, most clinical patients score between 60*T* and 69*T*. Third, the adverse consequences of a false negative are typically worse than those of a false positive. Therefore, despite the issues discussed above, it is a good idea to follow up even moderate elevations on *SUI*. For example, it may be useful to know if current ideation is associated to past attempts by an individual who appears to have learned more adaptive coping strategies, or an emerging tendency to see suicide as a solution to the individual's problems. Finally, *SUI* is a measure of suicidal ideation, not a measure of the probability of the occurrence of suicide. Although most individuals who commit suicide tend to ideate around the issue as well, the two factors are somewhat independent. Many individuals in clinical samples ideate without acting on their suicidal thoughts, whereas some suicides occur in individuals who apparently spent little time planning or ruminating about the act. The Suicide Potential Index (*SPI*) is composed of 20 configural indices of suicide potential independent of *SUI* (e.g., substance use, impulsivity) and was developed to increment *SUI* in the prediction of suicidal behavior. The Wang et al. (1997) and Rogers, Ustad, and Salekin (1998) studies cited earlier found significant positive correlations between suicidal behavior and other measures of suicidality, respectively. Morey (1996) recommends a score of 13 on the *SPI*, which is 1 standard deviation above the mean of the clinical standardization sample and 2 SDs above the mean of the community sample, as indicative of increased risk for suicide. However, the *SPI* may also be treated as a dimension on which greater obtained scores are associated with higher suicide risks. *NIM* appears to moderate the relation

between *SUI* and *SPI*, suggesting that an elevated *SPI* score with an *NIM* score in the normal range is worth particular attention (Morey, 2003).

Treatment Planning

The PAI includes two indices directly relevant to treatment planning. The Treatment Rejection Scale (*RXR*) is a measure of attitudes that could pose problems with motivation for treatment. Individuals achieving high scores on *RXR* may be unmotivated to change, may be unable to recognize problems or their responsibility in the generation or maintenance of problems, or may lack psychological mindedness or interest in self-improvement. It is quite rare for treatment-seeking individuals to achieve scores in the moderate range, as 50*T* on the PAI profile represents the average treatment resistance of community individuals, which is predictably quite high. Conversely, scores below 20*T* may represent desperation or negative dissimulation, neither of which is typically associated with smooth and productive treatment.

Motivation for treatment is not the sole indicator of good treatment response. The Treatment Process Index (*TPI*) is a configural index developed to predict treatment amenability (Morey, 1996). Indices including hostility, low insight, and limited social supports are harbingers of a difficult treatment process independent of motivation for change. High scores on *RXR* generally indicate resistance to beginning therapy. Individuals without *RXR* elevations but with *TPI* elevations may be motivated, at least initially, but the treatment is likely to be characterized by either early discontinuation or long length, reversals, and high levels of therapist frustration.

Limitations of the PAI in Forensic Settings

The validity data presented above suggest that the PAI is amenable to a wide variety of forensic assessment contexts. Any self-report instrument has unique advantages over other methods of forensic assessment, including efficiency, standardization, large normative samples, and the capacity to capture the phenomenology of the respondent from their point of view. The low reading requirements, breadth of coverage, relative brevity, and ease of interpretation due to recognizable scale names and relation to commonly used constructs represent advantages over other self-report methods of forensic assessment. The PAI also appears to stand up well against contemporary legal criteria (Morey, Warner, & Hopwood, 2007) and has several advantages over other multiscale instruments. Administrative advantages include short length, low required reading level, and scale names that represent the construct being measured (Rogers, 2003). Psychometric advantages include the fact that scales are composed of non-overlapping items, a strategy that enhances discriminant validity by reducing artifactual scale intercorrelations. Also, item response theory was used in item selection to ensure the appropriate depth of coverage across the full range of severity of each clinical construct.

Despite these reasons to recommend the PAI in forensic settings, several limitations of the PAI in such settings should be noted. These have to do with both its structure and the current lack of research on some of its specific uses. One structural limitation of any self-report instrument is that it requires the examinee to be literate. In the case of illiteracy, the items could be administered in interview format, though no research has been conducted thus far on the effects of this deviation from standardized administration on the validity of the test. Another feature of any self-report instrument is that response sets must be kept carefully in mind, and forensic decisions based on PAI data should always be incremented with data from other sources. Although the PAI is equipped with validity and clinical scales appropriate to assess the effects of disordered or compromised thinking on test validity, individuals with severe cognitive deficits or disordered thinking are likely

to produce a profile with significant distortion due to idiosyncratic or random responding. Additionally, the PAI does not measure every potentially relevant construct of clinical interest. For example, eating disorders are not directly assessed by the PAI, though these and other clinical disorders tend to produce characteristic profiles (Tasca et al., 2002). Finally, more research on the PAI in forensic contexts would be helpful. For example, research on the validity of DRG and SCZ has been mixed (e.g., Edens et al., 2001), and more work is necessary before firm conclusions (e.g., norm-specific cut scores) can be drawn about these scales.

Variations of the PAI in Forensic Settings

While the vast majority of the existing literature on the utility of the PAI in forensic settings has focused on the full-form PAI, it is worth noting that the PAI does have alternative versions that may be employed in certain circumstances. The Personality Assessment Screener (PAS; Morey, 1997) is a screening questionnaire consisting of 22 PAI items arranged into 10 separate areas ("elements") designed to represent distinct clinical domains. The PAS can be useful in situations that prevent the use of the full PAI (such as situations in which only a brief initial evaluation is needed before proceeding with more comprehensive evaluations), and can be accompanied by an observer-report version of the PAS in order to empirically compare self-report to informant report. The PAS is primarily useful as a method of predicting PAI elevations, with an optimal cutting score of 19, which yields 84.7% sensitivity and 78.7% specificity for this purpose (Morey, 1997). In addition, scores above 42 strongly suggest the presence of malingering, and scores below 13 strongly suggest the presence of defensiveness.

The PAI also has a short form (PAI-SF; Morey, 1991) which consists of the first 160 items of the full PAI. Because of the nature of the PAI's construction, the first 160 items also represent the items with the strongest item-scale correlations for each clinical scale. The PAI-SF assesses all the same constructs as the full PAI except for *ICN* and *STR*, which are unscorable, and the *AGG-A* subscale is likewise missing. The PAI-SF is a useful instrument in a variety of settings, such as when there is inadequate time for a full administration of the PAI or either when the respondent is unable to complete the second half of the PAI or there is evidence of back-half random responding. The PAI-SF has been shown to possess strong agreement with the full PAI (Sinclair, Antonius et al., 2009; Sinclair, Siefert et al., 2009). In addition, the PAI-SF has demonstrated good internal consistency in forensic and civil samples, as well as strong correlations with the full PAI across forensic and civil samples (Sinclair, Antonius et al., 2009).

Another important variation of the PAI is the Personality Assessment Inventory-Adolescent (PAI-A; Morey, 2007). The PAI-A is designed with all the same scales as the adult PAI, but is intended for use with adolescents between the ages of 12 and 18. The PAI-A community standardization sample comprised 707 adolescents aged 12–18 who were enrolled in junior high school, senior high school, or college. The clinical standardization sample comprised 1,160 adolescents being treated in clinical or correctional settings. The PAI-A was designed to match the PAI as closely as possible, although the number of items was cut from 344 to 264, and a small number of items were reworded in order to be more applicable to adolescents. Items were selected for the PAI-A using examinations of Differential Item Functioning between adolescent and adult samples to select items that were interpretively comparable across these samples, in an effort to assure continuity of interpretation across the two instruments. Thus, because the constructs and selection of items were designed to yield an instrument that would directly parallel the PAI, the PAI-A is scored and generally interpreted in the same manner as the PAI itself.

Although the research literature on the PAI-A is still developing, extant results tend to parallel those for the adult version of the instrument. For example, during the PAI-A validation studies

reported in the manual, simulation studies were conducted having adolescent participants feign a serious mental disorder or present a very favorable impression when completing the PAI-A. It was found that *NIM* was effective at detecting those in the fake-bad group with an optimal cutoff score of $\geq 77T$, and *PIM* was very effective at detecting those in the fake-good group with an optimal cutoff score of $\geq 65T$. More recently, a study was conducted examining the utility of the supplemental validity indicators of the PAI on the PAI-A (Meyer & Morey, 2012). This study, utilizing the same samples as the initial validation study, found that *DEF* was able to identify faking-good participants with an effect size of $d = 2.06$, *CDF* was able to identify faking good with an effect size of $d = 0.96$, *MAL* was able to identify faking bad with an effect size of $d = 2.83$, and *RDF* was able to identify faking bad with an effect size of $d = 2.26$. Thus, there is evidence to suggest that all the validity indicators of the PAI-A work similarly to the manner that they perform using the PAI itself, which supports their clinical utility in forensic settings in which the detection of malingering or defensiveness might be important considerations.

With respect to clinical scales relevant to forensic settings, *ANT* shows elevations in adolescents diagnosed with Antisocial Personality Disorder (those who are 18 years old), although this is not necessarily the case in adolescents diagnosed with Conduct Disorder. This suggests that the PAI-A may have utility as a tool for identifying those adolescents with a Conduct Disorder that may outlast their adolescence and develop into an Antisocial Personality Disorder. In addition, adolescents in a correctional setting are more likely to show elevations on *DRG* in addition to elevations on *ANT*, as well as milder elevations on *AGG-P*, *ALC*, and *ANT-A*. In adolescents identified as having substance abuse problems, elevations are apparent on *DRG* and *ALC* respective to the substance being abused. Cashel, Schember, Goodman, and Reed (2006) found that *AGG* was significantly related to prior violent offenses in addition to disciplinary infractions in adolescent females. *AGG-A* is most strongly associated with poor control over anger, with a similar correlation for *AGG-V*, although somewhat less associated with conduct problems. *AGG-P* was most strongly associated with indicators of conduct problems. The suicide scale, *SUI*, was found to be significantly related to suicide attempts and referrals (Cashel et al., 2006) in a juvenile detention sample.

As the PAI-A is still a relatively new instrument, empirical research on its use is still in its infancy. However, initial studies suggest that the PAI-A parallels the parent instrument closely. However, there continues to be a significant need for more research to determine whether other findings from the PAI research literature can be generalized to the adolescent population.

CASE EXAMPLE

Lionel was a single, 26-year-old African American male with 12 years of education, employed as a fry cook at the time of the evaluation. He was serving a three-year period of probation for writing checks with insufficient funds, which his probation officer suspected was related to his maintenance of a cocaine (crack) habit. Lionel denied any current or past problems with substance abuse, said that he had only experimented with drugs as a teenager, and generally complied with probation conditions. However, he was videotaped leaving the residence of a known crack dealer and tested positively for cocaine shortly thereafter. He was informed by his probation officer that he must either comply with a full substance abuse evaluation and a week of daily urine screens or be in violation of probation. He complied, but asserted that the positive drug test results were wrong and that his probation officer was "out to get him."

Upon presenting for the evaluation, Lionel completed the PAI and a clinical interview. During the interview, he reiterated that he felt he had no mental health or substance abuse problems requiring treatment. He did describe a long history of legal problems of an aggressive nature. He noted that his attitude has always been that “the only way to survive is to cover your back and look over your shoulder.” He had been fired from several jobs during the previous five years for poor performance. He declined to discuss his family history, but records indicated that he was the only child of separated parents from a low-income urban area, and that his parents had endured many legal problems of their own.

During the week-long series of daily drug screens that followed completion of the PAI, two of the tests were positive for cocaine. At this point, he was referred for outpatient substance abuse treatment. After two weeks of treatment, and two more positive screens following the evaluation, he finally admitted to the daily use of cocaine for nearly nine years. He also described, somewhat boastfully, how he had avoided detection. During the first six weeks of treatment, nearly all of the urine samples were positive for cocaine. Clinical staff described him as defiant and hostile during this time, and he was sent to a therapeutic community. However, he was removed after two months due to frequent aggressive behavior and lack of clinical improvement. He was then placed in a 28-day inpatient program, from which he graduated successfully. He returned to the original outpatient facility and graduated, then completed his probation with no further incident. He went on to take a job in landscaping and enroll in a community college part-time.

Lionel’s PAI full-scale and subscale profiles are presented in Figures 5.1 and 5.2, respectively, and his supplemental indices are shown in Table 5.3. The following subsections discuss various elements of the test results as they pertain to this case.

Validity of Test Results

Positive impression management is a high-base-rate phenomenon in many forensic evaluation settings, but is particularly under the conditions described as relating to Lionel. Lionel’s *PIM* and *Defensiveness Index* scores fall within normal limits, likely indicating his ability and willingness to admit to certain problems unrelated to substance use. However, the *Cashel Discriminant Function* was extremely high, indicating a high degree of similarity with the protocols of individuals who had intentionally positively dissimulated. In addition, his *ALC* and *DRG* predicted scores were far higher than his obtained scores, indicating a personality highly prone to substance use, as well as the likelihood that he was not forthcoming regarding substance use. Interestingly, Lionel’s elevated *Rogers Discriminant Function* was also somewhat elevated, suggesting some negative impression management, perhaps related to his expression of anxiety, trauma, or self-harm. In any case, his validity profile suggests efforts at distorting his self-presentation, and certain clinical scale elevations (e.g., the *ANT-E* elevation, and the high *DOM*/low *WRM* configuration) suggest an individual who may value his ability to mislead or deceive people. As a result, the remainder of the profile should be interpreted with these results in mind.

Clinical Features

Lionel’s clinical scale configuration depicts him as impulsive (*ANT-S*), labile (*BOR-A*), hostile (*PAR-R*), hypervigilant (*PAR-H*), egocentric (*ANT-E*), and aggressive (*AGG*). These traits may have helped him survive in a difficult early environment, as he suggested during the interview, but

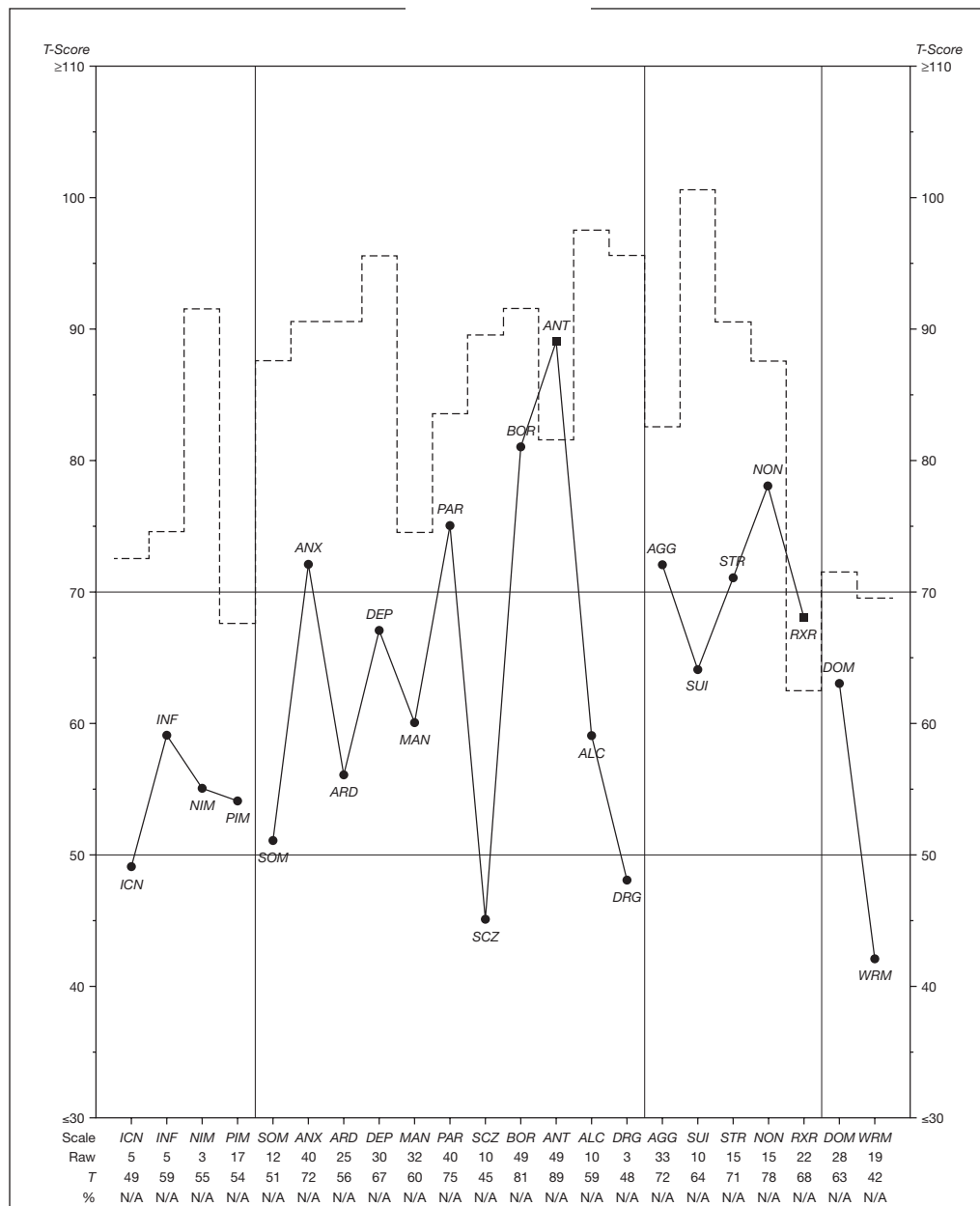


FIGURE 5.1 PAI Full Scale Profile for Case Example: Lionel.

they have led to interpersonal impoverishment, abuse, and mistrust (*NON*, *BOR*, *PAR*, *AGG*), and intrapersonal turmoil and dissatisfaction (*ANX-C*, *ANX-P*, *BOR-S*). Substance use is common among individuals with similar profiles (predicted *ALC* and *DRG* scores), and further complicates the clinical picture and treatment process.

Lionel's clinical scale profile suggests antisocial personality disorder; he is likely to be unreliable and irresponsible and, as his intake data suggest, has had difficulties in the past with

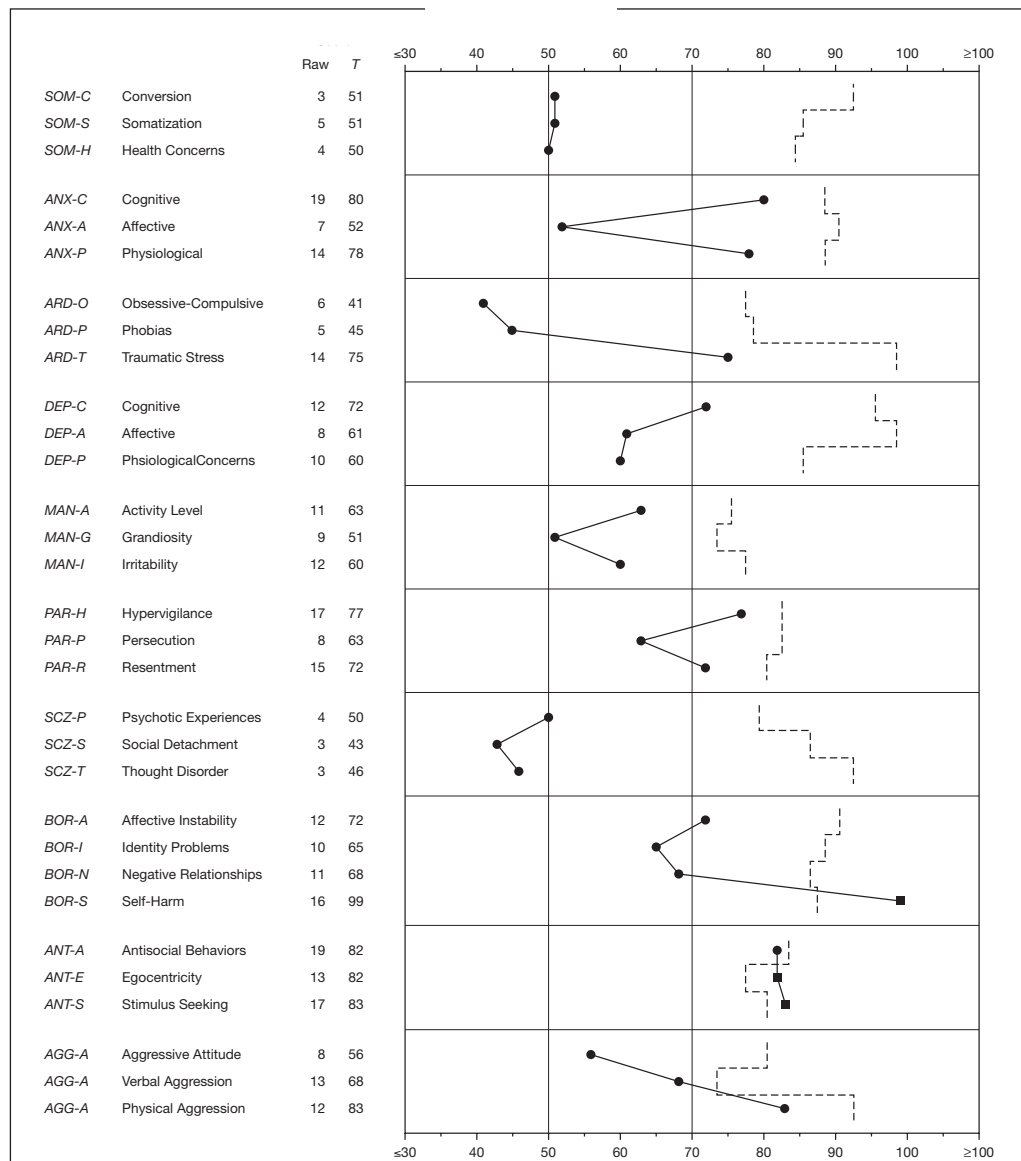


FIGURE 5.2 PAI Subscale Profile for Case Example: Lionel.

occupational or educational responsibilities. Also consistent with his intake, the data suggest a history of antisocial behavior that may have manifested as a conduct disorder during adolescence (*ANT-A*). His practice of manipulating his probation officer in order to maintain his drug use while passing urine tests is typical of such individuals. He is likely to be egocentric, with little regard for others or normative societal expectations (*ANT-E*). He is likely to be willing to do nearly anything to meet his own needs, and to experience little lasting remorse about his effect on others. His behavior is also likely to be reckless; he can be expected to entertain risks that are potentially dangerous to himself and to those around him (*ANT-S*).

TABLE 5.3 PAI Supplemental Indices for Case Example: Lionel

<i>Index</i>	<i>Value</i>	<i>T</i>
Defensiveness Index	3	51
Cashel Discriminant Function	192.69	87
Malingering Index	0	44
Rogers Discriminant Function	1.91	77
Suicide Potential Index	12	78
Violence Potential Index	11	93
Treatment Process Index	12	107
<i>ALC</i> Estimated Score	(25 <i>T</i> higher than <i>ALC</i>)	84
<i>DRG</i> Estimated Score	(41 <i>T</i> higher than <i>DRG</i>)	84

In addition to antisocial traits, he obtained significant clinical elevations across the *BOR* subscales. His emotional lability and mood swings (*BOR-A*) are likely to interfere significantly with his interpersonal life (*BOR-N*, *NON*), and to unpredictably manifest as anger (*AGG-P*). Associated with unstable affect is marked impulsivity (*BOR-S*); this may indicate his substance use or other masochistic behaviors involving spending, sex, or self-harm, particularly during times of affective turmoil. The combination of *BOR*, *ANT*, *ARD-T*, and *AGG* elevations, as well as his reluctance to discuss his family history, increases the probability that he experienced violence or abuse during his childhood.

Not surprisingly, given what is known about his development and the picture emerging from the clinical scales, he is likely to be a hypervigilant individual who often questions and mistrusts the motives of those around him (*PAR-H*). Lionel's interview and PAI data indicate that he believes others will reliably misuse him if given the opportunity, and he has decided to get to them first. He is probably extremely sensitive to perceived slights (*PAR-R*) and is likely to be cold, suspicious, and hostile in daily interactions, to expect the worst from others, and to have few (*NON*), heavily strained, relationships (*BOR-N*).

Lionel indicated some experience of overt physical signs of tension and stress, such as sweaty palms, trembling hands, complaints of irregular heartbeats, and shortness of breath (*ANX-P*), as well as some cognitive anxiety and depression (*ANX-C*, *DEP-C*). This pattern suggests the possibility that he occasionally experiences panic symptoms. However, he does not report a strong subjective experience of tension or major difficulties relaxing (*ANX-A*). This may represent the experience of a man who continues to rationalize a way of life that he is increasingly dissatisfied with, and may help clarify what appeared to be a relatively dramatic turnaround in the following months.

Finally, the low score on *DRG*, coupled with the high estimated score on *DRG*, suggests that the under-reporting of substance misuse is likely. The estimated *DRG* score of 89 *T* would place him in a range typical of individuals receiving a substance dependence diagnosis.

Self-Concept

Given the contrast between his attitude at intake and what is presumed to have been a subsequent turn toward a healthier lifestyle, Lionel's self-concept appears to have been shifting at the time of the evaluation. His posture was tough, cold, guarded, and he gave the impression

that he was uninterested in change (*RXR*), but underneath this veneer he was anxious and dissatisfied with his life (*DEP, ANX-P, ANX-C, STR*). Although he maintained his independence and attributed most of his problems to others (*PAR-R*), his inability to tolerate negative interpersonal transactions (*BOR-N*) likely had a strong influence on his self-concept, often causing him to lash out impulsively at others or himself (*AGG-P, BOR-S*) in response to interpersonal disappointment or frustration. This pattern may have contributed to Lionel's involvement with an impoverished social network with little prospects for improvement (*NON*) typical of a crack cocaine-using population, and Lionel probably considered solitary substance use a more satisfying alternative.

Interpersonal and Social Environment

Lionel's PAI profile suggests that he views relationships as a means to an end, rather than as a source of satisfaction, and anticipates others will do the same (*ANT-E, DOM*). He is not likely to be perceived by others as a warm and friendly person (*WRM*), although he is not necessarily lacking in social skills and he appears to have the capacity to be reasonably effective in social interactions. Those who know him well are likely to see him as shrewd, competitive, and independent. As discussed previously, Lionel was experiencing considerable turmoil at the time of the evaluation, largely as a result of environmental stressors, including legal, occupational, and financial problems (*STR*). A primary source of stress probably involves relationship issues, because he believes that his social relationships offer him little support; family relationships may be somewhat distant or ridden with conflict, and friends may not be available when needed (*NON*).

Treatment Considerations

Lionel's angry (*AGG-V*), resentful (*PAR-R*), and combative (*AGG-P*) style proved to be difficult for treatment staff. His responses suggest that he believes that he is generally in control of angry feelings and impulses, and expresses an angry outburst relatively infrequently. However, when he loses control of his anger he is likely to respond with more extreme displays of anger, including damage to property and threats to assault others (*ANT-A, AGG-P*). Some of these displays may be sudden and unexpected, as he may not display his anger readily when it is experienced. It is likely that those around him are intimidated by his temper and the potential for physical violence. It should also be noted that his risk for aggressive behavior is further exacerbated by the presence of a number of features, such as agitation (*ANX-C*), a limited capacity for empathy (*ANT-E*), impulsivity (*BOR-S, ANT-S*), and affective lability (*BOR-A*), that have been found to be associated with increased potential for violence.

On the basis of the PAI, Lionel would appear to be a very poor candidate for psychotherapy (*TPI, RXR*) at the time of the evaluation. His stated interest in and motivation for treatment is well below average in comparison to adults who are not being seen in a therapeutic setting, and his treatment motivation is substantially lower than is typical of individuals being seen in treatment settings (*RXR*). His responses suggest that he was willing to take little responsibility for the many difficulties in his life and he had no desire to make personal changes, despite the fact that several areas of his life did not seem to be going well. He seemed to put the responsibility for changing his life on treatment staff, but was openly hostile to them, indicating

an area of conflict that may have been useful in engaging him in treatment. However, suggesting such conflict directly with Lionel would likely be ineffective, given his affective lability and interpersonal mistrust. One would not predict from the PAI results that Lionel would seek therapy on his own initiative, and resistance would be considerable in any non-voluntary treatment. In addition, the elevated *TPI* indicates that the combination of problems that he reported portend that treatment would be an uphill struggle and that the treatment process would likely be arduous, with numerous reversals. These reversals were evident in his progression through three progressively more restrictive treatment programs; the *TPI* also suggests that his dramatic improvement during inpatient treatment should be regarded with some suspicion. Either he responded well to a heavily structured and restrictive environment, or he continued his pattern of manipulating the authorities in order to maintain his self-defeating habits. Follow-up data would be useful in tracking the effects of treatment and his apparent turnaround, as well as providing confirmatory evidence that he was not, in fact, still “fooling” the system by faking good.

Special Considerations

This case illustrates a number of points common in forensic settings, including efforts at impression management, substance misuse, poor treatment compliance, and impulse control problems. Lionel's apparent turnaround after progressively more intensive treatment efforts may suggest that removing an individual with an unstable and poorly articulated sense of self from a malignant environment (the culture of crack users) may have been a necessary first step in treatment. Typically, individuals with deep-seated character pathology such as is depicted in his PAI profile have very poor prognoses in psychotherapy, and when psychosocial treatment is successful, it is long and difficult. His intake profile indicates little interest in change, and a tendency to attribute responsibility for his problems to others. However, subsequent information suggests an individual who, once off the substances and away from the environment that got him into trouble, was able to take some responsibility for change and to succeed. The relationship between his personality and his drug use would be clarified by follow-up PAI data, which would also be very helpful in confirming this change. If his tendency to be suspicious, hostile, and cold were mostly a function of his environment and need to hide from the police, and if his affective lability and stress were the physiological and financial results of substance abuse, and his feelings of non-support were the result of a support network that had given up on him or that was genuinely untrustworthy, these problems might remit, given a clean lifestyle. However, in the presence of the prominent antisocial traits indicated by the PAI, it is reasonable to be skeptical about a full recovery and to anticipate and plan for eventual relapse.

References

- Alterman, A. I., Zaballero, A. R., Lin, M. M., Siddiqui, N., Brown, L. S. Jr., Rutherford, M. J., & McDermott, P. A. (1995). Personality Assessment Inventory (PAI) scores of lower-socioeconomic African American and Latino methadone maintenance patients. *Assessment*, 2, 91–100.
- Bagby, M. R., Nicholson, R. A., Bacchiochi, J. R., Ryder, A. G., & Bury, A. S. (2002). The predictive capacity of the MMPI-2 and PAI validity scales and indexes to detect coached and uncoached feigning. *Journal of Personality Assessment*, 78, 69–86.

- Bagge, C., Nickell, A., Stepp, S., Durrett, C., Jackson, K., & Trull, T. J. (2004). Borderline personality disorder features predict negative outcomes 2 years later. *Journal of Abnormal Psychology, 113*, 279–288.
- Ban, T. A., Fjetland, O. K., Kutcher, M., & Morey, L. C. (1993). CODE-DD: Development of a diagnostic scale for depressive disorders. In I. Hindmarch & P. Stonier (Eds.), *Human psychopharmacology. Vol. 4: Measures and methods* (pp. 73–85). New York: John Wiley.
- Beck, A. T., & Steer, R. A. (1987). *Beck Depression Inventory manual*. San Antonio, TX: Psychological Corporation.
- Beck, A. T., & Steer, R. A. (1988). *Beck Hopelessness Scale manual*. San Antonio, TX: Psychological Corporation.
- Bell, M. J., Billington, R., & Becker, B. (1985). A scale for the assessment of object relations: Reliability, validity, and factorial invariance. *Journal of Clinical Psychology, 42*, 733–741.
- Bell-Pringle, V. J., Pate, J. L., & Brown, R. C. (1997). Assessment of borderline personality disorder using the MMPI–2 and the Personality Assessment Inventory. *Assessment, 4*, 131–139.
- Bianchini, K. J., Etherton, J. L., & Greve, K. W. (2004). Diagnosing cognitive malingering in patients with work-related pain: Four cases. *Journal of Forensic Neuropsychology, 4*, 65–85.
- Blanchard, D. D., McGrath, R. E., Pogge, D. L., & Khadivi, A. (2003). A comparison of the PAI and MMPI–2 as predictors of faking bad in college students. *Journal of Personality Assessment, 80*, 197–205.
- Boccaccini, M. T., & Brodsky, S. L. (1999). Diagnostic test usage by forensic psychologists in emotional injury cases. *Professional Psychology: Research and Practice, 30*, 253–259.
- Boccaccini, M. T., Murrie, D. C., Hawes, S. W., Simpler, A., & Johnson, J. (2010). Predicting recidivism with the Personality Assessment Inventory in a sample of sex offenders screened for civil commitment as sexually violent predators. *Psychological Assessment, 22*, 142–148.
- Boyle, G. J., & Lennon, T. (1994). Examination of the reliability and validity of the Personality Assessment Inventory. *Journal of Psychopathology and Behavioral Assessment, 16*, 173–187.
- Buffington-Vollum, J., Edens, J. F., Johnson, D. W., & Johnson, J. K. (2002). Psychopathy as a predictor of institutional misbehavior among sex offenders: A prospective replication. *Criminal Justice and Behavior, 29*, 497–511.
- Butcher, J. N., Graham, J. R., Ben-Porath, Y. S., Tellegen, A., Dahlstrom, W. G., & Kaemmer, B. (2001). *The Minnesota Multiphasic Personality Inventory–2: Manual for administration, scoring, and interpretation* (rev. ed.). Minneapolis: University of Minnesota Press.
- Calhoun, P. S., Earnst, K. S., Tucker, D. D., Kirby, A. C., & Beckham, J. C. (2000). Feigning combat-related posttraumatic stress disorder on the Personality Assessment Inventory. *Journal of Personality Assessment, 75*, 338–350.
- Caperton, J. D., Edens, J. F., & Johnson, J. K. (2004). Predicting sex offender institutional adjustment and treatment compliance using the Personality Assessment Inventory. *Psychological Assessment, 16*, 187–191.
- Caruso, K. A., Benedek, D. M., Auble, P. M., & Bernet, W. (2003). Concealment of psychopathology in forensic evaluations: A pilot study of intentional and uninsightful dissimulators. *Journal of the American Academy of Psychiatry and Law, 31*, 444–450.
- Cashel, M. L., Rogers, R., Sewell, K., & Martin-Cannici, C. (1995). The Personality Assessment Inventory (PAI) and the detection of defensiveness. *Assessment, 2*, 333–342.
- Cashel, M. L., Schember, J., Goodman, C., & Reed, P. (2006, March). Concurrent and predictive validity of the PAI–A for incarcerated female adolescents: Strategies for risk assessment. Paper presented at the annual meeting of the Society for Personality Assessment, San Diego, CA.
- Cherepon, J. A., & Prinzhorn, B. (1994). Personality Assessment Inventory (PAI) profiles of adult female abuse survivors. *Assessment, 1*, 393–399.
- Clarkin, J. F., Hull, J. W., Yeomans, F., Kakuma, T., & Cantor, J. (1994). Antisocial traits as modifiers of treatment response in borderline inpatients. *Journal of Psychotherapy Practice and Research, 3*, 307–312.
- Costa, P. T., & McCrae, R. R. (1992). Normal personality in clinical practice: The NEO Personality Inventory. *Psychological Assessment, 4*, 5–13.
- Cull, J. G., & Gill, W. S. (1982). *Suicide Probability Scale manual*. Los Angeles: Western Psychological Services.
- Delsol, C., Margolin, G., & John, R. S. (2003). A typology of maritally violent men and correlates of violence in a community sample. *Journal of Marriage and Family, 65*, 635–651.
- Douglas, K. S., Guy, L. S., Edens, J. F., Boer, D. P., & Hamilton, J. (2007). The Personality Assessment Inventory as a proxy for the Psychopathy Checklist–Revised: Testing the incremental validity and cross-sample robustness of the antisocial features scale. *Assessment, 14*, 255–269.

- Douglas, K. S., Hart, S. D., & Kropp, P. R. (2001). Validity of the Personality Assessment Inventory for forensic assessments. *International Journal of Offender Therapy and Comparative Criminology*, 45, 183–197.
- Doyle, M., & Dolan, M. (2006). Evaluating the validity of anger regulation problems, interpersonal style, and disturbed mental state for predicting inpatient violence. *Behavioral Sciences and the Law*, 24, 783–798.
- Edens, J. F. (2009). Interpersonal characteristics of male criminal offenders: Personality, psychopathological, and behavioral correlates. *Psychological Assessment*, 21, 89–98.
- Edens, J. F., Cruise, K. R., & Buffington-Vollum, J. K. (2001). Forensic and correctional applications of the Personality Assessment Inventory. *Behavioral Sciences and the Law*, 19, 519–543.
- Edens, J. F., Hart, S. D., Johnson, D. W., Johnson, J. K., & Olver, M. E. (2000). Use of the Personality Assessment Inventory to assess psychopathy in offender populations. *Psychological Assessment*, 12, 132–139.
- Edens, J. F., Poythress, N. G., & Watkins-Clay, M. M. (2007). Detection of malingering in psychiatric unit and general population prison inmates: A comparison of the PAI, SIMS, and SIRS. *Journal of Personality Assessment*, 88, 33–42.
- Edens, J. F., & Ruiz, M. A. (2005). *PAI Interpretive Report for Correctional Settings (PAI-CS)*. Odessa, FL: Psychological Assessment Resources.
- Edens, J. F., & Ruiz, M. A. (2006). On the validity of validity scales: The importance of defensive responding in the prediction of institutional misconduct. *Psychological Assessment*, 18, 220–224.
- Edens, J. F., & Ruiz, M. A. (2008). Identification of mental disorders in an in-patient prison psychiatric unit: Examining the criterion-related validity of the Personality Assessment Inventory. *Psychological Services*, 5, 108–117.
- Edens, J. F., & Ruiz, M. A. (2009). Predicting institutional aggression by defensive prison inmates: An examination of the Personality Assessment Inventory treatment consideration scales. *International Journal of Forensic Mental Health*, 8, 178–185.
- Edwards, D. W., Scott, C. L., Yarvis, R. M., Paizis, C. L., & Panizzon, M. S. (2003). Impulsiveness, impulse aggression, personality disorder, and spousal violence. *Violence and Victims*, 18, 3–14.
- Evershed, S., Tennant, A., Boomer, D., Rees, A., Barkham, M., & Watson, A. (2003). Practice-based outcomes of dialectical behaviour therapy (DBT) targeting anger and violence, with male forensic patients: A pragmatic and non-contemporaneous comparison. *Criminal Behaviour and Mental Health*, 13, 198–213.
- Exner, J. E. (1993). *The Rorschach: A comprehensive system*. Asheville, NC: Rorschach Workshops.
- Fals-Stewart, W. (1996). The ability of individuals with psychoactive substance use disorders to escape detection by the Personality Assessment Inventory. *Psychological Assessment*, 8, 60–68.
- Fals-Stewart, W., & Lucente, S. (1997). Identifying positive dissimulation substance-abusing individuals on the Personality Assessment Inventory: A cross-validation study. *Journal of Personality Assessment*, 68, 455–469.
- Gaies, L. A. (1993). Malingering of depression on the Personality Assessment Inventory. Doctoral dissertation, University of South Florida, Tampa, 1993. *Dissertation Abstracts International*, 55, 6711.
- George, J. M., & Wagner, E. (1995). Correlations between the Hand Test Pathology score and Personality Assessment Inventory scales for pain clinic patients. *Perceptual and Motor Skills*, 80, 1377–1378.
- Greenstein, D. S. (1995). Relationship between frequent nightmares, psychopathology, and boundaries among incarcerated male inmates. Doctoral dissertation, Adler School of Professional Psychology. *Dissertation Abstracts International*, 55, 4119.
- Hamilton, M. (1960). A rating scale for depression. *Journal of Neurology, Neurosurgery, and Psychiatry*, 23, 56–62.
- Hanson, R. K., & Thornton, D. (1999). *Static-99: Improving actuarial risk assessments for sex offenders*. User Report 99-02. Ottawa: Department of the Solicitor General of Canada.
- Hare, R. D. (1985). Comparison of procedures for the assessment of psychopathy. *Journal of Consulting and Clinical Psychology*, 53, 7–16.
- Hare, R. D. (1991). *The Psychopathy Checklist-Revised*. Toronto: Multi-Health Systems.
- Hart, S. D., Cox, D. N., & Hare, R. D. (1995). *Psychopathy Checklist: Screening version*. Toronto: Multi-Health Systems.
- Hawes, S. W., & Boccaccini, M. T. (2009). Detection of overreporting of psychopathology on the Personality Assessment Inventory: A meta-analytic review. *Psychological Assessment*, 21, 112–124.
- Holden, R. R. (1989). Disguise and the structured self-report assessment of psychopathology: II. A clinical replication. *Journal of Clinical Psychology*, 45, 583–586.
- Holden, R. R., Book, A. S., Edwards, M. J., Wasylkiw, L., & Starzyk, K. B. (2003). Experimental faking in self-reported psychopathology: Unidimensional or multidimensional? *Personality and Individual Differences*, 35, 1107–1117.

- Holden, R. R., & Fekken, G. C. (1990). Structured psychopathological test item characteristics and validity. *Psychological Assessment*, 2, 35–40.
- Holmes, G. E., Williams, C. L., & Haines, J. (2001). Motor vehicle accident trauma exposure: Personality profiles associated with posttraumatic diagnoses. *Anxiety, Stress and Coping: An International Journal*, 14, 301–313.
- Holmes, T. H., & Rahe, R. H. (1967). The social readjustment rating scale. *Journal of Psychosomatic Research*, 11, 213–218.
- Jackson, D. N. (1970). A sequential system for personality scale development. In C.D. Spielberger (Ed.), *Current Topics in Clinical and Community Psychology* (Vol. 2, pp. 62–97). New York: Academic Press.
- Karlin, B. E., Creech, S. K., Grimes, J. S., Clark, T. S., Meagher, M. W., & Morey, L. C. (2005). The use of the Personality Assessment Inventory with individuals with chronic pain. *Journal of Clinical Psychology*, 61, 1571–1585.
- Kellogg, S. H., Ho, A., Bell, K., Schluger, R. P., McHugh, P. F., McClary, K. A., & Kreek, M. J. (2002). The Personality Assessment Inventory Drug Problems Scale: A validity analysis. *Journal of Personality Assessment*, 79, 73–84.
- Klonsky, E. D. (2004). Performance of Personality Assessment Inventory and Rorschach indices of schizophrenia in a public psychiatric hospital. *Psychological Services*, 1, 107–110.
- Kucharski, L. T., Toomey, J. P., Fila, K., & Duncan, S. (2007). Detection of malingering of psychiatric disorder with the Personality Assessment Inventory: An investigation of criminal defendants. *Journal of Personality Assessment*, 88, 25–32.
- Kurtz, J. E., & Morey, L. C. (2001). Use of structured self-report assessment to diagnose borderline personality disorder during major depressive episodes. *Assessment*, 8, 291–300.
- Kurtz, J. E., Morey, L. C., & Tomarken, A. (1992, March). The concurrent validity study of three self-report measures of borderline personality. Paper presented at the meeting of the Society for Personality Assessment, Washington, DC.
- Liljequist, L., Kinder, B. N., & Schinka, J. A. (1998). An investigation of malingering posttraumatic stress disorder on the Personality Assessment Inventory. *Journal of Personality Assessment*, 71, 322–336.
- Loevinger, J. (1957). Objective tests as instruments of psychological theory. *Psychological Reports*, 3, 635–694.
- Loranger, A. W. (1988). *Personality Disorders Examination Manual*. Yonkers, NY: DV Communications.
- Magyar, M. S., Edens, J. F., Lilienfeld, S. O., Douglas, K. S., Poythress, N. G. Jr., & Skeem, J. L. (2012). Using the Personality Assessment Inventory to predict male offenders' conduct during and progression through substance abuse treatment. *Psychological Assessment*, 24, 216–225.
- McLellan, A. T., Kushner, H., Metzger, D., Peters, R., Smith, I., Grissom, G., . . . Argeriou, M. (1992). The fifth edition of the Addiction Severity Index. *Journal of Substance Abuse Treatment*, 9, 199–213.
- Meyer, J. K., & Morey, L. C. (2012, March). Applying supplemental PAI validity indices to the PAI–A. Poster session presented at the annual meeting of the Society for Personality Assessment, Chicago.
- Montag, I., & Levin, J. (1994). The Five Factor Model and psychopathology in nonclinical samples. *Personality and Individual Differences*, 17, 1–7.
- Morey, L. C. (1991). *The Personality Assessment Inventory Professional Manual*. Odessa, FL: Psychological Assessment Resources.
- Morey, L. C. (1995). Critical issues in construct validation: Comment on Boyle and Lennon (1994). *Journal of Psychopathology and Behavioral Assessment*, 17, 393–401.
- Morey, L. C. (1996). *An interpretive guide to the Personality Assessment Inventory*. Odessa, FL: Psychological Assessment Resources.
- Morey, L. C. (1997). *Personality Assessment Screener professional manual*. Odessa, FL: Psychological Assessment Resources.
- Morey, L. C. (2003). *Essentials of PAI assessment*. Hoboken, NJ: John Wiley.
- Morey, L. C. (2007). *Personality Assessment Inventory—Adolescent professional manual*. Odessa, FL: Psychological Assessment Resources.
- Morey, L. C., & Hopwood, C. J. (2004). Efficiency of a strategy for detecting back random responding on the Personality Assessment Inventory. *Psychological Assessment*, 16, 197–200.
- Morey, L. C., & Lanier, V. W. (1998). Operating characteristics of six response distortion indicators for the Personality Assessment Inventory. *Assessment*, 5, 203–214.
- Morey, L. C., Warner, M. B., & Hopwood, C. J. (2007). The Personality Assessment Inventory: Issues in legal and forensic settings. In A. M. Goldstein (Ed.), *Forensic psychology: Emerging topics and expanding roles* (pp. 97–126). Hoboken, NJ: John Wiley.

- Morey, L. C., Waugh, M. H., & Blashfield, R. K. (1985). MMPI scales for DSM-III personality disorders: Their derivation and correlates. *Journal of Personality Assessment*, 49, 245–251.
- Mullen, K. L., & Edens, J. F. (2008). A case law survey of the Personality Assessment Inventory: Examining its role in civil and criminal trials. *Journal of Personality Assessment*, 90, 300–303.
- Newberry, M., & Shuker, R. (2012). Personality Assessment Inventory (PAI) profiles of offenders and their relationship to institutional misconduct and risk of reconviction. *Journal of Personality Assessment*, 94, 586–592. doi:10.1080/00223891.2012.669220
- Parker, J. D., Daleiden, E. L., & Simpson, C. A. (1999). Personality Assessment Inventory substance-use scales: Convergent and discriminant relations with the Addiction Severity Index in a residential chemical dependence treatment setting. *Psychological Assessment*, 11, 507–513.
- Patry, M. W., Magaletta, P. R., Diamond, P. M., & Weinman, B. A. (2010). Establishing the validity of the Personality Assessment Inventory drug and alcohol scales in a corrections sample. *Assessment*, 18, 50–59. doi:10.1177/1073191110368484
- Peebles, J., & Moore, R. J. (1998). Detecting socially desirable responding with the Personality Assessment Inventory: The Positive Impression Management Scale and the Defensiveness Index. *Journal of Clinical Psychology*, 54, 621–628.
- Peterson, G. W., Clark, D. A., & Bennett, B. (1989). The utility of MMPI subtle, obvious scales for detecting fake good and fake bad response sets. *Journal of Clinical Psychology*, 45, 575–583.
- Procidano, M. E., & Heller, K. (1983). Measures of perceived social support from friends and from family: Three validation studies. *American Journal of Community Psychology*, 11, 1–24.
- Riley, W. T., & Treiber, F. A. (1989). The validity of multidimensional self-report anger and hostility measures. *Journal of Clinical Psychology*, 45, 397–404.
- Rogers, R. (2003). Forensic uses and abuses of psychological tests: Multi-scale inventories. *Journal of Psychiatric Practice*, 9, 316–320.
- Rogers, R., Bagby, R. M., & Dickens, S. E. (1992). *Structured Interview of Reported Symptoms: Professional manual*. Odessa, FL: Psychological Assessment Resources.
- Rogers, R., Flores, J., Ustad, K., & Sewell, K. W. (1995). Initial validation of the Personality Assessment Inventory–Spanish version with clients from Mexican American communities. *Journal of Personality Assessment*, 64, 340–348.
- Rogers, R., Gillard, N. D., Wooley, C. N., & Ross, C. A. (2012). The detection of feigned disabilities: The effectiveness of the Personality Assessment Inventory in a traumatized inpatient sample. *Assessment*, 19, 77–88.
- Rogers, R., Ornduff, S. R., & Sewell, K. (1993). Feigning specific disorders: A study of the Personality Assessment Inventory (PAI). *Journal of Personality Assessment*, 60, 554–560.
- Rogers, R., Sewell, K. W., Cruise, K. R., Wang, E. W., & Ustad, K. L. (1998). The PAI and feigning: A cautionary note on its use in forensic-correctional settings. *Assessment*, 5, 399–405.
- Rogers, R., Sewell, K. W., Morey, L. C., & Ustad, K. L. (1996). Detection of feigned mental disorders on the Personality Assessment Inventory: A discriminant analysis. *Journal of Personality Assessment*, 67, 629–640.
- Rogers, R., Ustad, K. L., & Salekin, R. T. (1998). Convergent validity of the Personality Assessment Inventory: A study of emergency referrals in a correctional setting. *Assessment*, 5, 3–12.
- Ruiz, M. A., Dickinson, K. A., & Pincus, A. L. (2002). Concurrent validity of the Personality Assessment Inventory Alcohol Problems (ALC) Scale in a college student sample. *Assessment*, 9, 261–270.
- Ruiz, M. A., Douglas, K. S., Edens, J. F., Nikolova, N. L., & Lilienfeld, S. O. (2012). Co-occurring mental health and substance use problems in offenders: Implications for risk assessment. *Psychological Assessment*, 24, 77–87.
- Ruiz, M. A., & Ochshorn, E. (2010). Clinical applications of the PAI in criminal justice settings. In Blais, M. A., Baity, M. R., & Hopwood, C. J. (Eds.), *Clinical Applications of the Personality Assessment Inventory* (pp. 113–134). New York: Routledge.
- Salekin, R. T., Rogers, R., & Sewell, K. W. (1997). Construct validity of psychopathy in a female offender sample: A multitrait-multimethod evaluation. *Journal of Abnormal Psychology*, 106, 576–585.
- Salekin, R. T., Rogers, R., Ustad, K. L., & Sewell, K. W. (1998). Psychopathy and recidivism among female inmates. *Law and Human Behavior*, 22, 109–128.
- Schinka, J. A. (1995). PAI profiles in alcohol-dependent patients. *Journal of Personality Assessment*, 65, 35–51.
- Schinka, J. A., & Borum, R. (1993). Readability of adult psychopathology inventories. *Psychological Assessment*, 5, 384–386.

- Scragg, P., Bor, R., & Mendham, M. C. (2000). Feigning post-traumatic stress disorder on the PAI. *Clinical Psychology and Psychotherapy*, 7, 155–160.
- Sinclair, S. J., Antonius, D., Shiva, A., Siefert, C. J., Kehl-Fie, K., Lama, S., . . . Blais, M. A. (2009). The psychometric properties of the Personality Assessment Inventory–Short Form (PAI–SF) in inpatient forensic and civil samples. *Journal of Psychopathology and Behavioral Assessment*, 32, 406–415.
- Sinclair, S. J., Siefert, C. J., Shorey, H. S., Antonius, D., Shiva, A., Kehl-Fie, K., & Blais, M. A. (2009). A psychometric evaluation of the Personality Assessment Inventory–Short Form clinical scales in an inpatient psychiatric sample. *Psychiatry Research*, 170, 262–266.
- Skopp, N. A., Edens, J. F., & Ruiz, M. A. (2007). Risk factors for institutional misconduct among incarcerated women: An examination of the criterion-related validity of the Personality Assessment Inventory. *Journal of Personality Assessment*, 88, 106–117.
- Slick, D. J., Sherman, E. M. S., & Iverson, G. L. (1999). Diagnosing criteria for malingering neurocognitive dysfunction: Proposed standards for clinical practice and research. *Clinical Neuropsychologist*, 13, 545–561.
- Spielberger, C. D. (1988). *State-Trait Anger Expression Inventory*. Odessa, FL: Psychological Assessment Resources.
- Tasca, G. A., Wood, J., Demidenko, N., & Bissada, H. (2002). Using the PAI with an eating disordered population: Scale characteristics, factor structure and differences among diagnostic groups. *Journal of Personality Assessment*, 79, 337–356.
- Testa, S. M., Lesser, R. P., Krauss, G. L., & Brandt, J. (2011). Personality Assessment Inventory among patients with psychogenic seizures and those with epilepsy. *Epilepsia*, 52(8), e84–e88.
- Thomas, K. M., Hopwood, C. J., Orlando, M. J., Weathers, F. W., & McDevitt-Murphy, M. E. (2011). Detecting feigned PTSD using the Personality Assessment Inventory. *Psychological Injury and Law*. Advance Online Publication. doi:10.1007/s12207-011-9111-9116.
- Trapnell, P. D., & Wiggins, J. S. (1990). Extension of the Interpersonal Adjective Scale to include the big five dimensions of personality. *Journal of Personality and Social Psychology*, 59, 781–790.
- Trull, T. J. (1995). Borderline personality disorder features in nonclinical young adults: I. Identification and validation. *Psychological Assessment*, 7, 33–41.
- Trull, T. J., Useda, J. D., Conforti, K., & Doan, B. T. (1997). Borderline personality features in nonclinical young adults: Two year outcome. *Journal of Abnormal Psychology*, 106, 307–314.
- Wagner, M. T., Wymer, J. H., Topping, K. B., & Pritchard, P. B. (2005). Use of the Personality Assessment Inventory as an efficacious and cost effective diagnostic tool for nonepileptic seizures. *Epilepsy and Behavior*, 7, 301–304.
- Walters, G. D., Duncan, S. A., & Geyer, M. D. (2003). Predicting disciplinary adjustment in inmates undergoing forensic evaluation: A direct comparison of the PCL–R and the PAI. *Journal of Forensic Psychiatry and Psychiatry*, 14, 382–393.
- Walters, G. D., & Duncan, S. A. (2005). Use of the PCR–R and PAI to predict release outcome in inmates undergoing forensic evaluation. *Journal of Forensic Psychiatry and Psychology*, 16, 459–476.
- Wang, E. W., & Diamond, P. M. (1999). Empirically identifying factors related to violence risk in corrections. *Behavioral Sciences and the Law*, 17, 377–389.
- Wang, E. W., Rogers, R., Giles, C. L., Diamond, P. M., Herrington-Wang, L. E., & Taylor, E. R. (1997). A pilot study of the Personality Assessment Inventory (PAI) in corrections: Assessment of malingering, suicide risk, and aggression in male inmates. *Behavioral Sciences and the Law*, 15, 469–482.
- Wiggins, J. S. (1966). Substantive dimensions of self-report in the MMPI item pool. *Psychological Monographs*, 80, 22 (whole No. 630).
- Wiggins, J. S. (1979). A psychological taxonomy of trait-descriptive terms: The interpersonal domain. *Journal of Personality and Social Psychology*, 37, 395–412.
- Yeomans, F. E., Hull, J. W., & Clarkin, J. C. (1994). Risk factors for self-damaging acts in a borderline population. *Journal of Personality Disorders*, 8, 10–16.
- Yudofsky, S. C., Silver, J. M., Jackson, W., Endicott, J., & Williams, D. (1986). The Overt Aggression Scale for the objective rating of verbal and physical aggression. *American Journal of Psychiatry*, 143, 35–39.

6

THE MILLON CLINICAL MULTIAXIAL INVENTORY–III

Use in Forensic Settings

Robert J. Craig

Description and Purpose of the Instrument

The Millon Clinical Multiaxial Inventory (MCMI; Millon, 1983) as revised (MCMI–II and MCMI–III; Millon, 1987, 1994, 1997) is a clinical test of personality designed to measure personality disorders and major psychiatric syndromes. The personality disorders assessed by the instrument are not isomorphic with official nomenclature (e.g., that in the *Diagnostic and Statistical Manual of Mental Disorders* [DSM]; American Psychiatric Association [APA], 1983, 1987, 1994); rather, they emanate from Millon's (1990; 2011a) theory of the derivation of personality disorders. However, recent changes in the instrument bring it closer to the *DSM–IV* classification of personality disorders, although it is still not identical to it.

In its original version, the MCMI–I consisted of 11 personality scales and 9 clinical syndrome scales. When the *DSM–III* was revised (*DSM–III–R*; APA, 1987), the MCMI was also revised, in part to take account of additional personality disorders that were added to the official diagnostic classification system. Thus, the MCMI–II separated the Antisocial-Aggressive scale into two separate scales. Antisocial and Aggressive-Sadistic, and also added a Self-defeating (Masochistic) Personality Disorder scale, bringing the number of personality disorder scales to 13 with no change in the number of clinical syndrome scales. When the *DSM–IV* (APA, 1994) was published, the aggressive and self-defeating personality disorders were eliminated from consideration in official classification but a depressive personality disorder was added. The MCMI–II was then revised (Millon, 1994), in part to bring it closer to *DSM–IV*.

At the time of this writing, the *DSM–5* revision committee for personality disorders has proposed to significantly change the way personality disorders are diagnosed and to substantially reduced the number of official diagnosable personality disorders. The MCMI–III is presently being modified in line with what Millon terms theoretical advances (in his theory) as well as developments to be included in *DSM–5* (Millon, 2011b).

In its present form, the MCMI–III has 11 Clinical Personality Patterns (e.g., personality disorders) (Schizoid, Avoidant, Depressive, Dependent, Histrionic, Narcissistic, Antisocial, Aggressive (Sadistic), Compulsive, Passive-Aggressive, [Negativistic], and Self-Defeating), 3 Severe Personality Pathology scales (Schizotypal, Borderline, and Paranoid), 7 Clinical Syndrome scales (Anxiety Disorder, Somatoform Disorder, Bipolar: Manic Disorder, Dysthymic Disorder, Alcohol Dependence, Drug Dependence, and Post-Traumatic Stress), and 3 Severe Syndrome Scales (Thought Disorder, Major Depression, and Delusion Disorder).

The test also features three Validity scales, termed “modifier indices,” and a “Validity Index” consisting of two improbable statements which, if endorsed, call into question the validity of the instrument. An example of such an improbable item would be “I play quarterback for the Chicago Bears” (this is not an actual item on the test but is presented here as an illustration). The Disclosure scale assesses the degree to which a respondent is providing too little information and thus is essentially secretive and withholding, or, on the contrary, frank and self-revealing. The Desirability Index assesses whether the respondent endorsed items in an overly positive manner, and the Debasement Index assesses whether the respondent endorsed items to present an unfavorable picture of present functioning. Disclosure, Desirability, and Debasement are referred to as “modifier indices” rather than validity scales because high or low scores on these scales result in adjustment in scores on the other MCMI–III scales.

Administration time is approximately 30 minutes. The instrument was designed for use with adults who are seeking mental health services and have at least an eighth-grade education, and it is available in English and in Spanish in paper-and-pencil or computer-administration formats. Audiocassette recordings are also available for clients with special needs. It may be hand-scored, using templates available from the test’s distributor, or computer-scored. Two computer-narrative reports are available. One is published by Pearson Assessments (Millon, Millon, & Davis, 2003), and the other is marketed through Psychological Assessment Resources (Craig, 2006). Pearson Assessments also distributes an interpretive report specific to correctional settings that presents information on reaction to authority, response to crowding or isolation, risk of escape, malingering potential, suicidal tendencies, and amenability to mental health intervention and substance abuse treatment and rehabilitation. Most of these categorical assessments are clinical hypotheses and many lack empirical validation. Graduate training in psychodiagnostic assessment is required for test interpretation, and users of these interpretive reports should adhere to test ethics of the American Psychological Association pertaining to computer interpretive reports.

History of the Test’s Development

The *DSM–III* (APA, 1983) introduced the multiaxial format that required diagnostic information on five axes. Axis II required an evaluation to determine whether the patient had a personality disorder. Personality assessment achieved a new focus, since it became apparent that (1) an understanding of the patient’s personality was required to understand that patient’s clinical condition, (2) personality disorders affect the course and treatment of Axis I conditions, and (3) personality disorders could be the focus of treatment in their own right. Furthermore, the transition of the locus of care from inpatient to outpatient settings made the treatment of personality disorders more feasible. These developments stimulated new interest and research into personality and personality disorders in both psychiatry and psychology.

Psychiatry placed emphasis on its historical strength and developed a spate of structured and semi-structured clinical interviews to assess for Axis II disorders. These structured interviews were subsequently revised with the publication of *DSM–IV*. Concurrently, psychology relied on one of its historical strengths—knowledge of test construction—and developed a spate of tests for personality disorders, primarily using self-report methodology. The MCMI, as revised, is by far the most frequently used and researched instrument of this kind.

Millon argued that the structure of a clinical science should be based on *universal scientific principles* grounded in the laws of nature and consists of a *theory* that explained why nature takes the observed form, a *taxonomy* that classifies these observations into a coherent whole, ranging from a spectrum of normal to abnormal personality styles or disorders, *instrumentation* that is empirically grounded and measures these observations, as well as the theories, propositions, and hypotheses, and integrated

treatment *intervention* to remediate and modify problematic clinical characteristics. Thus, his theory of personality development and personality disorders preceded the development of instrumentation to measure them (Millon, 2011b).

Millon originally relied on biopsychosocial levels of explanation with which to begin theory construction but since has developed a bioevolutionary theory of the science, which he calls “personology.” He relied, in part, on knowledge of evolutionary biology to build a comprehensive evolutionary ecological model that has, at its core, three polarity structures that are considered universal motivating aims in both the animal and the plant world. These, in turn, are connected to principles found in studies of chemistry, biology, particle physics, and other natural sciences.

Millon argued that the first task of an organism is to *survive*. Once survival is assured, the next task is to *adapt* to the environment in which existence has been assured, and the final task is to *replicate* to ensure the preservation. These three survival aims or polarities have corollaries at the psychological level, namely *pain–pleasure* (a survival aim), *active–passive* (adaptational strategies), and *self–other* (replication polarity). (Millon has subsequently added a fourth polarity called “abstraction” but has not yet developed this into changes in instrumentation.)

Adaptation may be thought of as how we seek reinforcement (active or passive), and where we seek reinforcement is deemed a replication polarity. Millon theorizes that there are five such ways: (1) independent (self); (2) dependent (others); (3) ambivalent (self–other conflict); (4) discordant (a reversal of pain–pleasure gratification; and (5) detached (no pleasure).

This theory generated 15 different personality spectrum prototypes as depicted in Table 6.1. The first group is classified according to what he terms the “interpersonally imbalanced” spectra (dependent and histrionic as one subgroup and narcissistic and antisocial in another subgroup). The second group is classified as “intrapsychically conflicted” and includes an insecure and ambivalent subgroup (compulsive and negativistic) and what he terms a “paradoxically discordant” subgroup (masochistic and sadistic). The third group he termed an “emotionally extreme” spectrum and includes an “emotionally impassive” subgroup (schizoid and avoidant) and a subgroup termed “intensely expressive” disorders (depressive and manic). The fourth group of spectra disorders he terms “structurally defective” and includes borderline, schizotypal and paranoid. According to his full theory, personality can be manifest from the normal to the subclinical to the pathological. Each

TABLE 6.1 Millon’s 15 Different Personality Spectra of Normal and Subclinical Types (Moderate) and Clinical Disorders

Apathetic–Asocial–Schizoid
Eccentric–Schizotypal–Schizophrenic
Shy–Reticent–Avoidant
Deferential–Attached–Dependent
Ebullient–Exuberant–Turbulent
Sociable–Pleasuring–Histrionic
Confidant–Egoistic–Narcissistic
Mistrustful–Paranoid–Paraphrenic
Aggrandizing–Devious–Antisocial
Assertive–Denigrating–Sadistic
Forlorn–Doleful–Melancholic
Abused–Aggrieved–Masochistic
Discontented–Resentful–Negativistic
Unstable–Borderline–Cyclophrenic
Reliable–Constricted–Compulsive

spectrum continuum has three levels, graded according to severity: a normal style (mild), an abnormal (moderate) [subclinical] and a clinical disorder (severe).

From the theoretical model of how we adapt (active or passive) and how we seek reinforcement, Millon derived a 2 [multi] 5 basic personality styles. Table 6.2 depicts Millon's theory-derived models of personality and personality disorders. He believes that personality disorders emanate from environmental stress and/or compromised biology, but are basic extensions of the underlying style. Thus, by knowing the basic style we can predict the personality disorder that may ensue in times of psychological turmoil. It should be noted that Millon's theory *predicted* the presence of an avoidant personality disorder, which was subsequently added to the official classification in the *DSM-III*, and which is currently scheduled to be retained in the forthcoming *DSM-5*.

Once he had developed (and continues to develop) his theory and taxonomy, Millon pursued the development of instrumentation to assess syndromes predicted by the theory and observed in nature. One result was the MCMI-I, which was constructed as a measure of personality styles and personality disorders as well as major clinical syndromes.

Psychometric Characteristics

Millon used a self-report (questionnaire) methodology to develop the MCMI-I, but he is not tied to this approach to assess personality styles and personality disorders. A number of instruments have used the checklist method to operationalize Millon's personality model. The *Millon Personality Spectrometer* (Millon, Grossman, & Tringone, 2010) uses 120 descriptive phrases that assess the structural and functional domains of the personality disorders. Strack (1987) used a lexical (adjective checklist) approach to assess Millon's personal styles in a normal population, and Craig (2008) developed the *Personality Disorder Adjective Checklist* to screen for personality disorders based on Millon's theoretical model. This test screens for both personality styles and personality disorders. However, the self-report questionnaire method remains the most popular method in personality test development.

In constructing the MCMI-I (and the subsequent revisions), Millon used a three-stage validation process consisting of theoretical-substantive, internal-structural, and external criterion validation. In the first stage, test items are developed and then assessed according to the degree to which they fit the theory. In stage 2, the items are assessed to see how well they interrelate, and the psychometric properties of the test are determined. In stage 3, the convergent and discriminative validity of the test is assessed by correlating it with similar (or dissimilar) instruments.

For the MCMI-I, face-valid items were written, items were eliminated after clinician sorting and patient judging, and a split 1,100-item pool was divided into two equivalent forms. This represented the process involved in stage 1 validation (theoretical-substantive). For internal-structural validation, these two forms were given to clinical samples, and items were retained that had the highest item-total score correlations. Endorsement frequencies were determined and

TABLE 6.2 Millon's Bioevolutionary Characterization of Personality Styles and Personality Disorders

<i>Adaptation</i>	<i>Sources of Reinforcement</i>				
	<i>Independent</i>	<i>Dependent</i>	<i>Ambivalent</i>	<i>Discordant</i>	<i>Detached</i>
Active					
Normal	Unruly	Sociable	Sensitive	Forceful	Inhibited
Disordered	Antisocial	Histrionic	Passive-aggressive	Aggressive	Avoidant
Passive					
Normal	Confidant	Cooperative	Respectful	Defeatist	Introversive
Disordered	Narcissistic	Dependent	Compulsive	Self-defeating	Schizoid

item–scale correlations were run; items with endorsement frequencies less than .15 or greater than .85 were then eliminated. The remaining 440 items were then screened to ensure there was an adequate number of items for each scale, and the total number of items was reduced to 289. An external-criterion validation was then conducted in which 167 clinicians gave the test to their patients and completed a diagnostic form. On the basis of the results, the total number of items was reduced to 150, three scales were eliminated, and three scales were added. Then the entire process was repeated, resulting in a 175-item test.

The standardization sample eventually consisted of 1,591 clinical patients (58% males, 42% females) ranging in age from 18 to 66 and another 256 patients (57% males, 43% females) selected for cross-validation. MCMI–I test protocols were obtained from 233 clinicians at 108 hospitals and outpatient clinics, and 39 private practitioners from 27 states plus Great Britain. The non-clinical population consisted of 297 people from a variety of community settings (48% males, 52% females) and ranging in age from 18 to 66.

Millon revised the MCMI–I because new personality disorders were included in his theory, the *DSM* had been modified, and a comprehensive set of diagnostic criteria for personality disorders had been defined. Validation studies assisted in determining which items should be retained and which items should be eliminated. The MCMI–II standardization and validation process used the same procedures as those described for the MCMI–I. A provisional form of the MCMI–II was developed; it contained 368 items, including added items for a Sadistic and a Self-Defeating scale. A total of 45 items were subsequently changed in validation, and an item weighting system was introduced, with prototype items that were isomorphic with theory receiving the highest weight in scoring. The Disclosure, Desirability, and Debasement scales were added. The MCMI–I Antisocial Personality Disorder scale was divided into an Antisocial scale and an Aggressive Personality Disorder scale. The MCMI–II normative sample consisted of 1,292 clinical patients, 50% male and 50% female; 88% were White, 7% Black, 4% Hispanic, and 1% other races. The age range was similar to that for the MCMI–I.

The MCMI–III standardization sample consisted of 600 clinical patients and another 398 for cross-validation. The sample was 86% White, 9% Black, 3% Hispanic, and 2% other races, and the sample participants ranged in age from 18 to 88. Correctional inmates represented 8.5% of the standardization sample. Two new scales were added (Depressive Personality Disorder and Post-Traumatic Stress Disorder), up to 50% of the MCMI–II items were changed, and scales were reduced in length. Terms dealing with child abuse and eating disorders were added but not scored on any scale, and the prototype item weighting system was changed from a 3-point to a 2-point scale. Again, Millon emphasized and employed the three-stage validation process previously described.

More recently, Millon developed new norms for the MMI–III. Females ($N = 397$) now comprise 52.8% of the standardization sample, and males ($N = 355$) 47.2%. Whites comprise 76% of the sample, African Americans 11%, Hispanics 9%, and the remaining 4% were of other races. Also, there were increased prevalence rates for the scales Major Depression, Bipolar: Manic, and Drug Dependence, and decreased prevalence rates for the Masochistic personality disorder scale.

Furthermore, Seth Grossman has developed a set of facet subscales for the MCMI–III (Grossman, 2008; Grossman & del Rio, 2005), and these are routinely scored through the test's publisher. Theoretically, two individuals may have identical base rate (BR) scores on a given scale by differentially endorsing a different set of items within the scale. Facet subscales allow the assessor to determine the salient dimensions endorsed by the respondent, thereby refining the psychological assessment of that disorder. These facet subscales are somewhat analogous to the Harris–Lingoes subscales of the MMPI–2. However, these were rationally derived, whereas the Grossman facet subscales were developed by factoring the primary personality scales. Table 6.3 presents the most current iteration of the MCMI–III along with the Grossman Facet Scales.

TABLE 6.3 MCMI–III Scales and the Grossman Facet Scales

<i>Scale</i>	<i>Grossman Facet Scales</i>
Schizoid	Temperamentally Apathetic Interpersonally Unengaged Expressively Impassive
Avoidant	Interpersonally Aversive Alienated Self-Image Vexatious Representations
Depressive	Temperamentally Woeful Worthless Self-Image Cognitively Fatalistic
Dependent	Inept Self-Image Interpersonally Submissive Immature Representations
Histrionic	Gregarious Self-Image Interpersonally Attention-Seeking Expressively Dramatic
Narcissistic	Admirable Self-Image Cognitively Expansive Interpersonally Exploitive
Antisocial	Expressively Impulsive Acting-Out Mechanism
Aggressive (Sadistic)	Temperamentally Hostile Eruptive Organization Pernicious Representations
Compulsive	Cognitively Constricted Interpersonally Respectful Reliable Self-Image
Passive-Aggressive (Negativistic)	Temperamentally Irritable Expressively Resentful Discontented Self-Image
Self-Defeating	Discredited Representations Undeserving Self-Image Severe Personality Pathology
Schizotypal	Estranged Self-Image Cognitively Autistic Chaotic Representations
Borderline	Temperamentally Labile Interpersonally Paradoxical Uncertain Self-Image
Paranoid	Cognitively Mistrustful Expressively Defensive Projective Mechanism

While the facet scales may be clinically useful as an aid in personality description, emphasizing those facet subscales that are clinically elevated means that there are some troublesome problems with them. First, Grossman factor-analyzed the entire MCMI–III item pool to develop the facet scales. It might have been preferable to factor the individual parent personality disorder scale to develop subscales. Second, while BR scores were calculated, no information has been provided regarding how they were able to determine the prevalence for traits such as “Chaotic Representations” (a Grossman facet scale on the Schizotypal disorder scale). Furthermore, it seems inappropriate to use the BR scores since they are not linked with any diagnostic prevalence rate. Third, these subscales lack external criterion validity and lack any evidence of external (empirical) validity. Fourth, the BR metrics on the MCMI scales range from 0 to 115, whereas the BR scores on the facet scales range from 0 to 99. Perhaps these problems will be addressed in future research. Meanwhile, forensic psychologists are urged to use these facet scales with caution and not to rely on them to reach major conclusions, since at present they could not withstand close scrutiny upon cross-examination (Dyer, 2008).

Millon persuasively argued that personality disorders and clinical syndromes are not normally distributed in the general population. Therefore, transforming raw scores to a normalized distribution, using *T*-scores or standard scores, is not statistically appropriate. Millon then developed the *base rate* (BR) *score*, which addresses the issue of prevalence rates of the disorder being measured. Millon arbitrarily set a BR of 60 as representing the mean raw score of the clinical standardization sample on each scale and a BR score of 30 as the mean raw score of the nonclinical standardization sample. He then found that point in the distribution of scores where the patient had all of the traits and behaviors of the disorder or syndrome at the diagnostic level and assigned a BR score of 85 to that score. He interpolated BR scores between 75 and 84 for patients who had some of the features of the disorder or syndrome but not at the diagnostic label. A unique aspect of the instrument’s BR score is that *BR greater than 84 is that point in the distribution where the percentage of scores is equal to the rate of occurrence of that disorder within the population*. Ordinal values for other points in the distribution were assigned BR scores, and then conversion tables were added to the test manual.

Also, additional scoring adjustments are needed according to whether the respondent has endorsed an excessive number of items, resulting in the appearance of a favorable (Desirability scale) or an unfavorable (Debasement scale) impression. The Denial versus Complain adjustment alters scores upward for respondents who deny or accentuate problems. Their scores are adjusted upward if they obtained highest scores on the Personality Disorder scales of Histrionic, Narcissistic, and Compulsive, and their scores are lowered if they obtained high scores on Avoidant, and Self-Defeating. These adjustments are made to the Schizoid, Borderline, Paranoia, Anxiety, Dysthymia, and Somatoform scales. Scores on the Disclosure scale either increase or decrease scores on many of the MCMI–III scales, with the specific amount depending on the magnitude of the Disclosure score. Finally, adjustments are made for experiencing more acute psychological turmoil as manifested by high scores on Anxiety and Dysthymia and are also made depending on the setting in which the patient took the test (e.g., outpatient, inpatient, length of time as an inpatient). The test manual provides exact details on which scales are increased or decreased as a result of these multiple adjustments. As one can see, scoring the MCMI–III is a time-consuming, cumbersome, and error-prone process, and computer scoring of the test is definitely recommended, especially in forensic cases where testimony can be expected to be challenged.

Reliability

Information on the reliability and validity of the MCMI–I and MCMI–II has been extensively presented in the literature; similar information on the MCMI–III is not as substantial. Also, with

24 scales on the MCMI–III, the reliability of each scale would have to be determined by means of a multitrait–multimethod nomothetic network of studies among different populations. Details are presented here on the reliability of just one personality disorder scale—the Antisocial PD (scale 6A)—for illustrative purposes. This scale was chosen because the disorder is well described and had the highest reliability correlation in *DSM–IV* field trials among all personality disorders. Also, the disorder is quite prevalent in forensic (e.g., criminal) populations (Blackburn, Crellin, Morgan, & Tulloch, 1990).

Table 6.4 presents the research data on the test–retest correlations from all studies on scale 6A from all three versions of the test. There have been 15 studies on the stability and reliability of scale 6A, covering 20 clinical samples and non-clinical samples. For the MCMI–I, there are 11 data sets for clinical populations and 1 for a non-clinical sample. For the MCMI–II, there are 10 data sets from five studies. For the MCMI–III, there are 3 data sets. Stability estimates for a variety of clinical groups (inpatient and outpatient psychiatric samples, PTSD patients, and drug and alcohol abusers) ranged from .55 to .90 for clinical samples across a variety of test–retest intervals for scale 6 of MCMI–I, with a median of .79; for the MCMI–II scale 6A, stability estimates for clinical groups ranged from .63 to .84, with a median of .73, for up to 4 months. For the MCMI–III scale 6A, the median correlation was .76, based on three samples.

Internal consistency studies have also been published, although far fewer than test–retest reliability studies. Langevin, Paitich, Freeman, Mann, and Handy (1979) reported an internal

TABLE 6.4 Stability and Reliability of Scale 6A

<i>Authors</i>	<i>MCMI</i>	<i>Population</i>	<i>Sample Size</i>	<i>Interval</i>	<i>r'</i>
Millon (1983)	I	Psych. pts.	59	5–9 days	.90
			86	4–6 weeks	.83
McMahon, Flynn and Davidson (1985)	I	Alcoholics	96	4–6 weeks	.63
	I	Drug addicts	33	4–6 weeks	.72
	I	Drug abusers	33	3–5 weeks	.79
Piersma (1986)	I	Psych. inpts.	151	4–6 weeks	.55
Wheeler and Schwartz (1989)	I	College stds.	225	3 years	.60
Hyer et al. (1989)	I	PTSD vets	50	35 days	.79
Murphy et al. (1991)	I	Psych. inpts.	150	ca. 6 months	.64
Libb et al. (1990)	I	Depressed pts.	28	3 months	.62
Overholser (1990)	I	Psych. pts.	28	ca. 1 year	.85
		Major depress.	15	6 weeks	.79
Millon (1987)	II	Non-clinical	91	3–5 weeks	.88
		Psych. outpts.	37	3–5 weeks	.73
		Psych. inpts.	47	3–5 weeks	.64
Piersma (1989)	II	Psych. inpts.	98	21 days	.84
McMahon and Richards (1996)	II	Cocaine addicts	109	ca. 3 months	.79
			102	ca. 4 months	.63
			38	ca. 30 days	.74
Lenzenweger (1999)	II	College stds.	250	1 year	.74
				1 year	.70
				2 years	.70
Millon (1994)	III	Misc. psych.	87	5–14 days	.93
Piersma and Boes (1997)	III	Psych, inpts.	97	7–10 days	.76
Craig and Olson (1998)	III	Drug inpts.	35	ca. 6 months	.62

consistency estimate of .61 for MCMI–I scale 6, with a sample of 419 sex offenders. Millon (1987) reported an internal consistency estimate for MCMI–II scale 6A of .88. Sinha and Watson (2001) reported an internal median consistency estimate for MCMI–II 6A of .77 with university students. For MCMI–III scale 6A, Millon (1994) reported an internal consistency of .77, and an internal consistency for MCMI–III scale 6A of .73 has also been reported (Dyce, O'Connor, Parkins, & Janzen, 1977).

The data indicate that scale 6A of each version of the MCMI remains generally stable, suggesting that it is measuring rather enduring personality traits. The internal consistency of the scale appears good.

Validity

As with reliability, it would be difficult to describe the validity of the MCM, given the limited space. Again, within a multitrait–multimethod nomothetic network we would have to present data on the convergent and divergent validity on each scale of the MCMI–III for a variety of populations. For illustrative purposes, I present convergent validity data for scale 6A as well as for scale CC (Major Depression). The Antisocial Personality Disorder scale was selected for presentation because psychopathy is a frequent occurrence in many forensic determinations; the Major Depression scale was selected because depression is largely an observable behavior.

Table 6.5 presents the convergent validity data on MCMI scale 6A (all test versions). Researchers have most often correlated the MCMI Antisocial PD scale with the MMPI Antisocial scale (a personality disorder scale). Correlations ranged from $-.08$ to $.30$, with a median r' of $.15$, for the MCMI–I; correlations ranged from $.46$ to $.61$, with a median of $.57$, for MCMI–II scale 6A and the MMPI and MMPI–2 Antisocial scale (three studies); correlations for MCMI–III scale 6A and the MMPI–2 Antisocial scale ranged from $.57$ to $.70$, with a median of $.65$ (five studies). The data in Table 6.5 show that the median correlation of MCMI scale 6A (all versions) with similar constructs from structured clinical interviews was $.37$ and the median correlation with self-report inventories was $.45$. This suggests that method variance affects the convergent validity of these instruments, and a pattern of attaining higher correlations within methods (i.e., self-report inventories) than across methods (i.e., correlating a self-report measure with a structured clinical interview of that same construct) has been reported for many instruments that screen for personality disorders (Craig, 2003a) and has been well known for some time (Campbell & Fiske, 1959).

The Psychopathy Checklist (PCL) has become the standard measure for the diagnosis of antisocial personality disorder within a correctional setting. It shows the highest correlation with other measures of criminality (Salekin, Rogers, & Sewell, 1996). Only one study has correlated the PCL with MCMI scale 6A. Hart, Forth, and Hare (1991) reported a correlation of $.45$ between these two instruments. This brings us to an interesting point: The item content of scale 6A was theory driven and consists to type “types” of items. One group of items is consistent with the *DSM–IV* characterization of an antisocial disorder, emphasizing conduct-disordered behavior and criminality. However, the second group of items pertains to Millon’s *theory* as to what constitutes the essential behavior of the antisocial style. Millon believes that this personality type is motivated primarily by a desire to avoid being controlled by others and so acts in ways to dominate others before others dominate him or her. Given this motivational attribution, the MCMI, as revised, contains several items that tap this fiercely independent stance. This sense of independence is not contained in other measures of this style/disorder, and as a consequence the correlations between these instruments and scale 6A are lower than they otherwise would be.

Table 6.6 presents convergent validity data for the Major Depression scale (CC). The median correlation between MCMI–I scale CC was $.57$, based on eight studies with 10 data sets; the

TABLE 6.5 Convergent Validity of the MCMI–III Antisocial PD Scale with Similar Measures

<i>Author(s)</i>	<i>Instrument</i>	<i>MCMI</i>	<i>r'</i>
Marsh et al. (1988)	MMPI PD	I	-.08
Morey and Levine (1988)	MMPI PD	I	.30
Dubro and Wetzler (1989)	MMPI PD	I	.14
McCann (1989)	MMPI PD	I	.15
Zarella et al. (1990)	MMPI PD	I	.14
		I	.13
Schuler et al. (1994)	MMPI PD	I	.25
Wise (1994a)	MMPI PD	I	.09
Widiger and Sanderson (1987)	PIQ	I	.81
Hogg et al. (1990)	SIDP	I	.23
Klein et al. (1993)	Wisc DI	I	.11
Morey (1985)	ICL Agg/Sad	I	.45
Chick et al. (1993)	DSM–III–R Cklt	I	.04
Wise (1994b)	MBHI Forceful	I	.54
McCann (1991)	MMPI–PD	II	.57
Wise (1996)	MMPI–2 PD	II	.46
Wise (2001)	MMPI–2 PD	II	.61
Jones (2005)	MMPI–2 PD	II	.71
			.79
Turley et al. (1992)	SIDP	II	.47
Hart et al. (1991)	PCL	II	.45
Coolidge and Merwin (1992)	Coolidge	II	.53
Silberman et al. (1997)	Coolidge	II	.70
Soldz et al. (1993)	PDE	II	.37
Hart et al. (1993)	PDE sym. count	II	.41
Kennedy et al. (1995)	SCID–II	II	.17
Marlowe et al. (1997)	SCID–II	II	.49
Messina et al. (2001)	SCID–II	II	.27
Wierzbicki and Gorman (1995)	PDQ–R	II	.37
Bayon et al. (1996)	TCI Novelty Seeking	II	.40
Rossi, Haubin et al. (2003)	MMPI PD	III	.66
Hicklin and Widiger (2000)	MMPI PD	III	.57
			.70
Lindsay et al. (2000)	MMPI–2 PD	III	.57
Rossi, Haubin et al. (2003)	MMPI–2 PD	III	.65
Rossi, Van den Brande et al. (2003)	MMPI–2 PD	III	.65
Lindsay et al. (2000)	PDQ–4	III	.67

Note:

Coolidge = Coolidge Axis II Inventory; *DSM–III–R Cklt* = *DSM–III–R* Checklist; ICL Agg/Sad = Interpersonal Checklist for Aggression/Sadistic Personality Disorder; MBHI = Millon Behavioral Health Inventory Forceful scale; MMPI PD = Antisocial Personality Disorder scale from the Minnesota Multiphasic Personality Inventory; PCL = Psychopathy Check List; PDE = Personality Disorder Examination; PDE sym. count = Personality Disorder Examination symptom count; PDQ–4 = Personality Diagnostic Questionnaire, fourth edition; PIQ = Personality Inventory Questionnaire; SCID–II = Structured Clinical Inventory for *DSM–IV* Personality Disorders; SIDP = Structured Inventory for DSM Personality Disorders; RCI Novelty Seeking = Temperament and Character Inventory Novelty Seeking scale; Wisc PDI = Wisconsin Personality Disorder Inventory.

median correlation between MCMI–II scale *CC* and similar measures ($N = 4$) was .52; the median correlation between MCMI–III scale *CC* and other measures of major depression was .74, but the three studies in question were based on information from the test manual and have not been independently validated. Prior versions of the MCMI depression scale were criticized because they did not contain items pertaining to the vegetative signs of depression, and these symptoms are the hallmark of the disorder. The MCMI–III now includes these types of items in scale *CC*, and the correlations with similar instruments are now quite robust.

Unlike many other omnibus personality instruments, the MCMI (all versions) has determined precise cutting points to establish a diagnosis. This allows researchers to evaluate the accuracy of the MCMI diagnostic decisions using diagnostic power statistics (Gibertini, Brandenberg, & Retzlaff, 1986). This is particularly important in forensic evaluations because the *Daubert* standard for federal rules of evidence stipulates that the trier of fact (i.e., the judge or jury) must make a preliminary assessment as to whether an expert witness's testimony is scientifically valid. The assessment would consider the error rate, and whether the methodology was capable of being tested, had been subjected to peer reviews, and had achieved general acceptance in the field. The standard error-of-the-measurement statistic would provide this information, but there are no such data in any of the MCMI test manuals. Instead, researchers have reported on the test's diagnostic accuracy using diagnostic power statistics. Among these data, the test's positive predictive power (PPP) may be salient in expert testimony because that statistic indicates the test's ability to successfully detect a condition/disorder (i.e., the test indicates the condition or disorder is present and it actually is present). Of course, in some circumstances it would also be desirable to know the test's negative predictive power (NPP)—that is, its ability to successfully rule out a condition or disorder. Keep in mind that the MCMI–III has 24 scales and that the PPP and NPP would have

TABLE 6.6 Correspondence of the MCMI Major Depression Scale with Similar Measures

<i>Author(s)</i>	<i>Instrument</i>	<i>MCMI</i>	<i>r'</i>
Millon (1983)	MMPI D	I	.67
Smith et al. (1988)	MMPI D	I	.57
Millon (1983)	Wiggins Depress	I	.83
	SCL-90 Depress	I	.75
McMahon and Davidson (1985)	POMS Depress	I	.54
	POMS Confusion	I	.56
Goldberg et al. (1987)	Beck Depress Inv	I	.62
	Hamilton DRI	I	.45
Choca et al. (1988)	Hamilton DRI	I	.56
Holliman and Guthrie (1989)	CPI Well Being	I	–.58
McCann (1990)	MMPI D	II	.58
Blais et al. (1994)	MMPI-2 D	II	.46
Craig and Olson (1992)	16 PF Emotionally Stable	II	–.34
Bayon et al. (1996)	TCI Harm Avoid	II	.65
Millon (1994)	MMPI-2 D	III	.71
	SCL\CL-90 Depress	III	.74
	Beck Depress Inv	III	.74

Note:

Beck Depress Inv = Beck Depression Inventory; CPI Well Being = California Psychological Inventory Well Being; Hamilton DRI = Hamilton Depression Inventory; MMPI D = Minnesota Multiphasic Personality Inventory Depression (scale 2); POMS Depress = Profile of Mood States Depression; SL-90 Depress = Symptom Check List-90 Depression; 16 PF Emotionally Stable = Sixteen Personality Factors Questionnaire Emotionally Stable; TCI Harm Avoid = Temperament and Character Inventory Harm Avoidance; Wiggins Depress = MMPI Wiggins Content scale Depression.

to be determined for each one. The data are too voluminous to present here. Instead, I present diagnostic power data for the Borderline Personality Disorder in Table 6.7.

The diagnostic power of a test is dependent, in part, on the prevalence rate of the disorder within the researched sample as well as the criteria used to determine the presence or absence of a disorder. Although Millon has indicated that a BR score greater than 84 reflects the disorder at the diagnostic level, most researchers have also reported diagnostic power statistics at the BR greater than 74 level as well. The data indicate that the Borderline PD scale PPP ranges from .09 to .88, with a median value of .56, for all test versions across all criteria. The NPP ranges from .68 to .96, with a median of .94. This suggests that the Borderline PD scale is an excellent scale at ruling out a disorder and is able to detect the disorder slightly better than half the time, on average. This is not an unusual finding, for assessment measures have historically been better at ruling out a disorder than ruling it in. Of course, the forensic psychologist will need to know the diagnostic power statistics for any scale used as part of expert testimony. Most of the MCMI scales have a diagnostic power beyond chance when used to help establish an individual diagnosis (McCann, 2002).

Some dismiss the idea that diagnoses are relevant to many forensic issues that come before the court. A recent case illustrates this.

The examinee was a 42-year-old married housewife with a history of schizophrenia. She had been under psychiatric care for many years and had periods of remission; in fact, most recently she had been symptom-free. She had three children, and her husband was a successful businessman. Recent business had required him to travel to Europe during the summer for three months. He

TABLE 6.7 Diagnostic Power of the Borderline PD Scale

	<i>Classification</i>	<i>MCMI P</i>	<i>Prev</i>	<i>Sens</i>	<i>Spec</i>	<i>PPP</i>	<i>NPP</i>	<i>D×P</i>
(1)	BR>74	I	.20	.77	.92	.71	.94	.89
	BR>84		.13	.60	.95	.64	.94	.90
(2)	BR>75	I	.15	.64	.64	.25	.91	X
	BR>85		.33	.86	.30	.88	X	X
(3)	X	I	.14	.77	.93	.64	.96	X
(4)	BR>84	I	.08	.42	.80	.17	.96	X
	Clin Dx		.33	.60	.43	.35	.68	X
(5)	BR>74	I	.05	.40	.78	.09	.96	.76
	BR>84		.05	.20	.91	.10	.96	.87
(6)	Two highest scales	II	.15	.72	.93	.66	.95	.90
	Highest in code		.11	.67	.95	.60	.96	.92
(7)	BR>74	II	.65	.74	.61	.78	.71	X
	BR>84	II		.58	.83	.68	.76	.74
(8)	SCID DX	II	.17	.50	.91	.56	.89	.83
(9)	SCID-II	II	X	.70	.70	.49	.84	.70
(10)	BR>74	III	.27	.46	.77	.30	X	.70
	BR>84		.10	.20	.90	.11	X	.68
(11)	BR>74	II	X	.57	.66	.30	.85	.83
	BR>84		X	.48	.82	.41	.86	.75
(12)	Clin Dx	III	X	.71	X	.60	.X	X

Note:

Clin Dx = clinical diagnosis; Prev = prevalence; Sens = sensitivity; Spec = specificity; PPP = positive predictive power; NPP = negative predictive power; DP = overall diagnostic power.

Sources: (1) Gibertini, Brandenberg, and Retzlaff (1986); (2) Torgersen and Alnaes (1990); (3) Miller et al. (1992); (4) Patrick (1993); (5) Chick et al. (1993); (6) Millon (1987); (7) McCann et al. (1992); (8) Guthrie and Mobley (1994); (9) Hills (1995); (10) Millon (1994); (11) Marlowe et al. (1997); (12) Millon (1997).

periodically kept in touch with his wife, who told him things were going well. Because of his wife's past history, her husband had been managing their financial affairs, and he continued to do so while in Europe. However, upon his return home he found the house in disarray, the children emaciated and sickly, and his wife depressed, lethargic, and anergic with suicidal ideation. The children were temporarily removed from the home and placed out of state with near relatives. After receiving medical treatment, the wife was charged with neglect and jailed. The court ordered a psychological evaluation, and after reviewing the results of this examination her attorney hired an independent psychologist as expert witness. Both examiners agreed that the examinee was not experiencing any symptoms associated with schizophrenia, and hence this condition was ruled out as a reason for the neglect. Both examiners diagnosed major depression, single episode without psychotic features. However, although one expert testified that her depression was not severe enough to have caused the neglect and hence she should be held culpable, the other expert testified that her symptoms of depression were directly related to her inability to properly care for her children.

Hence, diagnosis is relevant in many assessments, and the diagnostic power of an assessment measure also becomes relevant in that assessment.

Clinical Uses and Limitations

All versions of the MCMI have been used primarily with clinical patients in both inpatient and outpatient settings. There is an especially large literature on using the MCMI with drug addicts (Craig, 2005c; Craig & Weinberg, 1992a), alcoholics (Craig, 2005c; Craig & Weinberg, 1992b), patients with post-traumatic stress disorder (Hyer, Brandsma, & Boyd, 1997; Hyer, Melton, & Gratton, 1993), and spouse abusers (Craig, 2003b), and these findings are briefly summarized here.

Substance Abuse

The MCMI (all versions) has been extensively used in the assessment of substance abusers (e.g., alcoholics, cocaine addicts, heroin addicts, and poly-substance abusers). This is because substance abuse is endemic in a variety of psychiatric settings and clinical disorders. Here I highlight some of the major findings. For an in-depth presentation and discussion of issues and findings pertaining to assessing substance abusers with the MCMI, see Craig (2005c), Craig and Weinberg (1992a; 1992b), Flynn, McCann, and Fairbank (1995), and McMahon and Diamond (2008).

Drug Addiction

MCMI-based studies have reported that the modal profile of drug addicts (users of heroin and cocaine) reflects antisocial and narcissistic traits along with elevations on the Paranoia (*P*), Drug Abuse (*T*), and Alcohol Abuse (*B*) syndrome scales. The elevations on scale *P* reflect not clinical paranoia but a wary alertness associated with the drug-using lifestyle. Although the MCMI has shown higher prevalence rates than other measures, almost all studies have reported that drug addicts have high rates of antisocial personality disorder (Craig, 2005c). Female addicts score higher than male addicts on the Self-Defeating PD scale (Haller, Miles, & Dawson, 2002). However, investigators have found up to six types of personality profiles upon typological analysis, including a within-normal-limits profile (Calsyn, Fleming, Wells, & Saxon, 1996; Craig, Bivens, & Olson, 1997). Drug addicts have been able to avoid detection of their substance abuse on the MCMI (Craig, Kuncel, & Olson, 1994; Fals-Stewart, 1995), and few studies have found detection rates for scale *T* greater than 50% (Calsyn, Saxon, & Daisy, 1990). However, a review of all published studies which reported BR scores for scale *T* in drug addict populations found that scale *T* was

elevated in clinical ranges in 85% of the studies (Choca, Rossini, & Craig, 2009). This finding is consistent with diagnostic power statistics for scale *T*. The difference may be attributed to differences in BR scores used in studies to establish detection rates. The MCMI–III continues to be used in the assessment of substance abusers (McMahon, Malow, Devieux, Rosenberg, & Jennings, 2008).

Alcoholism

A literature review suggested that the personality disorders of passive-aggressive (negativistic) and antisocial, with elevations on Anxiety, Depression, Alcohol Abuse, and Drug Abuse, predominated in the MCMI profiles of male alcoholics (Craig, 2005c; Craig & Weinberg, 1992b). Research also found this character type in cluster analysis, along with elevations on Avoidant, and within-normal-limit profiles reflecting narcissistic traits (Donat, 1994; Matano, Locke, & Schwartz, 1994; McMahon, Malow, & Penedo, 1998). Female alcoholics' types included those with passive-aggressive (negativistic) personality disorder elevations and within-normal-limits profile types with Compulsive and Histrionic/Narcissistic at subclinical elevations. Having an elevation on the Compulsive PD scale has been associated with higher social functioning, episodic rather than continuous drinking, and positive treatment outcomes. The MCMI–III continues to be a frequently used assessment instrument among alcohol researchers (Picci et al., 2012).

Post-traumatic Stress Disorder

The MCMI has been extensively used with Vietnam combat veterans (Craig & Olson, 1997; Hyer, Brandsma, & Boyd, 1997). The personality styles of schizoid/avoidant, passive/aggressive (negativistic), and compulsive/dependent are most prominent in the MCMI protocols of Vietnam veterans with PTSD. These findings led Hyer to conclude that the MCMI 8A2 (Passive-Aggressive/Avoidant) profile has appeared with such consistency in this population that it can be construed as a “traumatogenic” profile. However, this codetype has also appeared in populations without trauma (Craig, 1995). The personality styles of avoidant, passive-aggressive (negativistic), and borderline demonstrate conceptual closeness to the PTSD construct, and how people cope with traumatic memories is intrinsically related to their personality styles (Hyer et al., 1997). Research has begun to use the MCMI for investigating civilian trauma as well (Alexander et al., 1998; Allen, Coyne, & Huntoon, 1998; Lecic-Tosevski, Gavrilovic, Knezevic, & Priebe, 2003), but the data are insufficient to warrant any conclusions at this point concerning the pattern of scores.

Preliminary results on the MCMI–III PTSD scale (*R*) suggest good concurrent validity in distinguishing combat veterans with and without PTSD (Hyer, Boyd, Stanger, Davis, & Walters, 1997). Also, scale *R* was able to successfully discriminate between PTSD and non-PTSD patients, it was the best predictor of PTSD in a multiple regression equation, and its sensitivity and specificity were higher than those provided in the MCMI–III test manual (Craig & Olson, 1997). The sample was an inpatient veteran population. Use of the MCMI with a PTSD population is more extensively presented elsewhere (Hyer, Melton, & Gratton, 1993).

Domestic Violence

Personality disorders are quite prevalent among male domestic abusers, with antisocial, aggressive, and passive-aggressive (negativistic) disorders demonstrating higher prevalence rates in this population than other personality disorders. However, some male spouse-abusers produce a within-normal-limits MCMI profile, and their domestic violence may not be a product of a personality

disorder. MCMI test results suggest a relative absence of Axis I syndromes, except for substance abuse. Elevations on the Antisocial, Aggressive, and Passive-Aggressive (Negativistic) scales may constitute a psychological marker for risk of domestic violence and abuse, and may even be able to predict it, although no study has tested this latter hypothesis (Craig, 2003a). Female batterers were more likely than their male counterparts to score in the clinical range on Borderline, Somatoform, Major Depression, Bipolar Disorder, and Delusional Disorder syndrome scales (Henning, Jones, & Holdford, 2003). The MCMI–III continues to be a useful screening and assessment instrument for victims of partner violence (Pico-Alfonso, Echeburúa, & Martinez, 2008).

Forensic Uses and Limitations

In *People v. Stoll* (1989), the court upheld the admissibility of the MCMI in court testimony. McCann and Dyer (1996) found 22 cases through 1994 that have allowed MCMI-based testimony pertaining to a variety of issues, including child custody, personal injury, classification of sex offenders, domestic violence, determination of disability, employment discrimination, assessment of malingering and deception, and termination of parental rights.

The MCMI has an increasing literature covering forensic populations. It has been used with correctional inmates (Ahlmeyer, Kleinsasser, Stoner, & Retzlaff, 2003; Blackburn, 1998; Gudjonsson & Main, 2008; Retzlaff, Stoner, & Kleinsasser, 2002; Schwartz, Seeman, Buboltz, & Flye, 2004; Stoner, 2008; White, Ackerman, & Caraveo, 2001; Wise, 2001), as a predictor of institutional misconduct among incarcerated felons (Kelln, Dozois, & McKenzie, 1998), with sexual offenders (Ahlmeyer et al., 2003; Carpenter, Peed, & Eastman, 1995; Chantry & Craig, 1994a; 1994b; Cohen et al., 2002; Davis & Archer, 2010; Holt, Meloy, & Strack, 1999; Lehne, 2002; Oliver, Beech, Fisher, & Beckett, 2007), and in child custody (Lenny & Dear, 2009) and parental competency evaluations (Stredny, Archer, & Mason, 2006).

Nevertheless, use of the MCMI in forensic remains controversial (Lally, 2003; Rogers, 2003). Some have argued that the MCMI was standardized on a clinical population and did not include normals, and hence does not allow for discrimination between patients and normals (Butcher & Miller, 1999; Hess, 1998). This critique has been mitigated by current norms that include correctional samples. This issue may be especially salient in child custody evaluations, where parents are often free of personality disorders, compared with other types of forensic assessments (Otto & Butcher, 1995). In one survey, 201 psychologists from 39 states were questioned on multiple aspects of child custody evaluations. The MCMI was used by 34% of the sample (Ackerman & Ackerman, 1997). Surprisingly, 10 years before, the MCMI was not used at all for these purposes (Keilin & Bloom, 1986).

The MCMI is the second most frequently used personality test in civil (Boccaccini & Brodsky, 1999), criminal (Borum & Riso, 1995), and in child custody evaluations (Quinnell & Bow, 2001). Use of the MCMI in forensic work is beginning to receive more favorable comments by forensic psychologists (Camara, Nathan, & Ouate, 2000; Lampel, 1999; Schutte, 2001). However, attorneys may pick up the criticisms of the use of the MCMI and attempt to have MCMI-based testimony dismissed on the grounds that it does not meet the *Daubert* standards. In fact, arguments proffered by Rogers, Salekin, and Sewell (1999) were cited in an Alabama plaintiff brief that objected to MCMI-based testimony concerning the psychologist's evaluation of the plaintiff in an employment-related case. However, the trial judge rejected the motion, which sought to exclude anything related to the MCMI–II on the grounds that the test did not meet the *Daubert* standards (*Jeffrey v. Dillard's Department Stores and Dillard's Bel Air Mall*, 2000).

Also, it could be argued that in the case of forensic examinees the test does meet the criteria for usage because these examinees are in a setting in which they are being evaluated by a mental health

expert and hence could meet the definition of a “clinical” patient. Furthermore, the literature base of the MCMI–III continues to expand, and published normative data on the use of the MCMI–III in custody evaluations now exist (McCann et al., 2001).

One of the more serious assaults on the MCMI questioned whether the MCMI scales meet the *Daubert* standard as to admissibility based on scientific merit (Rogers, 2003; Rogers, Salekin, & Sewell, 1999). One argument was that only the Avoidant, Schizotypal, and Borderline scales (of the MCMI–II) have been the subject of sufficient convergent and discriminant validity studies to meet the admissibility criteria, whereas the other scales have not been. A second argument was that there are insufficient research data on which to base expert evaluations in many common forensic applications, such as insanity evaluations.

Dyer and McCann (2000) rebutted these arguments and concluded that “while the MCMI–II meets . . . [the] .80 criterion for all personality disorder scales and the MCMI–III meets it for most scales, other instruments do not fare as well. . . . The MMPI–2 has several scales . . . that have a reliability of less than .50” (p. 493).

In contrast, one rule of evidence is that the method is in frequent use in psychological practice—a criterion that is clearly met based on test usage surveys (Butcher & Rouse, 1996). Furthermore, no rules of evidence mandate that a test or method be without controversy before material based on that technique is admissible (Craig, 2005b). Still, the prudent forensic psychologist would be well served to be aware of the arguments surrounding this debate.

One potential problem area for a forensic psychologist using the MCMI–III as part of expert testimony pertains to the (mis)interpretation of BR scores in cross-examination. Suppose you render an opinion that an examinee is suffering with schizophrenia and that this conclusion was based, in part, on the examinee’s BR score on the Thought Disorder scale. During cross-examination, the attorney asks you if the MCMI–III test manual is an authoritative source on the test, and you assert that it is. The attorney points out that the PPP of scale SS (Thought Disorder) is .52 and asks if this means that the scale is able to detect the disorder 52% of the time, and you agree that this is what it means. Then the attorney suggests that this must mean that the test fails to diagnose the disorder 48% of the time, and again you agree. Finally, the attorney asks you, “Isn’t it fair to conclude that this test is no better than flipping a coin in making the diagnosis because in a coin flip we would be right 50% of the time?” What is your response?

The attorney is attempting to mislead the jury, or perhaps she or he does not really understand diagnostic power statistics as applied to the MCMI–III. Since the prevalence rate of patients with a thought disorder in the MCMI–III standardization sample was 10%, a coin flip would be incorrect 90% of the time and correct only 10% of the time, whereas the MCMI–III would achieve an immense improvement over the base rate (five times better).

Another criticism of the MCMI is that it overpathologizes. This contention was initially proposed by Dana and Cantrell (1988) and was voiced by Widiger (2001) in a review of the test. In settings where substantial clinical psychopathology would not be expected, such as college counseling or divorce mediation, it is argued that the MCMI–III overestimates the extent of personality disorders. If this were true, then the MCMI operating statistics would demonstrate weak specificity statistics across most scales. However, the opposite is true. Specificity statistics across MCMI scales have been excellent. This suggests that the test does not incorrectly label a person as having a particular disorder (Dyer, 2005).

There is potential for response bias to distort test results. This is not unique to the MCMI, but the scoring adjustments make it almost impossible to invalidate the test by symptom exaggeration (Morgan, Schoenberg, Dorr, & Burke, 2002).

Race and Gender Issues

Only two studies have explicitly researched the question of possible racial bias in the MCMI–I (Choca, Shanley, Peterson, & VanDenburg, 1990; Hamberger & Hastings, 1989), one has investigated racial bias in the MCMI–II (Munley, Vacha-Haase, Busby, & Paul, 1998), and six other studies have reported on differences between Blacks and Whites and between men and women on MCMI scales, although these differences were not the main focus of those studies (Davis, Greenblatt, & Pochyly, 1990; Dillon, 1988; Donat, Walters, & Hume, 1992; Gabrys, 1988; Matano et al., 1994; Piersma, 1986). The data indicated that Blacks scored higher than Whites on the Narcissistic, Paranoid, Drug, and Delusional Disorder scales; Whites scored higher than Blacks on the Dysthymia scale. Males scored higher on Antisocial, and females scored higher than men on Somatoform and Major Depression. Bornstein (1995) reported that women of all ages consistently score higher than men on dependency on all self-report and projective test measures of dependency. These data were all based on MCMI–I and MCMI–II research. The real (unanswered) question is whether these differences represent true differences between men and women and between Blacks and Whites or whether they are suggestive of racial and/or gender bias. One study concluded that some of the items on the MCMI–III had the *potential* for gender bias, primarily those on the Narcissistic scale (Lindsay & Widiger, 1995). However, although the BR scores on the aforementioned scales reached statistical significance, they would not have changed the *diagnostic decision* based on those scores.

Overview of Strengths and Weaknesses

The strengths of the MCMI include these: (1) test development was anchored to theory, (2) the scales are reliable and internally consistent, (3) there are precise cutting rules for the classification of personality disorders, and (4) there is more research on all versions of the MCMI than on all other self-report personality disorder assessment instruments combined.

Two interesting developments pertain to the effort to refine personality disorder assessment and make it more precise. The refinement began as theory-driven characterizations but eventuated in additional ways to assess personality disorder subtypes. Millon (2011b) theorized that there are subtypes of personality disorders and that each has the essential characteristics that define the parent disorder but contribute a unique aspect of the disorder. Roger Davis has presented ways in which the MCMI–III can be used to assess these theorized subtypes (Davis & Patterson, 2005).

The second development pertains to Millon's conceptual model in describing the prototypal features of each disorder. He categorized each personality disorder in terms of eight personologic domains that include both structural and functional domains: their self-image, regulatory mechanisms, cognitive style, object representations, interpersonal conduct, expressive behavior, morphological organization, and mood/temperament. Table 6.8 displays this theoretical model as it applies to Antisocial Personality Disorder.

Continuing problems with the test include these: (1) Research has not established how much of the prodigious MCMI–I and MCMI–II research findings can be applied to the MCMI–III. Some research suggests that applying previous research to the new version may be questionable practice. For example, the PTSD profile codes in the MCMI–I and MCMI–II differ from those in the MCMI–III, and possibly modal profile codes of drug addicts as well. (2) Forensic psychologists may be asked to evaluate the extent of possible trauma, but the literature on the use of the MCMI (all versions) with a traumatized civilian population is sparse. (3) The issue of whether or not the MCMI overpathologizes personality disorders continues to haunt this test, and additional research is needed with the MCMI–III to address this lingering question. (4) Some forensic psychologists continue to experience problems with inadequate knowledge of cutoff scores and an over-reliance on computer-generated interpretive reports (Bow, Flens, & Gould, 2010).

TABLE 6.8 Antisocial Personality Personologic Domains

<i>Personologic Domain</i>	<i>Description</i>
Behavioral level	
Expressive emotion	Impulsive
Interpersonal conduct	Irresponsible
Phenomenological level	
Cognitive style	Deviant
Self-image	Autonomous
Intrapsychic level	
Regulatory mechanisms	Acting-out
Morphologic organization	Unruly
Object representations	Debased
Mood/temperament	Callous

CASE EXAMPLE

The examinee was a 39-year-old married White male who was originally seen by me for an initial evaluation by an HMO after he requested help for an alcohol problem.

His presenting complaints in the diagnostic interview were “emotional stress and alcohol.” He reported a drinking history since age 16 and problematic drinking for about the past two years. He drank only on weekends and only a half-pint of vodka. He said that alcohol “acts like medicine” and calms him down. He reported an ability to go to a party and have only two or three drinks, but the next day he would feel the need to drink upon awakening, and he said he “binged” on Saturday and Sunday, again claiming that he did not drink more than a half-pint during each “binge.” He denied tolerance and said he thought he could no longer drink as much as previously. He denied a history of blackouts, DTs, memory problems, hepatitis, or diminished sexual activity or drive. He did report mild “shakes,” which dissipated with drinking. When asked if he was physically dependent on alcohol, he replied “Probably.” He reported early morning awakenings and “tossing and turning” in his sleep. He denied any depression, and said his appetite was good, and denied use of other drugs.

He had been treated once, in a 28-day inpatient alcohol treatment ward in a private hospital 1½ years previously. He had attended AA for about three months but discontinued it because he thought he did not need it. He was initially seen there by a psychologist, who had seen the examinee for four sessions. He also reported no other treatment history and said that a physical exam had found no problems. He had recently gone to his physician, who prescribed a tranquilizer, but the examinee did not fill the prescription, fearing he would become physically dependent on the drug.

He was the director of engineering at a moderately sized manufacturing facility. He did not believe that his peers or his boss knew about his drinking problem. He said that his performance evaluations were high, but his peers saw him sweating at work, and he had had some recent absenteeism, particularly during the early part of the week, which they were beginning to notice. He denied using vodka to avoid detection at work and seemed genuinely naïve about this.

He had been married for four years. It was the first marriage for both him and his wife. He said that his wife thought he drank too much, and he had been hiding his drinking because he did not want to get into fights with her. He reported no marital problems. He had no legal history and denied any driving under the influence (DUI) arrests, charges, or even driving after he had been drinking at all.

His family history indicated that his mother abstained and his father “drank too much” but was never treated for alcoholism and had no family problems due to drink. His sister did not drink.

When pressed about his “emotional stress,” which he said precipitated his drinking, he was unable to elaborate or specify. He complained about his long drive at night and some job pressures, which he later said were “not too great,” and he said he drank more during “good and happy times” (i.e., birthdays, holidays).

During the interview, the examinee sweated frequently, using tissue to wipe his face. No other symptoms were noticeable. He said that he had had his last drink two days ago: half a pint of vodka.

I diagnosed him as alcohol dependent and sent in my report with treatment recommendations. Several months later, he contacted me because his company had fired him for tardiness and absenteeism. He had retained an attorney and filed a civil suit, under Title 56, Chapter II of the Department of Human Rights section on handicap Discrimination in Employment Act, arguing that he was handicapped as a result of alcoholism and that, contrary to law, he was not given the chance to seek treatment prior to his termination. He retained me as an expert witness to testify that he did have alcoholism. I gave him another clinical interview and psychological testing. Table 6.9 presents the MCMI–III scores. The results suggest that he had a compulsive personality style. These traits would serve him well in his capacity as an engineer and manager. A compulsive style has also been shown to have positive prognostic

TABLE 6.9 MCMI–III Scores for the Case Example Examinee

Disclosure	15	Schizotypal	19
Desirability	78	Borderline	13
Debasement	12	Paranoid	31
Schizoid	38	Anxiety	27
Avoidant	11	Somatoform	40
Depressive	30	Bipolar	52
Dependent	64	Dysthymic	27
Histrionic	62	Alcohol	60
Narcissistic	49	Drug	35
Antisocial	32	PTSD	27
Aggressive	52	Thought Disorder	2
Compulsive	80	Major Depression	2
Passive-Aggressive	5	Delusional	37
Self-Defeating	5		

Grossman Facet Scales for Compulsive Personality Disorder

Cognitively Constricted	BR 66
Interpersonally Respectful	BR 86
Reliable Self-Image	BR 60

value in the treatment for alcoholism (McMahon, Davidson, & Flynn, 1986). The problem was that the Alcohol Scale (*B*) was in the normal range (*BR* = 60). (His *MAC-R*, *APS*, and *AAS* on the *MMPI-2* were also in the normal range.)

The examinee, in a clinical interview, was diagnosed as alcohol dependent because of the following: sleep disturbance with early-morning awakenings, sweating (early withdrawal), absenteeism, surreptitious drinking, and complaints by his wife about his drinking. On the disease model of alcoholism, he was in the early phases of alcoholism on a Jellinek chart and would probably be classified as a gamma alcoholic, although he showed some signs of the delta type. Why wasn't his substance abuse scale-elevated?

I had to do an item analysis, going over his responses to each item on the scale (and on the other substance abuse scales as well) to be able to ask him why he answered the questions in the manner he did. His explanations were quite revealing. For example, he doesn't have trouble controlling his drinking because he doesn't drink anymore. He doesn't experience any physical problems from drinking because he doesn't drink anymore. He doesn't have trouble stopping himself from using alcohol because he doesn't drink anymore and attends AA. He said his habit of using drugs didn't cause him to miss work in the past because he never used drugs and only missed work because of his use of alcohol.

In subsequent testimony, I presented the court with a copy of the Jellinek chart depicting the progressive nature of alcoholism, indicated the examinee's symptoms, and showed the judge where on the chart the examinee was functioning. I presented the results of my findings from psychological testing, cited the (negative) results from the substance abuse scales, and then detailed the examinee's individual answers to these items upon a detailed inquiry. The testimony was apparently so thorough that the defendant's attorney had few questions upon cross-examination.

Some guidelines for use of the MCMI in forensic work have been published (Craig, 1999; McCann, 2002). These guidelines include the following recommendations:

- Make sure that the MCMI is an appropriate instrument for the situation in which it is being used.
- Be prepared to cite evidence on the reliability and validity of the MCMI validity scales.
- Maintain knowledge on the reliability, validity, and other technical aspects, especially in the area of diagnostic power, of the scales on which you are basing expert testimony.
- Ensure that there are a sufficient number of criterion-related MCMI-III validity studies that address the issue before the court.
- Test the examinee individually and do not violate chain-of-custody procedures.
- Be prepared to explain the *BR* score, and use computer scoring to avoid scoring errors.
- Maintain currency on recent test developments and interpretations as they apply to certain MCMI-III scales (Craig, 2005b; Halon, 2001; Lampel, 1999).
- Be prepared to defend the applicability to the MCMI-III of any research done with the MCMI-I and -II.
- Be prepared to discuss in lay terms the meaning of diagnostic efficiency statistics and how they differ from decisions based on chance alone.
- Do not use a computer interpretive report in forensic cases but use a computer scoring report. This is avoid a situation in which your conclusions differ from those contained in the computer

report, placing you in disagreement with the “expert” who wrote the computer interpretation and hence putting you on the defensive.

- Finally, be prepared to accept and report the limitations of the test and use the test in a manner that can be supported by empirical evidence (McCann, 2002).

References

- Ackerman, M. J., & Ackerman, M. C. (1997). Custody evaluation practices: A survey of experienced professions revisited. *Professional Psychology: Research and Practice*, 28, 137–145.
- Ahlmeier, S., Kleinsasser, D., Stoner, J., & Retzlaff, P. (2003). Psychopathology of incarcerated sex offenders. *Journal of Personality Disorders*, 17, 306–318.
- Alexander, P. C., Anderson, C. L., Brand, B., Schaffer, C. M., Grelling, B. Z., & Kretz, L. (1998). Adult attachment and long-term effects in survivors of incest. *Child Abuse and Neglect*, 22, 45–61.
- Allen, J. G., Coyne, L., & Huntoon, J. (1998). Complex posttraumatic stress disorder in women from a psychometric perspective. *Journal of Personality Assessment*, 70, 277–298.
- American Psychiatric Association (1983). *Diagnostic and statistical manual of mental disorders* (3rd ed.). Washington, DC: Author
- American Psychiatric Association (1987). *Diagnostic and statistical manual of mental disorders* (3rd ed., revised). Washington, DC: Author
- American Psychiatric Association (1994). *Diagnostic and statistical manual of mental disorders* (4th ed.). Washington, DC: Author
- Bayon, C., Hill, K., Svrakic, D. M., Przybeck, T. R., & Cloninger, C. R. (1996). Dimensional assessment of personality in an outpatient sample: Relations of the systems of Millon and Cloninger. *Journal of Psychiatric Research*, 30, 341–352.
- Blackburn, R. (1998). Relationship of personality disorders to observer ratings of interpersonal style in forensic psychiatric patients. *Journal of Personality Disorders*, 12, 77–85.
- Blackburn, R., Crellin, M. C., Morgan, E. M., & Tulloch, R. M. (1990). Prevalence of personality disorders in a special hospital population. *Journal of Forensic Psychiatry*, 1, 43–52.
- Blais, M. A., Benedict, K., & Norman, D. (1994). Associations among MCMI–II clinical syndrome scales and the MMPI–2 clinical scales. *Assessment*, 1, 407–413.
- Boccaccini, M. T., & Brodsky, S. L. (1999). Diagnostic test usage by forensic psychologists in emotional injury cases. *Professional Psychology: Research and Practice*, 30, 253–259.
- Bornstein, R. F. (1995). Sex differences in objective and projective dependency tests: A meta-analytic review. *Assessment*, 2, 319–331.
- Borum, R., & Grisso, T. (1995). Psychological test use in criminal forensic evaluations. *Professional Psychology: Research and Practice*, 26, 465–473.
- Bow, J. N., Flens, J. R., & Gould, J. W. (2010). MMPI–2 and MCMI–III in forensic evaluations: A survey of psychologists. *Journal of Forensic Psychology Practice*, 10, 37–52.
- Butcher, J. N., & Miller, K. B. (1999). Personality assessment in personal injury litigation. In A. K. Hess & I. B. Weiner (Eds.), *The handbook of forensic psychology* (2nd ed., pp. 104–126). New York: John Wiley.
- Butcher, J. N., & Rouse, S. (1996). Clinical personality assessment. *Annual Review of Psychology*, 47, 87–111.
- Calsyn, D. A., Fleming, C., Wells, E. A., & Saxon, A. J. (1996). Personality disorder subtypes among opiate addicts in methadone maintenance. *Psychology of Addictive Behaviors*, 10, 3–8.
- Calsyn, D. A., Saxon, A. J., & Daisy, F. (1990). Validity of the MCMI drug abuse scale with drug abusing and psychiatric samples. *Journal of Clinical Psychology*, 46, 244–246.
- Camara, W., Nathan, J., & Ouate, A. (2000). Psychological test usage: Implications in professional use. *Professional Psychology: Research and Practice*, 31, 141–154.
- Campbell, D. T., & Fiske, D. W. (1959). Convergent and discriminant validation by the multitrait–multimethod matrix. *Psychological Bulletin*, 56, 81–105.
- Carpenter, D. R., Peed, S. F., & Eastman, B. (1995). Personality characteristics of adolescent sexual offenders: A pilot study. *Sexual Abuse*, 7, 195–202.
- Chantry, K., & Craig, R. J. (1994a). Psychological screening of sexually violent offenders with the MCMI. *Journal of Clinical Psychology*, 50, 430–435.

- Chantry, K., & Craig, R. J. (1994b). MCMI typologies of criminal sexual offenders. *Sexual Addiction and Compulsivity*, 1, 215–226.
- Chick, D., Schaeffer, C. I., & Goggin, W. C. (1993). The relationship between MCMI personality scales and clinician-generated DSM-III-R personality disorder diagnoses. *Journal of Personality Assessment*, 61, 264–276.
- Choca, J. P., Bresolin, L., Okonek, A., & Ostrow, D. (1988). Validity of the MCMI in the assessment of affective disorders. *Journal of Personality Assessment*, 52, 96–105.
- Choca, J., Rossini, E., & Craig, R. J. (2009, March 9). Advanced interpretation of the MCMI-III. Paper given at the Annual Meeting of the Society for Personality Assessment, Chicago.
- Choca, J. P., Shanley, L. A., Peterson, C. A., & VanDenburg, E. (1990). Racial bias and the MCMI. *Journal of Personality Assessment*, 54, 479–490.
- Cohen, L. J., Gans, S. W., McGeoch, P. G., Poznansky, O., Itskovich, Y., Galynker, I. I. (2002). Impulsive personality traits in male pedophiles versus healthy controls: Is pedophilia an impulsive-aggressive disorder? *Comprehensive Psychiatry*, 43, 127–134.
- Coolidge, F. L., & Merwin, M. M. (1992). Reliability and validity of the Coolidge AXIS II Inventory: A new inventory for the assessment of personality disorders. *Journal of Personality Assessment*, 59, 233–238.
- Craig, R. J. (1995). Clinical diagnosis and MCMI code types. *Journal of Clinical Psychology*, 51, 352–360.
- Craig, R. J. (1999). Testimony based on the Millon Clinical Multiaxial Inventory: Review, commentary, and guidelines. *Journal of Personality Assessment*, 73, 290–316.
- Craig, R. J. (2003a). Use of the Millon Clinical Multiaxial Inventory in the psychological assessment of domestic violence: A review. *Aggression and Violent Behavior*, 8, 235–243.
- Craig, R. J. (2003b). Prevalence of personality disorders among cocaine and heroin addicts. *Directions in Addiction Treatment and Prevention*, 7, 33–42.
- Craig, R. J. (2005a). Alternative interpretations for the Histrionic, Narcissistic, and Compulsive personality disorder scales of the Millon Clinical Multiaxial Inventory. In R. J. Craig (Ed.) *New directions in interpreting the Millon Clinical Multiaxial Inventory* (pp. 71–93). New York: John Wiley.
- Craig, R. J. (2005b). *Personality-guided forensic psychology*. Washington, DC: American Psychological Association.
- Craig, R. J. (2005c). *Assessing substance abusers with the Millon Clinical Multiaxial Inventory*. Springfield, IL: Charles C. Thomas.
- Craig, R. J. (2006). *MCMI-II/III computer narrative report*. Odessa, FL: Psychological Assessment Resources.
- Craig, R. J. (2008). *Personality Disorder Adjective Checklist*. Palo Alto, CA: Mind Garden.
- Craig, R. J., Bivens, A., & Olson, R. (1997). MCMI-III-derived typological analysis of cocaine and heroin addicts. *Journal of Personality Assessment*, 69, 583–595.
- Craig, R. J., Kuncel, R., & Olson, R. (1994). Ability of drug abusers to avoid detection of substance abuse on the MCMI-II. *Journal of Social Behavior and Personality*, 9, 95–106.
- Craig, R. J., & Olson, R. (1992). Relationship between MCMI-II scales and normal personality traits. *Psychological Reports*, 71, 699–705.
- Craig, R. J., & Olson, R. (1997). Assessing PTSD with the Millon Clinical Multiaxial Inventory-III. *Journal of Clinical Psychology*, 53, 943–952.
- Craig, R. J., & Olson, R. (1998). Stability of the MCMI-III in a substance-abusing inpatient sample. *Psychological Reports*, 83, 1273–1274.
- Craig, R. J., & Weinberg, D. (1992a). Assessing drug abusers with the Millon Clinical Multiaxial Inventory: A review. *Journal of Substance Abuse Treatment*, 9, 249–255.
- Craig, R. J., & Weinberg, D. (1992b). Assessing alcoholics with the Millon Clinical Multiaxial Inventory: A review. *Psychology of Addictive Behaviors*, 6, 200–208.
- Dana, R., & Cantrell, J. (1988). An update on the Millon Clinical Multiaxial Inventory (MCMI). *Journal of Clinical Psychology*, 44, 760–763.
- Davis, K. M., & Archer, R. P. (2010). A critical review of objective personality inventories with sex offenders. *Journal of Clinical Psychology*, 66, 1254–1280.
- Davis, R., & Patterson, M. P. (2005). Diagnosis of personality disorder subtypes. In R. J. Craig (Ed.), *New directions in interpreting the Millon Clinical Personality Inventory-III* (pp. 32–70). New York: John Wiley.
- Davis, W. E., Greenblatt, R. L., & Pochyly, J. M. (1990). Test of MCMI black norms for five scales. *Journal of Clinical Psychology*, 46, 175–178.

- Daubert v. Merrill Dow Pharmaceuticals, Inc.* 113 S. Ct. 2786 (1993).
- Dillon, S. K. (1988). Narcissism and embellishment of signature. *Psychological Reports*, 62, 152–154.
- Donat, D. C. (1994). Empirical groupings of perceptions of alcohol use among alcohol dependent persons: A cluster analysis of the Alcohol Use Inventory (AUI) scales. *Assessment*, 1, 103–110.
- Donat, D. C., Walters, J., & Hume, A. (1992). MCMI differences between alcoholics and cocaine abusers: Effects of age, sex, and race. *Journal of Personality Assessment*, 58, 96–104.
- Dubro, A. F., & Wetzler, S. (1989). An external validity study of the MMPI personality disorder scales. *Journal of Clinical Psychology*, 45, 570–575.
- Dyce, J. A., O'Connor, B. P., Parkins, S. Y., & Janzen, H. L. (1997). Correlational structure of the MCMI–III personality disorder scales and comparisons with other data sets. *Journal of Abnormal Psychology*, 69, 568–582.
- Dyer, F. J. (2005). Forensic applications of the MCMI–III in light of recent controversies. In R. J. Craig (Ed.), *New directions in interpreting the Millon Clinical Multiaxial Inventory* (pp. 201–226). New York: John Wiley.
- Dyer, F. J. (2008). Using the Millon inventories in forensic psychology. In T. Millon & C. Bloom (Eds.), *The Millon inventories: A practitioner's guide to personalized clinical assessment* (2nd ed., pp. 177–195). New York: Guilford Press.
- Dyer, F. J., & McCann, J. T. (2000). The Millon clinical inventories, research critical of their application, and *Daubert* criteria. *Law and Human Behavior*, 24, 487–497.
- Fals-Stewart, W. (1995). The effect of defensive responding by substance-abusing patients on the Millon Clinical Multiaxial Inventory. *Journal of Personality Assessment*, 64, 540–551.
- Flynn, P. M., McCann, J. T., & Fairbank, J. A. (1995). Issues in the assessment of personality disorder and substance abuse using the Millon Clinical Multiaxial Inventory (MCMI–III). *Journal of Clinical Psychology*, 51, 415–421.
- Gabrys, J. B., Utendale, K. A., Schumph, D., Phillips, N., Peters, K., Robertson, G. . . . Laye, R. C. (1988). Two inventories for the measurement of psychopathology: Dimensions and common factorial space on Millon's clinical and Eysenck's general personality scales. *Psychological Reports*, 62, 591–601.
- Gibertini, M., Brandenberg, N., & Retzlaff, P. (1986). The operating characteristics of the Millon Clinical Multiaxial Inventory. *Journal of Personality Assessment*, 50, 554–567.
- Goldberg, J. O., Shaw, B., & Segal, Z. V. (1987). Concurrent validity of the MCMI Depression scales. *Journal of Clinical and Consulting Psychology*, 55, 785–787.
- Grossman, S. (2008). The MCMI–III and MACI Grossman Facet Scales. In T. Millon & C. Bloom (Eds.), *The Millon inventories: A practitioner's guide to personalized clinical assessment* (pp. 112–134). New York: Guilford Press.
- Grossman, S., & del Rio, C. (2005). The MCMI–III Facet Subscales. In R. J. Craig (Ed.), *New directions in interpreting the Millon Clinical Multiaxial Inventory–III* (pp. 3–31). New York: John Wiley.
- Gudjonsson, G. H., & Main, N. (2008). How are personality disorders related to compliance? *Journal of Forensic Psychiatry and Psychology*, 19, 180–190.
- Guthrie, P. C., & Mobley, B. D. (1994). A comparison of the differential diagnostic efficiency of three personality disorder inventories. *Journal of Clinical Psychology*, 50, 656–665.
- Haller, D. L., Miles, D. R., & Dawson, K. S. (2002). Psychopathology influences treatment retention among drug-dependent women. *Journal of Substance Abuse Treatment*, 23, 431–436.
- Halon, R. L. (2001). The Millon Clinical Multiaxial Inventory–III: The normal quartet in child custody cases. *American Journal of Forensic Psychology*, 19, 57–75.
- Hamberger, L. K., & Hastings, J. E. (1989). Counseling male spouse abusers: Characteristics of treatment completers and dropouts. *Violence and Victims*, 4, 275–286.
- Hart, S. D., Dutton, D. G., & Newlove, T. (1993). The prevalence of personality disorder among wife assaulters. *Journal of Personality Disorders*, 7, 329–341.
- Hart, S. D., Forth, A. E., & Hare, R. D. (1991). The MCMI–II and psychopathy. *Journal of Personality Disorders*, 5, 318–327.
- Henning, K., Jones, A., & Holdford, R. (2003). Treatment needs of women arrested for domestic violence: A comparison with male offenders. *Journal of Interpersonal Violence*, 18, 839–856.
- Hess, A. K. (1998). Review of the Millon Clinical Multiaxial Inventory–III. In J. Mitchell (Ed.), *Mental measurements yearbook* (pp. 984–986). Lincoln: University of Nebraska Press.
- Hicklin, J., & Widiger, T. A. (2000). Convergent validity of alternative MMPI–2 personality disorder scales. *Journal of Personality Assessment*, 75, 502–518.

- Hills, H. A. (1995). Diagnosing personality disorders: An examination of the MMPI-2 and MCMI-II. *Journal of Personality Assessment*, 65, 21-34.
- Hogg, B., Jackson, H. J., Rudd, R. P., & Edwards, J. (1990). Diagnosing personality disorders in recent-onset schizophrenia. *Journal of Nervous and Mental Disease*, 179, 194-199.
- Holliman, N., & Guthrie, P. (1989). Comparison of the MCMI and the CPI in assessment of a nonclinical population. *Journal of Clinical Psychology*, 45, 373-382.
- Holt, S. E., Meloy, J. R., & Strack, S. (1999). Sadism and psychopathy in violent and sexually violent offenders. *Journal of the American Academy of Psychiatry and Law*, 27, 23-32.
- Hyer, L., Boyd, S., Stanger, E., Davis, H., & Walters, P. (1997). Validation of the MCMI-III PTSD scale among combat veterans. *Psychological Reports*, 80, 720-722.
- Hyer, L., Brandsma, J., & Boyd, S. (1997). The MCMI and posttraumatic stress disorder. In T. Millon (Ed.), *The Millon inventories: Clinical and personality assessment* (pp. 191-216). New York: Guilford Press.
- Hyer, L., Melton, M., & Graton, C. (1993). Post-traumatic stress disorders and MCMI-based assessment. In R. J. Craig (Ed.), *The Millon Clinical Multiaxial Inventory: A clinical research information synthesis* (pp. 159-171). Hillsdale, NJ: Lawrence Erlbaum.
- Hyer, L., Woods, M. G., Bruno, R., & Boudewyns, P. (1989). Treatment outcome of Vietnam veterans with PTSD and consistency of the MCMI. *Journal of Clinical Psychology*, 45, 547-552.
- Jeffrey v. Dillard's Department Stores and Dillard's Bel Air Mall*, United States District Court for the Southern District of Alabama, Southern Division, 2000.
- Jones, A. (2005). An examination of three sets of MMPI-2 personality disorder scales. *Journal of Personality Disorders*, 19, 370-385.
- Keilin, W. J., & Bloom, L. J. (1986). Child custody evaluation practices: A survey of experienced professionals. *Professional Psychology: Research and Practice*, 17, 338-346.
- Kelln, B. R., Dozois, D. J., & McKenzie, I. E. (1998). An MCMI-III discriminant functional analysis of incarcerated felons: Predictions of subsequent institutional misconduct. *Criminal Justice and Behavior*, 25, 177-189.
- Kennedy, S. H., Katz, R., Rockert, W., Mendlowitz, S., Ralevski, E., & Clewes, C. J. (1995). Assessment of personality disorders in anorexia nervosa and bulimia nervosa: A comparison of self-report and structured interview methods. *Journal of Nervous and Mental Disease*, 183, 358-364.
- Klein, M. H., Benjamin, L. S., Rosenfeld, R., Treece, C., Husted, J., & Greist, J. H. (1993). The Wisconsin Personality Disorders Inventory: Development, reliability, and validity. *Journal of Personality Disorders*, 7, 285-303.
- Lally, S. J. (2003). What tests are acceptable for use in forensic evaluations? A survey of experts. *Professional Psychology: Research and Practice*, 26, 54-60.
- Lampel, A. K. (1999). Use of the Millon Clinical Multiaxial Inventory-III in evaluating child custody litigants. *American Journal of Forensic Psychology*, 17, 19-31.
- Langevin, R., Paitich, D., Freeman, R., Mann, K., & Handy, L. (1979). Personality and sexual anomalies: An examination of the Millon Clinical Multiaxial Inventory. *Annals of Sex Research*, 1, 13-32.
- Lecic-Tosevski, D., Gavrilovic, G., & Priebe, S. (2003). Personality factors and posttraumatic stress: Associations in civilians one year after air attacks. *Journal of Personality Disorders*, 17, 537-549.
- Lehne, G. K. (2002). The NEO Personality Inventory and the Millon Clinical Multiaxial Inventory in forensic evaluations of sex offenders. In P. T. Costa & T. A. Widiger (Eds.), *Personality disorders and the five-factor model of Personality* (2nd ed., pp. 269-282). Washington, DC: American Psychological Association.
- Lenny, P., & Dear, G. E. (2009). Faking good on the MCMI-III and implications for child custody evaluations. *Journal of Personality Assessment*, 91, 553-559.
- Lenzenweger, M. F. (1999). Stability and change in personality disorder features. *Archives of General Psychiatry*, 56, 1009-1018.
- Libb, J. W., Stankovic, S., Sokol, A., Houck, C., & Switzer, O. (1990). Stability of the MCMI among depressed psychiatric outpatients. *Journal of Personality Assessment*, 55, 209-218.
- Lindsay, K. A., Sankis, L. M., & Widiger, T. A. (2000). Gender bias in self-report personality disorder inventories. *Journal of Personality Disorders*, 14, 218-232.
- Lindsay, K. A., & Widiger, T. A. (1995). Sex and gender bias in self-report personality disorder inventories: Item analysis of the MCMI-II, MMPI, and PDQ-R. *Journal of Personality Assessment*, 65, 1-20.

- Marlowe, D. B., Husband, S. D., Bonieske, L. M., Kirby, K. C., & Platt, J. J. (1997). Structured interview versus self-report test advantages for the assessment of personality pathology in cocaine dependence. *Journal of Personality Disorders, 11*, 177–190.
- Marsh, D. T., Stile, S. A., Stoughton, N. L., & Trout-Landen, B. I. (1988). Psychopathology among opiate addicts: Comparative data from the MMPI and MCMI. *American Journal of Drugs and Alcohol Abuse, 14*, 17–27.
- Matano, R. A., Locke, K. D., & Schwartz, K. (1994). MCMI personality subtypes for male and female alcoholics. *Journal of Personality Assessment, 63*, 250–264.
- McCann, J. T. (1989). MMPI personality disorder scales and the MCMI: Concurrent validity. *Journal of Clinical Psychology, 45*, 365–369.
- McCann, J. T. (1990). A multitrait–multimethod analysis of the MCMI–II clinical syndrome scales. *Journal of Personality Assessment, 55*, 465–476.
- McCann, J. T. (1991). Convergent and discriminant validity of the MCMKI–II and MMPI personality disorder scales. *Psychological Assessment, 3*, 9–18.
- McCann, J. T. (2002). Guidelines for forensic application of the MCMI–III. *Journal of Forensic Psychology Practice, 2*, 55–69.
- McCann, J. T., & Dyer, F. J. (1996). *Forensic assessment with the Millon inventories*. New York: Guilford Press.
- McCann, J. T., Flens, J. R., Campagna, V., Collman, P., Lazarro, T., & Connor, E. (2001). The MCMI–III in child custody evaluations: A normative study. *Journal of Forensic Psychology Practice, 1*, 27–44.
- McCann, J. T., Flynn, P. M., & Gersh, D. M. (1992). The MCMI–II diagnosis of borderline personality disorders: Base rate versus prototypic items. *Journal of Personality Assessment, 58*, 105–114.
- McMahon, R. C., & Davidson, R. S. (1985). An examination of the relationship between personality patterns and symptom/mood patterns. *Journal of Personality Assessment, 49*, 552–556.
- McMahon, R. C., Davidson, R. S., & Flynn, P. M. (1986). Psychological correlates and treatment outcomes for high and low social functioning alcoholics. *International journal of the Addictions, 21*, 819–835.
- McMahon, R., & Diamond, S. E. (2008). MCMI applications in alcohol and drug dependence. In T. Millon & C. Bloom (Eds.), *The Millon inventories: A practitioner's guide to personalized clinical assessment* (pp. 270–295). New York: Guilford Press.
- McMahon, R. C., Flynn, P. M., & Davidson, R. S. (1985). Stability of the personality and symptom scales of the Millon Clinical Multiaxial Inventory. *Journal of Personality Assessment, 49*, 231–234.
- McMahon, R. C., Malow, R. M., Devieux, J., Rosenberg, R., & Jennings, T. (2008). HIV risk and history of STDs in MCMI–III psychopathology subgroups of co-morbid substance abusers. *Journal of Drug and Alcohol Abuse, 34*, 329–337.
- McMahon, R. C., Malow, R. M., & Penedo, F. J. (1998). Substance abuse problems, psychiatric severity, and HIV risk in Millon Clinical Multiaxial Inventory–II personality subgroups. *Psychology of Addictive Behaviors, 12*, 3–13.
- McMahon, R. C., & Richards, S. K. (1996). Profile patterns, consistency, and change in the Millon Clinical Multiaxial Inventory–II in cocaine abusers. *Journal of Clinical Psychology, 52*, 75–79.
- Messina, N., Wish, E., Hoffin, J., & Nemes, S. (2001). Diagnosing antisocial personality disorder among substance abusers: The SCID–II versus the MCM–II. *American Journal of Drug and Alcohol Abuse, 27*, 699–717.
- Miller, H. R., Steiner, D. L., & Parkinson, A. (1992). Maximum likelihood estimates of the ability of the MMPI and MCMI personality disorder scales and the SIDP to identify personality disorders. *Journal of Personality Assessment, 59*, 1–13.
- Millon, T. (1983). *Millon Clinical Multiaxial Inventory manual*. New York: Holt, Rinehart & Winston.
- Millon, T. (1987). *Millon Clinical Multiaxial Inventory–II: Manual for the MCM–II*. Minneapolis, MN: Pearson Assessments.
- Millon, T. (1990). *Toward a new personology*. New York: John Wiley.
- Millon, T. (1994). *Millon Clinical Multiaxial Inventory–III: Manual*. Minneapolis, MN: Pearson Assessments.
- Millon, T. (1997). *Millon Clinical Multiaxial Inventory–III: Manual* (2nd ed.). Minneapolis, MN: Pearson Assessments.
- Millon, T. (2011a). Classifying personality disorders: An evolution-based alternative to an evidence-based approach. *Journal of Personality Disorders, 25*, 279–304.
- Millon, T. (2011b). *Disorders of personality: Introducing a DSM/ICD spectrum from normal to abnormal* (3rd ed.). New York: John Wiley.

- Millon, T., Grossman, S. D., & Tringone, R. (2010). The Millon Personality Spectrometer: A tool for personality spectrum analysis, diagnosis and treatments. In T. Millon, R. Krueger, & E. Simonson (Eds.), *Contemporary directions in psychopathology: Scientific foundations of the DSM-5 and ICD-11*. New York: Guilford Press.
- Millon, T., Millon, C., & Davis, R. (2003). *MCMI-III corrections report*. Minneapolis, MN: Pearson Assessments.
- Morgan, C. D., Schoenberg, M. R., Dorr, D., & Burke, M. J. (2002). Overreport on the MCMI-III: Concurrent validation with the MMPI-2 using a psychiatric inpatient sample. *Journal of Personality Assessment*, 78, 288-300.
- Morey, L. C. (1985). An empirical approach of interpersonal and DSM-III approaches to classification of personality disorders. *Psychiatry*, 48, 358-364.
- Morey, L. C., & Levine, D. J. (1988). A multitrait-multimethod examination of Minnesota Multiphasic Personality Inventory (MMPI) and Millon Clinical Multiaxial Inventory (MCMI). *Journal of Psychopathology and Behavioral Assessment*, 10, 333-344.
- Munley, P. H., Vacha-Haase, T., Busby, R. M., & Paul, B. D. (1998). The MCMI-II and race. *Journal of Personality Assessment*, 70, 183-189.
- Murphy, T. J., Greenblatt, R. I., Modziers, G. J., & Trimalas, K. A. (1991). Stability of the Millon Clinical Multiaxial Inventory among psychiatric inpatients. *Journal of Psychopathology and Behavioral Assessment*, 12, 143-150.
- Oliver, C. J., Beech, A. R., Fisher, D., & Beckett, R. (2007). A comparison of rapists and sexual murderers on demographic and selected psychometric measures. *International Journal of Offender Therapy and Comparative Criminology*, 51, 298-312.
- Otto, R. K., & Butcher, J. N. (1995). Computer-assisted psychological assessment in child custody evaluations. *Family Law Quarterly*, 29, 79-96.
- Overholser, J. C. (1990). Retest reliability of the Millon Clinical Multiaxial Inventory. *Journal of Personality Assessment*, 55, 202-208.
- Patrick, J. (1993). Validation of the MCMI-I borderline personality disorder scale with a well-defined criterion sample. *Journal of Clinical Psychology*, 49, 29-32.
- Picci, R. L., Vigna-Taglianti, F., Oliva, F., Mathis, F., Salmaso, S., Ostacoli, L., . . . Furlan, P. M. (2012). Personality disorders among patients accessing alcohol detoxification treatment: Prevalence and gender differences. *Comprehensive Psychiatry*, 53, 355-363.
- Pico-Alfonso, M., Echeburúa, E., & Martinez, M. (2008). Personality disorder symptoms in women as a result of chronic intimate male partner violence. *Journal of Family Violence*, 23, 577-588.
- Piersma, H. L. (1986). The Millon Clinical Multiaxial Inventory (MCM) as a treatment outcome measure for psychiatric inpatients. *Journal of Clinical Psychology*, 42, 493-499.
- Piersma, H. L. (1989). The stability of the MCMI-II for psychiatric inpatients. *Journal of Clinical Psychology*, 45, 781-785.
- Piersma, H. L., & Boes, J. L. (1997). The relationship between length of stay to MCMI-II and MCMI-III change scores. *Journal of Clinical Psychology*, 53, 535-542.
- Quinnell, F. A., & Bow, J. N. (2001). Psychological tests used in child custody evaluations. *Behavioral Sciences and the Law*, 19, 491-501.
- Retzlaff, P., Stoner, J., & Kleinsasser, D. (2002). The use of the MCMI-III in the screening and triage of offenders. *International Journal of Offender Therapy and Comparative Criminology*, 46, 319-332.
- Rogers, R. (2003). Forensic use and abuse of psychological tests: Multiscale inventories. *Journal of Psychiatric Practice*, 9, 316-320.
- Rogers, R., Salekin, R. T., & Sewell, K. W. (1999). Validation of the Millon Clinical Multiaxial Inventory for Axis II disorders: Does it meet the *Daubert* standard? *Law and Human Behavior*, 24, 501-506.
- Rossi, G., Hauben, C., Van den Brande, I., & Sloore, H. (2003). Empirical evaluation of the MCMI-III personality disorder scales. *Psychological Reports*, 92, 627-642.
- Rossi, G., Van den Brande, I., Sloore, H., & Hauben, C. (2003). Convergent validity of the MCMI-III personality disorder scales and the MMPI-2 scales. *Journal of Personality Disorders*, 17, 330-340.
- Salekin, R. T., Rogers, R., & Sewell, K. W. (1996). Review and meta-analysis of the Psychopathy Checklist and Psychopathy Checklist-Revised. *Clinical Psychology: Science and Practice*, 3, 203-215.
- Schwartz, J. P., Seemann, E., Buboltz, W. C., & Flye, A. (2004). Personality styles: Predictors of masculine gender role conflict in male prisoners. *Psychology of Men and Masculinity*, 5, 59-64.

- Schuler, C. E., Snibbe, J. R., & Buckwalter, J. G. (1994). Validity of the MMPI Personality Disorder scales (MMPI–Pd). *Journal of Clinical Psychology, 50*, 220–227.
- Schutte, J. W. (2001). Using the MCMI–III in forensic evaluations. *American Journal of Forensic Psychology, 19*, 5–20.
- Silberman, C. S., Roth, L., Segal, D. L., & Burns, W. J. (1997). Relationship between the Millon Clinical Multiaxial Inventory–II and Coolidge Axis II Inventory in chronically mentally ill older adults: A pilot study. *Journal of Clinical Psychology, 53*, 559–566.
- Sinha, B. K., & Watson, D. C. (2001). Personality disorder in university students: A multitrait–multimethod matrix study. *Journal of Personality Disorders, 15*, 235–244.
- Smith, D., Carroll, J. L., & Fuller, G. (1988). The relationship between the Millon Clinical Multiaxial Inventory and the MMPI in a private outpatient mental health clinic population. *Journal of Clinical Psychology, 44*, 165–174.
- Soldz, S., Budman, S., Demby, A., & Merry, J. (1993). Diagnostic agreement between the Personality Disorder Examination and the MCMI–II. *Journal of Personality Assessment, 60*, 486–499.
- Stoner, J. W. (2008). Using the MCMI in Correctional settings. In T. Millon and C. Bloom (Eds.), *The Millon inventories: A practitioner's guide to personalized clinical assessment* (pp. 196–243). New York: Guilford Press.
- Strack, S. (1987). Development and validation of an adjective check list to assess the Millon personality types in a normal population. *Journal of Personality Assessment, 51*, 572–587.
- Stredny, R. V., Archer, R. P., & Mason, J. A. (2006). MMPI–2 and MCMI–III characteristics of parental competency evaluations. *Journal of Personality Assessment, 87*, 113–115.
- Torgersen, S., & Alnaes, R. (1990). The relationship between the MCMI personality scales and DSM–III, Axis II. *Journal of Personality Assessment, 55*, 698–707.
- Turley, B., Bates, G. W., Edwards, J., & Jackson, H. J. (1992). MCMI–II personality disorders in recent-onset bipolar disorders. *Journal of Clinical Psychology, 48*, 320–329.
- Wheeler, D. S., & Schwartz, J. C. (1989). Millon Clinical Multiaxial Inventory (MCMI) scores with a collegiate sample: Long-term stability and self–other agreement. *Journal of Psychopathology and Behavioral Assessment, 11*, 339–352.
- White, R. J., Ackerman, R. J., & Caraveo, L. E. (2001). Self-identified alcohol abusers in a low-security federal prison: Characteristics and treatment implications. *International Journal of Offender Therapy and Comparative Criminology, 45*, 214–227.
- Widiger, T. (2001). Review of the Millon Clinical Multiaxial Inventory. In J. V. Mitchell Jr. (Ed.), *The ninth mental measurement yearbook* (Vol. 1, pp. 986–988). Lincoln: University of Nebraska Press.
- Widiger, T., & Sanderson, C. (1987). The convergent and discriminant validity of the MCMI as a measure of the DSM–III personality disorders. *Journal of Personality Assessment, 51*, 228–242.
- Wierzbicki, M., & Gorman, J. L. (1995). Correspondence between students' scores on the Millon Clinical Multiaxial Inventory–II and Personality Diagnostic Questionnaire–Revised. *Psychological Reports, 77*, 1079–1082.
- Wise, E. A. (1994a). Managed care and the psychometric validity of the MMPI and MCMI personality disorder scales. *Psychotherapy in Private Practice, 13*, 81–97.
- Wise, E. A. (1994b). Personality style codetype concordance between the MCMI and MBHI. *Journal of Clinical Psychology, 50*, 367–380.
- Wise, E. A. (1996). Comparative validity of MMPI–2 and MMPI–2 personality disorder scales with forensic examinees. *Journal of Personality Disorders, 15*, 275–279.
- Wise, E. A. (2001). The comparative validity of MCMI–II and MCMI–II personality disorder scales with forensic examinees. *Journal of Personality Disorders, 15*, 275–279.
- Zarella, K. L., Schuerger, J. M., & Ritz, G. H. (1990). Estimation of MCMI DSM–III Axis II constructs from MMPI scales and subscales. *Journal of Personality Assessment, 55*, 195–201.

7

THE RORSCHACH INKBLOT METHOD

Irving B. Weiner

The Rorschach Inkblot Method (RIM) is a relatively unstructured, performance-based personality assessment instrument. When properly administered and interpreted, the RIM provides dependable information about how people attend to and perceive their surroundings, how they form concepts and ideas, how they experience and express affect, how they manage stress, and how they view themselves and other people. Rorschach findings can thereby facilitate decision making whenever conclusions and recommendations concerning people are based at least in part on these personality characteristics. The contexts in which personality-based decisions are made include differential diagnosis, treatment planning, and outcome evaluation in clinical work and criminal, personal injury, and child custody determinations in forensic settings. The present chapter elaborates these clinical and forensic applications of Rorschach assessment, preceded by an introductory review of Rorschach procedures, the development of the instrument, and research findings bearing on its psychometric properties.

Rorschach Procedures

The Rorschach stimuli comprise 10 standard inkblots, 7 in shades of gray and black, 2 in shades of red as well as gray and black, and 3 in shades of various chromatic and achromatic colors. Respondents are shown these inkblots one at a time and asked, “What might this be?” After presenting the 10 cards in this way, examiners conduct an inquiry in which they show respondents the cards a second time and ask them to indicate for each of their percepts where in the inkblot they saw it and what made it look as it did (see Exner, 2003, chap. 4, for additional guidelines on standardized procedures used in Rorschach administration).

As elaborated by Weiner (2003a, chaps. 5–7), a Rorschach examination yields three kinds of data, usually referred to as *structural*, *thematic*, and *behavioral*. The structural data in a Rorschach protocol consist of codes for several aspects of how respondents perceive the inkblots. These codes categorize various objective features of a response, including its location (e.g., whether it is seen in the whole blot or in some blot detail), its determinants (e.g., whether it looks as it does because of its shape or because of its color), its content (e.g., whether it is seen as a human figure or an animal), and whether it is a Popular (defined empirically as a response given by one-third or more of non-psychotic persons).

In the Rorschach Comprehensive System (CS) developed by Exner (2003), response codes are tabulated and combined to yield over 100 summary scores and indices. These summary scores and

indices are interpreted on the basis of two assumptions: first, that the manner in which people perceive the inkblots is representative of how they generally perceive objects and events in their lives; and second, that how people perceive objects and events in their lives—that is, how they look at their world—reflects the kind of people they are, particularly with respect to their tendencies to think, feel, and act in certain ways.

Rorschach thematic data consist of the imagery with which most respondents embellish their reports of what the inkblots might be. Whereas the structural data derive from a perceptual process attuned to the stimulus properties of the inkblots (form, color, and shading), the thematic data emerge from an associative process in which characteristics not intrinsic to the blot stimuli are nevertheless attributed to them. For example, human figures may be seen as moving even though the blots are in fact static, and they may be described as being happy or sad even though the blots contain no objective indications to this effect. The interpretation of such attributions is based on the assumption that, because they are not based directly on what is in the blots, they must come from inside the person. Thematic imagery produced by associations to the blot stimuli is consequently likely to provide clues to a respondent's underlying needs, attitudes, conflicts, and concerns.

Behavioral data in Rorschach examinations consist of the manner in which respondents deal with the test situation, handle the test materials, relate to the examiner, and use language in expressing themselves. Interpretation of these behavioral data is based on the expectation that the way in which respondents approach the Rorschach task and structure their relationship with the examiner constitutes a representative sample of their problem solving and interpersonal style. As a relatively unstructured verbal interaction between the two parties, the Rorschach situation creates ample opportunity for such characteristic styles to become apparent.

Integrated interpretation of the structural, thematic, and behavioral data generated by a Rorschach examination leads to detailed descriptions of respondents' assets and limitations with respect to how they make decisions, solve problems, cope with the demands in their lives, and prefer to conduct themselves. These Rorschach-based descriptions of adaptive capacities, enriched by thematic clues to a respondent's underlying concerns, facilitate numerous applications of the instrument. As elaborated in the course of this chapter, these applications include clinical decision making, particularly with regard to differential diagnosis and treatment planning, and forensic determinations related to psycholegal issues in criminal, personal injury, and child custody cases.

Development of the Rorschach Inkblot Method

The Rorschach Inkblot Method was created over 90 years ago by Hermann Rorschach, a Swiss psychiatrist. In his youth, Rorschach had been adept at a popular parlor game of his day in which players competed to see who could generate the most interesting descriptions of inkblots made by dropping some ink in the middle of a piece of paper and folding it in half. Working in a large mental hospital after completion of his medical studies and psychiatric training, Rorschach became intrigued with the possibility that patients with different types of mental disorder might perceive inkblots differently from each other and from normally functioning people. He pursued this notion by developing a standard set of inkblots and showing them to a sample of 288 patients and 117 non-patient volunteers, asking the simple question, "What might this be?"

Rorschach formulated some basic codes for the previously mentioned location, determinant, and content features of the inkblot responses he obtained, and he then compared his subgroups with respect to their percentage frequencies for these codes (e.g., the proportion of responses given to the entire blot as opposed to blot details). From his findings he inferred numerous relationships between certain code frequencies and percentages, on the one hand, and particular mental disorders

and personality characteristics, on the other hand. Despite Rorschach's creativity in conceiving the notion of using inkblot percepts to identify personality characteristics, however, he was basically an empiricist rather than a theorist. His methods of coding and interpretation were guided by the data he collected and not by any particular conceptual framework. In his final paper, which became available after his death, he noted, "There has, however, been little progress in the development of the theory of the experiment" (Rorschach, 1921/1942, p. 186).

Rorschach published the findings from his research in *Psychodiagnostics*, the 1921 monograph just cited. The materials and methods he described in this monograph have remained basic to commonly practiced Rorschach procedures since that time. The 10 inkblots published with the original monograph are the standard Rorschach plates in use today, and Rorschach's delineation of test-inferred personality characteristics is a sophisticated treatise that can still be read with pleasure and profit. The 1921 monograph was only a preliminary work, however, and Rorschach died just one year after it was published, at the age of 37, leaving development of the instrument largely unfinished.

Over the 50 years following Rorschach's death, many different systems of administering, coding, and interpreting the RIM devolved from his monograph. Some of these systems, best exemplified in the United States by the work of Samuel Beck (Beck, Beck, Levitt, & Molish, 1961), continued, as Rorschach had done, to emphasize a structural approach to the method and an empirical basis for proposing coding refinements and interpretive strategies. Other systems, exemplified by the contributions of Bruno Klopfer (Klopfer, Ainsworth, Klopfer, & Holt, 1954) and Roy Schafer (1954), advocated embellishing Rorschach's approach with expanded attention to thematic imagery and recourse to theoretical formulations, primarily psychoanalytic in nature, for guidance in developing new coding and interpretive guidelines. Rorschach's life and the early development of the RIM are described in further detail by Ellenberger (1954), Exner (2003, chap. 1), Schwarz (1996), and Weiner (2003a, chap. 1).

By the 1970s, there were five major Rorschach systems being used in the United States, and several other systems enjoyed popularity elsewhere. In addition, most Rorschach clinicians were employing some individually preferred blend of these systems (e.g., a Beck coding for locations and a Klopfer coding for determinants). Although personalized embellishments of Rorschach procedures were serving clinical purposes in the individual case, this diversity of method made it difficult for Rorschach scholars to communicate effectively with each other and cumulate systematic data concerning the psychometric properties of the instrument. Exner addressed this problem by combining the clearest and most dependable aspects of the five U.S. systems into a standardized set of procedures for administration and coding that became the previously mentioned Comprehensive System (CS). First published in 1974, the CS has subsequently been modified over the years to take account of both new empirical data and contemporary conceptual formulations (see Exner, 1974, 2003).

Although focused primarily on the structural features of Rorschach responses, the CS approach includes interpretive strategies that call attention to thematic and behavioral data as well. Psychoanalytic perspectives on the interpretation of Rorschach imagery and behavior have also continued to flourish over the years, as exemplified by the work of Lerner (1991, 1998, 2007). Rorschach scholars and practitioners generally concur that the RIM realizes its full potential as a personality assessment instrument only when the interpretive process is guided by a judicious blend of structural and psychodynamic perspectives (Acklin, 1995; Acklin, Li, & Tyson, 2006; Smith, 1994; Weiner, 2003a; Weiner & Greene, 2008, chap. 11). The CS procedures for administering and coding the RIM became by far the most frequently used approach to Rorschach assessment in the United States, and the preferred approach in many other countries as well. Widespread adoption of these standardized procedures has fostered substantial advances in knowledge about

the RIM, including collection of large normative reference samples and systematic investigation of the psychometric properties of the instrument.

Psychometric Properties

The psychometric adequacy of Rorschach assessment has been a topic of debate throughout most of the instrument's history. The advent of Exner's Comprehensive System led to a lull in this storm during the 1980s, mainly by putting to rest concerns that the instrument lacked standardized procedures for administration and coding. Beginning in the mid-1990s, however, a new wave of psychometric challenges directed specifically at the CS washed over Rorschach assessment. In a series of influential articles, a group of critics contended (1) that the CS normative data were outdated, overpathologizing (i.e., likely to identify psychological disturbance where none is present), and insufficient to warrant using the RIM with culturally diverse populations; (2) that adequate intercoder and retest reliability had not yet been broadly established for CS variables; and (3) that for a broad range of purposes, Rorschach assessment lacked criterion and incremental validity (see Garb, Wood, Nezworski, Grove, & Stejskal, 2001; Lilienfeld, Wood, & Garb, 2000; Wood & Lilienfeld, 1999; Wood, Nezworski, Garb, & Lilienfeld, 2001).

These criticisms of Rorschach assessment included calling for a moratorium on Rorschach testing (Garb, 1999), referring to Rorschach assessment as "junk science" or "pseudoscience" (Lohr, Fowler, & Lilienfeld, 2002) and asserting that Rorschach testimony is insufficiently relevant and reliable to be admitted into evidence in a court of law (Grove & Barden, 1999). Some of the technical concerns raised by these critics of Rorschach assessment were well taken and had the salutary effect of stimulating expanded lines of research, including the collection of contemporary CS normative samples both in the United States and elsewhere. Abundant data now document that, contrary to the negative views expressed by some critics, Rorschach assessment with the CS is a psychometrically sound procedure with a broad and cross-cultural normative reference base, good to excellent intercoder agreement, substantial retest reliability, adequate validity when used properly for its intended purposes, and sufficient respectability to provide an acceptable basis for courtroom testimony.

Normative Reference Base

The development of the Rorschach CS included the compilation of descriptive statistics for each of its codes and summary scores for a sample of 600 nonpatient adults aged 19 to 69 ($M = 31.7$). As described by Exner (2003, chap. 12), this sample was randomly selected from a larger group of volunteer participants who were informed of the importance of what they were doing—namely, helping to standardize a widely used personality test—and who were examined by well-trained psychologists and research technicians. The sample was stratified to include an equal number of males and females and 120 persons from each of five geographic areas in the continental United States (Northeast, South, Midwest, Southwest, and West). This non-patient sample was also generally representative of the patterns of marital status, socioeconomic status, and urban–suburban–rural residence in the United States, and 18% of the respondents were African American, Hispanic, or Asian American.

Two-thirds of the 600 persons in the CS non-patient reference sample were recruited through their place of employment and were given time off from work to participate in the data collection. Another one-fourth were recruited through social or interest organizations (e.g., PTA groups, bowling leagues), and the remaining 8% through social service agencies. These 600 persons averaged 13.4 years of education, with 68% having attended college, and none of them had any

significant history of mental health problems. Consequently, this non-patient sample is representative of people who are relatively well educated and appear to be functioning reasonably well socially and vocationally. Reference data were also obtained from 1,390 non-patient young people aged from 5 to 16, and from three groups of adult psychiatric patients: 328 first-admission inpatients with schizophrenia, 279 patients hospitalized for depression, and 535 outpatients presenting a diversity of symptoms (Exner, 2001, chap. 11). Taken together, the CS reference data identify the frequency with which each of the Rorschach structural variables is likely to occur in non-patient adults and young people, and in persons with various kinds of psychological disorder.

Because these original CS reference data were collected between 1973 and 1986, it is reasonable to question whether they remain applicable a generation later. To answer this question, Exner in 1999 undertook a new normative data collection project to update the CS reference information. Following closely the procedures of the earlier work, participants were solicited to provide a demographically representative U.S. sample of 450 non-patient adults aged 19–86 ($M = 34.9$), who were administered the RIM by experienced examiners. With just a few minor exceptions, these new CS reference data closely resembled the older reference data and did not call for any major alterations in interpretive criteria (see Exner, 2007).

Also of note are the findings in an international collaborative study reported by Meyer, Erdberg, and Shaffer (2007) in which CS reference data were obtained from 21 samples of adults examined by local psychologists in 17 different countries (totaling 4,704 respondents, with a mean age of 36.5). These adult samples from many parts of the world were generally quite similar to each other, with average T -scores for the structural summary variables generally falling within a narrow range less than one-half a standard deviation from international composite mean for these variables. Meyer et al. (2007) concluded from the degree of cross-sample and cross-national similarity evident in these international data that “[u]sing these values in clinical practice should help ensure that inferences about functioning generalize across examiners, levels of skill in administration and scoring, testing context, language, and culture” (p. S213).

With respect to issues of cross-cultural diversity, moreover, studies with U.S. minority groups have found no substantial CS differences among them that would contraindicate cross-ethnic use of the RIM or require modifications in normative expectations. Presley, Smith, Hilsenroth, and Exner (2001) found a clinically significant difference on only 1 of 23 core Rorschach variables between 44 African Americans and 44 demographically matched European Americans in the CS non-patient reference sample. In a multicultural patient sample of 432 consecutive persons evaluated in a hospital-based psychological testing program, Meyer (2002) found no association between ethnicity and 188 Rorschach summary scores among demographically matched European American, African American, Hispanic American, Asian American, and Native American respondents. Meyer concluded from these findings that “the available data clearly support the cross-ethnic use of the Comprehensive System” (p. 127).

As for whether Rorschach assessment is likely to overpathologize, the old ($n = 600$) and new ($n = 450$) non-patient reference groups showed the following frequencies of elevations on the five main CS indices of psychopathology: for the Perceptual Thinking Index (PTI) > 3 , 0.16% and 0.22%, respectively (one person in each group); for the Depression Index ($DEPI$) > 5 , 5% and 4%; for the Coping Deficit Index (CDI) > 3 , 4% and 9%; and for the Hypervigilance Index (HVI) positive, 3% and 4%. These infrequent elevations on the Rorschach psychopathology scales in large, broadly representative, and carefully evaluated samples of non-patients indicate that there is little likelihood of the RIM suggesting psychological disorder when none is present.

Intercoder Agreement

Intercoder agreement for CS Rorschach variables, whether measured by percentage of agreement or by kappa and intraclass correlation coefficients (ICCs) that take account of chance agreement, has proved good to excellent in numerous studies. In one substantial instance of such findings, Meyer and colleagues (2002) examined four different samples and 219 protocols containing 4,761 responses and found a median ICC of .93 for intercoder agreement across 138 regularly occurring Rorschach variables, with 134 of these variables falling in the excellent range for chance-corrected agreement. In a similar study, Viglione and Taylor (2003) examined coder concurrence for 84 protocols with 1,732 responses and found a median ICC of .92 for 68 variables considered to be of central interpretive significance in the CS. Earlier meta-analytic reviews and studies with patient and non-patient samples identified mean kappa coefficients ranging from .79 to .88 across various CS coding categories, which for kappa coefficients is generally regarded as being in the good to excellent range (Acklin, McDowell, Verschell, & Chan, 2000; Meyer, 1997a; 1997b; Viglione & Taylor, 2003).

Research findings have also indicated that Rorschach practitioners in the field can achieve reasonably good agreement in their coding. In one of the samples in the Meyer et al. (2002) study and in a study reported by McGrath et al. (2005), patient protocols that had initially been coded in clinical practice, without any anticipation of their being used in a research project, were later coded independently for research purposes. In both cases, the obtained correlation coefficients for intercoder agreement were more than adequate to demonstrate the potential field reliability of Rorschach coding. Research by McGrath (2003) has indicated further that practicing clinicians can readily reduce their coding errors by attending to the coding criteria and guidelines provided in textbooks and workbooks (e.g., Exner, 2001; 2003; Viglione, 2002).

With further respect to the previously mentioned international applicability of Rorschach assessment, Sahly, Shaffer, Erdberg, and O'Toole (2011) examined the agreement between pairs of judges in nine different countries on 147 structural summary variables scored on 489 protocols. They found combined coefficients of agreement among these judges in the excellent range for 116 (79%) of these variables and in the good range for 25 (17%), with only four variables in the fair range and just two in the poor range.

Retest Reliability

Retest studies with both children and adults over intervals ranging from seven days to three years have demonstrated substantial reliability for Rorschach summary scores and indices that are conceptualized as relating to trait characteristics, which include almost all of the CS variables (see Exner, 2003, chap. 11; Grønnerød, 2003, 2006; Viglione & Hilsenroth, 2001; Viglione & Meyer, 2008). In adults, the short- and long-term stability of most CS variables exceed .75, and 19 core variables with major interpretive significance have shown one-year or three-year retest correlations of .85 or higher. The only Rorschach summary scores that show low retest correlations, even over brief intervals, are a few variables and combinations of these variables that are conceptualized as measuring situationally influenced state characteristics, most notably inanimate movement (*m*) and diffuse shading (*Y*).

Children show stability coefficients similar to those of adults when retested over brief intervals. When retested over two-year intervals between the ages of 8 and 16, young people fluctuate considerably in their Rorschach scores early on but then show steadily increasing long-term consistency as they grow older (Exner, Thomas, & Mason, 1985). The increasing long-term stability of Rorschach variables from age 8 to age 16 is consistent with the expected gradual consolidation

of personality characteristics that occurs during the developmental years, and hence speaks to the validity of the RIM in this regard.

Validity

In a carefully designed large-scale examination of Rorschach validity, Hiller, Rosenthal, Bornstein, Berry, and Brunell-Neuleib (1999) conducted a meta-analysis of a random sample of Rorschach and Minnesota Multiphasic Personality Inventory (MMPI) research studies published from 1977 to 1997 in which there was at least one external (i.e., non-test) variable and some reasonable basis for expecting associations between variables. In their analysis of these studies, which included 2,276 RIM and 5,007 MMPI protocols, they found the validity of the two measures, as indicated by their average effect sizes, to be virtually identical. The unweighted mean validity coefficients were .29 for RIM variables and .30 for MMPI variables, and there is no significant difference between these two validity estimates. Hiller et al. concluded that these obtained effect sizes are sufficiently large to warrant confidence in using both the RIM and the MMPI for their intended purposes and furthermore that the validity of both instruments “is about as good as can be expected for personality tests” (p. 291).

Hiller et al. also found some noteworthy differences between RIM and MMPI variables in the strength of their relationship with different types of dependent variables. On the average, Rorschach variables proved somewhat superior (mean validity coefficient = .37) to MMPI variables (mean validity coefficient = .20) in predicting behavioral outcomes, such as whether patients remain in or drop out of therapy. MMPI variables, on the other hand, showed higher effect sizes than Rorschach variables in correlating with psychiatric diagnosis and self-reports (.37 vs. .18). These differences probably reflect the particular sensitivity of the RIM to persistent behavioral dispositions, consistent with the primarily trait implications of most of its variables, and the self-report nature of the MMPI, which resembles both the methodology of other self-report measures and the typical method of gaining information on which to base psychiatric diagnoses (i.e., asking patients about their symptoms).

Extending well beyond the boundaries of this meta-analysis, an extensive research base has documented the validity of Rorschach assessment when the instrument is used for its intended purposes. As described in reviews by Bornstein and Masling (2005), Handler and Clemence (2005), Meyer (2004), Meyer and Archer (2001), Viglione and Hilsenroth (2001), Viglione and Meyer (2008), Weiner (2004a), and Weiner and Meyer (2009), this validating research has additionally demonstrated the incremental validity of Rorschach findings in contributing diagnostic and predictive information above and beyond what can be learned from other sources of information, including self-report inventories and clinical interviews (see Blais, Hilsenroth, Castlebury, Fowler, & Baity, 2001; Blasczyk-Schiep, Kazén, Kuhl, & Grygielski, 2011; Dao, Prevatt, & Horne, 2008; Grønnerød, 2004; Janson & Stattin, 2003; Meyer & Handler, 2000).

Admissibility into Evidence

By establishing the psychometric soundness of Rorschach assessment, the reference, reliability, and validity data just discussed refute contentions that Rorschach-based testimony is unlikely to be admitted into evidence in the courtroom. To the contrary, detailed rebuttals of these contentions by Hilsenroth and Stricker (2004), McCann and Evans (2008), and Ritzler, Erard, and Pettigrew (2002) include survey data indicating that the reliability and relevance of Rorschach findings are in fact seldom challenged in legal proceedings and hardly ever challenged successfully. On top of survey data, the likely admissibility into evidence of Rorschach-based testimony has been con-

firmed empirically by the regularity with which Rorschach assessment is in fact accepted in the courtroom. Meloy, Hansen, and Weiner (1997) reviewed the entire set of 247 cases in which Rorschach evidence was presented to a federal, state, or military court of appeals during the half-century from 1945 to 1995. They found that the admissibility and import of the Rorschach findings were questioned in only 10.5% of the hearings, and then mainly with reference to the interpretation of the test data, not use of the Rorschach method. In only two of these 247 appellate cases was there any challenge to the relevance and utility of Rorschach assessment.

In a follow-up study employing the same methodology, Meloy (2008) examined the entire set of 150 published cases in which Rorschach findings were cited in federal, state, or military appellate court proceedings between 1996 and 2005. He found a substantial decrease in the percentage of these court records that included criticisms of Rorschach testimony, from the 10.5% frequency during 1945–1995 to just 2% during 1996–2005. Grove and Barden (1999), in their previously mentioned assertion that Rorschach testimony would not be acceptable in the courtroom, alleged in particular that Rorschach assessment would not meet the standard for scientific respectability set by the *Daubert* test, which is one of the two most widely used guidelines for determining admissibility into evidence. As reported by Meloy (2008), however, the facts of the matter are quite different: “There has been no *Daubert* challenge to the scientific status of the Rorschach in any state, federal, or military court of appeal since the U.S. Supreme Court decision in 1993 set the federal standard for admissibility of scientific evidence” (p. 85).

These findings do not mean that forensic psychologists using the RIM, or any other assessment method for that matter, will be spared rigorous cross-examination on the witness stand. What they do mean is that adequately informed examiners will not be hard-pressed to defend either the psychometric soundness of Rorschach assessment or its history of admissibility into evidence in the courtroom. Examiners can also readily document that Rorschach assessment satisfies not only the *Daubert* test of scientific respectability, as just noted, but the other widely used standard for admissibility of evidence as well, the *Frye* test, which requires expert testimony to be based on procedures commonly used by practitioners in the relevant field (see Faigman & Monahan, 2009; Hess, 2006). In the field of personality assessment, the RIM has for many years been the second most frequently used assessment measure among clinical psychologists, following the MMPI (Hogan, 2005). Among recently surveyed APA members who were licensed psychologists and indicated having some interest or involvement in personality measurement, 72% reported using the Rorschach in their practice (Muscewicz, Marczyk, Knauss, & York, 2009). These usage frequency data demonstrate sufficiently common practice of Rorschach assessment in the relevant community to satisfy the *Frye* test for admissibility into evidence.

Clinical Uses and Limitations

Rorschach assessment contributes to clinical practice by helping to identify the presence and nature of psychological disorder and by elucidating a troubled person’s need for, amenability to, and extent of progress in treatments of various kinds. With respect to differential diagnosis, many of the personality characteristics identified by Rorschach variables are associated with particular patterns of psychopathology. For example, schizophrenia is usually defined to involve disordered thinking and impaired reality testing, which means that Rorschach indicators of these cognitive impairments (an elevated *WSum6* and a high *X*-%, respectively) increase the likelihood of a schizophrenia-spectrum disorder. Depressive disorder is suggested by Rorschach indices of dysphoria (elevated *C'*, presence of Color-Shading Blends) and negative self-attitudes (numerous Morbid responses, presence of Vista, low Egocentricity Ratio); obsessive-compulsive personality features are suggested by indices of pedantry and perfectionism (positive Obsessive Index); and so on (see Weiner, 2003a).

As a noteworthy limitation of the RIM in identifying maladaptive personality characteristics, however, Rorschach variables differ with respect to whether their interpretive significance is *bidirectional* or *unidirectional*. Bidirectional variables are interpretable when their values fall substantially above or below normative expectation. Unidirectional variables are noteworthy only when they are either unusually high or unusually low, but not both instances. Form level is a good example of a bidirectional variable, in that a low *X-%* usually indicates good reality testing and a high *X-%* impaired reality testing. By contrast, the *OBS* is a unidirectional variable, because it is likely to identify obsessive-compulsive characteristics when it is positive, but when negative says nothing at all about whether a respondent has these characteristics.

A unidirectional variable with particularly important implications in clinical decision making is the Suicide Constellation (*S-Con*). An elevated *S-Con* indicates risk for suicidal behavior (see Fowler, Piers, Hilsenroth, Holdwick, & Padawer, 2001; Ganellen, 2005), but a low *S-Con* does not contraindicate suicidal potential and should never be taken as sufficient basis for ruling out suicide risk. The Depression Index (*DEPI*) is similarly unidirectional, by virtue of capturing probable features of mood disorder when it is high but not negating the possibility of mood disturbance when it is low. Whether in identifying adjustment difficulties or personality strengths, then, bidirectional variables provide information both by their presence and by their absence. Unidirectional variables, on the other hand, help mainly to indicate when conditions and characteristics are present but provide little basis for inferring the absence of these conditions and characteristics. Proper application of Rorschach variables in research and practice requires taking adequate account of whether their interpretive significance is bidirectional or unidirectional, in the expectation that unilateral variables may frequently yield false negative findings (see Weiner, 2000, 2003a).

Moreover, as an instrument more attuned to personality traits than personality states, the RIM is of limited use in identifying the particular symptoms a disturbed person may be having. A respondent with Rorschach indices of dysphoric mood may be having crying spells, disturbed sleep, weight loss, or some other manifestations of depression; a person with Rorschach indications of an obsessive-compulsive personality style may be a compulsive handwasher, an obsessive prognosticator, or neither; someone with Rorschach signs of subjectively felt distress could be experiencing a phobic reaction, a generalized anxiety disorder, a post-traumatic stress disorder, or none of these conditions. The particular symptoms of psychological disorder that are in evidence—which is a central consideration in arriving at a *DSM* diagnosis—are better determined by observing people's behavior and asking them directly about their symptoms than by attempting to infer their symptoms from their Rorschach responses.

Similarly, with respect to determining whether respondents have conducted themselves in certain ways (e.g., abused alcohol or drugs) or had certain life experiences (e.g., been sexually abused), Rorschach data rarely provide dependable indications one way or the other. The RIM can generate dependable postdictions (i.e., what may have happened at some previous point in time) only when there is a substantial known correlation between specific personality characteristics and the likelihood of certain behaviors or experiences having occurred (see Weiner, 2003b). The predictive validity of Rorschach findings likewise depends on how strongly personality factors influence the occurrence of whatever future actions or events are being predicted.

As an instance in this latter regard, personality characteristics are known to influence whether and how people usually respond to various kinds of treatment, and Rorschach findings are accordingly pertinent to many of the decisions that therapists make during an intervention process. For example, the degree of disturbance or coping incapacity reflected in Rorschach responses can assist in determining whether a troubled person requires inpatient care or is functioning sufficiently well to be treated as an outpatient. The personality style and severity of distress or disorganization

revealed by the RIM can help to indicate whether a patient's treatment needs will be served best by a supportive approach oriented to relieving distress, a cognitive-behavioral approach designed to modify symptoms or solve problems, or an exploratory approach intended to enhance self-understanding. Whatever treatment approach is implemented, moreover, the maladaptive personality traits and the underlying concerns identified by Rorschach data can help therapists determine, in consultation with their patients, what the goals of the treatment should be and with what priority these treatment targets should be addressed (see Bram, 2010; Nygren, 2004; Sendin, 2010; Weiner, 2004b).

Predictive utility in treatment planning derives from the fact that certain personality characteristics measured by Rorschach variables are typically associated with being able to participate in and benefit from psychotherapy. These personality characteristics include being open to experience (*Lambda* not elevated), cognitively flexible (balanced active : passive ratio [*a : p*]), emotionally responsive (adequate *WSumC* and Affective Ratio [*Afi*]), interpersonally receptive (*T* > 0, ample human content [*SumH*]), and personally introspective (*FD* > 0), each of which facilitates engagement and progress in psychological treatment. By contrast, having an avoidant or guarded approach to experience, being set in one's ways, having difficulty recognizing and expressing one's feelings, being interpersonally aversive or withdrawn, and lacking psychological mindedness are often obstacles to progress in psychotherapy (see Clarkin & Levy, 2004; Groth-Marnat, Gotheil, Liu, Clinton, & Beutler, 2008; Harwood & Beutler, 2009; Joyce & McCallum, 2004).

In a research project relevant to the utility of the RIM in guiding therapist activity once treatment is under way, Blatt and Ford (1994) used Rorschach variables to assist in categorizing patients as having problems primarily with forming satisfying interpersonal relationships (*anaclitic* problems) or primarily with maintaining their own sense of identity, autonomy, and self-worth (*introjective* problems). In the course of their psychotherapy, the anaclitic patients studied by Blatt and Ford were more responsive than the introjective patients to their therapist's focusing on relational aspects of the treatment. The introjective patients, by contrast, were more influenced than the anaclitic patients by their therapist's interpretive activity.

By helping to identify treatment goals and targets, Rorschach assessment can also play a role in monitoring treatment progress and evaluating treatment outcome. Suppose that an RIM administered prior to beginning therapy identifies certain treatment targets in Rorschach terms, such as reducing subjectively felt distress (as in changing *D* < 0 to *D* = 0), increasing receptivity to emotional arousal (as in bringing up a low *Afi*), or promoting more careful problem solving (as in elevating a *Zd* < -3.0 to a *Zd* ≥ -3.0). Subsequent retesting can then provide quantitative indications of how much progress has been made toward achieving these goals and how much work remains to be done on them. Rorschach monitoring of the extent to which the goals of treatment have been achieved can also guide therapists in deciding whether and, if so, when termination is indicated. Later on, either at the point of termination or in a follow-up evaluation, comparison of Rorschach findings with findings obtained in a pretreatment evaluation can provide an objective measure of the effects of the treatment.

This utility of Rorschach assessment in evaluating treatment progress and outcome was documented in studies by Weiner and Exner (1991) and Exner and Andronikof-Sanglade (1992). In these studies, patients in long-term, short-term, and brief psychotherapy were examined at several points before, during, and after their treatment. The results of both studies showed significant positive changes over the course of therapy in numerous Rorschach variables considered to have implications for a person's level of adjustment. Of further significance, the amount of improvement in these patients was directly related to the length of their treatment, which corresponds to evidence from psychotherapy research that longer therapies are likely to produce more change than briefer therapies (Lambert & Ogles, 2004). Accordingly, these Rorschach studies

attest to both the effectiveness of psychotherapy in promoting positive personality change and the validity of the RIM in measuring such change.

More recent research demonstrating the utility of Rorschach assessment in predicting treatment outcome and monitoring treatment change is reported by Finn (2011), Fowler et al. (2004), and Stokes et al. (2003). Also of note is a meta-analysis by Grønnerød (2004) of 38 samples in which repeat Rorschach testing was used to monitor change during psychological treatment. Grønnerød found substantial effect sizes for numerous Rorschach variables as indicators of change, and, consistent with the earlier studies, the degree of change in his samples was positively associated with the length and intensity of the treatment received.

Forensic Uses and Limitations

Like clinical applications, forensic uses of the RIM derive from the relevance of personality characteristics to making various kinds of decisions. In clinical work, personality-based decisions concern differential diagnosis and treatment planning and outcome evaluation, as just discussed. In forensic work, personality characteristics are often relevant to resolving psycholegal issues that arise in criminal, personal injury, and child custody cases (see Craig, 2005). Rorschach assessors can assist the court in addressing these psycholegal issues by attending to Rorschach measures of personality characteristics that pertain to them. In so doing, forensic consultants can benefit from special advantages of Rorschach findings, including their applicability across diverse populations, the resistance of the RIM to impression management, and the objectivity of a structural emphasis in formulating Rorschach interpretations.

Criminal Cases

Psychologists consulting in criminal cases are most commonly asked to assess whether defendants are competent to proceed to trial and, if so, whether they should be held responsible for their alleged offense. Being competent to proceed is defined in legal terms as having a factual and rational understanding of the criminal acts with which one is charged and being able to participate effectively in one's defense. In more specific terms, these basic elements of competency are commonly considered to consist of defendants being able to appreciate the possible penalties as well as the charges they are facing, to comprehend the nature of the adversarial process and the roles of the key people in it, to disclose pertinent facts in their case to their attorney, to conduct themselves appropriately in the courtroom, and to testify relevantly on their own behalf (see Stafford, 2003; Zapf & Roesch, 2006).

With respect to personality characteristics, these components of competency ordinarily require defendants to be able to think logically and coherently and to perceive people and events realistically. If defendants lack these capabilities, disordered thinking and impaired reality testing may lead to poor judgment and peculiar behavior and thereby compromise their legal competency. Hence, the previously mentioned Rorschach indices of disordered thinking and impaired reality testing (elevated *WSum6* and high *X-%*) warrant some expectation that a defendant will have difficulty demonstrating competency. It is relevant to note in this regard that an elevation on the previously mentioned Perceptual Thought Index (*PTI*), which is based on the *WSum6* and *X-%* values, is well validated as an indication of psychotic disorder (see Dao & Prevatt, 2006; Hilsenroth, Eudell-Simmons, DeFife, & Chamas, 2007; Ilonen, Heinimaa, Korkeila, Svirskis, & Salokangas, 2010).

Because it is an indirect measure of personality characteristics, however, the RIM is limited in how much it can reveal about what a respondent actually knows or is able to do. As elaborated by Gray and Acklin (2008), Rorschach findings of disordered thinking and impaired reality testing do

not necessarily preclude a person's being legally competent. Whether defendants have an adequate grasp of courtroom procedures, for example, is better determined by asking them directly about courtroom procedures than by looking at their *X-%* or *WSum6*. On the other hand, when defendants cannot give an adequate account of the adversarial process or fall short in other ways of being able to demonstrate legal competence, Rorschach evidence of cognitive dysfunction can help an examiner inform the court of likely reasons for their inability to meet competency requirements.

Criminal responsibility in forensic cases hinges on whether defendants were legally sane when they committed an alleged offense. Insanity at the time of an offense is customarily defined by either a *cognitive* or a *volitional* incapacity. Cognitive incapacity in this context consists of being unable to recognize the criminality of one's illegal actions or to appreciate the wrongfulness of this conduct (i.e., telling right from wrong). Volitional incapacity consists being unable at the time of the offense to alter or refrain from the alleged criminal conduct (i.e., insufficient self-control). Jurisdictions vary with respect to whether insanity is defined solely by cognitive incapacity or can also be established by volitional incapacity (see Zapf, Golding, & Roesch, 2006; Goldstein, Morse, & Shapiro, 2003). The previously mentioned Rorschach indices of disordered thinking and impaired reality testing serve to translate the cognitive prong of legal insanity into the language of personality functioning. As for the volitional prong, susceptibility to losing self-control is suggested by Rorschach indications of acute or chronic stress overload ($D < -1$, $AdjD < -1$), which are commonly associated with limited frustration tolerance, intemperate outbursts of affect, and episodes of impulsive behavior.

However, because legal sanity is defined by a person's state of mind at the time of an alleged offense and not at the time of a later examination, Rorschach evidence of either cognitive or volitional incapacity cannot by itself sustain an insanity defense. The critical evidence in determining criminal responsibility comes from defendants' recollections of their mental state prior to, before, during, and following an alleged offense and from observers' reports of how a defendant was behaving at the time. Should evidence from these sources suggest cognitive or volitional incapacity while an offense was being committed, Rorschach indications of such incapacity would provide supplementary information that might strengthen an insanity plea, as would any mental health records showing a history of psychological disorder.

In identifying these functioning incapacities, examiners must take into account how the relevant Rorschach scores interact with other features of a person's protocol. None of the variables in a CS Structural Summary can be interpreted accurately without attention to other variables that may enhance or attenuate its interpretive significance. In the case of perceptual accuracy, for example, the implications of a high *X-%* for impaired reality testing vary with the levels of *XA%*, *WDA%*, and *Xu%*; with the number of Popular and S- responses in a record; and with any repetitive association of *FQ-* responses with particular content categories or embellishments.

Similarly, with regard to the implications of an elevated *WSum6* for disordered thinking, adequate Rorschach assessment of severity of thought disorder calls for considerations that go beyond the magnitude of this particular score. These considerations include the distribution of critical special scores among mild (*DV*, *INC*) and serious (*DR*, *FAB*, *CONTAM*) indicators of cognitive slippage, the frequency of bizarre (Level 2) as opposed to relatively unremarkable (Level 1) special scores, and the percentage of total *R* containing a critical special score. As for the *D* and *AdjD* scores in a record, their interpretive significance for self-control is a complex function of the *R* and *Lambda* in the record, the levels of the *EA* and *es*, and the extent to which the *EB* is balanced. Attention to these and other interactive influences, as elaborated in the texts by Exner and Erdberg (2005) and Weiner (2003a), is essential to adequate interpretation of Rorschach data.

Despite the limitation of the RIM with respect to describing a person's mental state at the prior time of an alleged offense—which is a limitation common to all personality assessment

instruments—test indices of chronicity and stability can sometimes guide an estimation of previous functioning capacity from presently obtained data. Evidence of chronicity and stability tends to increase the likelihood that an effectively functioning person has been well adjusted in the past and, correspondingly, that a psychologically disturbed person has probably had previous episodes of disorder. The key Rorschach finding in this regard is $D > 0$, which in an unguarded record is usually associated with consistency over time (even when the consistency involves being emotionally unstable), with little sense of needing to change or become a different kind of person, and with ego-syntonic as opposed to ego-alien symptom formation.

The chronicity and stability associated with $D > 0$ still does not warrant inferring legal insanity at the time of an offense from presently obtained test findings. On the other hand, should a defendant who appears to be functioning fairly well when examined be claiming temporary insanity at the earlier time of an alleged offense, Rorschach findings may lend themselves to a diathesis–stress perspective on this possibility. Specifically, the less stressful a defendant’s circumstances appear to have been at the time of an alleged offense and the more stable and effective the person’s coping resources as presently reflected in the test data, the less susceptible this person would have been at the previous time to a psychological breakdown involving loss of cognitive or volitional capacities. Conversely, the more limited the coping resources suggested by test findings and the more seemingly stressful the circumstances surrounding an alleged offense, the stronger the possibility that a person did in fact experience temporary insanity, as defined by the law.

Personal Injury Cases

Personality assessment becomes relevant in personal injury cases when plaintiffs complain of having become emotionally distressed or incapacitated as a consequence of irresponsible behavior by some person or entity (see Kane & Dvoskin, 2011; Stein, 2009; Walfish, 2006). The types of emotional distress or incapacitation alleged in such circumstances most commonly involve post-traumatic stress disorder or other anxiety reactions, but they may also include affective and cognitive features of depressive disorder and psychotic breakdowns in reality testing that are claimed to be preventing plaintiffs from meeting the demands of everyday living. Each of these conditions is likely to be reflected in patterns of Rorschach responses that help to identify their presence and severity.

Persons with an anxiety or stress disorder are likely to produce either a *flooded* Rorschach protocol, which is distinguished by pervasive indications of anxiety, or a *constricted* Rorschach protocol, which is notable for guardedness and evasion. When produced by persons who have experienced potentially traumatic events, flooded protocols are usually associated with such manifestations of stress disorder as hyperarousal and the reexperiencing of phenomena. With respect to hyperarousal, the utility of the D scores in assessing whether people are experiencing more demands than they can manage without becoming unduly upset by them—and are thereby experiencing a stress overload—has already been noted. In more specific terms, a combination of $D < -1$ with $AdjD \geq 0$ is likely to indicate a substantial acute or situational stress overload, and $D < -1$ combined with $AdjD < -1$ usually identifies a substantial and persistent stress overload.

Susceptibility to reexperiencing distressing events is suggested on the Rorschach by a high frequency of $FM + m$ responses, which generally serves as an index of intrusive ideation. Troubling preoccupations, particularly about incurring bodily harm, are also likely to be revealed by the Trauma Content Index (TCI ; Armstrong, 1991; 2002). The TCI consists of the number of Aggression, Anatomy, Blood, Morbid, and Sex contents in a record divided by the total number of responses, and $TCI \geq .30$ is commonly associated with having had a traumatic experience (see Kamphuis, Kugeares, & Finn, 2000).

A constricted Rorschach protocol given by persons who are suspected of having been traumatized suggests a stress disorder characterized not by hyperarousal or the reexperiencing of phenomena, but by defensive avoidance. Instead of becoming excitable, easily upset, and troubled by flashbacks, nightmares, and other types of reexperiencing, traumatized persons who are defensively avoidant do whatever they can to escape from thoughts, feelings, and situations that might precipitate or exacerbate an episode of distress. In the process of this escape, stress-avoidant people typically become withdrawn and emotionally numb. The constricted Rorschach protocols that such persons consequently produce usually bear such hallmarks of guardedness as low *R*, high *Lambda*, low *SumC*, low Affective Ratio (*Afr*), and bland or vague content. Research concerning these Rorschach correlates of anxiety and stress disorder is reviewed by Armstrong and Kaser-Boyd (2004), Luxenberg and Levin (2004), and Sloan, Arsenault, and Hilsenroth (2002).

Neither flooded nor constricted Rorschach protocols are specific to anxiety and stress disorder, however, and they do not provide conclusive evidence of the presence of such a disorder. Given historical and other clinical or test data suggesting an anxiety or stress disorder, they merely increase the likelihood of its presence (see Armon, Maoz, Gazit, & Klein, 2011; Kaser-Boyd & Evans, 2008). Moreover, as in the case of evaluating sanity, the results of a present personal injury examination are useful only when they are interpreted in the context of past events. Personal injury cases require examiners to determine whether a plaintiff's current distress or disability constitutes a decline in functioning capacity from some previously higher level of functioning prior to the defendant's alleged breach of duty or whether instead the distress or disability predated it (see Smith, 2008).

Two other considerations limit the application of these and other Rorschach data in evaluating possible traumatic stress in the individual case. First, as in the case of the previously mentioned *S-Con* and *DEPI*, Rorschach stress indices seldom yield false positives but may produce numerous false negatives. Hence, fearfulness, avoidance, and hyperarousal can be inferred with some confidence when the Rorschach index levels discussed in the preceding paragraphs occur, but the absence of these levels does not rule out the possibility of such phenomena. Of particular note in this regard, the type of defensive avoidance that leads to constriction in persons with stress disorder tends to constrain the emergence of the kinds of test responses that identify fearfulness and hyperarousal. Hence, defensive avoidance is likely to result in a bland and guarded Rorschach protocol that reveals little if any distress, despite indications from other sources that the respondent has in fact suffered a traumatic stress reaction.

Second, like flooded and constricted records in general, neither fearfulness, constriction, hyperarousal, nor any of the Rorschach index levels associated with them is specific to the experience of traumatic stress. Fearfulness is common among phobic and obsessive worriers, for example, constriction among depressed individuals, and hyperarousal among people with generalized anxiety disorder, to name just a few alternative explanations for a respondent's showing Rorschach patterns that are sometimes associated with stress disorder.

It is because most Rorschach variables neither rule out nor are specific to any one condition or previous life event that the RIM should not be used as the sole basis for deriving psycholegal inferences. The RIM shares this limitation with other personality assessment instruments, all of which, whether performance based or self-report in nature, work best when they are administered as part of a multifaceted test battery and interpreted in light of the respondent's personal history and sociocultural context. The scientific foundations and clinical utility of such integrative personality assessment are elaborated by Beutler and Groth-Marnat (2003), Weiner (2005a), and Weiner and Greene (2008).

Similar considerations apply in the assessment of depressive and psychotic features in plaintiffs seeking personal injury damages. Rorschach indices of dysphoric affect and negative cognitions are usually helpful in identifying these features of depression, but the absence of elevations on these

indices rarely warrants ruling out depressive disorder. When indications of depression do pervade a Rorschach protocol, forensic examiners must then draw on other sources of information to determine a causative timeline: Did the plaintiff's apparent depressive state emerge subsequent to the defendant's allegedly damaging actions or was the plaintiff already depressed? And if the depression began before the defendant's allegedly injurious actions, did the plaintiff become depressed in response to other events for which the defendant was not responsible?

When psychotic impairments of functioning are at issue, Rorschach findings can be helpful in ruling out as well as identifying this possibility. As previously mentioned, the critical variables in measuring such serious disturbance, $X-\%$ and $WSum6$, are bidirectional in nature and hence unlikely to generate false-negative findings in a record of at least average length. To the contrary, low scores on these two variables in an unguarded record ordinarily provide evidence of adequate reality testing and logical reasoning, and would accordingly challenge a plaintiff's claim of being psychotically impaired. When Rorschach findings do identify psychotic impairment, however, the timing of events once more becomes a crucial consideration. Present indications of psychosis would support a personal injury claim only if other reliable data (e.g., previous testing, mental health history) gave reason to believe that the plaintiff had not been psychotic prior to the alleged harmful conduct by the defendant or, if previously disturbed, had shown a demonstrable decline in functioning capacity subsequent to the conduct.

Child Custody Cases

Judges deciding how a child's time and supervision should be divided between separated or divorced parents often seek psychological opinions concerning the personality characteristics of the parties involved. Courts also commonly rely on test indications of the personality strengths and shortcomings or weaknesses of the parents to help determine whether their parental rights should be continued or terminated. No single set of personal qualities defines suitability to parent or makes one person better qualified than another to fill a parent's role. Nevertheless, there are some personality characteristics that are likely to affect how well parents can meet their children's needs. These characteristics include their general level of adjustment, the adequacy of their coping skills, and the extent of their interpersonal accessibility, each of which can be measured by Rorschach variables.

With respect to level of adjustment, serious psychological disturbance or incapacitation is likely to detract from the clear thinking, emotional stability, and impulse control usually necessary for effective parental functioning. Rorschach findings that help to identify such serious disturbance or incapacitation include substantial elevations on variables previously mentioned as measures of disordered thinking ($WSum6$), impaired reality testing ($X-\%$), negative mood states and cognitions (C' , $Col-Shd$ Blds, MOR , $3r+2/R$, V), and excessive anxiety ($D-minus$ and $AdjD-minus$).

In terms of coping skills, mental health professionals generally concur that sound judgment, careful decision making, flexible problem solving, and effective stress management facilitate good parenting (see Ackerman, 2011; Otto, Buffington-Vollum, & Edens, 2003). Conversely, parents who exercise poor judgment, make decisions carelessly, deal with problems in an inflexible manner, and become unduly upset in managing stressful situations are likely to perform relatively poorly as parents. Among Rorschach variables that provide clues to a respondent's skill levels in these respects are the $X-\%$ (the higher the value, the more likely the person is to exercise poor judgment, and conversely), the Zd score (large negative values indicate a propensity for hasty and careless decision making, positive values the opposite), the $a:p$ ratio (an imbalance is typically associated with cognitive rigidity and inflexibility, a balance with adequate ability to change mindset), and the D and $AdjD$ scores (with minus scores identifying limited stress tolerance and zero or plus scores indicating good stress tolerance).

As for interpersonal accessibility, the quality of care that children receive is usually enhanced when their parents are nurturant, caring, and empathic individuals who are genuinely interested in people, comfortable in interpersonal relationship, and sensitive to the needs and concerns of others. Parental effectiveness is likely to suffer, on the other hand, when parents are detached, self-absorbed, and insensitive persons with limited interest in being around other people or tending to their welfare. In Rorschach terms, then, good parenting is likely to be associated with an adequate *SumH* (an index of interest in people), more *H* than $Hd + (H) + (Hd)$ (an index of interpersonal comfort), a low Isolation Index (*ISOL*; an index of involvement with people), and the presence of Texture (*T*; an index of capacity to form attachments), Cooperative Movement (*COP*; an index of collaborative perspectives on interpersonal relationships), and accurate Human Movement responses (*M+*, *Mo*, *Mu*; an index of empathic capacity). Conversely, the interpersonal accessibility of parents who show a low *SumH*, fewer *H* than $Hd + (H) + (Hd)$, an elevated *ISOL*, $T = 0$ (especially when *HVI* is positive), $COP = 0$, and numerous inaccurate *M* responses (*M-*) is likely to be limited.

In drawing such inferences about interpersonal accessibility, examiners should keep in mind that these Rorschach findings are suggestive regarding how parents are likely to interact with their children, but they are not conclusive. The test data identify probable parental strengths or shortcomings in interpersonal accessibility, but the actual expression of these personality characteristics must be determined from direct observation or dependable reports of how parents are in fact relating to their children. Similarly, with respect to indications of adjustment level and coping skills, test findings may suggest or account for parental successes and shortcomings, and often prove useful on this basis, but they do not demonstrate either good or poor parenting unless affirmed by behavioral data. These and other substantive guidelines in Rorschach child custody evaluations are elaborated by Carstairs (2011), Erard (2005), Evans and Schutz (2008), and Weiner (2005b).

Special Advantages

Special advantages of using the RIM in forensic applications include its cross-cultural applicability, its resistance to impression management, and the objectivity of a structural interpretive focus. With respect to the examination of minority-group members in the United States, Rorschach assessors should be aware that African American, Hispanic American, and Asian American adults comprised 17% of the respondents in Exner's (2007) contemporary CS non-patient reference sample, which is just a slight under-representation of their population presence. However, although including minority groups in a normative sample is desirable, it does not guarantee that members of these groups can be expected to perform similarly to each other or to the majority population on an assessment measure. More pertinent in this regard are the previously mentioned studies by Presley et al. (2001) and Meyer (2002) showing that the Rorschach structural summaries of minority groups in the United States are virtually identical to those produced by Caucasian respondents. In a review of relevant research findings gathered in many different countries around the world, Ritzler (2004) concluded even more broadly that "the Rorschach is a major, culture-free assessment method that is likely to yield similar results across a wide range of cultures" (pp. 580–581). The previously mentioned international reference study reported by Meyer et al. (2007), in which the data patterns were basically similar across 17 countries, further confirms the broad applicability of Rorschach assessment to persons from many different backgrounds.

As for impression management, the RIM can be particularly helpful in forensic cases by virtue of its resistance to malingering and deception. This resistance derives from the direct relationship that usually exists between the face validity of a personality assessment measure and the ease with

which it can be faked (see Bornstein, Rossner, Hill, & Stepanian, 1994). The limited face validity of indirect, relatively unstructured, and performance-based measures like the RIM makes them difficult to fake, and a proper Rorschach administration gives respondents little guidance on how they should proceed with their task and few clues to what their responses might signify.

Faced with such uncertainty, respondents who are attempting to feign disturbance or disability—which is a distinct possibility among criminal defendants and personal injury plaintiffs—tend to produce inconsistent patterns of responses that exaggerate certain aspects of disorder or incapacity. In particular, respondents who are attempting to “fake bad” on the RIM tend to produce dramatic records with strikingly pathological content elaborations but often without correspondingly deviant or unusual structural features. Such differences between dramatic Rorschach content and unremarkable or only moderately deviant Rorschach structure have been documented in research studies in which participants were given disorder-relevant information and instructed to simulate schizophrenia, depression, or posttraumatic stress disorder (Caine, Frueh, & Kinder, 1995; Frueh & Kinder, 1994; Netter & Viglione, 1994; Perry & Kinder, 1992). In a real-world study (not an analog study), Ganellen, Wasylw, Haywood, and Grossman (1996) found similarly that the only Rorschach variable that differentiated groups of identified malingerers and honest responders was an ad hoc measure of dramatic content (see also Ganellen, 2008).

As for respondents who are attempting to deceive examiners by concealing their problems and thereby “looking good”—a test-taking attitude commonly encountered in child custody evaluations—these persons often give guarded records that reveal their effort to conceal but not much else about their personality characteristics. In the absence of guardedness, moreover, real-world research reported by Grossman, Wasylw, Benn, and Gyoerkoe (2002) indicates that criminal defendants who have good reason to conceal psychological disturbance on the RIM are likely to have difficulty doing so. In the Grossman et al. study, 74 men charged with sex offenses and presumably motivated to make a good impression were identified on the basis of their MMPI validity scales as “minimizers” ($n = 53$) or “non-minimizers” ($n = 21$). The minimizers in this sample scored significantly lower than the non-minimizers on most of the MMPI clinical scales, but on the RIM they were not able to appear better adjusted than the non-minimizers. To the contrary, these would-be minimizers showed abnormal elevations on several Rorschach indices of psychological dysfunction and were unsuccessful in deflecting attention away from various psychological problems they had.

Some malingering or deceptive respondents may have been “coached” on the kinds of Rorschach responses they should give, or they may have consulted manuals and websites that purport to give guidance on how to make a particular kind of Rorschach impression. Because of the multifaceted nature of the RIM, however, and the multidimensional and interactive manner in which responses are interpreted, such coaching and guidance barely scratch the surface of the knowledge necessary to produce a convincingly malingered or deceptive Rorschach protocol. Even knowledgeable Rorschach clinicians are unlikely to succeed in giving a certain kind of record, because of the difficulty they will encounter in keeping track of the summary score implications of their responses as they proceed through the 10 cards. Should knowledgeable clinicians take time to review and calculate in their heads how their summary scores are shaping up, their delayed answers will usually alert examiners that something is afoot other than a genuine and spontaneous set of Rorschach responses. The only certain way for an expert to mislead an examiner in some particular direction would be to write out a carefully designed set of responses in advance and then deliver these responses from memory.

With respect to interpretation in forensic cases, the advantage of a structural focus derives from some differences between the structural and thematic features of Rorschach data. The structural data are relatively objective and directly representative of what people are like or how they are

likely to behave, whereas the thematic data for the most part are suggestive of what may be on a person's mind, with or without the person's conscious awareness. For example, inaccurate perception of the inkblots as measured by the *X*-% (structure) is an objective index of perceptual distortion in how one looks at the world, and a high frequency or percentage of inaccurate Rorschach percepts identifies a propensity to misperceive situations in the real world. By contrast, describing a percept of a human figure as "a fat person who eats too much" (thematic) suggests several alternative possibilities—for example, dissatisfaction with one's body size or shape, concerns about one's eating habits, disparaging attitudes toward people who are obese or appear undisciplined—but the imagery in the response is not conclusive for any of these possibilities.

Because of this difference, Rorschach inferences based on structural data can be explained more easily and defended more effectively than inferences derived from thematic imagery, with the latter being better suited for raising alternative possibilities than for formulating conclusions with reasonable certainty. This observation should not be taken to indicate a necessarily lesser role for thematic than for structural interpretations in applications of Rorschach assessment. To the contrary, in clinical work the underlying attitudes and concerns suggested by Rorschach imagery often illuminate more clearly than structural data the sources of patients' adjustment difficulties and the treatment approaches that will benefit them. In giving expert witness testimony, however, forensic examiners can improve their prospects for being helpful to the court and faring well on the witness stand if they confine their testimony to conclusions they can present with reasonable certainty (see Weiner, 2008).

Nevertheless, pervasive content themes may occasionally speak for themselves with some authority. As an example, numerous *MOR* responses involving recurrent images of bloody body parts constitute compelling evidence of preoccupying concerns about experiencing bodily harm, and testimony to this effect would be likely to have considerable impact if such responses were given by an accident or assault victim claiming stress disorder. With allowance for such exceptions, however, forensic examiners are well advised to base their Rorschach interpretations primarily on the structural data in a protocol.

CASE EXAMPLE

Mr. Clark (fictitious name) is a 27-year-old Caucasian man who was evaluated in the county jail following an incident that led to his being charged with burglary and criminal mischief. According to police reports, he had boarded a 34-foot sailing vessel docked at a seaside marina, ordered the owner of the boat ashore, and attempted without success to cast off. Called by the boat owner from his cell phone, the police arrived on the scene and took Mr. Clark into custody. The Public Defender's Office, representing Mr. Clark and believing there was basis in his case for an insanity plea, requested a psychological evaluation.

When interviewed, Mr. Clark appeared to be in good spirits and talked freely about himself and his life. According to his report, his father was a disturbed man who lived an erratic lifestyle and had been hospitalized for manic-depressive disorder. About himself, he said that he had done poorly in school as a child and had been held back a few years. Nevertheless, he stated, he had eventually graduated from high school in an arts program with "the highest grade-point average ever recorded there." Asked about his employment history, he described working at some semi-skilled jobs before getting into "security work," and he added that "breaking codes is my specialty." He then volunteered the information that he had suffered a "mental

breakdown" five years earlier, been diagnosed as manic-depressive, and spent the next two years in and out of hospitals. Subsequent to his hospitalizations, he said, he was not able to persuade any employers to hire him, "because my reputation had been ruined by being diagnosed mentally ill." For the past three years, he has for the most part been a homeless person subsisting on welfare.

Mr. Clark has never married, but he is preoccupied with Gretchen (fictitious name), whom he identifies as his girlfriend and says he plans to marry. However, when asked further about his relationship with Gretchen, he said that he has spent very little time with her since meeting her three years ago, around the time of his hospital discharge, and that he was not sure where she was. Immediately after giving this information, he expressed a firm conviction that the CIA and "special police" in Germany have been conspiring to keep him and Gretchen apart, and he then explained that he was attempting to take the boat because "she must be in Germany, and I was going to sail there and find her."

Mr. Clark was pleasant and personable while being interviewed and said that he appreciated an opportunity to "tell my story." His speech was pressured as he did so, and his flow of ideas was frequently dissociated and circumstantial. He rarely remained focused on a topic for more than a few moments before wandering off into irrelevant, tangential, and sometimes seemingly delusional asides. When presented with a battery of psychological tests, he cooperated fully with the procedures and showed considerable interest in being examined. His Rorschach Sequence of Scores and Structural Summary are shown in Tables 7.1 and 7.2. Consistent with the previously mentioned guidelines for forensic interpretation of Rorschach data, the following discussion of Mr. Clark's protocol centers on structural features of the data that illustrate the use of the RIM in differential diagnosis and may have a bearing on his insanity defense.

Mr. Clark's Structural Summary indicates a valid Rorschach record with an above-average response total ($R = 29$) and indications of average range openness ($\Lambda = .61$). In accord with the interpretive search strategies developed for the CS (see Exner, 2003, chap. 13; Weiner, 2003a, chap. 3), the opening key variable is the presence of reflection responses, which identifies Mr. Clark's attitudes toward himself and other people as the most salient aspects of his current personality functioning. The relevant data in this regard suggest that he is a self-centered person who is inclined to overestimate his attributes and accomplishments, deny problems and shortcomings, externalize responsibility for difficulties, and feel a sense of entitlement ($Fr + rF = 3$, $3r + 2/R = .76$). Persons with these personality characteristics commonly show features of narcissism, psychopathy, or hypomania, with the last of these alternatives seeming most consistent with Mr. Clark's case history and interview behavior.

Along with a hypomanic person's typical social expansiveness and quest for an appreciative audience, however, this man also gives evidence of being suspicious and mistrustful in his relationships with people and consequently intent on maintaining his privacy and keeping his distance from others (positive Hypervigilance Index [HVI]). This finding is consistent with, and helps to confirm, the paranoid quality of his presentational style and thought content. Hence, the two main themes to pursue in Mr. Clark's record are his apparent hypomania and paranoia.

To elaborate first on Mr. Clark's moods, he seems inclined to express affect in a more intense and dramatic way than most adults ($FC:CF + F = 2:4$), which is consistent with hypomania. However, there are also indications in the record of dysphoria ($C' > 2$; $Col-Shd Blds > 0$), withdrawal from emotionally arousing situations ($Afr < .46$), and limited interest in collaborative interactions with others ($COP < 2$), which are depressive phenomena. Considered together with his hypomania, these findings suggest the possibility of a bipolar or cyclothymic disorder.

TABLE 7.1 Sequence of Scores for Mr. Clark

<i>Card</i>	<i>Resp. No</i>	<i>Location and DQ</i>	<i>Loc. No.</i>	<i>Determinant(s) and Form Quality</i>	<i>(2)</i>	<i>Content(s)</i>	<i>Pop</i>	<i>Z Score</i>	<i>Special Scores</i>
I	1	D+	1	Mpo	2	H,Cg,Ay		4.0	GHR
	2	Do	2	Fo	2	(A)			
	3	Do	3	Mp,FC',CF-		Ad			AG, ALOG, INC, PHR
III	4	D+	6	Mao	2	H		3.0	COP, GHR
	5	D+	9	Mp,FC',Fr+		H,Cg	P	3.0	AB
	6	W+	1	FCu	2	Cg,Id		5.5	GHR
IV	7	D+	9	Mpo	2	H,Id	P	3.0	FAB2, PHR
	8	DdS+	99	Mpo	2	H,An		4.5	DV, GHR
	9	W+	1	Mp,FD0		(H),Bt	P	4.0	PHR
V	10	Ddo	30	mpu		Hd,Sx		2.0	
	11	Wo	1	Fu		(A),Ay		1.0	
	12	Wo	1	Fo		A	P	1.0	PSV, DV
VI	13	Wo	1	Fo		A		1.0	MOR, DV
	14	Wo	1	Fo		Ad	P	2.5	PSV, DR2, PHR
	15	Ddo	24	F-	2	H			DV, PHR
VII	16	Ddo	31	Fu	2	Hd,Sx			AG, PHR
	17	Do	3	Mpu	2	(Hd)		2.5	DV, PHR
	18	W+	1	FC'-		H,Cg			
VIII	19	DSo	7	Fo		Cl			
	20	D+	1	Mp+	2	Hd,Cg	P	3.0	GHR
	21	W+	1	FMp,CF,Fro		A,Is	P	4.5	INC
IX	22	Ddo	99	F-	2	Ad			
	23	Wv/+	1	CF,Fro		Na		5.5	AB, PHR
	24	W+	1	Ma,CF,mp-		H,Cg,Id		5.5	PHR
X	25	DdSo	99	FC-	2	Hd		4.0	FAB, PHR
	26	D+	11	Mau	2	A,Id			
	27	Do	7	Fo		A			
	28	Do	10	FMau		A			
	29	Do	13	F-		An			DV2

TABLE 7.2a Structural Summary for Mr. Clark

Location Features Zf = 17 ZSum = 58.5 ZEst = 56.0 W = 10 (Wv = 0) D = 13 W+D = 23 Dd = 6 S = 3	Determinants Blends Single M.FC'.CF M = 7 M.FC'.Fr FM = 1 M.FD m = 1 FM.CF.Fr FC = 2 CF.Fr CF = 0 M.CF.m C = 0 Cn = 0 FC' = 1 C'F = 0 C' = 0 FT = 0 TF = 0 T = 0 FV = 0 VF = 0 V = 0 FY = 0 YF = 0 Y = 0 Fr = 0 rF = 0 FD = 0 F = 11 (2) = 13	Contents H = 8 (H) = 1 Hd = 4 (Hd) = 1 Hx = 0 A = 6 (A) = 2 Ad = 3 (Ad) = 0 An = 2 Art = 0 Ay = 2 Bl = 0 Bt = 1 Cg = 6 Cl = 1 Ex = 0 Fd = 0 Fi = 0 Ge = 0 Hh = 0 Ls = 1 Na = 1 Sc = 0 Sx = 2 Xy = 0 Idio = 4	S-Constellation ☐ FV+VF+V+FD > 2 ☒ Col-Shd Blends > 0 ☒ Ego < .31 or > .44 ☐ MOR > 3 ☐ Zd > ±3.5 ☐ es > EA ☒ CF + C > FC ☒ X+% < .70 ☐ S > 3 ☐ P < 3 or > 8 ☐ Pure H < 2 ☐ R < 17 ☐ Total Special Scores Lvl-1 Lvl-2 DV = 5 × 1 1 × 2 INC = 2 × 2 0 × 4 DR = 0 × 3 1 × 6 FAB = 1 × 4 1 × 7 ALOG = 1 × 5 CON = 0 × 7 Raw Sum6 = 12 Wgtd Sum6 = 33 AB = 2 GHR = 6 AG = 2 PHR = 10 COP = 1 MOR = 1 CP = 0 PER = 0 PSV = 2
DQ (FQ-) + = 12 (2) o = 16 (5) v/+ = 1 (0) v = 0 (0)	Form Quality FQx MQual W+D + = 2 2 2 o = 13 5 12 u = 7 2 5 - = 7 2 4 none = 0 0 0		

Further consideration of the paranoid elements in Mr. Clark's record brings to light some additional features of his condition. Of particular note are two findings commonly associated with delusion formation. First, he gives substantial evidence of having a thought disorder, with a predilection for dissociated ideation and illogical reasoning concerning relationships between events (*WSum* 6 = 33, with *Lvl*2 = 3). Second, he shows a prominent tendency to resort excessively to fantasy as a way of dealing with problems—that is, by imagining or daydreaming about how they will be solved (*Ma* : *Mp* = 3 : 8). Also noteworthy is the fact that Mr. Clark is an introversive person with a strong preference for dealing with his experience in an ideational and contemplative manner rather than by expressing feelings or taking action (*EB* = 11 : 5.0).

TABLE 7.2b Ratios, Percentages, and Derivatives

R = 29 L = 0.61			<i>Affect</i>	
EB = 11 : 5.0	EA = 16.0	EBPer = 2.2	FC:CF+C	= 2 : 4
eb = 4 : 3	es = 7	D = +3	Pure C	= 0
	Adj es = 6	Adj D = +3	SumC' : WSumC	= 3 : 5.0
FM = 2	SumC' = 3	SumT = 0	Afr	= 0.45
m = 2	SumV = 0	SumY = 0	S	= 3
			Blends:R	= 6 : 29
			CP	= 0

<i>Interpersonal</i>		<i>Ideation</i>	
COP =	AG = 2	a:p	= 4 : 11 Sum6 = 12
GHR:PHR	= 6 : 10	Ma:Mp	= 3 : 8 Lvl-2 = 3
a:p	= 4 : 11	2AB+(Art+Ay)	= 6 WSum6 = 33
Food	= 0	MOR	= 1 M- = 2
SumT	= 0		M none = 0
Human Content	= 14		
Pure H	= 8		
PER	= 0		
Isolation Index	= 0.21		

<i>Mediation</i>		<i>Processing</i>		<i>Self-perception</i>	
XA%	= 0.76	Zf	= 17	3r+(2)/R	= 0.76
WDA%	= 0.83	W:D:Dd	= 10:13:6	Fr+rF	= 3
X-%	= 0.24	W : M	= 10:11	SumV	= 0
S-	= 1	Zd	= +2.5	FD	= 1
P	= 7	PSV	= 2	An+Xy	= 2
X+%	= 0.52	DQ+	= 12	MOR	= 1
Xu%	= 0.24	DQv	= 0	H:(H)+Hd+(Hd)	= 8 : 6

PTI = 3	☒ DEPI = 5	☐ CDI = 2	S-CON = 4	☒ HVI = Yes	☐ OBS = No
---------	--	----------------------------------	-----------	---	-----------------------------------

Although being introversive does not preclude the possibility of bipolar disorder, manic-depressive individuals are typically more emotional than ideational in how they deal with experience and thereby likely to show an extratensive *EB*.

Hence, to the extent that inferring a *DSM* diagnosis may become relevant or required (which often is the case in expert witness testimony), the combination of Mr. Clark's thinking disorder and affective instability may indicate a schizoaffective disorder. Although Mr. Clark's affective instability and paranoia are the cornerstones of his clearly demonstrated psychological disturbance, the presence of these conditions, as noted earlier, is not sufficient in a court of law to support an insanity defense. With respect to being able to tell right from wrong, the critical psychodiagnostic question is whether the defendant was capable of distinguishing between reality and fantasy when the alleged offense occurred. The perceptual and attentional variables in the Structural Summary indicate that Mr. Clark was able when he was examined to recognize conventional modes of response ($P = 7$) but had some limitations in his reality

testing ($X\% = .24$). This level of reality testing is not in the psychotic range, but it does indicate a propensity on his part to misperceive events, show poor judgment, and fail to anticipate the consequences of his actions. Although insufficient to document impaired reality testing at the time of his offense, these findings were consistent with observers' reports of his strange behavior on that day and with the account he gave of his irrational basis for attempting to commandeer the vessel.

Significantly, however, Mr. Clark's $WDA\%$ is .83, with six Dd locations and four of his seven inaccurate percepts ($FQ-$) occurring in these Dd responses. This finding indicates that his reality testing is within normal limits when he keeps his attention focused on the central and obvious details of his experience, which would account for his being able much of the time to present himself as a reasonably "sane" person. On the other hand, when he becomes distracted by peripheral details and focuses on irrelevant aspects of situations—which he is more likely than most people to do—he tends to lose touch with reality and form distorted impressions of what is happening around him. Instances of such loss of reality sense could explain why at times he could break the law without appreciating the wrongfulness of his conduct.

However warranted this conclusion about Mr. Clark's susceptibility to episodes of legal insanity may be, it does not yet speak to his mental state at the time of the particular offense with which he is presently charged. To identify another finding that could be relevant in this regard, note that the Structural Summary data identify not only emotional instability but also a probability that Mr. Clark's disturbance is chronic rather than acute in nature. The pertinent indications of chronicity are the D and $AdjD$ scores of +3. D and $AdjD$ scores in the plus range usually signify that the personality characteristics suggested by a Rorschach protocol are stable and well entrenched, for better or worse, and likely to have been present at previous points in an adult person's life. Hence, these particular Rorschach findings give reason to believe the susceptibility to episodic loss of reality sense shown by Mr. Clark when he was tested was likely to have been present as well when he committed the offense with which he is charged.

Summary

As elaborated in this chapter, the Rorschach Inkblot Method (RIM) is a psychometrically sound personality assessment procedure that can facilitate decision making whenever personality characteristics participate in shaping conclusions and recommendations concerning people. In clinical practice, Rorschach findings help to identify the presence and nature of psychological disorder and a person's need for, amenability to, and progress in treatment of various kinds. In forensic work, Rorschach assessors can assist the court by identifying personality characteristics that are pertinent to resolving psycholegal issues in criminal, personal injury, and child custody cases. Also of value in forensic evaluations are the incremental validity of the RIM, which can add useful information to self-report and interview findings, and its sensitivity to malingering and deception. Forensic consultants should nevertheless be well schooled in the limitations as well as the utility of Rorschach assessment.

References

- Ackerman, M. J. (2011). Child custody evaluation practices: A 20-year follow-up. *Family Court Review*, 49, 618–628.

- Acklin, M. W. (1995). Integrative Rorschach interpretation. *Journal of Personality Assessment*, 64, 235–238.
- Acklin, M. W., McDowell, C. J., Verschell, M. S., & Chan, D. (2000). Interobserver agreement, intraobserver agreement, and the Rorschach Comprehensive System. *Journal of Personality Assessment*, 74, 15–57.
- Acklin, M. W., Li, S. H. S., & Tyson, J. (2006). Rorschach assessment of personality disorders: Applied clinical science and psychoanalytic theory. In S. K. Huprich (Ed.), *Rorschach assessment of the personality disorders* (pp. 423–444). Mahwah, NJ: Lawrence Erlbaum.
- Armon, Z., Maoz, G., Gazit, T., & Klein, E. (2011). Rorschach indicators of PTSD: A retrospective study. *Rorschachiana*, 32, 5–26.
- Armstrong, J. (1991). The psychological organization of multiple personality disordered patients as revealed in psychological testing. *Psychiatric Clinics of North America*, 14, 533–546.
- Armstrong, J. (2002). Deciphering the broken narrative of trauma: Signs of traumatic dissociation on the Rorschach. *Rorschachiana*, 25, 11–27.
- Armstrong, J., & Kaser-Boyd, N. (2004). Projective assessment of psychological trauma. In M. Hersen (Series Ed.) & M. Hilsenroth & D. Segal (Vol. Eds.), *Comprehensive handbook of psychological assessment*. Vol. 2: *Objective and projective assessment of personality* (pp. 500–512). Hoboken, NJ: John Wiley.
- Beck, S. J., Beck, A. G., Levitt, E. E., & Molish, H. B. (1961). *Rorschach's test: I. Basic processes* (3rd ed.). New York: Grune & Stratton.
- Beutler, L. E., & Groth-Marnat, G. (2003). *Integrative assessment of adult personality* (2nd ed.). New York: Guilford Press.
- Blais, M. A., Hilsenroth, M. J., Castlebury, F., Fowler, J. C., & Baity, M. R. (2001). Predicting DSM–IV Cluster B personality disorder criteria from MMPI–2 and Rorschach data: A test of incremental validity. *Journal of Personality Assessment*, 76, 150–168.
- Blasczyk-Schiep, S., Kazén, M., Kuhl, J., & Grygielski, M. (2011). Appraisal of suicidal risk among adolescents and young adults through the Rorschach test. *Journal of Personality Assessment*, 93, 518–526.
- Blatt, S. J., & Ford, R. Q. (1994). *Therapeutic change*. New York: Plenum.
- Bornstein, R. J., & Masling, J. M. (2005). The Rorschach Oral Dependency Scale. In R. F. Bornstein & J. M. Masling (Eds.), *Scoring the Rorschach: Seven validated systems* (pp. 135–158). Mahwah, NJ: Lawrence Erlbaum.
- Bornstein, R. F., Rossner, S. C., Hill, E. L., & Stepanian, M. L. (1994). Face validity and fakability of objective and projective measures of dependency. *Journal of Personality Assessment*, 63, 363–386.
- Bram, A. D. (2010). The relevance of the Rorschach and patient–examiner relationship in treatment planning and outcome assessment. *Journal of Personality Assessment*, 92, 91–115.
- Caine, S. L., Frueh, B. C., & Kinder, B. N. (1995). Rorschach susceptibility to malingered depressive disorders in adult females. In J. N. Butcher & C. D. Spielberger (Eds.), *Advances in personality assessment* (Vol. 10, pp. 165–173). Hillsdale, NJ: Lawrence Erlbaum.
- Carstairs, K. S. (2011). Rorschach assessment of parenting capacity: A case study. *Rorschachiana*, 32, 91–116.
- Clarkin, J. F., & Levy, K. N. (2004). The influence of client variables on psychotherapy. In Lambert, M. J. (Ed.), *Bergin and Garfield's handbook of psychotherapy and behavior change* (5th ed., pp. 194–226). Hoboken, NJ: John Wiley.
- Craig, R. J. (2005). *Personality-guided forensic psychology*. Washington, DC: American Psychological Association.
- Dao, T. K., & Prevatt, F. (2006). A psychometric evaluation of the Rorschach Comprehensive System's Perceptual Thinking Index. *Journal of Personality Assessment*, 86, 180–189.
- Dao, T. K., Prevatt, F., & Horne, H. L. (2008). Differentiating psychotic patients from nonpsychotic patients with the MMPI–2 and Rorschach. *Journal of Personality Assessment*, 90, 93–101.
- Ellenberger, H. F. (1954). The life and work of Hermann Rorschach (1884–1922). *Bulletin of the Menninger Clinic*, 18, 173–219.
- Erard, R. E. (2005). What the Rorschach can contribute to child custody and parenting time evaluations. *Journal of Child Custody*, 2, 119–142.
- Evans, F. B., & Schutz, B. M. (2008). The Rorschach in child custody and parenting plan evaluations: A new conceptualization. In C. B. Gacono & F. B. Evans (Eds.), *Handbook of forensic Rorschach assessment* (pp. 233–254). New York: Routledge.
- Exner, J. E. Jr. (1974). *The Rorschach: A comprehensive system*. New York: Wiley.
- Exner, J. E. Jr. (2001). *A Rorschach workbook for the Comprehensive System*. Asheville, NC: Rorschach Workshops.

- Exner, J. E. Jr. (2003). *The Rorschach: A comprehensive system*. Vol. 1: *Basic foundations and principles of interpretation* (4th ed.). Hoboken, NJ: John Wiley.
- Exner, J. E. Jr. (2007). A new US adult nonpatient sample. *Journal of Personality Assessment*, 89, S154–S158.
- Exner, J. E. Jr., & Andronikof-Sanglade, A. (1992). Rorschach changes following brief and short-term therapy. *Journal of Personality Assessment*, 59, 59–71.
- Exner, J. E. Jr., & Erdberg, P. (2005). *The Rorschach: A comprehensive system*. Vol. 2: *Advanced interpretation* (3rd ed.). Hoboken, NJ: John Wiley.
- Exner, J. E. Jr., Thomas, E. A., & Mason, B. (1985). Children's Rorschachs: Description and prediction. *Journal of Personality Assessment*, 49, 13–20.
- Faigman, D. L., & Monahan, J. (2009). Standards of legal admissibility and their implications for psychological science. In J. L. Skeem, K. S. Douglas, & S. O. Lilienfeld (Eds.), *Psychological science in the courtroom: Consensus and controversy* (pp. 3–25). New York: Guilford Press.
- Finn, S. E. (2011). Journeys through the valley of death: Multimethod psychological assessment and personality transformation in long-term psychotherapy. *Journal of Personality Assessment*, 93, 93–123.
- Fowler, J. C., Ackerman, S., Speanberg, S., Bailey, A., Blagys, M., & Conklin, A. (2004). Personality and symptom change in treatment-refractory inpatients: Evaluation of the phase model of change using Rorschach, TAT, and DSM-IV Axis V. *Journal of Personality Assessment*, 76, 333–351.
- Fowler, J. C., Piers, C., Hilsenroth, M. J., Holdwick, D. J. Jr., & Padawer, J. R. (2001). The Rorschach Suicide Constellation: Assessing various degrees of lethality. *Journal of Personality Assessment*, 76, 333–351.
- Frueh, B. C., & Kinder, B. N. (1994). The susceptibility of the Rorschach inkblot test to malingering of combat related PTSD. *Journal of Personality Assessment*, 62, 280–298.
- Ganellen, R. J. (2005). Rorschach contributions to assessment of suicidal risk. In R. I. Yufit & D. Lester (Eds.), *Assessment, treatment, and prevention of suicidal behavior* (pp. 93–119). Hoboken, NJ: John Wiley.
- Ganellen, R. J. (2008). Rorschach assessment of malingering and defensive response sets. In C. B. Gacono & F. B. Evans (Eds.), *The handbook of forensic Rorschach assessment* (pp. 89–119). New York: Routledge.
- Ganellen, R. J., Wasylwi, O. E., Haywood, T. W., & Grossman, L. S. (1996). Can psychosis be malingered on the Rorschach? An empirical study. *Journal of Personality Assessment*, 66, 65–80.
- Garb, H. N. (1999). Call for a moratorium on the use of the Rorschach Inkblot Test in clinical and forensic settings. *Assessment*, 6, 311–318.
- Garb, H. N., Wood, J. M., Nezworski, M., Grove, W. M., & Stejskal, W. J. (2001). Towards a resolution of the Rorschach controversy. *Psychological Assessment*, 13, 433–448.
- Goldstein, A. M., Morse, S. J., & Shapiro, D. L. (2003). Evaluation of criminal responsibility. In I. B. Weiner (Series Ed.) & A. M. Goldstein (Vol. Ed.), *Handbook of psychology*. Vol. 11: *Forensic psychology* (pp. 381–406). Hoboken, NJ: John Wiley.
- Gray, T., & Acklin, M. W. (2008). The use of the Rorschach Inkblot Method in trial competency evaluations. In C. B. Gacono & F. B. Evans (Eds.), *Handbook of forensic Rorschach assessment* (pp. 41–157). New York: Routledge.
- Grønnerød, C. (2003). Temporal stability in the Rorschach method: A meta-analytic review. *Journal of Personality Assessment*, 80, 272–293.
- Grønnerød, C. (2004). Rorschach assessment of changes following psychotherapy: A meta-analytic review. *Journal of Personality Assessment*, 83, 256–276.
- Grønnerød, C. (2006). Reanalysis of the Grønnerød (2003) Rorschach temporal stability meta-analysis set. *Journal of Personality Assessment*, 86, 222–225.
- Grossman, L., Wasylwi, O., Benn, A., & Gyoerkoe, K. L. (2002). Can sex offenders who minimize on the MMPI conceal psychopathology on the Rorschach? *Journal of Personality Assessment*, 78, 484–501.
- Groth-Marnat, G., Gotheil, E., Liu, W., Clinton, D. A., & Beutler, L. E. (2008). Personality and treatment planning for psychotherapy. In G. J. Boyle, G. E. Matthews, & D. H. Saklofske (Eds.), *SAGE handbook of personality theory and assessment*. Vol. 1: *Personality theory and assessment* (pp. 620–634). Thousand Oaks, CA: Sage.
- Grove, W. M., & Barden, R. C. (1999). Protecting the integrity of the legal system: The admissibility of testimony from mental health experts under *Daubert/Kumho* analyses. *Psychology, Public Policy, and Law*, 5, 224–242.
- Handler, L., & Clemence, A. J. (2005). The Rorschach Prognostic Rating Scale. In R. F. Bornstein & J. M. Masling (Eds.), *Scoring the Rorschach: Seven validated systems* (pp. 1–24). Mahwah, NJ: Lawrence Erlbaum.

- Harwood, T. M., & Beutler, L. E. (2009). Assessment of clients in pretreatment planning. In J. N. Butcher (Ed.), *Oxford handbook of personality assessment* (pp. 643–666). New York: Oxford University Press.
- Hess, A. K. (2006). Serving as an expert witness. In I. B. Weiner & A. K. Hess (Eds.), *Handbook of forensic psychology* (pp. 652–700). Hoboken, NJ: John Wiley.
- Hiller, J. B., Rosenthal, R., Bornstein, R. F., Berry, D. T. R., & Brunell-Neuleib, S. (1999). A comparative meta-analysis of Rorschach validity. *Psychological Assessment*, 11, 278–296.
- Hilsenroth, M. J., Eudell-Simmons, E. M., DeFife, J. A., & Chamas, J. W. (2007). The Rorschach Perceptual-Thinking Index (PTI): An examination of reliability, validity, and diagnostic efficiency. *International Journal of Testing*, 7, 269–291.
- Hilsenroth, M. J., & Stricker, G. (2004). A consideration of challenges to psychological assessment instruments used in forensic settings: Rorschach as exemplar. *Journal of Personality Assessment*, 83, 141–152.
- Hogan, T. P. (2005). 50 widely used psychological tests. In G. P. Koocher, J. C. Norcross, & S. S. Hill (Eds.), *Psychologists' desk reference* (2nd ed., pp. 101–104). New York: Oxford University Press.
- Ilonen, T., Heinimaa, M., Korkeila, J., Svirskis, T., & Salokangas, R. K. R. (2010). Differentiating adolescents at clinical high risk for psychosis from psychotic and non-psychotic patients with the Rorschach. *Psychiatry Research*, 179, 151–156.
- Janson, H., & Stattin, H. (2003). Predictions of adolescent and adult delinquency from childhood Rorschach ratings. *Journal of Personality Assessment*, 81, 51–63.
- Joyce, A. S., & McCallum, M. (2004). Assessing patient capacities for psychotherapy: Psychological-mindedness and quality of object relations. In D. P. Charman (Ed.), *Core processes in brief psychodynamic psychotherapy: Advancing effective practice* (pp. 69–100). Mahwah, NJ: Lawrence Erlbaum.
- Kamphuis, J. H., Kugeares, S. L., & Finn, S. E. (2000). Rorschach correlates of sexual abuse: Trauma content and aggression indexes. *Journal of Personality Assessment*, 75, 212–224.
- Kane, A. W., & Dvoskin, J. A. (2011). *Evaluation for personal injury claims*. New York: Oxford University Press.
- Kaser-Boyd, N., & Evans, F. B. (2008). Rorschach assessment of psychological trauma. In C. B. Gacono & F. B. Evans (Eds.), *The handbook of forensic Rorschach assessment* (pp. 255–277). New York: Routledge.
- Klopfer, B., Ainsworth, M. D., Klopfer, W. G., & Holt, R. R. (1954). *Developments in the Rorschach technique*. Vol. 1: *Technique and theory*. Yonkers-on-Hudson, NY: World Book.
- Lambert, M. J., & Ogles, B. M. (2004). The efficacy and effectiveness of psychotherapy. In M. J. Lambert (Ed.), *Bergin and Garfield's handbook of psychotherapy and behavior change* (5th ed., pp. 139–193). Hoboken, NJ: John Wiley.
- Lerner, P. M. (1991). *Psychoanalytic theory and the Rorschach*. Hillsdale, NJ: Analytic Press.
- Lerner, P. M. (1998). *Psychoanalytic perspectives on the Rorschach*. Hillsdale, NJ: Analytic Press.
- Lerner, P. M. (2007). On preserving a legacy: Psychoanalysis and psychological testing. *Psychoanalytic Psychology*, 24, 208–230.
- Lilienfeld, S. O., Wood, J. M., & Garb, H. N. (2000). The scientific status of projective techniques. *Psychological Science in the Public Interest*, 1, 27–66.
- Lohr, J. M., Fowler, K. A., & Lilienfeld, S. O. (2002). The dissemination and promotion of pseudoscience in clinical psychology: The challenge to legitimate clinical science. *Clinical Psychologist*, 55, 4–10.
- Luxenberg, T., & Levin, P. (2004). The role of the Rorschach in the assessment of trauma. In J. P. Wilson & T. M. Keane (Eds.), *Assessing psychological trauma and PTSD* (2nd ed., pp. 190–225). New York: Guilford Press.
- McCann, J. T., & Evans, F. B. (2008). Admissibility of the Rorschach. In C. B. Gacono & F. B. Evans (Eds.), *The handbook of forensic Rorschach assessment* (pp. 55–78). New York: Routledge.
- McGrath, R. E. (2003). Enhancing accuracy in observational test scoring: The Comprehensive System as a case example. *Journal of Personality Assessment*, 81, 104–110.
- McGrath, R. E., Pogge, D. L., Stokes, J. M., Cragolino, A., Zaccario, M., Hayman, J., . . . Wayland-Smith, D. (2005). Field reliability of Comprehensive System scoring in an adolescent inpatient sample. *Assessment*, 12, 199–209.
- Meloy, J. R. (2008). The authority of the Rorschach: An update. In C. B. Gacono and F. B. Evans (Eds.), *The handbook of forensic Rorschach assessment* (pp. 79–87). New York: Routledge.
- Meloy, J. R., Hansen, T., & Weiner, I. B. (1997). Authority of the Rorschach: Legal citations in the past 50 years. *Journal of Personality Assessment*, 69, 53–62.
- Meyer, G. J. (1997a). Assessing reliability: Critical corrections for a critical examination of the Rorschach Comprehensive System. *Psychological Assessment*, 9, 480–489.

- Meyer, G. J. (1997b). Thinking clearly about reliability: More critical corrections regarding the Rorschach Comprehensive System. *Psychological Assessment*, 9, 495–498.
- Meyer, G. J. (2002). Exploring possible ethnic differences and bias in the Rorschach Comprehensive System. *Journal of Personality Assessment*, 78, 104–129.
- Meyer, G. J. (2004). The reliability and validity of the Rorschach and Thematic Apperception Test (TAT) compared to other psychological and medical procedures: An analysis of systematically gathered evidence. In M. Hersen (Series Ed.) & M. J. Hilsenroth & D. L. Segal (Vol. Eds.), *Personality assessment*. Vol. 2: *Comprehensive handbook of psychological assessment* (pp. 315–342). Hoboken, NJ: John Wiley.
- Meyer, G. J., & Archer, R. P. (2001). The hard science of Rorschach research: What do we know and where do we go? *Psychological Assessment*, 13, 486–502.
- Meyer, G. J., Erdberg, P., & Shaffer, T. W. (2007). Toward international normative reference data for the Comprehensive System. *Journal of Personality Assessment*, 89(S 1), S201–S216.
- Meyer, G. J., & Handler, L. (2000). Incremental validity of the Rorschach Prognostic Rating Scale over the MMPI Ego Strength Scale and IQ. *Journal of Personality Assessment*, 74, 356–370.
- Meyer, G. J., Hilsenroth, M. J., Baxter, D., Exner, J. E. Jr., Fowler, J. C., Piers, C. C., & Resnick, J. (2002). An examination of the interrater reliability for scoring the Rorschach Comprehensive System in eight data sets. *Journal of Personality Assessment*, 78, 219–274.
- Muscewicz, J., Marczyk, G., Knauss, L., & York, D. (2009). Current assessment practices, personality measurement, and Rorschach usage by psychologists. *Journal of Personality Assessment*, 91, 453–461.
- Netter, B. E. C., & Viglione, D. J. (1994). An empirical study of malingering schizophrenia on the Rorschach. *Journal of Personality Assessment*, 62, 45–57.
- Nygren, M. (2004). Rorschach Comprehensive System variables in relation to assessing dynamic capacity and ego strength for psychodynamic psychotherapy. *Journal of Personality Assessment*, 83, 277–292.
- Otto, R. K., Buffington-Vollum, J. K., & Edens, J. F. (2003). Child custody evaluation. In I. B. Weiner (Series Ed.) & A. M. Goldstein (Vol. Ed.), *Handbook of psychology*. Vol. 11: *Forensic psychology* (pp. 179–208). Hoboken, NJ: John Wiley.
- Perry, G. G., & Kinder, B. N. (1992). Susceptibility of the Rorschach to malingering: A schizophrenia analogue. In C. D. Spielberger & J. N. Butcher (Eds.), *Advances in personality assessment* (Vol. 9, pp. 127–140). Hillsdale, NJ: Lawrence Erlbaum.
- Presley, G., Smith, C., Hilsenroth, M., & Exner, J. E. (2001). Rorschach validity with African Americans. *Journal of Personality Assessment*, 77, 491–507.
- Ritzler, B. (2004). Cultural applications of the Rorschach, apperception tests, and figure drawings. In M. Hersen (Series Ed.) & M. J. Hilsenroth & D. L. Segal (Vol. Eds.), *Comprehensive handbook of psychological assessment*. Vol. 2: *Personality assessment* (pp. 573–585). Hoboken, NJ: John Wiley.
- Ritzler, B., Erard, R., & Pettigrew, T. (2002). Protecting the integrity of Rorschach expert witnesses: A reply to Grove and Barden (1999) re: The admissibility of testimony under *Daubert/Kumho* analysis. *Psychology, Public Policy, and Law*, 8, 201–215.
- Rorschach, H. (1942). *Psychodiagnostics: A diagnostic test based on perception* (P. Lemkau & B. Kronenberg, Trans.). Berne, Switzerland: Hans Huber. (Original work published 1921)
- Sahly, J., Shaffer, T. W., Erdberg, P., & O'Toole, S. (2011). Rorschach intercoder reliability for protocol-level Comprehensive System variables in an international sample. *Journal of Personality Assessment*, 93, 592–596.
- Schafer, R. (1954). *Psychoanalytic interpretation in Rorschach testing*. New York: Grune & Stratton.
- Schwarz, W. (1996). Hermann Rorschach, M.D.: His life and work. *Rorschachiana*, 21, 6–17.
- Sendin, M. C. (2010). Rorschach usefulness in treatment planning. *Rorschachiana*, 31, 70–89.
- Sloan, P., Arsenault, L., & Hilsenroth, M. (2002). Use of the Rorschach in the assessment of war-related stress in military personnel. *Rorschachiana*, 25, 86–122.
- Smith, B. (1994). Object relations theory and the integration of empirical and psychoanalytic approaches to Rorschach interpretation. *Rorschachiana*, 19, 61–77.
- Smith, B. (2008). Rorschach assessment in tort and employment litigation. In C. B. Gacono & F. B. Evans (Eds.), *The handbook of forensic Rorschach psychology* (pp. 279–299). New York: Routledge.
- Stafford, K. P. (2003). Assessment of competence to stand trial. In I. B. Weiner (Series Ed.) & A. M. Goldstein (Vol. Ed.), *Handbook of psychology*. Vol. 11: *Forensic psychology* (pp. 359–380). Hoboken, NJ: Wiley.
- Stein, D. D. (2009). Personal injury: Consultation, evaluation, and the expert witness. In S. F. Bucky, J. E. Callan, & G. Stricker (Eds.), *Ethical and legal issues for mental health professionals in forensic settings* (pp. 21–38). New York: Routledge.

- Stokes, J., Pogge, D., Powell-Lunder, J., Ward, A., Bilginer, L., & DeLuca, V. (2003). The Rorschach Ego Impairment Index: Prediction of treatment outcome in a child psychiatric population. *Journal of Personality Assessment*, 81, 11–19.
- Viglione, D. J. (2002). *Rorschach coding solutions: A reference guide for the Comprehensive System*. San Diego, CA: Author.
- Viglione, D. J., & Hilsenroth, M. J. (2001). The Rorschach: Facts, fictions, and future. *Psychological Assessment*, 13, 452–471.
- Viglione, D. J., & Meyer, G. J. (2008). An overview of Rorschach psychometrics for forensic practice. In C. B. Gacono & F. B. Evans (Eds.), *The handbook of forensic Rorschach assessment* (pp. 21–54). New York: Routledge.
- Viglione, D. J., & Taylor, N. (2003). Empirical support for interrater reliability of Rorschach Comprehensive System coding. *Journal of Clinical Psychology*, 59, 111–121.
- Walfish, S. (2006). Conducting personal injury evaluations. In I. B. Weiner & A. K. Hess (Eds.), *Handbook of forensic psychology* (3rd ed., pp. 124–139). Hoboken, NJ: John Wiley.
- Weiner, I. B. (2000). Using the Rorschach properly in practice and research. *Journal of Clinical Psychology*, 56, 435–438.
- Weiner, I. B. (2003a). *Principles of Rorschach interpretation* (2nd ed.). Mahwah, NJ: Lawrence Erlbaum.
- Weiner, I. B. (2003b). Prediction and postdiction in clinical decision making. *Clinical Psychology: Science and Practice*, 10, 335–338.
- Weiner, I. B. (2004a). Rorschach assessment: Current status. In M. Hersen (Series Ed.) & M. J. Hilsenroth & D. L. Segal (Vol. Eds.), *Comprehensive handbook of psychological assessment*. Vol. 2: *Personality assessment* (pp. 343–355). Hoboken, NJ: John Wiley.
- Weiner, I. B. (2004b). Rorschach Inkblot Method. In M. Maruish (Ed.), *The use of psychological testing for treatment planning and outcome assessment* (3rd ed., Vol. 3, pp. 553–588). Mahwah, NJ: Lawrence Erlbaum.
- Weiner, I. B. (2005a). Integrative personality assessment with self-report and performance-based measures. In S. Strack (Ed.), *Personality and psychopathology* (pp. 317–331). Hoboken, NJ: John Wiley.
- Weiner, I. B. (2005b). Rorschach assessment in child custody cases. *Journal of Child Custody*, 2, 99–120.
- Weiner, I. B. (2008). Presenting and defending Rorschach testimony. In C. B. Gacono & F. B. Evans (Eds.), *The handbook of forensic Rorschach psychology* (pp. 121–137). New York: Routledge.
- Weiner, I. B., & Exner, J. E. Jr. (1991). Rorschach changes in long-term and short-term psychotherapy. *Journal of Personality Assessment*, 56, 453–465.
- Weiner, I. B., & Greene, R. L. (2008). *Handbook of personality assessment*. New York: John Wiley.
- Weiner, I. B., & Meyer, G. J. (2009). Personality assessment with the Rorschach Inkblot Method. In J. N. Butcher (Ed.), *Oxford handbook of personality assessment* (pp. 277–298). New York: Oxford University Press.
- Wood, J. M., & Lilienfeld, S. O. (1999). The Rorschach Inkblot Test: A case of overstatement. *Assessment*, 6, 341–349.
- Wood, J. M., Nezworski, M. T., Garb, H., & Lilienfeld, S. O. (2001). The misperception of psychopathology: Problems with the norms of the Comprehensive System. *Clinical Psychology: Science and Practice*, 8, 350–373.
- Zapf, P. A., Golding, S. L., & Roesch, R. (2006). Criminal responsibility and the insanity defense. In I. B. Weiner & A. K. Hess (Eds.), *Handbook of forensic psychology* (3rd ed., pp. 331–363). New York: John Wiley.
- Zapf, P. A., & Roesch, R. (2006). Competency to stand trial: A guide for evaluators. In I. B. Weiner & A. K. Hess (Eds.), *Handbook of forensic psychology* (3rd ed., pp. 305–331). New York: John Wiley.

8

THE PSYCHOPATHY CHECKLIST–REVISED

Forensic Applications and Limitations

Robert D. Hare, Pamela J. Black, and Zach Walsh

In 1941, Hervey Cleckley described a subgroup of patients who appeared to be sane and yet lacked remorse and empathy, and were very impulsive, deceptive, and manipulative. He suggested that these individuals suffered from “emotional poverty,” and described in detail a number of symptoms, subsequently operationalized for research purposes in the Hare Psychopathy Checklist (PCL; Hare, 1980). Many hundreds of studies using its successor, the Psychopathy Checklist–Revised (PCL–R; Hare, 1991, 2003), have established its position as the international standard for the assessment of psychopathy in research and in clinical and forensic practice (see, for example, Felthous & Sass, 2007; Gacono, 2000; Hare & Neumann, 2008, 2009; Hervé & Yuille, 2007; Patrick, 2005). The *Buros Mental Measurements Yearbook* described the PCL–R as “state of the art” (Fulero, 1995) and as “the gold standard for the assessment of psychopathy” (Acheson, 2005). In many jurisdictions, it is part of the “best practices” protocols (e.g., Khirya, Weaver, & Maden, 2009).

The PCL–R describes psychopathy as a constellation of interpersonal, affective, lifestyle, and antisocial traits and behaviors. Psychopaths are grandiose, deceptive, dominant, superficial, manipulative, affectively shallow, unable to form strong emotional bonds with others, and lacking in empathy, guilt, or remorse. They also are irresponsible and impulsive, and tend to ignore or violate social conventions and mores. A common genetic factor appears to account for substantial variance in these psychopathy domains (Larsson, Viding, & Plomin, 2008; Viding, Frick, & Plomin, 2007), providing evidence that they are part of the superordinate construct of psychopathy (Neumann, Hare, & Newman, 2007; Neumann, Vitacco, Hare, & Wupperman, 2005). As will be discussed later in the chapter, there is converging evidence for the strong association between psychopathy and serious repetitive crime, violent behavior and recidivism, instrumental and “cold-blooded” violence, and poor treatment prognosis.

The purpose of this chapter is to discuss the development, psychometric properties, strengths, and limitations of the PCL–R (Hare, 1991, 2003) in forensic settings. More general and detailed accounts of the role played by the PCL–R and its derivatives in basic and applied research on psychopathy—including cognitive/affective neuroscience, behavioral genetics and developmental psychopathology, treatment and management options, extensions to the community and to the corporate world, issues of legal culpability, and its explication in terms of general personality theory—are available elsewhere (e.g., Babiak & Hare, 2006; Babiak, Neumann, & Hare, 2010; Felthous & Sass, 2007; Frick & White, 2008; Gacono, 2000; Gregory et al., 2012;

Häkkinen-Nyholm & Nyholm, 2012; Harenski, Kiehl, & Hare, 2010; Hervé & Yuille, 2007; Kiehl, 2006; Koenigs, Baskin-Sommers, Zeier, & Newman, 2011; Lake, Baskin-Sommers, Li, Curtin, & Newman, 2011; Ly et al., 2012; Malatesti & McMillan, 2010; Mathieu, Hare, Jones, Babiak, & Hare, 2012; Newman, Hiatt, & MacCoon, in press; Olver & Wong, 2011; Patrick, 2005; Samuel, Lynam, Widiger, & Ball, 2012; Viding, Larsson, & Jones, 2008).

Description

The PCL-R (Hare, 1991, 2003) is a 20-item construct rating scale widely used in research, clinical, and forensic settings for the assessment of psychopathy in adults. The scale items measure personality traits and behaviors related to traditional conceptions of psychopathy (Berrios, 1996; Cleckley, 1976; Hare & Cox, 1978). Each item is rated on a three-point scale (0, 1, or 2) according to the extent to which an individual exhibits the trait or behavior being rated. The Total score varies from 0 to 40, reflecting the degree to which the individual matches the prototypical psychopath. This score can also be used to classify individuals for research purposes as psychopathic. Typically, a cut score of 30 is used, although other cut scores have been used periodically in research contexts (Hare, 2003; Hare & Neumann, 2008, 2010a). In the 1991 Manual this score appeared to have the best diagnostic efficiency with respect to clinical assessments of psychopathy in use prior to the inception of the precursor of the PCL-R, the PCL (Hare, 1980). A score of 30 was approximately 1 standard deviation above the mean for pooled samples of offenders. The use of cut scores will be discussed in more detail later in the chapter.

The standard procedure for administration of the PCL-R involves a semi-structured interview and a review of available file and collateral information (Hare, 2003). The interview typically covers education, employment and family background, relationships, substance use, and antisocial behaviors from adolescence on. It usually takes between 90 and 120 minutes, and may be spread over several sessions, allowing for a more representative sample of the individual's interactional style.

The interview itself has several purposes. First, certain historical information about the individual is required for scoring PCL-R items. Second, the interview often provides clues as to the individual's interpersonal style. It also gives the assessor a chance to evaluate consistency (and deception) within the interview and between the interview and collateral information. Finally, the use of an interview allows probes and challenges when further information is required, or inconsistencies are encountered. The latter purpose is facilitated if the file review is conducted prior to the interview, giving the interviewer bases for challenging or probing.

The second part of the assessment involves reviewing any collateral and/or file information that is available; this is a mandatory aspect of the assessment. The use of collateral information allows the rater to evaluate the truth and reliability of the information given in the interview, determine the representativeness of the interactional style during the interview, and gain access to primary data for scoring several of the PCL-R items. A collateral review typically takes about 60 minutes, but can be longer if file information is extensive and detailed.

Available collateral information will depend upon the setting in which the assessment is conducted. In correctional settings, files typically contain a variety of official records that may be helpful in scoring the items of the PCL-R, and usually are adequate for collateral review (Hare, 2003). In forensic psychiatric and pre-trial settings, raters may have access to official reports, but may also be able to examine reports of interviews with family members, friends, and employers, as well as the results of medical and psychological assessments. When dealing with non-offenders, it may be difficult to obtain sufficient collateral information, although there may be methods available for accessing some useful information, such as telephone interviews, work records, peer ratings,

and so forth. Regardless of the context in which the assessment is conducted, information should come from as many different sources as possible. Scores for items should not be calculated unless there is adequate collateral information.

History of the Instrument's Development

The PCL

The PCL is a result of concerns about assessment procedures available in the late 1970s, including clinical diagnosis and the use of self-report instruments, which did not demonstrate adequate reliability or validity (Hare, 1980; 2003; Hare & Neumann, 2010a). The first step in its development involved psychometric analyses of the features of psychopathy described by Cleckley (1976), along with a compilation of traits and behaviors Hare and his colleagues found useful in the assessment of psychopathy in a prison setting. This process resulted in a pool of more than 100 items. Those that were overly redundant, or would be next to impossible to rate reliably, were eliminated from the pool. Preliminary scoring criteria were developed for the remaining items. Items were scored by two investigators who used both interview and file information. Each item was scored on a three-point ordinal scale, as previously described. Analyses were conducted to ascertain which items were most psychometrically sound and were able to discriminate between inmates with low and high (seven-point) global ratings of psychopathy. After this procedure, only 22 items remained. Hare and Frazelle (1980) disseminated a mimeographed manual for other investigators to use. Some of the items were intended to measure complex behaviors and traits, and scoring these items required the use of clinical judgment. In spite of this, the items had acceptable interrater reliabilities (Hare, 1980). The Total scores, which could range from 0 to 44, also displayed high interrater reliability ($r = .93$) and internal consistency (Cronbach's $\alpha = .88$). Total scores were highly correlated with global ratings of psychopathy ($r = .83$). Numerous studies have indicated that the PCL was both reliable and valid (see Hare, 2003, for a review).

Revision: The PCL-R

Comments and concerns from fellow researchers indicated that it was important to make a number of improvements to the PCL (Hare, 2003). Two items were deleted because of scoring difficulties ("previous diagnosis as a psychopath or similar" and "drugs or alcohol not a direct cause of anti-social behavior"), and one item was altered to be more general: item 6 ("irresponsible behavior as a parent") was deemed to be too specific, and was changed to "irresponsibility" in general. Further, item content and scoring procedures were described in greater detail. Finally, the procedure for dealing with inadequate information was revised. Originally, a score of 1 could be assigned where there was not enough evidence or information to score an item with confidence. For the PCL-R, such an item would be omitted and the Total score would be prorated. A draft version of the PCL-R was created, based on these revisions (Hare, 1985), and was circulated throughout the research community. Subsequently, the draft was fine-tuned and the scoring criteria clarified in order to make the manual (and the instrument) easier for other investigators to use. These changes did not affect the actual scores assigned to an item. A formal manual was published as the *Hare Psychopathy Checklist-Revised* (Hare, 1991).

Although some commentators were concerned that formal measures of low trait anxiety and trait fearlessness were not included in the list of PCL-R items, recent research clearly indicates that the current items adequately reflect these two traits (Neumann, Hare, & Johansson, 2012).

The PCL-R Second Edition (2003)

Although no changes were made in the wording or scoring of the PCL-R items, a revision to the manual was deemed necessary for several reasons. First, Hare sought to minimize the misuse of the PCL-R, especially where it is used to guide adjudication and treatment decisions. One of the most important requirements for proper use of the PCL-R is familiarity with the current literature. However, it was apparent that many of those who produced psychological reports for the criminal justice system or who testified in court relied primarily on material published in the 1991 manual. The inclusion of more recent research in the 2003 manual was intended to provide users with more information about the PCL-R assessment of psychopathy. Nonetheless, it is still extremely important for users to keep abreast of the current literature, especially with regard to its implications for minority and legal issues. An up-to-date list of references now is available at <http://www.hare.org/references/>.

The second reason for the revision was the fact that a large amount of empirical data had been generated since publication of the original manual. The explosion in research provided a large amount of data for establishing comparison tables and for addressing issues concerning the factor structure, reliability, validity, and generalizability of the PCL-R. Psychometric analyses were conducted on 10,896 North American and European offenders and forensic patients, and are fully detailed in the manual (Hare, 2003).

Factor Structure, Reliability, and Generalizability

Factor Structure

Exploratory factor analyses of the data sets used in the 1991 manual yielded two correlated factors. Factor 1 reflected interpersonal and affective features (Items 1, 2, 4, 5, 6, 7, 8, and 16), whereas Factor 2 reflected social deviance features (Items 3, 9, 10, 12, 13, 14, 15, 18, 19). Items 11, 17, and 20 did not load on either of the factors (see F1 and F2 in Table 8.1). This two-factor structure formed the early basis for exploring the differential correlates of the components of psychopathy.

Hare (1991) described psychopathy in terms of a characteristic pattern of three dimensions: interpersonal, affective, and behavioral. Cooke and Michie (2001) performed a confirmatory factor analysis on a *selected* set of 13 PCL-R items and reported that the set was underpinned by three correlated factors reflecting the domains described by Hare (though with different labels; see Factors 1, 2, and 3, respectively, in Table 8.1). Cooke and Michie (2001) argued that excluded items were not essential to the psychopathy construct because they measured antisociality. However, the procedures used to select the items and to conduct the analyses can be criticized on statistical and conceptual grounds (see Hare, 2003; Hare & Neumann, 2005; Hill, Neumann, & Rogers, 2004; Vitacco, Neumann, & Jackson, 2005). Among other things, the decision to exclude items related to antisociality was based on subjective and inconsistent procedures. Cooke and Michie's (2001) assertion that Cleckley and other clinicians did not include antisociality in their descriptions of the essential features of psychopathy is simply incorrect. Hare and Neumann (2008) argued that antisociality was featured prominently in traditional clinical conceptions of psychopathy, and that it is difficult to understand how the defining traits of the construct could be measured without reference to antisocial behaviors. Similarly, in their critical review of the Psychopathic Personality Inventory-Revised (PPI-R; Lilienfeld & Widows, 2005), Lynam and Miller (2012) concluded that "Antisocial behavior [ASB] plays a clear and prominent role in psychopathy. . . . In fact, if there is an essential behavioral feature in common across the conceptualizations [of psychopathy], it is the presence of ASB. Any description of psychopathy is incomplete without ASB."

TABLE 8.1 Items and Factors in the Hare PCL Scales

<i>PCL-R</i>	<i>PCL:YV</i>	<i>PCL:SV</i>
F1	P1	
<i>Interpersonal</i>	<i>Interpersonal</i>	<i>Interpersonal</i>
1. Glibness/superficial charm	1. Impression management	1. Superficial
2. Grandiose sense of self-worth	2. Grandiose sense of self-worth	2. Grandiose
4. Pathological lying	4. Pathological lying	3. Deceitful
5. Conning/manipulative	5. Manipulation for personal gain	
<i>Affective</i>	<i>Affective</i>	<i>Affective</i>
6. Lack of remorse or guilt	6. Lack of remorse	4. Lacks remorse
7. Shallow affect	7. Shallow affect	5. Lacks empathy
8. Callous/lack of empathy	8. Callous/lack of empathy	6. Doesn't accept responsibility
16. Failure to accept responsibility	16. Failure to accept responsibility	
F2	P2	
<i>Lifestyle</i>	<i>Behavioral</i>	<i>Lifestyle</i>
3. Need for stimulation	3. Stimulation-seeking	7. Impulsive
9. Parasitic lifestyle	9. Parasitic orientation	9. Lacks goals
13. Lack of realistic, long-term goals	13. Lack of goals	10. Irresponsibility
14. Impulsivity	14. Impulsivity	
15. Irresponsibility	15. Irresponsibility	
<i>Antisocial</i>	<i>Antisocial</i>	<i>Antisocial</i>
10. Poor behavioral controls	10. Poor anger control	8. Poor behavioral controls
12. Early behavioral problems	12. Early behavior problems	11. Adolescent antisocial behavior
18. Juvenile delinquency	18. Serious criminal behavior	12. Adult antisocial behavior
19. Revocation of conditional release	19. Serious violations of release	
20. Criminal versatility	20. Criminal versatility	

Note:

The PCL-R, PCL: YV, and PCL: SV items are from Hare (1991; 2003), Forth, Kosson, and Hare (2003), and Hart, Cox, and Hare (1995), respectively. Note that the item titles cannot be scored without reference to the formal criteria contained in the published manuals. PCL-R items 11, Promiscuous sexual behavior, and 17, Many short-term marital relationships, contribute to the Total score but do not load on any factors. PCL: YV items 11, Impersonal sexual behavior, and 17, Unstable interpersonal relationships, contribute to the Total score but do not load on any factor. F1 and F2 are the original PCL-R factors, but with the addition of item 20. P1 and P2 are Parts 1 and 2 described in the PCL: SV Manual.

Recent exploratory and confirmatory factor analyses clearly indicate that a four-factor model of psychopathy that includes antisociality is statistically and conceptually defensible, not only for the PCL-R but for its derivatives, the PCL:SV and the PCL:YV (e.g., Forth, Kosson, & Hare, 2003; Hare, 2003; Hare & Neumann, 2008; Kosson et al., 2012; Mokros, Neumann, et al., 2011; Neumann et al., 2012; Neumann, Hare, & Newman, 2007; Vitacco, Rogers, Neumann, Harrison, & Vincent, 2005; Vitacco, Neumann, & Jackson, 2005; Žukauskienė, Laurinavičius, & Čėsniënė, 2010). Note that in these analyses, the items are properly treated as ordinal rather than as interval scales. These four first-order factors are listed in Table 8.1. The factors are intercorrelated, suggesting that they are indicators of a superordinate factor of psychopathy (Neumann et al., 2007). Further, depending on the pattern of intercorrelations among the four first-order factors, it is possible to model them to represent a higher-order two-factor model (see Hare & Neumann,

2008), with the Interpersonal and Affective factors loading onto a higher-order *Interpersonal/Affective* factor (e.g., a higher-order “F1” factor), and the Lifestyle and Antisocial factors loading onto a higher-order *Lifestyle/Antisocial* factor (e.g., a higher-order “F2” factor). However, this higher-order two-factor model, while conceptually similar to the traditional two-factor model, is not statistically isomorphic with the traditional two-factor model (i.e., where items 1, 2, 4, 5, 6, 7, 8, 16 loaded directly onto a first-order factor [F1] and items 3, 9, 10, 12, 13, 14, 15, 18, 19 loaded directly onto another first-order factor [F2]). As in the original factor structure, “Promiscuous sexual behavior” and “Many short-term marital relationships” contribute to the total PCL-R score but do not load on any factor. The terms Factor 1 (F1) and Factor 2 (F2) in this chapter refer to the broad PCL-R factors. The first-order factors in the four-factor model are referred to by name (i.e., Interpersonal, Affective, Lifestyle, Antisocial). Figures 8.1 and 8.2 depict the four-factor and two-factor models of the PCL-R, respectively. The advantage of the four correlated (first-order) factor model is that it provides four clearly unidimensional factors, which no doubt aid in interpretation when they are associated with a range of external correlates.

The most reliable and important information for the clinician is the Total score, to which all items contribute (Bolt, Hare, Vitale, & Newman, 2004). Inspection of the Interpersonal, Affective, Lifestyle, and Antisocial dimensions may help the clinician with the interpretation of an individual's PCL-R protocol. The publisher of the PCL-R Second Edition, Multi-Health Systems, provides QuikScore Forms that facilitate calculation of Total and factor scores.

Reliability

Many hundreds of studies confirm that the PCL-R is a reliable measure of the psychopathy construct in research and applied settings (for a full review, see Hare, 2003). Reliabilities for individual items are moderate and acceptable, with intra-class correlation (ICC) coefficients for male offenders varying from .41 for single ratings to .57 for averaged ratings. Similar values have been obtained with female offenders. Interrater reliability for Total scores based on standard and file reviews is high. The ICC is above .85 for a single rating and above .90 for the average of two ratings. The reliabilities of the PCL-R factors are adequate for research and clinical purposes. Finally, the standard error of measurement (SEM) of the PCL-R for the Total score is approximately 3 for a single rating and 2 for the average of two ratings. The SEM is the standard deviation of observed scores if the true score is held constant. This means that if 100 trained raters assessed the same subject at the same time, about 68% of the scores would fall within ± 1 SEM unit of the subject's obtained Total score, and about 95% would fall within ± 2 SEM units. The SEM can be used as an estimate of “the reasonable limits of the true score for (an individual) with any given obtained score” (Anastasi, 1982).

Most of the evidence on the reliability of the PCL-R is based on ratings made more or less concurrently with one another. Looman and Ismail (under review) obtained PCL-R scores for 175 male sex offenders first assessed at a Canadian federal reception center and again at a federal regional treatment center 2 to 3 years later. Mean Total scores for these two sets of independent ratings were virtually the same, and the ICC for the ratings was .90. Looman and Ismail concluded that PCL-R scores can be as reliable in applied settings as those in research contexts. They emphasize the importance of training, consultation, and following the guidelines of the manual.

The interrater reliability of the PCL-R may be lower in an adversarial context than when raters operate independently of allegiance pressures (see “Potential for Misuse,” p. 248).

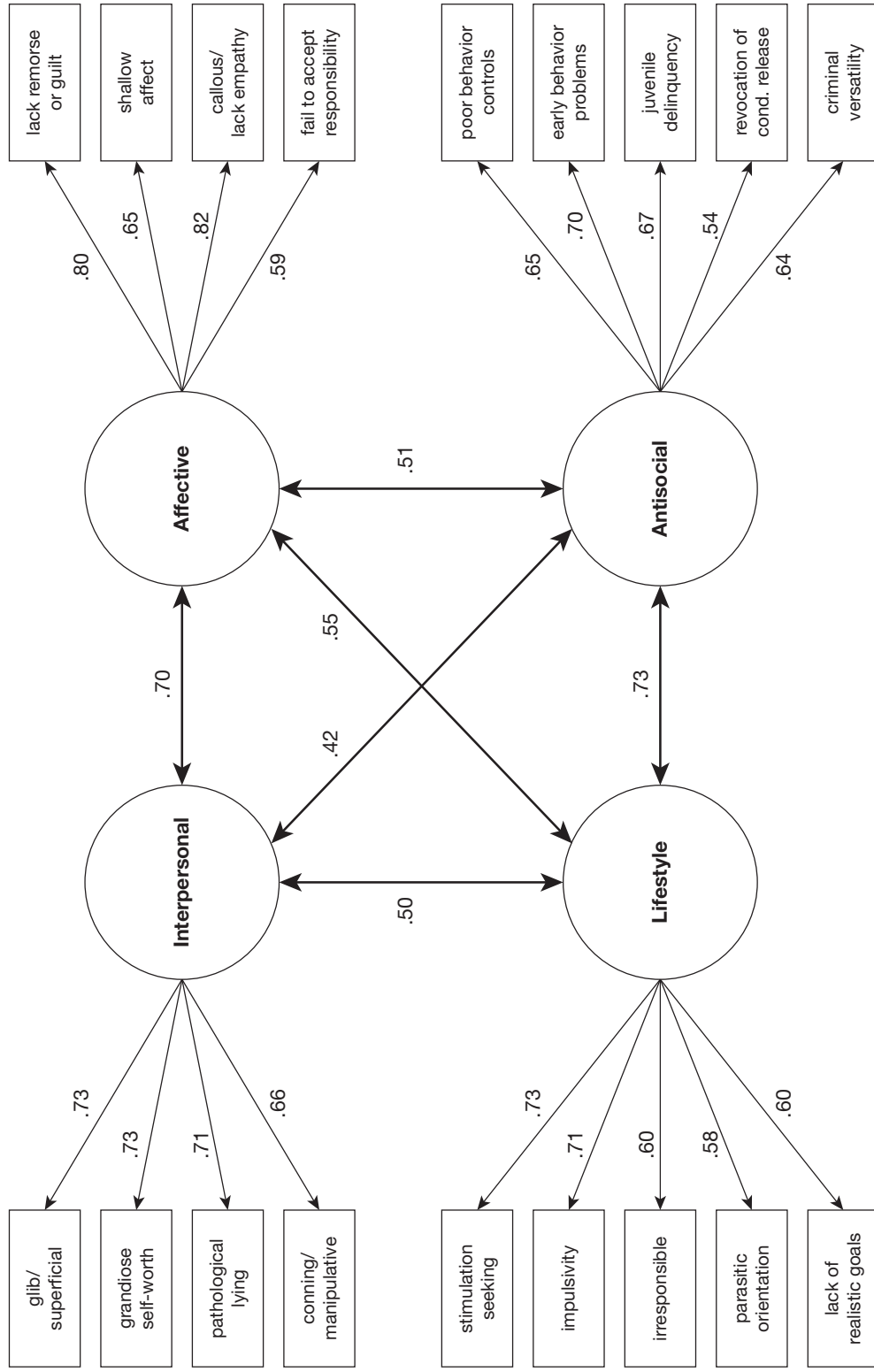


FIGURE 8.1 Four-Factor Structure of the PCL-R.

From Neumann, Hare, & Newman (2007). The super-ordinate nature of the Psychopathy Checklist-Revised. *Journal of Personality Disorders*, 21(2), 102–117. Reprinted with permission.

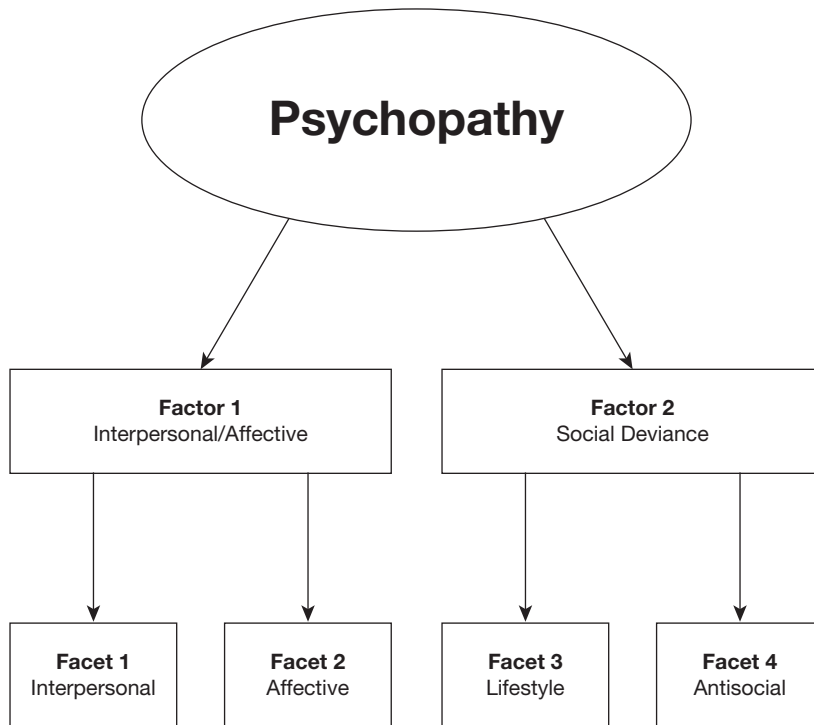


FIGURE 8.2 Two-Factor Structure of the PCL-R.

Generalizability

Although several of the samples described in the Second Edition of the Manual are representative of the forensic populations from which they were obtained, many were “samples of convenience” provided by researchers and institutions. However, the distributions of PCL-R scores in the samples that *were* representative were similar to those obtained from convenience samples. Moreover, the means, standard deviations, and other psychometric properties of samples described in the published literature generally are consistent across samples and similar to those in the Manual. Nonetheless, because the samples were not randomly selected from forensic populations, the percentile and *T*-score tables provided in the manual are meant only to be frameworks or guidelines (“comparison tables”) for evaluating the scores of inmates or patients similar to those for whom the tables were constructed.

Although PCL-R scores are based on samples that appear to be representative, possible moderating variables need to be mentioned and discussed. A number of variables could influence the scoring and interpretation of PCL-R scores, including type of sample, age, gender, method of assessment (standard vs. file review alone), culture, and ethnic background.

Type of Sample

Samples of sex offenders, substance-dependent offenders, and non-criminals are described in the Manual. Compared with offenders in general, child molesters score lower, and rapists and rapists/molesters higher, on the PCL-R. The differences between sex offenders and general offenders were not large enough to warrant the construction of separate percentile and *T*-score tables.

There is a considerable amount of research on the association between the PCL–R and substance use in prison inmates and forensic psychiatric inpatients and outpatients (Berger, Rotermund, Vieth, & Hohnhorst, 2012; Burns et al., 2011; Gustavson et al., 2007; Hare, 2003; Hicks, Vaidyanathan, & Patrick, 2010; Kennealy, Hicks, & Patrick, 2007; Pham & Saloppé, 2010; Richards, Casey, & Lucente, 2003; Schiffer et al., 2011). However, substance use in these populations is so extensive that there was little to be gained by providing in the Manual separate descriptive statistics or percentile and *T*-score tables for those who do and do not use and abuse alcohol and drugs. In general, substance use is strongly associated with PCL–R scores, though more so with Factor 2 than with Factor 1 (Hemphill, Hart, & Hare, 1994; Kennealy et al., 2007; Leistico, Salekin, DeCoster, & Rogers, 2008; Walsh, Allen & Kosson, 2007).

There has been little systematic research examining PCL–R scores in non-criminals, mostly because of problems in obtaining adequate collateral information to score the items with confidence. Weiler and Widom (1996) obtained a mean PCL–R score of 6.8 ($SD = 5.9$) for a sample of 489 adults who had been the control group for a study of abused children. Babiak and colleagues (2010) obtained a mean PCL–R score of 3.6 ($SD = 7.4$) for a sample of 203 corporate professionals selected by their companies to participate in management development programs. These studies, along with several other small-sample studies (Hare, 2003), suggest that most individuals in the general population would obtain a score of less than 4 or 5. A new instrument, the B-Scan 360 (Babiak & Hare, in preparation), may facilitate research in corporate settings by allowing personnel to rate others on psychopathic features. It has a four-factor structure consistent with the PCL–R (Mathieu et al., 2012). The 12-item PCL:SV is conceptually and empirically equivalent to the 20-item PCL–R (Cooke, Michie, Hart, & Hare, 1999; Guy & Douglas, 2006), making it possible to convert scores from one to the other. For example, a PCL:SV community sample score of 3 would be approximately equivalent to a PCL–R score of 5. The PCL:SV has been employed in a wide variety of samples, including college students (e.g. Forth, Brown Hart, & Hare, 1996; Hart et al., 1995), community samples (e.g., Coid, Yang, Ullrich, Roberts, & Hare, 2009), and civil psychiatric patients (e.g., Steadman et al., 2000; Swogger, Walsh, Homaifar, Caine, & Conner, 2012). PCL:SV scores in these samples generally are very low. For example, in the MacArthur Violence Risk Assessment Study community samples (Steadman et al., 2000) the mean PCL:SV score was 3.6 ($SD = 3.8$) for 197 males, and 2.2 ($SD = 3.2$) for 318 females. Similar scores were obtained in a large representative sample of the UK population (Coid et al., 2009). On the other hand, the mean PCL:SV score for a sample of outpatients with chronic social and behavioral problems was very high: 19.5 ($SD = 4.2$) for 19 males and 17.2 ($SD = 3.5$) for 31 females (de Oliveira-Souza, Ignácio, Moll, & Hare, 2008). Interestingly, patients from this sample displayed much the same structural brain anomalies as typically found with psychopathic offenders (de Oliveira-Souza et al., 2008).

Age

Cross-sectional analyses indicate that PCL–R Total scores obtained with the standard procedure are only weakly associated with the age of assessment among male offenders and male forensic psychiatric patients (Hare, 2003). Factor 1 scores in particular appear to be very stable, while Factor 2 scores decline slightly with age of assessment. For female offenders, there is a slight decrease in PCL–R Total, Factor 1, and Factor 2 scores with age of assessment. Because the association between age and PCL–R scores is weak, comparison tables based on age of assessment were not composed.

Gender

The PCL-R is a useful tool for assessing psychopathy in female offenders (Hare, 2003), and the descriptive properties and correlates are similar to those found with male offenders (e.g., Coid et al., 2009; Czar, Dahlen, Bullock, & Nicholson, 2011; Hare & Neumann, 2009; Kennealy et al., 2007; Leistico et al., 2008; Richards, Casey, & Lucente, 2003; Richards, Casey, Lucente, & Kafami, 2003; Vitale, Smith, Brinkley, & Newman, 2002; Warren et al., 2003). Mean PCL-R scores typically are several points lower in female than in male offenders (19.0 vs. 22.1). Separate percentile and T-score tables therefore are provided for female and male offenders (Hare, 2003).

Item response theory (IRT) provides a mathematical expression of the relationship between a score on an individual item or group of items (e.g., Total score) and the underlying construct or latent trait of psychopathy. IRT analyses (Bolt et al., 2004) indicate that PCL-R Total scores are related to the construct of psychopathy in much the same way in female as in male offenders, particularly at mid- and upper ranges of PCL-R scores. For example, a score of 28 reflects approximately the same level of psychopathy in female offenders as does a score of 30 in male offenders. IRT analysis indicates that not all PCL-R items function the same way in males and females, and it appears that the expression of some psychopathic features is sex related, in large part because of the influence of cultural and biological factors that influence gender-role expectations and behaviors.

Method of Assessment

The standard (and recommended) assessment procedure for the PCL-R involves a semi-structured interview and a review of collateral information (Hare, 2003). Assessments that do not include an interview typically yield lower scores than do standard assessments. The Second Edition of the Manual lists the mean score for male offenders assessed with the standard procedure as 22.1 ($SD = 7.9$) and for those assessed from file reviews as 16.5 ($SD = 8.3$). Part of this difference may be due to sampling issues, with many of the file reviews being based on sex offenders. Although IRT analyses indicate that a given PCL-R score obtained with either method appears to reflect much the same level of psychopathy (Bolt et al., 2004), the interview is important for scoring many of the interpersonal and affective items, and the standard procedure should be used when possible.

Culture

In recent years, the PCL-R has been used extensively in many countries outside of North America. PCL-R Total scores for male offenders and forensic psychiatric patients are somewhat lower in the United Kingdom and in several other European countries than in North America (Coid et al., 2009; Hare, 2003). The Second Edition provides separate percentile and T-score tables for the United Kingdom (samples from other countries were not large enough for this purpose). IRT analyses indicate that some of the items may not function the same in North America as in Britain (Hare, 2003) and in several other European countries (e.g. Sweden, Denmark, Germany, Belgium, Norway, Spain; Bolt, Hare, & Neumann, 2007; Cooke, Michie, Hart, & Clarke, 2005; Mokros, Statdland, Osterheider, & Nedopil, 2010). That is, there are cultural differences in the extent to which some items are discriminating of psychopathy. However, IRT analyses indicate that in the mid- to upper ranges a given Total score appears to have much the same meaning in Britain (Hare, 2003) and in the other European countries as in North America (Bolt et al., 2007; Hare & Neumann, 2009). That is, there is considerable metric equivalence of PCL-R scores (most notably for the Interpersonal and Affective factors) across countries and cultures. In addition to IRT

analyses, confirmatory factor analyses support the four-factor model of psychopathy in many different countries, including Sweden, Germany, Lithuania, Chile, Brazil, Korea, Japan, Spain, and the Netherlands (see Hare, Neumann, & Widiger, 2012). Nonetheless, more cross-cultural research on the PCL–R and its derivatives is needed.

Ethnic Background

Early data on the PCL–R were obtained predominantly from Caucasian samples. However, more recent research has begun to assess the applicability of the PCL–R across ethnic groups. While Lynn (2002) asserted that certain races or ethnic groups are more psychopathic than others, the scientific evidence, reviewed by Zuckerman (2003), suggests otherwise. In the Second Edition of the PCL–R Manual, the PCL–R was assessed in a large number of African American offenders and in smaller samples of Native American and Hispanic American offenders. The PCL–R scores did not appear to be influenced by ethnicity. Similar results were obtained in meta-analyses by Skeem, Edens, Camp, and Colwell (2004), and by McCoy and Edens (2006). Further, IRT analyses suggest that the PCL–R scores in the “high range” (above 30) have metric equivalence for Caucasian male and female offenders, European male offenders, Latino offenders, and African-American male offenders (Cooke, Kosson, & Michie, 2001; Jackson, Neumann, & Vitacco, 2007; Neumann & Hare, 2009; Skeem et al., 2004; Sullivan, Abramowitz, Lopez, & Kosson, 2006; Vachon, Lynam, Loeber, & Stouthamer-Loeber, 2012). Olver, Neumann, Wong, and Hare (2012) found that the factor structure and predictive correlates of the PCL–R were much the same in Canadian Aboriginal offenders as in non-Aboriginal offenders.

A recent study of European American and African American males that tested the cross-ethnic consistency of many theoretically important child and adult correlates of adult psychopathy reported that ethnic differences were scarce and small (Vachon et al., 2012). However, several issues remain to be addressed to more definitively determine the cross-ethnic equivalence of the psychopathy construct. For example, there are ethnic and cultural factors that may influence the expression of psychopathy, particularly as reflected in Factor 2 items. In addition, research indicates that some of the “deficits” observed in laboratory research with Caucasian offenders are not observed in African Americans (Doninger & Kosson, 2001; Lorenz & Newman, 2002; Newman & Schmitt, 1998). It is uncertain whether these differences are due to ethnic factors related to psychopathy or to the tasks involved in the research. More research is required to understand the nature and implications of the construct of psychopathy in different ethnic groups. At present, we have little information about potential rater bias when the rater and the individual being rated are from different ethnic backgrounds. Until more research is conducted to truly understand the way in which the construct of psychopathy varies across ethnicities, caution should be used when generalizing the existing psychopathy literature to non-Caucasian offenders.

Intellectually Disabled People

Although there are no exclusion criteria with respect to the populations for which the PCL–R and PCL:SV may be used, some groups of offenders and patients may pose scoring problems. Morrissey and her colleagues (Morrissey, 2003; Morrissey et al., 2005) have provided guidelines for the use of the PCL–R with intellectually disabled (ID) offenders. Gray, Fitzgerald, Taylor, MacCulloch, and Snowden (2007) found that the PCL:SV predicted general and violent recidivism at least as well in ID offenders as in other offenders. There is some evidence that the PCL–R is useful for selecting treatment options for ID offenders with intellectual deficiencies (Morrissey, Mooney, Hogue, Lindsay, & Taylor, 2007). Nonetheless, Morrissey and Hollin (2011) recommend that

clinicians exert caution in their use of the PCL-R and PCL:SV with ID offenders, particularly when evaluating risk for recidivism.

Conclusion

The available evidence indicates that the PCL-R and the construct it measures are generalizable across a variety of populations and contexts. Enough data are available for the use of separate percentile and *T*-score comparison tables for North American male offenders, male forensic psychiatric patients, female offenders, and British male offenders. The effects of other variables were either based on a small number of samples or yielded minimal differences. Therefore, separate tables were not deemed necessary or possible at this point.

Validity

Because of the strong association between the PCL-R and its predecessor, the PCL, evidence in support of the validity of the latter also is applicable to the former. Evidence for the validity of the PCL-R is extensive (see Hare, 2003), and comes from research examining its relationship to a variety of other instruments and scales, its association with behavioral data, such as recidivism and violence, and its role in basic laboratory research (the latter is not discussed here; see Hare, 2003).

Correlations with Other Measures

The PCL-R provides the anchor for delineating the nomological network that surrounds it. PCL-R scores are strongly correlated with other conceptually relevant clinical assessment tools, including global ratings of psychopathy (used to assess psychopathy prior to 1980), *DSM-III*, *DSM-III-R*, and *DSM-IV-TR* diagnoses of antisocial personality disorder (ASPD; American Psychiatric Association, 2000), prototypicality ratings of ASPD, and ASPD symptom counts (Hart & Hare, 1989; Hildebrand & de Ruiter, 2004). The PCL-R is also moderately correlated with various self-report measures of psychopathy, including the Psychopathic Personality Inventory and its revision (PPI and PPI-R, respectively; Copestake, Gray, & Snowden, 2011; J. D. Miller & Lynam, 2011), the Interpersonal Measure of Psychopathy (IMP; Zolondek, Lilienfeld, Patrick, & Fowler, 2006), Levenson's Self-Report Psychopathy Scale (SRPS; Brinkley, Schmitt, Smith, & Newman, 2001), and the Self-Report Psychopathy scale (SRP; Hare, 1985) and its most recent version, the SRP-IV (Paulhus, Hemphill, & Hare, in press; Williams, Paulhus, & Hare, 2007). Similarly, PCL-R scores are correlated with relevant scales in omnibus personality inventories, including the Five-Factor Model of Personality (Costa & Widiger, 2002; Decuyper, De Fruyt, & Buschman, 2008; Hicklin & Widiger, 2005; Lynam & Widiger, 2007; Pereira, Huband, & Duggan, 2008). Finally, psychopathy, as measured by the PCL-R, is correlated in a theoretically relevant manner with a wide variety of personality scales and with subscales on omnibus measures of psychopathology, including the Minnesota Multiphasic Personality Inventory (MMPI; Dahlstrom & Welsh, 1960), the Millon Clinical Multiaxial Inventory (MCMI; Millon, 1987) and its revisions (Millon, 1987; Millon, Davis, & Millon, 1997), and the Personality Assessment Inventory (PAI; Morey, 1991).

It is important to note, however, that the presence of significant correlations between the PCL-R and other instruments does not imply that they all measure the same construct (Hare, 1985; J. D. Miller & Lynam, 2011).

Relation with Criminal Behavior

The importance of the PCL–R in the criminal justice system is widely recognized, both by forensic clinicians (e.g., Archer, Buffington-Vollum, Stredny, & Handel, 2006) and by the courts (e.g., Walsh & Walsh, 2006). This is not surprising, given that many of the characteristics important for inhibiting antisocial and violent behavior—empathy, close emotional bonds, fear of punishment, guilt—are lacking or seriously deficient in psychopaths. Moreover, their egocentricity, grandiosity, sense of entitlement, impulsivity, general lack of behavioral inhibitions, and need for power and control constitute what might be described as the perfect prescription for the commission of antisocial and criminal acts (e.g., Porter & Porter, 2007). The PCL–R distinguishes a distinct subgroup within offender populations, a group that is at high risk for criminality across the lifespan (Lindberg et al., 2009; Porter, Birt, & Boer, 2001; Salekin, 2008), engages in a large number and variety of criminal acts (Blackburn & Coid, 1998; Coid & Yang, 2011; Czar et al., 2011; Douglas & Webster, 1999; Porter, ten Brinke, & Wilson, 2009; Reidy, Zeichner, & Seibert, 2011), is predatory in its aggression (Cornell et al., 1996; Declercq, Willemsen, Audenaert, & Verhaeghe, 2012; Woodworth & Porter, 2002), presents management problems within the institution (Hare, Clark, Grann, & Thornton, 2000; Langton, Hogue, Daffern, Mannion, & Howells, 2011; Reimann & Nussbaum, 2011; Vitacco, Gonsalves, Tomony, Smith, & Lishner, 2012; Walters & Heilbrun, 2010), and is more likely to engage in violence after release from an institution than are other offenders (Glover, Nicholson, Hemmati, Bernfeld, & Quinsey, 2002; Hemphill, Hare, & Wong, 1998; Kroner & Mills, 2001; see meta-analyses by Dolan & Doyle, 2000; Gendreau, Goggin, & Smith, 2002; Hemphill & Hare, 2004; Hemphill et al., 1998; Pederson, Kunz, Rasmussen, & Elsass, 2010; Salekin, Rogers, & Sewell, 1996; Wallinius, Nilsson, Hofvander, Anckarsäter, & Stålenheim, 2012; Walters & Heilbrun, 2010) (see also Swogger et al., 2012).

The predictive value of the PCL–R in adults applies not only to male but also to female offenders (Jackson & Richards, 2007; Verona & Vitale, 2006), forensic psychiatric patients, including those with schizophrenia (Dolan & Davies, 2006; Doyle, Dolan, & McGovern, 2002; Heilbrun et al., 1998; Hill et al., 2004; Hill, Rogers & Bickford, 1996; Lincoln & Hodgins, 2008; Rice, Harris, & Cormier, 1992; Tengström, Grann, Långström, & Kullgren, 2000; Tengström, Hodgins, Grann, Långström & Kullgren, 2004; Tengström et al., 2006), offenders with intellectual difficulties (Gray et al., 2007), and civil psychiatric patients (Steadman et al., 2000; Vitacco et al., 2005). Psychopathy also is increasingly being seen as an important factor in explaining domestic violence (Hilton, Harris, Rice, Houghton, & Eke, 2007; Swogger, Walsh, & Kosson, 2007; Walsh et al., 2010). In some cases, the predictive utility of the PCL–R and PCL:SV is at least as good as that of the purpose-built instruments, including those of which they are a part (Dahle, 2006; Dolan & Davies, 2006; Doyle et al., 2002; Edens, Skeem, & Douglas, 2006; Hare, 2003; Kroner, Mills, & Reddon, 2005; Pham, Ducro, Maghem, & Réveillère, 2005; Sjöstedt & Långström, 2002; Tengström, 2001). For example, in the MacArthur Risk Study (Monahan et al., 2001) the Violence Risk Appraisal Guide (VRAG; Quinsey, Harris, Rice, & Cormier, 2006) predicted violence in civil psychiatric patients, but the effect was due entirely to the inclusion in the VRAG of the PCL:SV (Edens, Marcus, Lilienfeld, & Poythress, 2006).

With regard to the cross-ethnic and cross-cultural stability of the association between psychopathy and criminal behavior, a review of European and North American studies reports positive associations between psychopathy and violence across nations (Hare et al., 2000). Comparisons of ethnic groups within nations also indicate a generally consistent pattern of association between psychopathy and violence; prospective and cross-sectional studies of U.S. inmates report equivalent associations with violence among European American and African American males (Walsh & Kosson, 2007; Walsh, Swogger, & Kosson, 2004), and a study that has

directly compared European American, African American, and Hispanic American men reports ethnic similarities in postdictive associations with violence (Sullivan et al., 2006). However, further research is required to more definitively determine the predictive validity of psychopathy among Hispanic Americans.

Psychopathic offenders with deviant sexual interests pose a high risk for sexual crimes (Hildebrand, de Ruiter, & de Vogel, 2004; Mokros, Osterheider, Hucker, & Nitschke, 2011; Porter et al., 2009; Rice & Harris, 1997; Saleh, Malin, Grudzinskas, & Vitacco, 2010; Serin, Mailloux, & Malcolm, 2001). Although PCL-R scores are somewhat lower in forensic psychiatric patients than in offender populations, the predictive power of the instrument is about the same in patients as in offenders (Crocker & Côté, 2009; Gray, McGleish, Timmons, MacCulloch, & Snowden, 2003; Rice et al., 1992; Tengström et al., 2000). The presence of psychopathic traits in forensic psychiatric patients greatly increases their risk for violent recidivism. DeLisi (2009, p. 256) has argued that psychopathy is so pervasive in criminality that it is “the unified theory of crime.”

The PCL-R was designed to measure the psychopathy construct, but because of its ability to predict criminal behavior, it commonly is used as a risk assessment tool, either on its own or as part of actuarial and structured clinical judgment risk scales. These include the Violence Risk Appraisal Guide (VRAG; Quinsey et al., 2006), the Sex Offender Risk Appraisal Guide (SORAG; Quinsey, Rice, & Harris, 1995), the Domestic Violence Risk Appraisal Guide (DVRAG; Hilton et al., 2008), the Historical-Clinical-Risk 20 (HCR-20; Webster, Douglas, Eaves, & Hart, 1997), and the Sexual Violence Risk 20 (SVR-20; Boer, Hart, Kropp, & Webster, 1997). In some cases, the predictive utility of the PCL-R and PCL:SV is at least as good as purpose-built risk instruments, including those of which they are a part. However, they should be used in conjunction with general risk evaluations. In addition to the instruments described above, there is increasing interest in the role of dynamic (changeable) risk factors in risk assessment (Lewis, Olver, & Wong, 2012; Quinsey, Jones, Book, & Barr, 2006).

Some evidence suggests that the predictive power of psychopathy for violence may be partially attributable to interactions between Factor 1 and Factor 2 (Lilienfeld, 1998), such that individuals with elevations on both factors are particularly prone to engage in violence. In an early study of interactive effects, Harpur and Hare (1991) found that interactions between factors were related to frequency of past violent behavior, extent of weapons use and violent and aggressive behavior in prison. Similarly, Kosson, Steuerwald, Forth, and Kirkhart (1997) found interactive factor associations with retrospective reports of fighting. In an explicit examination of interactive effects and violence, Walsh and Kosson (2008) reported factor interactions across two large prospective studies, such that the predictive relationship between Factor 2 and violence was attenuated at lower levels, and accentuated at higher levels, of Factor 1. However, further research is required to determine the robustness of interactive effects, as a recent meta-analysis of factor interactions concluded that that factor-level associations with violence were additive rather than interactive (Kennealy, Skeem, Walters, & Camp, 2010). Vitacco et al. (2012) reported that psychopathy, as measured by the PCL:SV, is a significant risk factor for both instrumental and reactive aggression in a male forensic inpatient sample. Notably, the interpersonal and affective components (Part 1; see Table 8.1) were somewhat more predictive of violence than were the lifestyle/antisocial components (Part 2).

In comparing the predictive validity of the PCL-R and its derivatives across different countries, the differences in the criminal justice systems involved should be considered. In Canada, there is only one criminal code. Offenders sentenced to less than two years serve their time in a provincial facility, whereas those with a sentence of two years or more are confined to a federal institution. Regardless of where they serve their time, it is relatively easy to determine their post-release recidivism because of a national integrated system of record keeping. It is largely because of this ability to access detailed institutional files and to do large-sample studies of post-release behavior

that the PCL–R and major risk assessment tools, such as the VRAG and the HCR–20, were developed in Canada. Many of the other countries that are active in research on psychopathy and risk, such as the United Kingdom, the Scandinavian nations, and other European countries, also have single national criminal codes. By contrast, the American judicial system consists of both federal and state criminal laws. This makes it difficult to do systematic, large-sample post-release research because offenders released from a federal or state prison could be “lost to the system” if they move to a different state.

Response to Treatment

Psychopathic offenders present a challenge to traditional treatments and interventions (Harris & Rice, 2006; Wong & Hare, 2005). Unlike most other offenders, psychopaths may see little wrong with their attitudes and behavior, and thus may participate in treatment only when it is in their best interest to do so, such as when seeking probation or parole. It is therefore not surprising that they typically derive little benefit from traditional prison programs, particularly those aimed at the development of empathy, conscience, and interpersonal skills (Abracen, Looman, & Langton, 2008; Harris & Rice, 2006). Programs that do not take into account the nature of psychopathic offenders, such as their attenuated ability to connect emotionally with others, which often hinders their ability to form a strong therapeutic alliance, are unlikely to be effective (Olver & Wong, 2009). Offenders with high PCL–R scores tend to drop out of treatment early (Berger et al., 2012; Olver, Stockdale, & Wormith, 2011; Olver & Wong, 2011).

There is some evidence that risk management and “harm reduction” programs that involve an integration of relapse-prevention techniques and risk/needs/responsivity principles (Andrews & Bonta, 2003) help to reduce the seriousness of post-release offending (Wong, Gordon, & Gu, 2007). There also is some evidence that therapeutic progress in cognitive-behavioral programs (Doren & Yates, 2008; Langton, Barbaree, Harkins & Peacock, 2006; Olver & Wong, 2009), as well as successful completion of such programs (Caldwell, Skeem, Salekin, & Van Rybroek, 2006; Catchpole & Gretton, 2003; Forth & Book, 2007; Olver & Wong, 2009), may be predictive of reduced recidivism rates among adolescent and adult offenders, including some with many psychopathic features. Thornton and Blud (2007) have discussed the implications of the four-factor model of psychopathy for working with offenders. Yet to be determined is the long-term efficacy of such programs and the extent to which their outcomes can be replicated and generalized to other jurisdictions and populations.

Wong and Hare (2005) have proposed that management of psychopathic offenders should be less concerned with “developing empathy and conscience” or effecting changes in personality than with convincing them that they alone are responsible for their behavior, and that there are more prosocial ways of using their strengths and abilities to satisfy their needs and wants. Similarly, Felthous (2011) has argued that it may be beneficial to target conditions associated with psychopathy, such as impulsive aggression. As part of an intense cognitive-behavior-oriented program for high-risk offenders (many of whom were psychopathic), Lewis, Olver, and Wong (2012) used the Violence Risk Scale (VRS) to target and treat dynamic (changeable) risk factors. They reported that the approach lowered the level of these risk factors for some offenders, and that for some of these offenders there was a significant reduction in post-release violent recidivism.

There is no evidence that treatment programs result in a change in the personality structure of psychopathic individuals. This is hardly surprising, given recent findings from behavioral genetics, developmental psychopathology, neuroscience, and evolutionary psychology that psychopathy is characterized by personality and behavioral propensities that are strongly entrenched and presumably difficult to change (see Book & Quinsey, 2003; Harenski et al., 2010; Harris & Rice,

2006; Larsson et al., 2008; Viding et al., 2008). In this respect, management may be a more appropriate goal than treatment, in the conventional clinical sense of the term.

Forensic Uses

Because of its demonstrated reliability and validity, the PCL-R is widely used in research and applied settings, including the mental health and criminal justice systems. Scoring is standardized, allowing for easy replication of the assessment protocol and communication of results. The comparison tables for the PCL-R are based on large samples (Hare, 2003), and scores on the instrument appear to have much the same explanatory and predictive value in a wide range of groups and contexts.

The number and variety of situations in which the PCL-R is applied are increasing (Walsh & Walsh, 2006). It has been employed in child custody hearings, risk assessments, parole evaluations, decisions about treatment suitability (specifically for violent offenses such as assault, murder, and sex offenses), transfer from juvenile to adult court decisions, civil commitment proceedings and death penalty hearings (in the United States), dangerous and long-term offender hearings (in Canada), and severe dangerous personality disorder evaluations (in the United Kingdom).

Two recent reviews suggest that the frequency with which the PCL-R is used in applied settings in the United States, such as civil commitment hearings or capital sentencing cases, increased dramatically in the early part of the 21st century (DeMatteo & Edens, 2006; Hare & Neumann, 2009; Walsh & Walsh, 2006). Similarly, a study found that psychopathy was mentioned in 63% of a random sample of dangerous offender and long-term offender cases in Canada (Clark & Forth, 2004). Examples of PCL-R being used in some controversial contexts are reviewed in what follows.

Child Custody Hearings

The PCL-R has been used in child custody hearings, including custody battles where the parents may lose custody to the state, and custody hearings that result from divorce. The most frequent use of the PCL-R in this context appears to be in helping to decide whether or not a parent poses a risk to the child (see Walsh & Walsh, 2006). For example, a father's high PCL-R score was treated as a risk variable and contributed to the judge's decision to bar the man from seeing his daughter because of the risk he posed for sexually offending against his daughter and her friends. In child custody hearings, the PCL-R may address the personality of an individual, and although we may reason that a high PCL-R score would increase risk of child neglect or abuse, there are as yet no empirical studies for the expert to draw upon.

Capital Sentencing Hearings

There is considerable debate regarding the use of the PCL-R in capital punishment hearings, in large part because it represents such a powerful aggravating factor (Cunningham & Reidy, 2001; Edens, Davis, Fernandez-Smith, & Guy, 2012). One of the important issues in capital cases is whether or not the offender presents a serious continuing threat to society (Cox, DeMatteo, & Foster, 2010; Ogloff & Chopra, 2004). Although the specific criteria vary across jurisdictions, in general an offender in a capital case who does not receive the death penalty may be sentenced to either life or a very long term, usually without parole. Therefore, some argue that in order to be useful in capital punishment trials, violence risk predictions should provide information regarding institutional adjustment or violence risk after a lengthy period of incarceration (Edens, Buffington-

Vollum, Keilen, Roskamp, & Anthony, 2005; Walsh & Walsh, 2006). To date, no research has been conducted to assess the predictive validity of the PCL-R with samples of death row inmates. However, in light of the very low base rate of institutional violence within this group of offenders, it is unlikely that any risk assessment scale would be able to accurately predict this outcome.

There is a moderate association between PCL-R scores and measures of institutional adjustment (Guy, Edens, Anthony, & Douglas, 2005; Buffington-Vollum, Edens, Johnson, & Johnson, 2002; Edens, Buffington-Vollum, Colwell, Johnson, & Johnson, 2002; Hare et al., 2000; Heilbrun et al., 1998; Hicks, Rogers, & Cashel, 2000; Hobson, Shine, & Roberts, 2000; Kroner & Mills, 2001; Walters, 2003; Walters & Heilbrun, 2010). However, institutional misconduct includes a range of behaviors from verbal infractions to assault with a weapon. Not all of these behaviors are likely to apply to the question of whether an offender presents a continuing threat. Nevertheless, experts are still called upon to address the issue of whether an offender presents a continuing danger to society. If the context to which this applies is an institutional setting, the expert should evaluate a range of variables that may be of help to the courts. These include the PCL-R and actuarial factors such as age, education, and type of sentence. Experts who use the PCL-R in their assessment of dangerousness should be explicit about the limitations of the research upon which their evaluations are based.

In some cases, the predictive value of the PCL-R and its derivatives apparently is not as high in American institutions as in Canadian, British, and European institutions (e.g., Guy et al., 2005). The reasons are unclear, but one possibility is that in many American prisons the rules of behavior are strongly influenced by the heavy presence of gangs, which may trump the potential effect that psychopathy has on institutional behaviors, including infractions.

Besides its predictive role in capital sentencing hearings, the PCL-R may be helpful in the evaluation of the capacity for empathy, guilt, and remorse, each of which might be considered as mitigating or aggravating factors in sentencing. In such a case, the clinician should be explicit about the reliability and measurement error involved in assessing these features.

Preventive Detention

As well as in capital sentencing cases, the PCL-R is frequently used in preventive detention hearings (H. A. Miller, Amenta, & Conroy, 2005; Schopp, Scalora, & Pearce, 1999). In the United States, many states have legislation that allows for the civil commitment of sexually violent predators (SVPs). For example, in Washington State an SVP is an offender convicted of a sexual offence who is judged likely to engage in future predatory sexual violence (Jackson & Richards, 2007; Washington Laws, 1990, 71.09.030). These offenders are committed for an indeterminate time until they no longer pose a threat to society (Washington Laws, 1990, 71.09.100). The specifics of these laws, including the need for a prior conviction and the length of commitment, vary across jurisdictions.

In the United Kingdom, programs have been developed to identify and treat people with dangerous and severe personality disorder (DSPD). The initiative was a joint effort by the Home Office, the Prison Service, and the Department of Health, and was aimed at people who are regarded as a risk for violence at the time of release from either a hospital or prison or those in the community who exhibit dangerous behavior (Home Office & Department of Health, 1999). The PCL-R played a major role in the determination, management, and treatment of DSPD (Burns et al., 2011; Howard, Khalifa, Duggan, & Lumsden, 2012). For this reason, special efforts were made to ensure that the administrators of the instrument are well trained. The Ministry of Justice recently introduced the Offender Personality Disorder Pathway, with emphasis on the *DSM-IV* diagnosis of ASPD (estimated at about 50% of the male prison population; <http://www.personalitydisorder>).

org.uk/criminal-justice/publications/). Perhaps one reason for this move is to improve the treatment outcome statistics, which are very poor for DSPD and no doubt will be better for ASPD.

In Canada, preventive detention laws are found in the Long Term Offender (LTO) and Dangerous Offender (DO) provisions of the criminal code. These offenders are dealt with through the criminal justice system. LTO provisions require a significant probability of future violence and involve an extended community supervision order (Criminal Code of Canada, 1985, C-46, s.753.1). Alternatively, a DO declaration requires the additional criteria of lack of prospects for community control and involves an indeterminate sentence (Criminal Code of Canada, 1985, C-46, s.753). A PCL-R assessment is often conducted with individuals who have been recommended for Dangerous Offender status (Lloyd, Clark, & Forth, 2010). In many respects, those committed as SVPs in the United States are similar to those committed as high-risk sex offenders in Canada (Wilson, Looman, Abracen, & Pake, 2012).

Properly used, the PCL-R provides useful information about risk and treatment options within each of these legislative frameworks. The Total score is the most important and most reliable information provided by a PCL-R assessment. However, it is possible to supplement this score with information about the clusters of items (Factors) that contribute to the Total score. In some cases, reference to Factor 1 and Factor 2 may suffice, whereas in others the user may wish to conduct a finer descriptive analysis of the individual by inspecting the scores on the Interpersonal, Affective, Lifestyle, and Antisocial factors (see Table 8.1). The Antisocial factor may be of particular importance as a predictor of general and violent recidivism (Kennealy et al., 2010; Sreenivasan, Walker, Weinberger, Kirkish, & Garrick, 2008; Swogger et al., 2012; Walters & Heilbrun, 2010; Walters, Knight, Grann, & Dahle, 2008). Felthous (2011) has argued that evidence for the treatability or untreatability of psychopathy is equivocal, and that it should not be a basis for civil or criminal commitment unless accompanied by potentially treatable co-morbid conditions, such as impulsive aggression.

Culpability

Users of the PCL-R should be aware of the potential implications of advances in cognitive/affective neuroscience for issues of legal culpability. Some clinical and behavioral features of psychopathy, such as impulsivity, poor response inhibition, and difficulty in processing emotional material, appear to be related to “anomalies” in certain brain functions and structures, including those related to moral decision making (e.g., de Oliveira-Souza, Hare et al., 2008; Gao, Glenn, Schug, Yang, & Raine, 2009; Glenn, Raine, & Schug, 2009; Harenski et al., 2010; Wahlund & Kristiansson, 2009). These findings increasingly will provide the basis for legal arguments that psychopathic offenders may not be completely responsible for their criminal acts. The extent to which these findings and arguments will influence the manner in which the courts evaluate psychopathic offenders remains uncertain. This and related issues are the subject of considerable research and debate by scientists, jurists, ethicists, and philosophers (Fox, Kvaran, & Fontaine, 2012; Malatesti & McMillan, 2010) and by the MacArthur Foundation’s Law and Neuroscience Project (<http://www.lawandneuroscienceproject.org>).

Several cautionary comments seem appropriate for clinicians who become involved in the application of the neuroscience of psychopathy to the law (see Hare & Neumann, 2010b, pp. 109–110). Because investigators often view psychopathy as a disorder, they tend to refer to neuroimaging findings in psychopathic persons as evidence of “deficits” or “dysfunctions.” But statistically significant differences do not necessarily imply a deficit or a function that falls outside of the “normal” range. In this respect, it is important to note that at present we know little about the variability in brain structure and function in the general population, and even less about how

such variability relates to differences in personality and behavior. Further, from an evolutionary psychology perspective, psychopathy is not a disorder but an adaptive life-strategy (Book & Quinsey, 2003; Harris, Rice, Hilton, Lalumière, & Quinsey, 2007; also see Buss, 2009, for a discussion of personality traits as forms of strategic individual differences in adaptation). Functional differences observed during performance of a task might reflect the use of different strategies in performance of the task, while structural differences might be a case of “use it or lose it.” The selection of participants for neuroimaging research typically is not random, and the tasks used for functional imaging studies generally have uncertain ecological validity. Measurement error in the assessment of psychopathy, methodological, measurement, and statistical problems in acquiring and interpreting neuroimaging data, and uncertainty about what such data tell us about underlying cognitive and affective processes, make it difficult to establish causal connections between brain function or structure and psychopathic behavior. Similar considerations have been raised about the implications of neuroscience for criminal culpability in general (e.g., Gazzaniga, 2008; Mobbs, Lau, Jones, & Frith, 2007).

Potential for Misuse

Although the research evidence for the reliability and validity of the PCL–R is extensive, this does not ensure that an individual assessment will be reliable or valid. In a research context, misuse of the PCL–R will have few negative consequences for the individual. However, when PCL–R scores are used in clinical and criminal justice contexts the implications of misuse are potentially very serious, especially if the scores are used to guide treatment or adjudication decisions (Campbell, 2007; Hare, 1998b; 2003). In addition, it is important when conducting an assessment to use all information available to provide a complete picture of the person. In each case, it behooves the user to ensure that the PCL–R is used properly and in accordance with the highest ethical and professional standards. Needless to say, the items must be scored in accordance with the criteria listed in the Manual, or not scored at all. Unfortunately, there have been cases in which a clinician has scored the PCL–R entirely from the item titles, without reference or access to the Manual (Hare, 1998b). Presumably the item titles were obtained from a journal article. Clinicians who use the PCL–R must be prepared to outline the information used to score the items and to explain and justify the manner in which they scored the items.

The PCL–R potentially can provide information that aids decision makers in a wide range of cases, but it is possible that some professionals will misuse or misinterpret the instrument. Edens (2001) provides two examples where this occurred in capital sentencing hearings in the United States. One expert interpreted a low PCL–R score to conclude that a man was not guilty of molesting his daughter. This is clear misuse of an instrument that provides evidence about risk of future behavior, not indications of guilt or innocence. In the second case, an expert testified that the offender was at a high-risk for violence in the institution because of his high PCL–R score. Although PCL–R scores may provide information regarding an offender’s risk for institutional violence, it is important to consider the predictive limitations of the PCL–R in this context, and to take into account other risk and protective factors. The evidence is not sufficient to provide strong statements regarding risk in institutional settings based solely on a PCL–R score.

An area in need of research is the effect that the personality, attitudes, and biases of the rater may have on scoring the PCL–R items (A. K. Miller, Rufino, Boccaccini, Jackson, & Murrie, 2011). Of particular concern is the possibility that in an adversarial context, some professionals provide psychological reports that are influenced by their “allegiance” to the side that hired them—that is, the prosecution or defense (Lloyd et al., 2010; Murrie et al., 2009). The problem is exacerbated by the possibility that each side has “shopped around” for an expert. Attempts to

minimize this allegiance effect should include ensuring that raters have adequate training and respect for professional standards, that each rater has access to much the same information about an individual, and by careful cross-examination of courtroom testimony.

Other Issues

Cut Scores

An important issue is the use of cut scores to “diagnose” psychopathy. The PCL-R provides a dimensional score that represents the extent to which a given individual is judged to match the “prototypical psychopath.” These dimensional ratings are more useful than categorical diagnoses in several respects. For example, they have superior psychometric properties, and they do not require that assumptions be made about whether the underlying construct is continuous or categorical. Moreover, there is recent evidence that the construct underlying the PCL-R is dimensional in nature (Guay, Ruscio, Knight, & Hare, 2007). This dimensionality may pose a problem for diagnosing or categorizing a person as a “psychopath,” a problem shared by other clinical disorders (e.g., ASPD, Marcus, Lilienfeld, Edens, & Poythress, 2006; Widiger & Mullins-Sweatt, 2009) that are described and treated as categorical but in fact may be dimensional. But, the dimensionality of a personality disorder does not preclude the use of “diagnostic” thresholds for making clinical decisions (Widiger & Mullins-Sweatt, 2009). The difficulty is to determine the most appropriate cut score to use for these purposes. For research purposes, a cut score of 30 has proven useful, for it provides researchers with a common working definition of psychopathy. Problems arise when clinicians and investigators treat a cut score of 30 as if it were a sharp dividing line between offenders and patients who are psychopaths (members of a taxon) and those who are not. The problems are compounded when the judicial system makes the same assumption. Hare (1998a) described a case in which a judge in a dangerous offender hearing questioned the prosecution psychiatrist’s PCL-R assessment, “scored” it himself, managed to get the score down to 29, and concluded that if 30 means that the defendant is a psychopath, 29 must mean that he is not a psychopath. Hare (1998a) provides several other examples of misuse of the PCL-R in forensic practice.

Labels

Those who wish to avoid diagnostic labels or categories may convert PCL-R and Factor scores to percentiles or *T*-scores, using the appropriate tables provided in the Second Edition of the Manual. An alternative strategy for reporting PCL-R scores would be to group them into several levels or descriptive categories, and then to determine the risk associated with a particular level or category. One such scheme is depicted in the Manual. In any case, the practical implications of a PCL-R score will depend on the population involved (e.g., offender, forensic psychiatric, sex of individual), the context in which the score is used (e.g., treatment options, conditional release, risk assessment, research), and its integration with other relevant information and variables.

Qualifications

Because of the potentially serious consequences associated with an assessment of psychopathy, the PCL-R has been subjected to psychometric and legal scrutiny that is more intense than is the case with most other psychological instruments (Hare, 2003). To ensure that the PCL-R is used as intended, individual raters must meet certain criteria prior to conducting assessments for clinical or forensic purposes. Hare (2003) suggests that clinicians who use the PCL-R (or who supervise its use) must possess an advanced degree and must have completed graduate courses in

psychopathology, statistics, and psychometric theory. This enables them to interpret the statistics and research given in the Manual and in the literature. Further, users of the PCL–R need to be familiar with the current clinical and empirical literature on psychopathy. This will help in the interpretation and practical implications of psychopathy scores. It is important for the user to have the appropriate professional credentials (e.g., registration, licensing), and to have experience with the populations in which PCL–R assessments will be conducted. That is, users must understand the principles and limitations of such assessments, and must ensure that they are conducting assessments in accordance with professional and legal standards for psychological testing.

Individuals wishing to use the PCL–R (in any context) must have adequate training and experience. Scoring the PCL–R involves clinical judgment, and specific training and experience are warranted (Campbell, 2000). Clinicians who conduct psychological assessments and write reports for the court or who testify in court should be aware that prosecutors and defense attorneys are becoming increasingly knowledgeable about the PCL–R, and that this knowledge will be reflected in detailed examinations of their credentials, assessments, and testimony.

Confusion with ASPD

Problems can arise when experts use their clinical judgment to diagnose psychopathy or confuse psychopathy with antisocial personality disorder (ASPD). Little is known about the validity and reliability of unstructured clinical diagnoses, but the association between the PCL–R and ASPD is well known (Hare & Neumann, 2009; Hare et al., 2012; Verona, Sprague, & Sadeh, 2012). Briefly, the PCL–R has an asymmetrical relationship with ASPD: the majority of offenders with high PCL–R scores meet the diagnostic criteria for ASPD, whereas the reverse is not true (Hare, 1996, 2003; Hare & Neumann, 2009; Pham & Saloppé, 2010). The PCL–R is able to target a subgroup within the forensic population for which there is a great deal of empirical evidence concerning implications for risk and for treatment suitability. If an offender with ASPD does not also have a high PCL–R score, it would be inappropriate to use the extensive literature on PCL–R psychopathy to draw inferences about the offender's risk or treatment options.

The forthcoming *DSM–5* criteria for ASPD may be closer to the PCL–R than currently is the case. An initial proposal of *DSM–5* was to rename ASPD as antisocial/psychopathic type. More recently, the label was changed to antisocial personality disorder/dissocial personality disorder (American Psychiatric Association, 2011). The two proposed item sets (Lack of empathy, Lack of Intimacy, Manipulativeness, Deceitfulness, Callousness; and Hostility, Irresponsibility, Impulsivity, Risk-taking), are conceptually similar to PCL–R Factors 1 and 2, respectively (see Table 8.1). However, the criteria for scoring each item are much less detailed and explicit (and likely to be less reliable) than are those for PCL–R items. On the basis of structural MRI findings, Gregory et al. (2012) argued that ASPD and psychopathy are not the same constructs, and that *hostility* is more characteristic of reactively violent men than of psychopathy. Hare et al. (2012) provide a critical analysis of the *DSM–5* proposal for ASPD.

Familiarity with the Literature

Experts clearly should keep up to date on the empirical literature regarding psychopathy, and they should not misinterpret or go beyond the literature in their testimony. For example, the treatment-refractory nature of criminality among high-psychopathy offenders has been exaggerated by putative experts; in fact, the treatability issue with respect to psychopathy is somewhat more complex and uncertain than these recommendations indicate (Abracen et al., 2011; Hare & Neumann, 2010a; Wong and Hare, 2005). In such instances, where testimony may have a profound

impact on individual lives, making broad statements about psychopathy without being familiar with current literature may have unfortunate consequences.

Research Context

The criteria for using the PCL-R in a research setting are less stringent than they are for clinical purposes. In research settings, individual PCL-R scores typically are kept confidential and not made available to correctional or institutional staff, parole boards, and so forth. However, like clinicians, researchers should possess an advanced degree in the social, medical, or behavioral sciences, have completed graduate-level courses in psychopathology, statistics, and psychometric theory, and accept responsibility for the supervision of raters with lesser qualifications (e.g., research assistants with an undergraduate degree in the social or behavioral sciences, counseling psychology, or social work, in addition to some experience in interviewing).

In sum, the PCL-R remains the international standard for the measure of psychopathy (Hare & Neumann, 2010a). When used in appropriate settings by psychologists who have received vigorous training, the PCL-R is a reliable and valid measure of the psychopathy construct.

CASE EXAMPLE

Brad is a 37-year-old white male who recently began a six-year sentence in a federal medium-security institution for fraud, theft, and assault causing bodily harm. The victim is a 63-year-old widow whom Brad met at a charity auction. The woman's husband had recently died, leaving her a considerable sum of money in the form of bank deposits, investments, and property. Brad used his considerable skills for manipulation to befriend the woman and to gain her confidence. He presented himself as having a degree in commerce and as being a registered money manager and consultant. Shortly thereafter, he took over management of her financial affairs, and over a period of several months diverted several million dollars from her accounts to his own accounts. When she finally realized what had been happening, she confronted him and in the ensuing argument he punched her several times in the face, breaking her nose and fracturing her orbital bone. Brad made no restitution of funds, claiming that he himself was a victim of bad luck and betrayal by colleagues.

On admission to prison, Brad underwent a period of evaluation to determine his security risk and to recommend appropriate management and treatment options. A staff psychologist administered and scored the PCL-R on the basis of a two-hour interview with Brad and on extensive probation, pre-sentencing, and institutional documents.

During the interview, Brad was forthcoming and answered all questions willingly. He discussed potentially painful or embarrassing events with little hesitation or visible affective disturbance. Although he appeared at times to be ingratiating and overtly deferent, he frequently interrupted the psychologist and tried to control the interview. Previous reports commented that he knows how to manage his impressions across situations, and is able to verbalize insight and to simulate self-awareness and a sense of responsibility and commitment. The psychologist noted the discrepancy between Brad's presentation and the reports he had read prior to the interview.

Brad is the oldest of three children from a middle-class family. His mother was a nurse who retired after the birth of her second child and who died four years ago. He said he loved his

mother but described her as timid and “smothering”. He did not attend her funeral. His father is a lawyer who has been investigated several times by the local law society but has not had any formal charges laid against him. Brad described his father as a “cool guy” who knew how to work the system, but also as a short-tempered man who drank a lot and who spent more time at work and with other women than with his own wife and family. He and his father never got along well, and by the time Brad was 12 years old, verbal confrontations were frequent and heated. Brad reported that when he reached his teen years he began to use threats and violence to defend his mother, his siblings, and himself against his father’s abuse. His sister, a nurse, and his brother, an engineer, describe their father as neglectful but deny experiencing physical abuse. They described Brad as skilled at lying and report that from a very early age he was able to talk himself out of trouble. Even when caught in the act, he would profess his innocence so vigorously and persuasively that some began to doubt what they had just witnessed. They also reported that he frequently stole money from his mother, who always seemed to accept his excuses and denials of guilt.

Brad’s behavior as a child and teenager was characterized by a cruel streak and a “short fuse” in which he would explode in seeming anger, then quickly settle down. He reportedly bullied his sister and tormented her pet dog. On one notable occasion, when he was 12, Brad reportedly tightened a cord around the dog’s neck until it fell unconscious, and pushed his sister down when she tried to intervene. She complained to her parents but Brad convinced his parents and others that his sister had exaggerated the event.

Brad reported that schoolwork came easily to him, but discipline, study, and hard work were not part of his repertoire. His early school years were punctuated by a series of expulsions from a number of private schools, some of which advertised their ability to deal with difficult children. His various counselors described him as a liar and a bully who was always creating mischief for others. In high school, he continued to be a discipline problem and his academic performance deteriorated badly. He was suspended several times for plagiarism, truancy, and disrupting class. In his senior year, he was disciplined for reportedly assaulting another student, and when the principal refused his protestations of innocence, he threatened her and was expelled without graduating.

Brad has an extensive history of substance use, beginning in adolescence and persisting to the time of assessment. His preferred substances are alcohol and cocaine, but he has “tried every drug available.” He was charged several times with driving while under the influence of alcohol, and stated that many of his physical altercations were committed when provoked after ingesting drugs or alcohol. Although he expressed regret for the injuries he has caused others, he appeared to enjoy relating the details of his physical exploits, and described them as “direct action at work.”

Brad reported that he first had sexual intercourse at 14 years old, and stated that he has had upwards of 50 different partners since that time. He has lived with approximately 16 different women for periods varying from a few weeks to a few months. He freely describes several of these relationships as a matter of convenience. Asked why his relationships with women have usually ended, he replied, “I like variety.” He has fathered five children by three different women and is indifferent as to the whereabouts and welfare of the children and mothers. He has never paid child support.

Brad appears intelligent and well-spoken, and is a convincing and extravagant liar. He is adept at charming both men and women. His seeming candor and air of self-assurance often

disarm even those who know him well into trusting him repeatedly, despite previous disappointments. Many seemingly perceptive and intelligent people have been repeatedly duped by Brad before taking steps to protect themselves from his blandishments and manipulation.

Brad's job history is marked by inconsistent effort and unreliability. Although he did not graduate from high school, he has claimed to have several degrees and professional certifications. He is described as consistently well dressed and often wearing expensive jewelry, but has difficulty accounting for the source of his resources. He reports having worked as a helicopter pilot, a truck driver, a registered financial adviser, and a real estate salesman. However, when attempts were made to pin him down about his activities and stories, he would become first evasive and then hostile. He did say, however, that he quit some jobs because they were boring and "beneath" him. In other cases, he said he was unfairly treated by employers or cheated by his business partners.

Brad had his first formal contact with the police at 13 years of age for shoplifting. He had repeated contacts with the police throughout his teenage years for a variety of offenses, including assault, theft, and reckless driving. However, he typically received probation, and spent only a few weeks in a juvenile detention center, where he was in frequent conflict with the staff and counselors. At the age of 18 years, he received his first adult conviction for street racing and assaulting the apprehending police officer. He was sentenced to six months' probation for this crime. The probation report described Brad as a convincing liar and manipulator who liked excitement and "pushing the limits."

As an adult, Brad has been charged and convicted for a variety of offenses, including reckless driving, domestic assault, fraud, living off the avails of prostitution, possession of drugs, probation violation, disturbing the peace, and several counts of assault. His ability to use and manipulate others and to convince them of his good intentions has contributed to several charges being dropped or reduced. He is known to have used several aliases, making it difficult to assess the full extent of his criminal activities. His longest previous sentence was 18 months for assault causing bodily harm; he punched a man who confronted him for parking his car inappropriately. "I tried to explain to him that I would be back in a few minutes," Brad said, "but he wouldn't let it go." His casual view of violence as "a matter of process" is notable.

In addition to his high level of comfort with violence, Brad has also engaged in diverse frauds, writing bad checks ranging from ones for several thousand dollars at a jewelry store to ones for 10 dollars at a corner grocery store. In his late 20s, Brad presented himself as a real estate developer and impressed several local businessmen, who arranged a guest membership at the local tennis club. He made many new contacts at this club and managed to persuade several club members to invest thousands of dollars each in his company, which turned out to be a scam. He was charged and convicted of fraud but placed on probation after he promised to reimburse his investors, a promise still unfulfilled.

His file includes reports of several institutional infractions during his times in custody, including fighting and possession of prohibited substances. He also has been suspected of being involved in drug distribution and of threatening inmates who owed him money. While incarcerated, he has a history of entering and dropping out of various prison programs. He has reported converting to "born-again" Christianity and has developed pen-pal relationships with women in Christian communities near the prison. In one case, this led to a romantic affair with a middle-aged woman, prison visitation rights, and church sponsorship for early release. Soon

after release, he was charged with burglarizing the home of one of his sponsors, and was returned to prison.

Brad has a very high opinion of himself and his abilities, and reports low motivation for behavioral change. His self-description is quite revealing:

I am what I want to be; I think everyone is. You don't whine about what happens to you, you don't step back from anyone. I have my own morals. I don't cheat myself, I don't lie to myself. I have no regrets. That would be whining.

Brad exhibits many of the traits associated with psychopathy. His score of 32 out of a possible 40 on the PCL-R places him at the 84th percentile among institutionalized male offenders in North America, meaning that 84% of other male offenders would receive a score similar to, or lower than, his score. The PCL-R is divided into four factors. The Interpersonal factor focuses on grandiosity and manipulativeness. The Affective factor measures callousness, lack of empathy, and remorseless use of others. The Lifestyle factor reflects impulsivity, irresponsibility, and lack of goals. The Antisocial factor is described by poor anger control and the early onset and high frequency of antisocial behaviors. Compared with other male offenders, Brad had notably elevated scores on all four factors: Interpersonal factor (90th percentile); Affective factor (87th percentile); Lifestyle factor (79th percentile); and Antisocial factor (70th percentile). Overall, Brad's score on the PCL-R is very high (exceeding the standard research threshold for psychopathy). On the basis of the PCL-R score, other clinical information, and previous reports and behaviors, the evaluator opined that Brad was at elevated risk for institutional infractions and violent recidivism, and a poor candidate for traditional emotion-focused treatment programs.

Note

This chapter is a revised and updated version of a chapter by Book, A. S., Clark, H. J., Forth, A. E., and Hare, R. D. (2006). The PCL-R and PCL:YV: Forensic applications and limitations In R. P. Archer (Ed.), *Forensic use of clinical assessment instruments*. (pp. 147–179). Mahwah, NJ: Lawrence Erlbaum. Robert D. Hare receives royalties from the sale of the PCL-R and its derivatives.

References

- Abracen, J., Looman, J., Ferguson, M., Harkins, L., & Mailloux, D. (2011). Recidivism among treated sexual offenders and comparison subjects: Recent outcome data from the Regional Treatment Centre (Ontario) high-intensity Sex Offender Treatment Programme. *Journal of Sexual Aggression*, 17, 142–152.
- Abracen, J., Looman, J., & Langton, C. M. (2008). Treatment of sexual offenders with psychopathic traits: Recent research developments and clinical implications. *Trauma, Violence, and Abuse*, 9, 144–166.
- Acheson, S. K. (2005). Review of the Hare Psychopathy Checklist-Revised, 2nd Edition. In R. A. Spies & B. S. Blake (Eds.), *The sixteenth mental measurements yearbook* (pp. 429–431). Lincoln, NE: Buros Institute of Mental Measurements.
- American Psychiatric Association. (2000). *Diagnostic and statistical manual of mental disorders* (4th ed.). Washington, DC: Author.
- American Psychiatric Association. (2011, June 21). *Personality disorders*. Retrieved from <http://www.dsm5.org/PROPOSEDREVISIONS/Pages/PersonalityandPersonalityDisorders.aspx> (accessed March 2012).
- Anastasi, A. (1982). *Psychological testing* (5th ed.). London: Collier Macmillan.
- Andrews, D. A., & Bonta, J. (2003). *The psychology of criminal conduct* (3rd ed.). Cincinnati, OH: Anderson.

- Archer, R. P., Buffington-Vollum, J. K., Stredny, R. V., & Handel, R. W. (2006). A survey of psychological test use patterns among forensic psychologists. *Journal of Personality Assessment*, 87, 84–94.
- Babiak, P., & Hare, R. D. (2006). *Snakes in suits: When psychopaths go to work*. New York: Regan Books/HarperCollins.
- Babiak, P., & Hare, R. D. (in preparation). The B-Scan 360 Manual.
- Babiak, P., Neumann, C. S., & Hare, R. D. (2010). Corporate psychopathy: Talking the walk. *Behavioral Sciences and the Law*, 28, 174–193.
- Berger, K., Rotermund, P., Vieth, E. R., & Hohnhorst, A. (2012). The prognostic value of the PCL-R in relation to the SUD treatment ending. *International Journal of Law and Psychiatry*, 35, 198–201.
- Berrios, G. E. (1996). *The history of mental symptoms: Descriptive psychopathology since the nineteenth century*. Cambridge: Cambridge University Press.
- Blackburn, R., & Coid, J. W. (1998). Psychopathy and the dimensions of personality disorders in violent offenders. *Personality and Individual Differences*, 25, 129–145.
- Boer, D. P., Hart, S. D., Kropp, P. R., & Webster, C. D. (1997). *Manual for the Sexual Violence Risk-20: Professional guidelines for assessing risk of sexual violence*. Vancouver: British Columbia Institute on Family Violence.
- Bolt, D. M., Hare, R. D., & Neumann, C. S. (2007). Score metric equivalence of the Psychopathy Checklist-Revised (PCL-R) across criminal offenders in North America and the United Kingdom: A critique of Cooke, Michie, Hart, and Clark (2005) and new analyses. *Assessment*, 14, 44–56.
- Bolt, D. M., Hare, R. D., Vitale, J. E., & Newman, J. P. (2004). A multigroup item response theory analyses of the Hare Psychopathy Checklist-Revised. *Psychological Assessment*, 16, 155–168.
- Book, A. S., & Quinsey, V. L. (2003). Psychopaths: cheaters or warrior-hawks? *Personality and Individual Differences*, 36, 33–45.
- Brinkley, C. A., Schmitt, W. A., Smith, S. S., & Newman, J. P. (2001). Construct validation of a self-report psychopathy scale: Does Levenson's Self-Report Psychopathy Scale measure the same constructs as Hare's Psychopathy Checklist Construct validation of a self-report psychopathy scale: Does Levenson's Revised? *Personality and Individual Differences*, 31, 1021–1038.
- Buffington-Vollum, J. K., Edens, J. F., Johnson, D. W., & Johnson, J. K. (2002). Psychopathy as a predictor of institutional misbehavior among sex offenders: A prospective replication. *Criminal Justice and Behavior*, 29, 497–511.
- Burns, T., Fazel, S., Fahy, T., Fitzpatrick, R., Rogers, R., Sinclair, J., . . . Yiend, J. (2011). Dangerous severe personality disordered (DSPD) patients: Characteristics and comparison with other high-risk offenders. *International Journal of Forensic Mental Health*, 10, 127–136.
- Buss, D. M. (2009). How can evolutionary psychology successfully explain personality and individual differences? *Perspectives on Psychological Science*, 4, 359–366.
- Caldwell, M., Skeem, J., Salekin, R., & Van Rybroek, G. (2006). Treatment response of adolescent offenders with psychopathy features: A 2-year follow-up. *Criminal Justice and Behavior*, 33, 571–596.
- Campbell, T. W. (2000). Sexual predator evaluations and phrenology: Considering issues of evidentiary reliability. *Behavioral Sciences and the Law*, 18, 111–130.
- Campbell, T. W. (2007). The validity of the Psychopathy Checklist-Revised in adversarial proceedings. *Journal of Forensic Psychology Practice*, 6, 43–53.
- Catchpole, R. E. H., & Gretton, H. M. (2003). The predictive validity of risk assessment with violent young offenders: A 1-year examination of criminal outcome. *Criminal Justice and Behavior*, 30, 688–708.
- Clark, H. J., & Forth, A. E. (2004). Psychopathy in the court. Unpublished manuscript. Department of Psychology, Carleton University.
- Cleckley, H. (1941). *The mask of sanity*. St. Louis, MO: C. V. Mosby.
- Cleckley, H. (1976). *The mask of sanity* (5th ed.). St. Louis, MO: C. V. Mosby.
- Coid, J., & Yang, M. (2011). The impact of psychopathy on violence among the household population of Great Britain. *Social Psychiatry and Psychiatric Epidemiology*, 46, 473–480.
- Coid, J., Yang, M., Ullrich, S., Roberts, A., & Hare, R. D. (2009). Prevalence and correlates of psychopathic traits in the household population of Great Britain. *International Journal of Law and Psychiatry*, 32, 65–73.
- Coid, J., Yang, M., Ullrich, S., Roberts, A., Moran, P., Bebbington, P., . . . Hare, R. D. (2009). Psychopathy among prisoners in England and Wales. *International Journal of Law and Psychiatry*, 32, 134–141.
- Cooke, D. J., Kosson, D. S., & Michie, C. (2001). Psychopathy and ethnicity: Structural, item and test generalizability of the Psychopathy Checklist Revised (PCL-R) in Caucasian and African-American participants. *Psychological Assessment*, 13, 531–542.

- Cooke, D. J., & Michie, C. (2001). Refining the construct of psychopathy: Towards a hierarchical model. *Psychological Assessment, 13*, 171–188.
- Cooke, D. J., Michie, C., Hart, S. D., & Clarke, D. (2005). Searching for the pan-cultural core of psychopathic personality disorder. *Personality and Individual Differences, 39*, 283–295.
- Cooke, D. J., Michie, C., Hart, S. D., & Hare, R. D. (1999). The functioning of the Clinical Version of the Psychopathy Checklist: An item response theory analysis. *Psychological Assessment, 11*, 3–13.
- Copestake, S., Gray, N. S., & Snowden, R. J. (2011). A comparison of a self-report measure of psychopathy with the Psychopathy Checklist–Revised in a UK sample of offenders. *Journal of Forensic Psychiatry and Psychology, 22*, 169–182.
- Cornell, D. G., Warren, J., Hawk, G., Stafford, E., Oram, G., & Pine, D. (1996). Psychopathy in instrumental and reactive violent offenders. *Journal of Consulting and Clinical Psychology, 64*, 3–19.
- Costa, P. T., & Widiger, T. A. (2002). *Personality disorder and the five-factor model of personality* (2nd ed.). Washington, DC: American Psychological Association.
- Cox, J., DeMatteo, D. S., & Foster, E. E. (2010). The effect of the Psychopathy Checklist–Revised in capital cases: Mock jurors' responses to the label of psychopathy. *Behavioral Sciences and the Law, 28*, 878–891.
- Crocker, A. G., & Côté, G. (2009). Evolving systems of care: Individuals found not criminally responsible on account of mental disorder in custody of civil and forensic psychiatric services. *European Psychiatry, 24*, 356–364.
- Cunningham, M. D., & Reidy, T. J. (2001). A matter of life or death: Special considerations and heightened practice standards in capital sentencing evaluations. *Behavioral Sciences and the Law, 19*, 473–490.
- Czar, K. A., Dahlen, E. R., Bullock, E. E., & Nicholson, B. C. (2011). Psychopathic personality traits in relational aggression among young adults. *Aggressive Behavior, 37*, 207–214.
- Dahle, K. P. (2006). Strengths and limitations of actuarial prediction of criminal reoffence in a German prison sample: A comparative study of LSI–R, HCR–20 and PCL–R. *International Journal of Law and Psychiatry, 29*, 431–442.
- Dahlstrom, W. G., & Welsh, G. S. (1960). *An MMPI handbook: A guide to use in clinical practice and research*. Minneapolis, MN: Minnesota Press.
- Declercq, F., Willemsen, J., Audenaert, K., & Verhaeghe, P. (2012). Psychopathy and predatory violence in homicide, violent, and sexual offences: Factor and facet relations. *Legal and Criminological Psychology, 17*, 59–74.
- Decuyper, M., De Fruyt, F., & Buschman, J. (2008). A five-factor model perspective on psychopathy and comorbid axis-II disorders in a forensic-psychiatric sample. *International Journal of Law and Psychiatry, 31*, 394–406.
- DeLisi, M. (2009). Psychopathy is the unified theory of crime. *Youth Violence and Juvenile Justice, 7*, 256–273.
- DeMatteo, D., & Edens, J. F. (2006). The role and relevance of the Psychopathy Checklist–Revised in court: A case law survey of U.S. courts (1991–2004). *Psychology, Public Policy, and Law, 12*, 214–241.
- de Oliveira-Souza, R., Hare, R. D., Bramati, I. E., Garrido, G. J., Ignácio, F. A., Tovar-Moll, F., & Moll, J. (2008). Psychopathy as a disorder of the moral brain: Fronto-temporo-limbic grey matter reductions demonstrated by voxel-based morphometry. *NeuroImage, 40*, 1202–1213.
- de Oliveira-Souza, R., Ignácio, R. A., Moll, J., & Hare, R. D. (2008). Psychopathy in a civil psychiatric outpatient sample. *Criminal Justice and Behavior, 35*, 427–437.
- Dolan, M., & Davies, G. (2006). Psychopathy and institutional outcome in patients with schizophrenia in forensic settings in the UK. *Schizophrenia Research, 81*, 277–281.
- Dolan, M., & Doyle, M. (2000). Violence risk prediction: Clinical and actuarial measures and the role of the Psychopathy Checklist. *British Journal of Psychiatry, 177*, 303–311.
- Doninger, N. A., & Kosson, D. S. (2001). Interpersonal construct systems among psychopaths. *Personality and Individual Differences, 30*, 1263–1281.
- Doren, D. M., & Yates, P. M. (2008). Effectiveness of sex offender treatment for psychopathic sexual offenders. *International Journal of Offender Therapy and Comparative Criminology, 52*, 234–245.
- Doyle, M., Dolan, M., & McGovern, J. (2002). The validity of North American risk assessment tools in predicting in-patient violent behaviour in England. *Legal and Criminological Psychology, 7*, 141–154.
- Douglas, K. S., & Webster, C. D. (1999). The HCR–20 violence risk assessment scheme: Concurrent validity in a sample of incarcerated offenders. *Criminal Justice and Behavior, 26*, 3–19.
- Edens, J. F. (2001). Misuses of the Hare Psychopathy Checklist–Revised in court: Two case examples. *Journal of Interpersonal Violence, 16*, 1082–1094.

- Edens, J. F., Buffington-Vollum, J. K., Colwell, K. W., Johnson, D. W., & Johnson, J. K. (2002). Psychopathy and institutional misbehavior among incarcerated sex offenders: A comparison of the Psychopathy Checklist-Revised and the Personality Assessment Inventory. *International Journal of Forensic Mental Health, 1*, 49–58.
- Edens, J. F., Buffington-Vollum, J. K., Keilen, A., Roskamp, P., & Anthony, C. (2005). Prediction of future dangerousness in capital murder trials: Is it time to “disinvent the wheel?” *Law and Human Behavior, 29*, 55–86.
- Edens, J. F., Davis, K. M., Fernandez-Smith, K., & Guy, L. S. (2012). No sympathy for the devil: Attributing psychopathic traits to capital murderers also predicts support for executing them. *Personality Disorders: Theory, Research, and Treatment*. doi:10.1037/a0026442.
- Edens, J. F., Marcus, D. K., Lilienfeld, S. O., & Poythress, N. G. Jr. (2006). Psychopathic, not psychopath: Taxometric evidence for the dimensional structure of psychopathy. *Journal of Abnormal Psychology, 115*, 131–144.
- Edens, J. F., Skeem, J. L., & Douglas, K. S. (2006). Incremental validity analyses of the Violence Risk Appraisal Guide and the Psychopathy Checklist: Screening Version in a civil psychiatric sample. *Assessment, 13*, 368–374.
- Felthous, A. R. (2011). The “untreatability” of psychopathy and hospital commitment in the USA. *International Journal of Law and Psychiatry, 34*, 400–405.
- Felthous, A. R., & Sass, H. (Eds.) (2007). *International handbook on psychopathic disorders and the law* (pp. 41–67). New York: John Wiley.
- Forth, A. E., & Book, A. S. (2007). Psychopathy in youth: A valid construct? In H. Hervé & J. Yuille, J. (Eds.), *The psychopath: Theory, research, and practice* (pp. 369–387). Mahwah, NJ: Lawrence Erlbaum.
- Forth, A. E., Brown, S. L., Hart, S. D., & Hare, R. D. (1996). The assessment of psychopathy in male and female noncriminals: Reliability and validity. *Personality and Individual Differences, 20*, 531–543.
- Forth, A. E., Kosson, D., & Hare, R. D. (2003). *The Hare PCL: Youth Version*. Toronto: Multi-Health Systems.
- Fox, A. R., Kvaran, T. H., & Fontaine, R. G. (2012). Psychopathy and culpability: How responsible is the psychopath for criminal wrongdoing? *Law and Social Inquiry*. doi: 10.1111/j.1747-4469.2012.01294.x.
- Frick, P. J., & White, S. F. (2008). Research review: The importance of callous-unemotional traits for developmental models of aggressive and antisocial behavior. *Journal of Child Psychology and Psychiatry, 49*, 359–375.
- Fulero, S. M. (1995). Review of the Hare Psychopathy Checklist-Revised. In J. C. Conoley & J. C. Impara (Eds.), *Twelfth mental measurements yearbook* (pp. 453–454). Lincoln, NE: Buros Institute.
- Gacono, C. (Ed.). (2000). *The clinical and forensic assessment of psychopathy: A practitioner's guide*. New York: Lawrence Erlbaum.
- Gao, Y., Glenn, A. L., Schug, R. A., Yang, Y., & Raine, A. (2009). The neurobiology of psychopathy: A neurodevelopmental perspective. *Canadian Psychiatric Association Journal / La Revue de l'Association des Psychiatres du Canada, 54*, 813–823.
- Gazzaniga, M. S. (2008). The law and neuroscience. *Neuron, 60*, 412–415.
- Gendreau, P., Goggin, C., & Smith, P. (2002). Is the PCL-R really the “unparalleled” measure of offender risk? A lesion in knowledge cumulation. *Criminal Justice and Behavior, 29*, 397–426.
- Glenn, A. L., Raine, A. A., & Schug, R. A. (2009). The neural correlates of moral decision-making in psychopathy. *Molecular Psychiatry, 14*, 5–6.
- Glover, A. J., Nicholson, D. E., Hemmati, T., Bernfeld, G. A., & Quinsey, V. L. (2002). A comparison of predictors of general and violent recidivism among high-risk federal offenders. *Criminal Justice and Behavior, 29*, 235–249.
- Gray, N. S., Fitzgerald, S., Taylor, J., MacCulloch, M. J., & Snowden, R. J. (2007). Predicting future reconviction in offenders with intellectual disabilities: The predictive efficacy of VRAG, PCL:SV and the HCR-20. *Psychological Assessment, 19*, 474–479.
- Gray, N. S., Hill, C., McGleish, A., Timmons, D., MacCulloch, M. J., & Snowden, R. J. (2003). Prediction of violence and self-harm in mentally disordered offenders: A prospective study of the efficacy of HCR-20, PCL-R, and psychiatric symptomatology. *Journal of Consulting and Clinical Psychology, 71*, 443–451.
- Gregory, S., ffytche, D., Simmons, A., Kumari, V., Howard, M., Hodgins, S., & Blackwood, N. (2012). The antisocial brain: Psychopathy matters. *Archives of General Psychiatry*. doi:10.1001/archgenpsychiatry.2012.222.

- Guay, J., Ruscio, J., Knight, R. A., & Hare, R. D. (2007). A taxometric analysis of the latent structure of psychopathy: Evidence for dimensionality. *Journal of Abnormal Psychology, 116*, 701–716.
- Gustavson, C., Ståhlberg, O., Sjödin, A., Forsman, A., Nilsson, T., & Anckarsäter, H. (2007). Age at onset of substance abuse: A crucial covariate of psychopathic traits and aggression in adult offenders. *Psychiatry Research, 153*, 195–198.
- Guy, L. S., & Douglas, K. S. (2006). Examining the utility of the PCL:SV as a screening measure using competing factor models of psychopathy. *Psychological Assessment, 18*, 225–230.
- Guy, L. S., Edens, J. F., Anthony, C., & Douglas, K. S. (2005). Does psychopathy predict institutional misconduct among adults? A meta-analytic investigation. *Journal of Consulting and Clinical Psychology, 73*, 1056–1064.
- Häkkinen-Nyholm, H., & Nyholm, J. O. (Eds.). (2012). *Psychopathy and the law: A practitioners' guide*. New York: Wiley Blackwell.
- Hare, R. D. (1980). A research scale for the assessment of psychopathy in criminal populations. *Personality and Individual Differences, 1*, 111–119.
- Hare, R. D. (1985). Comparison of the procedures for the assessment of psychopathy. *Journal of Consulting and Clinical Psychology, 53*, 7–16.
- Hare, R. D. (1991). *The Hare Psychopathy Checklist-Revised*. Toronto: Multi-Health Systems.
- Hare, R. D. (1996). Psychopathy and antisocial personality disorder: A case of diagnostic confusion. *Psychiatric Times, 13*, 39–40.
- Hare, R. D. (1998a). Psychopaths and their nature: Implications for the mental health and criminal justice systems. In T. Millon, E. Simonson, M. Burket-Smith, & R. Davis (Eds.), *Psychopathy: Antisocial, criminal, and violent behavior* (pp. 188–212). New York: Guilford Press.
- Hare, R. D. (1998b). The Hare PCL-R: Some issues concerning its use and misuse. *Legal and Criminological Psychology, 3*, 101–123.
- Hare, R. D. (2003). *The Hare Psychopathy Checklist-Revised* (2nd ed.). Toronto: Multi-Health Systems.
- Hare, R. D., Clark, D., Grann, M., & Thornton, D. (2000). Psychopathy and the predictive validity of the PCL-R: An international perspective. *Behavioral Sciences and the Law, 18*, 623–645.
- Hare, R. D., and Cox, D. N. (1978). Clinical and empirical correlates of psychopathy, and the selection of subjects for research. In R. D. Hare and D. Schalling (Eds.), *Psychopathic behaviour: Approaches to research* (pp. 1–21). Chichester, UK: John Wiley.
- Hare, R. D., & Frazelle, J. (1980). Some preliminary notes on the use of a research scale for the assessment of psychopathy in criminal populations. Unpublished manuscript, University of British Columbia.
- Hare, R. D., & Neumann, C. N. (2005). The PCL-R assessment of psychopathy: Development, structural properties, and new directions. In C. Patrick (Ed.), *Handbook of psychopathy*. New York: Guilford Press.
- Hare, R. D., & Neumann, C. S. (2008). Psychopathy as a clinical and empirical construct. *Annual Review of Clinical Psychology, 4*, 217–246.
- Hare, R. D., & Neumann, C. S. (2009). Psychopathy: Assessment and forensic implications. *Canadian Journal of Psychiatry, 54*, 791–802.
- Hare, R. D., & Neumann, C. S. (2010a). The role of antisociality in the psychopathy construct: Comment on Skeem and Cooke (2010). *Psychological Assessment, 22*, 446–454.
- Hare, R. D., & Neumann, C. S. (2010b). Psychopathy: Assessment and forensic implications. In L. Malatesti & J. McMillan (Eds.), *Responsibility and psychopathy: Interfacing law, psychiatry and philosophy* (pp. 93–123). New York: Oxford University Press.
- Hare, R. D., Neumann, C. S., & Widiger, T. A. (2012). Psychopathy. In T. A. Widiger (Ed.), *The Oxford handbook of personality disorders* (pp. 478–504). New York: Oxford University Press.
- Harenski, C. A., Hare, R. D., & Kiehl, K. A. (2010). Neuroimaging, genetics, and psychopathy: Implications for the legal system. In L. Malatesti & J. McMillan (Eds.), *Responsibility and psychopathy: Interfacing law, psychiatry, and philosophy* (pp. 125–154). Oxford: Oxford University Press.
- Harpur, T. J., & Hare, R. D. (1991, August). Psychopathy and violent behavior: Two factors are better than one. Paper presented at the 99th Annual Meeting of the American Psychological Association, August 1991.
- Harris, G. T., & Rice, M. E. (2006). Treatment of psychopathy: A review of empirical findings. In C. J. Patrick (Ed.), *Handbook of psychopathy* (pp. 555–572). New York: Guilford Press.
- Harris, G. T., Rice, M. E., Hilton, N. Z., Lalumière, M. L., & Quinsey, V. L. (2007). Coercive and precocious sexuality as a fundamental aspect of psychopathy. *Journal of Personality Disorders, 21*, 1–27.

- Hart, S. D., Cox, D. N., & Hare, R. D. (1995). *Manual for the Psychopathy Checklist: Screening Version (PCL:SV)*. Toronto: Multi-Health Systems.
- Hart, S. D., & Hare, R. D. (1989). Discriminant validity of the Psychopathy Checklist. *Psychological Assessment: A Journal of Consulting and Clinical Psychology*, 1, 211–218.
- Heilbrun, K., Hart, S. D., Hare, R. D., Gustafson, D., Nunez, C., & White, A. J. (1998). Inpatient and postdischarge aggression in mentally disordered offenders: The role of psychopathy. *Journal of Interpersonal Violence*, 13, 514–527.
- Hemphill, J. F., & Hare, R. D. (2004). Some misconceptions about the Hare PCL-R and risk assessment: A reply to Gendreau, Goggin, & Smith. *Criminal Justice and Behavior*, 31, 203–243.
- Hemphill, J. F., Hare, R. D., & Wong, S. C. P. (1998). Psychopathy and recidivism: A review. *Legal and Criminological Psychology*, 3, 139–170.
- Hemphill, J. F., Hart, S. D., & Hare, R. D. (1994). Psychopathy and substance use. *Journal of Personality Disorders*, 8, 169–180.
- Hervé, H., & Yuille, J. (Eds.). (2007). *The psychopath: Theory, research, and practice*. Mahwah, NJ: Lawrence Erlbaum.
- Hicklin, J., & Widiger, T. A. (2005). Similarities and differences among antisocial and psychopathic self-report inventories from the perspective of general personality functioning. *European Journal of Personality*, 19, 325–342.
- Hicks, M. M., Rogers, R., & Cashel, M. (2000). Predictions of violent and total infractions among institutionalized male juvenile offenders. *Journal of the American Academy of Psychiatry and the Law*, 28, 183–190.
- Hicks, B. M., Vaidyanathan, U., & Patrick, C. J. (2010). Validating female psychopathy subtypes: Differences in personality, antisocial and violent behavior, substance abuse, trauma, and mental health. *Personality Disorders: Theory, Research, and Treatment*, 1, 38–57.
- Hildebrand, M., & de Ruiter, C. (2004). PCL-R psychopathy and its relation to DSM-IV Axis I and Axis II disorders in a sample of male forensic psychiatric patients in the Netherlands. *International Journal of Law and Psychiatry*, 27, 233–248.
- Hildebrand, M., de Ruiter, C., & de Vogel, V. (2004). Psychopathy and sexual deviance in treated rapists: Association with sexual and non-sexual recidivism. *Sexual Abuse: A Journal of Research and Treatment*, 16, 1–24.
- Hill, C. D., Neumann, C. S., & Rogers, R. (2004). Confirmatory factor analysis of the Psychopathy Checklist: Screening Version (PCL:SV) in offenders with Axis I disorders. *Psychological Assessment*, 16, 90–95.
- Hill, C. D., Rogers, R., & Bickford, M. E. (1996). Predicting aggressive and socially disruptive behavior in a maximum security forensic psychiatric hospital. *Journal of Forensic Sciences*, 41, 56–59.
- Hilton, N. Z., Harris, G. T., Rice, M. E., Houghton, R. E., & Eke, A. W. (2008). An indepth actuarial assessment for wife assault recidivism: The Domestic Violence Risk Appraisal Guide. *Law and Human Behavior*, 32, 150–163.
- Hobson, J., Shine, J., & Roberts, R. (2000). How do psychopaths behave in a prison therapeutic environment? *Psychology, Crime, and Law*, 6, 139–154.
- Home Office & Department of Health. (1999). *Managing dangerous people with severe personality disorder*. London: Home Office and Department of Health. Retrieved from <http://www.homeoffice.gov.uk/docs/persdis.html>.
- Howard, R., Khalifa, N., Duggan, C., & Lumsden, J. (2012). Are patients deemed “dangerous and severely personality disordered” different from other personality disordered patients detained in forensic settings? *Criminal Behaviour and Mental Health*, 22, 65–78.
- Jackson, R. L., Neumann, C. S., & Vitacco, M. J. (2007). Impulsivity, anger, and psychopathy: The moderating effect of ethnicity. *Journal of Personality Disorders*, 21, 289–304.
- Jackson, R. L., & Richards, H. J. (2007). Diagnostic and risk profiles among civilly committed sex offenders in Washington State. *International Journal of Offender Therapy and Comparative Criminology*, 51, 313–323.
- Kennealy, P. J., Hicks, B. M., & Patrick, C. J. (2007). Validity of factors of the Psychopathy Checklist-Revised in female prisoners: Discriminant relations with antisocial behavior, substance abuse, and personality. *Assessment*, 14, 323–340.
- Kennealy, P. J., Skeem, J. L., Walters, G. D., & Camp, J. (2010). Do core interpersonal and affective traits of PCL-R psychopathy interact with antisocial behavior and disinhibition to predict violence? *Psychological Assessment*, 22, 569–580.

- Khiroya, R., Weaver, T., & Maden, T. (2009). Use and perceived utility of structured violence risk assessments in English medium secure forensic units. *Psychiatric Bulletin*, 33, 129–132.
- Kiehl, K. A. (2006). A cognitive neuroscience perspective on psychopathy: Evidence for paralimbic system dysfunction. *Psychiatry Research*, 142, 107–128.
- Koenigs, M., Baskin-Sommers, A., Zeier, J., & Newman, J. P. (2011). Investigating the neural correlates of psychopathy: A critical review. *Molecular Psychiatry*, 16, 792–799.
- Kosson, D., Neumann, C. S., Forth, A. E., Salekin, R. T., Hare, R. D., Krischer, M. K., & Sevecke, K. (2012). Factor structure of the Hare Psychopathy Checklist: Youth Version (PCL:YV) in adolescent females. *Psychological Assessment*.
- Kosson, D. S., Steuerwald, B. L., Forth, A. E., & Kirkhart, K. J. (1997). A new method for assessing the interpersonal behavior of psychopathic individuals: Preliminary validation studies. *Psychological Assessment*, 9, 89–101.
- Kroner, D. G., & Mills, J. F. (2001). The accuracy of five risk appraisal instruments in predicting institutional misconduct and new convictions. *Criminal Justice and Behavior*, 28, 471–489.
- Kroner, D. G., Mills, J. F., & Reddon, J. R. (2005). A coffee can, factor analysis, and prediction of antisocial behavior: The structure of criminal risk. *International Journal of Law and Psychiatry*, 28, 360–374.
- Lake, A. J., Baskin-Sommers, A. R., Li, W., Curtin, J. J., & Newman, J. P. (2011). Evidence for unique threat-processing mechanisms in psychopathic and anxious individuals. *Cognitive, Affective, and Behavioral Neuroscience*, 11, 451–462.
- Langton, C. M., Barbaree, H. E., Harkins, L., & Peacock, E. J. (2006). Sex offenders' response to treatment and its association with recidivism as a function of psychopathy. *Sexual Abuse: Journal of Research and Treatment*, 18, 99–120.
- Langton, C. M., Hogue, T. E., Daffern, M., Mannion, A., & Howells, K. (2011). Personality traits as predictors of inpatient aggression in a high-security forensic psychiatric setting: Prospective evaluation of the PCL-R and IPDE dimension ratings. *International Journal of Offender Therapy and Comparative Criminology*, 55, 392–415.
- Larsson, H., Viding, E., & Plomin, R. (2008). Callous-unemotional traits and antisocial behavior: Genetic, environmental, and early parenting characteristics. *Criminal Justice and Behavior*, 35, 197–211.
- Leistico, A. R., Salekin, R. T., DeCoster, J., & Rogers, R. (2008). A large-scale meta-analysis relating the Hare measures of psychopathy to antisocial conduct. *Law and Human Behavior*, 32, 28–45.
- Lewis, K., Olver, M. E., & Wong, S. C. P. (2012). The Violence Risk Scale: Predictive validity and linking changes in risk with violent recidivism in a sample of high-risk offenders with psychopathic traits. *Assessment*. doi:10.1177/1073191112441242.
- Lilienfeld, S. O. (1998). Methodological advances and developments in the assessment of psychopathy. *Behaviour Research and Therapy*, 36, 99–125.
- Lilienfeld, S. O., & Widows, M. R. (2005). *Professional manual for the Psychopathic Personality Inventory-Revised (PPI-R)*. Lutz, FL: Psychological Assessment Resources.
- Lincoln, T., & Hodgins, S. (2008). Is lack of insight associated with physically aggressive behavior among people with schizophrenia living in the community? *Journal of Nervous and Mental Disease*, 196, 62–66.
- Lindberg, N., Laajasalo, T., Holm, M., Putkonen, H., Weizmann-Henelius, G., & Häkkinen-Nyholm, H. (2009). Psychopathic traits and offender characteristics: A nationwide consecutive sample of homicidal male adolescents. *BMC Psychiatry*, 9 [full text].
- Lloyd, C. D., Clark, H. J., & Forth, A. E. (2010). Psychopathy, expert testimony, and indeterminate sentences: Exploring the relationship between Psychopathy Checklist-Revised testimony and trial outcome in Canada. *Legal and Criminological Psychology*, 15, 323–339.
- Looman, J., & Ismail, G. (under review). Field inter-rater reliability of the Psychopathy Checklist-Revised.
- Lorenz, A. R., & Newman, J. P. (2002). Do emotion and information processing deficiencies found in Caucasian psychopaths generalize to African-American psychopaths? *Personality and Individual Differences*, 32, 1077–1086.
- Ly, M., Motzkin, J. C., Philippi, C. L., Kirk, G. R., Newman, J. P., Kiehl, K. A., & Koenigs, M. (2012). Cortical thinning in psychopathy. *American Journal of Psychiatry*, 169, 743–749.
- Lynam, D. R., & Miller, J. D. (2012). Fearless dominance and psychopathy: A response to Lilienfeld et al. *Personality Disorders: Theory, Research, and Treatment*, 3, 341–353.
- Lynam, D. R., & Widiger, T. A. (2007). Using a general model of personality to identify the basic elements of psychopathy. *Journal of Personality Disorders*, 21, 160–178.

- Lynn, R. (2002). Racial and ethnic differences in psychopathic personality. *Personality and Individual Differences*, 32, 273–316.
- Malatesti, L., & McMillan, J. (Eds.). (2010). *Responsibility and psychopathy: Interfacing law, psychiatry and philosophy*. Oxford: Oxford University Press.
- Marcus, D. K., Lilienfeld, S. O., Edens, J. F., & Poythress, N. G. (2006). Is antisocial personality disorder continuous or categorical? A taxometric analysis. *Psychological Medicine*, 36, 1571–1581.
- Mathieu, C., Hare, R. D., Jones, D. N., Babiak, P., & Neumann, C. S. (2012). Factor structure of the B-Scan 360: A measure of corporate psychopathy. *Psychological Assessment*. doi: 10.1037/a0029262.
- McCoy, W. K., & Edens, J. F. (2006). Do Black and White youths differ in levels of psychopathic traits? A meta-analysis of the Psychopathy Checklist Measures. *Journal of Consulting and Clinical Psychology*, 74, 386–392.
- Miller, A. K., Rufino, K. A., Boccaccini, M. T., Jackson, R. L., & Murrie, D. C. (2011). On individual differences in person perception: Raters' personality traits relate to their Psychopathy Checklist-Revised scoring tendencies. *Assessment*, 18, 253–260.
- Miller, H. A., Amenta, A. E., & Conroy, M. A. (2005). Sexually violent predator evaluations: Empirical evidence, strategies for professionals, and research directions. *Law and Human Behavior*, 29, 29–54.
- Miller, J. D., & Lynam, D. R. (2011, August 8). An examination of the Psychopathic Personality Inventory's nomological network: A meta-analytic review. *Personality Disorders: Theory, Research, and Treatment*, 3, 305–326. doi:10.1037/a0024567
- Millon, T. (1987). *Millon Clinical Multiaxial Inventory-II manual*. Minneapolis, MN: National Computer Systems.
- Millon, T., Davis, R., & Millon, C. (1997). *MC MCMII-III manual* (2nd ed.). Minneapolis, MN: National Computer Systems.
- Mobbs, D., Lau, H. C., Jones, O. D., & Frith, C. D. (2007). Law, responsibility, and the brain. *PLoS Biology*, 5, 693–700. doi:10.1371/journal.pbio.0050103.t001
- Mokros, A., Neumann, C. S., Stadtland, C., Osterheider, M., Nedopil, N., & Hare, R. D. (2011). Assessing measurement invariance of PCL-R assessments from file reviews of North American and German offenders. *International Journal of Law and Psychiatry*, 34, 56–63.
- Mokros, A., Osterheider, M., Hucker, S. J., & Nitschke, J. (2011). Psychopathy and sexual sadism. *Law and Human Behavior*, 35, 188–199.
- Mokros, A., Stadtland, C., Osterheider, M., & Nedopil, N. (2010). Assessment of risk for violent recidivism through multivariate Bayesian classification. *Psychology, Public Policy, and Law*, 16, 418–450.
- Monahan, J., Steadman, H. J., Silver, E., Appelbaum, P. S., Robbins, P. C., Mulvey, E. P., . . . Banks, S. (2001). *Rethinking risk assessment: The McArthur study of mental disorder and violence*. New York: Oxford University Press.
- Morey, L. C. (1991). *Personality Assessment Inventory-professional manual*. Odessa, FL: Psychological Assessment Resources.
- Morrissey, C. (2003). The use of the PCL-R in forensic populations with learning disability. *British Journal of Forensic Practice*, 5, 20–24.
- Morrissey, C., Hogue, T. E., Mooney, P., Lindsay, W. R., Steptoe, L., Taylor, J., & Johnston, S. (2005). Applicability, reliability and validity of the Psychopathy Checklist-Revised in offenders with intellectual disabilities: Some initial findings. *International Journal of Forensic Mental Health*, 4, 207–220.
- Morrissey, C., & Hollin, C. (2011). Antisocial and psychopathic personality disorders in forensic intellectual disability populations: What do we know so far? *Psychology, Crime and Law*, 17, 133–149.
- Morrissey, C., Mooney, P., Hogue, T. E., Lindsay, W. R., & Taylor, J. L. (2007). Predictive validity of the PCL-R for offenders with intellectual disability in a high security hospital: Treatment progress. *Journal of Intellectual and Developmental Disability*, 32, 125–133.
- Murrie, D. C., Boccaccini, M. T., Turner, D. B., Meeks, M., Woods, C., & Tussey, C. (2009). Rater (dis)agreement on risk assessment measures in sexually violent predator proceedings: Evidence of adversarial allegiance in forensic evaluation? *Psychology, Public Policy, and Law*, 15, 19–53.
- Neumann, C. S., Hare, R. D., & Johansson, P. J. (2012, May 28). The Psychopathy Checklist-Revised (PCL-R), low anxiety, and fearlessness: A structural equation modeling analysis. *Personality Disorders: Theory, Research, and Treatment*. doi:10.1037/a0027886.
- Neumann, C. S., Hare, R. D., & Newman, J. P. (2007). The super-ordinate nature of the Psychopathy Checklist-Revised. *Journal of Personality Disorders*, 21, 102–117.

- Neumann, C. S., Vitacco, M. J., Hare, R. D., & Wupperman, P. (2005). Reconstructing the "reconstruction" of psychopathy: A comment on Cooke, Michie, Hart, and Clark. *Journal of Personality Disorders*, 19, 624–640.
- Newman, J. P., Hiatt, K. D., & MacCoon, D. G. (in press). Cognitive and affective neuroscience in disinhibitory psychopathology: Summary and integration. In D. Barch (Ed.), *Cognitive and affective neuroscience of psychopathology*. New York: Oxford University Press.
- Newman, J. P., & Schmitt, W. A. (1998). Passive avoidance in psychopathic offenders: A replication and extension. *Journal of Abnormal Psychology*, 107, 527–532.
- Ogloff, J. P. R., & Chopra, S. R. (2004). Stuck in the dark ages: Supreme Court decision-making and legal developments. *Psychology, Public Policy, and Law*, 10, 379–416.
- Olver, M. E., Neumann, C. S., Wong, S. C. P., & Hare, R. D. (2012). The structural and predictive properties of the Psychopathy Checklist-Revised in Canadian Aboriginal and non-Aboriginal offenders. *Psychological Assessment*. doi: 10.1037/a0029840.
- Olver, M. E., Stockdale, K. C., & Wormith, J. S. (2011). A meta-analysis of predictors of offender treatment attrition and its relationship to recidivism. *Journal of Consulting and Clinical Psychology*, 79, 6–21.
- Olver, M. E., & Wong, S. C. P. (2009). Therapeutic responses of psychopathic sexual offenders: Treatment attrition, therapeutic change, and long-term recidivism. *Journal of Consulting and Clinical Psychology*, 77, 328–336.
- Olver, M. E., & Wong, S. C. P. (2011). Predictors of sex offender treatment dropout: Psychopathy, sex offender risk, and responsivity implications. *Psychology, Crime and Law*, 17, 457–471.
- Patrick, C. J. (Ed.). (2005). *Handbook of psychopathy*. New York: Guilford Press.
- Paulhus, D. L., Hemphill, J. F., & Hare, R. D. (in press). *Manual for the Self-report Psychopathy Scale (SRP-IV)*. Toronto: Multi-Health Systems.
- Pedersen, L., Kunz, C., Rasmussen, K., & Elsass, P. (2010). Psychopathy as a risk factor for violent recidivism: Investigating the Psychopathy Checklist Screening Version (PCL:SV) and the Comprehensive Assessment of Psychopathic Personality (CAPP) in a forensic psychiatric setting. *International Journal of Forensic Mental Health*, 9, 308–315.
- Pereira, N., Huband, N., & Duggan, C. (2008). Psychopathy and personality: An investigation of the relationship between the NEO-Five Factor Inventory (NEO-FFI) and the Psychopathy Checklist-Revised (PCL-R) in a hospitalized sample of male offenders with personality disorder. *Criminal Behaviour and Mental Health*, 18, 216–223.
- Pham, T. H., Ducro, C., Maghem, B., & Réveillère, C. (2005). Évaluation du risque de récidive au sein d'une population de délinquants incarcérés ou internés en Belgique francophone / Prediction of recidivism among prison inmates and forensic patients in Belgium. *Annales Médico-psychologiques*, 163, 842–845.
- Pham, T. H., & Saloppé, X. (2010). PCL-R psychopathy and its relation to DSM Axis I and II disorders in a sample of male forensic patients in a Belgian security hospital. *International Journal of Forensic Mental Health*, 9, 205–214.
- Porter, S., Birt, A. R., & Boer, D. P. (2001). Investigation of the criminal and conditional release profiles of Canadian federal offenders as a function of psychopathy and age. *Law and Human Behavior*, 25, 647–661.
- Porter, S., & Porter, S. (2007). Psychopathy and violent crime. In H. Hervé & J. C. Yuille (Eds.), *The Psychopath: Theory, research, and practice* (pp. 287–300). Mahwah, NJ: Lawrence Erlbaum.
- Porter, S., ten Brinke, L., & Wilson, K. (2009). Crime profiles and conditional release performance of psychopathic and non-psychopathic sexual offenders. *Legal and Criminological Psychology*, 14, 109–118.
- Quinsey, V. L., Harris, G. T., Rice, M. E., & Cormier, C. (2006). *Violent offenders: Appraising and managing risk* (2nd ed.). Washington, DC: American Psychological Association.
- Quinsey, V. L., Jones, G. B., Book, A. S., & Barr, K. N. (2006). The dynamic prediction of antisocial behavior among forensic psychiatric patients. *Journal of Interpersonal Violence*, 21, 1539–1565.
- Quinsey, V. L., Rice, M. E., & Harris, G. T. (1995). Actuarial prediction of sexual recidivism. *Journal of Interpersonal Violence*, 10, 85–105.
- Reidy, D. E., Zeichner, A., & Seibert, L. A. (2011). Unprovoked aggression: Effects of psychopathic traits and sadism. *Journal of Personality*, 79, 75–100.
- Reimann, B. J., & Nussbaum, D. (2011). Predicting seclusion in a medium secure forensic inpatient setting. *International Journal of Forensic Mental Health*, 10, 150–158.
- Rice, M. E., & Harris, G. T. (1997). Cross-validation and extension of the Violence Risk Appraisal Guide for child molesters and rapists. *Law and Human Behavior*, 21, 231–241.

- Rice, M. E., Harris, G. T., & Cormier, C. A. (1992). An evaluation of a maximum security therapeutic community for psychopaths and other mentally disordered offenders. *Law and Human Behavior*, 16, 399–412.
- Richards, H. J., Casey, J. O., & Lucente, S. W. (2003). Psychopathy and treatment response in incarcerated female substance abusers. *Criminal Justice and Behavior*, 30, 251–267.
- Richards, H. J., Casey, J. O., Lucente, S. W., & Kafami, D. (2003). Differential association of Hare Psychopathy Checklist scores to HIV behaviors in incarcerated female substance abusers. *Individual Differences Research*, 2, 95–107.
- Saleh, F. M., Malin, H. M., Grudzinskas, A. J. Jr., & Vitacco, M. J. (2010). Paraphilias with co-morbid psychopathy: The clinical and legal significance to sex offender assessments. *Behavioral Sciences and the Law*, 28, 211–223.
- Salekin, R. T. (2008). Psychopathy and recidivism from mid-adolescence to young adulthood: Cumulating legal problems and limiting life opportunities. *Journal of Abnormal Psychology*, 117, 386–395.
- Salekin, R. T., Rogers, R., & Sewell, K. W. (1996). A review and meta-analysis of the Psychopathy Checklist-Revised: Predictive validity of dangerousness. *Clinical Psychology: Science and Practice*, 3, 203–215.
- Samuel, D. B., Lynam, D. R., Widiger, T. A., & Ball, S. A. (2012). An expert consensus approach to relating the proposed DSM-5 types and traits. *Personality Disorders: Theory, Research, and Treatment*, 3, 1–16.
- Schiffer, B., Müller, B. W., Scherbaum, N., Hodgins, S., Forsting, M., Wiltfang, J., . . . Leygraf, N. (2011). Disentangling structural brain alterations associated with violent behavior from those associated with substance use disorders. *Archives of General Psychiatry*, 68, 1039–1049.
- Schopp, R. F., Scalora, M. J., & Pearce, M. (1999). Expert testimony and professional judgment: Psychological expertise and commitment as a sexual predator after *Hendricks*. *Psychology, Public Policy, and the Law*, 5, 120–174.
- Serin, R. C., Mailloux, D. L., & Malcolm, P. B. (2001). Psychopathy, deviant sexual arousal, and recidivism among sexual offenders. *Journal of Interpersonal Violence*, 16, 234–246.
- Sjöstedt, G., & Långström, N. (2002). Assessment of risk for criminal recidivism among rapists: A comparison of four different measures. *Psychology, Crime and Law*, 8, 25–40.
- Skeem, J. L., Edens, J. F., Camp, J., & Colwell, L. H. (2004). Are there ethnic differences in levels of psychopathy? A meta-analysis. *Law and Human Behavior*, 28, 505–527.
- Sreenivasan, S., Walker, S. C., Weinberger, L. E., Kirkish, P., & Garrick, T. (2008). Four-facet PCL-R structure and cognitive functioning among high violent criminal offenders. *Journal of Personality Assessment*, 90, 197–200.
- Steadman, H. J., Silver, E., Monahan, J., Appelbaum, P. S., Robbins, P. C., Mulvey, E. P., . . . Banks, S. (2000). A classification tree approach to the development of actuarial violence risk assessment tools. *Law and Human Behavior*, 24, 83–100.
- Sullivan, E. A., Abramowitz, C. S., Lopez, M., & Kosson, D. S. (2006). Reliability and construct validity of the Psychopathy Checklist-Revised for Latino, European American, and African American male inmates. *Psychological Assessment*, 18, 382–392.
- Swogger, M. T., Walsh, Z., Homaifar, B. Y., Caine, E. D., & Conner, K. R. (2012). Predicting self- and other-directed violence among discharged psychiatric patients: The roles of anger and psychopathic traits. *Psychological Medicine: A Journal of Research in Psychiatry and the Allied Sciences*, 42, 371–379.
- Swogger, M. T., Walsh, Z., & Kosson, D. S. (2007). Domestic violence and psychopathic traits: Distinguishing the antisocial batterer from other antisocial offenders. *Aggressive Behavior*, 33, 1–8.
- Tengström, A. (2001). Long-term predictive validity of historical factors in two risk assessment instruments in a group of violent offenders with schizophrenia. *Nordic Journal of Psychology*, 55, 243–249.
- Tengström, A., Hodgins, S., Müller-Isberner, R., Jöckel, D., Freese, R., Özokuy, K., & Sommer, J. (2006). Predicting violent and antisocial behavior in hospital using the HCR-20: The effect of diagnoses on predictive accuracy. *International Journal of Forensic Mental Health*, 5, 39–53.
- Tengström, A., Grann, M., Långström, N., & Kullgren, G. (2000). Psychopathy (PCL-R) as a predictor of violent recidivism among criminal offenders with schizophrenia. *Law and Human Behavior*, 24, 45–58.
- Thornton, D., & Blud, L. (2007). The influence of psychopathic traits on response to treatment. In H. Hervé & J. C. Yuille (Eds.), *The Psychopath: Theory, research, and practice* (pp. 505–539). Mahwah, NJ: Lawrence Erlbaum.
- Vachon, D. D., Lynam, D. R., Loeber, R., & Stouthamer-Loeber, M. (2012). Generalizing the nomological network of psychopathy across populations differing on race and conviction status. *Journal of Abnormal Psychology*, 121, 263–269.

- Verona, E., Sprague, J., & Sadeh, N. (2012). Inhibitory control and negative emotional processing in psychopathy and antisocial personality disorder. *Journal of Abnormal Psychology, 121*, 498–510. doi:10.1037/a0025308
- Verona, E., & Vitale, J. (2006). Psychopathy in women: Assessment, manifestations, and etiology. In C. J. Patrick (Ed.), *Handbook of psychopathy* (pp. 415–436). New York: Guilford Press.
- Viding, E., Frick, P. J., & Plomin, R. (2007). Aetiology of the relationship between callous-unemotional traits and conduct problems in childhood. *British Journal of Psychiatry, 190* (Suppl. 49), s33–s38.
- Viding, E., Larsson, H., & Jones, A. P. (2008). Review: Quantitative genetic studies of antisocial behaviour. *Philosophical Transactions of the Royal Society B, 363*, 2519–2527 [special issue].
- Vitacco, M. J., Gonsalves, V., Tomony, J., Smith, B. E. R., & Lishner, D. A. (2012). Can standardized measures of risk predict inpatient violence? Combining static and dynamic variables to improve accuracy. *Criminal Justice and Behavior, 39*, 589–606.
- Vitacco, M. J., Neumann, C. S., & Jackson, R. L. (2005). Testing a four-factor model of psychopathy and its association with ethnicity, gender, intelligence, and violence. *Journal of Consulting and Clinical Psychology, 73*, 466–476.
- Vitacco, M. J., Rogers, R., Neumann, C. S., Harrison, K. S., & Vincent, G. (2005). A Comparison of factor models on the PCL–R with mentally disordered offenders: The development of a four-factor model. *Criminal Justice and Behavior, 32*, 526–545.
- Vitale, J. E., Smith, S. S., Brinkley, C. A., & Newman, J. P. (2002). The reliability and validity of the Psychopathy Checklist–Revised in a sample of female offenders. *Criminal Justice and Behavior, 29*, 202–231.
- Wahlund, K., & Kristiansson, M. (2009). Aggression, psychopathy and brain imaging: Review and future recommendations. *International Journal of Law and Psychiatry, 32*, 266–271.
- Wallinius, M., Nilsson, T., Hofvander, B., Anckarsäter, H., & Stålenheim, G. (2012). Facets of psychopathy among mentally disordered offenders: Clinical comorbidity patterns and prediction of violent and criminal behavior. *Psychiatry Research*. doi:10.1016/j.psychres.2012.01.005.
- Walsh, Z., Allen, L. C., & Kosson, D. S. (2007). Beyond social deviance: Substance use disorders and the dimensions of psychopathy. *Journal of Personality Disorders, 21*, 273–288.
- Walsh, Z., & Kosson, D. S. (2007). Psychopathy and violent crime: A prospective study of the influence of socioeconomic status and ethnicity. *Law and Human Behavior, 31*, 209–229.
- Walsh, Z., & Kosson, D. S. (2008). Psychopathy and violence: The importance of factor level interactions. *Psychological Assessment, 20*, 114–120.
- Walsh, Z., Swogger, M. T., & Kosson, D. S. (2004). Psychopathy, IQ, and violence in European American and African American county jail inmates. *Journal of Consulting and Clinical Psychology, 72*, 1165–1169.
- Walsh, Z., Swogger, M. T., O'Connor, B. P., Chatav Schonbrun, Y., Shea, M. T., & Stuart, G. L. (2010). Subtypes of partner violence perpetrators among male and female psychiatric patients. *Journal of Abnormal Psychology, 119*, 563–574.
- Walsh, T., & Walsh, Z. (2006). The evidentiary introduction of Psychopathy Checklist–Revised assessed psychopathy in U.S. courts: Extent and appropriateness. *Law and Human Behavior, 30*, 493–507.
- Walters, G. D. (2003). Predicting institutional adjustment and recidivism with the Psychopathy Checklist factor scores: A meta-analysis. *Law and Human Behavior, 27*, 541–558.
- Walters, G. D., & Heilbrun, K. (2010). Violence risk assessment and facet 4 of the Psychopathy Checklist: Predicting institutional and community aggression in two forensic samples. *Assessment, 17*, 259–268.
- Walters, G. D., Knight, R. A., Grann, M., & Dahle, K. (2008). Incremental validity of the Psychopathy Checklist facet scores: Predicting release outcome in six samples. *Journal of Abnormal Psychology, 117*, 396–405.
- Warren, J. I., Burnette, M., South, C. S., Chauhan, P., Bale, R., Friend, R., & Van Patten, I. (2003). Psychopathy in women: Structural modeling and co-morbidity. *International Journal of Law and Psychiatry, 26*, 223–242.
- Webster, C., Douglas, K., Eaves, D. & Hart, S. (1997). *HCR-20 assessing risk for violence: Version II*. Burnaby, BC: Mental Health, Law & Policy Institute, Simon Fraser University.
- Weiler, B. L., & Widom, C. S. (1996). Psychopathy and violent behaviour in abused and neglected young adults. *Criminal Behaviour and Mental Health, 6*, 253–271.
- Widiger, T. A., & Mullins-Sweatt, S. (2005). Categorical and dimensional models of personality disorder. In J. Oldham, A. Skodol, & D. Bender (Eds.), *Textbook of personality disorders* (pp. 35–53). Washington, DC: American Psychiatric Press.

- Williams, K. M., Paulhus, D. L., & Hare, R. D. (2007). Capturing the four-factor structure of psychopathy in college students via self-report. *Journal of Personality Assessment*, 88, 205–219.
- Wilson, R. J., Looman, J., Abracen, J., & Pake, D. R. Jr. (2012). Comparing sexual offenders at the Regional Treatment Centre (Ontario) and the Florida Civil Commitment Center. *International Journal of Offender Therapy and Comparative Criminology*. doi: 10.1177/0306624X11434918.
- Wong, S. C. P., Gordon, A., & Gu, D. (2007). Assessment and treatment of violence-prone forensic clients: An integrated approach. *British Journal of Psychiatry*, 190, s66–s74.
- Wong, S. C. P., & Hare, R. D. (2005). *Guidelines for a psychopathy treatment program*. Toronto: Multi-Health Systems.
- Woodworth, M., & Porter, S. (2002). In cold blood: Characteristics of criminal homicides as a function of psychopathy. *Journal of Abnormal Psychology*, 111, 436–445.
- Zolondek, S., Lilienfeld, S. O., Patrick, C. J., & Fowler, K. A. (2006). The interpersonal measure of psychopathy: Construct and incremental validity in male prisoners. *Assessment*, 13, 470–482.
- Zuckerman, M. (2003). Are there racial and ethnic differences in psychopathic personality? A critique of Lynn's (2002) racial and ethnic differences in psychopathic personality. *Personality and Individual Differences*, 35, 1463–1469.
- Žukauskienė, R., Laurinavičius, A., & Čėsniienė, I. (2010). Testing factorial structure and validity of the PCL:SV in Lithuanian prison population. *Journal of Psychopathology and Behavioral Assessment*, 32, 363–372.

9

THE HARE PSYCHOPATHY CHECKLIST–YOUTH VERSION

Angela S. Book, Adelle E. Forth, and Heather J. Clark

The Hare Psychopathy Checklist: Youth Version (PCL:YV; Forth, Kosson, & Hare, 2003) is designed to measure psychopathic traits in adolescents. Although not designed for any specific forensic purpose, because of the growing interest of the construct of psychopathy in many settings, it has increasingly been used in forensic contexts. The purpose of this chapter is to discuss the development of the PCL:YV, as well as its psychometric properties, utility, and limitations.

Description and Purpose of the Instrument

The publication of the Hare Psychopathy Checklist–Revised (PCL–R; Hare, 1991, 2003) marked the beginning of an emerging area of research. Today there is a substantial body of research and, as a result, we have a much clearer understanding of psychopathy. As interest in psychopathy grew, there was a desire to understand the development of psychopathy early in life. Psychopathy does not emerge suddenly in adulthood. Many researchers and clinicians believe that psychopathic traits are first manifested early in life (Salekin & Lynam, 2010). In order to understand how psychopathy manifests early in life, there was a need for an instrument to measure psychopathic traits in youth. Forth and colleagues (2003) adapted the PCL:YV from the PCL–R, modifying traits applicable only to adults (e.g., many short-term marital relationships) to more developmentally relevant indicators.

The PCL:YV is designed to measure psychopathic traits in adolescents aged 12–18 years. The PCL:YV emphasizes the need for information from several sources and domains in order to adequately assess 20 psychopathic traits. Each item is rated on a three-point scale (0 = the item definitely does not apply; 1 = the item applies to some degree; 2 = the item definitely applies). A total summed score can range from 0 to 40. Table 9.1 lists the items and factors in the PCL:YV. No exclusion criteria are used when scoring the items, a feature that permits the assessment of youth with different psychiatric diagnoses and co-morbid diagnoses. The PCL:YV user qualifications requires users to be familiar with adolescent development prior to use. There is no PCL:YV cut score for a diagnosis of psychopathy. The Manual provides *T*-scores and percentiles for Total and Factor scores for institutionalized male offenders, male probationers, and small samples of female offenders and community males. All samples were between the ages of 12 and 18 years. The rater should choose the most appropriate comparison sample and then obtain the *T*-score or percentile rank for the individual being assessed.

PCL:YV items are scored both from a semi-structured interview with the youth and a review of available file and collateral information. The PCL:YV interview takes approximately 90 minutes, may be spread over several sessions, and covers a variety of domains (educational, substance use, family and peer relationships, and antisocial history). The other major source of information used in the PCL:YV is a review of charts and collateral information. Depending on the amount of file information available, an interview with the youth's caregiver or guardian may be needed. For clinical assessments, it is required that the evaluator have access to collateral information in order to check the accuracy of the information given in the interview. Some researchers (Campbell, Porter, & Santor, 2004; Gretton, Hare, & Catchpole, 2004; Gretton, McBride, Hare, O'Shaughnessy, & Kumka, 2001; Marczyk, Heilbrun, Ladner, & DeMatteo, 2003; O'Neill, Lidz, & Heilbrun, 2003) have used extensive file information exclusively to generate PCL:YV scores (i.e., no interview was conducted). While not optimal, file-only PCL:YV scores are acceptable when it is impossible to conduct an interview or when an archival study is being conducted.

History of the Instrument's Development

A key decision made by the authors was to use the item content of the PCL–R to guide development of the PCL:YV. Thus, the PCL:YV is a downward extension of the PCL–R that assesses similar content domains. The first study to examine psychopathic traits in youth was by Forth, Hart, and Hare (1990). A number of modifications were made to the PCL–R in this study. Because adolescents have a limited work history and few marital relationships, PCL–R Items 9 (Parasitic lifestyle) and 17 (Many short-term marital relationships) were deleted. Furthermore, because adolescent offenders have had less opportunity than adult offenders to come into contact with the judicial system, it was necessary to modify two other items: item 18 (Juvenile delinquency) and item 20 (Criminal versatility). This original version of the scale has been referred to as the modified PCL–R or the 18-item PCL–R. Several studies used this version to study psychopathic traits in youth (e.g., Brandt, Kennedy, Patrick, & Curtin, 1997; Harvey, Stokes, Lord, & Pogge, 1996; Myers, Burket, & Harris, 1995).

Subsequently, additional modifications were made. First, raters are now required to compare the behavior of the youth they are assessing to same-age peers. This is done to ensure that the trait or behavior being rated is more extreme than what is normative for that age group. Second, items were modified to include descriptions of life domains more relevant to adolescents, such as peers, family, and school. Third, items 9 and 17 were modified to permit assessments of these characteristics in youth. Fourth, when scoring individual items the focus was placed on relatively enduring features of the individual, displayed across settings and situations since late childhood. This modified version of the PCL–R was named the *Psychopathy Checklist: Youth Version* (PCL:YV).

Generalizability, Factor Structure, and Reliability

Data from 2,438 youth in three countries (Canada, the United States, and the United Kingdom) across 19 different samples are described in the Manual. Research has also been conducted with institutionalized young offenders, young offenders on probation, psychiatric inpatient youth, and youth in the community.

Generalizability

PCL:YV total scores vary across samples, with institutionalized youth obtaining a relatively high score, followed closely by youth on probation. Although there have only been a few studies using the PCL:YV with community youth, the studies that have been conducted show very low scores. Within institutionalized and probation samples, there is a small negative correlation between PCL:YV total score and age at assessment. Younger adolescents scored slightly higher on the PCL:YV as compared to their older counterparts.

There have been an increasing number of studies assessing psychopathic traits in female adolescents. In the manual data set, both incarcerated and probation female adolescents were rated as slightly less psychopathic than males (about 1 point). Other researchers have reported no sex differences in the prevalence of youth scoring high on the PCL:YV (Campbell et al., 2004; Salekin, Neumann, Leistico, & Zalot, 2004). Table 9.1 provides PCL:YV scores for male and female adolescents across several recent studies. In most of these studies, there were again no sex differences observed.

Ethnicity is another important variable to consider (Sullivan & Kosson, 2006) especially considering the over-representation of minorities in the criminal justice system. Many of the data on ethnicity compare Caucasians with non-Caucasians. The strength of the correlation between PCL:YV total scores and ethnicity in the Manual data set ranges from .03 in probation samples to .17 in institutionalized samples, with non-Caucasians scoring from 1 to 3 points higher than Caucasians. A meta-analysis of a combined sample of 2,199 participants found that African-American youths scored significantly higher than Caucasian youths; however, the difference was on average 1.5 points and the effect was small in magnitude (Cohen $d = .20$; McCoy & Edens, 2006). Schmidt, McKinnon, Chattha, and Brownlee (2006) also reported no differences between White and Aboriginal young offenders on the PCL:YV. However, more research is needed to determine whether the PCL:YV is metrically equivalent across different ethnic and cultural groups.

TABLE 9.1 Hare Psychopathy Checklist: Youth Version (PCL:YV) Scores across Male and Female Adolescents

<i>Study</i>	<i>Population</i>	<i>Males: N, Mean (SD)</i>	<i>Females, N, Mean (SD)</i>	<i>Significance</i>
Sevecke et al. (2009)	Delinquent community, Germany	143, 26.9 (6.6) 99, 5.0 (6.5)	171, 19.7 (6.9) 94, 2.4 (3.5)	M > F** M > F**
Andershed et al. (2007)	Substance abusers, Sweden	70, 15.4 (7.6)	92, 12.4 (5.8)	M > F*
Campbell et al. (2004)	Incarcerated/remanded, Canada	188, 16.4 (6.1)	38, 15.5 (5.6)	ns
Vincent et al. (2008)	Incarcerated, Canada	201, 22.2 (6.7)	55, 20.6 (6.3)	ns
Penney & Moretti (2007)	Incarcerated, Canada	76, 24.1 (7.2)	66, 19.7 (6.8)	M > F*
Schmidt et al. (2011)	Incarcerated, Canada	80, 22.2 (6.7)	48, 21.8 (5.8)	ns
Cook et al. (2010)	Incarcerated, United States	50, 17.7 (6.2)	50, 18.0 (5.2)	ns
Stockdale et al. (2010)	Incarcerated, Canada	88, 22.4 (8.4)	73, 20.9 (6.6)	ns

Note:

Campbell et al. (2004), Cook et al. (2010), and Stockdale et al. (2010) used file-only ratings; Campbell et al. used an 18-item scale; ns = not significant.

* $p < .01$; ** $p < .001$.

Factor Analyses

Research on the PCL:YV shows that both the three- and the four-factor models provide acceptable fit (Forth et al., 2003; Jones, Cauffman, Miller, & Mulvey, 2006; Neumann, Kosson, Forth, & Hare, 2006; Salekin, 2006; Salekin, Brannen, Zalot, Leistico, & Neumann, 2006). Forth and colleagues (2003) and Neumann et al. (2006) have argued that for various statistical, methodological, and theoretical reasons, the four-factor model might be the preferred model. The items loading on the four-factor model are indicated in Table 9.2. Others have argued that the three-factor model is a better fit (Cauffman, Kimonis, Dmitrieva, & Monahan, 2009; Jones et al., 2006). The three-factor model consists of the first three factors and excludes the fourth factor. A recent study of the three- and four-factor models of the PCL:YV in a German sample of both male and female detainees and community youth found the four-factor model to be problematic in all samples, but the three-factor model to be adequate in incarcerated and community males (but not in females; Sevecke, Pukrop, Kosson, & Krischer, 2009). In a large-scale study of 646 female adolescents in North America and Europe, Kosson et al. (in press) reported an adequate fit for both the three- and the four-factor model in detained and community subsamples, but only the three-factor model achieved adequate fit in the incarcerated subsample. Determining which of these models is optimal remains an ongoing research question.

Reliability

Interrater reliability is high for total scores (ICC_1 of .90–.96; Forth et al., 2003). Other researchers have obtained good to excellent interrater reliability for total scores (ICCs range from .82 to .98; see Andershed, Hodgins, & Tengström, 2007; Cauffman et al., 2009; Das, de Ruiter, Doreleijers, & Hillege, 2009). In a recent meta-analysis, Olver and Stockdale (2010) reported very high interrater reliability for PCL:YV total score (.90) and good reliability for each of the PCL:YV factor scores (.75–.85). Despite these strong results, the authors note there was significant variability across samples. This variability indicates there are some situations in which reliability not as high. It is likely that the interrater reliability of PCL:YV total and factor scores will be lower in clinical and forensic settings, where youth may be more motivated to engage in impression management, where the amount of training is more variable, and where there may be more adversarial bias seen.

TABLE 9.2 Items and Factors in the PCL:YV

<i>Factor 1: Interpersonal</i>		<i>Factor 2: Affective</i>	
1.	Impression management	6.	Lack of remorse
2.	Grandiose sense of self-worth	7.	Shallow affect
4.	Pathological lying	8.	Callous/lack of empathy
5.	Manipulation for personal gain	16.	Will not accept responsibility
<i>Factor 3: Lifestyle</i>		<i>Factor 4: Antisocial</i>	
2.	Need for stimulation	10.	Poor anger control
9.	Parasitic orientation	12.	Early behavioral problems
13.	Lack of goals	18.	Serious criminal behavior
14.	Impulsivity	19.	Serious violations of release
15.	Irresponsibility	20.	Criminal versatility

Note:

Item 11, Impersonal sexual behavior, and Item 17, Unstable interpersonal relationships, contribute to the Total PCL:YV score but do not load on any factors.

There is good evidence for the internal consistency of the PCL:YV, with alpha coefficients ranging from .79 to .94 (Dolan & Rennie, 2006; Forth et al., 2003). Within the PCL:YV (Forth et al., 2003) Manual, the mean inter-item correlations were all above .20 across settings. There has been only one published study that has assessed the test–retest reliability of the PCL:YV in a sample of incarcerated youth. Skeem and Cauffman (2003) reported an ICC of .66 for the PCL:YV total score over a one-month follow-up.

Validity

In evaluating any measure, it is important to determine whether validity has been demonstrated. The import of this is further heightened by the possible consequences of a high score on the PCL:YV, including harsher sentences, the possible withholding of treatment, and the self-fulfilling prophecy (Forth & Book, 2010). The purpose of the following section is to review the validity of the PCL:YV as a measure of psychopathy in adolescents. First, we will briefly evaluate the content validity of the PCL:YV. We will then evaluate construct validity, determined by examining (1) the co-morbidity of psychopathic traits with internalizing and externalizing symptoms/behaviors, (2) the relationship between psychopathic traits and aggression, and (3) the utility of the PCL:YV as a predictive tool in assessing risk for recidivism.

Content Validity

A test demonstrates content validity if it adequately taps the construct under investigation (Carmines & Zeller, 1991). Researchers have examined the conceptualization of psychopathy in adolescence with samples of clinicians (Salekin, Rogers, & Machin, 2001) and juvenile justice personnel (Cruise, Colwell, Lyons, & Baker, 2003). In both studies, participants were asked to give ratings to prototypical psychopathic traits for both male and female adolescents. Interpersonal, affective, behavioral, and antisocial aspects were all found to be important indicators of psychopathy in adolescence, and all of them are tapped in the PCL:YV. These findings suggest that the PCL:YV meets the criteria of face validity, though the number of studies is small and findings should be replicated in future research.

Construct Validity

As already mentioned, it is also important to establish the construct validity, and this involves evaluating (1) concurrent validity (e.g., the measure is related to other measures of the same construct), and (2) convergent validity (e.g., the measure is related to appropriate criterion variables). We will begin by evaluating the concurrent validity of the PCL:YV, followed by convergent validity.

According to the PCL:YV Manual, scores tend to be moderately correlated with scores on screening measures for psychopathy that are based on the PCL–R (Forth et al., 2003). Further, PCL:YV scores tend to be moderately correlated with non-PCL measures, including the Antisocial Process Screening Device (APSD; Frick & Hare, 2001), the Childhood Psychopathy Scale (CPS; Lynam, 1997), the Youth Psychopathy Scale (YPS; Andershed, Gustafson, Kerr, & Stattin, 2002), and the Psychopathy Content Scale from the Millon Adolescent Clinical Inventory (PCS; Millon, 1993).

Several studies have examined the correlation between the PCL:YV and the APSD (Frick & Hare, 2001). For example, Lee, Vincent, Hart, and Corrado (2003) conducted a study on 100 incarcerated male adolescent offenders, examining the relationship between PCL:YV scores and

the self-report version of the APSD. The total scores for each scale were correlated at .40, which is a moderate to strong relationship. Factor scores were slightly less correlated (range = .20–.37), possibly because of the fact that the APSD was self-report while the PCL:YV was not. In another study of 130 juvenile offenders, Kuback and Salekin (2009) found that the totals were correlated at .37, and factor scores were correlated at around .30. Similarly, Murrie and Cornell (2002) found that APSD scores were correlated with the PCL:YV total score at .30 and .35 for self-report and staff rating respectively. Spain et al. (2004) also included the PCL:YV and APSD as measures in a study of 85 adjudicated delinquents, and while they did not report correlations between the measures, the PCL:YV and APSD demonstrated similar patterns of correlations with outcome measures (e.g., physical/verbal/administrative incidents). The authors did note that correlations with outcomes were higher for the APSD. Finally, Douglas, Epstein, and Poythress (2008) conducted a study on 83 delinquent youths. They found a moderate correlation ($r = .35$) between the self-report APSD and the PCL:YV. Factor scores were correlated between .04 and .29, but not all correlations were reported, and the range includes correlations between mismatched factors. From the studies above, it would seem that there is a moderate positive correlation between the APSD and the PCL:YV, supporting the concurrent validity of the PCL:YV. This is further supported by a review of 10 studies conducted by Frick and Hare (2001), who noted that the relationship was a moderate one.

Two studies have examined the relationship between the PCL:YV and the Child Psychopathy Scale (CPS; Lynam, 1997). First, Douglas, Epstein, and Poythress (2008) found a correlation of .27 for total scores, and a range from .05 to .34 for subscales. Again, it should be noted that this range of correlations includes relationships between subscales that are not supposed to be related. In a second study, Spain, Douglas, Poythress, and Epstein (2004) examined the relationships between these scales and institutional incidents. While correlations between the two measures were not provided, the PCL:YV and modified CPS were similarly correlated with incident reports, although the correlations were higher in the case of the CPS.

Another popular measure of psychopathic traits is the Youth Psychopathic Traits Inventory (YPI; Andershed et al., 2002), and several studies have examined the overlap between it and the PCL:YV. For example, Cauffman et al. (2009) found moderate correlations between the two total scores, and small to medium relationships for subscale scores in 1,170 serious juvenile offenders. Similarly, Skeem and Cauffman (2003) found moderate relationships (for total and relevant factor scores) in a sample of 160 serious adolescent offenders. This finding was mirrored by the results from Dolan and Rennie (2006), which included 115 male adolescents with conduct disorder. Finally, Andershed et al. (2007) found moderate correlations for totals and relevant factors in adolescents with substance misuse problems.

The last scale that has been compared to the PCL:YV is the Psychopathy Content Scale (a subscale of the Millon Adolescent Clinical Inventory; Millon, 1993). Murrie and Cornell (2002) examined this relationship and found that total scores were correlated at .49 in a sample of 113 juvenile offenders.

Overall, then, the PCL:YV demonstrates moderate relations with other measures of the same construct, and thus exhibits concurrent validity. This conclusion is not surprising, given that most of the comparison measures were developed and validated on the basis of their relationship with scores on the PCL:YV, and because the PCL:YV is the standard comparison for the other measures (Kotler & McMahon, 2010).

Divergent and Convergent Validity

Internalizing Symptoms and Disorders

The PCL:YV should be unrelated to variables that have been shown to be unrelated to psychopathy in adults, thus demonstrating divergent validity. We will examine the relationship between psychopathic traits (as measured by the PCL:YV) and internalizing behaviors and disorders, including anxiety, depression, self-harming behavior, and a history of trauma.

Within the literature, variables that should be theoretically unrelated to psychopathic traits include internalizing disorders, anxiety, depression, and negative affect. Psychopathy in adults (as measured by the PCL–R) has not been shown to be correlated with either anxiety or depression (Hare, 2003). Cleckley's (1941) original conceptualization of psychopathy includes a distinct lack of nervousness and neuroticism. While this conceptualization is still common (e.g., Hare, 2003), some researchers have suggested that there are two different types of psychopathic individuals, differentiated by their level of anxiety. Blackburn and Coid (1998) separated psychopaths into two types: primary (low anxiety) and secondary (high anxiety). In adolescents, a similar classification has been proposed: *undersocialized delinquents*, who are low in anxiety, and *neurotic delinquents*, who are high in anxiety (Quay, Routh, & Shapiro, 1987).

Is there any association between anxiety and psychopathic traits? Das, de Ruiter, and Doreleijers (2008) found that those scoring high on the PCL:YV had lower social avoidance and lower social anxiety. Further, Hodgins, Barbareschi, and Larsson (2011) found that youth with conduct disorder *and* an anxiety disorder did not have significantly different PCL:YV scores than those with conduct disorder alone. They were also similar to those with conduct disorder alone on substance use, delinquency, and aggression, suggesting that the existence of anxiety disorders is not mitigating or aggravating the tendency toward antisocial and externalizing behaviors.

The link between psychopathic traits and affective disorders has recently been examined by several researchers. As mentioned above, this relationship appears to be non-significant in adult populations (Hare, 2003). Similarly, Bauer, Whitman, and Kosson (2011) found no relationship between PCL:YV scores and depression and dysthymia in female adolescent offenders. Much of the research has examined the relationship between negative affectivity and psychopathic traits. This relationship in adults has been found to be non-significant in adult populations (Hare, 2003; Schmitt & Newman, 1999) for the most part. One study, though, did find a positive relationship in adults (Hale, Goldstein, Abramowitz, Calamari, & Kosson, 2004).

In adolescents, the findings have been mixed. Some researchers have found a negative relationship between PCL:YV scores and negative affectivity (Dolan & Rennie, 2007; Murrie & Cornell, 2000), some have found no relationship (Campbell et al., 2004; O'Neill et al., 2003), and still others have found positive associations (Bauer et al., 2011; Kosson, Cyterski, Steuerwald, Neumann, & Walker-Matthews, 2002; Schmidt et al., 2006).

One behavior that is often linked with affective disorders is self-directed aggression. Only one study has directly examined this issue, and a positive relationship with PCL:YV scores was found for girls but not for boys (Sevecke et al., 2009).

The nature of the relationship between psychopathic traits and affective disorders remains unclear for adolescent samples. Further, it is not clear whether any existing relationship with affective problems affects the relationships with other important outcome variables, such as aggression.

Some researchers have examined internalizing symptoms and behaviors more generally. For example, Cook, Barese, and Dicataldo (2010) found that high PCL:YV scores in female adolescents were correlated with both psychiatric hospitalizations and mental health needs. Similarly, Salekin, Neumann, Leistico, DiCicco, and Duros (2004) found that scores were positively related to

internalizing psychosocial problems ($r = .20$), and several researchers have found positive relationships with internalizing symptoms (e.g., Hemphälä & Tengström, 2010; Schmidt et al., 2006). On the other hand, Campbell et al. (2004) found no relationship between PCL:YV scores and internalizing symptoms, and Das et al. (2009) found negative relationships with social anxiety and introversion, especially for the Interpersonal dimension. As with affective disorders, then, the picture remains unclear.

One risk factor that has been identified in relation to affective disorders and internalizing symptoms, more generally, is childhood traumatization. Because adult psychopathy has been linked with childhood maltreatment (Marshall & Cooke, 1999) and severe victimization history (Lang, af Klinteberg, & Alm, 2002), it is logical to expect similar relationships in adolescents, and this assumption has been borne out in the literature. For example, Campbell et al. (2004) found that adolescent offenders scoring high on the PCL:YV were more likely to have a history of physical abuse, and a number of studies have found that the link between psychopathy and abuse is stronger in girls (Abram et al., 2004; Cauffman, Feldman, Waterman, & Steiner, 1998; Dixon, Howie, & Starling, 2004).

The research on internalizing symptoms and behaviors in relation to psychopathic traits in youth is mixed. Anxiety shows a negative or non-existent relationship, while there is evidence of childhood trauma in the histories of those with psychopathic traits. The evidence for affective disorders is mixed, while there is a positive relationship with self-directed aggression in girls in one study. The findings regarding general internalizing symptoms and behaviors have also been mixed, and thus there is no clear conclusion to be drawn at this point. Further research needs to be conducted in order to determine whether these symptoms are related to (or unrelated to) psychopathic traits in adolescents, and whether these relationships might affect the correlation between PCL:YV scores and important outcome variables.

Externalizing Symptoms, Disorders, and Behaviors

To demonstrate convergent validity, PCL:YV scores should be related to externalizing disorders, specifically because of the increased likelihood of antisocial and violent behavior that is typical of these disorders. Externalizing disorders that will be examined include Conduct Disorder (CD; American Psychiatric Association, 2000), Oppositional Defiant Disorder (ODD; American Psychiatric Association, 2000), Attention Deficit Hyperactivity Disorder (ADHD; American Psychiatric Association, 2000), substance abuse, and behavioral problems, including aggression.

Psychopathic traits include the presence of antisocial behaviors, and, therefore, there should be an inherent link with externalizing behaviors. In fact, Gretton et al. (2004) found that PCL:YV scores were positively correlated with externalizing behaviors, including aggression. This finding was replicated by Schmidt et al. (2006) in a community-based sample of 130 adjudicated youths. Further evidence comes from a study conducted by Hemphälä and Tengström (2010). In this sample of 180 adolescents from a substance abuse clinic, a positive correlation was once again observed between PCL:YV scores and externalizing behaviors. In this study, the behavioral dimension (Factor 2) of psychopathy was most important in predicting these behaviors.

This finding has also been replicated cross-culturally. Das et al. (2008) found that total and factor scores on the PCL:YV were associated with behavioral problems in a sample of 67 Dutch adolescents. In a second study on Dutch adolescents, Das et al. (2009) found significant positive correlations between PCL:YV scores (total and subscale) and MMPI–A scores measuring antisocial and externalizing behaviors, except for the content scale for school problems. In both of the above studies, it was, again, the behavioral dimensions that were most predictive of externalizing and antisocial behaviors (Lifestyle and Antisocial factors).

In general, then, the PCL:YV exhibits positive association with externalizing behaviors, including aggression (e.g., Grevet et al., 2004; Schmidt et al., 2006), and this appears to be true across sex (Hemphälä & Tengström, 2010). Further, it appears to be the behavioral dimension of the PCL:YV that is most predictive of externalizing behaviors (Hemphälä & Tengström).

Given the relationship between psychopathic traits and externalizing behaviors (e.g., Das et al., 2009), one would also expect a positive relationship between PCL:YV scores and CD, ODD, and, possibly, ADHD symptoms. Not surprisingly, a number of studies have found that PCL:YV scores are positively correlated with CD and ODD symptoms (Campbell et al., 2004; Forth & Burke, 1998; Grevet et al., 2001; Kosson et al., 2002; Mailloux, Forth, & Kroner, 1997; Salekin, Neumann, Leistico, DiCicco, & Duros, 2004). Lee, Salekin, and Iselin (2010) found the same relationships, as well as a significant positive relationship with ADHD.

Several studies have examined the relationship between psychopathic traits (as measured by the PCL:YV) and substance use or abuse. In an early study, Forth and Burke (1998) found that PCL:YV scores were positively correlated with both alcohol and substance use, and others have had similar results (Harvey et al., 1996; Lee et al., 2010; Roussy & Toupin, 2000; Salekin, Neumann, Leistico, DiCicco, & Duros, 2004), although one study found that this relationship was weaker in female adolescents (Bauer et al., 2011), and a couple of studies did not find a relationship, even in male participants (Das et al., 2009; O'Neill et al., 2003). In the Das et al. study, there was no relationship between PCL:YV scores and proneness to use alcohol or drugs, but there was a correlation with self-acknowledgment of a drug or alcohol problem.

In adults, psychopathic traits are also related to personality disorder symptomology, and thus this should also be the case in youth. Psychopathic traits should also be related to symptoms of personality disorders, as they are in adults (Hart & Hare, 1989; Huchzermeyer et al., 2007; Soderstrom, Nilsson, Sjödin, Carlstedt, & Forsman, 2005; Weizmann-Henelius, Viemerö, & Eronen, 2004). More specifically, adult psychopathy is related to symptoms of Antisocial Personality Disorder, Histrionic Personality Disorder, Narcissistic Personality Disorder, Borderline Personality Disorder, and Schizotypal Personality Disorder. Each of the above disorders is defined, in part, by the existence of negative personality traits. One study investigated negative personality traits and their relationship with psychopathic traits. Lee et al. (2010) found that PCL:YV scores were positively correlated with the number of negative personality traits in a sample of 94 male adolescent offenders. Further, Salekin, Neumann, Leistico, DiCicco, and Duros (2004) found that individuals with higher scores were more likely to have interpersonal problems, which is a defining feature of most personality disorders. There have not been any studies specifically examining the link between PCL:YV scores and personality disorders, although from the above studies it would seem that adolescents with higher PCL:YV scores show similar negative personality traits (Lee et al., 2010) and interpersonal problems (Salekin, Neumann, Leistico, DiCicco, & Duros, 2004).

In summary, PCL:YV scores show the expected relationships with externalizing behaviors, externalizing disorders (e.g., CD, ODD, ADHD), and substance use or abuse. This conclusion, combined with the fact that PCL:YV scores correlate at least moderately with other measures of psychopathic traits, suggests that the PCL:YV exhibits convergent validity.

Motivations for Crime and Aggression

A large literature exists that validates the strong relationship between psychopathy and aggression in adults (see Hare, 2003, for a review). In adolescents, studies have examined the link between PCL:YV scores and age of onset of criminal activity (e.g., Vincent, Vitacco, Grisso, & Corrado, 2003) and the frequency and versatility of general and violent offenses (e.g., Campbell et al., 2004). Research that has focused on age of onset for criminal activity suggests that, like adults, adolescents

with psychopathic traits (as measured by the PCL:YV) are more likely to begin criminal activity earlier. For example, Vincent et al. (2003) found that adolescent offenders who scored high on the PCL:YV were first convicted at an earlier age, and Brandt et al. (1997) report similar findings for age of first arrest. Salekin et al. (2006) and Kosson et al. (2002) also report negative correlations between PCL:YV scores and age of onset, but the findings were not significant. Overall, there does appear to be a negative relationship between PCL:YV scores and age of onset, which is consistent with the research on adult psychopaths (Hare, 2003). This relationship remains when items such as early behavioral problems and juvenile delinquency are removed from the PCL:YV prior to the analyses.

As mentioned above, psychopathic traits in adolescents are also related to the number and versatility of offenses. For example, Vincent et al. (2003) found that high scorers on the PCL:YV had more committed non-violent and violent offenses than lower scorers. In another study, Campbell et al. (2004) report a significant positive correlation between PCL:YV scores and self-reported delinquency, aggression, and versatility. However, scores were not correlated with official records of convictions. Similarly, Kosson et al. (2002) found moderate correlations between PCL:YV scores and violent, non-violent, and total number of charges. Murrie, Cornell, Kaplan, McConville, and Levy-Elkon (2004) also report moderate relationships between the PCL:YV and violent offenses.

Psychopathic traits in adults are clearly linked with aggressive behavior (Hare, 2003), and similar findings are available in the adolescent literature (e.g., Salekin et al., 2004). A meaningful distinction of classification of aggressive behavior is that into two subtypes: a premeditated or instrumental subtype (goal-oriented, planned, and “cold-blooded”) and an impulsive or reactive subtype (impulsive, unplanned, and emotion driven). Researchers have found that psychopathic adults tend to use instrumental aggression more often than reactive aggression (Walsh, Swogger, & Kosson, 2009; Woodworth & Porter, 2002). One would expect, then, that psychopathic traits in youth would also be related more strongly to instrumental aggression than to reactive aggression.

Several studies have examined the motivation for aggression in relation to psychopathic traits in adolescence. Research using a classification system developed by Cornell et al. (1996) has commonly found a relationship between psychopathic traits and instrumental aggression (Flight & Forth, 2007; Murrie et al., 2004; Vitacco, Neumann, Caldwell, Leistico, & Van Rybroek, 2006). Murrie et al. found a moderate correlation with instrumental aggression, while Flight and Forth found that PCL:YV scores were correlated with both instrumental and reactive aggression. Vitacco et al. (2006) found that the four-factor model of the PCL:YV accounted for 20% of the variance in instrumental aggression. Interestingly, Vitacco et al. found that the Interpersonal factor was positively related to instrumental aggression, while the correlation was negative with the Antisocial factor. No relationships were found for the Affective and Lifestyle factors. Overall, there is a clear relationship between psychopathic traits and instrumental aggression.

In conclusion, the PCL:YV demonstrates convergent validity in that scores are related, as expected, to externalizing disorders and behaviors (e.g., CD, ODD, ADHD), substance abuse, and aggression, especially instrumental aggression. The next subsection will focus on the predictive validity of the PCL:YV. That is, are scores predictive of important outcome variables, such as general antisocial behavior, offense history, and general and violent recidivism?

Predictive Validity

The PCL:YV has been studied in relation to a number of important outcomes. Scores on the PCL:YV are positively correlated with academic problems (Campbell et al., 2004; Ridenour, Marchant, & Dean, 2001), early onset of antisocial behavior (Corrado, Vincent, Hart, & Cohen,

2004; Forth & Burke, 1998), as well as the frequency and versatility of violent and non-violent offenses (Gretton et al., 2004; Kosson et al., 2002; Murrie et al., 2004). One study, though, found that PCL:YV scores were not better than using Antisocial Personality Disorder criteria in predicting violent or aggressive behavior (Rogers & Rogstad, 2010). Contrary to this, Salekin et al. (2004) found that the PCL:YV was a unique predictor of both violent and non-violent offenses, even when CD, ODD, and ADHD were accounted for.

A commonly used measure of conduct problems is the number of institutional infractions, and many researchers have found that the PCL:YV is able to predict institutional infractions (Murrie et al., 2004; Spain et al., 2004; Stafford & Cornell, 2003). For example, Caldwell (2011) collected data on a sample of 101 juvenile delinquent males as part of a treatment study. In this sample, the Interpersonal factor of the PCL:YV was positively correlated with admission behavior problems, while the other factors were not. Similarly, Kosson et al. (2002) used file information to code antisocial behaviors for adolescent males on probation and found that the incidence was predicted by PCL:YV scores. Edens and Campbell (2007) conducted a meta-analysis on 15 different studies (total sample size = 1,310) to examine this issue and found that weighted mean correlations were moderate, lending credence to the conclusion that psychopathic traits increase the likelihood of institutional misconduct.

One important outcome that has been examined in adolescent samples is recidivism. Findings with adolescent samples have reported that PCL:YV scores do predict recidivism, and that the PCL:YV often performs similarly to other specifically designed risk assessment measures.

Lee et al. (2010) found that adolescent offenders scoring higher on the PCL:YV were at greater risk for dangerousness. More specifically, PCL:YV scores have been shown to be predictive of violent and non-violent recidivism in adjudicated youths (Catchpole & Gretton, 2003; Corrado et al., 2004; Gretton et al., 2004; Vincent et al., 2003) and in juvenile sex offenders (Gretton et al., 2001). Schmidt et al. (2006) found the same relationships for violent and general recidivism in a community-based sample, and the prediction held true for male, Native Canadian, and Caucasian youths. Prediction was weaker in girls, and non-violent recidivism was not predicted by the PCL:YV in any of the subgroups. Another community study, by Stockdale et al. (2010), found that the PCL:YV was a significant predictor of general, violent, and non-violent recidivism over seven years. However, it was a stronger predictor of youth recidivism than adult recidivism. The Antisocial factor was the most predictive of adult outcomes. Leistico, Salekin, DeCoster, and Rogers (2008) found medium effects for violent and general recidivism, however, Edens and Cahill (2007) found that the PCL:YV was not predictive of general or violent reconviction over 10 years.

For the most part, there is agreement on the predictive utility of the PCL:YV in terms of violent recidivism, but findings regarding non-violent and general recidivism have been more mixed, though mostly positive. In a meta-analysis of 21 studies, Edens and Campbell (2007) found that the PCL:YV evidenced a moderate relationship with both violent and general recidivism, and a weak relationship with sexual recidivism. Findings were weaker for studies with female participants, non-Caucasian participants, and juvenile sex offenders.

A number of studies have compared the predictive utility of the PCL:YV with that of juvenile risk assessment instruments. For example, Parks and Bard (2006) conducted a longitudinal study on juvenile sex offenders ($N = 156$) and directly compared the PCL:YV to the Juvenile Sex Offender Assessment Protocol-II (JSOAP-II; Prentky & Righthand, 2003). Mixed offenders (those offending against both children and adults) had higher scores on the PCL:YV and the JSOAP-II as compared with non-mixed offenders. The Impulsive/Antisocial Behavior scale of the JSOAP-II and the Interpersonal and Antisocial factors of the PCL:YV were significant predictors of sexual recidivism, while the Behavioral and Antisocial factors of the PCL:YV predicted non-sexual recidivism.

Marshall, Egan, English, and Jones (2006) used the PCL:YV and the Youth Level of Service/Case Management Inventory (Hoge & Andrews, 2002) in a retrospective study of 94 institutionalized adolescents. Violent incidents within the institution, number of charges and convictions, and formally recorded assaults were coded. Results indicated that the PCL:YV was a better predictor of violence, and predictive validity was similar for charges, convictions, and assaults. Correlations for violence were higher for male than for female adolescents for both instruments.

Caldwell, Ziemke, and Vitacco (2008) compared the JSOAP–II, PCL:YV, three-state developed risk protocols, and the tier designation embedded in the federal Sex Offender Registration and Notification Act of 2006 (SORNA). All measures were coded from file information. The follow-up period was 71.6 months, and general, violent, and sexual offenses were coded. The only significant predictor of general, violent, and sexual reoffense charges was the PCL:YV.

While the above studies provide evidence for the predictive validity of the PCL:YV, other studies have not replicated these findings. For example, Cauffman et al. (2009) compared the PCL:YV, YPI, and the Revised NEO Personality Inventory (NEO:PIR; Costa & McCrae, 1992) in predicting short- and long-term recidivism (6 vs. 12 months). There was modest overlap between the measures, and all measures exhibited weak correlations with reoffending. Similarly, Dolan and Rennie (2008) looked at predictive validity of the Structured Assessment of Violence Risk in Youth (SAVRY; Borum, Bartel, & Forth, 2006) and the PCL:YV in 99 male adolescents who were followed for 12 months post-release. SAVRY risk total and rating were both moderately related to violent and general recidivism; however, the PCL:YV was not a significant predictor.

Douglas et al. (2008) conducted a prospective study with the PCL:YV, the APSD, and the CPS in relation to recidivism in 83 delinquent youth. The CPS had the most consistent relationships with most types of recidivism. Also, once covariates such as substance use, CD, age, and past property crime were controlled for, psychopathic traits were no longer predictive of recidivism.

Beyond a simple comparison of predictive utility, incremental validity controls for the effects of other diagnoses, risk assessment instruments, or other important predictors of recidivism. Three studies have examined incremental predictive validity in relation to risk of recidivism. Murrie et al. (2004) found that PCL:YV scores (with antisocial behavior items removed) showed incremental validity over previous violence and self-report psychopathy measures. Gretton et al. (2004) controlled for CD symptoms and found that PCL:YV scores predicted risk for violent reoffending, but this was not the case for non-violent or sexual recidivism. Similarly, Salekin (2008) found that psychopathy predicted recidivism even after controlling for variables typically linked with reoffending.

The available evidence indicates the PCL:YV scores are related to a variety of negative outcomes, and youth with many psychopathic traits should be targeted for intervention in order to decrease their risk for these outcomes.

Response to Treatment

The most significant benefit of understanding and measuring psychopathic traits in youth is the potential ability to truncate the antisocial career of a high-risk group of youth and intervene early when there is potential malleability to the characteristics of psychopathy (Frick, Cornell, Barry, Bodin, & Dane, 2003; Pardini & Loeber, 2008; Salekin & Lochman, 2008). Despite this benefit, very few published studies have examined the relationship between psychopathic traits and treatment.

O'Neill et al. (2003) assessed treatment response of youth who differ on the PCL:YV in a substance abuse treatment program ($N = 64$). Higher PCL:YV scores were related to poor attendance, low quality of participation, fewer clean urine samples, fewer consecutively clean urine

screens, and reduced clinical improvement. In addition, there was a significant positive relationship between PCL:YV and number of arrests after 12 months.

Caldwell and colleagues have provided the most significant contribution to our understanding of treating young offenders with psychopathic traits (Caldwell, 2011; Caldwell, McCormick, Umstead, & Van Rybroek, 2007; Caldwell, McCormick, Wolfe, & Umstead, 2012; Caldwell, Skeem, Salekin, & Van Rybroek, 2006). Caldwell and colleagues have consistently reported a positive treatment response of youth high on the PCL:YV in the Mendota Juvenile Treatment Center (MJTC), which houses a high-risk group of youth who have had problems in traditional institutions. Caldwell et al. (2006) compared youth who scored over 27 on the PCL:YV in the MJTC ($N = 56$) with those in a traditional correctional institution ($N = 85$). Results indicated that youth treated at the MJTC were significantly less likely to recidivate both non-violently and violently, and this effect remained significant when controlling for PCL:YV score. A significant treatment effect was also found in a subsequent study with a larger sample ($N = 248$; Caldwell, 2011). Caldwell et al. (2007) assessed the relationship between PCL:YV score, treatment outcome, and recidivism ($N = 86$). Overall, youth with high PCL:YV scores were more likely to receive negative behavior ratings both at baseline and at treatment completion compared to youth who scored lower on psychopathic traits. Yet although significant positive change from baseline to treatment completion was observed in youth high on PCL:YV score, the magnitude of the change was less than for youth who scored low on the PCL:YV. Examination of PCL:YV with recidivism found that although PCL:YV was initially associated with recidivism, when the effects of behavioral change were accounted for, this relationship disappeared.

In the most recent study, Caldwell and colleagues (2012) examined how amenable characteristics of psychopathy were to change during treatment in 72 youth at MJTC. However, a self-report measure of psychopathy was used rather than the PCL:YV. Youth were measured at baseline, 90 days, and then 180 days, and changes in scores were assessed at each time period. Results indicated a significant change in psychopathic traits, including measures of impulsivity, narcissism, and callous-unemotional traits. In addition, changes in psychopathic traits were associated with positive institutional behavior ratings.

Psychopathy is an important consideration when planning treatment, as youth with higher scores on the PCL:YV are less likely to respond to treatment or may have a reduced treatment response compared to youth who score low on the PCL:YV. However, it is clear that treatment does in fact reduce the offending of at least a subgroup of youth with psychopathic traits. Therapeutic change during treatment appears to be of paramount importance when assessing treatment gain. Caldwell et al. (2007) reported that although youth with psychopathic features were more likely to receive negative behavior ratings both pre- and post-treatment, at least a subgroup of youth made gains in behavioral change during treatment, and it was this change rather than PCL:YV score that was related to treatment.

Despite these encouraging results, there are a number of unanswered questions regarding treating youth with psychopathic traits. First, the relationship between psychopathic traits and treatment gain has not been tested in female youth. Also, the process and mechanisms that result in the reduced treatment response of youth with psychopathic traits is not well understood and therefore it is difficult to provide evidence-based guidance to treatment providers. Finally, it is a limitation that the vast majority of the research on psychopathic traits and treatment in youth comes from a single treatment facility and fairly modest sample size of youth. There is a need to attempt to replicate these results in other treatment settings.

Forensic Uses

Using a standardized assessment of psychopathic traits in adolescents is important if the results of the assessment are to be utilized in research, clinical, and forensic settings. Because the PCL:YV has established reliability and validity that is applicable across sex, age, and ethnicity (as already discussed), it has been used in all of these settings (Forth et al., 2003). However, given the limited number of longitudinal studies on psychopathy scores, it is difficult to reach conclusions regarding the stability of these traits or behaviors. Small differences were observed for sex and ethnicity, with boys scoring slightly higher than girls, and non-Caucasians scoring higher than Caucasians. These findings suggest that the PCL:YV has generalizability and utility. Further, any effect of age, ethnicity, or sex is quite small.

Viljoen, MacDougall, Gagnon, and Douglas (2010) examined 111 adolescent criminal cases in Canada and United States in which psychopathy was assessed. This review found that psychopathy has played a role in a diverse range of criminal cases, with the most common being juvenile transfer hearings and dispositions or sentencing planning. In the majority of cases, testimony regarding psychopathy was associated with an increased severity of disposition.

Given the relationship between psychopathic traits and negative outcomes, including recidivism (as discussed above), there is a clear need for identification of youth who might benefit from early intervention and treatment (Forth & Book, 2010). That being said, there is a definite negative connotation to the label “psychopathic.” In adults, psychopathy is much less amenable to treatment (e.g., Harris, Rice, & Cormier, 1991), and people may assume that youth with psychopathic traits are equally unamenable to intervention (Seagrave & Grisso, 2002). Here the issue of labeling becomes very important, in that such assumptions might result in the misuse of information gleaned from PCL:YV scores (e.g., an assumption that youth with psychopathic traits may be less amenable to treatment), resulting in differential treatment of youth with psychopathic traits (Forth & Book, 2010). This could include the institution of harsher sentences, making early release more difficult. It is even possible that in cases where there are limited resources, treatment could be denied to youth seen as not likely to benefit from intervention. Given that the whole point of using the PCL:YV is to target psychopathic youth for early intervention and appropriate treatments (Forth & Book, 2010), this would be counterproductive and would likely result in increased risk for recidivism.

The utility of the PCL:YV in various contexts (both research and applied) rests on three separate but related assumptions: (1) It is a good measure of psychopathic traits. (2) It demonstrates adequate predictive utility. (3) Scores may provide useful information for early intervention.

A good measure of any trait is reliable, valid, and has a standardized scoring method that uses multiple sources of information. These criteria are met by the PCL:YV. The reliability and validity of the PCL:YV have been well established (Forth et al., 2003) and are detailed in the previous sections. The PCL:YV is also a standardized assessment tool, providing comparison tables for several different populations (Forth et al., 2003). It is important to note that the standardized scoring procedures require training and experience to ensure that the instrument is used appropriately. Finally, the PCL:YV is not scored on the basis of information from any one domain. Raters must obtain information from several different domains and from a variety of sources (e.g., institutional files, school records, family reports).

In addition to the preceding requirements, an instrument should also have predictive value if it is to be a useful measure of a construct. The predictive validity of the PCL:YV has been demonstrated in several studies. According to Forth and Book (2010), the research literature consistently finds that the PCL:YV is related to a range of theoretically relevant correlates and outcome variables. These include, but are not limited to, academic problems (Ridenour et al.,

2001), a history of behavioral problems in school (Campbell et al., 2004), number of physical altercations (Salekin et al., 2004), overall aggression (Stafford & Cornell, 2003), problematic behaviors in institutions (Spain et al., 2004), frequency and versatility of criminal acts (Campbell et al., 2004; Kosson et al., 2002), risk assessment measures (Forth et al., 2003), and recidivism (Catchpole & Gretton, 2003; Corrado et al., 2004; Gretton et al., 2004; Toupin et al., 1995; Vincent et al., 2003).

Is the PCL:YV being used for risk assessment by practicing clinicians? A recent survey of 215 psychologists (85 doing juvenile risk assessments and 130 adult risk assessments) examined the use of psychopathy measures in risk assessments (Viljoen, McLachlan, & Vincent, 2010). Psychopathy was used more frequently in with adults (54% used it always or almost always) as compared to juveniles (34% used it always or almost always). There was also an interesting difference between survey respondents on whether or not they believed psychopathic traits should be assessed in individuals aged 17 and younger. The majority of psychologists conducting juvenile assessments (90%) felt that psychopathic traits should be assessed, whereas only 44% of psychologists conducting adult assessments believed this to be true.

One of the most important goals in using a measure of psychopathy with youth is to attempt to identify those who might benefit from early intervention and treatment programs. While psychopathic adults do not appear to benefit greatly from traditional treatment programs (Wong & Hare, 2005), it is possible that adolescents may be more responsive to intervention and treatment (Forth & Book, 2010; Forth et al., 2003). To date, some research, reviewed above, indicates that young offenders may be more “malleable” than adult offenders.

Overall, the PCL:YV has sound psychometric properties, good predictive utility, and may be useful in the development of early intervention programs.

Concerns and Limitations

It is the potential use (or misuse) of the PCL:YV in mental health and criminal justice settings that has raised concerns for some commentators (Edens, Skeem, Cruise, & Cauffman, 2001; Hart, Watt, & Vincent, 2002; Lyon & Ogloff, 2000; Seagrave & Grisso, 2002; Steinberg, 2002; Zinger & Forth, 1998). These concerns have focused on (1) the issue of labeling a youth as a psychopath; (2) the implications of the PCL:YV for classification, sentencing, and treatment; (3) the possibility that characteristics of psychopathy are common features of normally developing youth; and (4) the stability of psychopathic traits from late childhood to early adulthood.

A major concern is based on the possibility that PCL:YV scores will be used to diagnose adolescents as “psychopathic.” It is common for *researchers* to use cutoff scores on the PCL-R to diagnose psychopathy in adults, and similar cutoffs have been employed effectively with adolescent offenders for research purposes (e.g., Gretton et al., 2004; Gretton et al., 2001). However, it would be inappropriate to use the PCL:YV to render a clinical or forensic diagnosis of psychopathy for an adolescent (Forth et al., 2003). Such a diagnosis would require that adolescent psychopathy can be viewed as a taxon, or discrete category, and that we know what cut score to use for inclusion in the category. However, recent research with adolescents indicates that a continuum, not a taxon, underlies PCL:YV scores (Murrie, Boccaccini, McCoy, & Cornell, 2007).

Diagnostic issues are not of great concern in research contexts because the consequences for the participant are minimal at most. In forensic contexts, however, the consequences for an adolescent of labeling are potentially very serious. The authors of the PCL:YV note that psychopathy is a psychological term, not a legal one, and that the term has negative connotations for the general population, as well as for mental health and criminal justice professionals (Forth et al., 2003).

Boccaccini, Murrie, Clark, and Cornell (2008) have argued that although concerns about labeling are legitimate, such concerns about the possible negative consequences of assessing psychopathy in youth should be considered hypotheses to be empirically tested, rather than given truths. Researchers have examined the influence of diagnostic criterion (i.e., describing the personality traits associated with psychopathy), and diagnostic labels of psychopathy or conduct disorder. Researchers have found that describing a juvenile offender using either the psychopathy label or underlying traits sometimes results in more punitive decision making among undergraduates, juvenile probation officers, potential jurors, juvenile justice judges, and clinicians (Chauhan, Reppucci, & Burnette, 2007; Edens, Guy, & Fernandez, 2003; Jones & Cauffman, 2008; Murrie, Boccaccini, McCoy, & Cornell, 2007; Murrie, Cornell, & McCoy, 2005; Rockett, Murrie, & Boccaccini, 2007; Vidal & Skeem, 2007). More specifically, these studies have demonstrated that describing youth as psychopathic resulted in harsher sentence recommendations and higher ratings of risk for future violence.

The effects for treatment access or amenability have been mixed. Perhaps contrary to what most would expect, three studies have shown that the diagnosis of psychopathy assigned to defendants was associated with perceptions of an increased need for treatment among juvenile probation officers (Murrie et al., 2005), juvenile and family court judges (Murrie et al., 2007), and jury-eligible community participants (Boccaccini et al., 2008). Several major conclusions can be drawn from these studies: (1) Describing psychopathic traits produces stronger effects than does using the labels themselves. (2) There is evidence for a general labeling effect but not a specific labeling effect (there is no difference between a diagnosis of conduct disorder and a diagnosis of psychopathy). (3) The negative effects of psychopathic traits have been found for decisions regarding degree of risk for future offending and whether harsher sanctions are warranted. (4) Explicitly labeling young defendants as “psychopaths” led jurors to assume they posed greater risk for future crime (Boccaccini et al., 2008). Such results have important implications for clinical practice, and should alert practitioners to their use of language and diagnostic labels.

It appears from a recent survey that the majority of clinicians doing risk assessments with juveniles are not using the “psychopathic” label. Viljoen et al. (2010) asked psychologists how they presented information about their psychopathy assessments in their risk assessments reports. Only two (2.6%) of the juvenile psychologists provided a categorical diagnosis (“This youth is or is not a psychopath”), whereas 41% of the psychologists assessing adults do so. The most common method of communicating about the psychopathy assessment for juveniles was to describe the psychopathic traits but not to explicitly label them as psychopathic traits (46% of the juvenile psychologists did this).

As stated by Lyon and Ogloff (2000), “psychopathy is a powerful pejorative diagnostic label that can exert a profound influence over the legal decisions rendered in courts” (p. 139). There are many situations in which scores on the PCL:YV could be used to assist in making decisions in the legal and mental health arenas, including justification for harsher sanctions (e.g., pre-trial secure detention, longer sentences, an increased level of supervision after release). Further, with the recent trend toward limited treatment resources, it is possible that clinicians may feel pressure to exclude from treatment young offenders with high scores on the PCL:YV, in the belief that psychopaths do not benefit from treatment. This would be contrary to the intent of the PCL:YV, which is to identify psychopathic traits early enough to have a chance to develop and provide effective intervention, treatment, and management. As we have said, psychopathic and related features are more changeable in adolescents than in adults, and the PCL:YV may be useful in helping to identify targets for intervention. At this time, there is little systematic research on PCL:YV scores and treatment outcome, but several recent studies suggest that there may be reason for optimism.

Another concern has been that “scores on measures of psychopathy arguably may be inflated by general characteristics of adolescence” (Edens et al., 2001, p. 59). These authors contend that developmental issues may influence the measurement of interpersonal, affective, and social deviance features of psychopathy. Because of changes in perspective-taking ability and cognitive functions, developing an autonomous identity, and susceptibility to peer influences, youth may appear to be grandiose, lacking remorse or empathy, manipulative, sensation-seeking, impulsive, and lacking goals. Measures of youth psychopathy that fail to take into account normative adolescent patterns of behavior may well obtain scores that are spuriously high and overestimate the level of psychopathic traits in adolescent samples.

Although some adolescents may exhibit some features of psychopathy in certain contexts or for a limited time, in order to obtain a high score on the PCL:YV, evidence for the trait or behavior must be manifested across social contexts and over substantial time periods. High ratings of psychopathic traits are rare in community youth (80% of community males score less than 5 on the PCL:YV; Forth et al., 2003).

Adolescence is a period of substantial change in biological, psychological, and social systems (Arnett, 1999). Given the dynamic nature of adolescent development, the question is how much change occurs in personality traits during adolescence. Research measuring the stability of personality traits from childhood to adulthood has concluded that there is a moderate level of consistency and that most change occurs in young adulthood and not during adolescence (Caspi, Roberts, & Shiner, 2005). Caspi et al. (2005) concluded that during adolescence, “personality differences remain remarkably consistent during this period” (p. 467).

Researchers have underscored the need to assess whether youth with psychopathic traits develop into psychopathic adults to ensure that our measurement of psychopathy in adolescence is consistent with the construct in adulthood (Edens et al., 2001; Seagrave & Grisso, 2002). Several recent studies have reported that psychopathic traits demonstrate moderate stability throughout adolescence (Forsman, Lichtenstein, Andershed, & Larsson, 2008; Pardini & Loeber, 2008) and into early adulthood (Blonigen, Hicks, Krueger, Patrick, & Iacono, 2006; Lynam, Caspi, Moffitt, Loeber, & Stouthamer-Loeber, 2007; Lynam, Loeber, & Stouthamer-Loeber, 2008). However, there is only one published study that assessed psychopathic traits using the PCL:YV. Lee, Klaver, Hart, Moretti, and Douglas (2009) reported moderate stability of PCL:YV score over a six-month period.

Although there is moderate stability in samples, Pardini and Loeber (2008) reported significant individual variability with both increases and decreases in callousness over time. In addition, Lee et al. (2009) found that the affective factor of the PCL:YV demonstrated less stability than the other factors. Variability in traits over time may point to the impact of additional factors that impact the development of these traits. Several studies have highlighted how diverse moderators affect the stability of psychopathy during development: they demonstrate synergistic interactions, where risk factors combine in a more than additive fashion to produce offending (Cadoret, O’Gorman, Troughton, & Heywood, 1995; Cohen, Berliner, & Mannarino, 2003). Another line of research has investigated interaction effects between risk factors, where one variable reduces the impact of the other (the risk factor). An array of moderating, or buffering, factors seem to reduce commitment to an antisocial lifestyle, including IQ, nurturing relationship with adults, good academic performance, and prosocial peer groups (Fergusson & Lynskey, 1996; Quinton & Rutter, 1988; Werner & Smith, 1992).

The stability of personality traits is likely related to the strong genetic influence on personality. Behavioral genetics research has found evidence for a considerable genetic influence in personality traits (Bouchard & Loehlin, 2001). Researchers comparing monozygotic and same-sex dizygotic twins have also found evidence for a substantial genetic influence for psychopathic traits (Blonigen,

Carlson, Krueger, & Patrick, 2003; Larsson, Andershed, & Lichtenstein, 2006; Taylor, Loney Bobadilla, Iacono, & McGue, 2003; Viding, Blair, Moffitt, & Plomin, 2005)

Forth and Book (2010) suggested a number of ways to reduce assessment errors and misuse. First, individuals who use the PCL:YV for forensic purposes must be qualified to do so. Second, multiple sources of information should be considered in scoring PCL:YV items. Third, traits should be assessed over a substantial time period. Fourth, behavioral evidence should be considered relative to developmental norms, which change over the lifespan, and even within adolescence. Finally, do not use the PCL:YV to label a youth (e.g., no cutoff scores are given), do not use the PCL:YV score as a sole measure to justify harsher penalties, and do not use the PCL:YV score to justify the withholding of treatment. (These same cautions are listed in the PCL:YV Manual.)

CASE EXAMPLE

Jason is a 15-year-old white male adolescent offender who was recently convicted of reckless driving, automobile theft, and drug trafficking (all part of the same incident). He had been selling marijuana on a street corner that was well known for being a place to “score drugs.” The police began a raid on that particular street, and Jason took off before he could be caught. He saw a woman getting out of her car beside a parking meter. He pushed her aside, jumped into her car, and drove off. During ensuing police chase, Jason ran a red light and hit another car, seriously injuring the 62-year-old driver. Jason received minor injuries and was apprehended about two blocks from the scene of the incident.

An assessment of psychopathic traits was requested to aid in recommendations regarding risk for future criminal behavior and appropriate level of intervention. A forensic psychologist completed the PCL:YV in a detention facility on the basis of file information, an interview with Jason’s mother, stepfather, and probation officer, and a two-hour interview with Jason. During the interview, Jason was exceptionally talkative. He tried to control the interview by interrupting and questioning the qualifications of the rater. Jason reported that his father left his mother before his first birthday, and that he had been raised by his mother and stepfather, whom she had married when he was 5. Jason stated that his stepfather had never liked him, and that he was often punished physically, even in public. At around the same time, Jason began stealing toys and candy from the grocery store and from other children at school.

According to file information, Jason began acting out at an early age. He was first suspended from school at the age of 8 for stealing from other students’ lockers. His first contact with the local police was at age 9, also for theft. At the age of 12, he was convicted of his first crime: breaking and entering. He had entered a neighbor’s house and had stolen a wallet from the kitchen table. He took the cash from the wallet and threw the wallet in a nearby creek. Jason was expelled from school at age 13 for bringing a large kitchen knife to school. According to his mother’s report, he was hanging around with children who were a bad influence on him. In fact, from school reports, and from Jason himself, it seems that Jason was the bad influence.

Academically, Jason has done quite poorly. He rarely completed homework assignments, was consistently absent, and was suspended numerous times, beginning in elementary school. While his academic performance was poor, he was always sent on to the next grade. Jason completed grade 9 and then dropped out of school.

Jason had a minimal employment history in the community. When he was 14, he had a paper route, although he was consistently reprimanded, and eventually fired, for throwing the newspapers in the trash rather than delivering them.

Jason has an extensive history of substance use, ranging from alcohol and marijuana to heroin, LSD, and ecstasy. He states that he enjoys using drugs, and that when he doesn't use them, the boredom is unbearable. During the current assessment, he stated that he had started using drugs and alcohol at the age of 13. He also stated that drugs or alcohol preceded most of his criminal activities. He is, however, very proud of not being addicted to anything.

Jason reports having first experienced sexual intercourse when he was 12 years old, and that he couldn't count the number of times since. He has not had any serious girlfriends or even dating partners. He states that having a girlfriend would be too much of a hassle.

His previous convictions include breaking and entering (two counts), car theft (three counts), and assault causing bodily harm. The assault causing bodily harm occurred when he was 14 and he received a six-month sentence. While at a party, Jason got into a verbal argument with a 16-year-old boy. Jason left the party and returned with a baseball bat and waited outside the party for the victim. Jason hit the victim three times with the bat before being restrained by other youth.

Jason has had problems within the institution. He is consistently rude and sarcastic with the staff, and has on occasion been physically aggressive toward younger offenders. During his previous incarceration for the assault causing bodily harm, he refused to participate in treatment, stating that therapy is for crazy people.

Jason has been on probation for the break and enters and car thefts. He violated probation once. He missed meetings with his probation officer, and his probation was revoked. He pleaded guilty at his trial (to all charges). However, when asked about the consequences of his crime, he stated that no one was hurt, and that insurance would cover the losses. This statement completely ignored the physical harm to the man whose car he hit. When asked about this, he said, "That part was an accident. As if I should have been charged with that."

Jason exhibits many psychopathic traits. Among institutionalized male adolescent offenders, he would score at the 64th percentile, meaning that 64% of other male adolescent offenders would receive a score similar to, or lower than, his score. Among male adolescent probationers, he would score at the 86th percentile, indicating that he has many more psychopathic features than most offenders in such a setting. The PCL:YV is divided into four factors. Factor 1 focuses on interpersonal features, such as grandiosity and manipulateness. Factor 2 measures affective features, such as callousness and remorseless use of others. Factor 3 reflects features such as impulsivity, irresponsibility, and lacking goals. Factor 4 represents poor anger control and the early onset and high frequency of antisocial behaviors. Compared with other institutionalized young male offenders, he had elevated scores on Factors 2, 3 and 4. He had some of the interpersonal (42nd percentile) and affective (68th percentile) psychopathic features. His score on the behavioral factor was very high at the 82nd percentile and his score on the antisocial factor was at the 66th percentile. Jason's score on the PCL:YV is within the range of scores associated with institutional adjustment problems, treatment non-compliance, and an increased risk of future criminal behavior.

The evaluator recommended interventions to improve parental management and an intensive cognitive-behavioral program focused on the following treatment targets: self-management of negative affect and impulse control, substance use, negative attitudes, and antisocial peers.

References

- Abram, K. M., Teplin, L. A., Charles, D. R., Longworth, S. L., McClelland, G. M., & Dulcan, M. K. (2004). Posttraumatic stress disorder and trauma in youth in juvenile detention. *Archives of General Psychiatry*, 61, 403–410.
- American Psychiatric Association. (2000). *Diagnostic and statistical manual of mental disorders* (4th ed., text rev.). Washington, DC: Author.
- Andershed, H. A., Gustafson, S. B., Kerr, M., & Stattin, H. (2002). The usefulness of self-reported psychopathy-like traits in the study of antisocial behaviour among non-referred adolescents. *European Journal of Personality*, 16, 383–402.
- Andershed, H., Hodgins, S., & Tengström, A. (2007). Convergent validity of the Youth Psychopathic Traits Inventory (YPI): Association with the Psychopathy Checklist: Youth Version (PCL:YV). *Assessment*, 14, 144–154.
- Arnett, J. J. (1999). Adolescent storm and stress, reconsidered. *American Psychologist*, 54, 317–326.
- Bauer, D. L., Whitman, L. A., & Kosson, D. S. (2011). Reliability and construct validity of the Psychopathy Checklist: Youth Version scores among incarcerated adolescent girls. *Criminal Justice and Behavior*, 38, 965–987.
- Blackburn, R., & Coid, J. W. (1998). Psychopathy and the dimensions of personality disorder in violent offenders. *Personality and Individual Differences*, 25, 129–145.
- Blonigen, D. M., Carlson, S. R., Krueger, R. F., & Patrick, C. J. (2003). A twin study of self-reported psychopathic traits. *Personality and Individual Differences*, 35(1), 179–197.
- Blonigen, D. M., Hicks, B. M., Krueger, R. F., Patrick, C. J., & Iacono, W. G. (2006). Continuity and change in psychopathic traits as measured via normal-range personality: A longitudinal-biometric study. *Journal of Abnormal Psychology*, 115, 85–95.
- Boccaccini, M. T., Murrie, D. C., Clark, J. W., & Cornell, D. G. (2008). Describing, diagnosing, and naming psychopathy: How do youth psychopathy labels influence jurors? *Behavioral Sciences and the Law*, 26, 487–510.
- Borum, R., Bartel, P., & Forth, A. E. (2006). *Manual for the structured assessment of violent risk in youth (SAVRY)*. Odessa, FL: Psychological Assessment Resources.
- Bouchard, T. J., & Loehlin, J. (2001). Genes, evolution, and personality. *Behavioral Genetics*, 31, 243–273.
- Brandt, J. R., Kennedy, W. A., Patrick, C. J., & Curtin, J. J. (1997). Assessment of psychopathy in a population of incarcerated adolescent offenders. *Psychological Assessment*, 9, 429–435.
- Cadoret, R. J., O’Gorman, T. W., Troughton, E., & Heywood, E. (1995). Alcoholism and antisocial personality. *Archives of General Psychiatry*, 42(2), 161–167.
- Caldwell, M. (2011). Treatment-related changes in behavioral outcomes of psychopathy facets in adolescent offenders. *Law and Human Behavior*, 35, 275–287.
- Caldwell, M. F., McCormick, D. J., Umstead, D., & Van Rybroek, G. J. (2007). Evidence of treatment progress and therapeutic outcomes among adolescents with psychopathic features. *Criminal Justice and Behavior*, 34, 573–587.
- Caldwell, M. F., McCormick, D., Wolfe, J., & Umstead, D. (2012). Treatment-related changes in psychopathy features and behavior in adolescent offenders. *Criminal Justice and Behavior*, 39, 144–155.
- Caldwell, M. F., Skeem, J., Salekin, R., & Van Rybroek, G. (2006). Treatment response of adolescent offenders with psychopathy features: A 2-year follow-up. *Criminal Justice and Behavior*, 33, 571–596.
- Caldwell, M. F., Ziemke, M. H., & Vitacco, M. J. (2008). An examination of the sex offender registration and notification as applied to juveniles: Evaluating the ability to predict sexual recidivism. *Psychology, Public Policy, and Law*, 14, 89–114.
- Campbell, M. A., Porter, S., & Santor, D. (2004). Psychopathic traits in adolescent offenders: An evaluation of criminal history, clinical, and psychosocial correlates. *Behavioral Sciences and the Law*, 22, 23–47.
- Carmines, E. G., & Zeller, R. A. (1991). *Reliability and validity assessment*. Newbury Park, CA: Sage.
- Caspi, A., Roberts, B. W., & Shiner, R. L. (2005). Personality development: Stability and change. *Annual Review of Psychology*, 56, 453–484.
- Catchpole, R. E. H., & Gretton, H. M. (2003). The predictive validity of risk assessment with violent young offenders: A 1-year examination of criminal outcome. *Criminal Justice and Behavior*, 30, 688–708.
- Cauffman, E., Feldman, S. S., Waterman, J., & Steiner, H. (1998). Posttraumatic stress disorder among female juvenile offenders. *Journal of the American Academy of Child and Adolescent Psychiatry*, 37, 1209–1216.

- Cauffman, E., Kimonis, E. R., Dmitrieva, J., & Monahan, K. C. (2009). A multimethod assessment of juvenile psychopathy: Comparing the predictive utility of the PCL:YV, YPI, and NEO PIR. *Psychological Assessment, 21*, 528–542.
- Chauhan, P., Reppucci, N. D., & Burnette, M. L. (2007). Application and impact of the label psychopath to juveniles. *International Journal of Forensic Mental Health, 6*, 3–14.
- Cleckley, H. (1941). *The mask of sanity*. St. Louis, MO: C. V. Mosby.
- Cohen, J. A., Berliner, L., & Mannarino, A. P. (2003). Psychosocial and pharmacological interventions for child crime victims. *Journal of Traumatic Stress, 16*, 175–186.
- Cook, N. E., Barese, T. A., & Dicataldo, F. (2010). The confluence of mental health and psychopathic traits in adolescent female offenders. *Criminal Justice and Behavior, 37*, 119–135.
- Cornell, D. G., Warren, J., Hawk, G., Stafford, E., Oram, G., & Pine, D. (1996). Psychopathy in instrumental and reactive violent offenders. *Journal of Consulting and Clinical Psychology, 64*, 3–19.
- Corrado, R. R., Vincent, G. M., Hart, S. D., & Cohen, I. M. (2004). Predictive validity of the Psychopathy Checklist: Youth Version for general and violent recidivism. *Behavioral Sciences and the Law, 22*, 5–22.
- Costa, P. T., & McCrae, R. R. (1992). *NEO-PI-R professional manual*. Odessa, FL: Psychological Assessment Resources.
- Cruise, K. R., Colwell, L. H., Lyons, P. M., & Baker, M. D. (2003). Prototypical analysis of adolescent psychopathy: Investigating the juvenile justice perspective. *Behavioral Sciences and the Law, 21*, 829–846.
- Das, J., de Ruiter, C., & Doreleijers, T. (2008). Reliability and validity of the Psychopathy Checklist: Youth Version in Dutch female adolescents. *International Journal of Law and Psychiatry, 31*, 219–228.
- Das, J., de Ruiter, C., Doreleijers, T., & Hillege, S. (2009). Reliability and construct validity of the Dutch Psychopathy Checklist: Youth Version: Findings from a sample of male adolescents in a juvenile justice treatment institution. *Assessment, 16*, 88–102.
- Dixon, A., Howie, P., & Starling, J. (2004). Psychopathology in female juvenile offenders. *Journal of Child Psychology and Psychiatry, 45*, 1150–1158.
- Dolan, M., & Rennie, C. (2006). Psychopathy Checklist: Youth Version and Youth Psychopathic Traits Inventory: A comparison study. *Personality and Individual Differences, 41*, 779–789.
- Dolan, M., & Rennie, C. (2007). Is juvenile psychopathy associated with low anxiety and fear in conduct-disordered male offenders? *Journal of Anxiety Disorders, 21*, 1028–1038.
- Dolan, M., & Rennie, C. (2008). The Structured Assessment of Violence Risk in Youth as a predictor of recidivism in a United Kingdom cohort of adolescent offenders with conduct disorder. *Psychological Assessment, 20*, 35–46.
- Douglas, K. S., Epstein, M. E., & Poythress, N. G. (2008). Criminal recidivism among juvenile offenders: Testing the incremental and predictive validity of three measures of psychopathic features. *Law and Human Behavior, 32*, 423–438.
- Edens, J. F., & Cahill, M. A. (2007). Psychopathy in adolescence and criminal recidivism in young adulthood. *Assessment, 14*, 57–64.
- Edens, J. F., & Campbell, J. S. (2007). Identifying youths at risk for institutional misconduct: A meta-analytic investigation of the psychopathy checklist measures. *Psychological Services, 4*, 13–27.
- Edens, J. F., Guy, L. S., & Fernandez, K. (2003). Psychopathic traits predict attitudes toward a juvenile capital murderer. *Behavioral Sciences and the Law, 21*, 807–828.
- Edens, J. F., Skeem, J. L., Cruise, K. R., & Cauffman, E. (2001). Assessment of “juvenile” psychopathy and its association with violence: A critical review. *Behavioral Sciences and the Law, 19*, 53–80.
- Fergusson, D. M., & Lynskey, M. T. (1996). Adolescent resiliency to family adversity. *Journal of Child Psychology and Psychiatry and Allied Disciplines, 37*, 281–292.
- Flight, J. I., & Forth, A. E. (2007). Instrumentally violent youths: The roles of psychopathic traits, empathy, and attachment. *Criminal Justice and Behavior, 34*, 739–751.
- Forsman, M., Lichtenstein, P., Andershed, H., & Larsson, H. (2008). Genetic effects explain the stability of psychopathic personality from mid- to late adolescence. *Journal of Abnormal Psychology, 117*, 606–617.
- Forth, A. E., & Book, A. S. (2010). Psychopathic traits in children and adolescents: The relationship with antisocial behaviors and aggression. In R. T. Salekin and D. R. Lynam (Eds.), *Handbook of child and adolescent psychopathy* (pp. 251–283). New York: Guilford Press.
- Forth, A. E., & Burke, H. C. (1998). Psychopathy in adolescence: Assessment, violence and developmental precursors. In D. J. Cooke, A. E. Forth, and R. D. Hare (Eds.), *Psychopathy: Theory, research, and implications for society* (pp. 205–229). New York: Kluwer Academic.

- Forth, A. E., Hart, S. D., & Hare, R. D. (1990). Assessment of psychopathy in male young offenders. *Psychological Assessment*, 2, 342–344.
- Forth, A. E., Kosson, D., & Hare, R. D. (2003). *The Hare PCL: Youth Version*. Toronto: Multi-Health Systems.
- Frick, P. J., Cornell, A. H., Barry, C. T., Bodin, S. D., & Dane, H. E. (2003). Callous-unemotional traits and conduct problems in the prediction of conduct problem severity, aggression, and self-report of delinquency. *Journal of Abnormal Child Psychology*, 31(4), 457–470.
- Frick, P. J., & Hare, R. D. (2001). *The Antisocial Process Screening Device*. Toronto: Mental Health Systems.
- Gretton, H. M., Hare, R. D., & Catchpole, R. E. H. (2004). Psychopathy and offending from adolescence to adulthood: A 10-year follow-up. *Journal of Consulting and Clinical Psychology*, 72, 636–645.
- Gretton, H. M., McBride, M., Hare, R. D., O'Shaughnessy, R., & Kumka, G. (2001). Psychopathy and recidivism in adolescent sex offenders. *Criminal Justice and Behavior*, 28, 427–449.
- Hale, L. R., Goldstein, D. S., Abramowitz, C. S., Calamari, J. E., & Kosson, D. S. (2004). Psychopathy is related to negative affectivity but not to anxiety sensitivity. *Behaviour Research and Therapy*, 42, 697–710.
- Hare, R. D. (1991). *The Hare Psychopathy Checklist–Revised*. Toronto: Multi-Health Systems.
- Hare, R. D. (2003). *The Hare Psychopathy Checklist–Revised* (2nd ed.). Toronto: Multi-Health Systems.
- Harris, G. T., Rice, M. E., & Cormier, C. A. (1991). Psychopathy and violent recidivism. *Law and Human Behavior*, 15, 625–637.
- Hart, S. D., & Hare, R. D. (1989). Discriminant validity of the Psychopathy Checklist. *Psychological Assessment: A Journal of Consulting and Clinical Psychology*, 1, 211–218.
- Hart, S. D., Watt, K., & Vincent, G. (2002). Commentary on Seagrave and Grisso: Impressions of the state of the art. *Law and Human Behavior*, 26, 241–245.
- Harvey, P. D., Stokes, J. L., Lord, J., & Pogge, D. L. (1996). Neurocognitive and personality assessment of adolescent substance abusers: A multidimensional approach. *Assessment*, 3, 241–253.
- Hemphälä, M., & Tengström, A. (2010). Associations between psychopathic traits and mental disorders among adolescents with substance use problems. *British Journal of Clinical Psychology*, 49, 109–122.
- Hodgins, S., Barbareschi, G., & Larsson, A. (2011). Adolescents with conduct disorder: Does anxiety make a difference? *Journal of Forensic Psychiatry and Psychology*, 22, 669–691.
- Hoge, R. D., & Andrews, D. A. (2002). *Youth Level of Service/Case Management Inventory: YLS/CMI interview guide*. Toronto: Multi-Health Systems.
- Huchzermeier, C., Geiger, F., Bruß, E., Godt, N., Köhler, D., Hinrichs, G., & Aldenhoff, J. B. (2007). The relationship between DSM–IV cluster B personality disorders and psychopathy according to Hare's criteria: Clarification and resolution of previous contradictions. *Behavioral Sciences and the Law*, 25, 901–911.
- Jones, S., & Cauffman, E. (2008). Juvenile psychopathy and judicial decision making: An empirical analysis of an ethical dilemma. *Behavioral Sciences and the Law*, 26, 151–165.
- Jones, S., Cauffman, E., Miller, J. D., & Mulvey, E. (2006). Investigating different factor structures of the Psychopathy Checklist: Youth Version: Confirmatory factor analytic findings. *Psychological Assessment*, 18, 33–48.
- Kosson, D. S., Cyterski, T. D., Steurwald, B. L., Neumann, C. S., & Walker-Matthews, S. (2002). The reliability and validity of the Psychopathy Checklist: Youth Version (PCL:YV) in nonincarcerated adolescent males. *Psychological Assessment*, 14(1), 97–109.
- Kosson, D. S., Neumann, C. S., Forth, A. E., Hare, R. D., Salekin, R. T., Krischer, M. K., & Sevecke, K. (in press). Factor structure of the Hare Psychopathy Checklist: Youth Version (PCL:YV) in adolescent females. *Psychological Assessment*.
- Kotler, J. S., & McMahon, R. J. (2010). Assessment of child and adolescent psychopathy. In R. T. Salekin & D. R. Lynam (Eds.) *Handbook of child and adolescent psychopathy* (pp. 79–109). New York: Guilford Press.
- Kuback, F. A., & Salekin, R. T. (2009). Psychopathy and anxiety in children and adolescents: New insights on developmental pathways to offending. *Journal of Psychopathology and Behavioral Assessment*, 31, 271–284.
- Lang, S., af Klinteberg, B., & Alm, P.-O. (2002). Adult psychopathy and violent behavior in males with early neglect and abuse. *Acta Psychiatrica Scandinavica*, 106, 93–100.
- Larsson, H., Andershed, H., & Lichtenstein, P. (2006). A genetic factor explains most of the variation in the psychopathic personality. *Journal of Abnormal Psychology*, 115, 221–230.
- Lee, Z., Klaver, J. R., Hart, S. D., Moretti, M. M., & Douglas, K. S. (2009). Short-term stability of psychopathic traits in adolescent offenders. *Journal of Clinical Child and Adolescent Psychology*, 38, 595–605.
- Lee, Z., Salekin, R. T., & Iselin, A.-M. (2010). Psychopathic traits in youth: Is there evidence for primary and secondary subtypes? *Journal of Abnormal Child Psychology*, 38, 381–393.

- Lee, Z., Vincent, G. M., Hart, S. D., & Corrado, R. R. (2003). The validity of the Antisocial Process Screening Device as a self-report measure of psychopathy in adolescent offenders. *Behavioral Sciences and the Law*, 21, 771–786.
- Leistico, A. R., Salekin, R. T., DeCoster, J., & Rogers, R. (2008). A large-scale meta-analysis relating the Hare measures of psychopathy to antisocial conduct. *Law and Human Behavior*, 32, 28–45.
- Lynam, D. R. (1997). Pursuing the psychopath: Capturing the fledgling psychopath in a nomological net. *Journal of Abnormal Psychology*, 106, 425–438.
- Lynam, D. R., Caspi, A., Moffitt, T. E., Loeber, R., & Stouthamer-Loeber, M. (2007). Longitudinal evidence that psychopathy scores in early adolescence predict adult psychopathy. *Journal of Abnormal Psychology*, 116, 155–165.
- Lynam, D. R., Loeber, R., & Stouthamer-Loeber, M. (2008). The stability of psychopathy from adolescence into adulthood: The search for moderators. *Criminal Justice and Behavior*, 35, 228–243.
- Lyon, D. R., & Ogloff, J. R. P. (2000). Legal and ethical issues in psychopathy assessment. In C. B. Gacono (Ed.), *The clinical and forensic assessment of psychopathy: A practitioner's guide* (pp. 139–173). Mahwah, NJ: Lawrence Erlbaum.
- Mailloux, D. L., Forth, A. E., & Kroner, D. G. (1997). Psychopathy and substance use in adolescent male offenders. *Psychological Reports*, 81, 529–530.
- Marczyk, G. R., Heilbrun, K., Lander, T., & DeMatteo, D. (2003). Predicting juvenile recidivism with the PCL:YV, MAYSI, and YLS/CMI. *International Journal of Forensic Mental Health*, 2, 7–18.
- Marshall, J., Egan, V., English, M., & Jones, R. M. (2006). The relative validity of psychopathy versus risk/needs-based assessments in the prediction of adolescent offending behaviour. *Legal and Criminological Psychology*, 11, 197–210.
- Marshall, L. A., & Cooke, D. J. (1999). The childhood experiences of psychopaths: A retrospective study of familial and societal factors. *Journal of Personality Disorders*, 13, 211–225.
- McCoy, W. K., & Edens, J. F. (2006). Do Black and White youths differ in levels of psychopathic traits? A meta-analysis of the Psychopathy Checklist measures. *Journal of Consulting and Clinical Psychology*, 74, 386–392.
- Millon, T. (1993). *Millon Adolescent Clinical Inventory manual*. Minneapolis, MN: National Computer Systems.
- Murrie, D. C., Boccaccini, M. T., McCoy, W., & Cornell, D. G. (2007). Diagnostic labeling in juvenile court: How do descriptions of psychopathy and conduct disorder influence judges? *Journal of Clinical Child and Adolescent Psychology*, 36, 228–241.
- Murrie, D. C., & Cornell, D. G. (2000). The Millon Adolescent Clinical Inventory and psychopathy. *Journal of Personality Assessment*, 75, 110–125.
- Murrie, D. C., & Cornell, D. G. (2002). Psychopathy screening of incarcerated juveniles: A comparison of measures. *Psychological Assessment*, 14, 390–396.
- Murrie, D. C., Cornell, D. G., Kaplan, S., McConville, D., & Levy-Elkon, A. (2004). Psychopathy scores and violence among juvenile offenders: A multi-measure study. *Behavioral Sciences and the Law*, 22, 49–67.
- Murrie, D. C., Cornell, D. G., & McCoy, W. K. (2005). Psychopathy, conduct disorder, and stigma: Does diagnostic labeling influence juvenile probation officer recommendations? *Law and Human Behavior*, 29, 323–342.
- Murrie, D. C., Marcus, D. K., Douglas, K. S., Lee, Z., Salekin, R. T., & Vincent, G. (2007). Youth with psychopathy features are not a discrete class: A taxometric analysis. *Journal of Child Psychology and Psychiatry*, 48, 714–723.
- Myers, W. C., Burket, R. C., & Harris, E. H. (1995). Adolescent psychopathy in relation to delinquent behaviors, conduct disorder, and personality disorders. *Journal of Forensic Sciences*, 40(3), 436–440.
- Neumann, C. S., Kosson, D. S., Forth, A. E., & Hare, R. D. (2006). Factor structure of the Hare Psychopathy Checklist: Youth Version in incarcerated adolescents. *Psychological Assessment*, 18, 142–154.
- Olver, M. E., & Stockdale, K. C. (2010). Psychopathy and youth violence: Research, controversies, and clinical utility. *British Journal of Forensic Practice*, 12(2), 3–13.
- O'Neill, M. L., Lidz, V., & Heilbrun, K. (2003). Predictors and correlates of psychopathic characteristics in substance abusing adolescents. *International Journal of Forensic Mental Health*, 2, 35–45.
- Pardini, D. A., & Loeber, R. (2008). Interpersonal callousness trajectories across adolescence: Early social influences and adult outcomes. *Criminal Justice and Behavior*, 35, 173–196.
- Parks, G. A., & Bard, D. E. (2006). Risk factors for adolescent sex offender recidivism: Evaluation of predictive factors and comparison of three groups based upon victim type. *Sex Abuse*, 18, 319–342.

- Penney, S. R., & Moretti, M. M. (2007). The relation of psychopathy to concurrent aggression and antisocial behavior in high-risk adolescent girls and boys. *Behavioral Sciences and the Law*, 25, 21–41.
- Prentky, R. A., & Righthand, S. (2003). *Juvenile Sex Offender Assessment Protocol–II (J–SOAP–II)*. Washington, DC: U.S. Department of Justice, Office of Justice Programs, Office of Juvenile Justice and Delinquency Prevention.
- Quay, H. C., Routh, D. K., & Shapiro, S. K. (1987). Psychopathology of childhood: From description to validation. *Annual Review of Psychology*, 38, 491–532.
- Quinton, D., & Rutter, M. (1988). *Parenting breakdown*. Aldershot, UK: Avebury.
- Ridenour, T. A., Marchant, G. J., & Dean, R. S. (2001). Is the Psychopathy Checklist Revised clinically useful for adolescents? *Journal of Psychoeducational Assessment*, 19, 227–238.
- Rockett, J. L., Murrie, D. C., & Boccaccini, M. (2007). Diagnostic labeling in juvenile justice settings: Do psychopathy and conduct disorder findings influence clinicians? *Services*, 4(2), 107–122.
- Rogers, R., & Rogstad, J. E. (2010). Psychopathy and APD in non-forensic patients: Improved predictions or disparities in cut scores? *Journal of Psychopathology and Behavioral Assessment*, 32, 353–362.
- Roussy, S., & Toupin, J. (2000). Behavioral inhibition deficits in juvenile psychopaths. *Aggressive Behavior*, 26, 413–424.
- Salekin, R. T. (2006). Psychopathy in children and adolescents. In C. J. Patrick (Ed.), *Handbook of psychopathy* (pp. 389–414). New York: Guilford Press.
- Salekin, R. T. (2008). Psychopathy and recidivism from mid-adolescence to young adulthood: Cumulating legal problems and limiting life opportunities. *Journal of Abnormal Psychology*, 117, 386–395.
- Salekin, R. T., Brannen, D. N., Zalot, A. A., Leistico, A. M., & Neumann, C. S. (2006). Factor structure of psychopathy in youth: Testing the applicability of the new four-factor model. *Criminal Justice and Behavior*, 33, 135–157.
- Salekin, R. T., & Lochman, J. E. (2008). Child and adolescent psychopathy: The search for protective factors. *Criminal Justice and Behavior*, 35, 159–172.
- Salekin, R. T., & Lynam, D. R. (2010). *Handbook of child and adolescent psychopathy*. New York: Guilford Press.
- Salekin, R. T., Neumann, C. S., Leistico, A. M., DiCicco, T. M., & Duros, R. L. (2004). Psychopathy and comorbidity in a young offender sample: Taking a closer look at psychopathy's potential importance over disruptive behavior disorders. *Journal of Abnormal Psychology*, 113, 416–427.
- Salekin, R. T., Neumann, C. S., Leistico, A. R., & Zalot, A. A. (2004). Psychopathy in youth and intelligence: An investigation of Cleckley's hypothesis. *Journal of Clinical Child and Adolescent Psychology*, 33, 731–742.
- Salekin, R. T., Rogers, R., & Machin, D. (2001). Psychopathy in youth: Pursuing diagnostic clarity. *Journal of Youth and Adolescence*, 30, 173–194.
- Schmidt, F., Campbell, M. A., & Houlding, C. (2011). Comparative analysis of the YLS/CMI, SAVRY, and PCL:YV in adolescent offenders: A 10-year follow-up into adulthood. *Youth Violence and Juvenile Justice*, 9, 23–42.
- Schmidt, F., McKinnon, L., Chattha, H. K., & Brownlee, K. (2006). Concurrent and predictive validity of the Psychopathy Checklist: Youth Version across gender and ethnicity. *Psychological Assessment*, 18, 393–401.
- Schmitt, W. A., & Newman, J. P. (1999). Are all psychopathic individuals low-anxious? *Journal of Abnormal Psychology*, 108, 353–358.
- Seagrave, D., & Grisso, T. (2002). Adolescent development and the measurement of juvenile psychopathy. *Law and Human Behavior*, 26, 219–239.
- Sevecke, K., Pukrop, R., Kosson, D. S., & Krischer, M. K. (2009). Factor structure of the Hare Psychopathy Checklist: Youth Version in German female and male detainees and community adolescents. *Psychological Assessment*, 21, 45–56.
- Skeem, J. L., & Cauffman, E. (2003). View of the downward extension: Comparing the Youth Version of the Psychopathy Checklist with the Youth Psychopathic Traits Inventory. *Behavioral Sciences and the Law*, 21, 737–770.
- Soderstrom, H., Nilsson, T., Sjodin, A., Carlstedt, A., & Forsman, A. (2005). The childhood-onset neuropsychiatric background to adult psychopathic traits and personality disorders. *Comprehensive Psychiatry*, 46, 111–116.
- Spain, S. E., Douglas, K. S., Poythress, N. G., & Epstein, M. (2004). The relationship between psychopathic features, violence, and treatment outcome: The comparison of three youth measures of psychopathic features. *Behavioral Sciences and the Law*, 22, 85–102.

- Stafford, J. E., & Cornell, D. (2003). Psychopathy scores predict adolescent inpatient aggression. *Assessment*, 10, 102–112.
- Steinberg, L. (2002). The juvenile psychopath: Fads, fictions, and facts. *National Institute of Justice Perspectives on Crime and Justice: 2001 Lecture Series* (Vol. 5, pp. 35–64). Rockville, MD: National Criminal Justice Reference Service.
- Stockdale, K. C., Olver, M. E., & Wong, S. C. (2010). The Psychopathy Checklist: Youth Version and adolescent and adult recidivism: Considerations with respect to gender, ethnicity, and age. *Psychological Assessment*, 22, 768–781.
- Sullivan, E. A., & Kosson, D. S. (2006). Ethnic and cultural variations in psychopathy. In C. J. Patrick (Ed.), *Handbook of psychopathy* (pp. 437–458). New York: Guilford Press.
- Taylor, J., Loney, B. R., Bobadilla, L., Iacono, W. G., & McGue, M. (2003). Genetic and environmental influences on psychopathy trait dimensions in a community sample of male twins. *Journal of Abnormal Child Psychology*, 31(6), 633–645.
- Toupin, J., Mercier, H., Dery, M., Côté, G., & Hodgins, S. (1995). Validity of the PCL–R for adolescents. *Issues in Criminological and Legal Psychology*, 24, 143–145.
- Vidal, S., & Skeem, J. L. (2007). Effect of psychopathy, abuse, and ethnicity on juvenile probation officers' decision-making and supervision strategies. *Law and Human Behavior*, 31, 479–498.
- Viding, E., Blair, R. J. R., Moffitt, T. E., & Plomin, R. (2005). Evidence for substantial genetic risk for psychopathy in 7-year-olds. *Journal of Child Psychology and Psychiatry*, 46, 592–597.
- Viljoen, J. L., MacDougall, A. E. M., Gagnon, N. C., & Douglas, K. S. (2010). Psychopathy evidence in legal proceedings involving adolescent offenders. *Psychology, Public Policy, and Law*, 16, 254–283.
- Viljoen, J. L., McLachlan, K., & Vincent, G. M. (2010). Assessing violence risk and psychopathy in juvenile and adult offenders: A survey of clinical practices. *Assessment*, 17, 377–395.
- Vincent, G. M., Odgers, C. L., McCormick, A. V., & Corrado, R. R. (2008). The PCL:YV and recidivism in male and female juveniles: A follow-up into young adulthood. *International Journal of Law and Psychiatry*, 31, 287–296.
- Vincent, G. M., Vitacco, M. J., Grisso, T., & Corrado, R. R. (2003). Subtypes of adolescent offenders: Affective traits and antisocial behavior patterns. *Behavioral Sciences and the Law*, 21, 695–712.
- Vitacco, M. J., Neumann, C. S., Caldwell, M. F., Leistico, A. M., & Van Rybroek, G. J. (2006). Testing factor models of the Psychopathy Checklist: Youth Version and their association with instrumental aggression. *Journal of Personality Assessment*, 87, 74–83.
- Walsh, Z., Swogger, M. T., & Kosson, D. S. (2009). Psychopathy and instrumental violence: Facet level relationships. *Journal of Personality Disorders*, 23, 416–424.
- Weizmann-Henelius, G., Viemerö, V., & Eronen, M. (2004). Psychopathy in violent female offenders in Finland. *Psychopathology*, 37, 213–221.
- Werner, E., & Smith, R. (1992). *Overcoming the odds: High-risk children from birth to adulthood*. Ithaca, NY: Cornell University Press.
- Wong, S., & Hare, R. D. (2005). *Guidelines for a psychopathy treatment program*. Toronto: Multi-Health Systems.
- Woodworth, M., & Porter, S. (2002). In cold blood: Characteristics of criminal homicides as a function of psychopathy. *Journal of Abnormal Psychology*, 111(3), 436–445.
- Zinger, I., & Forth, A. E. (1998). Psychopathy and Canadian criminal proceedings: The potential for human rights abuses. *Canadian Journal of Criminology*, 40, 237–276.

10

FORENSIC NEUROPSYCHOLOGICAL ASSESSMENT

Eric A. Zillmer and Christopher M. King¹

Psychology in the United States has become increasingly characterized by the development of applied specialty areas (Drum & Blom, 2001). The emergence of neuropsychological assessment within legal contexts is one illustration of this trend (Heilbrun et al., 2003). Forensic psychology is a specialty area within psychology broadly concerned with the various intersections among law and psychology (Heilbrun, Grisso, & Goldstein, 2009); forensic neuropsychological assessment is one of these juncture points. Forensic neuropsychological assessment generally focuses upon the relationship between cognitive parameters, such as memory or decision-making ability, and legal issues. It is one of the fastest-growing areas within clinical neuropsychology, with an increasing number of neuropsychologists presenting and evaluating the results of neuropsychological assessments in court. In turn, this activity has also led to a definable area of research activity (e.g., Sweet, Condit, & Nelson, 2008; Sweet, Ecklund-Johnston, & Malina, 2007; Sweet, King, Malina, Bergman, & Simmons, 2002). In fact, it has become common even for non-forensic neuropsychologists to be periodically confronted with forensic issues throughout their careers.

As neuropsychologists have broadened their approach to the study of behavior, they have become increasingly able to offer relevant clinical and scientific data to legal professionals on topics ranging from disabilities stemming from a traumatic brain injury to Miranda rights comprehension (see *Miranda v. Arizona*, 1966, and its progeny). The expertise offered by neuropsychologists can be of immense value for a wide range of forensic issues; indeed, the challenge facing forensic neuropsychologists is the question of how to continue expanding their roles to meet the complex needs of the legal system while also refining the areas already established. For the interested reader considering work in the forensic arena, there are several excellent compendiums on forensic psychological assessment (Grisso, 2003; Heilbrun, 2001; Melton, Petrila, Poythress, & Slobogin, 2007) and forensic neuropsychology (Denney & Sullivan, 2008; Heilbrunner, 2005; Larrabee, 2012).

With this chapter, we highlight major research, practice areas, and concepts pertaining to neuropsychologists who employ neuropsychological assessment practices to aid the legal system. Chapter coverage includes principles of forensic neuropsychological assessment, test selection considerations, and the assessment of suspect neurocognitive effort or feigning. The future of forensic neuropsychological assessment is also forecast and further readings are recommended throughout the chapter.

Principles of Forensic Neuropsychological Assessment

Heilbrun et al. (2003) described 29 principles of forensic mental health assessment (from Heilbrun, 2001)—grouped in the sequential domains of preparation, data collection and interpretation, and communication—and demonstrated that they can be readily adapted to forensic neuropsychological assessment (Table 10.1). While important differences exist between psychological assessments and neuropsychological assessments in forensic contexts, necessitating specialized training in forensic neuropsychology, Heilbrun et al. (2003) highlighted that the two specialty areas actually share much common ground. Application of these principles can be used to improve one's forensic neuropsychological evaluations and also to review the quality of others'.

TABLE 10.1 Application of Principles of Forensic Mental Health Assessment to Forensic Neuropsychological Assessment

<i>Principle</i>	<i>Special Application to Forensic Neuropsychology</i>
Preparation	
1. Identify relevant forensic issues.	Describe neuropsychological aspects of forensic issues.
2. Accept referrals only within area of expertise.	Ensure that curriculum vitae includes information documenting training and experience with particular populations, forensic issues, and legal questions similar to those in present case. Highlight information about neuropsychological assessments performed in clinical and forensic contexts.
3. Decline the referral when evaluator impartiality is unlikely.	Monitor own personal and professional reactions to particular questions to be assessed in present case (e.g., questions concerning criminal responsibility and compensation for personal injury in those who have neuropsychological deficits).
4. Clarify the evaluator's role with the attorney.	Same.
5. Clarify financial arrangements.	Same.
6. Obtain appropriate authorization.	Same.
7. Avoid playing the dual role of therapist and forensic evaluator.	Same.
8. Determine the particular role to be played within the forensic assessment if the referral is accepted.	Same.
9. Select the most appropriate model to guide data gathering, interpretation, and communication.	Interpret <i>mental disorder</i> and <i>symptoms</i> broadly to include components of neuropsychological functioning that contribute to defining the forensic issues and help to inform the court on the legal question(s).
Data Collection	
10. Use multiple sources of information for each area being assessed.	Obtain neuropsychological testing data, in addition to self-report, third-party interviews, and collateral records data.
11. Use relevance and reliability (validity) as guides for seeking information and selecting data sources.	Provide theoretical and empirical justification for use of particular neuropsychological measures for assessing specific functions.
12. Obtain relevant historical information.	Add historical information regarding nature and date(s) of brain injury and functioning in relevant areas prior to and following injury.
13. Assess clinical characteristics in relevant, reliable, and valid ways.	Define <i>clinical characteristics</i> broadly to include aspects of functioning that are potentially relevant to functional demands

TABLE 10.1 Continued

<i>Principle</i>	<i>Special Application to Forensic Neuropsychology</i>
14. Assess legally relevant behavior.	in specific case (e.g., cognitive, executive, attentional, linguistic). Focus primarily on functional demands, using neuropsychological functioning data to inform assessment of such demands.
15. Ensure that conditions for evaluations are quiet, private, and distraction-free.	May be particularly important to ensure such conditions when testing is timed and assesses cognitive capacities such as attention/concentration.
16. Provide appropriate notification of purpose and/or obtain appropriate authorization before beginning.	Same.
17. Determine whether the individual understands the purpose of the evaluation and the associated limits on confidentiality.	When neuropsychological deficits are being assessed, more detailed documentation of examinee's lack of understanding and possible reasons for such may be indicated.
Data Interpretation	
18. Use third-party information in assessing response style.	Anticipate problem of potential feigning of neuropsychological deficits in forensic contexts and be prepared to use third-party information as independent source of information on functioning.
19. Use testing when indicated in assessing response style.	Employ specialized measures to assess neurocognitive feigning.
20. Use case-specific (idiographic) evidence in assessing clinical condition, functional abilities, and causal connection.	Compare examinee's functioning to his or her previous level (e.g., prior to sustaining a brain injury).
21. Use nomothetic evidence in assessing causal connection between clinical condition, functional abilities, and causal connection.	Compare examinee levels of neuropsychological functioning to normative levels gauged from validated measures.
22. Use scientific reasoning in assessing causal connection between clinical condition and functional abilities.	Describe explanations for neuropsychological capacities and functional abilities that have the most supporting evidence and the least disconfirming evidence. When evidence is mixed or competing explanations seem comparably well supported, say so.
23. Do not answer the ultimate legal question directly.	Same.
24. Describe findings and limits so that they need little further elucidation under cross-examination.	Same.
Communication	
25. Attribute information to sources.	Same.
26. Use plain language; avoid technical jargon.	It is particularly important to translate, define, and give examples, considering the high density of technical language used in neuropsychological assessment.
27. Write the report in sections, according to model and procedures.	Include sections on referral information, sources of information, relevant history, neuropsychological functioning, relevant functional legal capacities, and conclusions. Describe

TABLE 10.1 Continued

<i>Principle</i>	<i>Special Application to Forensic Neuropsychology</i>
	causal relationship between neuropsychological deficits and functional legal capacities (which may be pre–post comparisons in some evaluations).
28. Base testimony on the results of the properly performed forensic mental health assessment.	Same.
29. Testify effectively.	Same.

Note:

Adapted with permission from “Principles of forensic mental health assessment: Implications for neuropsychological assessment in forensic contexts,” by K. Heilbrun, G. Marczyk, D. DeMatteo, E. Zillmer, J. Harris, & T. Jennings, 2003, *Assessment*, 10, pp. 334–336.

The most important distinction between traditional clinical and forensic neuropsychological assessments is that the latter require the examiner to relate neuropsychological abilities to functional legal capacities. In some types of cases, the forensic neuropsychologist may describe a causal relationship between neuropsychological deficits and functional legal capacities (e.g., competency to proceed), whereas in others (e.g., traumatic brain injury evaluations), pre–post comparisons of neurocognitive functioning may be highlighted. Additionally, the assessment of neurocognitive feigning emerges as particularly important in forensic contexts.

Test Selection

Neuropsychological assessment entails studying the brain’s behavioral product to make inferences regarding structural and physiological functioning. Like traditional psychological assessments, neuropsychological evaluations employ standardized tests that provide sensitive indices of brain–behavior relationships. Neuropsychological tests have been used on an empirical and clinical basis in various medical and psychiatric settings. They are responsive to the integrity of the cerebral hemispheres and can often pinpoint specific neurocognitive deficits (Zillmer, 2003a). The objective and quantitative nature of neuropsychological assessment has become a valuable asset in the courtroom by providing data to attorneys, judges, and jury members regarding the determination, effects, and prognosis of claimed brain dysfunction.

The neuropsychological evaluation is an objective, comprehensive assessment of a wide range of cognitive and behavioral areas of functioning. Different neuropsychological instruments test the functioning of various areas of the brain. In using such tests, clinical neuropsychologists focus on identifying, quantifying, and describing changes in behavior that relate to the integrity and functioning of the brain.

No single neuropsychological test assesses all aspects of brain functioning or is able to answer every referral question. It is thus important that an appropriate combination of tests be selected on a case-by-case basis to examine aspects of neurocognitive functioning relevant to each referral question. Most neuropsychological assessments cover multiple domains, including intellectual ability; academic achievement; language production and comprehension; memory; problem-solving ability; abstract reasoning ability; constructional ability; motor speed, strength, and coordination; and personality (Zillmer, Spiers, & Culbertson, 2008). One way to approach the construction of a forensic neuropsychology battery is to include at least one instrument that tests each of the different functional areas of the brain. Listed hierarchically from lower to higher

cognitive functions, these functional domains are orientation/arousal, sensation and perception, attention/concentration, motor skills, verbal functions/language, visual-spatial organization, memory, judgment/problem solving, and adaptive functioning. Worth noting is that higher cognitive functions depend to a large degree on intact lower cognitive functions.

Psychometric Considerations

As a scientist-practitioner, the forensic expert must keep abreast of new scientific techniques and research in the field of neuropsychology. Familiarity with up-to-date research is important because it provides the practicing forensic neuropsychologist with the necessary scientific data and reasoning to pursue his or her work with precision. And because forensic issues tend to be multifaceted and complex, and evaluations closely scrutinized, neuropsychological practice in the legal arena typically requires using a higher degree of accuracy in the assessment process, which ultimately mandates scrupulous attention to ethical and scientific processes on the part of the forensic neuropsychologist (Zillmer, 2003b).

Forensic neuropsychologists must be well informed about the different forms of reliability and validity of the neuropsychological tests they employ. Although some jurisdictions still apply the sometimes controversial *Frye* general acceptance test in determining the admissibility of expert testimony (*Frye v. United States*, 1923; see also Melton et al., 2007, pp. 19–20), a majority of American states and the U.S. federal government now apply the arguably more conservative *Daubert* reliability standard (see *Daubert v. Merrell Dow Pharmaceuticals*, 1993, and its progeny; see also Melton et al., 2007, pp. 20–23). In order for expert testimony to be admissible under the *Daubert* standard, the methods and procedures used by the expert must be *reliable* (the use of the term “reliability” by the *Daubert* court has come to be understood as implying both reliability and validity). Under *Daubert*, the trial judge is assigned a gatekeeping function; in making a reliability determination, the judge is tasked with considering the following factors (among others): (1) whether the theory or method employed is testable or falsifiable; (2) whether it has been subjected to peer review and published; (3) whether it has a known or potential error rate; (4) whether there exist standards and controls for its operation; and (5) whether it is generally accepted within the relevant scientific community.

Forensic neuropsychologists should not underestimate the significance of the *Daubert* criteria in choosing the assessment tools they employ. Any neuropsychological procedure employed by a forensic neuropsychologist should have been subjected to empirical examination, peer review and publication, and standardization; its error rates must be known and it should have gained common acceptance among forensic neuropsychologists (although differences of opinion by forensic neuropsychologists over the acceptability of an instrument are not necessarily fatal to its admissibility). Forensic neuropsychologists must be able to readily articulate how the tools they use satisfy these criteria. In the context of an overall assessment battery, forensic neuropsychologists should avoid including one or a few neuropsychological assessment tools with poor reliability and/or validity that may place into question the entire assessment battery.

Traumatic Brain Injury in the Context of Violence Risk Assessment

Given that this volume includes coverage of the Psychopathy Checklist–Revised and mentions risk assessment periodically throughout, we briefly discuss the value of neuropsychological data for violence risk assessment. Because aggression and agitation often stem from traumatic brain injuries (Granacher, 2008; Hesdorffer, Rauch, & Tamminga, 2009), forensic neuropsychologists can add valuable insights to violence risk assessments and risk management recommendations (Merbitz, Jain,

Good, & Jain, 1995; Miller, 1999; Sarapata, Herrmann, Johnson, & Aycock, 1998) with examinees with suspected histories of head trauma. Indeed, a history of head trauma that resulted in a concussion has been validated as a risk factor for violence among psychiatric inpatients (Monahan et al., 2001). Typically, psychologists who engage in violence risk assessments commonly utilize actuarial instruments or structured professional judgment tools to forecast an examinee's risk of engaging in future violent behavior. Each of these empirically supported approaches uses the presence of nomothetically validated risk factors in the individual case to arrive at a final risk estimate (for reviews, see Heilbrun, 2009; Otto & Douglas, 2010). One violence risk assessment instrument, the Classification of Violence Risk, was developed on a large sample of individuals discharged from mental health facilities, and includes prior brain injury with loss of consciousness among the risk factors for violence that it measures (see Monahan, 2010, for an overview). Anamnestic assessment, or functional analysis of prior violent behavior to identify idiographic risk factors for violence, also calls upon the examiner to consider the impact of a prior brain injury on an examinee's aggression potential, especially if violent behavior first occurred subsequent to a sustained head injury.

Reported correlates of aggression in various traumatic brain injury samples include frontal cortex injuries, heavy use of alcohol pre- and post-injury, young age, and post-injury depression (Granacher, 2008; Hesdorffer et al., 2009). Conversely, neither the size of a brain injury lesion nor the presence of post-injury seizures has been found to be correlated with aggression (Granacher, 2008). In males, sustaining a traumatic brain injury increases one's likelihood of engaging in domestic violence (Granacher, 2008). High rates of traumatic brain injuries have also been found among male, female, and juvenile prisoners; forensic psychiatric patients; and sexual offenders (Viljoen, Viljoen, Nicholls, & Brink, 2011).

Detection of Feigned Neurocognitive Presentations

The past two decades have seen significant research strides made in the detection of fabricated or exaggerated neurocognitive impairments (Berry & Schipper, 2008), termed *malinger* when the clinician concludes that the examinee is being motivated by external incentives to deceive, or *feigning* when no such assumptions about the goals of deception are made (Rogers, 2008a). Slick, Sherman, & Iverson (1999) proposed a popular diagnostic framework for malingered neurocognitive dysfunction (but for critiques, see Boone, 2007a; Rogers & Bender, 2003). Their proposal consisted of four diagnostic criteria: the presence of substantial external incentives; testing evidence of feigning; self-report discrepancies; and plausible alternative explanations ruled out. In addition, three levels of diagnostic certainty were set forth: *definite* (presence of external incentives, significant below-chance performance on a symptom validity test, and absence of plausible alternative explanations); *probable* (same requirements as definite, except that the failure is either on two tests using detection strategies other than symptom validity testing or one test coupled with a self-reported discrepancy); and *possible* (presence of external incentives and self-report discrepancies, and plausible alternative explanations ruled out). There remains a need for the development and validation of detection strategies for the converse of neurocognitive feigning: *defensiveness*, or the denial or minimization of impairment (Rogers, 2008d). Such a presentation may arise, for example, in personnel screening contexts.

Given the high prevalence estimates of feigned neurocognitive impairments in forensic contexts (see Ardolf, Denney, & Houston, 2007; Larrabee, 2003; Mittenberg, Patton, Canyock, & Condit, 2002; Slick, Tan, Strauss, & Hultsch, 2004), which have been reported to be around 40–50% (Berry & Schipper, 2008; Larrabee, 2007c), consideration of the possibility of neurocognitive feigning, and the use of objective strategies to detect it, is now a recommended component of standard forensic neuropsychological practice (Bush et al., 2005; Sweet, Condit, & Nelson, 2008; Van Gorp & Hassenstab, 2009). Indeed, several excellent review texts are available on the subject (Boone,

2007b; Larrabee, 2007a; Morgan & Sweet, 2009; see also pertinent chapters in Rogers, 2008b). Given the importance and complexity of assessing feigned neuropsychological presentations, practitioners are encouraged to review these and other pertinent sources (see, for example, sources cited in Bender, 2008, pp. 69, 81–82).

Forensic neuropsychologists are called upon to assess feigning in a variety of neurocognitive domains, including memory, executive functioning, intellectual functioning, and motor/sensory functioning. The pervasiveness of feigning across different neuropsychological domains (as well as across psychiatric and somatic domains) must also be determined (e.g., Granacher & Berry, 2008). Alleged deficits in neurocognitive functioning arise in both civil (e.g., disability, personal injury, and personnel evaluations) and criminal contexts (e.g., legal responsibility; competency to stand trial, to receive the death penalty, and to be executed; trial strategy) with both adult and child examinees. Underlying examinee claims of impairment may be alleged organic or psychogenic causes, including traumatic brain injury; delirium, dementia, and amnesic disorders; attention deficit disorders; learning disorders; developmental disorders; mental retardation; post-traumatic stress disorder; dissociative disorders; pain disorders, fatigue disorders, neurological disorders; electrical injury; and exposure to toxic substances, among others.

A variety of assessment techniques are available to the forensic neuropsychologist who suspects that an examinee may be feigning neurocognitive symptoms. The most exotic entail neuroimaging, electrophysiological measures, and electroencephalography to detect possible markers of deception in the peripheral and central nervous systems (Kingery & Schretlen, 2007; Ricker, 2012). Neuroimaging is more typically employed in neurological versus neuropsychological examinations (see Albers & Schiffer, 2007).

More common, however, is the use of detection strategies embedded in standard neuropsychological tests and specialized measures of feigned cognitive impairment. It is recommended practice to use multiple validated measures of cognitive feigning—both embedded and specialized, given the strengths and drawbacks of each—that tap a range of distinct abilities (Bender, 2008). Rogers (2008c; 2008d, pp. 24–25, 395) provides particularly helpful tables summarizing many popular embedded scales and stand-alone measures, the cognitive abilities they target, and which detection strategy they utilize.

Observational, historical, and collateral data are also useful (for example, if the clinician notices discrepancies between claimed impairments and actual behavior, or that the reported course or symptomatology of the supposed underlying disorder or injury is implausible; Bender, 2008; Slick et al., 1999). Neuropsychologists should pay specific attention to the examinee's documented history and self-report, third-party information, cultural factors, the pattern of test results, and demand characteristics of the testing situation. Referral to a neurologist or other physician may also be necessary to rule out organic causes, especially in the case of suspect somatic and medical symptomatology (Albers & Schiffer, 2007; Granacher & Berry, 2008; Rogers, 2008c).

Given the nature of the present book, we do not herein cover specialized measures; the interested reader is again directed to the reference texts we have cited. Nor do we attempt to describe what is known about the utility of all neuropsychological tests (besides that of the two most commonly employed Wechsler batteries) to detect cognitive feigning. Instead, Table 10.2 lists many of these neuropsychological measures, along with references to sources that summarize or contribute to the deception literature as it pertains to the instrument. Also, in-depth discussions of the application of some of these instruments to specific examinee types (e.g., minority persons; individuals claiming to be mentally retarded) can be found in chapters collected in Boone (2007b) and Morgan and Sweet (2009).

Because of their inclusion elsewhere in this volume, we briefly discuss what is known about the utility of the Minnesota Multiphasic Personality Inventory–2 (MMPI–2), the Millon Clinical

Multiaxial Inventory–III (MCMI–III), the Personality Assessment Inventory (PAI), and the Rorschach Inkblot Method (RIM) for detecting feigned cognitive impairments. We also discuss detection strategies embedded in two commonly employed Wechsler batteries: the Wechsler Adult Intelligence Scale–Revised, –Third Edition, and –Fourth Edition (WAIS–R, WAIS–III, and WAIS–IV) and the Wechsler Memory Scale–Revised, –Third Edition, and –Fourth Edition (WMS–R, WMS–III, and WMS–IV). Benefits of using these instruments as simultaneous measures of genuine and feigned cognitive functioning include the following:

- It is efficient to do so. The measures are often already employed and so the examiner benefits from the tests serving a dual purpose.
- It permits an evaluator to retrospectively review past examinations (in which these measures were likely employed) for possible feigning.
- Embedded indicators may rely on different detection strategies than stand-alone measures, some of which might be less obvious to an examinee who is privy to detection techniques via independent research or coaching (Bender, 2008; Berry & Schipper, 2008).

TABLE 10.2 Commonly Employed Neuropsychological Instruments and References Discussing Their Utility for Detecting Neurocognitive Feigning

<i>Test Domain</i>	<i>Summary References</i>
Global Batteries	
Halstead-Reitan Battery	Arnold and Boone (2007); Bender (2008); Berry and Schipper (2008); Granacher and Berry (2008);
Category Test and Booklet Category Test	Greiffenstein (2007); Greve and Bianchini (2007);
Controlled Oral Word Association	Larrabee (2007b); Suhr and Barrash (2007); Sweet and Nelson (2007)
Speech Sounds Perception Test	Larrabee (2007b)
Luria-Nebraska Neuropsychological Battery	Larrabee (2007b)
Intelligence, Attention, and Executive Functioning	
Boston Naming Test	Granacher and Berry (2008)
Brief Test of Attention	Granacher and Berry (2008)
d2 Test of Attention	Brickenkamp and Zillmer (1998)
Raven's Progressive Matrices	Greve and Bianchini (2007)
Ruff 2 and 7 Selection Attention Test	Granacher and Berry (2008)
Ruff Figural Fluency Test	Sweet and Nelson (2007)
Stroop Test	Sweet and Nelson (2007)
TOL-DX Tower of London—Drexel University	Culbertson & Zillmer (2005)
Trail Making Test	Berry and Schipper (2008); Granacher and Berry (2008); Suhr and Barrash (2007); Sweet and Nelson (2007)
Wechsler Adult Intelligence Scale (WAIS–R and WAIS–III)	Babikian and Boone (2007); Bender (2008); Berry and Schipper (2008); Granacher and Berry (2008);
Digit Span (and Reliable Digit Span)	Larrabee (2007b); Suhr and Barrash (2007); Sweet, Condit, and Nelson (2008); Van Gorp and Hassenstab (2009)
Working Memory Index	Granacher and Berry (2008)
Processing Speed Index	Granacher and Berry (2008)
Wechsler Test of Adult Reading	Berry and Schipper (2008); Granacher and Berry (2008); Greve and Bianchini (2007); Sweet and Nelson (2007)
Wide Range Achievement Test 3	
Wisconsin Card Sorting Test	

TABLE 10.2 Continued

<i>Test Domain</i>	<i>Summary References</i>
Memory (and Other Instruments Used to Detect Feigned Memory Impairments)	
California Verbal Learning Test (CVLT and CVLT-II)	Bender (2008); Lu, Rogers, and Boone (2007); Suhr and Barrash (2007); Sweet, Condit, and Nelson (2008); Van Gorp and Hassenstab (2009)
Memory Assessment Scales	Larrabee (2007b)
Millon Clinical Multiaxial Inventory-III	Berry and Schipper (2007)
Minnesota Multiphasic Inventory-2 and Berry (2008); Greiffenstein, Fox, and Lees-Haley (2007); Sweet, Condit, and Nelson (2008)	Bender (2008); Berry and Schipper (2007); Granacher Bender (2008); Berry and Schipper (2007)
Personality Assessment Inventory	
Recognition Memory Test	Berry and Schipper (2008); Suhr and Barrash (2007)
Rey Auditory Verbal Learning Test	Berry and Schipper (2008); Lu, Rogers, and Boone (2007); Suhr and Barrash (2007)
Rey-Osterrieth Complex Figure Test	Greiffenstein (2007); Lu, Rogers, and Boone (2007); Suhr and Barrash (2007)
Warrington Recognition Memory Test	Lu, Rogers, and Boone (2007)
Wechsler Memory Scale (WMS-R and WMS-III)	Berry and Schipper (2008); Granacher and Berry (2008); Larrabee (2007b); Lu, Rogers, and Boone (2007); Sweet, Condit, and Nelson (2008); Van Gorp and Hassenstab (2009)
Motor and Sensory	
Bender Judgment of Line Orientation	Greiffenstein (2007)
Bender Visual-Motor Gestalt Test	Greiffenstein (2007)
Benton Visual Form Discrimination Test	Greiffenstein (2007)
Finger Tapping Measures	Arnold and Boone (2007); Berry and Schipper (2008); Granacher and Berry (2008); Greiffenstein (2007)
Grip Strength Measures	Arnold and Boone (2007); Granacher and Berry (2008); Greiffenstein (2007)
Grooved Pegboard	Arnold and Boone (2007); Granacher and Berry (2008); Greiffenstein (2007)
Knox Cube Test	Greiffenstein (2007)
Portland Digit Recognition Test	Greiffenstein (2007)
Reitan-Klove Sensory-Perceptual Examination	Greiffenstein (2007)
Rorschach Inkblot Method	Greiffenstein (2007)
Static Steadiness Test	Arnold and Boone (2007)
Tactile Finger Recognition and Finger Trip	Arnold and Boone (2007)
Number Writing	

The Minnesota Multiphasic Personality Inventory-2

While the “F-family” of MMPI-2 validity scales (*F*: Infrequency; *Fb*: Infrequency-Back; *Fp*: Infrequency-Psychopathology; and *F-K*: Gough Dissimulation Index) have demonstrated strong utility in detecting feigned psychiatric complaints (especially psychotic symptomatology), they have proven less useful in the detection of feigned emotional disorders of a non-psychotic variety, somatic complaints, and other disabilities often at issue in civil litigation (Berry & Schipper, 2007;

Greiffenstein, Fox, & Lees-Haley, 2007; see also Sweet, Condit, & Nelson, 2008). The Fake Bad Scale (*FBS*) was thus developed to detect feigning in these later domains (Lees-Haley, English, & Glenn, 1991), and subsequent studies have positioned the *FBS* as valid for this contextual purpose. For instance, a meta-analysis by Nelson, Hoelzlea, Sweet, Arbisiab, and Demakise (2010) examining the performance of MMPI-2 validity scales in distinguishing between over-reporting and comparison groups reported a large omnibus effect size ($d = .95$) for the *FBS*. Even larger *FBS* effect sizes were found in studies of individuals evincing (known) insufficient cognitive effort ($d = 1.16$) or presenting with claims of traumatic brain injury ($d = 1.28$). Greiffenstein, Fox, and Lees-Haley (2007, p. 229) summarized the strengths of the *FBS*, which

include validation in diverse clinical and litigated samples; consistent demonstrations of sensitivity to symptom implausibility and disability seeking; proven association with malingered cognitive deficit; discriminant validity illustrated by superiority over the MMPI-2 F-family in natural groups; acceptable specificity dependent on cut-score; large samples; and convergent validity demonstrated by correlations with measures of nonpsychotic malingering.

However, two points should be kept in mind about the *FBS*. First, certain presentations (e.g., women with psychiatric histories; examinees with genuine co-morbid injuries and medical problems) require the use of higher cut-scores. And second, because the *FBS* has not been well studied in criminal contexts (e.g., mental state at the time of offense), its use in that setting has been cautioned. Also worth noting are the criticisms directed at the *FBS* and corresponding rebuttals, summarized by Berry and Schipper (2007) and Greiffenstein, Fox, and Lees-Haley (2007).

In addition to the *FBS*, the Response Bias Scale (*RBS*) was recently developed by Gervais, Ben-Porath, Wygant, and Green (2007) to detect insufficient cognitive effort. In a sample of non-head-injured disability claimants, the scale evinced excellent specificity but poor sensitivity with a cut-off score of 17 (Bender, 2008). The Henry-Heilbronner Index (*HHI*) is another recently developed scale, sensitive to exaggerated neurocognitive complaints among personal injury litigants and disability claimants (Henry, Heilbronner, Mittenberg, & Enders, 2006). Recent studies comparing the *FBS*, *RBS*, and *HHI* in medicolegal samples have found all three to evince discriminative utility (Dionysus, Denney, & Halfaker, 2011; Jones & Ingram, 2011; Tsushima, Geling, & Fabrigas, 2011).

The Millon Clinical Multiaxial Inventory–III

Although the MCMI–III includes five response-style indicators (*V*: Invalidity; *W*: Inconsistency; *X*: Disclosure; *Y*: Desirability; and *Z*: Debasement; “Product Summary,” n.d.), the scales have been subjected to little research (Berry & Schipper, 2007). However, Aguerrevereab, Greveab, Bianchiniab, and Jonathan (2011) recently reported promising results supporting the utility of the three modifier indices—*X* (degree of openness), *Y* (faking good), and *Z* (faking bad)—to distinguish feigning from non-feigning patients in a sample of traumatic brain injury referrals. The *Z* scale was found to be the most sensitive among the three. Alone and in combination, the three indices were able to detect approximately half of the feigning patients, while misclassifying very few (e.g., 4%) to none of the non-feigners.

The Personality Assessment Inventory

Four validity scales are embedded in the PAI (Inconsistency [*INC*], Infrequency [*INF*], Positive Impression Management [*PIM*], and Negative Impression Management [*NIM*]), and two additional

faking-bad research scales have been developed: the Malingering Index (*MAL*; Morey, 1996) and the Rogers Discriminant Function (*RDF*; Rogers, Sewell, Morey, & Ustad, 1996). However, little research has examined the PAI with neurologic patients or in the context of neuropsychological functioning (see Aikman & Souheaver, 2008, and Demakis et al., 2007, for two exceptions). The few studies that have examined the utility of the PAI to detect insufficient cognitive effort (Kurtz, Shealy, & Putnam, 2007; Sumanti, Boone, Savodnik, & Gorsuch, 2006; Whiteside et al., 2010) have consistently revealed distinct elevations on the Somatic Complaints (*SOM*) scale in individuals suspected of exaggerating cognitive symptoms. Also, Sumanti et al. (2006) reported correlations between the PAI validity scales and test of cognitive effort that ranged from nearly non-existent (*RDF*) to modest (*NIM* and *MAL*) in a sample of stress claimants seeking worker's compensation. Boone (2008) concluded that "[a]t present, the usefulness of the PAI with feigned TBI and other feigned neurocognitive deficits has not been established" (p. 80). Given that only a very small number of studies have been published on the topic since that time, his conclusion appears to remain true.

The Rorschach Inkblot Method

Conceptualizing the RIM as a unique test of visuospatial perception, Greiffenstein (2007) performed a search in 2005 for studies examining its ability to detect feigned neurocognitive impairment. He uncovered no such studies. Our search of the literature in 2012 likewise failed to uncover any studies involving the RIM and neuropsychological feigning. Greiffenstein (2007) did, however, reference two articles from the 1940s discussing the performance of military personnel with neurological and neuropsychological problems on the RIM (Benton, 1945; cited in Brussel, Grassi, & Melniker, 1942; Lezak, Howieson, & Loring, 2004).

The Wechsler Adult Intelligence Scale (various editions)

Both atypical-pattern analysis (resulting in a *discriminant function*, or particular pattern of performance suggestive of feigning; Mittenberg et al., 2001; Mittenberg, Theroux-Fichera, Zielinski, & Heilbronner, 1995) and improbably poor subtest performance (i.e., the floor-effect detection strategy; Greiffenstein, Baker, & Gola, 1994) have been used to derive embedded WAIS indices capable of detecting suspect cognitive effort (Berry & Schipper, 2008; Larrabee, 2007b; Van Gorp & Hassenstab, 2009). Regarding the first approach, Mittenberg et al. (1995; 2001) reported two discriminant functions, one involving the Digit Span, Vocabulary, Arithmetic, Comprehension, Similarities, Picture Completion, and Digit Symbol subtests, and the other the Vocabulary minus Digit Span scaled score. A review of subsequent studies by Berry & Schipper (2008) found a median *d* of .97 for the seven-subtest discriminant function, and a median *d* of .44 for the Vocabulary minus Digit Span index.

The discriminatory ability of individual subtests has also been examined, most notably the Digit Span subtest (e.g., Iverson & Tulsky, 2003). The utility of the Processing Speed Index has also been examined (Suhr & Barrash, 2007). The Digit Span subtest (which is also included in the WMS) has shown the most utility because many feigners mistakenly believe it to measure memory rather than attention. That is, feigners fail to realize that "Digit Span is robust and relatively insensitive to many forms of brain damage and dysfunction" (Sweet, Condit, & Nelson, 2008, p. 227, citing Axelrod, Fichtenberg, Millis, & Wertheimer, 2006). A meta-analysis by Jasinski, Berry, Shandera, and Clark (2011) reported large effect sizes for both the Digit Span age-corrected scaled score and the Reliable Digit Span (*RDS*) scoring variant (Greiffenstein et al., 1994) in detecting cognitive feigning (average weighted *d* values of 1.08 and 1.34, respectively).

The Wechsler Memory Scale (various editions)

Two approaches have been used to derive indices of cognitive feigning on the WMS–R and WMS–III (Berry & Schipper, 2008; Larrabee, 2007b; Suhr & Barrash, 2007). The first approach relies on atypical-pattern analysis across multiple subtests (i.e., discriminant function analysis; Bernard, McGrath, & Houston, 1993; Lange, Iverson, Sullivan, & Anderson, 2006; Mittenberg, Azrin, Millsaps, & Heilbronner, 1993; Mittenberg, Patton, & Legler, 2003, as cited in Larabee, 2007b). The second approach focuses on individual subtest performance (e.g., Killgore and DellaPietra, 2000; Langeluddecke & Lucas, 2003). For example, Killgore and DellaPietra (2000) utilized the floor-effect strategy in developing the Rarely Missed Index (*RMI*). The *RMI* consists of six weighted items from the Logical Memory Delayed Recognition (*LMDR*) subtest that are answered correctly by most respondents who have not been exposed to the *LMDR* stories (i.e., above chance levels). The assumption underlying the index is that brain-injured examinees *exposed* to the *LMDR* stories should likewise perform above chance levels on the six items comprising the *RMI*. Therefore, notably poor performance is thought to be indicative of the possibility of feigning.

In combining the results from three discriminant function and four *RMI* studies, Berry and Schipper (2008) concluded that the discriminant function index has evinced slightly more promise than has the *RMI* for detecting cognitive feigning with the WMS–III (largely because of the discriminant function index having demonstrated better sensitivity). The WMS–IV, released in 2008, retained some subtests, dropped others, and added still others. To date, only two studies have examined the utility of the WMS–IV to detect cognitive feigning (Miller et al., 2011; Young, Caron, Baughman, & Sawyer, 2012). Results from each of the studies were promising.

Conclusion

Looking forward, the continued development of neurocognitive models that attempt to account for the cognitive operations of criminal offenders—including perspective taking and impulsivity—will allow for a fuller understanding of the relationship between brain processes and criminal behavior. The same is true for aggressive behavior stemming from traumatic brain injuries. This line of work has the potential to inform and influence policymakers on legal issues of utmost importance. Consider, for example, the influential role that neuropsychological research has played in reshaping death penalty and sentencing law—specifically in relation to offenders with mental retardation and juvenile offenders (see *Atkins v. Virginia*, 2002; *Roper v. Simmons*, 2010; *Graham v. Florida*, 2010; *Miller v. Alabama*, 2012). This research might also help refine contemporary approaches to offender treatment and tailoring of interventions (see Andrews & Bonta, 2010; Andrews, Bonta, & Hoge, 1990). In addition, the development of new specialized forensic neuropsychological tools, as has been seen in the areas of suspect cognitive effort, competency to proceed, and Miranda rights comprehension, will benefit the field by alleviating the over-reliance on neuropsychological tests developed for non-forensic purposes that offer less insight for some forensic issues encountered by forensic neuropsychologists (e.g., civil capacities in persons with dementia).

Forensic issues and applications are some of the fastest-growing and most actively researched topics within the field of neuropsychological assessment. Resultantly, clinical neuropsychologists are offering their expertise in legal settings with increasing frequency. The burgeoning sophistication of forensic neuropsychology research and practice attests to the field's continued maturation. It is the uniqueness and circumscribed nature of each legal case, coupled with the meaningful, sometimes dispositive role that neuropsychological assessment results can play, that makes the practice of forensic neuropsychology a stimulating and meaningful subspecialty of applied psychology.

CASE EXAMPLE

I (E.A.Z.) examined J.W., age 25, at a maximum-security prison facility to provide information that might be relevant to capital sentencing. J.W. was found guilty of murder stemming from a drug transaction, qualifying him for the death penalty. During the sentencing phase of the trial, at which I was to testify, neuropsychological evidence was to be presented as to whether he should be put to death for his crime or receive a sentence of life imprisonment instead. I was tasked with determining the existence of neuropsychological mitigating factors that might tilt the scales in J.W.'s favor with the jury, who might elect to spare him from the death penalty. At the time of testing, J.W. had been in prison for 20 months awaiting trial.

My role as a defense-retained forensic neuropsychologist was to determine whether J.W. lacked a neurological "democracy" with respect to his ability to act—that is, whether J.W. lacked a "free won't." This could be related to neuropsychological disinhibition, emotional disturbances that are biological in origin, or other neurological and neuropsychological issues. Should the testing support such a position, I would argue in court that J.W. is not independent of his biology, that a slight change in the balance of one's brain chemistry can cause large changes in behavior, and that neuropsychological research suggests that not everyone is equally free to make socially appropriate choices. Consequently, J.W. may be less blameworthy. This area of biological mitigation falls into the new intersection of neuroscience and law. It does not aim to excuse J.W. from his crimes on the basis of neuropsychological factors, but rather seeks to explain why he may be more at risk for engaging in criminal behavior. Ultimately, the issue of culpability or degree of blameworthiness is a legal and moral one that the jury will have to decide.

J.W. cooperated during the neuropsychological evaluation, and the data from the TOMM and MMPI-2 showed no clear indication that he was trying to distort the results. Thus, the findings of the evaluation were assumed to be reliable and valid.

Evidence of a Learning Disability

My findings suggested that J.W.'s educational background and neuropsychological testing were consistent with a learning disability diagnosis. Learning disabilities are defined by significant differences from the norm in the direction of deficits in different domains of intellectual functioning and achievement. Given that J.W.'s intelligence quotient on the WAIS-IV was in the low average range (full scale IQ = 82), his academic achievement and verbal intellectual abilities were relatively low in comparison to the average person.

On the Boston Naming Test, for example, his Word Finding subtest score suggested significantly below average ability for word finding. His WAIS subtest score for Vocabulary (corresponding to the fifth percentile) was his lowest among all of the 10 IQ subtest scores. His basic academic attainment as tested by a standardized test (WRAT-4) ranged from the fifth- to eighth-grade level. At the high end were Word Reading (grade equivalent = 8.9) and Spelling (grade equivalent = 8.9). J.W. had more pronounced deficits in Sentence Comprehension (grade equivalent = 7.0) and Math Computation (grade equivalent = 5.4).

Psychosocial data were consistent with this pattern of testing, revealing that J.W. had struggled academically all of his life and had dropped out of high school. His educational records revealed varying performance marks, ranging from As to Fs. He also moved frequently

from school to school, stating, "I moved schools and it was not a positive transition." J.W. also reported that "around [age] 15, [he] started having academic problems," although school records indicate problems occurring even earlier. J.W. further stated, "I do believe I got a learning disability. In an SAT test, for example, I feel like I won't get the score I deserve because I didn't read everything. If I got more time to read it at least twice, I'd do better."

The consequences of learning disabilities are many, including poor self-esteem and self-confidence. Individuals who have learning disabilities may be less observant of their social environment, may misinterpret the social behavior of others, and may not learn as easily from experiences or social cues as do their peers. They often show poor judgment and impulse control, poorly anticipate the consequences of their behavior, and demonstrate low frustration tolerance that often results in disruptive behavior. Many children with learning disabilities act out inappropriately, and this seems to have been the case for J.W. In fact, it appears that his behavioral impulsivity was a major issue throughout much of his schooling, and may be why his academic problems were never addressed. His academic success depended, as it does with many learning-disabled children, on the structure and support of his school environment. Unfortunately, he was not diagnosed with a learning disability while in school.

Evidence of Head Injuries

J.W. reports having incurred several head injuries and concussions due to fighting, a car accident, and fainting or falling. Testing was consistent with this report. On the Trail Making Test A and B, he performed in the impaired range of functioning. On the Tower of London, a neuropsychological assessment that evaluates executive planning and problem solving, J.W. approached the task impulsively, but scored overall in the average range. On the d2 Test of Attention, a timed test of selective attention and concentration, J.W.'s performance was low average for his age, in terms of his speed of processing information and his ability to focus on relevant (versus irrelevant) stimuli. He made very few errors, however.

On a measure requiring the recall of digits (WAIS-4), J.W.'s performance was consistent with low average functioning. J.W. also performed in the average range of functioning on an additional visual-spatial problem-solving task, the Benton Facial Recognition Test. Finally, he scored in the low average to average range on the Working Memory (WMI = 83, 13th percentile), Processing Speed (PSI = 86, 18th percentile) and Perceptual Reasoning (PRI = 92, 30th percentile) indices of the WAIS-IV. Taken together, his ability to conceptualize change, respond objectively, generate and select alternatives, and sustain attention varied from significantly impaired to average depending on the characteristics of the task. Such inconsistency in performance could be related to cognitive deficits associated with a history of concussions.

Evidence of Marijuana Abuse

J.W. had a significant history of marijuana abuse. He reported, "I was a bad child. . . . [M]y weed habit was bad. Smoked joints, plural, a day. More than five joints a day." His self-report also revealed that he used marijuana as a form of self-medication to calm down. When asked about his experience with marijuana and how it altered his personality, J.W. replied, "I was the coolest dude in the world. It took me to a place where I was comfortable. It obviously was a

problem if my PO [(probation officer)] said, 'don't do it or you're gonna get locked up again' and I keep doing it. Like I said it made me cool down."

Marijuana's psychoactive effects on the brain have been well studied. In adults, the long-lasting cognitive effects of prolonged marijuana use demonstrate subtle deficits in attention, executive function, and memory. Additionally, frequent marijuana use during adolescence may negatively influence neuromaturation and cognitive development. Both phenomena are tenable in J.W.'s presentation.

Evidence of Abuse-Type Trauma

J.W. strongly felt that he was a "victim of abuse and child neglect." He moved frequently as a child and reported being physically abused. Additionally, he was concerned about his mother having been diagnosed with systemic lupus erythematosus (lupus) and his dad being "practically paralyzed" and unable to work.

When asked about his physical abuse, J.W. stated:

For things I've done wrong, I got it bad. Naked, belt, ya feel me? I got beat by mom, pop, grandmother, uncle, when I did things wrong. I was a knucklehead growing up and I was an only child. I was spoiled. I got suspended a lot . . . but they felt bad about it [the beatings], so they'd give me presents after.

J.W.'s father was reportedly the primary authoritarian in the household. J.W. relayed that his father would pick him up from school and threaten him with physical punishment, which would be meted out when he got home.

When asked to elaborate on the beatings, J.W. said, "They hit me on my heinie [buttocks] with the belt. I was the type of child that wasn't gonna let them hit me so they had to overpower me." When asked how old he was when these beatings occurred, he replied, "Beginning probably first grade, six years old until they saw it didn't phase me anymore." He continued:

They were bare-hand spanks. I ain't go to school with black eyes. Everything was below the waist. Dad [would] say, I'm trying to make ends meet, you're running around, you're spoiled. You're gonna be beat. I was crying. I knew it was coming. It hurt.

Abuse was not limited to J.W.'s home life; his peers also bullied him throughout childhood and adolescence. It is common for children who are abused by their caregivers to not report physical abuse or to downplay it. One possible explanation for this is that it may be difficult for a child to integrate the two parts of a parent's personality, one loving and the other punishing. J.W. is similar to many other abuse victims in that he reported that he actually felt he deserved the physical punishment.

J.W. was administered the MMPI-2. The test results were clearly consistent with the presence of psychological distress and impulsivity, including resentment of authority. Several of the clinical scales were elevated to a degree that suggests the presence of psychological disturbances. The most common psychological explanation for this type of profile is that the test-taker had a chronic maladjustment to life. J.W.'s appears to be a long-standing and chronic personality style, and may be associated with a history of psychological and physical abuse.

Conclusions

Throughout his life, J.W. experienced numerous events that likely impacted the integrity of his brain and his neurocognitive functioning. His presentation entails a verbal learning disability, sequelae associated with a history of concussions, significant drug abuse, and a history of psychological trauma. Each one of these experiences often breeds specific neuropsychological and psychological sequelae. Cumulatively, they indicate significant neuropsychological trauma and alteration in the normal development of the brain.

Particularly significant is that J.W. exhibits symptoms consistent with the diagnosis of a verbal learning disability, which adversely affects the ability of children to communicate and meet the challenges of education. Children with learning disabilities are one of the largest groups referred for neuropsychological services. The primary emphasis of neuropsychological research concerning dyslexia has focused on the language cortex of the left hemisphere of right-handed individuals, and the planum temporale of the left posterior temporal lobe is one of the neural regions specifically implicated. There is at present no cure for learning disabilities, since they are problems that affect the brain's ability to receive, process, analyze, and store information.

In addition, J.W. had experienced several concussions during his lifetime, reported a history of physical abuse, and had been a significant drug abuser. Many individuals with similar histories act out inappropriately, demonstrate behavioral impulsivity, and have a problem with processing and analyzing information. They are less observant as to their social environment, often misinterpret those who occupy it, and may not learn as easily from experiences with it. They frequently evince poor judgment and impulse control, do not anticipate the consequences of their behavior, and demonstrate low frustration tolerance, resulting in disruptive behavior.

The causes of these behaviors were thought to be cumulative and hardwired (i.e., related to neurological processes in the brain) to the extent ascertainable with contemporary neuropsychological testing technologies. J.W. appeared to have been a victim of various neuropsychological-relevant events, and as a result his behaviors were most likely mediated to some extent by these experiences. In particular, the biopsychosocial factors outlined above (i.e., presence of learning disability, concussion sequelae, abuse, and drug history) can be related to aggression and violence both individually and cumulatively and thus represent mitigating factors in the capital sentencing paradigm.

After deliberation, the jury voted 11 to 1 to recommend a life sentence for J.W., rejecting invocation of the death penalty.

Note

- 1 Please address correspondence to Eric A. Zillmer, Carl R. Pacifico Professor of Neuropsychology, Drexel University, zillmer@drexel.edu.

References

- Aguerrevereab, L. E., Greveab, K. W., Bianchiniab, K. J., & Jonathan, S. O. (2011). Classification accuracy of the Millon Clinical Multiaxial Inventory–III modifier indices in the detection of malingering in traumatic brain injury. *Journal of Clinical and Experimental Neuropsychology*, *33*, 497–504.
- Aikman, G. G., & Souheaver, G. T. (2008). Use of the Personality Assessment Inventory (PAI) in neuropsychological testing of psychiatric outpatients. *Applied Neuropsychology*, *15*, 176–183.

- Albers, J. W., & Schiffer, R. (2007). Features of the neurological evaluation that suggest noncredible performance. In G. J. Larrabee (Ed.), *Assessment of malingered neuropsychological deficits* (pp. 312–333). New York: Oxford University Press.
- Andrews, D. A., & Bonta, J. (2010). *The psychology of criminal conduct* (5th ed.). New Providence, NJ: LexisNexis Matthew Bender.
- Andrews, D. A., Bonta, J., & Hoge, R. D. (1990). Classification for effective rehabilitation: Rediscovering psychology. *Criminal Justice and Behavior*, 17, 19–52.
- Ardolf, B. R., Denney, R. L., & Houston, C. M. (2007). Base rates of negative response bias and malingered neurocognitive dysfunction among criminal defendants referred for neuropsychological evaluation. *Clinical Neuropsychologist*, 21, 899–916.
- Arnold, G., & Boone, K. B. (2007). Use of motor and sensory tests as measures of effort. In K. B. Boone (Ed.), *Assessment of feigned cognitive impairment: A neuropsychological perspective* (pp. 178–209). New York: Guilford Press.
- Atkins v. Virginia*, 536 U.S. 304 (2002).
- Axelrod, B. N., Fichtenberg, N. L., Millis, S. R., & Wertheimer, J. C. (2006). Detecting incomplete effort with digit span from the Wechsler Adult Intelligence Scale–Third Edition. *Clinical Neuropsychologist*, 20, 513–523.
- Babikian, T., & Boone, K. B. (2007). Intelligence tests as measures of effort. In K. B. Boone (Ed.), *Assessment of feigned cognitive impairment: A neuropsychological perspective* (pp. 103–127). New York: Guilford Press.
- Bender, S. D. (2008). Malingered traumatic brain injury. In R. Rogers (Ed.), *Clinical assessment of malingering and deception* (3rd ed., pp. 69–86). New York: Guilford Press.
- Bernard, L. C., McGrath, M. J., & Houston, W. (1993). Discriminating between simulated malingering and closed-head injury on the Wechsler Memory Scale–Revised. *Archives of Clinical Neuropsychology*, 8, 539–551.
- Berry, T. R., & Schipper, L. J. (2007). Detection of feigned psychiatric symptoms during forensic neuropsychological examinations. In G. J. Larrabee (Ed.), *Assessment of malingered neuropsychological deficits* (pp. 226–263). New York: Oxford University Press.
- Berry, T. R., & Schipper, L. J. (2008). Assessment of feigned cognitive impairment using standard neuropsychological tests. In R. Rogers (Ed.), *Clinical assessment of malingering and deception* (3rd ed., pp. 237–252). New York: Guilford Press.
- Boone, K. B. (2007a). A reconsideration of the Slick et al. (1999) criteria for malingered neurocognitive dysfunction. In K. B. Boone (Ed.), *Assessment of feigned cognitive impairment: A neuropsychological perspective* (pp. 29–49). New York: Guilford Press.
- Boone, K. B. (Ed.) (2007b). *Assessment of feigned cognitive impairment: A neuropsychological perspective*. New York: Guilford Press.
- Brickenkamp, R., & Zillmer, E. A. (1998). *d2 Test of Attention*. Göttingen, Germany: Hogrefe & Huber.
- Brussel, J. A., Grassi, J. R., & Melniker, A. A. (1942). The Rorschach method and postconcussion syndrome. *Psychiatric Quarterly*, 16, 707–743.
- Bush, S. S., Ruff, R. M., Tröster, A. I., Barth, J. T., Koffler, S. P., Pliskin, N. H., . . . Silver, C. H. (2005). Symptom validity assessment: Practice issues and medical necessity [NAN position paper by NAN Policy & Planning Committee]. *Archives of Clinical Neuropsychology*, 20, 419–426.
- Culbertson, W., & Zillmer, E. A. (2005). *TOL–DX Tower of London–Drexel University* (2nd ed.). Toronto: Multi-Health Systems.
- Daubert v. Merrell Dow Pharmaceuticals*, 509 U.S. 579 (1993).
- Demakis, G. J., Hammond, F., Knotts, A., Cooper, D. B., Clement, P., Kennedy, J., & Sawyer, T. (2007). The Personality Assessment Inventory in individuals with traumatic brain injury. *Archives of Clinical Neuropsychology*, 22, 123–130.
- Denney, R. L., & Sullivan, J. P. (Eds.). (2008). *Clinical neuropsychology in the criminal forensic setting*. New York: Guilford Press.
- Dionysus, K. E., Denney, R. L., & Halfaker, D. A. (2011). Detecting negative response bias with the Fake Bad Scale, Response Bias Scale, and Henry–Heilbronner Index of the Minnesota Multiphasic Personality Inventory–2. *Archives of Clinical Neuropsychology*, 26, 81–88.
- Drum, D. J., & Blom, B. E. (2001). The dynamics of specialization in professional psychology. *Professional Psychology: Research and Practice*, 32, 513–521.
- Frye v. United States*, 293 F. 1013 (D.C. Cir. 1923).

- Gervais, R. O., Ben-Porath, Y. S., Wygant, D. B., & Green, P. (2007a). Development and validation of a response bias scale (RBS) for the MMPI-2. *Assessment*, 14, 196–208.
- Graham v. Florida*, 130 S. Ct. 2011 (2010).
- Granacher, R. P. (2008). *Traumatic brain injury: Methods for clinical and forensic neuropsychiatric assessment* (2nd ed.). Boca Raton, FL: CRC Press.
- Granacher, R. P., & Berry, D. T. R. (2008). Feigned medical presentations. In R. Rogers (Ed.), *Clinical assessment of malingering and deception* (3rd ed., pp. 145–156). New York: Guilford Press.
- Greiffenstein, M. F. (2007). Motor, sensory, and perceptual-motor pseudoabnormalities. In G. J. Larrabee (Ed.), *Assessment of malingered neuropsychological deficits* (pp. 100–130). New York: Oxford University Press.
- Greiffenstein, M. F., Baker, W. J., & Gola, T. (1994). Validation of malingered amnesia measures with a large clinical sample. *Psychological Assessment*, 6, 218–240.
- Greiffenstein, M. F., Fox, D., & Lees-Haley, P. R. (2007). The MMPI-2 Fake Bad Scale in detection of noncredible brain injury claims. In K. B. Boone (Ed.), *Assessment of feigned cognitive impairment: A neuropsychological perspective* (pp. 210–235). New York: Guilford Press.
- Greve, K. W., & Bianchini, K. J. (2007). Detection of cognitive malingering with tests of executive function. In G. J. Larrabee (Ed.), *Assessment of malingered neuropsychological deficits* (pp. 171–225). New York: Oxford University Press.
- Grisso, T. (2003). *Evaluating competencies: Forensic assessments and instruments* (2nd ed.). New York: Kluwer Academic.
- Heilbrunner, R. L. (Ed.). (2005). *Forensic neuropsychology casebook*. New York: Guilford Press.
- Heilbrun, K. (2001). *Principles of forensic mental health assessment*. New York: Kluwer Academic.
- Heilbrun, K. (2009). *Evaluation for risk of violence in adults*. New York: Oxford University Press.
- Heilbrun, K., Grisso, T., & Goldstein, A. (2009). *Foundations of forensic mental health assessment*. New York: Oxford University Press.
- Heilbrun, K., Marczyk, G., DeMatteo, D., Zillmer, E., Harris, J., & Jennings, T. (2003). Principles of forensic mental health assessment: Implications for neuropsychological assessment in forensic contexts. *Assessment*, 10, 329–343.
- Henry, G. K., Heilbrunner, R. L., Mittenberg, W., & Enders, C. (2006). The Henry–Heilbrunner index: A 15-item empirically derived MMPI-2 subscale for identifying probable malingering in personal injury litigants and disability claimants. *Clinical Neuropsychologist*, 20, 786–797.
- Hesdorffer, D. C., Rauch, S. L., & Tamminga, C. A. (2009). Long-term psychiatric outcomes following traumatic brain injury: A review of the literature. *Journal of Head Trauma Rehabilitation*, 24, 452–459.
- Iverson, G. L., & Tulsky, D. S. (2003). Detecting malingering on the WAIS-III: Unusual Digit Span performance patterns in the normal population and in clinical groups. *Archives of Clinical Neuropsychology*, 18, 1–9.
- Jasinski, L. J., Berry, D. T. R., Shandera, A. L., & Clark, J. A. (2011). Use of the Wechsler Adult Intelligence Scale Digit Span subtest for malingering detection: A meta-analytic review. *Journal of Clinical and Experimental Neuropsychology*, 33, 300–314.
- Jones, A., & Ingram, M. V. (2011). A comparison of selected MMPI-2 and MMPI-2-RF validity scales in assessing effort on cognitive tests in a military sample. *Clinical Neuropsychologist*, 25, 1207–1227.
- Kilgore, W. D., & DellaPietra, L. (2000). Using the WMS-III to detect malingering: Empirical validation of the Rarely Missed Index. *Journal of Clinical and Experimental Neuropsychology*, 28, 294–305.
- Kingery, L. R., & Schretlen, D. J. (2007). Functional neuroimaging of deception and malingering. In K. B. Boone (Ed.), *Assessment of feigned cognitive impairment: A neuropsychological perspective* (pp. 13–25). New York: Guilford Press.
- Kurtz, J. E., Shealy, S. E., & Putnam, S. H. (2007). Another look at paradoxical severity effects in head injury with the Personality Assessment Inventory. *Journal of Personality Assessment*, 88, 66–73.
- Lange, R. T., Iverson, G. L., Sullivan, K., & Anderson, D. (2006). Suppressed working memory on the WMS-III as a marker for poor effort. *Journal of Clinical and Experimental Neuropsychology*, 22, 761–771.
- Langeluddecke, P. M., & Lucas, S. K. (2003). Quantitative measures of memory malingering on the Wechsler Memory Scale (3rd ed.) in mild head injury litigants. *Archives of Clinical Neuropsychology*, 18, 18–197.
- Larrabee, G. J. (2003). Detection of malingering using atypical performance patterns on standard neuropsychological tests. *Clinical Neuropsychologist*, 17, 410–425.
- Larrabee, G. J. (Ed.). (2007a). *Assessment of malingered neuropsychological deficits*. New York: Oxford University Press.

- Larrabee, G. J. (2007b). Identification of malingering by pattern analysis on neuropsychological tests. In G. J. Larrabee (Ed.), *Assessment of malingered neuropsychological deficits* (pp. 80–99). New York: Oxford University Press.
- Larrabee, G. J. (2007c). Introduction: Malingering, research designs, and base rates. In G. J. Larrabee (Ed.), *Assessment of malingered neuropsychological deficits* (pp. 3–13). New York: Oxford University Press.
- Larrabee, G. J. (Ed.). (2012). *Forensic neuropsychology: A scientific approach* (2nd ed.). New York: Oxford University Press.
- Lees-Haley, P. R., English, L. T., & Glenn, W. J. (1991). A Fake Bad Scale on the MMPI–2 for personal injury claimants. *Psychological Reports*, 68, 203–210.
- Lezak, M. D., Howieson, D. B., & Loring, D. W. (2004). *Neuropsychological assessment* (4th ed.). New York: Oxford University Press.
- Lu, P. H., Rogers, S. A., & Boone, K. B. (2007). Use of standard memory tests to detect suspect effort. In K. B. Boone (Ed.), *Assessment of feigned cognitive impairment: A neuropsychological perspective* (pp. 128–151). New York: Guilford Press.
- Melton, G. B., Petrila, J., Poythress, N. G., & Slobogin, C. (2007). *Psychological evaluations for the courts: A handbook for mental health professionals and lawyers* (3rd ed.). New York: Guilford Press.
- Merbitz, C., Jain, S., Good, G. L., & Jain, A. (1995). Reported head injury and disciplinary rule infractions in prison. *Journal of Offender Rehabilitation*, 22, 11–19.
- Miller v. Alabama*, 132 S. Ct. 2455 (2012).
- Miller, E. (1999). Head injury and offending. *Journal of Forensic Psychiatry and Psychology*, 10, 157–166.
- Miller, J. B., Millis, S. R., Rapport, L. J., Bashem, J. R., Hanks, R. A., & Axelrod, B. N. (2011). Detection of insufficient effort using the Advanced Clinical Solutions for the Wechsler Memory Scale, Fourth Edition. *Clinical Neuropsychologist*, 25, 160–172.
- Miranda v. Arizona*, 384 U.S. 436 (1966).
- Mittenberg, W., Azrin, R., Millsaps, C., & Heilbrunner, R. (1993). Identification of malingered head injury on the Wechsler Memory Scale–Revised. *Psychological Assessment*, 5, 34–40.
- Mittenberg, W., Patton, C., Canyock, E. M., & Condit, D. C. (2002). Base rates of malingering and symptom exaggeration. *Journal of Clinical and Experimental Neuropsychology*, 24, 1094–1102.
- Mittenberg, W., Patton, C., & Legler, W. (2003, October). Identification of malingered head injury on the Wechsler Memory Scale, 3rd edition. Paper presented at the meeting of the National Academy of Neuropsychology, Dallas, TX.
- Mittenberg, W., Theroux, S., Aguila-Puentes, G., Bianchini, K., Greve, K., & Rayls, K. (2001). Identification of malingered head injury on the Wechsler Adult Intelligence Scale–3rd edition. *Clinical Neuropsychologist*, 15, 440–445.
- Mittenberg, W., Theroux-Fichera, S., Zielinski, R. E., & Heilbrunner, R. L. (1995). Identification of malingered head injury on the Wechsler Adult Intelligence Scale–Revised. *Professional Psychology: Research and Practice*, 26, 491–498.
- Monahan, J. (2010). The Classification of Violence Risk. In R. K. Otto & K. S. Douglas (Eds.), *Handbook of violence risk assessment* (pp. 187–198). New York: Routledge.
- Monahan, J., Steadman, H. J., Silver, E., Appelbaum, P. S., Robbins, P. C., Mulvey, E. P., . . . Banks, S. (2001). *Rethinking risk assessment: The MacArthur study of mental disorder and violence*. New York: Oxford University Press.
- Morey, L. C. (1996). *An interpretive guide to the Personality Assessment Inventory (PAI)*. Tampa, FL: Psychological Assessment Resources.
- Morgan, J. E., & Sweet, J. J. (Eds.). (2009). *Neuropsychology of malingering casebook*. New York: Psychology Press.
- Nelson, N. W., Hoelzlea, J. B., Sweet, J. J., Arbisab, P. A., & Demakise, G. J. (2010). Updated meta-analysis of the MMPI–2 symptom validity scale (FBS): Verified utility in forensic practice. *Clinical Neuropsychologist*, 24, 701–724.
- Otto, R. K., & Douglas, K. S. (Eds.). (2010). *Handbook of violence risk assessment*. New York: Routledge.
- Product Summary. (n.d.). Retrieved from <http://www.pearsonassessments.com/HAIWEB/Cultures/en-us/Productdetail.htm?Pid=PAg505> (accessed March 1, 2012).
- Ricker, J. (2012). Functional neuroimaging in forensic neuropsychology. In G. J. Larrabee (Ed.), *Forensic neuropsychology: A scientific approach* (2nd ed., pp. 160–178). New York: Oxford University Press.
- Rogers, R. (2008a). An introduction to response styles. In R. Rogers (Ed.), *Clinical assessment of malingering and deception* (3rd ed., pp. 237–252). New York: Guilford Press.

- Rogers, R. (Ed.). (2008b). *Clinical assessment of malingering and deception* (3rd ed.). New York: Guilford Press.
- Rogers, R. (2008c). Current status of clinical methods. In R. Rogers (Ed.), *Clinical assessment of malingering and deception* (3rd ed., pp. 391–410). New York: Guilford Press.
- Rogers, R. (2008d). Detection strategies for malingering and defensiveness. In R. Rogers (Ed.), *Clinical assessment of malingering and deception* (3rd ed., pp. 14–35). New York: Guilford Press.
- Rogers, R., & Bender, S. D. (2003). Evaluation of malingering and deception. In A. Goldstein (Ed.), *Handbook of psychology: Forensic psychology* (Vol. 11, pp. 109–129). Hoboken, NJ: John Wiley.
- Rogers, R., Sewell, K. W., Morey, L. C., & Ustad, K. L. (1996). Detection of feigned mental disorders on the Personality Assessment Inventory: A discriminant analysis. *Journal of Personality Assessment*, 67, 629–640.
- Roper v. Simmons*, 543 U.S. 551 (2005).
- Sarapata, M., Herrmann, D., Johnson, T., & Aycock, R. (1998). The role of head injury in cognitive functioning, emotional adjustment and criminal behaviour. *Brain Injury*, 12, 821–842.
- Slick, D. J., Sherman, E. M. S., & Iverson, G. L. (1999). Diagnostic criteria for malingered neurocognitive dysfunction: Proposed standards for clinical practice and research. *Clinical Neuropsychologist*, 13, 545–561.
- Slick, D. J., Tan, J. E., Strauss, E., & Hultsch, D. (2004). Detecting malingering: A survey of experts' practices. *Archives of Clinical Neuropsychology*, 19, 465–473.
- Suhr, J. A., & Barrash, J. (2007). Performance on standard attention, memory, and psychomotor speed tasks as indicators of malingering. In G. J. Larrabee (Ed.), *Assessment of malingered neuropsychological deficits* (pp. 131–170). New York: Oxford University Press.
- Sumanti, M., Boone, K. B., Savodnik, I., & Gorsuch, R. (2006). Noncredible psychiatric and cognitive symptoms in a workers' compensation "stress" claim sample. *Clinical Neuropsychologist*, 20, 754–765.
- Sweet, J. J., Condit, D. C., & Nelson, N. W. (2008). Feigned amnesia and memory loss. In R. Rogers (Ed.), *Clinical assessment of malingering and deception* (3rd ed., pp. 218–236). New York: Guilford Press.
- Sweet, J. J., Ecklund-Johnston, E., & Malina, A. (2008). Overview of forensic neuropsychology. In J. Morgan & J. Ricker (Eds.), *Textbook of clinical neuropsychology* (pp. 869–890). New York: Taylor & Francis.
- Sweet, J. J., King, J. H., Malina, A. C., Bergman, M. A., & Simmons, A. (2002). Documenting the prominence of forensic neuropsychology at national meetings and in relevant professional journals from 1990 to 2000. *Clinical Neuropsychologist*, 16, 481–494.
- Sweet, J. J., & Nelson, N. W. (2007). Validity indicators within executive function measures: Use and limits in detection of malingering. In K. B. Boone (Ed.), *Assessment of feigned cognitive impairment: A neuropsychological perspective* (pp. 152–177). New York: Guilford Press.
- Tsushima, W. T., Geling, O., & Fabrigas, J. (2011). Comparison of MMPI–2 validity scale scores of personal injury litigants and disability claimants. *Clinical Neuropsychologist*, 25, 1403–1414.
- Van Gorp, W. G., & Hassenstab, J. (2009). Why questions regarding effort and malingering are always raised in forensic neuropsychological evaluations. In J. E. Morgan & J. J. Sweet (Eds.), *Neuropsychology of malingering casebook* (pp. 9–20). New York: Psychology Press.
- Viljoen, S., Viljoen, H., Nicholls, T., & Brink, J. (2011, March). Investigating neuropsychological, psychological and risk profiles of forensic psychiatric patients with a history of head injury. Paper presented at the meeting of the American Psychology–Law Society/Fourth International Congress of Psychology and the Law, Miami, FL.
- Whiteside, D., Clinton, C., Diamonti, C., Stroemel, J., White, C., Zimmeroff, A., & Waters, D. (2010). Relationship between suboptimal cognitive effort and the clinical scales of the Personality Assessment Inventory. *Clinical Neuropsychologist*, 24, 315–325.
- Young, J. C., Caron, J. E., Baughman, B. C., & Sawyer, R. J. (2012). Detection of suboptimal effort with Symbol Span: Development of a new embedded index. *Archives of Clinical Neuropsychology*, 27, 159–164.
- Zillmer, E. A. (2003a). Introduction to a special issue on psychological and neuropsychological assessment in the forensic arena: Art or science? [Guest editor's introduction]. *Assessment*, 10, 318–320.
- Zillmer, E. A. (2003b). Sports-related concussions [Introduction]. *Applied Neuropsychology*, 10, 1–3.
- Zillmer, E. A., Spiers, M. V., & Culbertson, W. C. (2008). *Principles of neuropsychology* (2nd ed.). Belmont, CA: Thomson Wadsworth.

11

THE ACHENBACH SYSTEM OF EMPIRICALLY BASED ASSESSMENT (ASEBA)

Applications in Forensic Contexts

Thomas M. Achenbach and Leslie A. Rescorla

Basic Description

The Achenbach System of Empirically Based Assessment (ASEBA) includes standardized instruments for assessing behavioral, emotional, and social problems and adaptive functioning from the age of 1½ to 90-plus years. The purpose of the instruments is to assess a broad spectrum of problems and adaptive functioning, as seen from multiple perspectives, including self-reports and reports by people who know the person being assessed, designated as “collaterals.” The instruments include dimensional items that are scored on scales for empirically based syndromes and on *DSM*-oriented scales. In addition to dimensional items and scales, the instruments also obtain clinically useful, individualized qualitative information.

If it is suspected that a respondent may have difficulty completing an instrument because of poor reading skills or for other reasons, the instrument can be administered orally by an interviewer, who need not be a trained clinician. In such cases, the following procedure is recommended in order to maintain standardization: The interviewer hands the respondent a copy of the instrument while keeping a second copy. The interviewer then says, “I’ll read you the questions on this form and I’ll write down your answers.” Respondents whose reading skills are adequate typically start answering the questions without waiting for them to be read. However, even for respondents who need to have the questions read by the interviewer, seeing the instrument helps them follow along. For respondents who cannot read English but can read another language, translations are available in over 90 languages, which are listed at <http://www.aseba.org>.

ASEBA Instruments

The ASEBA self-report and collateral-report instruments take about 10–20 minutes to complete. They are available as paper-and-pencil forms and also via web-based applications that enable respondents to interactively complete the instruments on any web-connected computer. In addition to enabling interactive entry at a computer, one of the applications, Web-Link, also enables users to have paper forms printed by any web-connected computer.

One of the most widely used ASEBA instruments is the *Child Behavior Checklist for Ages 6 to 18* (CBCL/6–18; Achenbach & Rescorla, 2001). The CBCL/6–18 is completed by parents and parent surrogates to describe and rate children’s competencies and problems. Figure 11.1 displays the

competencies reported for 15-year-old Wayne Webster on the CBCL/6-18 completed by his mother (all personal details in this chapter are fictitious). By looking at Figure 11.1, you can see that the CBCL/6-18 first requests demographic information about the child, plus information about the person completing the instrument. Thereafter, it requests information about the child's

ASEBA  Please print **CHILD BEHAVIOR CHECKLIST FOR AGES 6-18** For office use only ID #

CHILD'S FULL NAME First Middle Last <u>Wayne Andrew Webster</u>			PARENTS' USUAL TYPE OF WORK, even if not working now. (Please be specific — for example, auto mechanic, high school teacher, homemaker, laborer, lathe operator, shoe salesman, army sergeant.) FATHER'S TYPE OF WORK <u>computer consultant</u> MOTHER'S TYPE OF WORK <u>computer programmer</u>		
CHILD'S GENDER ☒ Boy ☐ Girl	CHILD'S AGE <u>15</u>	CHILD'S ETHNIC GROUP OR RACE <u>African Amer. white</u>	THIS FORM FILLED OUT BY: (print your full name) <u>Alice N. Webster</u>		
TODAY'S DATE Mo. <u>04</u> Date <u>04</u> Yr. <u>01</u>		CHILD'S BIRTHDATE Mo. <u>03</u> Date <u>03</u> Yr. <u>86</u>		Your gender: ☐ Male ☒ Female Your relation to the child: ☒ Biological Parent ☐ Step Parent ☐ Grandparent ☐ Adoptive Parent ☐ Foster Parent ☐ Other (specify) _____	
GRADE IN SCHOOL <u>9</u>	Please fill out this form to reflect your view of the child's behavior even if other people might not agree. Feel free to print additional comments beside each item and in the space provided on page 2. Be sure to answer all items.				
NOT ATTENDING SCHOOL ☐					

I. Please list the sports your child most likes to take part in. For example: swimming, baseball, skating, skate boarding, bike riding, fishing, etc.

☐ None

a. Basketball

b. _____

c. _____

Compared to others of the same age, about how much time does he/she spend in each?				Compared to others of the same age, how well does he/she do each one?			
Less Than Average	Average	More Than Average	Don't Know	Below Average	Average	Above Average	Don't Know
☐	☒	☐	☐	☐	☒	☐	☐
☐	☐	☐	☐	☐	☐	☐	☐
☐	☐	☐	☐	☐	☐	☐	☐

II. Please list your child's favorite hobbies, activities, and games, other than sports. For example: stamps, dolls, books, piano, crafts, cars, computers, singing, etc. (Do not include listening to radio or TV.)

☐ None

a. Computers

b. _____

c. _____

Compared to others of the same age, about how much time does he/she spend in each?				Compared to others of the same age, how well does he/she do each one?			
Less Than Average	Average	More Than Average	Don't Know	Below Average	Average	Above Average	Don't Know
☐	☐	☒	☐	☐	☐	☒	☐
☐	☐	☐	☐	☐	☐	☐	☐
☐	☐	☐	☐	☐	☐	☐	☐

III. Please list any organizations, clubs, teams, or groups your child belongs to.

☒ None

a. _____

b. _____

c. _____

Compared to others of the same age, how active is he/she in each?			
Less Active	Average	More Active	Don't Know
☐	☐	☐	☐
☐	☐	☐	☐
☐	☐	☐	☐

IV. Please list any jobs or chores your child has. For example: paper route, babysitting, making bed, working in store, etc. (Include both paid and unpaid jobs and chores.)

☐ None

a. cutting lawn

b. _____

c. _____

Compared to others of the same age, how well does he/she carry them out?			
Below Average	Average	Above Average	Don't Know
☐	☒	☐	☐
☐	☐	☐	☐
☐	☐	☐	☐

Be sure you answered all items. Then see other side.

FIGURE 11.1 Pages 1 and 2 of the Child Behavior Checklist completed for Wayne Webster by his mother (from Achenbach & Rescorla, 2001).

involvement in sports, other kinds of activities, organizations, jobs and chores, and friendships. It also asks about the child's relationships with significant other people, how well the child plays and works alone, and the child's functioning in school, or, if not attending school, the reason for not attending. Open-ended items request information about disabilities and illnesses, what concerns the respondent most about the child, and the best things about the child.

Please print. Be sure to answer all items.

V. 1. About how many close friends does your child have? (Do not include brothers & sisters)

☐ None ☒ 1 ☐ 2 or 3 ☐ 4 or more

2. About how many times a week does your child do things with any friends outside of regular school hours?

(Do not include brothers & sisters)

☐ Less than 1 ☒ 1 or 2 ☐ 3 or more

VI. Compared to others of his/her age, how well does your child:

	Worse	Average	Better	
a. Get along with his/her brothers & sisters?	☒	☐	☐	☐ Has no brothers or sisters
b. Get along with other kids?	☒	☐	☐	
c. Behave with his/her parents?	☒	☐	☐	
d. Play and work alone?	☐	☒	☐	

VII. 1. Performance in academic subjects.

☐ Does not attend school because _____

Check a box for each subject that child takes	Failing	Below Average	Average	Above Average
a. Reading, English, or Language Arts	☒	☐	☐	☐
b. History or Social Studies	☐	☐	☒	☐
c. Arithmetic or Math	☐	☐	☐	☒
d. Science	☐	☒	☐	☐
e. <u>computer course</u>	☐	☐	☐	☒
f. _____	☐	☐	☐	☐
g. _____	☐	☐	☐	☐

Other academic subjects—for example: computer courses, foreign language, business. Do not include gym, shop, driver's ed., or other nonacademic subjects.

2. Does your child receive special education or remedial services or attend a special class or special school?

☒ No ☐ Yes—kind of services, class, or school: _____

3. Has your child repeated any grades? ☐ No ☒ Yes—grades and reasons: 2nd grade

Was immature

4. Has your child had any academic or other problems in school? ☐ No ☒ Yes—please describe: Seems

to have lost interest in some subjects, Failing English,
When did these problems start? 6 months ago
Have these problems ended? ☒ No ☐ Yes—when? problems with anger

Does your child have any illness or disability (either physical or mental)? ☒ No ☐ Yes—please describe: _____

What concerns you most about your child? Loss of interest in school, falling grades, anger, conflicts with family, bad attitude

Please describe the best things about your child. Very bright, talent for computers. Can be very friendly sometimes

FIGURE 11.2 Page 3 of the Child Behavior Checklist completed for Wayne Webster (from Achenbach & Rescorla, 2001).

For assessing responses to interventions (RTIs) and other changes over periods of days, weeks, or months, the ASEBA includes the Behavior Problem Monitor (BPM; Achenbach, McConaughy, Ivanova, & Rescorla, 2011). Parent, teacher, and youth self-report versions (BPM-P, T, Y) are completed in 1–2 minutes at user-selected intervals. The BPM is scored on normed Internalizing, Attention Problems, Externalizing, and Total Problems scales. ASEBA software enables users to display cross-informant comparisons of item and scale scores, plus trajectories of scale scores over 10 intervals. For forensic purposes, the BPM is especially useful for evaluating children and youths in residential settings.

Table 11.1 summarizes the ASEBA self-report and collateral-report instruments, the age ranges for these instruments, who completes them, and the types of scales on each instrument.

TABLE 11.1 ASEBA Self-Report and Collateral Report Forms

<i>Form</i>	<i>Ages</i>	<i>Completed by</i>	<i>Types of Scales^a</i>
Self-report forms			
Brief Problem Monitor (BPM-Y) ^b	11–18	Youths	Internalizing, Attention Problems, Externalizing, Total Problems
Youth Self-Report (YSR) ^c	11–18	Youths	Competence, Syndromes, <i>DSM</i> -oriented
Adult Self-Report (ASR) ^d	18–59	Emancipated youths, adults	Adaptive, Syndromes, <i>DSM</i> -oriented
Older Adult Self-Report (OASR) ^e	60–90+	Older adults	Adaptive, Syndromes, <i>DSM</i> -oriented
Collateral report forms			
Brief Problem Monitor (BPM-P, BPM-T) ^b	6–18	Parent figures, educational personnel, residential staff	Internalizing, Attention Problems, Externalizing, Total Problems
Child Behavior Checklist/1½–5 (CBCL/1½–5) ^e	1½–5	Parent figures	Language, Syndromes, <i>DSM</i> -oriented
Child Behavior Checklist/6–18 (CBCL/6–18) ^b	6–18	Parent figures, residential staff	Competence, Syndromes, <i>DSM</i> -oriented
Caregiver-Teacher Report Form (C-TRF) ^c	1½–5	Teachers, daycare staff	Syndromes, <i>DSM</i> -oriented
Teacher's Report Form (TRF) ^c	6–18	Educational personnel	Academic, Adaptive, Syndromes, <i>DSM</i> -oriented
Adult Behavior Checklist (ABCL) ^d	18–59	People who know the subject	Adaptive, Syndromes, <i>DSM</i> -oriented
Older Adult Behavior Checklist (OABCL) ^e	60–90+	People who know the subject	Adaptive, Syndromes, <i>DSM</i> -oriented

Notes:

- a Forms also include Total Problems, Internalizing, and Externalizing scales (except the OASR and OABCL). “Internalizing” refers to problems that are mainly within the self, such as anxiety, depression, and physical complaints without apparent medical cause. “Externalizing” refers to problems that involve conflicts with other people and with social mores, such as fighting, stealing, and lying.
- b Achenbach, McConaughy, Ivanova, and Rescorla (2011).
- c Achenbach and Rescorla (2001).
- d Achenbach and Rescorla (2003).
- e Achenbach, Newhouse, and Rescorla (2004).
- f Achenbach and Rescorla (2000).

These references provide details of the forms, including their applications, reliability, validity, and norms.

In addition to self-report and collateral-report instruments, the ASEBA includes the *Semistructured Clinical Interview for Children and Adolescents* (SCICA; McConaughy & Achenbach, 2001), which is administered by trained interviewers to obtain a wealth of standardized and individualized observational and self-report data from 6- to 18-year-olds. ASEBA instruments for obtaining observational data in other contexts include the *Direct Observation Form* (DOF; McConaughy & Achenbach, 2009), which is completed by paraprofessionals to document child and adolescent behavior in group settings, and the *Test Observation Form* (TOF; McConaughy & Achenbach, 2004), which is completed by psychological examiners to document problems observed during the administration of ability and achievement tests.

Profiles for Scoring the ASEBA Instruments

Data obtained via online completion of ASEBA instruments are automatically scored by ASEBA software. Data obtained via completion of paper ASEBA forms can be entered and scored by clerical workers in 2–5 minutes using ASEBA computer software and can also be scored by hand. The scores are displayed on profiles of adaptive functioning and problem scales. The total score for each scale is obtained by summing the ratings of all the items on the scale. The profiles display the scale scores in relation to norms for large samples of people of the same gender and age range as the person who is being assessed.

ASEBA Competence Profiles

Figure 11.3 shows a hand-scored profile of competency scales scored from pages 1 and 2 of the CBCL/6–18 completed for Wayne Webster by his mother. In this figure, there are two broken lines printed across the graphic display. Scale scores above the top broken line are in the normal range, because they are high enough to be in the upper 93% of scores obtained by a nationally representative normative sample of U.S. youths who had not received mental health or drug abuse services in the preceding 12 months. Scores below the bottom broken line are below the 3rd percentile, which is considered to be in the clinical range because it reflects very low levels of competence. And scores between the two broken lines are between the 3rd and 7th percentiles, which constitute the borderline clinical range. Scale scores in the borderline clinical range are low enough to be of concern but not so low as to be clinically deviant in competencies measured by the scale.

Figure 11.3 shows that Wayne's CBCL scores indicate a borderline clinical level of competence on the Activities scale, a clinically low level of competence on the Social scale, and a low-normal level of competence on the School scale, compared to a U.S. normative sample of adolescent boys. Scores for these three scales are summed to provide a Total Competence score, for which Wayne's *T*-score was 26, as indicated to the right of the profile.

ASEBA Profiles of Empirically Based Syndromes

Ratings of problem items (Figure 11.2) provide scores for empirically based syndrome scales. These scales reflect actual patterns of co-occurring problem items identified by factor-analyzing parent ratings, teacher ratings, and self-ratings on over 12,000 ASEBA forms (Achenbach & Rescorla, 2001). Figure 11.4 shows a computer-scored profile of Wayne's scores on the eight syndromes derived by factor-analyzing parent ratings, teacher ratings, and self-ratings. The profile in Figure 11.4 was scored from Wayne's self-ratings on the YSR (Achenbach & Rescorla, 2001). Note that, for the syndromes, scores *above the top* broken line are in the *clinical range*, because they are higher



CBCCL PROFILE FOR BOYS — COMPETENCE SCALES

%ile	Age: 6-11	12-18	6-11	12-18	6-11	12-18	6-11	12-18	T	Name
≥93	14.5-15.0 14.0 13.5	14.5-15.0 14.0 13.5	12.5-14.0 12.0 11.5	13.0-14.0 12.5 12.0	12.5-14.0 12.0 11.5	13.0-14.0 12.5 12.0	12.5-14.0 12.0 11.5	13.0-14.0 12.5 12.0	65	Wayne A. Webster
84	13.0	13.0	11.0	11.5	11.0	11.5	11.0	11.5	60	230/25/405
69	12.5	12.5	10.5	10.5	10.0	10.5	10.0	10.5	55	15
50	12.0	12.0	9.5	9.5	9.0	9.5	9.0	9.5	50	Date CBCCL filled out 4-4-01
31	11.5	11.5	8.5	8.5	8.0	8.5	8.0	8.5	45	CBCCL filled out by Alice N. Webster
16	11.0	11.0	7.5	7.5	7.0	7.5	7.0	7.5	40	Total Competence Score
7	10.5	10.5	6.5	6.5	6.0	6.5	6.0	6.5	35	Activities 7.0
	10.0	10.0	5.5	5.5	5.0	5.5	5.0	5.5	30	Social 3.0
	9.5	9.5	5.0	5.0	4.5	5.0	4.5	5.0	25	School + 3.0
	9.0	9.0	4.5	4.5	4.0	4.5	4.0	4.5	20	Total 13.0
	8.5	8.5	4.0	4.0	3.5	4.0	3.5	4.0	15	Total T from table 26
	8.0	8.0	3.5	3.5	3.0	3.5	3.0	3.5	10	
	7.5	7.5	3.0	3.0	2.5	3.0	2.5	3.0	5	
	7.0	7.0	2.5	2.5	2.0	2.5	2.0	2.5	0	
	6.5	6.5	2.0	2.0	1.5	2.0	1.5	2.0		
	6.0	6.0	1.5	1.5	1.0	1.5	1.0	1.5		
	5.5	5.5	1.0	1.0	0.5	1.0	0.5	1.0		
	5.0	5.0	0.5	0.5	0.0	0.5	0.0	0.5		
	4.5	4.5	0.0	0.0	0.0	0.0	0.0	0.0		
	4.0	4.0								
	3.5	3.5								
	3.0	3.0								
	2.5	2.5								
	2.0	2.0								
	1.5	1.5								
	1.0	1.0								
	0.5	0.5								
	0.0	0.0								

ACTIVITIES

I. A. # sports
B. Mean of participation and skill in sports

1
1

2

II. A. # of other activities
B. Mean of participation and skill in activities

1
1

2

IV. A. # of jobs
B. Mean job quality

1
1

7.0

Total

SOCIAL

III. A. # of organizations
B. Mean of participation in organizations

0
0

3.0

V. 1. # of friends
2. Frequency of contacts with friends

1
1

3.0

VI. A. Behavior with others
B. Behavior alone

0
1

3.0

Total

SCHOOL

VII. 1. Mean performance
2. Special class
3. Repeated grade
4. School problems

1.8
0
0
2.8

2.8

Total

Broken lines =
borderline clinical range

FIGURE 11.3 Hand-scored Competence Profile for Wayne Webster's Child Behavior Checklist (from Achenbach & Rescorla, 2001).

YSR/11-18 – Syndrome Scale Scores for Boys

ID: 2301251405-003
 Name: Wayne Webster
 Clinician: Dr. Barrett

Gender: Male
 Age: 15

Date Filled: 04/08/2001
 Birth Date: 03/03/1986
 Agency: CMHC
 Verified: Scanned

Informant: Self
 Relationship: Self

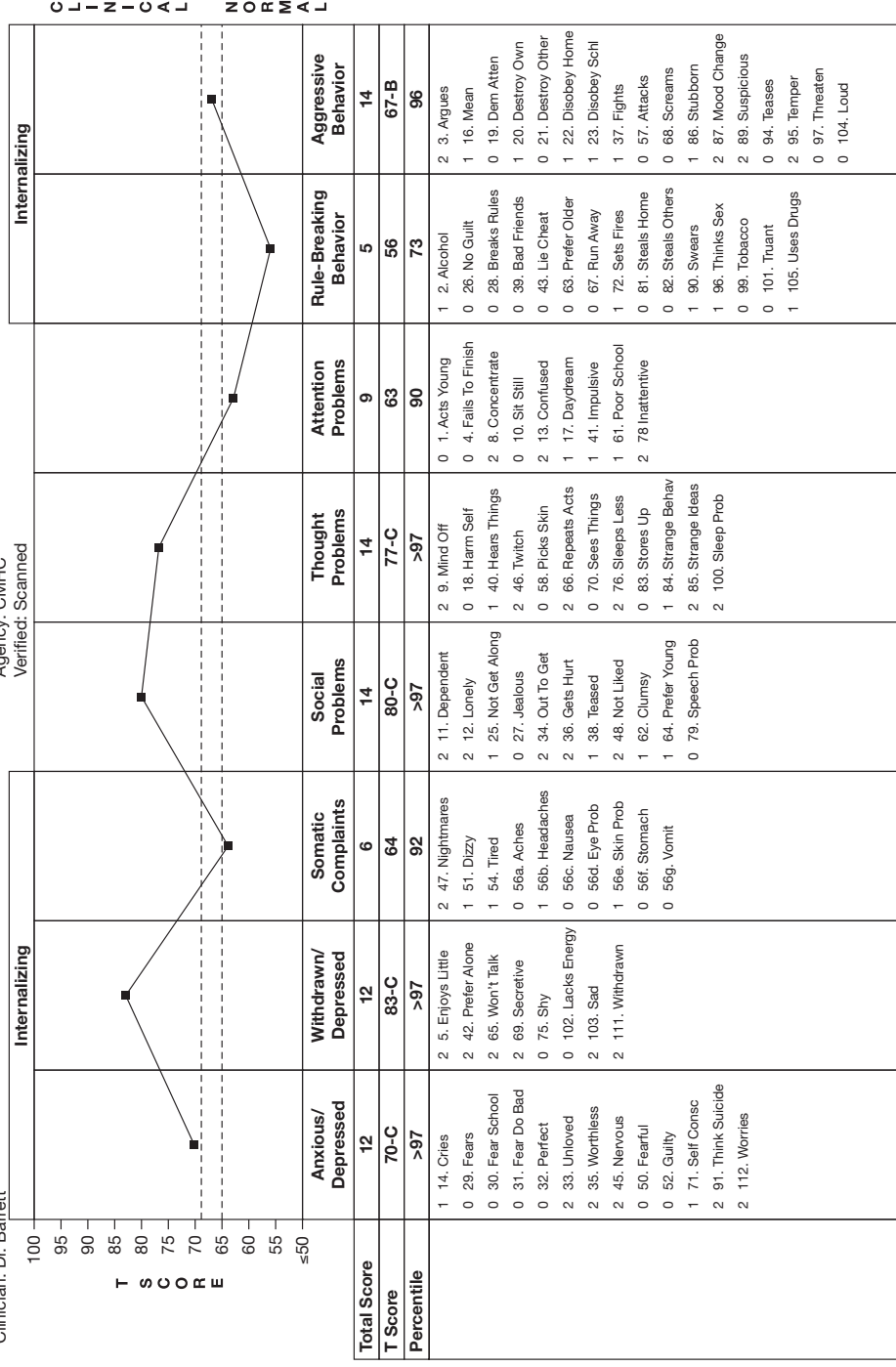


FIGURE 11.4 Computer-scored Syndrome Profile for the Youth Self-Report completed by Wayne Webster (from Achenbach & Rescorla, 2001).

than the problem scores obtained by 97% of the normative sample. Scores *between* the two broken lines are in the *borderline clinical range* (93rd to 97th percentiles). And scores *below the bottom* broken line are in the *normal range*.

ASEBA Profiles of DSM-Oriented Scales

In addition to being scored on empirically based syndrome scales, ASEBA problem items are scored on *DSM*-oriented scales. The *DSM*-oriented scales consist of problem items identified by experts from 16 cultures as being very consistent with particular diagnostic categories of the American Psychiatric Association's (1994) *Diagnostic and Statistical Manual of Mental Disorders*, fourth edition (*DSM-IV*). Like the competence and syndrome scales, the *DSM*-oriented scales are scored on profiles in relation to age-, gender-, and informant-specific norms. The *DSM*-oriented scales will be revised on the basis of *DSM-V* criteria when these are finalized.

Figure 11.5 shows a profile of *DSM*-oriented scales scored by hand from the TRF (Achenbach & Rescorla, 2001) completed by one of Wayne Webster's teachers. Table 11.2 summarizes the scales scored from the ASEBA self-report and collateral-report forms.

Cross-informant Comparisons

ASEBA instruments that are completed by different people to assess the same individual enable users to quickly and economically compare multiple reports about the individual. Documentation of cross-informant comparisons is especially valuable for assessment in forensic contexts, as different parties to a case are apt to have conflicting views and motivations. In child custody cases, for example, a mother and father may genuinely see their child differently, as well as being forensic adversaries. It is therefore important to have each parent complete a separate CBCL/6–18 (or CBCL/1½–5 for a preschool child) and to have others who know the child complete the relevant instruments such as the TRF or the *Caregiver-Teacher Report Form for Ages 1½ to 5* (C-TRF; Achenbach & Rescorla, 2000). A clinical interviewer can also administer the SCICA (McConaughy & Achenbach, 2001). Adolescents should complete the YSR.

The ASEBA software prints side-by-side comparisons of the scores obtained from different respondents. These comparisons enable users to quickly see where informants agree and disagree on particular items and scales. The cross-informant comparisons are also useful for forensic evaluations of adolescents. For example, Figure 11.6 shows bar graphs of syndrome scale scores obtained by Wayne Webster from CBCL ratings by his mother and father, YSR ratings by Wayne, and TRF ratings by three teachers. Figure 11.6 makes it clear that most or all informants reported borderline or clinical elevations of problems on the Anxious/Depressed, Withdrawn/Depressed, Social Problems, and Aggressive Behavior syndromes. By contrast, all informants' ratings were in the normal range for the Somatic Complaints and Rule-Breaking Behavior syndromes. The picture was more mixed for the Attention Problems syndrome. As is illustrated in the case example at the end of this chapter, marked differences between reports by one or two informants versus reports by the other informants may shed light on characteristics of outlier informants and how they portray and interact with the individual being assessed.

Because ASEBA instruments are available for assessing adults as well as children, it is useful to have the adult parties to custody cases complete self-report instruments and instruments describing their spouse or partner. When possible, it is also helpful to have additional adult collaterals complete collateral report instruments to describe the relevant adults. The results can be viewed in terms of profiles and bar graphs of scale scores like those shown for Wayne Webster in Figures 11.3–11.6.



FIGURE 11.5 Hand-scored *DSM-Oriented* Profile scored from the Teacher's Report Form completed by one of Wayne Webster's teachers (from Achenbach & Rescorla, 2001).

TABLE 11.2 Scales Scored from ASEBA Self-Report and Collateral Report Instruments

<i>Form</i>	<i>Competence and Adaptive</i>	<i>Syndromes</i>	<i>DSM-Oriented Scales</i>
Ages 1½–5			
CBCL, C-TRF	Language Development Survey ^a Length of Phrases ^a Vocabulary ^a	Emotionally Reactive	Affective Problems
		Anxious/Depressed	Anxiety Problems
		Somatic Complaints	Pervasive Developmental Problems
		Withdrawn	Attention Deficit/Hyperactivity Problems
		Sleep Problems ^a	Oppositional Defiant Problems
Ages 6–18 BPM, CBCL, TRF, YSR	Activities ^b Social ^b School ^b Total Competence ^b Academic ^c Adaptive Functioning ^c	Attention Problems	
		Aggressive Behavior	
		Anxious/Depressed	Affective Problems
		Withdrawn/Depressed	Anxiety Problems
		Somatic Complaints	Somatic Problems
Ages 18–59^f ASR, ABCL	Friends Spouse/Partner Family ^e Job ^e Education ^e Mean Adaptive ^e	Social Problems	Attention Deficit/Hyperactivity Problems ^d
		Thought Problems	Oppositional Defiant Problems
		Attention Problems ^d	Conduct Problems
		Rule-Breaking Behavior	
		Aggressive Behavior	
Ages 60–90+ OASR, OABCL	Friends Family ^e Spouse/Partner	Anxious/Depressed	Depressive Problems
		Withdrawn	Anxiety Problems
		Somatic Complaints	Somatic Problems
		Thought Problems	Avoidant Personality Problems
		Attention Problems	Attention Deficit/Hyperactivity Problems ^d
		Aggressive Behavior	Antisocial Personality Problems
		Rule-Breaking Behavior	
		Intrusive	
		Anxious/Depressed	Depressive Problems
		Functional Impairment	Anxiety Problems
		Irritable/Disinhibited	Somatic Problems
		Memory/Cognition Problems	Dementia Problems
		Somatic Complaints	Psychotic Problems
		Thought Problems	Antisocial Personality Problems
		Worries	

Notes: Table 11.1 provides full names of instruments, plus the four BPM scales. All instruments are also scored for Total Problems, and all, except the OASR and OABCL, are scored for Internalizing and Externalizing. Critical Items scales are scored from forms for ages 18–90+.

a CBCL/1½–5 only.

b CBCL/6–18 and YSR only (on YSR mean score for academic performance substitutes for the CBCL/6–18 School scale).

c TRF only.

d TRF, ASR, and ABCL scales have subscales for Inattention and Hyperactivity-Impulsivity.

e Self-report forms only.

f Tobacco, Alcohol, Drugs, and mean Substance Use sales are scored from ASR and ABCL. Substance use is assessed by specific items but not by normed scales on CBCL/6–18, TRF, YSR, OASR, and OABCL.

Cross-Informant Comparison – CBCL/TRF/YSR Syndrome Scale T Scores

ID: 2301251405		Name: Wayne Webster			Gender: Male		Birth Date: 03/03/1986			Comparison Date: 04/13/2001		
Form	Eval ID	Age	Relationship	Informant Name	Date	Form	Eval ID	Age	Informant Name	Relationship	Date	
CBCL1	001	15	Biological Mother	Alice N. Webster	04/04/2001	TRF5	005	15	Carmen Hernandez	Classroom Teacher (F)	04/11/2001	
CBCL2	002	15	Biological Father	Ralph. F. Webster	04/05/2001	TRF6	006	15	Charles Dwyer	Classroom Teacher (M)	04/12/2001	
YSR3	003	15	Self	Self	04/08/2001							
TRF4	004	15	Classroom Teacher (M)	George Jackson	04/10/2001							

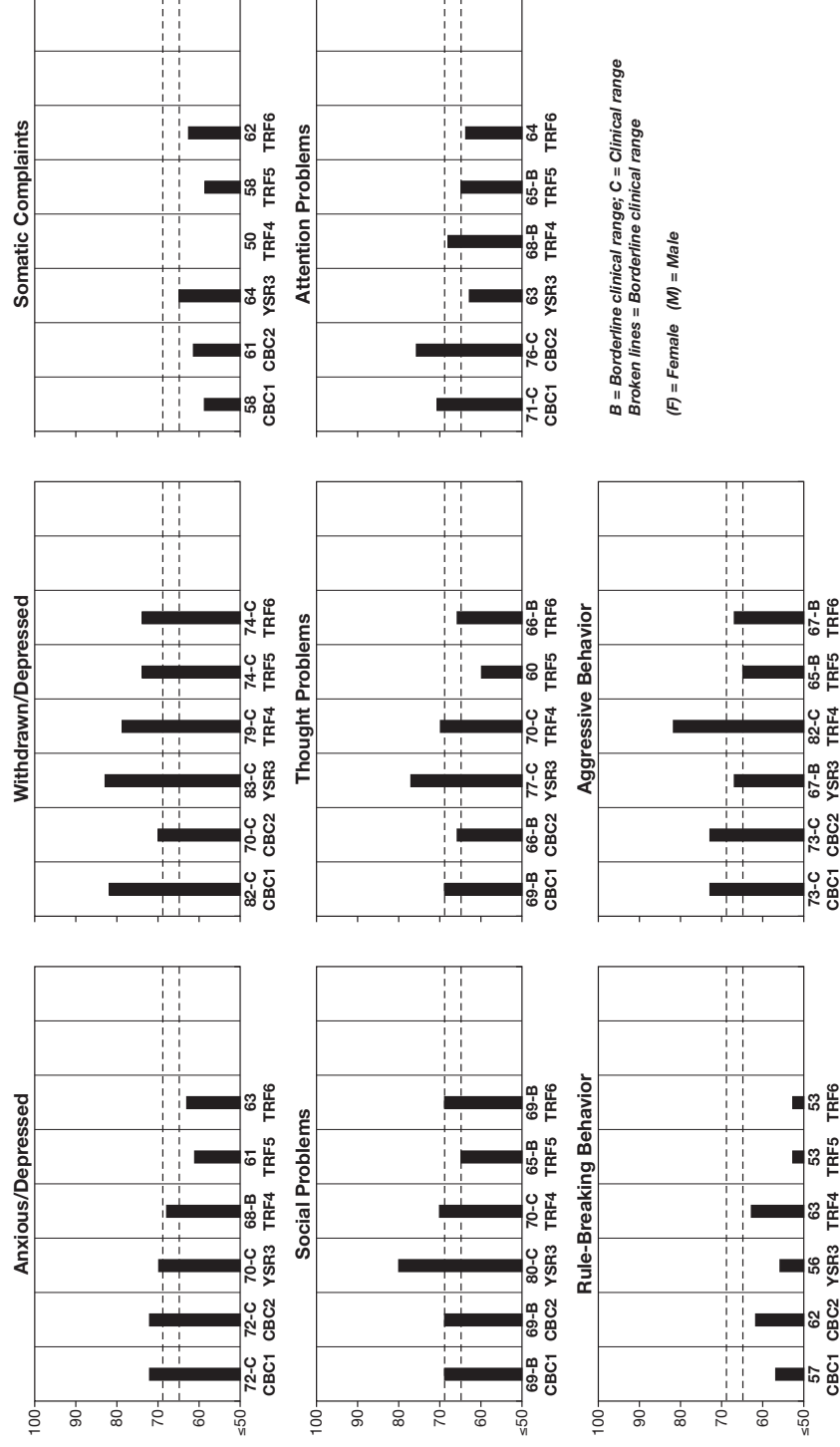


FIGURE 11.6 Cross-informant comparison of syndrome scores obtained from instruments completed by Wayne Webster, his mother, father, and three teachers (from Achenbach & Rescorla, 2001).

Narrative Reports and Critical Items

In addition to displaying profiles and side-by-side comparisons of item and scale scores, the ASEBA software also prints narrative reports of the results obtained from each ASEBA form. As an example, Figure 11.7 displays a narrative report of the results obtained from the YSR completed by Wayne Webster. At the bottom of the narrative report is a box in which Wayne's self-ratings of critical

YSR/11–18 – Narrative Report & Critical Items

ID: 2301251405-003
Name: Wayne Webster
Age: 15

Birth Date: 03/03/1986
Gender: Male

Date Filled: 04/08/2001
Informant: Self
Clinician: Dr. Barrett

The Youth Self-Report (YSR) was completed by Wayne to obtain his perceptions of his competencies and problems. Wayne reported that he participates in one sport and that he has interests in one hobby. He belongs to no social organizations, teams or clubs. Wayne reported that he has one job or chore. Wayne's responses indicate that he has one close friend and that he sees friends three or more times a week outside of regular school hours. Wayne rated his school performance as failing in language arts, below average in social studies, above average in math, and below average in science. He rated his performance in one additional subject as above average.

Wayne's Total Competence score was in the clinical range below the 10th percentile for self-reports by boys aged 11 to 18. His score on the Activities scale was in the borderline clinical range (3rd to 7th percentiles), and his score on the Social scale was in the clinical range below the 3rd percentile.

On the YSR problem scales, Wayne's Total Problems and Internalizing scores were both in the clinical range above the 90th percentile for boys ages 11 to 18. His Externalizing score was in the borderline clinical range (84th to 90th percentiles). His scores on the Somatic Complaints, Attention Problems, and Rule-Breaking Behavior syndromes were in the normal range. His scores on the Anxious/Depressed, Withdrawn/Depressed, Social Problems, and Thought Problems syndromes were in the clinical range above the 97th percentile. His score on the Aggressive Behavior syndrome was in the borderline clinical range (93rd to 97th percentiles). These results indicate that Wayne reported more problems than are typically reported by boys aged 11 to 18, particularly problems of anxiety or depression, withdrawn or depressed behavior, problems in social relationships, thought problems, and problems of an aggressive nature.

On the DSM-oriented scales, Wayne's scores on the Somatic Problems, Attention Defecit/Hyperactivity Problems, and Conduct Problems scales were in the normal range. His scores on the Affective Problems and Oppositional Defiant Problems scales were in the clinical range (above the 97th percentile). His score on the Anxiety Problems scale was in the borderline clinical range (93rd to 97th percentiles). These results suggest that the DSM should be consulted to determine whether Wayne meets diagnostic criteria for affective disorders and Oppositional Defiant Disorder. Wayne's score in the borderline clinical range suggests that the DSM should be consulted to determine whether Wayne might meet diagnostic criteria for disorders characterized by problems included on that scale.

Critical Items

In addition to the scale scores, it is important to consider scores on the individual problem items. Because they may raise particular challenges for management, it is especially important to note the problems listed below that were reported with scores of 1 or 2. Look at comments made by the informant on the form in relation to these problems to obtain more information about risks associate with the problems and the contexts in which the problems occur.

Score	Problem Item	Score	Problem Item
0	18. Harm Self	0	70. Sees Things
1	40. Hears Things	1	72. Sets Fires
0	57. Attacks	2	91. Think Suic
0	67. Run Away	1	105. Uses Drugs

FIGURE 11.7 Narrative Report summarizing Youth Self-Report results for Wayne Webster (from Achenbach & Rescorla, 2001).

items are displayed. These are items that clinicians have identified as potentially presenting particular risks and challenges for management. The critical items listed in the box are abbreviated versions of the items actually printed on the YSR. For example, the complete version of item 40 on the YSR is “I hear sounds or voices that other people think aren’t there (describe).” Wayne gave item 40 a rating of 1, which means “somewhat or sometimes true,” and he wrote “Sometimes I think I hear people talking about me.”

The ASEBA software also generates narrative reports and critical item scores from ASEBA instruments completed by other respondents, such as parents, teachers, and interviewers.

History of the ASEBA

The development of the ASEBA can be summarized by highlighting the main advances in several stages spanning five decades.

Analyses of Psychiatric Case Records

The ASEBA originated in the 1960s with efforts to determine whether children’s problems formed more differentiated patterns than were implied by the only two diagnostic categories for childhood disorders provided by the psychiatric nosology of that time (*DSM-I*; American Psychiatric Association, 1952). The two categories were Adjustment Reaction of Childhood and Schizophrenic Reaction, Childhood Type. By factor-analyzing problems reported in child psychiatric case records, Achenbach (1966) found many more syndromes than were implied by the two *DSM-I* categories. The term *syndrome* is used here to designate empirically identified patterns of co-occurring problems without implying assumptions about whether the etiologies are biological, environmental, or blends of both.

In addition to the numerous syndromes, the factor analyses identified two broad groupings of problems for which Achenbach (1966) coined the terms “Internalizing” and “Externalizing.” Internalizing problems are those that are primarily within the self, such as anxiety, depression, withdrawal, and somatic complaints without apparent physical cause. Externalizing problems are those that primarily involve conflict with other people and with social mores, such as aggressive behavior and delinquent behavior (now called “rule-breaking behavior”; Achenbach & Rescorla, 2001).

Normed Scales Derived from Parent, Teacher, Youth, and Observer Reports

Although numerous syndromes were derived from factor analyses of case record data, case records are compiled by mental health professionals who may not be fully aware of the problems and adaptive functioning that could potentially be reported by various informants or by the individual being assessed. To systematically obtain informants’ reports and self-reports, the CBCL, TRF, and YSR were developed to be completed by parents, teachers, and youths, respectively. Successive pilot editions of the CBCL, TRF, and YSR were tested and refined via data and feedback obtained from large samples of parents, teachers, and youths. The Direct Observation Form (DOF) was developed to enable paraprofessionals, such as teacher aides and staff in residential settings, to record observations of behavior in group contexts such as classrooms and recreational activities.

Data for large samples of clinically referred children were factor-analyzed to derive syndromes and Internalizing–Externalizing groupings of problems separately for different age groups of each gender on each instrument. In addition, data for large samples of non-referred children were used to construct norms for each problem scale, as well as for scales assessing competencies and adaptive

functioning (Achenbach & Edelbrock, 1983; 1986; 1987). Hand-scored profiles and computer software were developed to facilitate use of the ASEBA in forensic, clinical, educational, research, and other contexts.

Cross-Informant Syndromes

To test whether similar syndromes could be identified in parent ratings, teacher ratings, and self-ratings, and to facilitate cross-informant assessment, new analyses of much larger and more diverse clinical and normative samples were conducted (Achenbach, 1991). Eight syndromes were identified as having clear enough counterparts in factor analyses of parent ratings, teacher ratings, and self ratings to provide a basis for *cross-informant syndrome constructs*. Hand-scored and computer-scored profiles were constructed for displaying these eight syndromes in relation to gender- and age-specific norms, separately for the CBCL, TRF, and YSR. Equally important, to facilitate cross-informant comparisons, procedures like those illustrated earlier were developed to display item and scale scores obtained from multiple informants' ratings of the individual being assessed (Achenbach, 1991). Thereafter, ASEBA assessment forms, syndromes, and scoring profiles were published for parents' ratings of 2- and 3-year-olds (Achenbach, 1992), for caregivers' and teachers' ratings of 2- to 5-year-olds (Achenbach, 1997a), and for self-ratings and collateral ratings of 18- to 30-year-old adults (Achenbach, 1997b).

Revised Instruments, Profiles, and DSM-Oriented Scales for Ages 1½–18

In 2000 and 2001, slightly revised versions of the instruments for ages 1½–18 were published. The revisions were undertaken for the following reasons: (1) to capitalize on experience accumulated with the previous editions, (2) to take advantage of advances in multivariate statistics and in Windows® software, (3) to incorporate norms from new U.S. national probability samples, and (4) to add *DSM*-oriented scales. The revisions involved replacing a few problem items with more effective items and improving the wording of a few other items. Large new clinical and normative samples were factor-analyzed using cutting-edge combinations of exploratory factor analysis (EFA) and confirmatory factor analysis (CFA). For the CBCL/6–18, TRF, and YSR, the eight syndromes and the Internalizing–Externalizing groupings that were published in 1991 were replicated with minor changes. Correlations between scores on the 1991 syndromes and their 2001 counterparts ranged from .87 to 1.00 (Achenbach & Rescorla, 2001). The 2001 counterpart of the 1991 Delinquent Behavior syndrome was renamed the “Rule-Breaking Behavior syndrome” because some of its constituent behavior problems would not necessarily be cause for adjudication and could be manifested by children too young to be adjudicated. The 2001 counterpart of the 1991 Withdrawn syndrome was renamed as “Withdrawn/Depressed” because certain problems indicative of depression were found to load on the same factor as problems of withdrawal. Other problems indicative of depression were found to load on a factor with problems of anxiety, comprising a syndrome designated “Anxious/Depressed.”

A major innovation of the 21st-century ASEBA instruments is the addition of the *DSM*-oriented scales described earlier. By using international experts to identify items for *DSM*-oriented scales scored from the same rating instruments as the syndromes derived via factor analyses, we combined the “top-down” expert judgment approach with the “bottom-up” empirically based approach to scoring ASEBA instruments. Users can thus view problem ratings in terms of both top-down *DSM*-oriented scales and bottom-up syndrome scales.

Instruments for Ages 18 to 59 and 60 to 90-plus

As was mentioned earlier, ASEBA young adult instruments for ages 18–30 were published in 1997 (Achenbach, 1997b). To extend the assessment of adults through age 59, the young adult instruments were substantially revised, tested through several pilot editions with many 18- to 59-year-olds, and used to assess thousands of clinically referred adults as well as a U.S. national probability sample of adults assessed in home interviews. Following procedures like those used for the age 1½–18 instruments, combinations of EFA and CFA were used to derive empirically based syndromes from the instruments for ages 18–59, which are designated as the *Adult Self-Report* (ASR) and *Adult Behavior Checklist* (ABCL; Achenbach & Rescorla, 2003). *DSM*-oriented scales were constructed by having international experts in adult mental health identify ASR and ABCL items that they judged to be very consistent with particular *DSM-IV* diagnostic categories.

In response to the need for broad-spectrum, empirically based, and *DSM*-oriented assessment of the elderly, a geriatric psychiatrist, Professor Paul Newhouse, collaborated with the authors of this chapter to develop the *Older Adult Self-Report* (OASR) and *Older Adult Behavior Checklist* (OABCL; Achenbach, Newhouse, & Rescorla, 2004). The OASR and OABCL are scored in terms of empirically based syndromes and *DSM*-oriented scales developed according to the procedures used for the adult forms. Norms for the scales are based on large samples of 60- to 90-year-olds, but the research for deriving and testing the scales included participants as old as 102 years.

Multicultural Versions of ASEBA Instruments

As the United States and other countries become increasingly multicultural, forensic professionals must be prepared to assess people from cultures very different from their own. Many immigrants become involved with the courts, for a wide variety of reasons. ASEBA instruments are used in diverse cultures, with translations in over 90 languages and over 8,000 published studies from 80 societies and cultural groups (Bérubé & Achenbach, 2013). If respondents cannot read or understand English but can read languages for which ASEBA translations are available, they can complete the translated ASEBA forms. Because the item numbers and scoring are the same for the translated forms, clerical workers need not know each language in order to score the translated forms via English-language hand-scored or computer-scored profiles.

To enable users to evaluate scale scores based on ratings by respondents from particular societies, norms are needed for these societies. To obtain norms, researchers in many societies have used ASEBA instruments to assess large samples of 1½- to 59-year-olds. CFAs of data from each society have been used to test the degree to which the ASEBA syndrome models fit the data. In nearly every case, the ASEBA syndrome models met criteria for adequate or good fit to the data. The findings thus indicate that the ASEBA syndromes validly reflect patterns of ratings for 1½- to 59-year-olds in very diverse societies, such as Australia, China, Ethiopia, Iran, Peru, Russia, Thailand, Turkey, and most European countries, as well as the United States (Achenbach & Rescorla, 2007, 2010, 2014). Based on ASR and ABCL data from many societies, multicultural norms for ages 18–59 are scheduled for release in 2014.

Data from 55 societies have been used to construct multicultural norms for ASEBA syndromes, *DSM*-oriented scales, Internalizing, Externalizing, Total Problems, and other problem scales for ages 1½–59 (Achenbach & Rescorla, 2014). The mean scale scores for many societies were within 1 standard deviation (SD) of the *omnicultural mean*, which was calculated by averaging the mean Total Problems scores for all societies for which ratings on a particular instrument (e.g., the CBCL/6–18) were available. However, the mean Total Problems score for some societies was >1 SD below the omnicultural mean, while the mean Total Problems score for some other societies

was >1 SD above the omnicultural mean. Because the distribution of mean Total Problems scores on each instrument could be effectively divided into three groups of societies, norms have been incorporated into ASEBA software for societies with mean Total Problems scores >1 SD below the omnicultural mean (Group 1), mean Total Problems scores between -1 and $+1$ SD from the omnicultural mean (Group 2), and mean Total Problems scores >1 SD above the omnicultural mean (Group 3).

Users can elect to have scales scored from a particular instrument (e.g., CBCL/6–18) in relation to norms for a particular society. If ASEBA normative data are not available from a particular society, the user can select Group 1, 2, or 3 norms or can display scale scores in relation to each set of norms in turn. The user can thus see whether a child's scores would be deviant according to none of the norms, some of the norms, or all three sets of norms.

As an illustration, for a forensic evaluation of 18-year-old Anna, whose father came from Puerto Rico, scales scored from the Latin American Spanish CBCL/6–18 completed by her father would be displayed on a profile in relation to the Group 3 norms appropriate for CBCL/6–18 ratings by parents in Puerto Rico. If Anna's mother came from Peru, scales scored from the Latin American Spanish CBCL/6–18 completed by her mother would be displayed on a profile in relation to the Group 2 norms appropriate for Peru. If Anna were attending school in the United States, scales scored from TRFs completed by her teachers would be displayed in relation to Group 2 norms for the United States. And if Anna had grown up in the United States and completed the English-language YSR, her YSR scores would also be displayed in relation to the Group 2 norms appropriate for the United States. For each instrument, the profiles—like those illustrated in Figure 11.4—would thus show how Anna compared with norms for adolescent girls rated by a particular kind of informant (parent, teacher, self) from a particular society (Puerto Rico, Peru, the United States). Furthermore, each bar in the cross-informant bar graphs—like those illustrated in Figure 11.6—would be standardized for adolescent girls rated by each kind of informant from the relevant society. Users can thus identify scores that are in the normal, borderline, or clinical range in relation to norms appropriate for the youth's age and gender, the type of informant, and the relevant society.

Psychometric Characteristics

The ASEBA manuals provide extensive details of the norming procedures, norms, and psychometric characteristics of the instruments, as summarized in the following sections.

Norming Procedures and Norms

The norms for the self-report and collateral report instruments for ages $1\frac{1}{2}$ to 90-plus were constructed primarily from data obtained in the National Survey of Children, Youths, and Adults (Achenbach & Rescorla, 2000; 2001; 2003). Conducted in 1999 and 2000, the National Survey was carried out by having interviewers visit homes selected by multistage probability sampling in 40 American states and the District of Columbia. The homes were selected to be representative of the continental United States with respect to geographical region, socioeconomic status, ethnicity, and urbanization.

To obtain data on children and youths, the interviewers initially administered the CBCL/1½–5 or CBCL/6–18 to parents or surrogates. For 1½- to 5-year-olds who attended daycare or educational programs, the parents' consent was requested to permit interviewers to send the C-TRF to the daycare provider or teacher who knew the child best. For 6- to 18-year-olds who attended school, the parents' consent was requested to permit interviewers to send the TRF to the

teacher who knew the child best. For 11- to 18-year-olds, consent was requested from the parent and youth to have the youth complete the YSR. To obtain data on adults, the interviewers initially administered the ASR or OASR. They then requested consent to visit an informant who knew the adult participant well. If the participant and informant both consented, the interviewers administered the ABCL or OABCL to the informant.

Completion rates for the initial assessments ranged from 89.7% for ages 60 to 90-plus to 94.4% for ages 1½ to 5. In order to base norms on what epidemiologists call “healthy samples,” we excluded from the normative samples people who had received mental health or substance abuse services in the preceding 12 months. Table 11.3 displays the size and demographic characteristics of the normative samples for the ASEBA self-report and collateral report forms. To include older adults who would not be represented in probability samples of private dwellings, the normative samples for ages 60 to 90-plus included people drawn from senior centers and living facilities for the elderly, as well as random probability samples from private dwellings (Achenbach et al., 2004).

Percentiles and *T*-Scores

The distributions of scale scores in the normative samples provided the data for assigning percentiles and normalized *T*-scores to each raw score. The percentiles and *T*-scores were assigned to each gender separately within particular age ranges for each instrument. Thus, for example, percentiles and *T*-scores for the CBCL/6–18 scales were based on the distributions of scores obtained by boys and by girls aged 6 to 11 and 12 to 18. Similarly, percentiles and *T*-scores for the TRF were based on the distributions of scores obtained on the TRF by boys and by girls aged 6 to 11 and 12 to 18. And the percentiles and *T*-scores for the YSR were based on the distributions of scores obtained by boys and by girls on the YSR. The profiles, percentiles, and *T*-scores shown in Figures 11.3–11.6 thus indicate how a child’s scores compare with scores for normative samples of peers of the same gender and age range as assessed by a particular kind of informant.

TABLE 11.3 National Normative Samples for ASEBA Self-Report and Collateral Report Instruments^a

<i>Form</i>	<i>N</i>	<i>% Male</i>	<i>% Non-Latino White</i>	<i>% African American</i>	<i>% Latino</i>	<i>% Mixed, Other</i>
CBCL/1½–5	700	52	56	21	13	10
C-TRF/1½–5	1,192	49	48	36	8	9
CBCL/6–18	1,753	52	60	20	9	12
TRF/6–18	2,319	48	72	14	7	7
YSR/11–18	1,057	52	60	20	8	11
ABCL/18–59	1,435	41	69	17	10	4
ASR/18–59	1,767	42	63	20	11	6
OABCL/60–90+ ^b	822	42	79	13	5	3
OASR/60–90+ ^b	1,397	37	79	14	5	2

Notes:

- a Random samples from private dwellings, excluding people who had received mental health or substance abuse services in the preceding 12 months, as detailed by Achenbach and Rescorla (2000, 2001, 2003) and Achenbach, Newhouse, and Rescorla (2004).
- b Includes people drawn from senior centers and living facilities for the elderly, as well as random samples from private dwellings.

Reliability of ASEBA Scales

The ASEBA manuals present extensive data on the test–retest correlations (r_s) and Cronbach alpha coefficients for each scale of each instrument. For brevity, Table 11.4 summarizes the mean test–retest r_s and alphas for each type of scale from the self-report and collateral report instruments for ages 1½ to 90-plus. The *Bibliography of Published Studies Using the ASEBA* (Bérubé & Achenbach, 2013) lists dozens of additional studies that report on the reliability of ASEBA scales.

Validity of ASEBA Items and Scales

The ASEBA manuals report validity data for all ASEBA items in terms of significant discrimination between demographically similar clinically referred and non-referred samples, significant loadings on syndromes derived from factor analyses, and/or identification by international panels of experts as being very consistent with *DSM-IV* diagnostic categories (Achenbach et al., 2004; Achenbach & Rescorla, 2000, 2001, 2003; McConaughy & Achenbach, 2001, 2004, 2009). In addition, criterion-related validity for the ASEBA scale scores is supported by their ability to discriminate significantly between clinically referred and non-referred samples, with the effects of demographic

TABLE 11.4 Psychometric Characteristics of ASEBA Scales

Scale	CBCL/ 1½–5	C-TRF	CBCL/ 6–18	TRF	YSR	ABCL	ASR	OABCL	OASR
Competence, Adaptive									
Test–retest r	.99	NA	.90	.90	.88	.83	.79	.94	.89
Alpha	.99	NA	.70	.90	.66	.72	.65	.77	.73
Syndromes									
Test–retest r	.82	.80	.89	.83	.79	.84	.85	.94	.86
Alpha	.76	.78	.83	.85	.79	.82	.78	.84	.82
DSM-oriented									
Test–retest r	.82	.76	.88	.78	.78	.85	.83	.93	.88
Alpha	.75	.82	.82	.84	.76	.79	.78	.82	.76
Internalizing									
Test–retest r	.90	.77	.91	.86	.80	.80	.89	NA	NA
Alpha	.89	.89	.90	.90	.90	.92	.93	NA	NA
Externalizing									
Test–retest r	.87	.89	.92	.89	.89	.92	.91	NA	NA
Alpha	.92	.96	.94	.95	.90	.93	.89	NA	NA
Total Problems									
Test–retest r	.90	.88	.94	.95	.87	.92	.94	.95	.95
Alpha	.95	.97	.97	.97	.95	.97	.97	.97	.96
Substance Use									
Test–retest r	NA	NA	NA	NA	NA	.91	.93	NA	NA
Alpha	NA	NA	NA	NA	NA	NA	NA	NA	NA

Notes:

NA = not applicable.

a Mean r or alpha for relevant scales. Test–retest r for 7- to 16-day intervals. For psychometric characteristics of individual scales, see Achenbach and Rescorla (2000, 2001, 2003) and Achenbach et al. (2004).

characteristics partialled out. The validity of the clinical and borderline clinical *T*-score cutpoints on the ASEBA scales is supported by their significant discrimination between clinical and non-clinical samples in odds ratios and chi squares.

Many kinds of evidence for construct validity have been reported in the ASEBA manuals and in thousands of studies by thousands of authors, as listed in the *Bibliography of Published Studies Using the ASEBA* (Bérubé & Achenbach, 2013). Examples include significant associations with DSM diagnoses (e.g., Achenbach & Rescorla, 2001; Edelbrock & Costello, 1988; Kasius, Ferdinand, van den Berg, & Verhulst, 1997; Kazdin & Heidish, 1984; Weinstein, Noam, Grimes, Stone, & Schwab-Stone, 1990). As summarized by Achenbach and Rescorla (2000, 2001, 2003) and Achenbach et al. (2004), additional examples include significant correlations with other assessment instruments, such as the Beck Anxiety Inventory and Beck Depression Inventory (Beck & Steer, 1990), Conners' (1997) parent and teacher rating scales, the Behavior Assessment System for Children (BASC; Reynolds & Kamphaus, 1992), the Infant–Toddler Social and Emotional Assessment (ITSEA; Briggs-Gowan & Carter, 1998), the Mini-Mental State Exam (MMSE; Folstein, Folstein, & McHugh, 1975), the Minnesota Multiphasic Personality Inventory–Second Edition (MMPI–2, Butcher, Dahlstrom, Graham, Tellegen, & Kaemmer, 1989), the Neuropsychiatric Inventory (NPI; Cummings et al., 1994), the Richman Behavior Checklist (BCL; Richman, Stevenson, & Graham, 1982), the SCL–90–R (Derogatis, 1994), and the Toddler Behavior Screening Inventory (TBSI; Mouton-Simien, McCain, & Kelley, 1997).

Other kinds of validity evidence include longitudinal findings of significant predictions of outcomes, including substance abuse, trouble with the law, referral for mental health services, suicidal behavior, and psychiatric diagnoses over periods as long as 24 years (Achenbach, Howell, McConaughy, & Stanger, 1995, 1998; Ferdinand, Blüm, & Verhulst, 2001; Ferdinand et al., 2003; Ferdinand & Verhulst, 1994; Hofstra, van der Ende, & Verhulst, 2001, 2002a, 2002b; Reef, Diamantopoulou, van Meurs, Verhulst, & van der Ende, 2009). Table 11.5 summarizes validity data for ASEBA self-report and collateral report instruments.

Psychometric Strengths and Limitations

In terms of reliability, internal consistency, and multiple kinds of validity, the psychometric strengths of ASEBA instruments have been well documented in thousands of studies (Bérubé & Achenbach, 2013). Some might regard the low to moderate correlations between scores typically obtained from different kinds of informants as a psychometric limitation. These cross-informant correlations are sometimes incorrectly equated with “interrater reliability.” However, the cross-informant correlations reflect agreement (and disagreement) between informants whose roles differ vis-à-vis the person being assessed, who interact differently with the person being assessed, and whose knowledge of the person's functioning differs. By contrast, interrater reliability concerns the level of agreement between trained raters who observe the same samples of behavior.

For the DOF, which uses trained nonparticipant observers to assess specific samples of behavior, the interrater reliability *rs* have averaged .79 across all problem scales and .97 for on-task scores (McConaughy & Achenbach, 2009). However, cross-informant correlations for reports and ratings of psychopathology have averaged from .21 between TRF and YSR ratings to .75 between mothers' and fathers' CBCL/6–18 ratings (Achenbach et al., 2004; Achenbach & Rescorla, 2000, 2001, 2003). Although the TRF $\times$ YSR correlation of .21 may seem small, meta-analyses have shown that it is typical of correlations between teachers' ratings of students and the students' self-ratings (Achenbach, McConaughy, & Howell, 1987). Correlations between other combinations of informants, such as parents $\times$ their children, teachers $\times$ teachers, adults $\times$ collaterals, and clinicians $\times$ adult clients, are generally small to medium, as shown by meta-analyses of cross-

TABLE 11.5 Validity Data for ASEBA Self-Report and Collateral Report Instruments

<i>Forms</i>	<i>Validity Data</i>
CBCL/1½–5	1. All scales discriminate between referred and non-referred at $p < .01$. 2. Significant correlations with Behavior Checklist (Richman, 1977), Toddler Behavior Screening Inventory (Mouton-Simien et al., 1997), Infant–Toddler Social and Emotional Assessment (Briggs-Gowan & Carter, 1998), and <i>DSM</i> criteria (Arend et al., 1996; Keenan & Wakschlag, 2000).
C-TRF	1. All scales discriminate between referred and non-referred at $p < .01$.
CBCL/6–18	1. All scales discriminate between referred and non-referred at $p < .01$. 2. Significant correlations with Conners (1997) and BASC (Reynold & Kamphaus, 1992) parent rating scales, plus concurrent and predictive associations with many other variables (Bérubé & Achenbach, 2013).
TRF	1. All scales discriminate between referred and non-referred at $p < .01$. 2. Significant correlations with Conners (1997) and BASC (Reynolds & Kamphaus, 1992) teacher rating scales, plus concurrent and predictive associations with many other variables (Bérubé & Achenbach, 2013).
YSR	1. All scales discriminate between referred and non-referred at $p < .01$. 2. Over periods of 3, 4, and 10 years, YSR scores predicted adult ASEBA scores, signs of disturbance, and <i>DSM</i> diagnoses (Achenbach et al., 1995; Achenbach et al., 1998; Ferdinand et al., 1995; Hofstra et al., 2001), and in addition they had concurrent and predictive associations with many other variables (Bérubé & Achenbach, 2013).
ASR	1. All scales discriminate between referred and non-referred at $p < .01$. 2. Significant correlations with SCL–90–R, MMPI–2, Beck Depression Inventory, and Beck Anxiety Inventory (Beck & Steer, 1990); discriminated between referred and non-referred Dutch adults significantly better than General Health Questionnaire (Goldberg, 1972) and non-significantly better than the SCL–90–R; significantly predicted <i>DSM–IV</i> diagnoses over 6 years; significant correlations with GAF in American and Dutch samples; significantly predicted signs of disturbance over 2 years, including referral for mental health services, police contacts, alcohol abuse, and suicidal behavior and ideation.
ABCL	1. All scales discriminate between referred and non-referred at $p < .01$ except Intrusive. 2. Significant associations with diagnoses of major depressive disorder made 3 years earlier; significant prediction by scores on the corresponding CBCL scales obtained 8 years earlier in American clinical sample and 10.5 years earlier in Dutch clinical sample.
OASR	1. All problem scales discriminate between referred and non-referred at $p < .01$. 2. Significant associations with diagnoses, Mini Mental State Exam, cognitive measures, and the Neuropsychiatric Inventory.
OABCL	1. All problem scales discriminate between referred and non-referred at $p < .01$. 2. Significant associations with diagnoses, Mini Mental State Exam, cognitive measures, and the Neuropsychiatric Inventory.

Note:

Many other validity data are presented in the manual for each instrument and in thousands of studies listed in the *Bibliography of Published Studies Using the ASEBA* (Bérubé & Achenbach, 2013).

informant correlations for adults (Achenbach, Krukowski, Dumenci, & Ivanova, 2005) as well as for children (Achenbach et al., 1987).

Because different informants have different knowledge of the person being assessed and also bring different perspectives, motives, memories, and interpretations to assessment tasks, it is simply a fact of life that their reports often differ. Some instruments include “validity,” “social desirability,”

and “lie” scales that are intended to detect informant biases. However, research has shown that use of these scales to “correct” scores on substantive scales actually reduced the validity of the substantive scale scores (Archer, Fontaine, & McCrae, 1998; McGrath, Mitchell, Kim, & Hough, 2010; Piedmont, McCrae, Riemann, & Angleitner, 2000). In fact, Piedmont et al. concluded that “the best evidence on protocol validity, and the best alternative to the use of validity scales, comes from the comparison of self-report scores with independent assessments, on a case-by-case basis” (p. 590). Thus, as Meyer (2002) has argued, rather than viewing differences among reports by different informants as psychometric defects, psychologists should make use of the differences to improve assessment over what can be obtained with data from single sources. ASEBA instruments explicitly document both the similarities and the differences between reports by different informants.

Clinical Uses and Limitations

ASEBA instruments are designed to be used in diverse mental health, medical, educational, counseling, forensic, foster care, training, and research contexts. Because the self-report and collateral report instruments are completed by the people being assessed and by those who know them, no specific training is needed to obtain the assessment data. Even when ASEBA instruments are read to respondents who cannot complete them independently, the person who reads the instrument and records the answers need not have clinical training, because no clinical probing or interpretation is involved. However, to administer the SCICA, interviewers should be trained to conduct clinical interviews with children. The TOF is to be completed by people who are trained to administer individual ability or achievement tests. And the DOF requires that observers, who may be paraprofessionals, be trained to use it. Proper interpretation of ASEBA profiles and integration with other data require training in standardized assessment of at least the master’s degree level in psychology or equivalent levels of training in other fields.

For decades, ASEBA instruments have been used routinely to assess children and adolescents in many mental health, education, forensic, medical, training, and research contexts. Since the publication of instruments for ages 18 to 30 in 1997 (Achenbach, 1997b), their extension to age 59 (Achenbach & Rescorla, 2003), and the addition of instruments for ages 60 to 90-plus (Achenbach et al., 2004), ASEBA instruments have been increasingly used to assess adults as well. Their use with adults includes assessment of parents of children who are assessed for clinical and forensic purposes.

When the ASEBA is used routinely in mental health and medical settings, parents and foster parents of child clients are typically asked to complete the CBCL/1½–5 or CBCL/6–18 as part of the intake process. Other adults who are involved in the care of the child, such as grandparents, other relatives, and non-parental partners, may also be asked to complete the CBCL to provide their perspectives on the child’s functioning. If the child attends school or day care, teachers and/or daycare providers should complete the C-TRF or TRF, with parental permission. In the case of a youth aged 11 to 18, parental permission for the youth to complete the YSR is typically requested, unless the youth is being evaluated in a context where parental permission is not required, such as some forensic settings.

To make it easy to obtain data without requiring respondents to come to providers’ offices or requiring forms to be mailed back and forth, web-based applications enable users to send instruments electronically to any web-connected computer. The instrument can be printed at the remote computer or can be displayed on the monitor to enable respondents to key in their responses. The data can then be electronically transmitted back to the user’s computer for scoring and cross-informant comparisons. The user can also transmit scored profiles and cross-informant comparisons electronically to other web-connected computers.

In school settings, teachers may complete TRFs as part of referral and evaluation procedures for special education or accommodation. School psychologists often use TRF data to decide whether and, if so, what further assessments are needed, such as CBCLs completed by parents, YSRs completed by students, TOFs and SCICAs completed by school psychologists, DOFs completed by school psychologists or by paraprofessionals such as teacher aides, and other procedures. The ASEBA data can be used to support decisions about whether students meet criteria for particular services or whether other remedies, such as changes in teachers, are warranted.

Strengths and Limitations for Clinical Use

ASEBA instruments are employed in very diverse settings and cultures for very diverse purposes by diverse users without specialized training in any single theory or profession. Consequently, they provide a common data language for communication among people of very different backgrounds. For record compliance purposes, the profiles can be used to concisely and concretely document functioning at intake and subsequent time points. If practitioners deem it appropriate, the profiles and cross-informant comparisons can be shown to parents, teachers, and others involved in a case.

When an individual is initially evaluated in a particular setting, the profiles scored from ASEBA forms provide baseline measures of functioning at that point. Thereafter, ASEBA forms can be completed periodically to monitor the course and outcome of problems and adaptive functioning. After termination of services, the forms can be completed again to provide follow-up data. By comparing scores and profiles from one administration to another, users can identify areas in which functioning improves, remains stable, or worsens.

If the person who is assessed needs to be evaluated in other settings, the profiles obtained in one setting can be sent, with appropriate permissions, to practitioners in the other settings. If they elect to do so, these practitioners can request that the ASEBA forms be completed again. The resulting profiles can then be compared with the previous profiles to identify stability and change, as seen by each respondent.

Numerous published studies have reported use of ASEBA instruments for the assignment of cases to particular treatments, for identifying problems qualifying for particular educational services, and for evaluation of outcomes. The *Bibliography of Published Studies Using the ASEBA* (Bérubé & Achenbach, 2013) lists over 700 publications related to treatment and over 800 related to outcomes.

Although the ASEBA is designed to assess diverse aspects of functioning under diverse conditions, additional assessment procedures are often needed. These include standardized tests of ability and achievement; medical and neuropsychological assessment procedures when indicated; and histories of development, illness, medications, and past functioning. Customized assessment procedures may also be needed to answer the specific questions arising in particular cases. Not only the ASEBA instruments but other assessment procedures can be employed according to empirically based principles, as documented in a special section of the *Journal of Clinical Child and Adolescent Psychology* organized by Mash and Hunsley (2005).

Forensic Uses and Limitations

Many individuals who are assessed for forensic purposes are also assessed for other purposes, either before forensic assessment or after forensic assessment. For example, in child abuse and child custody cases the children in question may have been previously assessed by medical, mental health, and/or education professionals. In some cases, findings from these assessments may lead to litigation. The results of such assessments may be employed by one or more parties in litigation. The professionals who did the assessments may be called upon to consult, to serve as witnesses, and to perform further

assessments. Depending on the disposition of the cases, the same or different professionals may be asked to do additional assessments as a basis for helping the children and for monitoring their progress. In child abuse and custody cases, assessment of parents and other adults may also be needed.

When juveniles are arrested, previous psychological assessments are often considered in formulating dispositions. If no previous assessments are available or if arrested juveniles appear to have significant psychological problems, assessments are likely to be warranted. In reviewing studies of psychopathology among incarcerated juveniles, Grisso (2005) concluded that 60–70% of youths in detention centers and correction facilities have diagnosable mental disorders. Furthermore,

at least 15,000 youths annually are incarcerated upon arrest not because they are especially dangerous, but because of their acute mental disorders. They are locked up in juvenile detention centers until someone can find psychiatric or community mental health resources to provide them treatment.

(Grisso, Vincent, & Seagrave, 2005, p. xi)

Serious mental disorders should receive extensive evaluation and treatment. However, the realities of juvenile justice systems dictate that for mental health assessments to be done at all, they must typically begin with brief, standardized, structured procedures that can be administered at low cost by people who do not have advanced degrees in any of the mental health professions (Grisso, 2005). Even if mental health specialists eventually perform more extensive evaluations in the most severe cases, subsequent reassessments of youths who remain in the justice system are likely to require brief, low-cost procedures that can be used by less specialized personnel.

In adult criminal cases, psychological assessments are also most likely to be done if non-specialists can use procedures that are inexpensive, brief, standardized, and structured. Although some adults in the criminal justice system may be subsequently evaluated by mental health specialists, initial decisions can seldom wait for such evaluations. Furthermore, assessment instruments that can be applied at low cost to many adults in the system can help to determine which ones will receive more extensive evaluations and can also provide baseline data for comparison with findings from subsequent assessments.

In other kinds of adult forensic cases, including those requiring evaluations of the competency of elderly or infirm individuals, it is also valuable to have brief, standardized, structured procedures that can be routinely applied at low cost by people without advanced mental health training. As is true for other forensic applications, such assessments of the elderly and infirm can help to determine when more extensive evaluations should be done and can provide baseline data for comparison with subsequent assessments.

Applications of ASEBA Instruments in Forensic Contexts

Because ASEBA instruments are widely used by people from diverse disciplines and cultures, they can provide exceptional continuity between assessments done in different contexts for different purposes. It is especially helpful to be able to compare the results of forensic assessments with assessments done previously with the same instruments under less duress. Because ASEBA instruments are self-administered in 10–20 minutes and can be scored by clerical workers by hand or computer, they can be applied quickly, at low cost, and by non-specialists. For respondents who cannot complete them independently, ASEBA instruments can be read aloud and the answers can be recorded by anyone who is familiar with them. Forensic personnel can easily learn to identify deviant scale scores and can use the single-page narrative reports to quickly identify strengths, deviance, and critical items reported for the person being assessed.

The parallel ASEBA instruments completed by different informants and the systematic cross-informant comparisons provided by ASEBA software make it easy to identify similarities and differences between reports by different informants. This is especially valuable in forensic contexts, where conflicting perceptions and motives often lead to very different reports of problems and strengths. Because validity, social desirability, and lie scales cannot resolve such discrepancies (Piedmont et al., 2000), it is necessary to explicitly document consistencies and inconsistencies between reports by the person being evaluated and various people who know that person.

As an example, suppose that ASEBA instruments completed by several informants yield high scores on scales such as Rule-Breaking Behavior, Aggressive Behavior, Conduct Problems, or Antisocial Personality Problems, but an ASEBA self-report form yields low scores on these scales. This would be evidence that the person being assessed is not being candid or is unaware of his or her problem behaviors. On the other hand, if a very deviant Somatic Complaints syndrome score is obtained from a self-report form but not from any collateral-report forms, this would suggest possible malingering.

As another example, if a CBCL completed by a parent whose parental rights are in question yields scores that are very different from scores yielded by ASEBA forms completed by others who know the child, this would be evidence that the parent's report is not an accurate picture of the child's functioning.

Because ASEBA instruments for ages 1½ to 18 have been available much longer than those for adults, most of the published studies pertain to children and adolescents. As of this writing, the published studies include over 600 on physical and sexual abuse, 200 on substance abuse, 200 on antisocial conduct, 90 on divorce, 110 on foster care, and 20 on custody (Bérubé & Achenbach, 2013). Because ASEBA instruments are used for so many purposes in so many contexts, the data obtained with them can be relevant to many facets and stages of forensic assessment, as outlined earlier. The availability of forms for ages 18 to 59 and 60 to 90-plus as well as for ages 1½ to 18 enables users to do multigenerational assessments when indicated. For forensic cases involving abuse and custody issues, this means that ASEBA instruments may also be used to periodically reassess children and relevant adults in order to monitor their functioning after initial decisions are made.

As an example, if a child has been placed with a parent, relatives, or foster parents, the child's subsequent progress should be assessed. Such assessments should include data from multiple informants, such as the adults with whom the child is placed, family members, teachers, daycare providers, and the child. As with initial assessments, informants' reports should be compared to identify similarities and differences as a basis for determining whether certain problems are limited to particular personal relationships or to particular situations such as school. If significant problems are reported by only one informant, then that informant's relationship with the child should be examined to determine the degree to which the informant's perceptions of the child or behavior toward the child or contexts of their interactions contribute to reports of elevated problem levels. If multiple informants report especially high or low levels of problems in a particular situation, such as school or day care, then that situation should be examined to determine why the child is seen as behaving worse or better in that situation than elsewhere. Professionals who conduct continuing and follow-up evaluations of this sort may be especially able to promote the best interests of the child by identifying and helping to ameliorate adverse dynamics among the relevant parties.

Examples of relevant research

Because the relevant research is so vast and diverse, it cannot be succinctly summarized. Instead, we illustrate some relevant research applications.

Maltreatment of children and placement of children in the custody of child protection agencies raise multiple forensic issues. In addition, children who are maltreated and those in the custody of protective agencies may be at especially high risk for subsequent problems in adolescence and adulthood, including delinquency, criminal behavior, and psychopathology. McGee, Wolfe, and Wilson (1997) tested associations between different kinds of maltreatment and subsequent problems in 11- to 17-year-old wards of a child protection agency. To assess current problems, the youths completed the YSR, and their primary caretakers completed the CBCL. The complex associations between kinds and degrees of maltreatment, gender of the youths, sources of data, and types of problems were illustrated by significant interactions such as the following:

- For both genders, YSR scores for Internalizing problems increased with the severity of physical maltreatment experienced by the youths. However, there was a strong interaction with gender such that, among youths who had not experienced physical maltreatment, girls reported fewer Internalizing problems than boys, but the opposite gender difference was found for youths who had experienced moderate to severe physical maltreatment.
- For youths whose case records were rated as indicating severe psychological maltreatment, YSR scores for Internalizing problems were much higher among those who self-reported moderate to severe psychological maltreatment than among those who self-reported none to mild psychological maltreatment. However, for youths whose case records were rated as indicating no psychological maltreatment, YSR Internalizing scores were slightly lower among those who self-reported moderate to severe psychological maltreatment than among those who self-reported none to mild maltreatment.
- For youths whose case records were rated as indicating severe sexual abuse, caretakers' CBCL ratings of Externalizing problems were higher for those who self-reported none to mild sexual abuse than for those who self-reported moderate to severe sexual abuse. However, for youths whose case records were rated as indicating no sexual abuse, caretakers' CBCL ratings of Externalizing problems were lower for those who self-reported none to mild abuse than for those who reported moderate to severe abuse.

As with so many aspects of forensic research and practice, these findings indicate important variations between reports by different sources, such as youths, caretakers, and agencies. To advance knowledge and practice, it is therefore necessary to use multiple data sources and to respect rather than ignore the complexity that they may reveal.

In a study of adjudicated delinquents, the YSR and CBCL were used to test the hypothesis that females would be more deviant than males with respect to psychological symptoms (McCabe, Lansing, Garland, & Hough, 2002). Using *T*-scores based on the age-, gender-, and informant-specific U.S. normative samples for the YSR and CBCL, McCabe et al. found higher *T*-scores for delinquent girls than boys on the Total Problems, Internalizing, and Externalizing scales of both the YSR and the CBCL. However, on both instruments the gender differences were statistically significant for Total Problems and for Externalizing, but not for Internalizing. This indicated that, compared to delinquent boys, delinquent girls were especially deviant with respect to Externalizing Problems, which in turn contributed to their significantly higher Total Problems *T*-scores.

Multicultural Aspects of Assessment

People who are assessed for forensic purposes are often of different ethnic and cultural backgrounds than the people who assess them. For example, the people being assessed may be from native-born

minority groups or may be immigrants from other cultures, whereas the people doing the assessment are from native-born majority groups. In these and other cases where cultural issues may arise, it is important to use assessment instruments whose multicultural applications are well supported by research. As indicated earlier, translations of ASEBA forms are available in over 90 languages. Furthermore, confirmatory factor analyses have shown that patterns of problems reported for children and adults in dozens of societies fit the ASEBA syndromes (Achenbach & Rescorla, 2014). ASEBA scores from these societies have been used to construct multicultural norms that are available in the software for scoring the CBCL, C-TRF, TRF, YSR, ASR, BPM, and ABCL. The user can thus display scale scores in relation to multicultural norms selected by the user as being appropriate for ratings by each informant.

Analyses of ASEBA scores have shown that differences among native-born ethnic groups did not exceed chance expectations when one controls for socioeconomic status in U.S. national and clinical samples (Achenbach et al., 2004; Achenbach & Rescorla, 2000, 2001, 2003). These findings and findings from more than 150 published studies of ethnicity and more than 2,800 published cross-cultural and multicultural studies (Bérubé & Achenbach, 2013) support multicultural applications of ASEBA instruments.

Administrative and Interpretive Adjustments for Forensic Use

Because ASEBA self-report and collateral report forms are brief, standardized, structured, and self-administered, forensic use does not typically require special administrative adjustments. However, as detailed earlier, ASEBA forms can be read aloud by interviewers who write down or key enter the respondents' answers. ASEBA forms can also be completed by respondents at remote sites via web applications, as described earlier. And self-report and collateral-report forms can be completed by respondents via ASEBA client-entry software in users' offices.

Most people who do forensic assessments are accustomed to the stressful and adversarial aspects. They also understand that the people being assessed may not be candid or may not be mentally competent. Because stress, conflict, lack of candor, and mental incompetence are common in forensic assessment, the need for reports by as many informants as possible is even more compelling than in assessment for non-forensic purposes. Depending on the goal of a particular forensic assessment, the consistencies and inconsistencies between reports by different informants contribute evidence regarding such questions as veracity, distortion, reality testing, fitness for parenting, and candidacy for restrictive settings, such as incarceration, residential treatment, or nursing homes. Although systematic comparison of reports by multiple informants is a key ingredient of appropriate assessment, the findings should be compared with findings from other sources of data, such as interviews, observations, and documents, before final conclusions are drawn.

Strengths and Limitations for Forensic Use

As outlined in the preceding sections, ASEBA instruments have the advantage of being applicable to very diverse cases under very diverse conditions. ASEBA instruments can be completed by a variety of respondents for comparison with U.S. national norms for ages 1½ to 90-plus years and with multicultural norms for ages 1½ to 59. ASEBA software provides systematic comparisons between data obtained from self-reports and multiple collateral reports. Thousands of published studies report findings for people from many backgrounds and cultures. Because they can be self-administered or administered by non-clinician interviewers in 10–20 minutes, ASEBA instruments can be quickly applied at low cost. They can also be readministered periodically, and the results obtained in non-forensic and forensic assessment contexts can be compared. With appropriate

consents and protection of confidentiality, ASEBA profiles, cross-informant comparisons, and narrative reports can be presented as evidence in forensic cases.

No single approach can be expected to provide everything that is needed for comprehensive forensic evaluations. Although ASEBA instruments obtain data on diverse aspects of functioning from diverse perspectives, comprehensive evaluation involves comparison and supplementation of ASEBA findings with data from interviews and histories as well as from reports regarding the forensic issues. Cognitive, neuropsychological, and personality tests may also be needed to assess aspects of functioning not assessed by ASEBA instruments.

CASE EXAMPLE

Dr. Lorraine Mercer was hired to conduct a custody evaluation for 4-year-old Lisa Gordon in the divorce case of her parents, David and Cynthia Gordon, aged 26 and 22 respectively. Two years prior to the custody evaluation, the Gordons had separated. David had taken a job about three hours' drive away, and Cynthia had moved with their 2-year-old daughter Lisa back to her mother's home. Under the separation agreement, Lisa had been spending alternate weekends with her father.

As David became increasingly concerned about Cynthia's parenting of Lisa, he decided to sue for physical custody as part of the divorce proceedings. Because Cynthia also wished to have custody of Lisa, the family court recruited Dr. Mercer to evaluate Lisa and her parents in order to render an opinion on what custody arrangements would be in the child's best interests. Dr. Mercer made appointments to see Lisa and her parents and sent each parent several forms to complete: a developmental/family history form, a CBCL/1½–5 to complete about Lisa, an ASR, two ABCLs for relatives or friends to complete about each parent, and one ABCL for each parent to complete about the other. Dr. Mercer also obtained consent from both parents to send C-TRFs to two teachers at Lisa's daycare center.

ASEBA Findings for Lisa

As can be seen from Table 11.6, the C-TRF completed by one of Lisa's daycare teachers yielded scores in the clinical range on Total Problems, Internalizing, Externalizing, the Emotionally Reactive and Anxious/Depressed syndromes, and the *DSM*-oriented Affective, Anxiety, and Oppositional Defiant Problems scales. Scores on the Withdrawn and Aggressive Behavior syndromes were in the borderline clinical range. The C-TRF completed by Lisa's other teacher yielded scores in the borderline or clinical range on most of these same scales.

The CBCL completed by David had above-average cross-informant correlations with both teachers' C-TRFs. As shown in Table 11.6, David's CBCL yielded scores in the borderline or clinical range on most of the same scales as the C-TRFs did, with the exception of Aggressive Behavior, on which David scored Lisa in the normal range. David's CBCL also yielded a score in the borderline clinical range on the Sleep Problems syndrome, which is not scored from the C-TRF. In contrast, as shown in Table 11.6, Cynthia's CBCL yielded scores in the normal range on all scales with the exception of borderline clinical range scores on the Aggressive Behavior syndrome and the *DSM*-oriented Oppositional Defiant Problems scale. Her CBCL had below-average cross-informant correlations with the forms completed by Lisa's teachers and by David.

TABLE 11.6 *T*-Scores for Lisa Based on Ratings by Four Informants

<i>Scale</i>	<i>Teacher 1</i>	<i>Teacher 2</i>	<i>Father</i>	<i>Mother</i>
Emotionally Reactive	72 – C ^a	70 – C	66 – B	54
Anxious/Depressed	71 – C	68 – B	70 – C	53
Somatic Complaints	53	52	51	51
Withdrawn	67 – B ^b	64	60	57
Sleep Problems	NA	NA	66 – B	59
Attention Problems	55	56	52	54
Aggressive Behavior	66 – B	65 – B	53	67 – B
Internalizing	71 – C	68 – C	64 – B	51
Externalizing	72 – C	70 – C	58	53
Total Problems	72 – C	70 – C	67 – C	54
DSM Affective Problems	72 – C	68 – B	66 – B	52
DSM Anxiety Problems	71 – C	67 – B	65 – B	51
DSM PDD Problems	53	52	50	50
DSM ADH Problems	54	55	51	50
DSM ODD Problems	74 – C	71 – C	66 – B	68 – B

Notes:

a C = clinical range ($T \geq 70$ for syndromes and *DSM* scales, ≥ 64 for I, E, and TP).

b B = borderline range ($T \geq 65$ – 69 for syndromes and *DSM* scales, ≥ 60 – 63 for I, E, and TP).

In response to the C-TRF's open-ended questions, Lisa's teachers wrote that Lisa was often angry or withdrawn. They reported that she often cried at morning drop-off but then sometimes refused to leave at evening pickup. They noted that Lisa was not fully toilet trained; some days she came to school in underpants and other days she wore pull-ups. They also wrote that she sometimes arrived at day care without having eaten breakfast, in inappropriate or soiled clothing, or with uncombed hair. On the C-TRF item regarding the best things about the child, Lisa's teachers noted that she was a bright and verbal little girl with good imaginative play skills who could be lively and engaging when in a good mood.

In her responses to the open-ended questions on the CBCL, Cynthia described Lisa as being very demanding, stubborn, and a picky eater. She also reported that Lisa resisted going to sleep and staying in her own bed, using the potty, and having her hair washed and brushed. However, on the CBCL item regarding the best things about the child, Cynthia wrote that Lisa was smart, pretty, funny, and good at singing, dancing, and drawing.

In response to the CBCL question about what concerned him most about Lisa, David expressed concern that Cynthia's erratic practices with regard to meals, bedtime, hygiene, and toilet training were putting Lisa at risk and preventing her from developing important skills. He added that although Cynthia could be affectionate and responsive to Lisa at times, she was often either overindulgent, neglectful, or angry. In response to the CBCL question regarding the best things about Lisa, he wrote that she was a smart, charming, pretty little girl who could be very loving and cooperative when she was in a good mood.

ASEBA Findings for Lisa's Father, David Gordon, Age 26

Based on David's self-reports on the ASR, his scores were in the clinical range on the Spouse/Partner scale, in the borderline clinical range on the Friends scale, and in the normal range on

the Family, Job, and Education scales. He obtained scores in the borderline range on the Somatic Complaints syndrome and the *DSM*-oriented Anxiety Problems scale. David described himself as performing adequately in his job as an assistant computer systems manager in a health insurance company. He wrote that he lived near both his parents and his older sister, that he saw them frequently, and that they provided emotional support. However, he wrote that he was preoccupied about the custody battle, was not sleeping well or exercising enough, and was having trouble concentrating at work. On the ASR questions about substance use, David reported that he typically had two beers at night. He also reported that he had occasional headaches and stomach pain.

David asked his mother and older sister to complete the ABCLs sent by Dr. Mercer. Ratings by his mother yielded scores in the borderline clinical range on the Anxious/Depressed syndrome and on the *DSM*-oriented Anxiety Problems and Avoidant Personality Problems scales. His mother's responses to open-ended questions on the ABCL indicated that she admired her son for his struggle to manage his job, take care of his daughter, and deal with his divorce. In response to the ABCL question about her greatest concerns about her son, she referred to his agitated and depressed state of mind.

The ABCL completed by David's sister Amanda yielded scores in the normal range on all scales except borderline clinical range scores on the Anxious/Depressed syndrome and on the *DSM*-oriented Anxiety Problems scale. On the ABCL item regarding the best things about her brother, Amanda wrote that she was impressed with how hard he was trying to be a good father to Lisa. Amanda also wrote that David regularly brought Lisa over to play with her two children and would often ask her for tips on how to handle Lisa or things to do with her. Amanda added that she provided home-based childcare services for several young children and that Lisa sometimes joined this group when David was at work.

ASEBA Findings for Lisa's Mother, Cynthia Gordon, Age 22

Cynthia's ASR yielded scores in the normal range on all scales. The only problems she endorsed as very true were worrying, headaches, and sleep problems. Items she endorsed as somewhat or sometimes true included being nervous, forgetful, disorganized, poor at details, late, argumentative, stubborn, impatient, easily bored, and jealous. In response to the ASR open-ended item regarding her greatest concerns, Cynthia wrote that she felt dissatisfied with her life, worried about her future, and concerned about losing custody of Lisa. On the ASR item regarding the best things about herself, she wrote that she was enthusiastic, fun-loving, generous, and attractive.

The ABCL completed by Cynthia's mother also yielded scores in the normal range on all scales. Cynthia's mother rated as very true some of the problems Cynthia rated as somewhat true, such as arguing with others, getting along badly with family members, rushing into things, and being disorganized, poor at details, forgetful, and stubborn. In response to the ABCL question regarding what concerned her most about Cynthia, Cynthia's mother listed her dropping out of college when she became pregnant, her frustration with her job, her quick temper, and her disorganization. On the ABCL item regarding the best things about Cynthia, her mother wrote that she was warm-hearted, fun to be with, lively, and generous. She added that Cynthia loved Lisa and did many nice things with her.

Cynthia also asked her Aunt Rosalie to complete an ABCL. Rosalie's ABCL yielded scores in the clinical range for Cynthia on the Spouse/Partner scale and on the Externalizing scale and

in the borderline clinical range on the Attention Problems, Aggressive Behavior, and Rule-Breaking Behavior syndromes and the Total Problems, Friends, and Alcohol scales. In response to the ABCL item about what concerned her most about her niece, Rosalie listed Cynthia's immaturity, lack of independence, quick temper, and impulsivity. She added that Cynthia seemed to be struggling in her attempt to work, care for her daughter, and have some life of her own. Her comments about her niece's best qualities included that she was pretty, friendly, artistic, and creative.

Information Obtained from Clinical Interviews

Dr. Mercer interviewed David and Cynthia Gordon separately as well as together. She observed Lisa at home, at school, and in the office with both parents. She also spoke by phone with David's mother and his sister and with Cynthia's mother and aunt, in order to follow up on information they had provided on the ABCL.

On the basis of these contacts, Dr. Mercer learned that David and Cynthia had met at a fraternity party when he was a senior and she was a freshman. They had been dating only a few months when Cynthia became pregnant with Lisa, despite having told David she was taking birth control pills. Although the couple initially agreed that an abortion was the best option, Cynthia kept missing appointments for the procedure until it became too late to have it. At this point, David proposed that they get married. Cynthia completed her freshman year and then dropped out to care for their daughter, and David started work in the computer center at the university where he had obtained his degree.

Dr. Mercer learned that the couple soon started arguing about money, housework, child care, and use of leisure time. David was dissatisfied with his job and critical of his wife's management of the apartment, their daughter, and finances. David was especially concerned with how angry Cynthia became at Lisa, and he worried that she might lose control and harm their daughter. On the other hand, Cynthia felt isolated staying home all day with Lisa, complained that David never liked to have fun anymore, and accused him of being unsympathetic.

After the couple separated and Cynthia returned to her mother's, her mother insisted that she find a job so that she could help with expenses. Cynthia found a job as a receptionist in a local doctor's office and enrolled Lisa in a daycare center. Cynthia's mother reported that having Cynthia back home was challenging at times but that she liked having contact with her little granddaughter. She reported that she sometimes babysat for Lisa when Cynthia went out with friends but that she often worked evenings and weekends.

Dr. Mercer learned that David and Cynthia had many conflicts about how to parent Lisa and that disagreement about basic issues such as diet, bedtime, toileting, medical care, discipline, and visitation was frequent. David said that Lisa had told him twice that Cynthia had slapped and scratched her. He also reported that Cynthia made it difficult for him to talk to his daughter on the phone, which he tried to do every night before she went to bed. Cynthia complained that David lectured her and made what she considered to be unreasonable demands. She added that David did not realize how infuriating Lisa could be because he was not with her all the time.

Custody Recommendations

Dr. Mercer interpreted the ASEBA results as indicating that David's perceptions of Lisa and of his own functioning were more realistic than Cynthia's perceptions of Lisa and her own

functioning. She observed that Lisa's attachment to her mother was strong but conflicted, whereas her attachment to her father was less intense but also less ambivalent. Dr. Mercer noted that David appeared to understand what was developmentally appropriate for Lisa and knew when to ask for advice, whereas Cynthia did not appear knowledgeable about Lisa's needs. Dr. Mercer suspected that Cynthia might use excessive physical force with Lisa at times, although her evaluation did not substantiate any physical abuse.

Dr. Mercer also interpreted the ASEBA results as indicating that Lisa behaved more age-appropriately when in her father's care. She was less volatile and oppositional, and she cooperated more with eating, bathing, hygiene, and sleep routines. Lisa also appeared to be less demanding and bossy when with her father than when with her mother.

Citing the ASEBA findings and her own observations, Dr. Mercer reported that David displayed more maturity in balancing his work, leisure time, and care of Lisa than did Cynthia, who seemed resentful of childcare responsibilities and frustrated that she could not have more free time. Dr. Mercer noted that both parents had full-time jobs and thus required child care for Lisa during the workday. Although both David and Cynthia had family nearby, Dr. Mercer concluded that David's parents and sister would be more likely to provide helpful backup and support for Lisa's care than Cynthia's mother and aunt.

Dr. Mercer concluded that David could provide a more stable, predictable, and wholesome environment for Lisa than Cynthia could. Dr. Mercer also expressed concern about Cynthia's lack of insight into her own issues, her impulsive lifestyle, and her volatile temper. Dr. Mercer therefore recommended that David have physical custody and that Cynthia have visitation privileges on alternate weekends, for half of the year's major holidays, and for two weeks during the summer.

Summary

This chapter presented the ASEBA and its applications in clinical and forensic contexts. The ASEBA comprises a family of self-report and collateral report instruments for assessing adaptive and maladaptive functioning for ages 1½ to 90-plus years. It also includes instruments for completion by clinical interviewers, psychological examiners, and observers of behavior in group settings. The instruments are scored on scales that include adaptive functioning, empirically based syndromes, *DSM*-oriented scales, and Internalizing, Externalizing, and Total Problems. Scores are displayed on profiles in relation to age-, gender-, and informant-specific norms. Multicultural norms are available for ages 1½ to 59. ASEBA software prints profiles, cross-informant comparisons, and narrative reports. ASEBA forms are available in over 90 languages, and findings have been reported in over 8,000 publications from 80 societies and cultural groups.

Psychometric data, evidence for validity, strengths, and limitations were presented. Forensic applications were illustrated in a case example in which a child, her mother, and her father were each assessed via ASEBA instruments completed by multiple informants. Because ASEBA instruments can be self-administered in 10–20 minutes under diverse conditions and can be automatically scored online or quickly scored by clerical workers, they can be routinely used for clinical assessment and for initial forensic assessments and follow-up assessments after forensic decisions have been made.

References

- Achenbach, T. M. (1966). The classification of children's psychiatric symptoms: A factor-analytic study. *Psychological Monographs*, 80 (No. 615).
- Achenbach, T. M. (1991). *Integrative guide for the 1991 CBCL/4-18, YSR, and TRF profiles*. Burlington: University of Vermont, Department of Psychiatry.
- Achenbach, T. M. (1992). *Manual for the Child Behavior Checklist/2-3 and 1992 Profile*. Burlington: University of Vermont, Department of Psychiatry.
- Achenbach, T. M. (1997a). *Guide for the Caregiver-Teacher Report Form for Ages 2-5*. Burlington: University of Vermont, Department of Psychiatry.
- Achenbach, T. M. (1997b). *Manual for the Young Adult Self-Report and Young Adult Behavior Checklist*. Burlington: University of Vermont, Department of Psychiatry.
- Achenbach, T. M., & Edelbrock, C. (1983). *Manual for the Child Behavior Checklist and Revised Child Behavior Profile*. Burlington: University of Vermont, Department of Psychiatry.
- Achenbach, T. M., & Edelbrock, C. (1986). *Manual for the Teacher's Report Form and Teacher Version of the Child Behavior Profile*. Burlington: University of Vermont, Department of Psychiatry.
- Achenbach, T. M., & Edelbrock, C. (1987). *Manual for the Youth Self-Report and Profile*. Burlington: University of Vermont, Department of Psychiatry.
- Achenbach, T. M., Howell, C. T., McConaughy, S. H., & Stanger, C. (1995). Six-year predictors of problems in a national sample: III. Transitions to young adult syndromes. *Journal of the American Academy of Child and Adolescent Psychiatry*, 34, 658-669.
- Achenbach, T. M., Howell, C. T., McConaughy, S. H., & Stanger, C. (1998). Six-year predictors of problems in a national sample: IV. Young adult signs of disturbance. *Journal of the American Academy of Child and Adolescent Psychiatry*, 37, 718-727.
- Achenbach, T. M., Krukowski, R. A., Dumenci, L., & Ivanova, M. Y. (2005). Assessment of adult psychopathology: Meta-analyses and implications of cross-informant correlations. *Psychological Bulletin*, 131, 361-382.
- Achenbach, T. M., McConaughy, S. H., & Howell, C. T. (1987). Child/adolescent behavioral and emotional problems: Implications of cross-informant correlations for situational specificity. *Psychological Bulletin*, 101, 213-232.
- Achenbach, T. M., McConaughy, S. H., Ivanova, M. Y., & Rescorla, L. A. (2011). *Manual for the Brief Problem Monitor (BPM)*. Burlington: University of Vermont, Research Center for Children, Youth, and Families.
- Achenbach, T. M., Newhouse, P. A., & Rescorla, L. A. (2004). *Manual for the ASEBA Older Adult Forms and Profiles*. Burlington: University of Vermont, Research Center for Children, Youth, and Families.
- Achenbach, T. M., and Rescorla, L. A. (2000). *Manual for the ASEBA Preschool Forms and Profiles*. Burlington: University of Vermont, Research Center for Children, Youth, and Families.
- Achenbach, T. M., & Rescorla, L. A. (2001). *Manual for the ASEBA School-Age Forms and Profiles*. Burlington: University of Vermont, Research Center for Children, Youth, and Families.
- Achenbach, T. M., & Rescorla, L. A. (2003). *Manual for the ASEBA Adult Forms and Profiles*. Burlington: University of Vermont, Research Center for Children, Youth, and Families.
- Achenbach, T. M., & Rescorla, L. A. (2007). *Multicultural Supplement to the Manual for the ASEBA School-Age Forms & Profiles*. Burlington: University of Vermont, Research Center for Children, Youth, and Families.
- Achenbach, T. M., & Rescorla, L. A. (2010). *Multicultural Supplement to the Manual for the ASEBA Preschool Forms & Profiles*. Burlington: Research Center for Children, Youth, and Families.
- Achenbach, T. M., & Rescorla, L. A. (2014). *Multicultural Guide for the ASEBA Forms and Profiles for Ages 1½-59* (3rd ed.). Burlington: University of Vermont, Research Center for Children, Youth, and Families.
- American Psychiatric Association. (1952). *Diagnostic and statistical manual of mental disorders* (1st ed.). Washington, DC: Author.
- American Psychiatric Association. (1994). *Diagnostic and statistical manual of mental disorders* (4th ed.). Washington, DC: Author.
- Archer, R. P., Fontaine, J., & McCrae, R. R. (1998). Effects of two MMPI-2 validity scales on basic scale relations to external criteria. *Journal of Personality Assessment*, 70, 87-102. doi:10.1207/s15327752jpa7001.
- Arend, R., Lavigne, J. V., Rosenbaum, D., Binns, H. J., & Christoffel, K. K. (1996). Relation between taxonomic and quantitative diagnostic systems in preschool children: Emphasis on disruptive disorders. *Journal of Clinical Child Psychology*, 25, 388-397.

- Beck, A. T., & Steer, R. A. (1990). *Beck Anxiety Inventory Manual*. San Antonio, TX: Psychological Corporation.
- Bérubé, R. L., & Achenbach, T. M. (2013). *Bibliography of published studies using the Achenbach System of Empirically Based Assessment (ASEBA)*. Burlington: University of Vermont, Research Center for Children, Youth, and Families.
- Briggs-Gowan, M. J., & Carter, A. S. (1998). Preliminary acceptability and psychometrics of the Infant-Toddler Social and Emotional Assessment (ITSEA): A new adult-report questionnaire. *Infant Mental Health Journal*, 19, 422–445.
- Butcher, J. N., Dahlstrom, W. G., Graham, J. R., Tellegen, A., & Kaemmer, B. (1989). *Minnesota Multiphasic Personality Inventory (MMPI-2): Manual for administration and scoring*. Minneapolis: University of Minnesota Press.
- Conners, C.K. (1997). *Conners' Rating Scales-Revised Technical Manual*. North Tonawanda, NY: Multi-Health Systems.
- Cummings, J. L., Mega, M., Gray, K., Rosenberg-Thompson, S., Carusi, D. A., & Gornbein, J. (1994). The Neuropsychiatric Inventory: Comprehensive assessment of psychopathology in dementia. *Neurology*, 44, 2308–2314.
- Derogatis, L. R. (1994). *SCL-90-R: Administration, scoring, and procedures manual*. Minneapolis, MN: National Computer Systems.
- Edelbrock, C., & Costello, A. J. (1988). Structured psychiatric interviews for children. In M. Rutter, A. H. Tuma, & I. S. Lann (Eds.), *Assessment and diagnosis in child psychopathology* (pp. 87–112). New York: Guilford Press.
- Ferdinand, R. F., Blüm, M., & Verhulst, F. C. (2001). Psychopathology in adolescence predicts substance use in young adulthood. *Addiction*, 96, 861–870.
- Ferdinand, R. F., Hoogerheide, K. N., van der Ende, J., Heijmans Visser, J. H., Koot, H. M., Kasius, M. C., & Verhulst, F. C. (2003). The role of the clinician: Three-year predictive value of parents', teachers', and clinicians' judgment of childhood psychopathology. *Journal of Child Psychology and Psychiatry*, 44, 867–876.
- Ferdinand, R. F., & Verhulst, F. C. (1994). The prediction of poor outcome in young adults: Comparison of the Young Adult Self-Report, the General Health Questionnaire, and the Symptom Checklist. *Acta Psychiatrica Scandinavica*, 89, 405–410.
- Ferdinand, R. F., Verhulst, F. C., & Witnitzer, M. (1995). Continuity and change of self-reported problem behaviors from adolescence into young adulthood. *Journal of the American Academy of Child and Adolescent Psychiatry*, 34, 680–690.
- Folstein, M. F., Folstein, S. E., & McHugh, P. R. (1975). "Mini-mental state": A practical method for grading the cognitive state of patients for the clinician. *Journal of Psychiatry Research*, 12, 189–198.
- Goldberg, D. P. (1972). *The detection of psychiatric illness by questionnaire*. London: Oxford University Press.
- Grisso, T. (2005). Why we need mental health screening and assessment in juvenile justice programs. In T. Grisso, G. Vincent, & D. Seagrave (Eds.), *Mental health screening and assessment in juvenile justice* (pp. 3–21). New York: Guilford Press.
- Grisso, T., Vincent, G., & Seagrave, D. (2005). *Mental health screening and assessment in juvenile justice*. New York: Guilford Press.
- Hofstra, M. B., van der Ende, J., & Verhulst, F. C. (2001). Adolescents' self-reported problems as predictors of psychopathology in adulthood: 10-year follow-up study. *British Journal of Psychiatry*, 179, 203–209.
- Hofstra, M. B., van der Ende, J., & Verhulst, F. C. (2002a). Child and adolescent problems predict DSM-IV disorders in adulthood: A 14-year follow-up of a Dutch epidemiological sample. *Journal of the American Academy of Child and Adolescent Psychiatry*, 41, 182–189.
- Hofstra, M. B., van der Ende, J., & Verhulst, F. C. (2002b). Pathways of self-reported problem behaviors from adolescence into adulthood. *American Journal of Psychiatry*, 159, 401–407.
- Kasius, M. C., Ferdinand, R. F., van den Berg, H., & Verhulst, F. C. (1997). Associations between different diagnostic approaches for child and adolescent psychopathology. *Journal of Child Psychology and Psychiatry*, 38, 625–632.
- Kazdin, A. E., & Heidish, I. E. (1984). Convergence of clinically derived diagnoses and parent checklists among inpatient children. *Journal of Abnormal Child Psychology*, 12, 421–435.
- Keenan, K., & Wakschlag, L. S. (2000). More than the terrible twos: The nature and severity of behavior problems in clinic-referred preschool children. *Journal of Abnormal Child Psychology*, 28, 33–46.
- Mash, E. J., & Hunsley, J. (2005). Evidence-based assessment of child and adolescent disorders: Issues and challenges. *Journal of Clinical Child and Adolescent Psychology*, 34, 362–379.

- McCabe, K. M., Lansing, A. E., Garland, A., & Hough, R. (2002). Gender differences in psychopathology, functional impairment, and familial risk factors among adjudicated delinquents. *Journal of the American Academy of Child and Adolescent Psychiatry*, 41, 860–867.
- McConaughy, S. H., & Achenbach, T. M. (2001). *Manual for the Semistructured Clinical Interview for Children and Adolescents* (2nd ed.). Burlington: University of Vermont, Research Center for Children, Youth, and Families.
- McConaughy, S. H., & Achenbach, T. M. (2004). *Manual for the Test Observation Form for Ages 2–18*. Burlington: University of Vermont, Research Center for Children, Youth, and Families.
- McConaughy, S. H., & Achenbach, T. M. (2009). *Manual for the ASEBA Direct Observation Form*. Burlington, VT: University of Vermont, Research Center for Children, Youth, and Families.
- McGee, R. A., Wolfe, D. A., & Wilson, S. K. (1997). Multiple maltreatment experiences and adolescent behavior problems: Adolescents' perspectives. *Development and Psychopathology*, 9, 131–149.
- McGrath, R. E., Mitchell, M., Kim, B. H., & Hough, L. (2010). Evidence for response bias as a source of error variance in applied assessment. *Psychological Bulletin*, 136, 450–470.
- Meyer, G. J. (2002). Implications of information gathering methods for a refined taxonomy of psychopathology. In L. E. Beutler & M. L. Malik (Eds.), *Rethinking the DSM: A psychological perspective* (pp. 69–105). Washington, DC: American Psychological Association.
- Mouton-Simien, P., McCain, A. P., & Kelley, M. L. (1997). The development of the Toddler Behavior Screening Inventory. *Journal of Abnormal Child Psychology*, 25, 59–64.
- Piedmont, R. L., McCrae, R. R., Riemann, R., & Angleitner, A. (2000). On the invalidity of validity scales: Evidence from self-report and observer ratings in volunteer samples. *Journal of Personality and Social Psychology*, 78, 582–593.
- Reef, J., Diamantopoulou, S., van Meurs, I., Verhulst, F., & van der Ende, J. (2009). Child to adult continuities of psychopathology: A 24-year follow-up. *Acta Psychiatrica Scandinavica*, 120, 230–238.
- Reynolds, C. R., & Kamphaus, R. W. (1992). *Behavior Assessment System for Children*. Circle Pines, MN: American Guidance Service.
- Richman, N., Stevenson, J., & Graham, P. J. (1982). *Pre-school to school: A behavioural study*. London: Academic Press.
- Weinstein, S. R., Noam, G. G., Grimes, K., Stone, K., & Schwab-Stone, M. (1990). Convergence of DSM-III diagnoses and self-reported symptoms in child and adolescent inpatients. *Journal of the American Academy of Child and Adolescent Psychiatry*, 29, 627–634.

12

THE FORENSIC USES AND LIMITATIONS OF THE PARENTING STRESS INDEX

Richard R. Abidin, William G. Austin, and James R. Flens

Introduction

While not originally designed for forensic purposes, the Parenting Stress Index is increasingly being used in forensic contexts. The originator of the PSI (Abidin, 1998) has acknowledged that he is not a forensic psychologist and that he never envisioned that the PSI would be used for forensic purposes. Nevertheless, the PSI, like other psychological measures, has had its utility extended through the research and clinical efforts of others. The purpose of this chapter is to inform the reader about the nature of the PSI, the types of information it generates, its core reliability and validity, and its possible uses in forensic work. The writing team for this chapter represents a collaboration between the PSI's developer and two forensic experts with extensive experience using the PSI. It is the goal of the authors that potential users of the PSI will be better able to judge its utility and validity for their purposes, and to recognize its limitations, after reading this chapter.

Basic Description of the Parenting Stress Index

The purposes of the Parenting Stress Index are to identify parents in need of guidance and support, to identify potential dysfunctional parent–child relationships, and to identify children at risk for emotional and behavioral developmental problems. Over time, the PSI has been found also to be useful as a treatment planning tool, and as a means of evaluating both individual and group interventions. The treatment planning function involves identifying problem areas using the Domain and subscale scores and focusing interventions in areas that are most stressful for the parent.

The 101 items of the PSI focuses on two Domains: the Child's Characteristics, and the Parent Characteristics/family context. There is also an optional 19-item life events stress scale which is designed to provide an overview of the context of the parent–child relationship. The Child Characteristics Domain consists of six subscales (Table 12.1) that are generated from 47 items. Four of these subscales address child characteristics are related to both temperament and learned behaviors. These child characteristics are recognized as ones that make parenting more challenging. The other two child characteristic subscales relate to the parents' cognitive/affective responses to their child.

The Parent Characteristics Domain consists of a Total Domain score and seven subscales (Table 12.2) derived from 54 items. The seven subscales cover characteristics of the parents, and the parents' perception of their social support for parenting.

TABLE 12.1 Child Characteristic Domain Subscales

Adaptability	Difficulties adjusting to changes, inflexible
Distractibility/Hyper	ADHD type behaviors
Demandingness	Demands requiring accommodation or attention
Mood	Moodiness, crying, displays of unhappiness
Acceptability	Behaviors that do not match parent's expectations, hoped-for child
Reinforces Parent	Parent does not experience positive reinforcement from interactions with their child

TABLE 12.2 Parent Characteristics Domain Subscales

Competence	Sense of competency in the parenting role
Isolation	Lack of social support for their role as parent
Health	Impact of physical health on parenting
Role Restriction	Impact of the restrictions parenting places on their choices and freedom
Depression	Impact of depression and feelings of guilt on their parenting behavior
Spouse	Help and emotional support from child's other parent

It is important to note that while some of the PSI subscale names are similar to concepts measured by other instruments, the items relate to the impact of each concept on the individual's perception of their ability to parent and the distress they experience in the parenting role. This is an important distinction since it relates to the PSI's score's validity in relation to parenting. For example, while a general measure of depression will likely have a stronger association to the diagnosis of clinical depression in a parent, the PSI Depression subscale has a stronger association to parental behavior. The PSI Depression subscale has correlations with general measures of depression ranging from .40 to .76. This issue of the relation of the subscales of the PSI to important clinical variables is addressed more fully in the section on the PSI's validity.

The PSI is a self-report measure that may be completed by a parent or parent surrogate in relation to a specific child between the ages of 3 months and 12 years. The items are responded to using a five-point Likert scale with anchor points ranging from "strongly agree" to "strongly disagree." The PSI requires approximately 20 minutes to complete and requires a fourth- to fifth-grade reading level. The PSI may be administered three ways: by using a test booklet and a self-scoring answer sheet, by a computer program, and by internet PARiConnect that administers, scores, and writes an interpretive report. The computer program not only interprets the results but provides recommendations, at-risk-for diagnoses, profile comparisons to known clinical samples, and a topically arranged research reference list of validity studies. There is also an upward extension of the PSI called the Stress Index for Parents of Adolescents (SIPA; Sheras, Abidin, & Konold, 1998) for use in relation to adolescents ages 11–19. The SIPA is a separate measure and is mentioned for information purposes but is not part of this presentation.

There are at present official translations of the PSI into 42 languages, which are available from the publisher, Psychological Assessment Resources, Inc. (PAR; website <http://www4.parinc.com>). There are norms for non-English-speaking respondents and a professional manual in their language for Chinese, Dutch, French Canadian, German, Italian, Japanese, Korean, and Portuguese.

History of the PSI's Development

The Parenting Stress Index was originally designed for use as a screening and triage tool to be used in pediatric medical practices by psychologists and pediatricians. Dr. Abidin in 1965 was working at Wilford Hall USAF Hospital, San Antonio, Texas, and consulting with pediatricians, parents, and teaching pediatric residents in an elective course of "behavioral pediatrics." A recurrent problem for the pediatricians was the limited time they had to devote to parents for anticipatory guidance and/or the management of minor behavioral problems. The pediatric staff and residents expressed the need for a valid system that would identify situations in need of referral and that did not consume much of their time. The challenge was: "Can you give us three or four questions which we can ask parents that would help us distinguish those in need of additional attention from those parents who are okay?" All of the PSI's development followed from that question, with the key issue being to find a way to efficiently and validly identify parent-child systems that are either currently dysfunctional, at-risk for dysfunction, or where the child's development has gone awry and the parents need support and consultation.

It quickly became apparent that a three- or four-item solution would not work, and that some sort of framework, or theoretical model, would be needed to guide the effort to create an early identification screening tool that was empirically based and that was informed by the research and theorizing of others. One primary source of data came from Abidin's observations and interactions with parents, and those of other clinicians who worked with them on parenting issues in the Pediatric Center of Wilford Hall USAF Hospital, and the Parenting Clinic of Pediatric Associates of Charlottesville, VA. The result of these observations was the creation of the theoretical model that was instrumental in guiding the development of the PSI (Figure 12.1).

PSI Model Assumptions

The PSI model assumes:

- that all stresses in the parenting system, regardless of source, are multidimensional and are summative;
- that stress results from important transactions between parent and child, and the parent's social environment;
- that all these transactions are cognitively appraised by the parent and given either a positive or negative stress value (stress valance);
- that life events provide a context which may add to the stress load of a parent;
- that parents generally act as buffers for their children and that the most proximal sources of parenting stress will have the greatest impact on the parenting behavior exhibited;
- that the total level of parenting stress is the best predictor of dysfunctional parenting, and of that portion of a child's development that is linked to parent-child transactions.

The PSI items were developed by using four sources of information: (1) Abidin's and other clinicians' experience working with parents in pediatric contexts, and parent education groups; (2) reviews of parents' daily stress logs regarding interactions with their child and their performance in their parenting role; (3) interviews with 25 prominent clinical child developmental psychologists and child psychiatrists regarding what they perceived to be the primary sources of parenting stress; and (4) a wide-ranging review of the research literature regarding child and parent characteristics associated with parenting stress, dysfunctional parenting, and negative child outcomes. The information gathered using these methods was then used to guide the writing of each questionnaire item. The primary structure of the questions was to ask about a child or parent characteristic, an

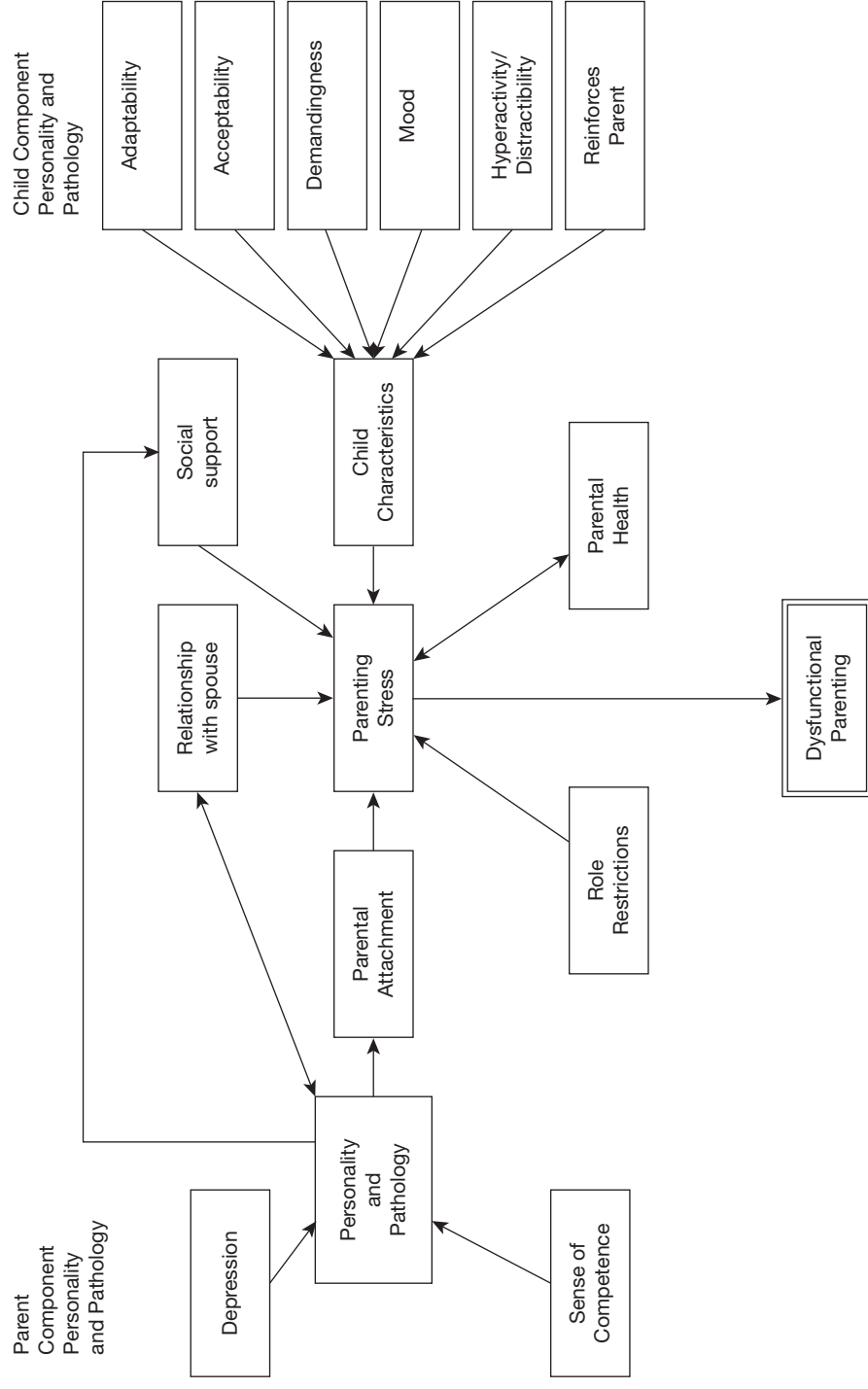


FIGURE 12.1 Theoretical Model for the PSI's Development (from Abidin & Burke, 1976).

event, or the parenting context so that there was a specific referent to be connected to the level of distress the parent experienced: for example, “My child is so active that it exhausts me,” “I often feel my child’s needs control my life,” “Since having my child, my spouse (male/female friend) has not given me as much help and support as I expected.”

The initial question item pool was administered to a series of samples of parents and subjected to revision using item analysis, criteria for inclusion, alpha reliability estimates, exploratory and confirmatory factor analysis. The current edition of the PSI is the sixth version (Burke, 1978; Hauenstein, Scarr, & Abidin, 1987; Solis & Abidin, 1991).

The PSI Short Form (PSI-SF)

In the early 1990s, as researchers and some clinicians began inquiring about and asking for an abbreviated version of the PSI, the “three or four questions” idea arose again. Abidin resisted the idea of a short form until it became apparent that users in a variety of ways were shortening the PSI and calling their version the PSI. Questions about the validity of these PSI variants, and of the published works using the variants, and reported clinical experiences with the different short forms required a response, namely the creation of the PSI-SF. All of the items of the PSI-SF came from the 101 items of the PSI parent and child domains of the regular PSI. The decision was made to subject the original PSI item data pool to a three-factor solution.

The rationale for this decision was that some researchers reported that when they exposed their PSI data sets to exploratory factor analysis, a third weak factor could be justified on the basis of the eigen values of the factor analysis. Examination of the item’s loadings on the three factors suggested that if the 12 strongest items were retained, a three-factor solution would result with factor loadings of .40 or greater for all items and with only two of the 36 items having a double loading with another factor. The newly created PSI-SF was then submitted to a replication using confirmatory factors analysis with a new sample. The replication produced the same factor structure with two double loadings, and very similar factor loading weights. Independent replications of the factors suggest that the three-factor solution is stable, although some researchers suggest that there are higher-order factors imbedded in the subscales that could be used for more refined data interpretation (Reitman, Currier, & Stickle, 2002).

On the basis of the item content, the three factors were labeled (1) Difficult Child, (2) Parental Distress, and (3) Parent–Child Dysfunctional Interaction. The former two are reflections of the Child Characteristics and Parent Characteristics Domains of the full PSI, and should be interpreted as one would those Domain scores. The third factor score’s interpretation relates to parents’ perceptions of their transactions with and expectations of their child and as such addresses the quality of the parent–child relationship.

Psychometric Characteristics

Norms and Generalization

In 2010, the decision was made to undertake a review and re-norming of the PSI based on the most recent census demographic of the United States, the cultural changes in parenting, and the fact that fathers were not a part of the PSI-3 norm sample. The PSI-4 would address those primary issues while also attempting to improve the psychometric properties of the PSI. The PSI-4 was standardized on a sample of 1,056 adults, of whom 534 were mothers and 522 were fathers. Attempts were made to collect equal proportions of mother and father protocols in terms of child gender and across all child age bands (e.g., less than 1 year to 12 years old). The standardiza-

tion sample included individuals from 17 states in the Southern, Northeastern, Western, and Midwestern regions of the United States. One goal was to create a sample matching the overall proportions for education level and ethnicity reported by the U.S. Census (2007) in order to represent the very diverse population and range of education levels present in the United States. The mothers and fathers in the standardization sample were very similar in terms of demographic characteristics. The mean age did not differ significantly for mothers ($M = 33.24$ years, $SD = 7.81$) and fathers ($M = 34$ years, $SD = 7.46$), nor did their highest grade completed in school.

In contrast to the norms for the PSI-3, the PSI-4 reports norms by both parent's gender and the child's age. Further, the normative scores are presented both as percentiles and as T -scores. Calculation of percentile scores is based on the frequency distribution of raw scores within the standardization sample. A percentile represents the proportion of individuals within a particular normative group who obtained the specified raw score or lower. For example, if an individual's raw score is equivalent to a percentile of 85, 85% of the individuals within the normative group obtained that raw score or lower. The advantage of percentile scores is that they allow for rapid comparison of an individual's score to scores obtained by similar individuals in the standardization sample. The limitation of percentile scores is that they do not allow for comparison of scores across scales because the distribution of scores is different for each scale. In contrast with percentiles, T -scores allow the examiner to compare scores across scales using a standardized metric. Additionally, T -scores allow for the comparison of scores across assessment measures. A more detailed description of the PSI-4 normative sample may be found in the PSI-4 *Professional Manual* (Abidin, 2012).

One important consideration in the use of the PSI, as with all other evaluation measures and procedures, has to do with its applicability to different populations. One crude method of addressing this question is to use a sampling procedure that is designed to match all segments of society, e.g., men versus women, various racial and ethnic groups, various income and educational levels, varying geographic groups. The problem is that the endless subdivisions would require samples in the hundreds of thousands, and even if given groups were included in the sample, that fact would not indicate the validity of the measure for that particular subgroup. The most powerful test of the robustness of a measure and its likely utility and validity is the demonstration of its construct integrity and predictive validity when applied to populations that are vastly different from the original normative sample. The research literature on the transcultural use of the PSI and other studies of diverse U.S. populations that are vastly different from the normative sample indicates that the PSI is a robust and valid measure for use with diverse populations both within and outside of the United States. These studies required the PSI to hold its construct and predictive validity after surviving a language translation and application to a different racial and cultural group. The following is a sampling of studies that address the issue of generalizability of the PSI.

PSI Generalization and Norm Limitations

(All the studies referenced in this subsection contain normative data for the specified population.)

Supplemental Norm Samples from the United States

Hutcheson and Black (1996) examined the psychometric properties of the PSI in a sample of 191 low-income, urban (Baltimore, MD) African American mothers of infants and toddlers. They reported that

despite differences from the normative sample in ethnicity, socioeconomic class, family structure, and child age, internal consistency coefficients ranged from acceptable to excellent

and were similar in magnitude to those reported for both Hispanic (Solis and Abidin, 1991) and Caucasian samples (Abidin, 1990). . . . The results of the confirmatory factor analysis were consistent with Abidin's (1983) original child and parent domains.

(p. 396)

Solis and Abidin (1991) conducted a psychometric study of the Spanish version of the PSI based on a sample of 223 urban (New York City) Hispanic mothers. They reported alpha reliabilities comparable to those reported for the original normative sample. Confirmatory factor analysis of the subscales supported the original two-factor solution; nevertheless, Solis and Abidin suggested that a three-factor solution based on the subscale factor analysis may facilitate interpretation of the PSI for Hispanic parents.

Innocenti, Huh, and Boyce (1992) reported on a PSI data set of 725 families collected across the United States as a normative study for parents of children with disabilities. The children in the sample had a wide array of disabilities, ranged in age from 4 to 60 months, and 76% were Caucasian. Family income and parental education approximated the national statistics. Innocenti et al. reported one-year test-retest reliabilities between .67 and .76, which is in the range reported in the PSI manual. Mean comparisons by age between the norms in the PSI manual and the current sample confirmed information in the manual that parents of disabled children have significantly more stress related to the child's characteristics and that no significant differences occur in the Parent domain. A norm table at the Domain level is presented to facilitate interpretations with parents of preschool-aged disabled children.

International Cross-Validation Norm Studies

Bigras, LaFreniere, and Abidin (1996) described the translation and norming of the PSI on a sample of 377 French-speaking parents from Quebec. The results obtained from a confirmatory factor analysis closely approximated the original factor analysis report by Abidin (1995), and in fact the subscale loadings were generally higher than those reported by Abidin. The alpha reliabilities for the Total and Domain scores were essentially identical with those reported by Abidin. At the subscale level, however, the reliabilities were lower, clustering around .75. The norms generated for the French Canadian sample closely approximated those for the U.S. population with the exception of two Child Domain subscales, Adaptability and Mood, which were slightly higher for the French Canadian sample. On the basis of the norms and the French Canadian validity studies reported, Bigras et al. recommend the use of the same cutoff scores for interpretation of the French Canadian parents' scores as those reported in the PSI manual for the U.S. population. They also report that high PSI scores were predictive of eight known risk factors in their sample. These findings suggest that the PSI is robust enough to survive translation and maintain its validity in application to a population that is markedly different from the original norming group.

Abidin and Santos (2003) reported on a PSI translation, a factor analytic and normative study ($N = 600$) conducted in Lisbon. The results confirm the United States norm group's two-factor solution, with stress levels in the child and parent domains comparable to those of the U.S. normative sample.

A study by Tam, Chan, and Wong (1994) entitled "Validation of the Parenting Stress Index among Chinese Mothers in Hong Kong" is probably as different a sample from the original PSI norm group as any group one could find in the United States. In a sample of 248 mothers, Tam et al. reported a factor structure that duplicated that of the U.S. sample. To validate the PSI, they used interviews, self-reports about parenting difficulties, and mothers' social service status, and independently created high- and low-stress parent groups of 34 and 96 parents respectively.

Univariate analysis of the PSI Domain and subscale scores found that 14 of the 16 significantly discriminated between groups at $p < .001$, and the other two, Mood and Parent Attachment, at $p < .01$. The data were subsequently submitted to a discriminant analysis that resulted in a 93.1% correct classification. Only 4 of the 34 high-stress group members were classified as low stress, and 5 of the 96 low-stress parents were classified as high stress.

Evidence of the robust nature of the PSI in transcultural use may be found in the professional manuals published in other countries (Tables 12.3 and 12.4).

Limitation of the Original PSI Norms

For information relevant to the differential interpretation of mothers' and fathers' PSI scores, users need to be familiar with the research literature regarding mothers' and fathers' responses to the PSI. Some brief abstracts of a sampling of relevant studies follow.

Parental Gender Differences on the PSI

Deater-Deckard and Scarr (1996), using the standard PSI Short Form in a study of 589 married couples, examined the relation of parenting stress to parental gender. Their sample was predominately

TABLE 12.3 PSI and PSI-SF Publishers and Approved Translations

<i>Test</i>	<i>Language</i>	<i>Publisher</i>	<i>Contact Information</i>
PSI	Chinese	Psychological Publishing Co. Ltd.	www.psy.com.tw
PSI	Dutch	Pearson	www.psychcorp.co.uk
PSI	French-Canadian	Multi-Health Systems Inc.	www.mhs.com
PSI	German	Hogrefe Verlag	www.hogrefe.de
PSI-SF	Italian	Giunti Organizzazioni Speciali	www.giuntios.it
PSI	Japanese	Koyomondai-Kenkyukai	www.koyoerc.or.jp
PSI, PSI-SF	Korean	Hakjisa Publisher	www.hakjisa.co.kr
PSI	Portuguese	Cegoc Tea Lda.	www.cegoc.pt

TABLE 12.4 Availability of PSI and PSI-SF Translations

<i>PSI Translations</i>		<i>PSI-SF Translations</i>
Afrikaans	Indonesian	Amharic
Arabic	Italian	Bengali
Chinese	Japanese	Brazilian Portuguese
Croatian	Korean	Cambodian
Danish	Lithuanian	Creole
Dutch	Norwegian	French
Farsi	Portuguese	Kanada
Finnish	Puerto Rican	Malay
Greek	Russian	Polish
Hebrew	Slovene	Romanian
Hindi	Spanish	Serbian
Hmong	Swedish	Sotho
Hungarian	Thai	Turkish
Icelandic	Vietnamese	Zulu

college-educated and Caucasian (90%), with a mean child age of 2.6 years. Gender differences were described as “negligible,” with mothers’ Parental Distress score being statistically significantly higher than that for fathers at $p < .05$ but not clinically significant, given that the mean was only one-tenth of a standard deviation higher. There was no difference in the Parent–Child Dysfunctional Interaction subscale or the Difficult Child score. No ethnic group or child gender differences were found. Using LISREL modeling, the relation of PSI scores to parental discipline and child behavior was examined and no gender differences in the overall model were found. In more fine-grained analyses, differences in the size of the associations of a few of the PSI scores to criteria were found between mothers and fathers.

M. W. Krauss (1993), in a study of 121 families of toddlers with disabilities, examined the similarities and differences in the parenting stress reported by mothers and fathers. No difference was found for the Parent Domain; however, for the Child Domain, fathers reported higher stress (mean scores of 106) than mothers (101). The source of the difference was that fathers were more distressed, as noted by elevation of the Adaptability, Mood, and Reinforces Parent subscales on the Child Domain. At the subscale level of Parent Domain, fathers had significantly more stress on Attachment, while mothers had significantly more stress on Restriction of Role, Relation with Spouse, and Health. These differences appear logically related to the parenting roles traditionally performed by mothers and fathers.

Beckman (1991), in a study of 54 families, examined the effect of young children’s disabilities upon the stress experienced by the mothers and fathers. For both the disabled and the non-disabled samples, the mothers reported higher levels of stress than fathers on the Parenting Domain and Total Score. The differences were approximately 0.05 SD higher. At the subscale level for the disabled sample, mothers experienced greater stress on Child Demandingness, Parental Depression, Restriction of Role, Sense of Competence, and Health. Fathers reported greater stress associated with the Child Domain and in relation to their Attachment to their child. Social support was significantly associated with parent stress for both mothers ($r = .65$) and fathers ($r = .47$).

D. B. Baker (1994), in an exploratory study, examined the level of parenting stress of 20 married couples with a child diagnosed as having ADHD (child mean age = 9 years). Mothers reported significantly more stress in the Child Domain, but not in the Parent Domain or the Total Score. Of the 13 subscales, the only significant difference ($p < .05$) was that fathers reported feeling less attached to their child.

Frank et al. (1991) explored the relation of child illness and the quality of the parenting alliance to mothers’ and fathers’ perceptions of parenting stress. The sample consisted of 56 couples who were parents of 3- or 4-year-old children, 86% Caucasian, and of middle-class socioeconomic status. Gender differences were noted for both the parenting alliance and the level of parenting stress. Child illness was linked to mothers’, but not fathers’, experience of parenting stress. Fathers’ experience of parent stress was associated with the strength of their parenting alliance with their spouse for both the Child and the Parent Domains of the PSI and all their subscales. Mothers’ perceptions of the quality of their parenting alliance were not associated with the Parent Domain but were associated with the Child Domain and indicated the strongest association with the Acceptability of Child Scores. No interaction (buffering) effects were found for the quality of parenting alliance and child illness in predicting maternal stress. However, the quality of the parenting alliance does seem to buffer fathers from the stresses of parenting a sick child.

Webster-Stratton (1988) compared the PSI responses of 85 mother-and-father couples in a study of children (mean age 4.4) with conduct problems. Fathers reported significantly less stress on the Child Domain ($p < .001$, 10 raw score points) and less stress in the Parent Domain ($p < .01$, 9 raw score points). The only subscale on which fathers reported significantly more stress than mothers was Attachment, which relates to the fathers feeling less connected to their child. Both mothers’

and fathers' PSI scores were found to be significantly correlated to their perception of the child's adjustment as measured by the CBCL and the Eyberg Child Behavior Inventory. The magnitude of these associations varied between parents. The most interesting difference was that the CBCL Hyperactivity scale was correlated $r = 0.60$ with the PSI Parent Domain, $p < .001$ for mothers but was not significantly correlated ($r = 0.17$) for fathers.

PSI Interpretation: Cautions and Supporting Data

Users of the PSI are cautioned to remember that the interpretation of the PSI proceeds from the global to the specific. For the Total Score, the alpha reliability is .95, which the Child Domain score is .90 and the Parent Domain score is .93. These measures of internal consistency suggest that these components of the PSI may be interpreted as coherent. However, when the user proceeds to the interpretation of specific subscales, more caution may be needed. In this regard, the alpha reliabilities of the PSI subscales range from .70 for the shortest Health subscale (five items) to .84 for Depression. Caution should also be taken not to overinterpret the PSI clinical profiles presented in the manual because their sample sizes are small for the most part. These profiles need to be considered as starting points for relating a particular case to the profiles presented, and to the research on the population in question. For example, while the manual provides three profiles of samples that related to child abuse, familiarity with the abuse research literature, such as that presented below, is necessary for the proper interpretation.

The question raised about the PSI norms as to why norms are not reported by the age of the child and the child's gender was corrected for PSI-4. The results of an ANOVA across the groups indicated that a statistically significant age effect was present across the groups.

While the PSI professional manual describes the test norms and procedures used to develop the PSI, it needs to be recognized that a number of different reference samples and norms have been developed by researchers both in the United States and in other countries (Portuguese: Abidin & Santos, 2003; African American: Hutcheson & Black, 1996; French: Lacharité, Éthier, & Piché, 1992; Chinese [Hong Kong]: Chan, 1994; Chinese [Taiwan]: Tam, Chan, & Wong, 1994; PSI-SF Head Start: Reitman, Currier, & Stickle, 2002; Dutch: de Brock, Vermulst, Gerris, & Abidin, 1992). This body of research indicates that the PSI's factor structure, construct, and predictive validity are very robust, since it has been replicated internationally and survived language translations and cultural differences from the original norm groups. Given this evidence, it is highly likely that the PSI would be valid for the broad spectrum of parents in the United States.

One caution that must be noted is that the PSI does not have as large a research base supporting its use with fathers. There are, however, a number of research studies indicating that the PSI is as valid a predictor of dysfunctional parenting in fathers as it is with mothers. Unfortunately, this circumstance requires the forensic user to be familiar with a literature that is published in a wide array of journals (Bagner & Eyberg, 2003; Burbach, Fox & Nicholson, 2004; Calzada, Eyberg, Rich, & Querido, 2004; Deater-Deckard & Scarr, 1996; Judge, 2003; McBride, 1989; Schiller, 2003; Webster-Stratton, 1988). Work with the PSI-SF also suggests that there are not distinct gender differences in perceived parenting stress for the three subscales of the PSI-SF (B. L. Baker et al., 2003; Schiller, 2003).

Clinical Uses and Limitations

The PSI and the PSI-SF have both been used clinically in three primary ways: (1) screening and triage of large populations; (2) as a first-gate intake measure; and (3) as part of an individual clinical assessment.

Screening and Triage (Predictive Validity Research)

As a screening and triage measure, the PSI is most commonly used in pediatric practices, public health outreach programs, and preventive programs such as Head Start. The objective when it is used in this manner is to help identify parent–child systems that are under sufficient stress to lead to dysfunctional parenting and to problems in child development. Given that there are never sufficient resources to provide guidance, consultation, and treatment to all parents in any given population, the PSI helps to identify those under the greatest degree of distress and in need of services, and is predictive of future child adjustment (Abidin, Jenkins, & McGaughey, 1992; Endriga, Jordan, & Speltz, 2003; Florsheim et al., 2003; Goldberg et al., 1997; Saylor, Boyce, & Price, 2003).

Two examples of such uses follow. First, the PSI is used as a component of a screening battery that is periodically given to mothers on the island nation of Bermuda as a part of National Health screening. The battery typically identifies 10–15% of parents who are referred to the Child Development Center for services. Second, the PSI and the PSI-SF have also been used by a number of Head Start programs to identify parents who would benefit from supportive guidance and consultation (Grossman & Shigaki, 1994; Reitman, Currier, & Stickle, 2002; Seo, 2004). Critical evaluations of the use of the PSI and the PSI-SF with such low-income, minority, and single parents have indicated that the PSI and PSI-SF are useful and valid predictors of parental dysfunction and problem child behaviors (Bendell, Stone, Field, & Goldstein, 1989; Bhavnagri, 1999; Hutcheson & Black, 1996; Kelley, 1993; LeCuyer-Maus, 2003; Moss, Cyr, & Dubois-Comtois, 2004; Reitman, Currier, & Stickle, 2002). Studies have replicated both the factor structure and the construct validity of the PSI and PSI-SF in low-socioeconomic-status, minority populations (Breen & Barkley, 1988; Cuccaro, Holmes, & Wright, 1993; Reitman, Currier, & Stickle, 2002; Wakschlag & Keenan, 2001).

First-Gate or Intake Measure

The use of the PSI as a first-gate or intake measure involves administering the PSI as the first psychometric measure in a clinical assessment. This is typically handled in one of two ways. In the first, the PSI can be sent to the parent along with other forms to be completed and returned prior to the first face-to-face contact. This allows the clinician to score and interpret the results so that he or she can structure and focus both the interview and the selection of other psychometric components of the assessment. The PSI subscales cover the major child characteristics, parent characteristics, and situational variables that relate parenting difficulties. These subscales are not diagnosis oriented but rather identify issues and possible areas in need of intervention or more in-depth examination. The second concern procedure is to have the parent take the PSI by computer administration immediately prior to the initial interview. The PSI report is then reviewed by the clinician prior to the clinical interview, and sometimes the generated report is used as part of the assessment interactions with the parent.

Routine Clinical Assessment

PSI Scores and Dysfunction–Validity Studies

The PSI and PSI-SF are often used as a part of the assessment of a child and family system. Their contribution to the assessment is to help the clinician determine the level and sources of parenting stress. The PSI scores are related to a number of critical diagnostic and intervention issues.

Premature Treatment Termination and Non-compliance

PSI scores are associated with parental compliance with treatment expectations, and continuation in treatment. High PSI Total scores have been found to be associated with premature termination of psychological treatment and non-compliance with medical treatment (Calam, Bolton, & Roberts, 2002; Friars & Mellor, 2007; Gerson, Furth, Neu, & Fivush, 2004; Hipke, Wolchik, Sandler, & Braver, 2002; Kazdin, 1990; 1995; Kazdin, Mazurick, & Bass, 1993; Mellins, Kang, Leu, Havens, & Chesney, 2003). There is also evidence that high Stress scores are linked to excessive and inappropriate medical services utilization (Abidin & Wilfong, 1989; Waisbren et al., 2003).

Child Attachment and Quality of Parent–Child Transactions

The PSI scores are also associated with insecure attachment relationships, (Jarvis & Creasey, 1991; Teti, Nakagawa, Das, & Wirth, 1991) which is related the quality of the parent's perception and cognitions about their child, and the quality of parent-child interactions. Typically highly stressed parents show reduced sensitivity to their child's need and utilize more intrusive and aggressive interactions with their child (Barkley, Anastopoulos, Guevremont, & Fletcher, 1992; Beck, Young, & Tarnowski, 1990; Bigras & LaFreniere, 1994; Diener, Nievar, & Wright, 2003; Hitchcock, 2003; Ispa et al., 2004; Jarvis & Creasey, 1991; Laganière, Tessier, & Nadeau, 2003; Levendosky & Graham-Bermann, 1998; Mash, Johnston, & Kovitz, 1983; Moore, Probst, Tompkins, Cuffe, & Martin, 2007). PSI scores are also associated with harsh punishment (Burbach et al., 2004; Florsheim et al., 2003; Jackson, Gyamfi, Brooksgunn, & Blake 1998) and they typically had the strongest correlations of all the family variables examined. Elevated Parenting Stress is in general a predictor of harsh punishment, and this is particularly problematic when the PSI subscales of Child Adaptability, Reinforces Parent, Parental Competence, and Parental Attachment are elevated (Coyle, Roggman, & Newland, 2002; Lyons-Ruth, Lyubchik, Wolfe, & Bronfman, 2002; Repetti & Wood, 1997).

The level of parenting stress as measured by the PSI Total score and the Parent Domain and Subscales has been shown to be predictive of a wide variety of observed inappropriate and dysfunctional parenting behaviors (Bigras, LaFreniere, & Dumas, 1996; Calkins, Hungerford, & Dedmon, 2004; Degroat, 2003; Diener et al., 2003; Feinfeld & Baker, 2004; Fite, Greening, & Stoppelbein, 2008; Hall & Marteau, 2003; Moore et al., 2007; Moran, Pederson, Pettit, & Krupka, 1992; Onufrak, Saylor, Taylor, Eyberg, & Boyce, 1995; LeCuyer-Maus, 2003; Rodriguez & Green, 1997; Teti, Nakagawa, Das, & Wirth, 1991; Wilfong, Saylor, & Elksnin, 1991).

Parental Depression

Depression, a variable that has been linked to dysfunctional parenting behavior, is highly related to stress in the Parent Domain of the PSI and to the subscale Depression (Anastopoulos, Guevremont, Shelton, & DuPaul, 1992; Frankel & Harmon, 1996; Gelfand, Teti, & Fox, 1992; Milgrom & McCloud, 1996; Willner & Goldstein, 2001). The PSI Depression subscale more directly relates to parenting behavior than the more commonly broad-band measures of depression and has been shown to be a better predictor of dysfunctional parenting behavior and problems in child adjustment (Breen & Barkley, 1988; Calzada et al., 2004; Donenberg & Baker, 1993; Kazdin et al., 1993; Webster-Stratton, 1988). High levels of total parenting stress and elevations of the Depression, Social Isolation, and Attachment scores are related to a variety of mental health problems in both mothers and fathers (LeCuyer-Maus, 2003; Wong, Lam, & Kwoak, 2003).

Child Adjustment

The PSI Child Domain and its various subscales have, in research studies, been found to be predictive of children's current and future level of psychosocial adjustment (Goldberg et al., 1997; Éthier & LaFreniere, 1993; Nelson, Stage, Duppong-Hurley, Synhorst, & Epstein, 2007). The pattern of subscales provides the clinician with an indication of likely types of problematic behavior (Anastopoulos, Shelton, DuPaul, & Guevremont, 1993; Bagley & Mallick, 1997; Bigras et al., 1996; Breen & Barkley, 1988; Irvin, Carter, & Briggs-Gowan, 2002; Karp, Serbin, Stack, & Schwartzman, 2004; Murphy & Barkley, 1996; Tomanik, Harris, & Hawkins, 2004; Weiss, Sullivan, & Diamond, 2003). PSI scores have been used as outcome measures to document changes resulting from treatments that target children's behavior problems (Barkley, Edwards, Laneri, Fletcher, & Metevia, 2001; Kazdin & Wassell, 2000; Sanders & McFarland, 2000).

Children's academic success in school is moderated by the quality of their home environment, in particular by the support and direction they receive from their parents. Parenting Stress Index scores relate to those parenting behaviors and are predictive of a child's problem-solving skills (Hughes, Brestan, Christens, Klinger, & Valle, 2004) and of their achievement scores, particularly in the early years of school (Bramlett, Hall, Barnett, & Rowell, 1995; Bramlett, Rowell, & Mandenberg, 2000; Chazan-Cohen et al., 2009). It appears that when parents are stressed, they tend to deal with their children's academic issues more harshly and punitively than when they are not stressed. Typically, highly stressed parents report increased inattention in their children, a known risk factor for underachievement. In contrast, parents who are more supportive about their children's learning are more likely to have children who display less inattention and higher academic success. Rogers, Wiener, Marton, and Tannock (2009) report that the relationship between the quality of parental involvement and children's achievement was not explained by children's hyperactive and impulsive symptoms, but rather by the parent's level of parenting stress.

Marital Relationship and Child Functioning

Marital conflict and the quality of the parenting alliance are major variables impacting the quality of parenting in a family and a child's adjustment. The Parent Domain score of the PSI and the "Relationship with Spouses" subscales have been shown to have a significant association with child adjustment (Frank et al., 1991; Ha, Oh, & Kim, 1999; Kazui, Muto, & Sonoda, 1996). Intimate partner violence has been shown to raise parenting stress and increase the risk for harsh and or abusive parenting of the child (Taylor, Guterma, Lee, & Rathouz, 2009).

Likewise social support as measured by the social isolation subscale is linked to the level of parenting stress and to the quality of parenting and child adjustment (Chan, 1994; Feldman, Varghese, Ramsay, & Rajska, 2002; Jackson et al., 1998; Trute, 2003). Studies of families with children who have special needs have shown that supportive social networks buffer the effects of stress (Feldman et al., 2002). Social network support also buffers men's stress during the transition into parenthood, thus positively influencing paternal involvement (Fagan, Bernd, & Whiteman, 2007). Black, Schuler, and Nair (1993), using the PSI, compared the caregiving environments of infants prenatally exposed to cocaine to those of drug-free infants. Both groups of parents used social support to offset parenting stress, but the relationship was more robust among mothers of drug-free infants. Family support was the most effective form of social support, accounting for 11% of the variance in parenting stress. In a study of mothers of developmentally delayed children, Guralnick, Hammond, Neville, and Connor (2008) found similar results: when social support was high, stress levels were low. Particularly important was the finding that when mothers experience high levels of support from their spouse, the mothers were more reinforcing and experienced less role restriction as measured by the PSI.

Forensic Application of the PSI: Child Custody Evaluation

Main Forensic Contexts for Using the PSI

The PSI yields data on potential sources of stress in the family system through a parent's perceptions of his or her functioning in the parental role, issues in the parent–child relationship, child developmental issues, life stress, and degree of support from the other parent. One reviewer indicated, “The potential uses of the PSI are wide and varied. The PSI will be helpful for anyone interested in the evaluation or assessment of children and parents within a systems context” (Allison, 1998, p. 3). It would appear there is potential for the PSI as a screening tool and as a complement to other data sources whenever there is a forensic context that is examining parent and child relationships and the impact of psychosocial stressors in the family system.

It is hard to imagine any population that is experiencing more stress than families involved in a child custody evaluation. This group also is unique in that the parents rightly perceive they are being put under the microscope concerning the quality of the parent–child relationships. In this section, we will focus on the use of the PSI in child custody evaluations (CCEs) as one forensic area where the relevance of the PSI is readily apparent. One survey of child custody evaluators revealed that 41% of evaluators use the PSI in their investigations and forensic assessments, and, on average, in about two-thirds of their evaluations (Bow & Quinnell, 2001). Another survey found that 27% of evaluators used the PSI in almost every case (Bow, Gould, Flens, & Greenhut, 2006).

One other potential application would include cases of suspected child maltreatment. Stress in the family system and issues of parenting capacity are often the focus of social service and mental health evaluations in dependency and neglect cases. Two texts on parental capacity evaluations for cases involving child maltreatment and legal proceedings for the termination of parental rights both mention the PSI as a potentially useful source of data (Condie, 2003; Dyer, 1999). One of the authors of this chapter has found the PSI useful in dozens of such “termination” cases.

Child Custody Evaluations

Overview

The main concerns in these forensic evaluations are variously labeled as issues of child custody, physical custodial arrangements, parenting time, visitation, access, parenting plans, and parental responsibility/decision making. They occur in the context of child custody and parenting time litigation following marital separation and pending dissolution, or, in a pre-decree context, where the custody evaluator's role is to help the court craft a parenting plan that will be in the best interests of the children. CCEs are also conducted when one parent wants to modify an existing parenting, or in a post-decree context. With increasing frequency, evaluators are called upon to examine the needs of children whose parents never married but are in conflict over a parenting plan (Insabella, Williams, & Pruett, 2003). This would be called a “paternity action,” but it involves the same issues of parenting time, access, and allocation of decision making by the court.

In many American states, the parents, through their attorneys, will agree to or stipulate the choice of a mental health professional to conduct an evaluation to help with the resolution of the dispute, through either settlement or a court hearing. The evaluator is usually a psychologist. This stipulation then may be incorporated into a court order. The most common practice would be for the court to directly appoint the evaluator to conduct an evaluation after one of the parties filed a motion for a parenting evaluation. The evaluator would be expected to conduct an objective, neutral evaluation as the “court's expert.” The attorneys often would nominate the names of one or more evaluators. Usually the evaluators are private practitioners; in a few states, there will be

public, court-based services for providing evaluations. States often allow for a second evaluation when one party is dissatisfied with the results of the first evaluation, but almost always both parties will need to agree on the choice of the second evaluator, and usually the court will appoint the evaluator. The practice of one party unilaterally hiring their own evaluator is almost universally disapproved, but this does occur in rare instances in some states. The perceived objectivity of such an evaluator would be low and one parent might refuse to cooperate.

Comprehensive and Brief-Focused Evaluation

CCEs are usually expected to be comprehensive in nature, with the evaluator thoroughly investigating all of the issues that are salient in the case. Often, the court's order of appointment for the evaluator will delineate the legal issues to be addressed such as parenting time, decision making, or relocation. The evaluation may take several months to complete. The court usually expects the evaluator to formulate opinions and to make specific recommendations on these ultimate issues before the court. It is helpful if the court's order also addresses salient behavioral issues that are known, such as allegations of domestic violence or substance abuse. Psychological testing almost always would be used in comprehensive evaluations. Professional guidelines and model standards of practice for comprehensive CCEs have recently been revised and updated (American Psychological Association, 2010; Association of Family and Conciliation Courts, 2006; Martindale et al., 2007).

Courts sometimes issue orders for a brief-focused evaluation (BFE) when there is a specific issue to be addressed and assessed (e.g., school choice, summer parenting time schedule). The court then wants a specific recommendation to resolve the issue. California allows for a BFE by court rules. It is expected that the evaluation would be performed in a short time frame. In such evaluations it would be rare for psychological testing to be conducted. Professional guidelines exist for BFEs (Association of Family and Conciliation Courts, 2009).

Criticisms of Custody Evaluations

While there have been vocal critics of both the concept and the practice of child custody evaluations to assist the courts in the difficult task of creating parenting plans for the children of divorcing parents, the practice is accepted in all of the states, often codified by statute (Colorado Revised Statutes, 2012) or court rules (e.g., California Rules of Court, 2005; Florida Family Law Rules of Procedure, 2005). Custodial/parenting evaluations are also used in Canada, Australia, and England. Critics have asserted there is not a sufficient scientific basis for the methodology or substance for a family study that would provide a reasonable basis for making predictions about the child's future welfare (Emery, Otto, & O'Donohue, 2005; Melton, Petrila, Poythress, & Slobogin, 2007; O'Donohue & Bradley, 1999). The field of child custody has debated whether there is sufficient scientific basis for evaluators formulating ultimate issue opinions and recommendations on type of custody, parenting time, and decision making (Gould & Martindale, 2005; Tippins & Wittman, 2005), but as a practical matter, most judges expect opinions on the central issues, and specific recommendations for a parenting plan.

Others have suggested the legal concept of the best interests of the child is too broad, has not been specifically defined by mental health professionals, and therefore cannot be operationalized in the context of a child custody evaluation (Shuman, 2002; see also Jameson, Ehrenberg, & Hunter, 1997; D. A. Krauss & Sales, 2000). In contrast, other authorities have asserted that the global concept and legal standard of best interests of the child can be defined and measured, just like any general variable, and efforts should be directed at assessing the costs and benefits associated

with child custody evaluations within the family court system (R. F. Kelly & Ramsey, 2009). Many states have a list of best interests factors in their statutes for courts and custody evaluators to address. The issue of quality of the forensic work product by evaluators is a legitimate concern, and attorneys often retain experts to review the custody report and evaluator, which serves as a form of checks and balances in the system and functions as forensic quality control for the welfare of children (Austin, 2009).

Scientific Grounding of Custody Evaluations

Forensic theorists and practitioners have more recently attempted to systematically describe the scientific basis for child custody evaluations (Galatzer-Levy, Kraus, & Galatzer-Levy, 2009; Gould, 2006; Gould & Martindale, 2007; Kuehnle & Drozd, 2012) and presented research-based forensic evaluation models for topics that may surface within parenting evaluations. These evaluation models address issues of child maltreatment (Kuehnle, Coulter, & Firestone, 2000), partner violence (Austin, 2001), child sexual abuse (Kuehnle, 1996; 1998), relocation of parent with a child (Austin, 2008; 2012), and parental gatekeeping (Austin, 2011; Austin, Fieldstone, & Pruett, 2012).

The “forensic approach” to custody evaluation (Martindale & Gould, 2004) encourages custody evaluators to strive to make their data gathering and analysis as scientifically grounded as possible by using reliable and valid procedures and avoiding instruments known to be unreliable (e.g., human figure drawings; TAT) and approaches (e.g., ASPECT). Evaluators are able to use “research hypotheses” on important issues for examining the forensic data and making parenting plan recommendations (Austin, 2008; 2012). Progress has been made in summarizing the research findings relevant to virtually all of the substantive issues or problems that custody evaluators encounter (Kuehnle & Drozd, 2012; Galatzer-Levy et al., 2009; Gould, 2006; Gould & Martindale, 2007). Some of these issues affecting children’s adjustment following divorce include the following:

- parental conflict;
- attachment in parent–child relationships;
- paternal involvement;
- parental alienating behaviors and child alienation;
- intimate partner violence;
- relocation;
- gay and lesbian parenting;
- shared parenting arrangements;
- risk and protective factors following divorce.

Multi-method, Multi-source Approach and Psychological Testing

Forensic psychologists, including custody evaluators, are encouraged to employ multiple methods, or forensic procedures, and to gather interview data from multiple sources, or to use a multi-method, multi-source approach to data gathering (Heilbrun, 2001; Martindale & Gould, 2004). Psychological testing is one of the methods. It will be the most scientific source of data in the dataset, but needs to be integrated into the overall forensic assessment.

It is easy for psychologists to overemphasize the importance of testing data in forensic evaluations (Brodzinsky, 1993; Otto & Collins, 1995), but it has a potentially important role in custody evaluation. The PSI can provide very useful testing data for the evaluator concerning parent–child relationships and child development concerns, and also by comparing PSI results with personality testing data, as the case vignette described on pp. 370–371 will illustrate. However, because of the

defensiveness and other response biases that are elicited by the custody evaluation context, testing data may provide little incremental value for the evaluator. Psychological testing data from personality tests, such as the MMPI-2 and MCMI-III, are used by evaluators to generate hypotheses about the psychological functioning of the parents and children. PSI data may create hypotheses about parent-child relationships and child development. Data from interviews, observations, and documents concerning past behaviors and events are juxtaposed with testing data to address the issues in the case. Testing data can also stand on their own. With highly elevated scale scores and configurations, it is difficult not to draw conclusions about a person's psychological functioning, for example 3-6 or 4-9 on the MMPI-2, especially with comparisons to norms for custody litigants. Because the typical custody litigant does not produce subscale elevations on the PSI, when there are very high elevations it is a bright red flag that needs to be investigated.

The child custody evaluator will always interview parents, interview children if they are old enough, observe the children both alone and together with the parents either in a home and/or office setting, conduct psychological testing with all persons in a parenting role, conduct collateral interviews to obtain third-party information, and review relevant documents and records (Austin, 2002; Austin & Kirkpatrick, 2004; Heilbrun, 2001). The evaluator strives for a diverse and thorough database in order to achieve an understanding of the family system and dynamics and to accurately address the questions in the case. In some cases, the evaluator is asked to make recommendations about the distribution of parenting time and decision making between two fit and very capable parents who simply cannot reach agreement about a parenting plan. More often, there will be considerable conflict between the parents, and allegations are asserted by both parents that call into question the parenting effectiveness of the other parent and even potential harm to the children. There may be allegations of past partner violence, child maltreatment, or child sexual abuse. Alcohol or substance abuse is not an infrequent issue. One parent may be seeking to relocate with the child to a different state. The case confronting the evaluator is generally quite complex and conflicted. Child custody evaluations are likely to be the most complicated and time-consuming of all forensic mental health evaluations (Flens, 2004; Otto, Edens, & Barcus, 2000).

The Forensic Approach to Parenting Evaluations

The forensic approach to custody evaluation requires the evaluator to articulate hypotheses on the salient issues in the case. It requires the evaluator to be vigilant in being objective in his or her analysis and to be mindful of potential sources of confirmatory bias that might compromise the fair consideration of alternative hypotheses (Martindale, 2005). One survey study found that the greatest concern of practicing attorneys about custody evaluation work product was the perceived lack of objectivity by the evaluation (Bow & Quinnell, 2004).

The evaluator needs to acknowledge that there are factors the court may deem relevant to a custody determination that lie outside of the competency and appropriate consideration by an evaluator, such as which parent can provide the best opportunity for religious training, which environment is most conducive for a child's moral upbringing, and the moral fitness of the parents. For this reason, the evaluator needs to provide an analysis of the child's *psychological* best interests (American Psychological Association, 1994; Jameson et al., 1997; D. A. Krauss & Sales, 2000).

Forensic theorists and practitioners have described an emerging evaluation paradigm for child custody, variously referred to as the forensic method (Martindale & Gould, 2004) or the forensic scientist-practitioner approach (Austin & Kirkpatrick, 2004; Kuehnle, 1998). This approach proposes the universal use of multiple methods of data collection and the need for utilizing multiple sources for information about the parents and children. The goal is to create a comprehensive and diverse data set on the family. It is acknowledged that parents will be motivated to provide

positively biased data about themselves and negatively biased against the other parent, and evaluators possess no special skills at deciphering the truth (Ekman & O'Sullivan, 1991; Meissner & Kassen, 2002), so collateral interview data and documentary data are a necessary part of forensic mental health evaluations, including child custody evaluations (Austin, 2002; Austin & Kirkpatrick, 2004; Heilbrun, Warren, & Picarello, 2003).

The evaluator structures the evaluation around alternative hypotheses on each salient issue, and the data collection and analysis then become a process of ruling out alternative rival hypotheses (Heilbrun, 2001). Testing data are useful in generating hypotheses relevant to parenting and the child's development. The evaluator looks for support for hypotheses from multiple sources so as to achieve convergent validity (Austin, 2002). This process of data interpretation often requires careful investigation by the evaluator in order to obtain critical data and goes well beyond the superficial data that may be presented by the parents or their representatives (Austin & Kirkpatrick, 2004). The evaluator in the end will usually make specific recommendations to the court on custodial parental arrangements and parenting time (Gould, 2006), though there may be instances where the evaluator needs to stop short of addressing the ultimate issue on custody and parenting time because only the court can know the "threshold of harm" that is at issue (Austin, 2000).

Finally, the evaluator needs to be cognizant of issues of admissibility of expert testimony. While child custody experts have largely been exempted from application of the increasingly higher standards for showing reliability and helpfulness to the trier of fact (Shuman, 2002; D. A. Krauss & Sales, 2000), it is to be expected that this will change (Ramsey & Kelley, 2004). In the case of testimony about psychological testing data, it has been proposed there should be a higher standard for reliability in the forensic context (Goodman-Delahanty, 1997). When custody evaluators utilize unreliable instruments and approaches, then they should expect to have their work product reviewed and scrutinized (Kirkpatrick, Austin, & Flens, 2011).

The reliability and validity data on the PSI, its publication in peer-reviewed journals, and the known error rate of the instrument in different contexts suggest it would meet standards for admissibility as a scientific instrument (i.e., *Daubert v. Merrell Dow Pharmaceuticals*, 1993; Bow et al., 2006). The recent publication of the PSI-4 corrects previous limitations of the instrument and the vulnerability for attack as not meeting criteria for admissibility, for example the lack of males in the clinical normative sample. There now is a need to generate custody litigant norms for the PSI-4.

Population of Custody Litigants and Demand Characteristics

Custody Litigants

The population of child custody litigants represents a statistically extreme group. It is estimated that about 90% of all custody cases reach a settlement and do not require formal litigation, and about 90% of the remaining cases do not go to trial (Ash & Guyer, 1986). So, it is a fairly extreme group that ends up with an evaluator and an even rarer event that the case goes to trial after the completion of an evaluation. Later in the chapter, we present normative data on the PSI-3 for the population of child custody litigants. It is generally accepted by evaluators that the custody litigant and evaluation population includes a high percentage of individuals with personality disturbances that will be exacerbated by litigation and conflict.

Demand Characteristics of Litigation

It is important to understand that this population is seen by the custody evaluator as the result of a court order and after attempts at dispute resolution have failed. Conflict is at a high level and trust

may be minimal between two former intimate partners who face the arduous task of co-parenting their child in the aftermath of a custody and gatekeeping dispute. By the time the litigants are seen by an evaluator, each side often has levied multiple allegations against the other parent, who responds by upping the ante and responding with even more allegations. The situational and contextual variables must be taken into account when interpreting psychological tests.

Response Bias

A review of the content from many of the common psychological tests used in custody cases will quickly show how transparent the items really are. This fact lends itself to identifying when parents are trying to present themselves in the best possible light. Many tests have response-style indicators that will identify the approach used by the litigant to answer the questions. The *defensive* individual is likely to approach psychological tests by developing a response style consisting of two interwoven elements: (1) the assertion of mentally healthy attributes that are inaccurate; and (2) the denial of characteristics that are accurate but less than desirable. Nichols and Greene (1997), for example, noted that the defensive test-taker “may endorse items not merely to conceal symptoms and maladjustment, but to assert a degree of soundness, virtue, prudence, strength and well-being that is, if anything, superior to ‘normal’ levels” (p. 254). More specifically, these authors added, “In child custody, the evaluator is likely to see underreporting by a mix of trying to conceal any indication of psychological distress (dissimulation) and trying to assert superior adjustment) simulation” (p. 254). The defensive individual will use a combination of dissimulation—“concealing what one believes to be true” (p. 252)—and simulation—“displaying what one believes to be false” (p. 252). Exner, McDowell, Pabst, Stackman, and Kirk (1963) noted more than 40 years ago that psychological tests rely heavily on the test-taker’s willingness to answer the items honestly:

Thus the total usefulness of the device as a diagnostic instrument, particularly when used for screening or in situations where social desirability is an important element, seems to be dependent largely upon the honesty of the subject or on the ability of the interpreter to detect willful attempts at dishonest responding.

(p. 91)

Between the contextual variables in a child custody evaluation and the item content of the tests, it is not surprising that several common profiles emerge on the commonly used tests. Discussing the contextual variables associated with child custody evaluation allows that evaluator to understand the mindset of the parent who responds to the items on any given test battery. The following are frequently seen in child custody evaluations:

- Parents in a stressful divorce challenge each other for custody of their child or children.
- The contextual variables (described above) generate defensive postures in the parents.
- These parents are forced to take tests with “loaded” item content.
- The evaluator is faced with prototypical defensive test profiles.

Given this sequence, however, the typical rules of test interpretation simply may not apply. The American Psychological Association’s Ethics Code (APA, 2002) addresses the issue of situational and contextual variables as a threat to the reliability of test data and interpretations:

9.06 Interpreting Assessment Results

When interpreting assessment results, including automated interpretations, psychologists take into account the purpose of the assessment as well as the various test factors, test-taking

abilities, and other characteristics of the person being assessed, such as situational, personal, linguistic, and cultural differences, that might affect psychologists' judgments or reduce the accuracy of their interpretations. They indicate any significant limitations of their interpretations. (See also Standards 2.01b and c, Boundaries of Competence, and 3.01, Unfair Discrimination.)

(p. 24)

This is the point where it is important to have a thorough understanding of the psychometric properties of the tests. It is also essential to be aware of the effects on the test data that result from a defensive response style. Defensive test-taking attitudes affect each test differently. It is the responsibility of the prudent evaluator to understand and consider these effects when interpreting tests in the child custody context.

Goodness of Fit between the PSI and the Child Custody Context

The PSI has unique potential to contribute to the psychological testing data in a parenting evaluation because the variables being measured are similar to several statutory factors that most American states require to be considered in these evaluations: child's developmental status, adjustment, and psychological functioning; quality of parent-child relationships; parents' mental health; and level of support between the parents. Variables that are measured by evaluators generally revolve around family relationship factors which make the use of the PSI a logical application. Competently conducted evaluations will assess the child's development, parenting effectiveness, parent psychological disturbance or health, parent-child relationships, and interparental conflict. Some variations of these factors are often found in state statutes that have incorporated the Uniform Marriage and Divorce Act criteria (1979). PSI data are logically relevant to this combination of factors. PSI data are also correlated with a number of useful concepts and research literature that are addressed by custody evaluators: security of child to parent attachment; emotional security; parent insightfulness and attunement; depression; and child temperament, adaptability, and resilience.

Researchers have examined the role of stress and parental coping ability in studying the adjustment of children to separation and divorce. The majority of children show adequate or normal development and functioning as adults (Emery & Forehand, 1994), but a significant percentage, perhaps 25%, do not. Divorce stands as a general developmental risk factor for children. Evaluators need to uncover risk factors for adjustment problems as well as strengths and resources available to children that would increase the likelihood of successful coping with the stress of divorce and finding a normal developmental course. Evaluators need to follow the lead of researchers on issues of risk and resiliency facing children (J. B. Kelly & Emery, 2003).

Research has shown that long-term stresses from divorce produce on average less academic achievement, poorer self-concept, and generally higher rates of psychological maladjustment compared to children who did not experience parental separation and divorce (Amato, 2001). Children who grow up with single mothers after divorce show higher rates of high school dropout, poorer academic achievement, and higher rates of teen pregnancy (McLanahan & Sandefur, 1994). School-age children and adolescents from divorced families show a higher frequency of problematic behaviors and appear on average to be off of a normal developmental track in terms of well-being, social interactions, and academic achievement (Amato, 2001; Guidubaldi, Perry, & Nastasi, 1987; McLanahan & Teitler, 1999). The effects of divorce on very young children have been less extensively studied (Pruett, Williams, Insabella, & Little, 2003), but studies have shown lower academic achievement among primary school children (Amato, 2001), lower level of cognitive

ability scores (Clarke-Stewart, Vandell, McCartney, Owen, & Booth, 2000), and insecurities about attachment relationships with both parents and physical and emotional safety (Pruett & Pruett, 1999).

Research has shown that parental dysfunction and interparental conflict place children at risk following divorce by the mere presence of the greater levels of stress reported by divorced and separated parents compared to married adults (Kitson & Morgan, 1991). When dysfunctional parental emotional states diminish warm and authoritative parenting, then children show more externalizing and internalizing symptoms (DeGarmo & Forgatch, 1999; J. B. Kelly, 2000).

Parental depression and anxiety interfere with parenting capacity and are correlated with more negative child adjustment (Pett, Wamold, Turner, & Vaughn-Cole, 1999). Numerous studies have established the link between parental distress, their diminished well-being, and children's adjustment (Hetherington, Bridges, & Insabella, 1998). This finding is most true at the time surrounding the marital separation and the subsequent two years (Hetherington & Kelly, 2002), a time when parents are most likely to be participating in a custody evaluation. One of the most reliable empirical findings in the divorce effects research is that children's adjustment is negatively correlated with greater levels of interparental conflict (Lamb, 2002), and these families are more likely to litigate, which means that greater levels of child adjustment problems might be expected to be observed and reported in the context of custody evaluations.

The state legal standards for child custody determinations are always some variant of a Best Interests of the Child standard (BIS). When issues of harm or detriment surface, then these are treated as another dimension of BIS issues (*In re Marriage of Martin*, 2002) and parenting time arrangements may be discussed in terms of the Least Detrimental Alternative (LDA) residential parenting plan. LDA is the conceptual obverse to BIS (Austin, 2000). The role of the evaluator is to operationalize these general legal concepts in behavioral terms and gather meaningful data so that helpful recommendations can be made to the court on what parenting arrangements will be most likely to maximize the child's positive developmental outcomes and insulate the child from harm. The PSI holds potential in identifying sources of stress in the family system that raise concerns about potential harm to the child. In the context of cases where there are issues of partner violence, child maltreatment, child alienation, and major parental dysfunction, such as with a major mental illness or substance abuse, then test data may help ferret out the potential effects on the child and the parent-child relationship.

A Mindful Approach to Psychological Testing in the Forensic Context

Hypothesis Generation

Commentators have not been kind in their appraisal of how psychological testing is often used in custody evaluations (Brodzinsky, 1993; Melton et al., 1997; Otto & Collins, 1995). The main criticism is that testing data are often used in a direct application to an individual and not in conjunction with other sources of data (Clark & Clark, 1991; Heilbrun, 1992). It is strongly urged that testing data only be used as sources for hypotheses (Brodzinsky, 1993) and not as confirmatory of any hypotheses. When data from several tests and other methods are consistent, then there may be a basis for hypothesis confirmation (Heilbrun, 2001), or creating convergent validity (Heilbrun, 2001; Austin, 2002). Unfortunately, when reviewing the forensic work of evaluators, it is not uncommon to find one-to-one interpretations of a parent's psychological functioning based on psychological testing, and then sometimes with verbatim quotes from a computer-generated report (Bow et al., 2006). In a recent case, a prominent expert witness placed extraordinary weight on results from a Rorschach Ink Blot test to opine about the ultimate issue of relocation, to the obvious consternation of the judge.

The rationale for not employing test data as confirmatory or treating results as isomorphic with an individual's psychological functioning is that the interpretation of a test profile is made by comparison to group average scores, or aggregate data in the clinical, normative sample for the test. To then make the reductionistic interpretive leap to the individual is to commit what statisticians call the "ecological fallacy" (Ostroff, 1993; Robinson, 1950). That is, the individual data cannot be explained by the aggregate data norm and need to be supported by other sources of data specific to the individual.

Emphasis on Function

Grisso (1986) cautions on the use of psychological testing in the custody context, largely on the grounds that the instruments were not designed specifically to measure aspects of parental capacity or the legal competencies before the court. Grisso's message should alert evaluators to be sensitive to the issue of what the test is measuring, in what context, and to translate the testing data to functional capacities. In the custody context, this means the evaluator needs to uncover to what extent the instruments are tapping elements of parenting effectiveness, quality of or stress within parent-child relationships, and child development.

Response Bias and Impression Management

Evaluators routinely need to be sensitive to issues of response bias, defensiveness, and impression management in an individual's approach to test taking when interpreting the data. As noted above, a general rule of thumb suggests that parties will be somewhat defensive on psychological testing and trying to put forth a favorable impression in the custody context. In our normative data set on the PSI, we found male and female custody litigants to have essentially the same scores on the domains, subscales, and total scores (using $p > 0.001$ significance levels; see Table 12.5). Interestingly, only the Defensive Indicator scores showed a statistical—and clinical—difference between the male and female caregivers (i.e., $p = 0.0002$). As noted in Table 12.5, the average Defensive Indicator score was 27.3 ($SD = 6.7$). Both male and female caregivers obtained scores in the non-defensive range. Female caregivers obtained defensive scores of 28.9 ($SD = 7.0$) and males obtained scores of 25.6 ($SD = 6.1$). This suggests that male caregivers in the custody sample were slightly more defensive than female caregivers.

Normative Comparisons and Extrapolation

When interpreting test data gathered in a forensic context to the normative data in a test manual, one needs to be cautious because of differences between the original normative sample and the population represented by the forensic examinee. Researchers have gathered normative data on special populations who take instruments in forensic settings, especially the MMPI-2. Such special population norms are available for the MMPI-2 for custody litigants (Bathurst, Gottfried, & Gottfried, 1997) and the MCMI-III (McCann et al., 2001). Authorities have recommended utilizing such norms as a way of providing enhanced contextual interpretation and as a supplement to the normal profile interpretation using the original normative sample for the instrument (Otto, 2002). Preliminary norms for custody litigants taking the PSI are presented on p. 368.

TABLE 12.5 Preliminary Descriptive PSI–3 Data for Use in the Child Custody Context

Scales/Subscales	Mean ^a	SD	Confidence Interval		
			SEM	±68%	±95%
Total Stress Score	185.5	33.2	8.2	±8.2	±16.1
Child Domain	87.4	17.5	6.0	±6.0	±11.8
Distractibility	20.9	4.6	2.0	±2.0	±3.9
Adaptability	23.2	5.2	2.8	±2.8	±5.5
Reinforces Parent	7.9	2.5	1.2	±1.2	±2.4
Demandingness	15.4	4.3	2.4	±2.4	±4.7
Mood	8.8	2.7	1.6	±1.6	±3.1
Adaptability	11.2	3.7	1.6	±1.6	±3.1
Parent Domain	98.1	19.2	6.5	±6.5	±12.7
Competence	21.2	5.1	2.5	±2.5	±4.9
Isolation	10.2	3.0	1.6	±1.6	±3.1
Attachment	10.1	2.6	1.6	±1.6	±3.1
Health	10.0	3.0	1.9	±1.9	±3.7
Role Restriction	14.1	3.7	2.4	±2.4	±4.7
Depression	15.1	4.1	2.2	±2.2	±4.3
Spouse	17.5	5.3	2.2	±2.2	±4.3
Life Stress	16.8	10.2	—	—	—
Defensive Indicator	27.3	6.7	3.4	±3.4	±6.7

Note:

a Total sample of custody litigants = 214 (male = 107, female = 107).

Descriptive Data and Item Analysis

Psychological tests with a high degree of “face validity,” such as the PSI, allow for content analysis that is relevant to the working hypotheses for the case. Examination of individual test items may overlap and supplement actual questions asked during the forensic interviews. The evaluator may want to follow up test responses with specific interview questions for clarification. In this type of “critical item analysis,” a better understanding of the parent’s perceptions can be achieved. Other measures that may lack proven predictive validity, such as the *Parenting Alliance Measure* (Abidin & Konold, 1999) and *Parenting Satisfaction Survey* (Guidubaldi & Cleminshaw, 1994), can also be used in this way as a source of descriptive data on the quality of the parents’ co-parenting relationship.

Interpretation of PSI Data in a Custody Evaluation

A few preliminary comments are in order. First, when using the PSI, or any other test, it is essential to read thoroughly the accompanying test manual, including the psychometric characteristics of the test. This may seem like a given; however, the authors have reviewed cases where the evaluator had little actual knowledge of the content of the manual, and in some cases did not even own the manual, or current version of the manual. It is also important to know the literature regarding the test. A good source of information is reviews contained in the *Buros Mental Measurements Yearbook*. In addition, it is often prudent to have both supportive and critical literature regarding the test (Flens, 2004). Learning the critical aspects of the test on the witness stand could be an unpleasant experience for the unprepared evaluator.

It is important to use the standardized instructions when administering the PSI. Doing so allows the evaluator to rely on the standardization sample to make interpretive comparisons. One slight

modification may be in order when using the PSI in a custody setting. The Spouse subscale contains questions regarding the test-taker's partner. Since many custody evaluations involve modification of either existing custody and/or visitation arrangements or relocation issues, the test-taker may have re-partnered to someone they find to be supportive of their parenting of the child at issue. For this reason, it may be appropriate to ask the test-taker to consider the other parent in those questions that address issues associated with other parent of the child, rather than their current partner. On theoretical grounds, this appears to be an appropriate alteration of the standardized instructions. Currently, however, no data exist regarding this issue. Should the evaluator choose to use this altered instruction, a caveat should be included in the report, which is consistent with the Ethics Code of the American Psychological Association (hereinafter APA; see, for example, 9.06 Interpreting Assessment Results; APA, 2002).

The author of the PSI now recommends taking into account the context of the test administration in giving instructions on how to answer the items referring to the "spouse." In the context of a custody evaluation, the other spouse should refer to the other parent of the child. When a remarried or re-partnered parent is completing the PSI in a non-forensic setting, then "spouse" should refer to the current spouse or partner. A limitation of the PSI in custody evaluation also concerns the Spouse subscale. An examination of the seven items shows that some of them are worded in a manner that assumes the parents are still together. For example, one item is "Since having my child, my spouse and I don't do as many things together." So, the theoretical basis of using this subscale seems weak and evaluators who interpret it must specify what construct is being measured. Items on measures such as the Parenting Alliance Measure have more face validity on the process of supporting the other parent-child relationship and their view of the other parent's parenting ability.

Interpretation of the PSI in a custody case begins with a review of the response style the caregiver used when responding to the test items. Both Heilbrun (1992) and Otto et al. (2000) have recommended using tests in forensic evaluations that contain response-style indicators. The PSI includes the Defensive Responding indicator, which consists of 15 items scattered throughout the Parenting Domain subscales. The breakdown of items is as follows: Isolation (4 items), Depression (4 items), Role Restriction (3 items), Spouse (3 items) and Competence (1 item). The Attachment and Health subscales do not contain any items on the Defensive Responding indicator. A PSI profile is considered "defensive" when the Defensive Responding score is ≤ 24 . Roughly one-third of the profiles in the preliminary data were considered "defensive" (see Table 12.5). Lower scores on the Parent Domain may indicate a defensive profile, since all the Defensive Responding items are on the Parent Domain scales. As would be expected, a defensive profile should be interpreted with caution—an important caveat mentioned in the PSI manual.

Research regarding the issue of response styles has indicated two different factors account for the "fake good" presentation (Bagby & Marshall, 2004; Friedman, Lewak, Nichols, & Webb, 2001; Lanyon, 2004; Strong, Green, Hoppe, Johnston, & Olesen, 1999; Strong, Greene, & Kordinak, 2002). The first involves a conscious and deliberate attempt to present a favorable impression. As noted earlier, this can include asserting favorable, but inaccurate, characteristics and denying unfavorable, but accurate, characteristics. The parent in this case is trying to fool the evaluator into believing he or she is more virtuous and mentally healthy than may actually be the case. The second factor involves a tendency toward self-deceptive overconfidence in one's own strengths and abilities (namely, Self-Deceptive Enhancement). This response style is a more subtle characteristic in which the individual actually believes they embody the favorable image presented. It could be said that the individual in this case is actually fooling themselves, or engaging in "self-spin" in producing the responses. A parent who presents with Self-Deceptive Enhancement can be described as having narcissistic qualities and a lack of insight, and is easily angered when confronted. The parent in

this case believes that her or his responses are accurate and justified. The analysis of our PSI data suggests that the Defensive Responding is strongly related to the second factor, Self-Deceptive Enhancement, and almost completely unrelated to the first factor, Impression Management. This would suggest that caregivers who score in the defensive range have a narcissistic overconfidence in their abilities and a lack of insight into their own behavior, and are easily angered when confronted (see, for example, Paulhus, 1998), and further, that it may be advisable for evaluators not to describe such individuals as responding in a “defensive” manner.

The prudent evaluator would do well to make affirmative statements regarding the response-style issues and how these affect the psychological confidence and certainty that can be placed in the PSI interpretation. This is also consistent with the ethics codes of the APA (see, for example, 9.06 Interpreting Assessment Results; APA, 2002). Finally, after addressing the response-style issues the evaluator can address any domain and subscale elevations. To aid in the interpretation, caregivers’ profiles can be plotted on the same profile form, allowing comparisons of the various domains and subscales. It may also be appropriate to compare the resultant profiles with the context-specific normative data described in Table 12.5. A cautionary statement is appropriate at this point. The context-specific normative data do not supplant the standardization data presented in the PSI manual. These data simply allow for a comparison with other caregivers going through similar evaluation contexts. The clinical sample for the PSI-4 is a major step forward because of its sample size and equal proportions of males and females. When a custody litigant normative sample is available for the PSI-4, then the evaluator will have access to both percentile and *T*-score interpretations for custody norms.

Interpretation of the individual domain and subscales is straightforward. A domain or subscale is considered elevated at or above the 85th percentile. The addition of *T*-score transformations in the PSI-4 will allow for interscale quantitative comparisons with the average for the clinical sample set at 50 and each standard deviation equal to 10 points; for example, $T = 70$ means the score is at the 98th percentile. The manual describes the issues and hypotheses associated with the individual domains and subscales. As noted above, interpretations from the PSI should be used as hypotheses, corroborated or refuted by other sources of data. It is important that the evaluator not consider elevations on the domains and subscales as stand-alone data sources, but rather as hypotheses to be confirmed—or not—by other forensically relevant data collection methods (i.e., interview, observation, testing, collateral interview, record review). As the appropriate focus is the parent–child match, the PSI is particularly relevant to child custody evaluations, because of the emphasis on attributes of the child and parent that may lead to dysfunctional parenting (Abidin, 1995).

CASE STUDY: THE COERCIVE, CONTROLLING FATHER

The parents of two children were divorcing. Their daughter was 9 years old and their son 11 years. The mother had been a primary caregiver, a stay-at-home mother. The father was an extremely successful and wealthy financial investor. His level of pre-separation involvement in parenting had been fairly minimal. Data from multiple interview sources showed that the mother was an exceptional mother and mentally healthy individual who was highly attuned to her children’s needs. The mother’s MMPI-2 data showed defensive response bias and no significant clinical scale elevation, so there were no meaningful personality interpretations to be made. The father’s MMPI-2 testing data painted him as an extremely controlling and self-focused individual. Collateral data were very consistent with these testing hypotheses. After

the father's verbal abuse and controlling behaviors escalated into physical aggression and domestic violence, the parents separated. The last incident of intimate partner violence occurred while the mother was recovering from cancer surgery. The PSI-3 testing data showed no elevations for the mother within either the Child or the Parent Domains, but multiple elevations within both domains for the father's PSI-3 data for both children.

A power struggle developed between the parents over the issue of whether the son had genuine developmental problems in the area of intersensory integration disorder. The father lobbied for extensive evaluations and therapy. Many professional providers became involved and the issue became the central issue in the custody litigation. An objective assessment of all of the various evaluations and reports showed this was a highly functioning boy with an IQ of 150 and who probably had nothing more than a developmental issue with handwriting. The judge concurred, and also stated that the father had been coercive, controlling, and intrusive in his dealings with the wife both before and after the separation. Extensive data from multiple sources revealed a narcissistic man with limited capacity for empathy or parental warmth.

This case illustrates the utility of the PSI and for comparing test data configurations on the PSI and MMPI-2. Despite the constraining demand characteristics of the custody context, the father's data were highly revealing of his personality, stress in his relationship with his children, and how negatively he viewed their development. While the mother's PSI data showed only one scale elevation, RE in the Child Domain (and this was consistent with the reports of other observers), the father viewed the child as developmentally impaired, with multiple very high subscale elevations. Four Child Domain subscales were extremely elevated (three scales at the 100th percentile) and one in the Parent Domain (HE). The father's PSI data could have been plausibly interpreted as suggesting Munchausen's by Proxy or Factitious Disorder by Proxy. He seemed to have a need to see his gifted son as highly disturbed, or as "damaged goods," perhaps as a way of gaining control of the parenting, since he viewed the mother as an ineffective parent with questionable judgment. The father's MMPI-2 results were revealing, consistent with other data sources, and consistent with the PSI data. The highly elevated 3-6 profile generated the hypotheses on personality functioning (Scale 3: *T*-score = 79; Scale 6: *T*-score = 72).

Individuals with this negativistic and rigid outlook on life would be expected to be at risk for showing parenting and co-parenting deficiencies. The father's negative view of others and the world was directed at his son and ex-partner. The implications for making parenting plan recommendations were straightforward. The judge awarded the mother sole custody with control over decision making; the father's parenting time was limited and made conditional on receiving and benefiting from mental health treatment.

References

- Abidin, R. R. (1983). *Parenting Stress Index manual*. Charlottesville, VA: Pediatric Psychology Press.
- Abidin, R. R. (1990). *Parenting Stress Index test manual* (3rd ed.). Charlottesville, VA: Pediatric Psychology Press.
- Abidin, R. R. (1995). *Parenting Stress Index: Professional manual* (3rd ed.). Odessa, FL: Psychological Assessment Resources.
- Abidin, R. R. (1998, August). Parenting Stress Index: Its empirical validation. Paper presented at the Symposium on Child Custody, APA Annual Conference, San Francisco.
- Abidin, R. R. (2012). *PSI-4 professional manual*. Psychological Assessment Resources, Lutz, FL.
- Abidin, R. R., & Burke, W. T. (1978). Parenting Stress Index—draft manual. Unpublished manuscript, Department of Foundations of Education, University of Virginia, p. 6.

- Abidin, R. R., Jenkins, C. L., & McGaughey, M. C. (1992). The relationship of early family variables to children's subsequent behavior adjustment. *Journal of Clinical Child Psychology, 21*, 60–69.
- Abidin, R. R., & Konold, T. R. (1999). *Parenting alliance measure: Professional manual*. Sarasota, FL: Psychological Assessment Resources.
- Abidin, R. R., & Santos, S. V. (2003). *Índice de Stress Parental*. Lisbon: CEGOC-TEA.
- Abidin, R. R., & Wilfong, E. (1989). Parenting stress and its relationship to child health care. *Children's Health Care, 18*, 114–117.
- Allison, J. A. (1998). *Parenting stress index* (3rd ed.). In J. C. Impara & B. S. Plake (Eds.), *The thirteenth mental measurements yearbook* (pp. 1–8). Lincoln, NE: Buros Institute of Mental Measurements.
- Amato, P. R. (2001). Children and divorce in the 1990s: An update of the Amato and Keith (1991) meta-analysis. *Journal of Family Psychology, 15*, 355–370.
- American Psychological Association. (1994). Guidelines for child custody evaluation in divorce proceedings. *American Psychologist, 49*, 677–680.
- American Psychological Association. (2002). Ethical principles of psychologists and code of conduct. *American Psychologist, 57*(12), 1060–1073.
- American Psychological Association. (2010). Guidelines for child custody evaluations in family law proceedings. *American Psychologist, 65*(9), 863–867.
- Anastopoulos, A. D., Guevremont, D. C., Shelton, T. L., & DuPaul, G. J. (1992). Parenting stress among families of children with attention deficit hyperactivity disorder. *Journal of Abnormal Child Psychology, 20*(5), 503–520.
- Anastopoulos, A. D., Shelton, T. L., DuPaul, G. J., & Guevremont, D. C. (1993). Parent training for attention-deficit hyperactivity disorder: Its impact on parent functioning. *Journal of Abnormal Child Psychology, 21*(5), 581–596.
- Ash, P., & Guyer, M. (1986). Psychiatry and the law: The functions of psychiatric evaluation in contested custody and visitation cases. *Journal of the American Academy of Child Psychiatry, 25*, 554–561.
- Association of Family and Conciliation Courts. (2006, May). *Model standards of practice for child custody evaluation*. Madison, WI: Author.
- Association of Family and Conciliation Courts. (2009). *Guidelines for brief focused assessment*. Madison, WI: Author.
- Austin, W. G. (2000). Relocation law and the threshold of harm: Integrating legal and behavioral perspectives. *Family Law Quarterly, 34*, 63–82.
- Austin, W. G. (2001). Partner violence and risk assessment in child custody evaluations. *Family Court Review, 39*, 483–496.
- Austin, W. G. (2002). Guidelines for utilizing collateral sources of information in child custody evaluations. *Family Court Review, 40*, 177–184.
- Austin, W. G. (2008). Relocation, research, and forensic evaluation: Part II: Research support for the relocation risk assessment model. *Family Court Review, 46*(2), 347–365.
- Austin, W. G. (2009). Responding to the call to child custody evaluators to justify the reason for their professional existence: Some thoughts on Kelly and Ramsey (2009). *Family Court Review, 47*(3), 544–551.
- Austin, W. G. (2011). Parental gatekeeping in custody disputes. *American Journal of Family Law, 25*(4), 148–153.
- Austin, W. G. (2012). Relocation, research, and child custody disputes. In K. Kuehnle & L. Drozd (Eds.), *Parenting plan evaluations: Applied research for the family court* (pp. 540–559). New York: Oxford University Press.
- Austin, W. G., Fieldstone, L., & Pruett, M. K. (2012). *Bench book on parental gatekeeping in parenting disputes: Understanding the dynamics of gate-closing and opening for the best interests of children*. Miami: Authors.
- Austin, W. G., & Kirkpatrick, H. D. (2004). The investigation component in forensic mental health evaluations: Considerations in the case of parenting time evaluations. *Journal of Child Custody, 1*, 23–43.
- Bagby, R. M., & Marshall, M. B. (2004). Assessing underreporting response bias on the MMP-2. *Assessment, 11*(2), 115–126.
- Bagley, C., & Mallick, K. (1997). Temperament, CNS problems and maternal stressors: Interactive predictors of conduct disorder in 9-yr.-olds. *Perceptual and Motor Skills, 84*(2), 617–618.
- Bagner, D. M., & Eyberg, S. M. (2003). Father involvement in parent training: When does it matter? *Journal of Clinical Child and Adolescent Psychology, 32*(4), 599–605.

- Baker, B. L., McIntyre, L. D., Blacher, J., Crnic, K., Edelbrock, C., & Low, C. (2003). Pre-school children with and without developmental delay: Behavior problems and parenting stress over time. *Journal of Intellectual Disability Research*, 47(4-5), 217-230.
- Baker, D. B. (1994). Parenting stress and ADHD: A comparison of mothers and fathers. *Journal of Emotional and Behavior Disorders*, 2, 46-50.
- Barkley, R. A., Anastopoulos, A. D., Guevremont, D. C., & Fletcher, K. E. (1992). Adolescents with attention deficit hyperactivity disorder: Mother-adolescent interactions, family beliefs and conflicts, and maternal psychopathology. *Journal of Abnormal Child Psychology*, 20(3), 263-288.
- Barkley, R. A., Edwards, G., Laneri, M., Fletcher, K., & Metevia, L. (2001). The efficacy of problem-solving communication training alone, behavior management training alone, and their combination for parent-adolescent conflict in teenagers with ADHD and ODD. *Journal of Consulting and Clinical Psychology*, 69, 926-941.
- Bathurst, K., Gottfried, A., & Gottfried, A. (1997). Normative data for the MMPI-2 in child custody litigation. *Psychological Assessment*, 9, 205-211.
- Beck, S. J., Young, G. H., & Tarnowski, K. J. (1990). Maternal characteristics and perceptions of pervasive and situational hyperactives and normal controls. *Journal of the American Academy of Child and Adolescent Psychiatry*, 29(4), 558-565.
- Beckman, P. J. (1991). Comparison of mothers' and fathers' perceptions of the effect of young children with and without disability. *American Journal of Mental Retardation*, 95(5), 585-595.
- Bendell, R. D., Stone, W. L., Field, T. M., & Goldstein, S. (1989). Children's effects on parenting stress in a low income, minority population. *Topics in Early Childhood Special Education*, 8(4), 58-71.
- Bhavnagri, N. P. (1999). Low income African-American mothers' parenting stress and instructional strategies to promote peer relationships in preschool children. *Early Education and Development*, 10(4), 551-571.
- Bigras, M., & LaFreniere, P. J. (1994). Influence of psychosocial risk, marital conflicts and parental stress on the quality of mother-son and mother-daughter interactions. *Canadian Journal of Behavioural Science*, 26(2), 280-297.
- Bigras, M., LaFreniere, P. J., & Abidin, R. R. (1996). *L'indice de stress parental*. Toronto: Multi-Health Systems.
- Bigras, M., LaFreniere, P. J., & Dumas, J. E. (1996). Discriminant validity of the parent and child scales of the Parenting Stress Index. *Early Education and Development*, 7(2), 167-178.
- Black, M., Schuler, M., & Nair, P. (1993). Prenatal drug exposure: Neurodevelopmental outcome and parenting environment. *Journal of Pediatric Psychology*, 18(5), 605-620.
- Bow, J. N., Gould, J. W., Flens, J. R., & Greenhut, D. (2006). Testing in child custody evaluations: Selection, usage, and admissibility: A survey of psychologists. *Journal of Forensic Psychology Practice*, 6(2), 17-38.
- Bow, J. N., & Quinnell, F. A. (2001). Psychologists' current practices and procedures in child custody evaluations: Five years after American Psychological Association guidelines. *Professional Psychology: Research and Practice*, 32(3), 261-268.
- Bow, J. N., & Quinnell, F. A. (2004). Critique of child custody evaluations by the legal profession. *Family Court Review*, 42(1), 115-127.
- Bramlett, R. K., Hall, J. D., Barnett, D. W., & Rowell, K. (1995). Child developmental/educational status in kindergarten and family coping as predictors of parenting stress: Issues for parent consultation. *Journal of Psychoeducational Assessment*, 13(2), 157-166.
- Bramlett, R. K., Rowell, R. K., & Mandenberg, K. (2000). Predicting first grade achievement from kindergarten screening measures: A comparison of child and family predictors. *Research in the Schools*, 7(1), 1-9.
- Breen, M. J., & Barkley, R. A. (1988). Child psychopathology and parenting stress in girls and boys having attention deficit disorder with hyperactivity. *Journal of Pediatric Psychology*, 13(2), 265-280.
- Brodzinsky, D. M. (1993). On the use and misuse of psychological testing in child custody evaluations. *Professional Psychology: Research and Practice*, 24, 213-219.
- Burbach, A. D., Fox, R. A., & Nicholson, B. C. (2004). Challenging behaviors in young children: The father's role. *Journal of Genetic Psychology*, 165(2), 169-183.
- Burke, W. T. (1978). Parenting stress. Doctoral dissertation, University of Virginia, Charlottesville.
- Cal. R. Ct. 5.220. Court-ordered child custody evaluations. Retrieved from http://www.courtinfo.ca.gov/cms/rules/index.cfm?title=five&linkid=rule5_220.
- Calam, R., Bolton, C., & Roberts, J. (2002). Maternal expressed emotion, attributions and depression and entry into therapy for children with behavioural problems. *British Journal of Clinical Psychology*, 41, 213-216.

- Calkins, S. D., Hungerford, A., & Dedmon, S. E. (2004). Mothers' interactions with temperamentally frustrated infants. *Infant Mental Health Journal*, 25(3), 219–239.
- Calzada, E. J., Eyberg, S. M., Rich, B., & Querido, J. G. (2004). Parenting disruptive preschoolers: Experiences of mothers and fathers. *Journal of Abnormal Child Psychology*, 32(2), 203–213.
- Chan, Y. C. (1994). Parenting stress and social support of mothers who physically abuse their children in Hong Kong. *Child Abuse and Neglect*, 18(3), 261–269.
- Chazan-Cohen, R., Raikes, H., Brooks-Gunn, J., Ayoub, C., Pan, B. A., Kisker, E. E., . . . Fuligni, A. S. (2009). Low-income children's school readiness: Parent contributions over the first five years. *Early Education and Development*, 20(6), 958–977.
- Clark, B. K., & Clark, C. R. (1991). Psychological testing in child forensic evaluations. In D. Schetsky & E. Benedek (Eds.), *Clinical handbook of child psychiatry and the law* (pp. 34–52). Baltimore, MD: Williams & Wilkins.
- Clarke-Stewart, K. A., Vandell, D. L., McCartney, K., Owen, M. T., & Booth, C. (2000). Effects of parental separation and divorce on very young children. *Journal of Family Psychology*, 14, 304–326.
- Colorado Revised Statutes § 14–10–124(1.5). Dissolution of Marriage—Parental Responsibilities. Retrieved from <http://www.coleg.co.us>.
- Condie, L. O. (2003). *Parenting evaluations for the court: Care and protection matters*. New York: Kluwer Academic/Plenum.
- Coyle, D. D., Roggman, L. A., & Newland, L. A. (2002). Stress, maternal depression, and negative mother–infant interactions in relation to infant attachment. *Infant Mental Health Journal*, 23, 145–163.
- Cuccaro, M. L., Holmes, G. R., & Wright, H. H. (1993). Behavior problems in preschool children: A pilot study. *Psychological Reports*, 72(1), 121–122.
- Daubert v. Merrell Dow Pharmaceuticals, Inc.*, 509 U.S. 579 (1993).
- Deater-Deckard, K., & Scarr, S. (1996). Parenting stress among dual-earner mothers and fathers: Are there gender differences? *Journal of Family Psychology*, 10(1), 45–59.
- de Brock, A. J. L. L., Vermulst, J. R. M., Gerris, J. R. M., & Abidin, R. R. (1992). *Nijmeegse Ouderlijke Stress Index*. Lisse, Netherlands: Swets & Zeitlinger.
- DeGarmo, D. S., & Forgatch, M. S. (1999). Contexts as predictors of changing maternal parenting practices in diverse family structures: A social interactional perspective of risk and resilience. In E. M. Hetherington (Ed.), *Coping with divorce, single parenting, and remarriage* (pp. 227–252). Mahwah, NJ: Lawrence Erlbaum.
- Degroat, J. S. (2003). Parental stress and emotion attributions as correlates of maternal positive affect and sensitivity during interaction with young children. *Dissertation Abstracts International: Section B: The Sciences and Engineering*, 64(5-B), 2383.
- Diener, M. L., Nievar, M. A., & Wright, C. (2003). Attachment security among mothers and their young children living in poverty: Associates with maternal, child and contextual characteristics. *Merrill-Palmer Quarterly*, 49(2), 154–182.
- Donenberg, G., & Baker, B. L. (1993). The impact of young children with externalizing behaviors on their families. *Journal of Abnormal Child Psychology*, 21(2), 179–198.
- Dyer, F. J. (1999). *Psychological consultation in parental rights cases*. New York: Guilford Press.
- Ekman, P., & O'Sullivan, M. (1991). Who can catch a liar? *American Psychologist*, 46, 913–920.
- Emery, R. E., & Forehand, R. (1994). Parental divorce and children's well-being: A focus on resilience. In R. J. Haggerty, L. R. Sherrod, N. Garnezy, & M. Rutter (Eds.), *Stress, risk, and resiliency in children and adolescents* (pp. 64–99). Cambridge: Cambridge University Press.
- Emery, R. E., Otto, R. K., & O'Donohue, W. T. (2005). A critical assessment of child custody evaluations. *Psychological Science in the Public Interest*, 6(1), 1–29.
- Endriga, M. C., Jordan, J. R., & Speltz, M. L. (2003). Emotion self-regulation in preschool-aged children with and without orofacial clefts. *Journal of Developmental and Behavioral Pediatrics*, 24(5), 336–344.
- Éthier, L. S., & LaFreniere, P. J. (1993). Le stress des mères monoparentales en relation avec l'agressivité de l'enfant d'âge préscolaire. *Journal International de Psychologie*, 38, 273–289.
- Exner, J. E., McDowell, E., Pabst, J., Stackman, W., & Kirk, L. (1963). On the detection of willful falsifications in the MMPI. *Journal of Consulting Psychology*, 27(1), 91–94.
- Fagan, J., Bernd, E., & Whiteman, V. (2007). Adolescent fathers' parenting stress, social support, and involvement with infants. *Journal of Research on Adolescence*, 17(1), 1–22.
- Feinfield, K. A., & Baker, B. L. (2004). Empirical support for a treatment program for families of young children with externalizing problems. *Journal of Clinical Child and Adolescent Psychology*, 33(1), 182–195.

- Feldman, M. A., Varghese, J., Ramsay, J., & Rajska, D. (2002). Relationships between social support, stress and mother-child interactions in mothers with intellectual disabilities. *Journal of Applied Research in Intellectual Disabilities*, 15(4), 314-323.
- Fite, P. J., Greening, L., & Stoppelbein, L. (2008). Relation between parenting stress and psychopathic traits among children. *Behavioral Sciences and the Law*, 26(2), 239-248.
- Fla. Fam. L.R.P. Rule 12.363 Evaluation of Minor Child (West, 2005).
- Flens, J. R. (2004, October). *Advanced Institute: Psychological Testing in Child Custody*. Paper presented at the Association of Family and Conciliation Courts, 6th International Symposium on Child Custody Evaluations. Nashville, TN.
- Florsheim, P., Sumida, E., McCann, C., Winstanley, M., Fukui, R., Seefeldt, T., & Moore, D. (2003). The transition to parenthood among young African-American and Latino couples: Relational predictors of risk for dysfunction. *Journal of Family Psychology*, 17(1), 65-79.
- Frank, S. J., Olmsted, C. L., Wagner, A. E., Lamb, C. C., Freeark, K., Breitzer, G. M., & Peters, J. M. (1991). Child illness, the parenting alliance, and parenting stress. *Journal of Pediatric Psychology*, 16(3), 361-371.
- Frankel, K. K., & Harmon, R. J. (1996). Depressed mothers: They don't always look as bad as they feel. *Journal of the American Academy of Child and Adolescent Psychiatry*, 35(3), 289-298.
- Friars, P. M., & Mellor, D. J. (2007). Drop out from behavioral management training programs for ADHD: A prospective study. *Journal of Child and Family Studies*, 16(3), 427-441.
- Friedman, A. F., Lewak, R., Nichols, D. S., & Webb, J. T. (2001). *Psychological assessment with the MMPI-2*. Mahwah, NJ: Lawrence Erlbaum.
- Galatzer-Levy, R. M., Kraus, L., & Galatzer-Levy, J. (Eds.). (2009). *The scientific basis of child custody decisions* (2nd ed.). New York: John Wiley.
- Gelfand, D. M., Teti, D. M., & Fox, C. E. (1992). Sources of parenting stress for depressed and nondepressed mothers of infants. *Journal of Clinical Child Psychology*, 21, 262-272.
- Gerson, A. D., Furth, S. L., Neu, A. M., & Fivush, B. A. (2004). Assessing associations between medication adherence and potentially modifiable psychosocial variables in pediatric kidney transplant recipients and their families. *Pediatric Transplantation*, 8, 543-550.
- Goldberg, S., Janus, M., Washington, J., Simmons, R. J., MacLusky, I., & Fowler, R. S. (1997). Prediction of preschool behavioral problems in healthy and pediatric samples. *Journal of Developmental and Behavioral Pediatrics*, 18(5), 304-313.
- Goodman-Delahunty, J. (1997). Forensic psychological expertise in the wake of *Daubert*. *Law and Human Behavior*, 21, 121-140.
- Gould, J. W. (2006). *Conducting scientifically crafted child custody evaluations* (2nd ed.). New York: Guilford Press.
- Gould, J. W., & Martindale, D. A. (2007). *The art and science of child custody evaluations*. New York: Guilford Press.
- Gould, J. W., & Martindale, D. A. (2005). A second call for clinical humility and judicial vigilance. *Family Court Review*, 43, 246-252.
- Grisso, T. (1986). *Evaluating competencies: Forensic assessments and instruments*. New York: Plenum.
- Grossman, J., & Shigaki, I. S. (1994). Investigation of familial and school-based risk-factors for Hispanic Head Start children. *American Journal of Orthopsychiatry*, 64(3), 456-467.
- Guidubaldi, J., & Cleminshaw, H. K. (1994). *Parenting satisfaction scale*. San Antonio, TX: Psychological Corporation.
- Guidubaldi, J., Perry, J. D., & Nastasi, B. K. (1987). Growing up in a divorced family: Initial and long-term perspective on children's adjustment. *Annual Review of Applied Social Psychology*, 7, 202-237.
- Guralnick, M. J., Hammond, M. A., Neville, B., & Connor, R. T. (2008). The relationship between sources and functions of social support and dimensions of child- and parent-related stress. *Journal of Intellectual Disability Research*, 52(12), 1138-1154.
- Ha, E. H., Oh, K. J., & Kim, E. J. (1999). Depressive symptoms and family relationship of married women: Focused on parenting stress and marital dissatisfaction. *Korean Journal of Clinical Psychology*, 18(1), 79-93.
- Hall, S., & Marteau, T. M. (2003). Causal attributions and blame: associations with mothers' adjustment to the birth of a child with Down syndrome. *Psychology, Health and Medicine*, 8(4), 415-423.
- Hauenstein, E., Scarr, S., & Abidin, R. R. (1987). Detecting children at-risk for developmental delay: Efficacy of the Parenting Stress Index in a non-American culture. Unpublished manuscript, University of Virginia, Charlottesville.

- Heilbrun, K. (1992). The role of psychological testing in forensic assessment. *Law and Human Behavior*, 16, 257–272.
- Heilbrun, K. (2001). *Principles of forensic mental health assessment*. New York: Kluwer Academic/Plenum Publishers.
- Heilbrun, K., Warren, J., & Picarello, K. (2003). Third party information in forensic assessment. In A. Goldstein (Ed.), *Handbook of psychology*. Vol. 11: *Forensic psychology* (pp. 69–86). Hoboken, NJ: John Wiley.
- Hetherington, E. M., Bridges, M., & Insabella, G. M. (1998). What matters? What does not? Five perspectives on the association between marital transitions and children's adjustment. *American Psychologist*, 53, 167–184.
- Hetherington, E. M., & Kelly, J. (2002). *Divorce reconsidered: For better or for worse*. New York: W. W. Norton.
- Hipke, K. N., Wolchik, S. A., Sandler, I. N., & Braver, S. L. (2002). Predictors of children's intervention-induced resilience in a parenting program for divorced mothers. *Family Relations*, 51, 121–129.
- Hitchcock, D. L. (2003). A construct validity study of the Marschank Interaction Method Rating System with adolescent mother-child dyads. *Dissertation Abstracts International: Section B: The Sciences and Engineering*, 64(6-B), 2977.
- Hughes, J. C., Brestan, E. V., Christens, B. D., Klinger, L. J., & Valle, L. A. (2004). Problem-solving interactions between mothers and children. *Child and Family Behavior Therapy*, 26(1), 1–16.
- Hutcheson, J. J., & Black, M. M. (1996). Psychometric properties of the Parenting Stress Index in a sample of low-income African-American mothers of infants and toddlers. *Early Education and Development*, 7(4), 381–400.
- Innocenti, M. S., Huh, K., & Boyce, G. C. (1992). Families of children with disabilities: Normative data and other considerations of parenting stress. *Topics in Early Childhood Special Education*, 12(3), 403–427.
- Insabella, G. M., Williams, T., & Pruett, M. K. (2003). Individual and coparenting differences between divorcing and unmarried fathers: Implications for family court services. *Family Court Review*, 41, 290–306.
- Irvin, J. R., Carter, A. S., & Briggs-Gowan, M. J. (2002). The social-emotional development of "late-talking" toddlers. *Journal of the American Academy of Child and Adolescent Psychiatry*, 41(11), 1324–1332.
- Ispa, J. M., Fine, M. A., Halgunseth, L. C., Harper, S., Robinson, J., Boyce, L., . . . Brady-Smith, C. (2004). Maternal intrusiveness, maternal warmth, and mother-toddler relationship outcomes: Variations across low-income ethnic and acculturation groups. *Child Development*, 75(6), 1613–1631.
- Jackson, A. P., Gyamfi, P., Brooks-Gunn, J., & Blake, M. (1998). Employment status, psychological well-being, social support, and physical discipline practices of single black mothers. *Journal of Marriage and the Family*, 60(4), 894–902.
- Jameson, B. J., Ehrenberg, M. F., & Hunter, M. A. (1997). Psychologists' ratings of the best-interests-of-the-child custody and access criterion: A family systems assessment model. *Professional Psychology: Research and Practice*, 28(3), 253–262.
- Jarvis, P. A., & Creasey, G. L. (1991). Parental stress, coping, and attachment in families with an 18-month-old infant. *Infant Behavior and Development*, 14(4), 383–395.
- Judge, S. (2003). Determinants of parental stress in families adopting children from Eastern Europe. *Family Relations: Interdisciplinary Journal of Applied Family Studies*, 52(3), 241–248.
- Karp, J., Serbin, L. A., Stack, D. M., & Schwartzman, A. E. (2004). An observational measure of children's behavioural style: Evidence supporting a multi-method approach to studying temperament. *Infant and Child Development*, 13(2), 135–158.
- Kazdin, A. E. (1990). Premature termination from treatment among children referred for antisocial behavior. *Journal of Child Psychology and Psychiatry and Allied Disciplines*, 31(3), 415–425.
- Kazdin, A. E. (1995). Child, parent and family dysfunction as predictors of outcome in cognitive-behavioral treatment of antisocial children. *Behaviour Research and Therapy*, 33(3), 271–281.
- Kazdin, A. E., & Wassell, G. (1999). Barriers to treatment participation and therapeutic change among children referred for conduct disorder. *Journal of Clinical Child Psychology*, 38, 1051–1062.
- Kazdin, A. E., Mazurick, J. L., & Bass, D. (1993). Risk for attrition in treatment of antisocial children and families. *Journal of Clinical Child Psychology*, 22(1), 2–16.
- Kazui, M., Muto, T., & Sonoda, N. (1996). The roles of marital quality and parenting stress in mother-preschooler relationships. [Japanese]. *Japanese Journal of Developmental Psychology*, 7(1), 31–40.
- Kelley, S. J. (1993). Caregiver stress in grandparents raising grandchildren. *Journal of Nursing Scholarship*, 25(4), 331–337.
- Kelly, J. B. (2000). Children's adjustment in conflicted marriage and divorce: A decade of review of research. *Journal of the American Academy of Child and Adolescent Psychiatry*, 39, 963–973.

- Kelly, J. B., & Emery, R. (2003). Children's adjustment following divorce: Risk and resilience perspectives. *Family Relations*, 52, 352–362.
- Kelly, R. F., & Ramsey, S. H. (2009). Child custody guidelines: The need for systems-level outcome assessments. *Family Court Review*, 47(2), 286–303.
- Kirkpatrick, H. D., Austin, W. G., & Flens, J. R. (2011). Psychological and legal considerations in reviewing the work product of a colleague in child custody evaluation. *Journal of Child Custody: Research, Issues, and Practices*, 8(1), 103–123.
- Kitson, G. C., & Morgan, L. A. (1991). The multiple consequences of divorce: A decade review. *Journal of Marriage and the Family*, 52, 913–924.
- Krauss, D. A., & Sales, B. D. (2000). Legal standards, expertise, and experts in the resolution of contested child custody cases. *Psychology, Public Policy and Law*, 6, 843–879.
- Krauss, M. W. (1993). Child-related and parenting stress: Similarities and differences between mothers and fathers of children with disabilities. *American Journal of Mental Retardation*, 97(4), 393–404.
- Kuehnle, K. (1996). *Assessing allegations of child sexual abuse*. Sarasota, FL: Professional Resources Press.
- Kuehnle, K. (1998). Child sexual abuse evaluations: The scientist-practitioner model. *Behavioral Sciences and the Law*, 16, 5–20.
- Kuehnle, K., Coulter, M., & Firestone, G. (2000). Child protection evaluations: The forensic stepchild. *Family and Conciliation Courts Review*, 38, 368–391.
- Kuehnle, K., & Drozd, L. (Eds.) (2012). *Parenting plan evaluations: Applied research for the family court*. New York: Oxford University Press.
- Lacharité, C., Éthier, L., & Piché, C. (1992). Le stress parental chez les mères d'enfants d'âge préscolaire: Validation et norms québécoises pour l'Inventaire de Stress Parental. *Santé Mentale au Québec*, 17(2), 183–203.
- Laganière, J., Tessier, R., & Nadeau, L. (2003). Mother–infant attachment and prematurity: A link mediated by maternal perceptions. *Enfance*, 55(2), 101–117.
- Lamb, M. E. (2002). Placing children's interests first. *Virginia Journal of Social Policy and the Law*, 10, 98–119.
- Lanyon, R. I. (2004). Favorable self-presentation on psychological inventories: An analysis. *American Journal of Forensic Psychology*, 22(1), 53–65.
- LeCuyer-Maus, E. A. (2003). Stress and coping in high-risk mothers: Difficult life circumstances, psychiatric-mental health symptoms, education, and experiences in their families of origin. *Public Health Nursing*, 20(2), 132–145.
- Levendosky, A. A., & Graham-Bermann, S. A. (1998). The moderating effects of parenting stress on children's adjustment in woman-abusing families. *Journal of Interpersonal Violence*, 13(3), 383–397.
- Lyons-Ruth, K., Lyubchik, A., Wolfe, R., & Bronfman, E. (2002). Parental depression and child attachment: Hostile and helpless profiles of parent and child behavior among families at risk. In S. H. Goodman and I. H. Gotlib (Eds.), *Children of depressed parents: Mechanisms of risk and implications for treatment* (pp. 89–120). Washington, DC: American Psychological Association.
- Martin, *In re Marriage of*. 42 P.3d 75 (Colo. App. 2002).
- Martindale, D. A. (2005). Confirmatory bias and confirmatory distortion. *Journal of Child Custody*, 2, 33–50.
- Martindale, D. A., & Gould, J. W. (2004). The forensic model: Ethics and scientific methodology applied to custody evaluations. *Journal of Child Custody*, 1, 1–22.
- Martindale, D. M., Martin, L., Austin, W. G. et al. (2007). Model standards of practice for child custody evaluation. *Family Court Review*, 45(1), 70–91.
- Mash, E. J., Johnston, C., and Kovitz, K. (1983) A comparison of the mother–child interactions of physically abused and non-abused children during play and task situations. *Journal of Clinical Child Psychology*, 12, 337–346.
- McBride, B. (1989). Stress and fathers' parental competence: Implications for family life and parent educators. *Family Relations*, 38, 385–389.
- McCann, J. T., Flens, J. R., Campagna, V., Collman, P., Lazzaro, T., & Connor, E. (2001). The MCMI–III in child custody evaluations: A normative study. *Journal of Forensic Psychology Practice*, 1, 27–44.
- McLanahan, S., & Sandefur, G. (1994). *Growing up with a single parent: What hurts, what helps*. Cambridge, MA: Harvard University Press.
- McLanahan, S., & Teitler, J. (1999). The consequences of father absence. In M. E. Lamb (Ed.), *Parenting and child development in nontraditional families* (pp. 83–102). Mahwah, NJ: Lawrence Erlbaum.

- Meissner, C. A., & Kassen, S. M. (2002). "He's guilty!": Investigator bias in judgments of truth and deception. *Law and Human Behavior*, 26, 469–480.
- Mellins, C. A., Kang, E., Leu, C. S., Havens, J. F., & Chesney, M. A. (2003). Longitudinal study of mental health and psychosocial predictors of medical treatment adherence in mothers living with HIV disease. *AIDS Patients Care and STDs*, 7(8), 407–416.
- Melton, G. B., Petrila, J., Poythress, N. G., & Slobogin, C. (1997). *Psychological evaluations for the courts* (34d ed.). New York: Guilford Press.
- Milgrom, J., & McCloud, P. (1996). Parenting stress and postnatal depression. *Stress Medicine*, 12(3), 177–186.
- Moore, C. G., Probst, J. C., Tompkins, M., Cuffe, S., & Martin, A. B. (2007). The prevalence of violent disagreements in US families: Effects of residence, race/ethnicity, and parental stress. *Pediatrics*, 119, 68–76.
- Moran, G., Pederson, D. R., Pettit, P., & Krupka, A. (1992). Maternal sensitivity and infant mother attachment in a developmentally delayed sample. *Infant Behavior and Development*, 15(4), 427–442.
- Moss, E., Cyr, C., & Dubois-Comtois, K. (2004). Attachment at early school age and developmental risk. *Developmental Psychology*, 40(4), 519–532.
- Murphy, K. R., & Barkley, R. A. (1996). Parents of children with attention-deficit hyperactivity disorder: Psychological and attentional impairment. *American Journal of Orthopsychiatry*, 66(1), 93–102.
- Nelson, J. R., Stage, S., Duppong-Hurley, K., Synhorst, L., & Epstein, M. H. (2007). Risk factors predictive of the problem behavior of children at risk for emotional and behavioral disorders. *Exceptional Children*, 73(3), 367–379.
- Nichols, D. S., & Greene, R. L. (1997). Dimensions of deception in personality assessment: The example of the MMPI-2. *Journal of Personality Assessment*, 68, 251–266.
- O'Donohue, W., & Bradley, A. R. (1999). Conceptual and empirical issues in child custody evaluations. *Clinical Psychology: Science and Practice*, 6, 310–322.
- Onufrak, B., Saylor, C. F., Taylor, M. J., Eyberg, S. M., & Boyce, G. C. (1995). Determinants of responsiveness in mothers of children with intraventricular hemorrhage. *Journal of Pediatric Psychology*, 20(5), 587–599.
- Ostroff, C. (1993). Comparing correlations based on individual-level and aggregate data. *Journal of Applied Psychology*, 78, 569–582.
- Otto, R. K. (2002). Use of the MMPI-2 in forensic settings. *Journal of Forensic Psychology Practice*, 2, 71–91.
- Otto, R. K., & Collins, R. P. (1995). Use of the MMPI-2/MMPI-A in child custody evaluations. In Y. S. Ben-Porath, J. R. Graham, G. C. N. Hall, R. D. Hirschman, & M. S. Zaragoza (Eds.), *Forensic applications of the MMPI-2* (pp. 222–252). Thousand Oaks, CA: Sage.
- Otto, R. K., Edens, J. F., & Barcus, E. H. (2000). The use of psychological testing in child custody evaluations. *Family and Conciliation Courts Review*, 38(3), 312–340.
- Paulhus, D. L. (1998). *Paulhus Deception Scales (PDS): The Balanced Inventory of Desirable Responding-7*. North Tonawanda, NY: MHS.
- Pett, M. A., Wampold, B. E., Turner, C. W., & Vaughan-Cole, B. (1999). Paths of influence of divorce and preschool children's psychosocial adjustment. *Journal of Family Psychology*, 13, 145–164.
- Pruett, K. D., & Pruett, M. K. (1999). "Only God decides": Young children's perceptions of divorce and the legal system. *Journal of the American Academy of Child and Adolescent Psychiatry*, 38, 1544–1550.
- Pruett, M. K., Williams, T. Y., Insabella, G., & Little, T. D. (2003). Family and legal indicators of child adjustment to divorce among families with young children. *Journal of Family Psychology*, 17, 169–180.
- Ramsey, S. H., & Kelley, R. F. (2004). Social science knowledge in family law cases: Judicial gate-keeping in the *Daubert* era. *University of Miami Law Review*, 59, 1–81.
- Reitman, D., Currier, O. R., & Stickle, T. R. (2002). A critical evaluation of the Parenting Stress Index-Short Form (PSPI-SF) in a Head Start population. *Journal of Clinical Child and Adolescent Psychology*, 31(3), 384–392.
- Repetti, R. L., & Wood, J. (1997). Families accommodating to chronic stress: Unintended and unnoticed processes. In B. H. Gottlieb (Ed.), *Coping with chronic stress* (pp. 191–220). New York: Plenum.
- Robinson, W. S. (1950). Ecological correlations and the behavior of individuals. *American Sociological Review*, 15, 351–357.
- Rodriguez, C. M., & Green, A. J. (1997). Parenting stress and anger expression as predictor of child abuse potential. *Child Abuse and Neglect*, 21(4), 367–377.
- Rogers, M. A., Wiener, J., Marton, I., & Tannock, R. (2009). Supportive and controlling parental involvement as predictors of children's academic achievement: Relations to children's ADHD symptoms and parenting stress. *School Mental Health*, 1, 89–102.

- Sanders, M. R., & McFarland, M. (2000). Treatment of depressed mothers with disruptive children: A controlled evaluation of cognitive behavioral family intervention. *Behavior Therapy*, 31, 89–112.
- Saylor, C. F., Boyce, G. C., & Price, C. (2003). Early predictors of school-age behavior problems and social skills in children with intraventricular hemorrhage (IVH). *Child Psychiatry and Human Development*, 33(3), 175–192.
- Schiller, E. W. (2003). Gender differences in coping and stress among parents of children with complex congenital heart disease (CHD). Dissertation. *University Microfilm International Section B: The Sciences and Engineering*, 64(5-B), 2404.
- Seo, S. (2004). Maternal self-efficacy, quality of parenting, and child development outcome among mothers with young children from early head start. Dissertation. University Microfilm International.
- Sheras, P. L., Abidin, R. R., & Konold, T. R. (1998). *Stress Index for Parents of Adolescents, professional manual*. Lutz, FL: Psychological Assessment Resources.
- Shuman, D. W. (2002). The role of mental health experts in custody decisions: Science, psychological tests, and clinical judgment. *Family Law Quarterly*, 36, 135–162.
- Solis, M. L., & Abidin, R. R. (1991). The Spanish version Parenting Stress Index: A psychometric study. *Journal of Clinical Child Psychology*, 20(4), 372–378.
- Strong, D. R., Greene, R. L., Hoppe, C., Johnston, T., & Olesen, N. (1999). Taxometric analysis of impression management and self-deception on the MMPI-2 in child-custody litigation. *Journal of Personality Assessment*, 73(1), 1–18.
- Strong, D. R., Greene, R. L., & Kordinak, S. T. (2002). Taxometric analysis of impression management and self-deception in college student and personnel evaluation settings. *Journal of Personality Assessment*, 78(1), 161–175.
- Tam, K.-K., Chan, Y.-C., and Wong, C.-K. M. (1994). Validation of the Parenting Stress Index among Chinese mothers in Hong Kong. *Journal of Community Psychology*, 22(3), 211–223.
- Taylor, C. A., Guterma, N. B., Lee, S. J., & Rathouz, P. J. (2009). Intimate partner violence, maternal stress, nativity, and risk for maternal maltreatment of young children. *American Journal of Public Health*, 99, 175–183.
- Teti, D. M., Nakagawa, M., Das, R., & Wirth, O. (1991). Security of attachment between preschoolers and their mothers: Relations among social interaction, parenting stress, and mother's sorts of the Attachment Q-Set. *Developmental Psychology*, 27(3), 440–447.
- Tippins, T., & Wittman, J. (2005). Empirical and ethical problems with custody recommendations. *Family Court Review*, 43, 193–222.
- Tomanik, S., Harris, G. E., & Hawkins, J. (2004). The relationship between behaviours exhibited by children with autism and maternal stress. *Journal of Intellectual and Developmental Disability*, 29(1), 16–26.
- Trute, B. (2003). Grandparents of children with developmental disabilities: Intergenerational support and family well-being. *Families in Society*, 84(1), 119–126.
- Uniform Marriage and Divorce Act. (1979). *Uniform Laws Annotated*, 9A.
- U.S. Census Bureau. (2007). Current population survey, March 2007. Washington, DC: U.S. Department of Commerce.
- Waisbren, S. E., Albers, S., Amato, S., Ampola, M., Brewster, T. G., Demmer, L., . . . Levy, H. L. (2003). Effect of expanded newborn screening for biochemical genetic disorders on child outcomes and parental stress. *JAMA: Journal of the American Medical Association*, 290(19), 2564–2572.
- Wakschlag, L. S., & Keenan, K. (2001). Clinical significance and correlates of disruptive behavior in environmentally at-risk preschoolers. *Journal of Clinical Child Psychology*, 30(1), 262–275.
- Webster-Stratton, C. (1988). Mothers' and fathers' perceptions of child deviance: Roles of parent and child behaviors and parent adjustment. *Journal of Consulting and Clinical Psychology*, 56(6), 909–915.
- Weiss, J. A., Sullivan, A., & Diamond, T. (2003). Parent stress and adaptive functioning of individuals with developmental disabilities. *Journal of Developmental Disabilities*, 10(1), 129–135.
- Wilfong, E. W., Saylor, C., & Elksnin, N. (1991). Influences on responsiveness: Interactions between mothers and their premature infants. *Infant Mental Health Journal*, 12, 31–40.
- Willner, P., & Goldstein, R. C. (2001). Mediation of depression by perceptions of defeat and entrapment in high-stress mothers. *British Journal of Medical Psychology*, 74(4), 473–485.
- Wong, D. F., Lam, D. O., & Kwoak, S. Y. (2003). Stresses and mental health of fathers with younger children in Hong Kong: Implications for social work practices. *International Social Work*, 46(1), 103–119.
- To obtain a more complete topically arranged PSI research reference list readers are encouraged to access http://www4.parinc.com/webuploads/productinfo/PSI_Bibliography.pdf.

AUTHOR INDEX

- Abidin, R. R. 10, 17, 346, 347, **349**, 350, 351, 352, 355, 356, 357, 368, 370
- Abracen, J. 244, 247, 250
- Abram, K. M. 273
- Abramowitz, C. S. 240, 243, 272
- Achenbach, T. M. 10, 17, 315, 316, 319, 324, 325, 326, 327, 328, 329, 330, 331, 332, 333, 335, 337
- Acheson, S. K. 230
- Ackerman, M. C. 9
- Ackerman, M. J. 9, 216
- Ackerman, R. J. 189
- Ackerman, S. 212
- Acklin, M. W. 204, 207, 212
- Adams, N. 35
- Adams, R. E. 79
- af Klinteberg, B. 273
- Aguerrevereab, L. E. 300
- Aguila-Puentes, G. 301
- Ahlmeyer, S. 189
- Aiduk, R. 28, 29
- Aikman, G. G. 301
- Ainsworth, M. D. 204
- Akenson, S. T. 75, 92
- Akerlind, I. 29
- Albaugh, B. 34
- Albers, J. W. 297
- Albers, S. 357
- Aldenhoff, J. B. 274
- Alexander, P. C. 188
- Allain, N. 31
- Allan, A. 31
- Allan, M. M. 31
- Allen, J. G. 188
- Allen, L. C. 238
- Allgeier, L. M. 75, 89
- Allison, J. A. 359
- Alm, P.-O. 273
- Almagor, M. 24, 28
- Alnaes, R. **186**
- Alperin, J. J. 116, 120, 121
- Alterman, A. I. 146, 153, 158
- Amato, P. R. 365
- Amato, S. 357
- Amenta, A. E. 246
- Ampola, M. 357
- Anastasi, A. 235
- Anastopoulos, A. D. 357, 358
- Anckarsäter, H. 238, 242
- Andershed, H. A. 268, 269, 270, 271, 282, 283
- Anderson, C. L. 188
- Anderson, D. 302
- Anderson, G. L. 114
- Anderson, J. L. 17, 38, 75, 89
- Anderson, R. W. 30
- Anderson, S. V. 41
- Anderson, T. W. 79
- Andrews, D. A. 244, 277, 302
- Andrews, P. 80
- Andronikof-Sanglade, A. 211
- Anestis, J. C. 34
- Angleitner, A. 332, 335
- Angold, A. 124
- Anthony, C. 245–6
- Antonius, D. 162
- Appelbaum, P. S. 238, 242, 296
- Arbisi, P. A. 22, 23, 26, 27, 28, 29, 30, 32, 33, 34, 43, 75, 78, 88, 92
- Arbisiab, P. A. 300
- Archer, E. M. 75, 92
- Archer, R. P. 7, 8, 9, 10, 13, 27, 28, 29, 30, 39, 75, 87, 92, 110, 111, 112–13, 114, 115–16, 117, 120, 121, 122, 123, 124, 125, 126, 127, 128, 130, 132, 134, 208, 332

- Ardolf, B. R. 32
 Arend, R. **331**
 Argeriou, M. 153
 Armon, Z. 215
 Armstrong, J. 214, 215
 Arnau, R. C. 27
 Arnett, J. J. 282
 Arnold, G. 298, **299**
 Arsenault, L. 215
 Arvey, R. D. 30
 Ash, P. 363
 Atlis, M. M. 23, 33
 Auble, P. M. 148, 155
 Audenaert, K. 242
 Austin, K. D. 13
 Austin, W. G. 360, 361, 362, 363, 366
 Avery, R. D. 30
 Axelrod, B. N. 301, 302
 Aycock, R. 295
 Ayearst, L. E. 75
 Ayoub, C. 358
 Azrin, R. 302
- Baade, L. E. 89
 Babiak, P. 230, 231, 238
 Babikian, T. **298**
 Bacchiochi, J. R. 149, 150
 Baer, R. A. 22, 27, 28, 33, 117
 Bagby, M. R. 38, 149, 150
 Bagby, R. M. 32, 33, 39, 75, 77, 87, 88, 90, 92, 149, 369
 Bagby, R. 41
 Bagge, C. 152
 Bagley, C. 358
 Bagner, D. M. 355
 Bailey, A. 212
 Baity, M. R. 208
 Baker, B. L. 355, 357
 Baker, D. B. 354
 Baker, E. M. 121, 126
 Baker, M. D. 270
 Baker, W. J. 301
 Bale, R. 239
 Balfour, J. F. 29
 Ball, J. C. 127
 Ball, S. A. 231
 Ballenger, J. 117
 Ban, T. A. 151
 Banks, S. 238, 242, 296
 Bansal, A. 34, 35
 Barbaree, H. E. 244
 Barbareschi, G. 272
 Barcus, E. H. 7
 Bard, D. E. 276
 Barden, R. C. 205, 209
 Barefoot, J. C. 29
 Barendregt, M. 37
- Barese, T. A. **268**, 272
 Barkham, M. 152
 Barkley, R. A. 124, 356, 357, 358
 Barnett, D. W. 358
 Barnum, R. 5
 Barr, K. N. 243
 Barrash, J. 298, 299, 301, 302
 Barry, C. T. 277
 Bartel, P. 10, 277
 Barth, J. T. 296
 Barthlow, D. L. 28, 34
 Bartol, C. R. 30
 Bashem, J. R. 302
 Baskin-Sommers, A. R. 231
 Baskin-Sommers, A. 231
 Bass, D. 357
 Bassett, G. R. 30
 Bates, G. W. **184**
 Bathurst, K. 39
 Bauer, D. L. 272, 274
 Baughman, B. C. 302
 Baum, L. J. 13, 39, 92
 Baxter, D. 207
 Bayon, C. 184, **185**
 Bebbington, P. 238
 Beck, A. G. 204
 Beck, A. T. 10, 151, 154, 330
 Beck, S. J. 204, 357
 Becker, B. 152
 Beckett, R. 189
 Beckham, J. C. 149, 150, 151
 Beckman, P. J. 354
 Beech, A. R. 189
 Belevich, J. K. S. 117
 Bell, K. 153
 Bell, M. J. 152
 Bellak, L. 9
 Bellak, S. S. 9
 Bell-Pringle, V. J. 152
 Ben-Porath, Y. S. 9, 10, 22, 23, 24, 26, 27, 28, 29, 30, 32, 33, 34, 35, 38, 39, 40, 41, 42, 43, 63, 69, 70, 71, 72, 73, 74, 75, 77, 80, 81, 83, 85, 86, 87, 88, 90, 91, 92, 101, 113, 114, 116, 120, 121, 123, 124, 132, **133**
 Bendell, R. D. 356
 Bender, L. 7
 Bender, S. D. 296, 297, 298, 299
 Benedek, D. M. 148, 155
 Benedict, K. **185**
 Benjamin, L. S. **184**
 Benn, A. 218
 Bennett, B. 144
 Benton, A. L. 301
 Berg, E. A. 7
 Berger, K. 238, 244
 Bergman, M. A. 291
 Berliner, L. 282

- Bernard, L. C. 302
 Bernd, E. 358
 Bernet, W. 148, 155
 Bernfeld, G. A. 242
 Bernstein, I. H. 30
 Berrios, G. E. 231
 Berrol, S. 79
 Berry, D. R. 27
 Berry, D. T. R. 22, 33, 75, 89, 117, 208, 297, 298, 299, 301
 Berry, T. R. 296, 298, 299, 300, 301, 302
 Bertinetti, J. F. 117
 Beutler, L. E. 30, 211, 215
 Bhavnagri, N. P. 356
 Bianchini, K. J. 33, 89, 157, **298**
 Bianchini, K. 301
 Bianchiniab, K. J. 300
 Bickford, M. E. 242
 Bigras, M. 352, 357
 Bilginer, L. 212
 Billington, R. 152
 Binder, L. M. 32
 Binns, H. J. **331**
 Birt, A. R. 242
 Bissada, H. 151, 162
 Bivens, A. 187
 Bjurulf, P. 29
 Blacher, J. 355
 Black, J. D. 27, 41
 Black, K. 111
 Black, M. M. 351–2, 355, 356
 Black, M. S. 41
 Black, M. 358
 Black, P. J. 17
 Blackburn, R. 182, 189, 242
 Blackwood, N. 230, 250
 Blagys, M. 212
 Blair, G. 35
 Blair, R. J. R. 283
 Blais, M. A. 162, 185, 208
 Blake, M. 357, 358
 Blanchard, D. D. 150
 Blasczyk-Schiep, S. 208
 Blashfield, R. K. 152
 Blatt, S. J. 211
 Blüm, M. 330
 Block, A. 83
 Blom, B. E. 291
 Blonigen, D. M. 282–3, 282
 Blood, L. 13
 Bloom, L. J. 189
 Blud, L. 244
 Blumenthal, J. A. 29
 Bobadilla, L. 283
 Boccacini, M. T. 8, 9, 30, 155, 160, 248, 280, 281
 Bodin, S. D. 277
 Boer, D. P. 152, 242, 243
 Boes, J. L. **182**
 Bohn, M. J. 39, 41
 Bolinskey, P. K. 120, 121, 124, 125, **132**
 Bolt, D. M. 235, 239
 Bolton, C. 357
 Bonieske, L. M. 184, **186**
 Bonnie, R. J. 37
 Bonta, J. 244, 302
 Book, A. S. 17, 148, 243, 244, 248, 270, 279, 280, 283
 Boomer, D. 152
 Boone, K. B. 296, 297, 298, 299, 301
 Booth, C. 366
 Boothby, J. L. 8
 Bor, R. 149, 150
 Bornstein, R. F. 191, 208, 218
 Bornstein, R. J. 208
 Borum, R. 7, 8–9, 10, 145, 189, 277
 Bosshardt, M. J. 30
 Bouchard, T. J. 282
 Boudewyns, P. **182**
 Boutacoff, L. I. 29
 Bow, J. N. 10, 14, 189, 191, 359, 362
 Bownas, D. A. 30
 Boyce, G. C. 352, 356, 357
 Boyce, L. 357
 Boyd, S. 187, 188
 Boyle, G. J. 146, 147, 151, 153, 158
 Bradley, A. R. 360
 Brady-Smith, C. 357
 Bram, A. D. 211
 Bramati, I. E. 247
 Bramlett, R. K. 358
 Brand, B. 188
 Brandenburg, N. 185, **186**
 Brandsma, J. 187, 188
 Brandt, J. R. 267, 275
 Brandt, J. 151
 Brannen, D. N. 269, 275
 Braver, S. L. 357
 Breen, M. J. 356, 357, 358
 Breitzer, G. M. 354, 358
 Brenowitz, L. H. 125
 Bresolin, L. **182**
 Brestan, E. V. 358
 Brewster, T. G. 357
 Brickenkamp, R. **298**
 Bricklin, B. 9
 Bridenstine, M. P. 29
 Bridges, M. 366
 Briggs, P. F. 110, 111, 120, 124
 Briggs-Gowan, M. J. 330, 358
 Brink, J. 296
 Brinkley, C. A. 239, 241
 Brockman, C. J. 89
 Brodsky, S. L. 9
 Brody, P. 125

- Brodzinsky, D. M. 361, 366
 Bronfinan, E. 357
 Brooks-Gunn, J. 357, 358
 Brousseau, J. D. 30
 Brown, J. S. 34
 Brown, L. S. Jr. 146, 153, 158
 Brown, R. C. 152
 Brown, S. L. 238
 Brownlee, K. 268, 272, 273, 274, 276
 Bru, E. 274
 Bérubé, R. L. 326, 329, 330, 331, 333, 335, 337
 Brunell-Neuleib, S. 208
 Bruno, R. **182**
 Brussel, J. A. 301
 Bryant, F. B. 75
 Buboltz, W. C. 189
 Buckwalter, J. G. **184**
 Budman, S. **184**
 Buffington-Vollum, J. K. 7, 8, 10, 13, 140, 216, 245–6
 Buis, T. 32, 33, 39
 Bullock, E. E. 239, 242
 Burbach, A. D. 355, 357
 Burchett, D. L. 91
 Burke, H. C. 274, 276
 Burke, M. J. 190
 Burke, W. T. **349**, 350
 Burket, R. C. 267
 Burlew, A. 41
 Burnette, M. L. 239, 281
 Burnette, M. 239
 Burns, T. 238, 246
 Burns, W. J. **184**
 Bury, A. S. 41, 149, 150
 Busby, R. M. 191
 Buschman, J. 241
 Bush, S. S. 296
 Buss, D. M. 248
 Butcher, J. N. 7, 9, 10, 13, 21, 23, 24, 25, 26, 27, 28, 29, 30, 33, 35, 69, 72, 113, 114, 116, 123, 126, **133**, 189
 Butler, R. J. 30

 Cadoret, R. J. 282
 Cahill, M. A. 276
 Caillouet, B. A. 30
 Caine, E. D. 238, 242, 247
 Caine, S. L. 218
 Calam, R. 357
 Calamari, J. E. 272
 Caldwell, A. 34
 Caldwell, M. F. 275, 277, 278
 Caldwell, M. 276, 278
 Calhoun, G. B. 125
 Calhoun, P. S. 149, 150, 151
 Calkins, S. D. 357
 Calsyn, D. A. 187

 Calzada, E. J. 355, 357
 Camara, W. J. 21, 32
 Camara, W. 189
 Camp, J. 240, 243, 247
 Campagna, V. 190
 Campbell, D. T. 183
 Campbell, J. S. 276
 Campbell, M. A. 267, 268, **268**, 272, 273, 274, 275, 280
 Campbell, T. W. 248, 250
 Cantor, J. 153
 Cantrell, J. 190
 Canyock, E. M. 32, 296
 Caperton, J. D. 154
 Capwell, D. F. 31, 123
 Caraveo, L. E. 189
 Carbonell, J. L. 15, 41
 Carlson, S. R. 282–3
 Carlstedt, A. 274
 Carmines, E. G. 270
 Caron, J. E. 302
 Carpenter, D. R. 189
 Carroll, J. L. **185**
 Carstairs, K. S. 217
 Carter, A. S. 330, 358
 Carusi, D. A. 330
 Caruso, K. A. 148, 155
 Casey, J. O. 238, 239
 Cashel, M. L. 124, 148, 149, 163
 Casillas, A. 28
 Caspi, A. 282
 Castagna, N. 9
 Castlebury, F. 208
 Castro, Y. 34
 Catchpole, R. E. H. 244, 267, 274, 276, 280
 Cauffman, E. 269, 270, 271, 273, 277, 280, 281
 Čěsnienė, I. 234
 Chafetz, M. D. 32
 Chakraborty, D. 33
 Chamas, J. W. 212
 Chan, D. 207
 Chan, Y. C. 352, 355, 358
 Chantry, K. 189
 Charles, D. R. 273
 Chatav Schonbrun, Y. 242
 Chattha, H. K. 268, 272, 273, 274, 276
 Chauhan, P. 239, 281
 Chazan-Cohen, R. 358
 Cherepon, J. A. 151
 Chesney, M. A. 357
 Chibnall, J. T. 28, 30, 34
 Chick, D. 184, **186**
 Choca, J. P. **182**, 188, 191
 Choney, S. K. 35
 Chopra, S. R. 245
 Christens, B. D. 358
 Christoffel, K. K. **331**

- Cicerone, K. 79
 Clark, B. K. 366
 Clark, C. R. 366
 Clark, D. A. 144
 Clark, D. 242, 246
 Clark, E. 75
 Clark, H. J. 17, 245, 247, 248
 Clark, J. A. 301
 Clark, J. W. 281
 Clark, L. A. 28, 69
 Clark, M. E. 22
 Clark, T. S. 151
 Clarke, D. 239
 Clarke-Stewart, K. A. 366
 Clarkin, J. C. 152
 Clarkin, J. F. 153, 211
 Cleckley, H. 230, 231, 232
 Clemence, A. J. 208
 Clement, P. 301
 Clemminshaw, H. K. 368
 Clewes, C. J. **184**
 Clinton, C. 301
 Clinton, D. A. 211
 Cloninger, C. R. 184, **185**
 Clopton, J. R. 15
 Coates, G. D. 116, 120, 121
 Cohen, I. M. 275, 276, 280
 Cohen, J. A. 282
 Cohen, L. J. 189
 Coid, J. W. 242
 Coid, J. 238, 242
 Colligan, R. C. 109
 Collins, R. P. 40
 Collman, P. 190
 Colwell, K. W. 246
 Colwell, L. H. 240, 270
 Condie, L. O. 359
 Condit, D. C. 32, 291, 296, 298, 299, 300, 301
 Conforti, K. 152
 Conklin, A. 212
 Conner, K. R. 238, 242, 247
 Connors, C. K. 10
 Connor, E. 190
 Connor, R. T. 358
 Conroy, M. A. 246
 Cook, N. E. **268**, 272
 Cooke, D. J. 233, 238, 239, 240, 273
 Cooke, G. 36, 39
 Coolidge, F. L. **184**
 Cooper, D. B. 301
 Cooper, V. G. 9
 Copestake, S. 241
 Cormier, C. A. 242, 243, 279
 Cormier, C. 242
 Cornell, A. H. 277
 Cornell, D. G. 32, 242, 271, 272, 275, 276, 277, 280, 281
 Cornell, D. 276, 280
 Corrado, R. R. **268**, 270, 274, 275, 276, 280
 Corrigan, S. K. 33
 Costa, P. T. 151, 152, 153, 241
 Costello, A. J. 330
 Costello, R. M. 30
 Coulter, M. 361
 Cox, D. N. 10, 231, 238
 Cox, J. 245
 Coyle, D. D. 357
 Coyne, L. 188
 Cragolino, A. 207
 Craig, R. J. 176, 178, 183, 185, 187, 188, 189, 190, 194
 Creasey, G. L. 357
 Creech, S. K. 151
 Crellin, M. C. 182
 Crnic, K. 355
 Crocker, A. G. 243
 Crovitz, E. 30
 Cruise, K. R. 140, 151, 270, 280
 Côté, G. 243, 280
 Cuccaro, M. L. 356
 Cuffe, S. 357
 Culbertson, W. C. 294
 Culbertson, W. **298**
 Cull, J. G. 154
 Cummings, J. L. 330
 Cunningham, M. D. 245
 Currier, O. R. 350, 355, 356
 Curtin, J. J. 231, 267, 275
 Cyr, C. 356
 Cyterski, T. D. 272, 274, 275, 276, 280
 Czar, K. A. 239, 242

 Daffern, M. 242
 Dahle, K. P. 242, 247
 Dahlen, E. R. 239, 242
 Dahlstrom, L. E. 27, 31, 34, 127
 Dahlstrom, W. G. 7, 25, 27, 29, 31, 33, 34, 116, 127
 Daisy, F. 187
 Daleiden, E. L. 153
 Dana, R. 190
 Dandy, K. L. 27
 Dane, H. E. 277
 Dao, T. K. 208, 212
 Das, J. 269, 273, 274
 Das, R. 357
 Davidson, R. S. **185**, 194
 Davies, G. 242
 Davis, H. 188
 Davis, K. M. 39, 245
 Davis, Q. 128
 Davis, R. D. 7, 30
 Davis, R. 10, 176, 191
 Davis, W. E. 191

- Dawson, K. S. 187
 de Beuers, E. 37
 de Brock, A. J. L. L. 355
 De Fruyt, F. 241
 de Mey, H. R. 29
 de Oliveira-Souza, R. 238, 247
 de Ruiter, C. 241, 243, 269, 273, 274
 de Vogel, V. 243
 Dean, R. S. 275, 279
 Dear, G. E. 189
 Deater-Deckard, K. 353, 355
 Declercq, F. 242
 DeCoster, J. 238, 239
 Decuyper, M. 241
 Dedmon, S. E. 357
 DeFife, J. A. 212
 DeGarmo, D. S. 366
 Degroat, J. S. 357
 del Rio, C. 179
 DeLisi, M. 243
 DellaPietra, L. 302
 Delsol, C. 153, 154
 DeLuca, V. 212
 Demakis, G. J. 23, 27, 40, 301
 Demakise, G. J. 300
 DeMatteo, D. S. 16, 245, 267, **293–4**
 Demby, A. **184**
 Demers, J. 30
 Demidenko, N. 151, 162
 Demmer, L. 357
 Denney, R. L. 32, 75, 291
 Derksen, J. J. L. 29
 Derogatis, L. R. 330
 Dery, M. 280
 Detrick, P. 28, 30, 34
 Devieux, J. 188
 Diamantopoulou, S. 330
 Diamond, P. M. 150, 153, 154, 155, 159, 160
 Diamond, S. E. 187
 Diamond, T. 358
 Diamonti, C. 301
 Dicataldo, F. **268**, 272
 DiCicco, T. M. 272, 274, 275
 Dickens, S. E. 37, 149
 Dickinson, K. A. 153
 Diener, M. L. 357
 Dillon, S. K. 191
 Dionysus, K. E. 75
 Dixon, A. 273
 Dmitrieva, J. 269, 271, 277
 Doan, B. T. 152
 Doebbeling, B. N. 28
 Dolan, M. 155, 242, 270, 271, 272, 277
 Donat, D. C. 188, 191
 Donenberg, G. 357
 Doninger, N.A. 240
 Doreleijers, T. 269, 273, 274
 Doren, D. M. 244
 Dorfman, W. J. 41
 Dorrr, D. 190
 Douglas, K. S. 38, 39, 152, 154, 155, 159, 238, 242, 246, 271, 276, 277, 279, 280, 281, 282, 296
 Doyle, M. 155, 242
 Dozois, D. J. 189
 Dragon, W. R. 22
 Drazkowski, J. F. 75, 81, 82
 Drerup, L. C. 75, 83
 Drozd, L. 361
 Drum, D. J. 291
 Dubois-Comtois, K. 356
 Dubro, A. F. **184**
 Ducro, C. 242
 Duggan, C. 241, 246
 Duker, J. 25, 27, 29
 Dulcan, M. K. 273
 Dumas, J. E. 357
 Dumenci, L. 331
 Duncan, S. A. 153, 154
 Duncan, S. 33, 75, 88, 149
 Dunn, J. T. 7, **8**
 Dunnette, M. D. 30
 DuPaul, G. J. 357, 358
 Duppong-Hurley, K. 358
 Duros, R. L. 272, 274, 275
 Durrett, C. 152
 Dutton, D. G. **184**
 Dvoskin, J. A. 214
 Dyce, J. A. 183
 Dyer, F. J. 13, 181, 190, 359
 Earnst, K. S. 149, 150, 151
 Eastman, B. 189
 Eaves, D. 39
 Echeburúa, E. 189
 Ecklund-Johnston, E. 291
 Edelbrock, C. 325, 330, 355
 Edens, J. F. 7, 37, 140, 143–4, 146, 147, 148, 151, 152, 153, 154, 155, 156, 159, 160, 216, 240, 242, 245–6, 248, 249, 276, 280, 281
 Edwards, D. W. 153
 Edwards, G. 358
 Edwards, J. 184, **184**
 Edwards, M. J. 148
 Efendov, A. A. 41
 Egan, V. 277
 Ehrenberg, M. F. 360, 362
 Eisenber, M. 37
 Eke, A. W. 242, 243
 Ekman, P. 363
 Elhai, J. D. 41
 Elion, V. H. 35
 Elkins, D. E. 28, 114, 115, 116, 117, 120, 121, 124, **132**

- Elksnin, N. 357
 Ellenberger, H. F. 204
 Elsass, P. 242
 Emery, R. E. 4, 360, 365
 Emery, R. 365
 Enders, C. 300
 Endicott, J. 154
 Endriga, M. C. 356
 English, L. T. 22
 English, M. 277
 Epstein, M. E. 271, 277
 Epstein, M. H. 358
 Epstein, M. 271, 276, 280
 Erard, R. E. 217
 Erard, R. 208
 Erbaugh, J. K. 10
 Erbes, C. R. 75, 78
 Erdberg, P. 206, 207, 213, 217
 Erez, E. 39, 92
 Eron, L. D. 124
 Eronen, M. 274
 Etherton, J. L. 157
 Éthier, L. S. 358
 Éthier, L. 355
 Eudell-Simmons, E. M. 212
 Evans, F. B. 208, 215, 217
 Evans, T. M. 40
 Evershed, S. 152
 Exner, J. E. Jr. 202, 204, 205, 206, 207, 211, 213, 217
 Exner, J. E. 10, 152, 206, 217, 364
 Eyberg, S. M. 355, 357
 Ezzo, F. R. 40, 92

 Fabriga, L. 75
 Fabrigas, J. 40
 Fabry, J. J. 117
 Fagan, J. 358
 Fahy, T. 238, 246
 Faigman, D. L. 209
 Fairbank, J. A. 31, 41, 187
 Fals-Stewart, W. 148, 153, 159, 187
 Farris, K. L. 120, 121, 124, 125, 132
 Fawcett, R. C. 85
 Fazel, S. 238, 246
 Feinfield, K. A. 357
 Fekken, G. C. 144
 Feldman, M. A. 358
 Feldman, S. S. 273
 Felthous, A. R. 230, 244, 247
 Ferdinand, R. F. 330, **331**
 Ferguson, M. 250
 Fergusson, D. M. 282
 Fernandez, K. 281
 Fernandez-Smith, K. 245
 Ferrier-Auerbach, A. G. 79, 80
 Ffytche, D. 230, 250

 Fichtenberg, N. L. 301
 Fidler, B. J. 39
 Field, T. M. 356
 Fieldstone, L. 361
 Fila, K. 149
 Fine, M. A. 357
 Finn, J. A. 38, 39
 Finn, S. E. 30, 212, 214
 Firestone, G. 361
 First, M. B. 88
 Fischler, G. L. 28, 30, 34
 Fisher, D. 189
 Fiske, D. W. 183
 Fite, P. J. 357
 Fitzgerald, S. 240, 242
 Fitzpatrick, R. 238, 246
 Fivush, B. A. 357
 Fjetland, O. K. 151
 Fleming, C. 187
 Flens, J. R. 10, 14, 17, 190, 191, 362, 363, 368
 Fletcher, K. E. 357
 Fletcher, K. 358
 Flight, J. I. 275
 Flores, J. 146, 147
 Florsheim, P. 356, 357
 Flye, A. 189
 Flynn, P. M. 186, 187, **194**
 Folstein, M. F. 330
 Folstein, S. E. 330
 Fontaine, J. L. 121
 Fontaine, J. 332
 Fontaine, R. G. 247
 Forbey, J. D. 13, 26, 34, 41, 42, **43**, 114, 116, 123, 124, 126, **132**
 Ford, R. Q. 211
 Fordyce, W.E. 29
 Forehand, R. 365
 Forgatch, M. S. 366
 Forsman, A. 238, 274
 Forsman, M. 282
 Forsting, M. 238
 Forth, A. E. 10, 17, 183, 184, 234, 238, 243, 244, 245, 247, 248, 267, 269, 270, 274, 275, 276, 277, 279, 280, 283
 Forth, A. 10
 Foster, E. E. 245
 Fowler, J. C. 207, 208, 210, 212
 Fowler, K. A. 205, 241
 Fowler, R. S. 356, 358
 Fox, A. R. 247
 Fox, C. E. 357
 Fox, D. D. 33
 Fox, D. 33, 299, 300
 Fox, R. A. 355, 357
 Frank, J. D. 69
 Frank, S. J. 354, 358
 Frankel, K. K. 357

- Frazelle, J. 232
 Frederick, R. I. 8
 Freeark, K. 354, 358
 Freeman, D. B. 40, 75, 90
 Freeman, R. 182
 Freese, R. 242
 Friars, P. M. 357
 Frick, P. J. 230, 270, 271, 277
 Friedman, A. F. 369
 Friend, R. 239
 Frith, C. D. 248
 Frueh, B. C. 218
 Fukui, R. 356, 357
 Fulero, S. M. 230
 Fuligni, A. S. 358
 Fuller, G. **185**
 Furlan, P. M. 188
 Furth, S. L. 357

 Gabrys, J. B. 191
 Gacono, C. . 230
 Gagnon, N. C. 279, 281
 Gaies, L. A. 149, 150
 Gaines, J.A. 30
 Galatzer-Levy, J. 361
 Galatzer-Levy, R. M. 361
 Galinsky, M. D. 121
 Gallagher, R. W. 32
 Gallen, R. T. 27
 Galynker, I. I. 189
 Ganellen, R. J. 210, 218
 Gans, S. W. 189
 Gao, Y. 247
 Garb, H. N. 205
 Garb, H. 205
 Garcia, N. 75
 Garetz, F. K. 30
 Garland, A. 336
 Garrick, T. 247
 Garrido, G. J. 247
 Gartland, D. M. 80
 Gartland, D. 42
 Gartner, A. 31
 Gartner, J. 31
 Gavrilovic, G. 188
 Gazit, T. 215
 Gazzaniga, M. S. 248
 Geiger, F. 274
 Gelfand, D. M. 357
 Geling, O. 40, 75
 Gendreau, P. 242
 George, J. M. 151
 Gerris, J. R. M. 355
 Gersh, D. M. **186**
 Gerson, A. D. 357
 Gervais, R. O. 33, 40, 75, 90, 91, 92
 Geyer, M. D. 154

 Giannetti, R. A. 114
 Gibbon, M. 88
 Gibertini, M. 185, **186**
 Gilberstadt, H. 25, 27, 29
 Giles, C. L. 150, 154, 155, 160
 Gill, W. S. 154
 Gillard, N. D. 4, 75, 89, 150
 Gillis, J. R. 37
 Girona, R. J. 22, 35
 Glaser, B. A. 125
 Glenn, A. L. 247
 Glenn, W. J. 22
 Glover, A. J. 242
 Glynn, G. 89
 Gómez, F. C. 128
 Goddard, K. M. 75, 88
 Godt, N. 274
 Goggin, C. 242
 Goggin, W. C. 184, **186**
 Gola, T. 301
 Gold, S. N. 41
 Goldberg, D. P. **331**
 Goldberg, J. O. 185
 Goldberg, S. 356, 358
 Golden, C. J. 10
 Golding, S. L. 213
 Goldman, D. J. 34, 79, 80
 Goldstein, A. M. 213
 Goldstein, A. 291
 Goldstein, D. S. 272
 Goldstein, R. C. 357
 Goldstein, S. 356
 Gonsalves, V. 242, 243
 Good, G. L. 295
 Goodman, C. 163
 Goodman, G. S. 4
 Goodman-Delahunty, J. 363
 Gordon, A. 244
 Gordon, K. H. 34
 Gordon, R. A. 114
 Gorman, J. L. **184**
 Gornbein, J. 330
 Gorsuch, R. 301
 Gotheil, E. 211
 Gottdiener, W. H. 37
 Gottesman, I. I. 121, 128
 Gottfried, A. E. 39
 Gottfried, A. W. 39
 Gould, J. W. 7, 10, 14, 191, 360, 361, 362, 363
 Graham, J. R. 7, 9, 10, 21, 22, 23, 24, 25, 26, 27,
 28, 29, 30, 33, 34, 35, 36, 38, 41, 43, 46, 68,
 69, 113, 114, 116, 117, 123, 124, 130, 132, **133**
 Graham, P. J. 330
 Graham-Bermann, S. A. 357
 Granacher, R. P. 75, 89, 295, 296, 297, 298, 299
 Grann, M. 242, 243, 246, 247
 Grassi, J. R. 301

- Gratton, C. 187, 188
 Gray, K. 330
 Gray, N. S. 240, 241, 242, 243
 Gray, T. 212
 Gray-Little, B. 121
 Green, A. J. 357
 Green, P. 40
 Greenberg, S. A. 7, 16
 Greenblatt, R. L. 68, **182**, 191
 Greene, G. L. 26
 Greene, R. L. 25, 33, 34, 35, 36, 127, 204, 215, 364, 369
 Greenhut, D. 10, **14**
 Greening, L. 357
 Greenstein, D. S. 151
 Greenway, P. 116
 Gregory, C. 39, 92
 Gregory, S. 230, 250
 Greiffenstein, M. F. 33, 298, 299, 300, 301
 Greist, J. H. **184**
 Grelling, B. Z. 188
 Gretton, H. M. 244, 267, 273, 274, 276, 280
 Greve, K. W. 33, 89, 157, **298**
 Greve, K. 301
 Greveab, K. W. 300
 Griffin, R. 28, 29
 Grimes, J. S. 151
 Grimes, K. 330
 Grisso, T. 5, 7, 8–9, 122, 189, 274, 275, 276, 279, 280, 282, 291, 334
 Grissom, G. 153
 Grønnerød, C. 207, 208, 212
 Grossman, J. 356
 Grossman, L. S. 32, 218
 Grossman, S. D. 178
 Grossman, S. 179, 180, 181
 Groth-Marnat, G. 85, 211, 215
 Grove, W. M. 205, 209
 Grudzinskas, A. J. Jr. 243
 Grygielski, M. 208
 Gu, D. 244
 Guay, J. 249
 Gudjonsson, G. H. 189
 Guevremont, D. C. 357, 358
 Guidubaldi, J. 365, 368
 Guralnick, M. J. 358
 Gustafson, D. 242, 246
 Gustafson, S. B. 270
 Gustavson, C. 238
 Guterman, N. B. 358
 Guthrie, G. M. 27
 Guthrie, P. C. **186**
 Guthrie, P. **185**
 Guy, L. S. 152, 238, 245, 246, 281
 Guyer, M. 363
 Guzman-Cavazos, L. 117
 Gyamfi, P. 357, 358
 Gynther, M. D. 127
 Gyoerkoe, K. L. 218
 Ha, E. H. 358
 Hagan, L. D. 75, 92
 Hagen, E. P. 10
 Hagen, M. A. 9
 Haines, J. 151
 Häkkänen-Nyholm, H. 231, 242
 Hakstian, A. R. 38
 Hale, L. R. 272
 Halfaker, D. A. 75
 Halfaker, D. 75, 92
 Halgunseth, L. C. 357
 Hall, G. N. C. 34, 35
 Hall, J. D. 358
 Hall, S. 357
 Hallam, R. A. 27
 Haller, D. L. 110, 187
 Halon, R. L. 194
 Halstead, W. C. 7
 Hamberger, L. K. 191
 Hamilton, J. 152
 Hamilton, M. 151
 Hammeke, T. A. 10
 Hammond, F. 301
 Hammond, M. A. 358
 Handel, R. W. 7, 8, 10, 13, 17, 22, 27, 75, 87, 92, 114, 115, 116, 121, 124, **132**
 Handler, L. 208
 Handy, L. 182
 Haney, T. L. 29
 Hanks, R. A. 302
 Hansen, T. 209
 Hanson, R. K. 121, 160
 Hare, R. D. 10, 17, 38, 183, 184, 231, 232, 233, 234, 235, 236, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251, 267, 269, 270, 271, 273, 274, 276, 280
 Hare, R. 238
 Harenski, C. A. 231, 244, 247
 Harkins, L. 244, 250
 Harkness, A. R. 24, 26, 27, 28, 29, 38, 39, 47, 71, 114, 120
 Harmon, R. J. 357
 Harper, S. 357
 Harpur, T. J. 38, 243
 Harrington, D. E. 79
 Harris, E. H. 267
 Harris, G. E. 358
 Harris, G. T. 10, 242, 243, 244, 248, 279
 Harris, J. **293–4**
 Harris, M. J. 22
 Harris, R. 23
 Harrison, K. S. 234
 Hart, S. D. 10, 39, 152, 155, 183, 184, 238, 239, 241, 242, 243, 246, 267, 270, 275, 276, 280, 282

- Hart, S. 38
 Hartlage, L. C. 13
 Hartman, B. J. 30
 Harvey, P. D. 267, 274
 Harwood, T. M. 211
 Hassenstab, J. 296, 298, 299, 301
 Hastings, J. E. 191
 Hathaway, S. R. 21, 24–5, 27, 31, 41, 72, 108, 109, 110, 123, 124
 Hauben, C. **184**
 Hauenstein, E. 350
 Haugaard, J. J. 4
 Havens, J. F. 357
 Hawes, S. W. 155, 160
 Hawk, G. L. 32
 Hawk, G. 242
 Hawkins, J. 358
 Hayman, J. 207
 Hays, J. R. 30
 Haywood, T. W. 218
 Hebben, N. 85
 Heibrun, K. 267
 Heidish, I. E. 330
 Heijmens Visser, J. H. 330
 Heilbronner, R. L. 40, 75, 90, 291, 300, 301, 302
 Heilbrun, A. B. 38
 Heilbrun, K. 4, 6, 16, 38, 242, 246, 247, 267, 272, 274, 277, 291, 292, **293–4**, 296, 363, 366, 369
 Heinimaa, M. 212
 Heinrichs, R. J. 89
 Heller, K. 154
 Helmes, E. 69
 Hemmati, T. 242
 Hemphill, J. F. 238, 241, 242
 Hemphälä, M. 273, 274
 Henning, K. 189
 Heinrichs, T. F. 29
 Henry, G. K. 300
 Herrington-Wang, L. E. 150, 154, 155, 160
 Herrmann, D. 295
 Hervé, H. 231
 Hesdorffer, D. C. 295, 296
 Hess, A. K. 189, 209
 Hetherington, E. M. 366
 Heywood, E. 282
 Hiatt, K. D. 231
 Hicklin, J. 184, 241
 Hicks, B. M. 238, 239, 282
 Hicks, M. M. 124
 Hildebrand, M. 241, 243
 Hill, C. D. 233, 242
 Hill, E. L. 218
 Hill, J. S. 35
 Hill, K. 184, **185**
 Hillary, B. E. 123
 Hillege, S. 269, 273, 274
 Hiller, J. B. 208
 Hills, H. A. **186**
 Hilsenroth, M. J. 16, 207, 208, 210, 212
 Hilsenroth, M. 206, 215, 217
 Hilton, N. Z. 242, 243, 248
 Hinrichs, G. 274
 Hipke, K. N. 357
 Hitchcock, D. L. 357
 Hjemboe, S. 24
 Ho, A. 153
 Hobson, J. 246
 Hodgins, S. 230, 238, 242, 250, 268, 269, 271, 272, 280
 Hoelzle, J. B. 23, 79, 80
 Hoelzlea, J. B. 300
 Hoffman, J. **184**
 Hofstra, M. B. 330
 Hofvander, B. 242
 Hogan, J. D. 2
 Hogan, T. P. 209
 Hoge, R. D. 277, 302
 Hogg, B. **184**
 Hogue, T. E. 240, 242
 Hohnhorst, A. 238, 244
 Holcomb, W. R. 35
 Holden, R. R. 144, 148
 Holdford, R. 189
 Holdwick, D. J. Jr. 210
 Holi, M. 242
 Holliman, N. G. 124
 Holliman, N. **185**
 Hollin, C. 240
 Holmes, G. E. 151
 Holmes, G. R. 356
 Holmes, T. H. 154
 Holt, R. R. 204
 Holt, S. E. 189
 Homaifar, B. Y. 238, 242, 247
 Hoogerheide, K. N. 330
 Hopp, G. A. 89
 Hoppe, C. 369
 Hopwood, C. J. 64, 147, 150, 155
 Horabin, I. 64
 Horne, H. L. 208
 Hornquist, J. O. 29
 Horton, A. M. Jr. 13
 Hostetler, K. 114
 Houck, C. **182**
 Hough, L. 90, 91, 92
 Hough, R. 336
 Houghton, R. E. 242, 243
 Houlding, C. **268**
 Houston, C. M. 32
 Houston, W. 302
 Howard, M. 230, 250
 Howard, R. 246
 Howell, C. T. 330, 331
 Howells, K. 242

- Howie, P. 273
 Howieson, D. B. 301
 Huband, N. 241
 Huchzermeier, C. 274
 Hucker, S. J. 38, 243
 Huesmann, L. R. 124
 Hughes, J. C. 358
 Huh, K. 352
 Hull, J. W. 152, 153
 Hultsch, D. F. 8
 Hume, A. 191
 Hume, M. P. 125
 Humm, D. G. 109
 Hungerford, A. 357
 Hunsley, J. 333
 Hunter, M. A. 360, 362
 Huntoon, J. 188
 Hurt, S. W. 31
 Hurst, D. F. 75, 81, 82
 Husband, S. D. 41, 184, 186
 Huse, M. N. 30
 Husted, J. **184**
 Hutcheson, J. J. 351–2, 355, 356
 Hyer, L. **182**, 187, 188

 Iacono, W. G. 282, 283
 Ignácio, F. A. 247
 Ignácio, R. A. 238
 Ilonen, T. 212
 Imhof, E. A. 110, 115
 Ingram, M. V. 300
 Innocenti, M. S. 352
 Insabella, G. M. 359, 366
 Insabella, G. 365
 Inwald, R. E. 30
 Irvin, J. R. 358
 Iselin, A.-M. 274, 276
 Ismail, G. 235
 Ispa, J. M. 357
 Itskovich, Y. 189
 Ivanova, M. Y. 315, **315**, 331
 Iverson, G. L. 89, 301, 302

 Jackson, A. P. 358
 Jackson, D. N. 69, 140
 Jackson, H. J. 184, **184**
 Jackson, K. 152
 Jackson, R. L. 6, 233, 234, 240, 242, 246, 248
 Jackson, W. 154
 Jacobson, J. M. 116, 117
 Jain, A. 295
 Jain, S. 295
 Jameson, B. J. 360, 362
 Janson, H. 208
 Janus, M. 356, 358
 Janzen, H. L. 183
 Jarvis, P. A. 357
 Jasinski, L. J. 301
 Jöckel, D. 242
 Jenkins, C. L. 356
 Jennings, T. 188, **293–4**
 Johansen, J. 121
 Johansson, P. J. 232, 234
 John, R. S. 153, 154
 Johnson, D. W. 152, 246
 Johnson, J. K. 152, 154, 246
 Johnson, J. 160
 Johnson, R. 124, 128
 Johnson, T. 295
 Johnston, C. 357
 Johnston, S. 240
 Johnston, T. 369
 Joiner, T. R. 34
 Jonathan, S. O. 300
 Jones, A. P. 231, 245
 Jones, A. 184, 189, 300
 Jones, D. N. 231, 238
 Jones, G. B. 243
 Jones, O. D. 248
 Jones, R. M. 277
 Jones, S. 269, 281
 Jordan, J. R. 356
 Joyce, A. S. 211
 Judge, S. 355

 Kaemmer, B. 7, 9, 10, 23, 25, 26, 28, 29, **43**, 113, 114
 Kafami, D. 239
 Kakuma, T. 153
 Kamphaus, R. W. 330
 Kamphuis, J. H. 30, 214
 Kane, A. W. 214
 Kang, E. 357
 Kaplan, S. 275, 276, 277
 Karlin, B. E. 151
 Karp, J. 358
 Kaser-Boyd, N. 126, 215
 Kasius, M. C. 330
 Kassen, S. M. 363
 Kastl, A. J. 15
 Katz, R. C. 125
 Katz, R. **184**
 Kawachi, I. 29
 Kay, T. 79
 Kazdin, A. E. 330, 357, 358
 Kazén, M. 208
 Kazui, M. 358
 Keane, T. M. 31, 41
 Keenan, K. 356
 Kehl-Fie, K. 162
 Keilen, A. 245–6
 Keilin, W. J. 189
 Keller, J. W. 21
 Kelley, M. L. 330

- Kelley, R. F. 363
 Kelley, S. J. 356
 Kelln, B. R. 189
 Kellogg, S. H. 153
 Kelly, J. B. 365, 366
 Kelly, J. 366
 Kelly, P. H. 29
 Kelly, R. F. 361
 Kelly, W. L. 30
 Kennealy, P. J. 238, 239, 243, 247
 Kennedy, J. 301
 Kennedy, S. H. **184**
 Kennedy, W. A. 125, 267, 275
 Kerr, M. 270
 Khadivi, A. 150
 Khalifa, N. 246
 Khiroya, R. 230
 Köhler, D. 274
 Kiehl, K. A. 231, 244, 247
 Kilgore, W. D. 302
 Kim, B. H. 90, 91, 92
 Kim, E. J. 358
 Kimonis, E. R. 269, 271, 277
 Kinder, B. N. 149, 151, 218
 King, C. M. 17, 293
 King, J. H. 13, 291
 Kingery, L. R. 297
 Kirby, A. C. 149, 150, 151
 Kirby, K. C. 184, **186**
 Kirk, G. R. 231
 Kirk, L. 364
 Kirkhart, K. J. 243
 Kirkish, P. 30, 247
 Kirkpatrick, H. D. 362, 363
 Kirlin, K. A. 75, 81, 82
 Kisker, E. E. 358
 Kitaëff, J. 85
 Kitson, G. C. 366
 Klaver, J. R. 282
 Klein, E. 215
 Klein, M. H. **184**
 Kleinsasser, D. 189
 Klinger, L. J. 358
 Klonsky, E. D. 152
 Klopfer, B. 204
 Klopfer, W. G. 204
 Knauss, L. 209
 Knight, R. A. 247, 249
 Knotts, A. 301
 Kobos, J. 30
 Koenigs, M. 231
 Koffler, S. P. 296
 Kong, Y. H. 29
 Konold, T. R. 347, 368
 Koot, H. M. 330
 Kordinak, S. T. 369
 Korkeila, J. 212
 Koss, M. P. 116
 Kosson, D. S. 238, 240, 242, 243, 268, 269, 272, 274, 275, 276, 280
 Kosson, D. 10, 234
 Kotler, J. S. 271
 Kovitz, K. 357
 Kraus, L. 361
 Krauss, D. A. 360, 362, 363
 Krauss, G. L. 151
 Krauss, M. W. 354
 Kreek, M. J. 153
 Kretz, L. 188
 Krischer, M. K. **268**, 269, 272
 Krishnamurthy, R. 111, 114, 115, 117, 119, 120, 123, 127, 128
 Kristiansson, M. 247
 Kroll, L. S. 117
 Kroner, D. G. 242, 246, 274
 Kropp, P. R. 155, 243
 Krueger, R. F. 38, 24, 282–3
 Krukowski, R. A. 331
 Krupka, A. 357
 Kuback, F. A. 271
 Kubazansky, L. D. 29
 Kucharski, L. T. 75, 88, 149
 Kucharski, L. 33
 Kuehnle, K. 361, 362
 Kugeares, S. L. 214
 Kuhl, J. 208
 Kullgren, G. 242, 243
 Kumari, V. 230, 250
 Kumka, G. 267, 273, 274, 276, 280
 Kuncel, R. 187
 Kunz, C. 242
 Kurtz, J. E. 152, 301
 Kushner, H. 153
 Kushner, S. 75
 Kutcher, M. 151
 Kvaran, T. H. 247
 Kwoak, S. 357
 Laajasalo, T. 242
 Lacey, K. 35
 Lachar, D. 30, 127, 128
 Lacharité, C. 355
 LaFreniere, P. J. 352, 357, 358
 Laganière, J. 357
 Lainhart, J. E. 75
 Lake, A. J. 231
 Lally, S. J. 10, 11–12
 LaLone, L. 26
 Lalumière, M. L. 248
 Lam, D. O. 357
 Lama, S. 162
 Lamb, C. C. 354, 358
 Lamb, M. E. 366
 Lambert, M. J. 211

- Lampel, A. K. 189, 194
 Lander, T. 267
 Laneri, M. 358
 Lang, S. 273
 Lange, R. T. 302
 Langeluddecke, P. M. 302
 Langevin, R. 182
 Långström, N. 242, 243
 Langton, C. M. 242, 244
 Lanier, V. W. 148, 149
 Lansing, A. E. 336
 Lanyon, R. I. 369
 Larrabee, G. J. 22, 33, 85, 296, 297, 298, 299, 301, 302
 Larsen, R. M. 21, 108
 Larsson, A. 272
 Larsson, H. 230, 231, 245, 282, 283
 Lau, H. C. 248
 Laurinavičius, A. 234
 Lavigne, J. V. **331**
 Laye, R. C. 191
 Lazarro, T. 190
 Lecci, L. 92
 Lecic-Tosevski, D. 188
 LeCuyer-Maus, E. A. 356, 357
 Lee, K. L. 29
 Lee, S. J. 358
 Lee, T. C. 33
 Lee, T. T. C. 92
 Lee, Z. 270, 274, 276, 282
 Lees-Haley, P. R. 7, 8, 22, 31, 33, 89, 299, 300
 Lefkowitz, M. M. 124
 Legler, W. 302
 Lehne, G. K. 189
 Leistico, A. M. 269, 272, 274, 275
 Leistico, A. R. 238, 239, 268
 Lennon, T. 146, 147, 151, 153, 158
 Lenny, P. 189
 Lenzenweger, M. F. **182**
 Lerner, P. M. 204
 Lesser, R. P. 151
 Leu, C. S. 357
 Levendosky, A. A. 357
 Levin, J. 151
 Levin, P. 215
 Levine, D. J. **184**
 Levitt, E. E. 204
 Levy, H. L. 357
 Levy, K. N. 211
 Levy-Elkon, A. 275, 276, 277
 Lewak, R. 369
 Lewis, D. E. 30
 Lewis, K. 243, 244
 Leygraf, N. 238
 Lezak, M. D. 301
 Li, S. H. S. 204
 Li, W. 231
 Liao, H. 30
 Libb, J. W. **182**
 Lichtenstein, P. 282, 283
 Lidz, V. 267, 272, 274, 277
 Lilienfeld, S. O. 38, 80, 154, 155, 159, 205, 233, 241, 242, 243, 249
 Liljequist, L. 149, 151
 Lin, M. M. 146, 153, 158
 Lincoln, T. 242
 Lindberg, N. 242
 Lindsay, K. A. **184**, 191
 Lindsay, W. R. 240
 Lingo, J. 23
 Lishner, D. A. 242, 243
 Little, T. D. 365
 Litz, B. T. 40
 Liu, W. 211
 Lloyd, C. D. 247, 248
 Lochman, J. E. 277
 Locke, D. E. C. 75, 81, 82
 Locke, D. E. 81, 82
 Locke, K. D. 188, 191
 Loeber, R. 240, 277, 282
 Loehlin, J. 282
 Loevinger, J. 68, 69, 140
 Lohr, J. M. 205
 Loney, B. R. 283
 Long, A. C. 7, 16
 Long, K. A. 26
 Longworth, S. L. 273
 Looman, J. 235, 244, 247, 250
 Lopez, I. R. 34, 35
 Lopez, M. 240, 243
 Loranger, A. W. 152
 Lord, J. 267, 274
 Lorenz, A. R. 240
 Loring, D. W. 301
 Low, C. 355
 Lowman, J. 121
 Lu, P. H. **299**
 Lubin, B. 21, 108
 Lucas, S. K. 302
 Lucente, S. W. 238, 239
 Lucente, S. 153
 Lumsden, J. 246
 Luxenberg, T. 215
 Ly, M. 231
 Lynam, D. R. 231, 233, 240, 241, 266, 270, 282
 Lynch, K. D. 121
 Lynn, R. 240
 Lynskey, M. T. 282
 Lyon, D. R. 280, 281
 Lyons, P. M. 270
 Lyons-Ruth, K. 357
 Lyubchik, A. 357

- MacAndrew, C. 24
 MacCoon, D. G. 231
 MacCulloch, M. J. 240, 242, 243
 MacDougall, A. E. M. 279, 281
 Machin, D. 270
 MacLusky, I. 356, 358
 Maden, T. 230
 Magaletta, P. R. 159
 Maghem, B. 242
 Magyar, M. S. 154, 155
 Mailloux, D. L. 243, 274
 Mailloux, D. 250
 Main, N. 189
 Malatesti, L. 231, 247
 Malcolm, P. B. 243
 Malec, J. 79
 Malin, H. M. 243
 Malina, A. C. 291
 Mallick, K. 358
 Malloy, P. F. 31, 41
 Malow, R. M. 188
 Mandenberg, K. 358
 Mann, K. 182
 Mannarino, A. P. 282
 Mannion, A. 242
 Maoz, G. 215
 Marchant, G. J. 275, 279
 Marcus, D. K. 242, 249
 Marczyk, G. R. 16, 267, 209, **293–4**
 Marek, R. J. 83
 Margolin, G. 153, 154
 Marion, B. E. 87, 88
 Markon, K. E. 24, 38
 Marks, P. A. 25, 27, 29, 110, 111, 120
 Marlowe, D. B. 184, **186**
 Marquette, J. 125
 Marsh, D. T. **184**
 Marshall, J. 277
 Marshall, L. A. 273
 Marshall, M. B. 369
 Marteau, T. M. 357
 Martin, A. B. 357
 Martin, L. 360
 Martin, M. A. 6, 23, 27, 31
 Martin-Cannici, C. 148, 149
 Martindale, D. A. 360, 361, 362
 Martindale, D. M. 360
 Martinez, M. 189
 Marton, I. 358
 Maruish, M. 110
 Marx, B. P. 40
 Mash, E. J. 333, 357
 Masling, J. M. 208
 Mason, B. 207
 Mason, J. A. 75, 92, 114, 115, 116, 124, 132
 Mason, J. 75, 92
 Matano, R. A. 188, 191
 Matarazzo, J. D. 21, 108
 Mathieu, C. 231, 238
 Mathis, F. 188
 Mattson, C. A. 75, 92
 Maxson, L. S. 37
 Mazurick, J. L. 357
 McBride, B. 355
 McBride, M. 267, 273, 274, 276, 280
 McCabe, K. M. 336
 McCain, A. P. 330
 McCallum, M. 211
 McCann, C. 356, 357
 McCann, J. T. 13, 184, 185, 186, 187, 190, 194, 195, 208
 McCartney, K. 366
 McClary, K. A. 153
 McClelland, G. M. 273
 McClinton, B. K. 114
 McCloud, P. 357
 McConaughy, S. H. 315, 316, 319, 329, 330, 331
 McConville, D. 275, 276, 277
 McCord, D. M. 75, 83
 McCormick, A. V. **268**
 McCormick, D. J. 278
 McCormick, D. 278
 McCoy, W. K. 240, 281
 McCoy, W. 280, 281
 McCrae, R. R. 151, 152, 153, 332, 335
 McCully, E. 28
 McCumber, S. 116
 McDermott, P. A. 146, 153, 158
 McDevitt-Murphy, M. E. 150
 McDowell, C. J. 207
 McDowell, E. 364
 McFarland, M. 358
 McGaughey, M. C. 356
 McGee, R. A. 336
 McGeoch, P. G. 189
 McGleish, A. 243
 McGovern, J. 242
 McGrath, M. J. 302
 McGrath, R. E. 90, 91, 92, 117, 150, 207
 McGue, M. 283
 McGuire, K. A. 79, 80
 McHugh, P. F. 153
 McHugh, P. R. 330
 McIntyre, L. D. 355
 McKee, G. R. 37
 McKenna, T. 24, 29
 McKenzie, I. E. 189
 McKinley, J. C. 24–5, 21, 72, 108, 109
 McKinnon, L. 268, 272, 273, 274, 276
 McLachlan, K. 10
 McLanahan, S. 365
 McLellan, A. T. 153
 McMahon, R. C. **182, 185**, 188, 194
 McMahon, R. J. 271

394 Author Index

- McMahon, R. 187
McMain, S. 37
McMillan, J. 231, 247
McNulty, J. L. 23, 24, 26, 27, 28, 29, 30, 33, 34, 38, 39, 41, 43, 69, 71, 114, 120
McNulty, J. 28, 34
Meagher, M. W. 151
Medoff, D. 3, 15
Meehl, P. E. 25, 27, 68, 69, 109
Meeks, M. 248
Mega, M. 330
Megargee, E. I. 15, 31, 35, 39, 41, 125
Meissner, C. A. 363
Mellins, C. A. 357
Mellor, D. J. 357
Melniker, A. A. 301
Meloy, J. R. 189, 209
Melton, G. B. 4–5, 7, 16, 37, 366
Melton, M. 187, 188
Mendelsohn, A. 31
Mendelson, M. 10
Mendham, M. C. 149, 150
Mendlowitz, S. **184**
Merbitz, C. 295–6
Mercier, H. 280
Merry, J. **184**
Merwin, M. M. **184**
Messina, N. **184**
Metevia, L. 358
Metzger, D. 153
Meyer, G. J. 206, 207, 208, 217
Meyer, J. E. Jr. 39, 41
Meyer, J. K. 163
Michie, C. 233, 238, 239, 240
Milberg, W. 85
Miles, D. R. 187
Milgrom, J. 357
Miller, A. K. 248
Miller, E. 295
Miller, H. A. 246
Miller, H. R. **186**
Miller, J. B. 302
Miller, J. D. 233, 241, 269
Miller, J. 28, 33
Miller, K. B. 189
Millis, S. R. 301, 302
Millon, C. 7, 10, 176
Millon, T. 7, 10, 111, 123, 175, 176, 177, 178, 183, 185, 186, 191
Mills, J. F. 242, 246
Millsaps, C. 302
Milne, L. C. 116
Mitchell, M. 90, 91, 92
Mitchell, W. E. 32
Mittenberg, W. 296, 300, 301, 302
Mittenburg, W. 32
Mobbs, D. 248
Mobley, B. D. **186**
Mock, J. E. 10
Modzierz, G. J. **182**
Moffitt, T. E. 282, 283
Mokros, A. 234, 239, 243
Molish, H. B. 204
Moll, J. 238, 247
Monachesi, E. D. 27, 31, 41, 110, 123, 124
Monahan, J. 37, 38, 39, 209, 238, 242, 296
Monahan, K. C. 269, 271, 277
Montag, I. 151
Mooney, P. 240
Moore, C. G. 357
Moore, D. 356, 357
Moore, R. J. 148, 149
Moran, G. 357
Moran, P. 238
Moretti, M. M. **268**, 282
Morey, L. C. 10, 17, 64, 111, 140, 147, 148, 149, 150, 151, 152, 153, 155, 156, 159, 160, 161, 162, 163, 184
Morgan, C. D. 190
Morgan, E. M. 182
Morgan, J. E. 297
Morgan, L. A. 366
Morgan, R. D. 15
Morrissey, C. 240
Morse, S. J. 213
Morton, T. L. 120, 121, 124, 125, 132
Moss, E. 356
Motzkin, J. C. 231
Mouton-Simien, P. 330
Mullen, K. L. 155
Müller, B. W. 238
Muller, E. 37
Müller-Isberner, R. 242
Mullins-Sweatt, S. 249
Mulvey, E. 269
Mulvey, E. P. 238, 242, 296
Munley, P. H. 191
Murphy, K. R. 124, 358
Murphy, T. J. **182**
Murray, H. A. 9
Murrie, D. C. 160, 248, 271, 272, 275, 276, 277, 280, 281
Muscewicz, J. 209
Mussio, S. J. 30
Muto, T. 358
Myers, W. C. 267

Nadeau, L. 357
Nair, P. 358
Nakagawa, M. 357
Nastasi, B. K. 365
Nathan, J. S. 21, 32
Nathan, J. 189
Nedopil, N. 234, 239

- Nelson, J. R. 358
 Nelson, N. W. 23, 27, 40, 75, 79, 80, 291, 296, 298, **298**, 299, 300, 301
 Nemes, S. **184**
 Netter, B. E. C. 218
 Neu, A. M. 357
 Neukrug, E. S. 85
 Neumann, C. N. 233
 Neumann, C. S. 230, 231, 232, 233, 234, 236, 238, 239, 240, 242, 245, 247, 250, 251, 268, 269, 272, 274, 275, 276, 280
 Neuringer, C. 37
 Neville, B. 358
 Newberry, M. 159
 Newhouse, P. A. 326, 328, 329, 330, 332, 337
 Newland, L. A. 357
 Newlove, T. **184**
 Newman, J. P. 230, 231, 234, 235, 236, 239, 240, 241, 272
 Newsom, C. R. 126
 Nezworski, M. T. 205
 Nezworski, M. 205
 Nicholls, T. 296
 Nichols, D. S. 364, 369
 Nicholson, B. C. 239, 242, 355, 357
 Nicholson, D. E. 242
 Nicholson, R. A. 32, 33, 37, 39, 149, 150
 Nickell, A. 152
 Nievar, M. A. 357
 Nijman, H. 37
 Nikolova, N. L. 159
 Nilsson, T. 238, 242, 274
 Nitschke, J. 243
 Noam, G. G. 330
 Noe, K. H. 75, 81, 82
 Norman, D. **185**
 Norman, W. 68, 69
 Nunez, C. 242, 246
 Nussbaum, D. 242
 Nutting, S. M. 30
 Nygren, M. 211
 Nyholm, J. O. 231

 Ochshorn, E. 140
 O'Connell, C. O. 114, 116, 124, **132**
 O'Connor, B. P. 122, 183, 242
 Odgers, C. L. **268**
 O'Donohue, W. T. 360
 O'Donohue, W. 360
 Offord, K. P. 109
 Ogles, B. M. 211
 Ogloff, J. P. R. 245
 Ogloff, J. R. P. 36, 280, 281
 O'Gorman, T. W. 282
 Oh, K. J. 358
 Okazaki, S. 35
 Okonek, A. **182**

 Olesen, N. 369
 Oliva, F. 188
 Oliver, C. J. 189
 Olmsted, C. L. 354, 358
 Olson, R. 185, 187, 188
 Olver, M. E. 152, 231, 240, 243, 244, 268, 269, 276
 O'Neill, M. L. 267, 272, 274, 277
 Onufrak, B. 357
 Oram, G. 242
 Orlando, M. J. 150
 Ornduff, S. R. 149
 Osborne, D. 75, 81, 82
 O'Shaughnessy, R. 267, 273, 274, 276, 280
 Ostacoli, L. 188
 Osterheider, M. 234, 239, 243
 Ostroff, C. 367
 Ostrow, D. **182**
 OSullivan, M. 363
 O'Toole, S. 207
 Otto, R. K. 4, 6, 7, 31, 37, 40, 189, 216, 296, 360
 Otto, R. L. 7, 16
 Ouente, A. 189
 Overduin, L. Y. 8
 Overholser, J. C. **182**
 Owen, M. T. 366
 Özokyay, K. 242
 Ozonoff, S. 75

 Pabst, J. 364
 Pace, T. M. 35
 Padawer, J. R. 210
 Paine, C. 108
 Paitich, D. 182
 Paizis, C. L. 153
 Pake, D. R. Jr. 247
 Pan, B. A. 358
 Pancoast, D. L. 110
 Panizzon, M. S. 153
 Paolo, A. M. 27
 Pardini, D. A. 277, 282
 Parker, J. D. 153
 Parkins, S. Y. 183
 Parkinson, A. **186**
 Parks, G. A. 276
 Partyka, D. J. 125
 Pate, J. L. 152
 Patrick, C. J. . 230, 231
 Patrick, C. J. 282–3, 38, 80, 125, 238, 239, 241, 267, 275, 282
 Patrick, J. **186**
 Patry, M. W. 159
 Patterson, M. P. 191
 Patton, C. 32, 296, 302
 Paul, B. D. 191
 Paulhus, D. L. 241, 370

- Payne, G. 30
 Peacock, E. J. 244
 Pearce, M. 246
 Peck, C. P. 89
 Pedersen, L. 242
 Pederson, D. R. 357
 Peebles, J. 148, 149
 Peed, S. F. 189
 Pena, L. M. 125
 Penedo, F. J. 188
 Penney, S. R. **268**
 Pennuto, T. O. 13, 122
 Pereira, N. 241
 Perry, G. G. 218
 Perry, J. D. 365
 Peters, J. M. 354, 358
 Peters, K. 191
 Peters, R. 153
 Peterson, C. A. 191
 Peterson, G. W. 144
 Peterson, M. 125
 Petrla, J. 4–5, 7, 16, 37, 366
 Petrocelli, J. V. 125
 Petroskey, L. J. 28
 Pett, M. A. 366
 Pettigrew, T. 208
 Pettit, P. 357
 Pham, T. H. 238, 242, 250
 Philippi, C. L. 231
 Phillips, N. 191
 Piacentini, T. 117
 Picarello, K. 363
 Picci, R. L. 188
 Piché, C. 355
 Pico-Alfonso, M. 189
 Piedmont, R. L. 332, 335
 Piers, C. C. 207
 Piers, C. 210
 Piersma, H. L. **182**, 191
 Pike, P. L. 35
 Pincus, A. L. 153
 Pine, D. 242
 Pinsoneault, T. B. 40, 92
 Piotrowski, C. 21, 110
 Pirelli, G. 37
 Plante, T. G. 85
 Platt, J. J. 184, **186**
 Platt, J. R. 41
 Pliskin, N. H. 296
 Plomin, R. 230, 245, 283
 Pochlyly, J. M. 191
 Podboy, J. W. 15
 Pogge, D. L. 117, 150, 207, 267, 274
 Pogge, D. 212
 Polusny, M. A. 75, 78
 Pope, H. S. 126
 Pope, K. S. 13
 Porter, S. 242, 243, 267, 268, 272, 273, 274, 275, 280
 Pothast, M. D. 31
 Powell-Lunder, J. 212
 Powers, B. K. 75, 92
 Poythress, N. G. Jr. 154, 155, 242
 Poythress, N. G. 4–5, 7, 16, 37, 151, 249, 271, 276, 277, 280, 366
 Poythress, N. 37
 Poznansky, O. 189
 Prentky, R. A. 276
 Prescott, C. A. 128
 Presley, G. 206, 217
 Prevatt, F. 208, 212
 Price, C. 356
 Priebe, S. 188
 Prinzhorn, B. 151
 Pritchard, P. B. 151, 157
 Probst, J. C. 357
 Procidano, M. E. 154
 Pruett, K. D. 366
 Pruett, M. K. 359, 361, 365, 366
 Przybeck, T. R. 184, **185**
 Puente, A. E. 21, 32
 Pukrop, R. **268**, 269, 272
 Purdon, S. E. 75, 88
 Purisch, A. O. 10
 Purser, S. M. 75, 88
 Putkonen, H. 242
 Putnam, S. H. 301
 Quay, H. C. 272
 Querido, J. G. 355, 357
 Quilty, L. C. 38, 75, 77
 Quinnell, F. A. 189, 359, 362
 Quinsey, V. L. 10, 242, 243, 244, 248
 Quinton, D. 282
 Radovanovic, H. 39
 Rahe, R. H. 154
 Raikes, H. 358
 Raine, A. A. 247
 Raine, A. 247
 Rajska, D. 358
 Ralevski, E. **184**
 Ramsay, J. 358
 Ramsey, S. H. 361, 363
 Rapier, J. L. 75, 89
 Rapport, L. J. 302
 Rasmussen, K. 242
 Rathouz, P. J. 358
 Rauch, S. L. 295, 296
 Rayls, K. 301
 Rector, N. A. 33
 Reddon, J. R. 69, 242
 Reddy, M. K. 75, 78
 Reed, P. 163

- Reef, J. 330
 Rees, A. 152
 Reidy, D. E. 242
 Reidy, T. J. 245
 Reimann, B. J. 242
 Reitan, R. M. 10
 Reitman, D. 350, 355, 356
 Rennie, C. 270, 271, 272, 277
 Repetti, R. L. 357
 Reppucci, N. D. 281
 Rescorla, L. A. 10, 17, 315, 319, 326, 327, 328, 329, 330, 332, 337
 Resendes, J. 92
 Resnick, J. 207
 Retzlaff, P. 185, 186, 189
 Rey, A. 8
 Reynolds, C. R. 330
 Reynolds, S. K. 27, 33
 Reynolds, W. M. 21
 Rice, M. E. 10, 242, 243, 244, 248, 279
 Rich, B. 355, 357
 Richards, H. J. 238, 239, 242, 246
 Richards, S. K. **182**
 Richman, N. 330
 Ricker, J. 297
 Ridenour, T. A. 275, 279
 Riemann, R. 332, 335
 Righthand, S. 276
 Riley, W. T. 159
 Rinaldo, J. 117
 Ritz, G. H. **184**
 Ritzler, B. 208, 217
 Robbins, B. 125
 Robbins, P. C. 238, 242, 296
 Robbins, R. R. 35
 Roberts, A. 238
 Roberts, B. W. 282
 Roberts, J. 357
 Roberts, R. 246
 Robertson, G. 191
 Robin, R. W. 34
 Robinson, J. 357
 Robinson, W. S. 367
 Rock, R. C. 75, 92
 Rockert, W. **184**
 Rockett, J. L. 281
 Rodriguez, C. M. 357
 Roesch, R. 212, 213
 Rogers, M. A. 358
 Rogers, R. 4, 6, 16, 23, 27, 32, 33, 37, 75, 89, 124, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 159, 160, 183, 189, 190, 233, 234, 238, 239, 242, 246, 270, 276, 296, 297
 Rogers, S. A. **299**
 Roggman, L. A. 357
 Rogstad, J. E. 276
 Roman, D. D. 32
 Rorschach, H. 202–3, 204, 205 206, 207, 208, 209, 211, 212, 213, 214, 215, 216, 217, 218, 219
 Rosenbaum, D. **331**
 Rosenberg, R. 188
 Rosenberg-Thompson, S. 330
 Rosenfeld, R. **184**
 Rosenthal, R. 208
 Roskamp, P. 245–6
 Ross, C. A. 150
 Rossi, G. **184**
 Rossini, E. 188
 Rossner, S. C. 218
 Rosso, M. 28, 30, 34
 Rostow, C. D. 30
 Rotermund, P. 238, 244
 Roth, L. **184**
 Rouse, S. V. 21, 27
 Roussy, S. 274
 Routh, D. K. 272
 Rowell, K. 358
 Rowell, R. K. 358
 Rudd, R. P. **184**
 Ruff, R. M. 296
 Rufino, K. A. 248
 Ruiz, M. A. 140, 143–4, 146, 147, 148, 151, 152, 153, 156, 159, 160
 Rupp, W. M. 29
 Ruscio, J. 249
 Rutherford, M. J. 146, 153, 158
 Rutter, M. 282
 Réveillère, C. 242
 Ryan, J. J. 27
 Ryba, N. L. 9
 Ryder, A. G. 149, 150

 Sadeh, N. 250
 Sahly, J. 207
 Saleh, F. M. 243
 Salekin, R. T. 75, 92, 151, 152, 153, 154, 159, 160, 183, 189, 190, 234, 238, 239, 242, 266, 268, 269, 270, 271, 272, 274, 275, 276, 277
 Salekin, R. 278
 Sales, B. D. 360, 362, 363
 Salmaso, S. 188
 Salokangas, R. K. R. 212
 Saloppé, X. 238, 250
 Samuel, D. B. 231
 Sandefur, G. 365
 Sanders, M. R. 358
 Sanderson, C. **184**
 Sandler, I. N. 357
 Sankis, L. M. **184**
 Santor, D. 267, 268, 272, 273, 274, 275, 280
 Santos, S. V. 352, 355
 Sarapata, M. 295
 Sass, H. 230
 Sattler, J. M. 10

398 Author Index

- Savodnik, I. 301
 Sawyer, R. J. 302
 Sawyer, T. 301
 Saxon, A. J. 187
 Saylor, C. F. 356, 357
 Saylor, C. 357
 Scalora, M. J. 246
 Scarr, S. 350, 353, 355
 Schaeffer, C. I. 184, **186**
 Schafer, R. 204
 Schaffer, C. M. 188
 Schare, M. L. 123
 Schember, J. 163
 Schenk, P. W. 28, 29
 Scherbaum, N. 238
 Schiffer, B. 238
 Schiffer, R. 297
 Schiller, E. W. 355
 Schinka, J. A. 26, 145, 146, 149, 151
 Schipper, L. J. 296, 298, 299, 300, 301, 302
 Schluger, R. P. 153
 Schmidt, F. 268, **268**, 272, 273, 274, 276
 Schmitt, W. A. 240, 241, 272
 Schneider, S. L. 30
 Schoenberg, M. R. 190
 Schoenfeld, L. S. 30
 Schopp, R. F. 246
 Schretlen, D. J. 297
 Schroeder, R. W. 89
 Schuerger, J. M. **184**
 Schug, R. A. 247
 Schuler, C. E. **184**
 Schuler, M. 358
 Schumph, D. 191
 Schutte, J. W. 189
 Schutz, B. M. 217
 Schwab-Stone, M. 330
 Schwartz, J. C. **182**
 Schwartz, J. P. 189
 Schwartz, K. 188, 191
 Schwartzman, A. E. 358
 Schwarz, W. 204
 Scogin, F. 30
 Scott, C. L. 153
 Scott, R. 123
 Scragg, P. 149, 150
 Seagrave, D. 279, 280, 282, 334
 Seefeldt, T. 356, 357
 Seelen, J. 13, 126
 Seeman, M. V. 33
 Seeman, W. 25, 27, 29, 110
 Seemann, E. 189
 Seever, M. F. 108
 Segal, D. L. **184**
 Segal, Z. V. 185
 Seibert, L. A. 242
 Sellbom, M. 23, 28, 29, 30, 33, 34, 36, 38, 39, 40,
 41, 43, 75, 80, 87, 88, 89, 90, 91, 92, 114, 116,
 124, 132
 Sellers, A. H. 41
 Sendin, M. C. 211
 Seo, S. 356
 Serbin, L. A. 358
 Serin, R. C. 243
 Sevecke, K. 234, **268**, 269, 272
 Sewell, K. W. 4, 6, 23, 27, 124, 146, 147, 150,
 151, 152, 153, 154, 159, 183, 189, 190
 Sewell, K. 148, 149
 Shaffer, T. W. 206, 207, 217
 Shandera, A. L. 301
 Shanley, L. A. 191
 Shapiro, D. L. 213
 Shapiro, S. K. 272
 Shaw, B. 185
 Shea, M. T. 242
 Shealy, S. E. 301
 Shelton, T. L. 357, 358
 Sheras, P. L. 347
 Sherman, E. M. S. 89
 Sherwood, N. E. 24, 113
 Shields, S. M. 38, 39
 Shigaki, I. S. 356
 Shine, J. 246
 Shiner, R. L. 282
 Shiva, A. 162
 Shondrick, D. D. 35
 Shorey, H. S. 162
 Shuker, R. 159
 Shuman, D. W. 360, 363
 Siddiqui, N. 146, 153, 158
 Siefert, C. J. 162
 Silberman, C. S. **184**
 Silver, C. H. 296
 Silver, E. 238, 242, 296
 Silver, J. M. 154
 Sim, H. A. 79, 80
 Simmons, A. 230, 250, 291
 Simmons, R. J. 356, 358
 Simms, L. J. 28, 30
 Simonds-Bisbee, E. C. 114, 115, 116, 124, **132**
 Simpler, A. 160
 Simpson, C. A. 153
 Sinclair, J. 238, 246
 Sinclair, S. J. 162
 Singles, J. M. 114
 Sinha, B. K. 183
 Sink, F. 39, 41
 Sjdin, A. 238
 Sjödin, A. 274
 Sjöstedt, G. 242
 Skeem, J. L. 38, 39, 154, 155, 240, 242, 243, 247,
 270, 271, 280, 281
 Skeem, J. 278
 Skopp, N. A. 160

- Slick, D. J. 8, 89
 Sliger, G. L. 41
 Sloan, P. 215
 Slobogin, C. 4–5, 7, 16, 37, 366
 Sloore, H. **184**
 Smart, C. M. 75
 Smith, B. E. R. 242, 243
 Smith, B. 204, 215
 Smith, C. 206, 217
 Smith, D. R. 64
 Smith, D. **185**
 Smith, H. H. 7, **8**
 Smith, I. 153
 Smith, M. 64
 Smith, P. 242
 Smith, R. 282
 Smith, S. S. 239, 241
 Snibbe, J. R. **184**
 Snowden, R. J. 240, 241, 242, 243
 Sobel, H. J. 29
 Soderstrom, H. 274
 Sokol, A. **182**
 Soldz, S. **184**
 Solis, M. L. 350, 352
 Solway, K. S. 30
 Sommer, J. 242
 Sonoda, N. 358
 Souheaver, G. T. 301
 South, C. S. 239
 Spain, S. E. 271, 276, 280
 Sparrow, D. 29
 Speanberg, S. 212
 Speltz, M. L. 356
 Spielberger, C. D. 154
 Spiers, M. V. 294
 Spiro, A. 29
 Spitzer, R. L. 88
 Sponheim, H. R. 79, 80
 Sprague, J. 250
 Sreenivasan, S. 247
 Stack, D. M. 358
 Stackman, W. 364
 Stadtland, C. 234
 Stafford, E. 242
 Stafford, J. E. 276, 280
 Stafford, K. P. 28, 35, 36, 37, 38, 40, 75, 80, 90, 212
 Stage, S. 358
 Stahl, P. M. 7
 Stålenheim, G. 242
 Stanger, C. 330
 Stanger, E. 188
 Stankovic, S. **182**
 Starling, J. 273
 Starzyk, K. B. 148
 Stadtland, C. 239
 Stattin, H. 208, 270
 Steadman, H. J. 238, 242, 296
 Steenkamp, M. M. 40
 Steer, R. A. 10, 151, 154, 330
 Steffan, J. S. 15
 Stein, D. D. 214
 Stein, L. A. R. 34, 114, 117
 Steinberg, L. 280
 Steiner, D. L. **186**
 Steiner, H. 273
 Stejskal, W. J. 205
 Stenner, A. J. 64
 Stepanian, M. L. 218
 Stepp, S. 152
 Steptoe, L. 240
 Steuerwald, B. L. 243, 272, 274, 275, 276, 280
 Steurwald, B. L. 275
 Stevenson, J. 330
 Stevenson, M. 75
 Ståhlberg, O. 238
 Stickle, T. R. 350, 355, 356
 Stiening, J. A. 30
 Stile, S. A. **184**
 Stockdale, K. C. 244, 268, 269, 276
 Stokes, J. L. 267, 274
 Stokes, J. M. 207
 Stokes, J. 212
 Stone, D. 123
 Stone, K. 330
 Stone, L. A. 30
 Stone, W. L. 356
 Stoner, J. W. 189
 Stoner, J. 189
 Stoppelbein, L. 357
 Storm, A. 30
 Storm, J. 33
 Stoughton, N. L. **184**
 Stouthamer-Loeber, M. 240, 282
 Strack, S. 178, 189
 Strauss, E. H. 8
 Strauss, E. 89
 Stredney, R. V. 7, 8, 10, 13, 75, 92
 Strenger, V. E. 29
 Stricker, G. 16
 Stroemel, J. 301
 Strong, D. R. 369
 Stuart, G. L. 242
 Sturgeon, J. 75
 Sue, S. 35
 Suhr, J. A. 298, 299, 301, 302
 Sullivan, A. 358
 Sullivan, E. A. 240, 243, 268
 Sullivan, J. P. 291
 Sullivan, K. 302
 Sumanti, M. 301
 Sumida, E. 356, 357
 Sumpter, J. C. 27
 Sundberg, D. 21

- Sutker, P. B. 31
 Svirskis, T. 212
 Svrakic, D. M. 184, **185**
 Sweet, J. J. 13, 23, 27, 40, 75, 291, 296, 297, 298, **298**, 299, 300, 301
 Switzer, O. **182**
 Swogger, M. T. 238, 242, 247, 275
 Synhorst, L. 358
- Tam, K.-K. 352, 355
 Tamminga, C. A. 295, 296
 Tan, J. E. 8
 Tannock, R. 358
 Tarnowski, K. J. 357
 Tasca, G. A. 151, 162
 Taylor, C. A. 358
 Taylor, E. R. 150, 154, 155, 160
 Taylor, J. L. 240
 Taylor, J. 240, 242, 283
 Taylor, M. J. 357
 Taylor, N. 207
 Teitler, J. 365
 Tellegen, A. 7, 9, 10, 23, 25, 26, 28, 29, 30, 33, 34, 43, 63, 69, 70, 71, 72, 73, 74, 77, 81, 83, 85, 86, 87, 88, 101, 113, 114, 121
 ten Brinke, L. 242, 243
 Tengström, A. 242, 243, 268, 269, 271, 273, 274
 Tennant, A. 152
 Teplin, L. A. 273
 Tessier, R. 357
 Testa, S. M. 151
 Teti, D. M. 357
 Theroux, S. 301
 Theroux-Fichera, S. 301
 Thomas, E. A. 207
 Thomas, K. M. 150
 Thomas, M. L. 75, 81, 82
 Thompson, G. B. 89
 Thorndike, R. L. 10
 Thornton, D. 160, 242, 246
 Thornton, D. 244
 Thuras, P. 75, 78
 Tillbrook, C. E. 6
 Timbrook, R. E. 26, 27, 34, 69
 Timmons, D. 243
 Tippins, T. 360
 Tirrell, C. A. 120
 Tolin, D. F. 40
 Tomanik, S. 358
 Tomarken, A. 152
 Tombaugh, T. N. 8
 Tomony, J. 242, 243
 Tompkins, M. 357
 Toomey, J. A. 33, 75, 88
 Toomey, J. P. 149
 Topping, K. B. 151, 157
 Torgersen, S. **186**
- Toupin, J. 274, 280
 Tovar-Moll, F. 247
 Trapnell, P. D. 155
 Treece, C. **184**
 Treiber, F. A. 159
 Trimalas, K. A. **182**
 Tringone, R. 178
 Troughton, E. 282
 Trout-Landen, B. I. **184**
 Tröster, A. I. 296
 Trull, T. J. 152
 Trute, B. 358
 Tsai, D. C. 35
 Tsushima, W. T. 29, 40, 75
 Tucker, D. D. 149, 150, 151
 Tuley, M. R. 32
 Tulloch, R. M. 182
 Tulskey, D. S. 301
 Turley, B. **184**
 Turner, C. W. 366
 Turner, D. B. 248
 Tussey, C. 248
 Tyson, J. 204
- Ullrich, S. 238
 Umstead, D. 278
 Useda, J. D. 152
 Ustad, K. L. 150, 151, 152, 153, 154, 159, 160
 Ustad, K. 146, 147
 Utendale, K. A. 191
- Vaccaro, T. P. 2
 Vacha-Haase, T. 191
 Vachon, D. D. 240
 Vaidyanathan, U. 238
 Valle, L. A. 358
 Vandenburg, E. 191
 Vandenberg, H. 330
 van den Brande, I. **184**
 van der Ende, J. 330
 van Gorp, W. G. 296, 298, 299, 301
 van Meurs, I. 330
 van Patten, I. 239
 van Rybroek, G. J. 275, 278
 van Rybroek, G. 278
 Vandell, D. L. 366
 Varela, J. G. 30
 Varghese, J. 358
 Vaughan-Cole, B. 366
 Velásquez, R. J. 128
 Veltri, C. O. C. 38, 114, 116, 124, **132**
 Vendrig, A. A. 29
 Verhaeghe, P. 242
 Verhulst, F. C. 330, **331**
 Vermulst, J. R. M. 355
 Verona, E. 242, 250
 Verschell, M. S. 207

- Vidal, S. 281
 Viding, E. 230, 231, 245, 283
 Viemerö, V. 274
 Vieth, E. R. 238, 244
 Viglione, D. J. 207, 208, 218
 Vigna-Taglianti, F. 188
 Viljoen, H. 296
 Viljoen, J. L. 10, 279, 281
 Viljoen, S. 296
 Villanueva, M. R. 32
 Vincent, G. M. 10, **268**, 270, 274, 275, 276, 280
 Vincent, G. 234, 280, 334
 Vitacco, M. J. 23, 27, 230, 233, 234, 240, 242, 243, 274, 275, 276, 277, 280
 Vitale, J. E. 235, 239
 Vitale, J. 242
 Vokonas, P. S. 29
 VonDran, E. J. 89

 Wadsworth, G. W. 109
 Wagner, A. E. 354, 358
 Wagner, E. 151
 Wagner, M. T. 151, 157
 Wahlund, K. 247
 Waisbren, S. E. 357
 Wakschlag, L. S. 356
 Walfish, S. 214
 Walker, S. C. 247
 Walker-Matthews, S. 272, 274, 275, 276, 280
 Wallinius, M. 242
 Wallis, R. R. 108
 Walsh, T. 242, 245, 246
 Walsh, Z. 17, 238, 242, 243, 245, 246, 247, 275
 Walters, G. D. 153, 154, 242, 243, 246, 247
 Walters, J. 191
 Walters, P. 188
 Wampold, B. E. 366
 Wang, E. W. 150, 151, 153, 154, 155, 160
 Ward, A. 212
 Ward, C. H. 10
 Warner, M. B. 64
 Warren, J. I. 239
 Warren, J. 242, 363
 Washington, J. 356, 358
 Wassell, G. 358
 Wasylwi, O. E. 32, 218
 Wasylwi, O. 218
 Wasylkiw, L. 148
 Waterman, J. 273
 Waters, D. 301
 Waters, W. F. 29
 Watkins-Clay, M. M. 151
 Watson, A. 152
 Watson, C. 75, 77
 Watson, D. C. 183
 Watson, D. 28, 69
 Watt, K. 280

 Watts, D. 27
 Waugh, M. H. 152
 Wayland-Smith, D. 207
 Weathers, F. W. 150
 Weaver, T. 230
 Webb, J. T. 369
 Webster, B. K. 89
 Webster, C. D. 38, 39, 242, 243
 Webster-Stratton, C. 354, 355, 357
 Wechsler, D. 7, 9–10
 Weed, N. C. 24, 28, 29, 113
 Weiler, B. L. 238
 Weinberg, D. 187, 188
 Weinberger, L. E. 247
 Weiner, I. B. 13, 17, 204, 208, 209, 210, 211, 215, 217, 219
 Weinman, B. A. 159
 Weinstein, S. R. 330
 Weiss, J. A. 358
 Weiss, S. T. 29
 Weizmann-Henelius, G. 242, 274
 Wells, E. A. 187
 Welsh, G. S. 24, 27, 28, 31, 33, 34
 Werner, E. 282
 Wershba, R. 75
 Wertheimer, J. C. 301
 Wetter, M. W. 27, 33, 117
 Wetzler, S. **184**
 Whalen, R. E. 29
 Wheeler, D. S. **182**
 White, A. J. 242, 246
 White, C. 301
 White, R. J. 189
 White, R. S. 114, 116, 124, **132**
 White, S. F. 230
 Whiteman, V. 358
 Whiteside, D. 301
 Whitman, L. A. 272, 274
 Widiger, T. A. 27, 184, 191, 231, 240, 241, 249, 250
 Widiger, T. **184**, 190
 Widom, C. S. 238
 Widows, M. R. 233
 Wiener, J. 358
 Wierzbicki, M. **184**
 Wiggins, J. S. 151, 155
 Wiggins, J.S. 24
 Wilfong, E. W. 357
 Wilfong, E. 357
 Willemsen, J. 242
 Williams, C. L. 9, 10, 24, 113, 114, 120, 123, 130, **133**, 151
 Williams, C. W. 7, **8**
 Williams, D. 154
 Williams, J. B. W. 88
 Williams, K. M. 241
 Williams, R. B. 29

402 Author Index

- Williams, T. Y. 365
Williams, T. 359
Willner, P. 357
Wilkinson, R. A. 41
Wilson, K. 242, 243
Wilson, R. J. 247
Wilson, S. K. 336
Wiltfang, J. 238
Winstanley, M. 356, 357
Wirt, R. D. 124
Wirth, O. 357
Wise, E. A. 184, 189
Wish, E. **184**
Witnitzer, M. **331**
Wittman, J. 360
Wolchik, S. A. 357
Wolfe, D. A. 336
Wolfe, J. 278
Wolfe, R. 357
Wong, C.-K. M. 352, 355
Wong, D. F. 357
Wong, S. C. P. 231, 240, 242, 243, 244, 268, 276
Wood, J. M. 205
Wood, J. 151, 162, 357
Woods, C. 248
Woods, M. G. **182**
Woodworth, M. 242
Wooley, C. N. 150
Worden, W. 29
Wormith, J. S. 244
Wright, C. 357
Wright, H. H. 356
Wrobel, N. H. 128
Wupperman, P. 230
Wygant, D. B. 13, 29, 36, 37, 40, 75, 80, 88, 89, 90, 91
Wymer, J. H. 151, 157
Yang, M. 238, 242
Yang, Y. 247
Yarvis, R. M. 153
Yates, P. M. 244
Yeomans, F. E. 152
Yeomans, F. 153
Yiend, J. 238, 246
York, D. 209
Yossef, S. 27, 32
Young, G. H. 357
Young, J. C. 302
Young, R. W. 22
Youngjohn, J. R. 75
Yudofsky, S. C. 154
Yuille, J. 231
Zaballero, A. R. 146, 153, 158
Zaccario, M. 117, 207
Zalot, A. A. 268, 269, 275
Zapf, P. A. 9, 37, 212, 213
Zarella, K. L. **184**
Zeichner, A. 242
Zeier, J. 231
Zeller, R. A. 270
Zieben, M. 30
Zielinski, R. E. 301
Ziemke, M. H. 277
Zillmer, E. A. **293–4**, 294, 295, 298
Zimberoff, A. 301
Zinger, I. 280
Zinn, S. 116
Zolondek, S. 241
Zuckerman, M. 240
Žukauskienė, R. 234

SUBJECT INDEX

Locators in **bold** refer to figures/tables

Abbreviations used in subheadings

ASEBA = Achenbach System of Empirically Based Assessment

CBCL/6–18 = Child Behavior Checklist for Ages 6 to 18

MCMI = Millon Clinical Multiaxial Inventory

MMPI = Minnesota Multiphasic Personality Inventory

MMPI–A = Minnesota Multiphasic Personality Inventory–Adolescent

MMPI–2 = Minnesota Multiphasic Personality Inventory–2

MMPI–2–RF = Minnesota Multiphasic Personality Inventory–2–Restructured Form

PAI = Personality Assessment Inventory

PCL = Psychopathy Checklist: R=Revised; SV= Screening Version; YV=Youth Version

PSI = Parenting Stress Index

RIM = Rorschach Inkblot Method

AAS (Addiction Admission Scale) 24

ABCL (Adult Behavior Checklist) **315**, 326, **328**, **329**

Aberrant Experiences Scale, MMPI–2 23, 43

abstract reasoning 294

abuse, childhood 305

academic achievement 275, 365–6

academic journals 4

acculturation, ethnic minorities 35

Achenbach System of Empirically Based Assessment (ASEBA) 17, 311, 342; case example 338–42, **339**; clinical uses/limitations 332–3; cross-informant comparisons 319, **322**; forensic uses/limitations 333–8; history of development 324–7; instruments 311–15, **312**, **313**, **314**; multicultural versions 326–7; narrative reports/critical items 323–4, **323–4**; norms 324–5; psychometric characterization 324–5, 327–32, **329**, **331**; scales **321**; scoring profiles 316–19, **317**

acquiescent response style 117

ACRM (American Congress of Rehabilitation Medicine) 79

acting crazy/hurt exaggerations 33

acting out 31; and psychopathy 283

Activation Scale, MMPI–2–RF 77, 78

adaptation 311; psychopathy as 248, 280; scientific basis 177–8

adaptive competence, ASEBA **329**. *see also* Achenbach System of Empirically Based Assessment

Addiction Admission Scale (AAS) 24

ADHD (Attention Deficit Hyperactivity Disorder) 273

adjustment reaction of childhood diagnostic category 324

adjustment scales: MMPI–2 29; MMPI–2–RF 74; PSI 358

administration/interpretative issues. *see* interpretation

admissibility into evidence: PSI 363; RIM 208–9

Adolescent Interpretive System, MMPI–A 130

- adolescents, psychopathy 280, 282. *see also* Hare Psychopathy Checklist
- Adult Behavior Checklist (ABCL) **315**, 326, **328**, **329**
- Adult Self-Report Scale **315**, 326, **321**
- Aesthetic-Literary Interests Scale, MMPI-2-RF 72
- affective scales: PCL-R **234**, 235; PCL:YV **269**, 272
- African Americans: MMPI-2 34-5, 42-51, **44-52**; PCL:YV 268. *see also* ethnic minorities
- age: CBCL 311-14, **312**, **314**, **321**, 324-5, 327-30, **328**, **329**, 332; PCL-R 238; PCL:YV 266-7
- aggression: impulse control problems 244; personality style 188; and psychopathy 274-5; self-directed 272; and traumatic brain injury 295-6, 302. *see also* hostility; violence
- aggression scales: ASEBA 334; MMPI-A 126, 130; MMPI-2 31, 42; PAI 153, 154, 159-60; PCL:YV 273, 274-5
- airline pilots, employment screening 30
- alcohol abuse, case example 192-5, **193**. *see also* substance abuse
- alcohol abuse scales: MCMI 188; MMPI-2 39; MMPI-A 134; PAI 145, 153, 158-9, 164; PAI-A 163
- alienation 123
- American Board of Forensic Psychology diplomates 4, 9-10, **11**
- American Congress of Rehabilitation Medicine (ACRM) 79
- American Pain Society 8
- American Psychiatric Association, DSM. *see* *Diagnostic and Statistical Manual of Mental Disorders*
- American Psychological Association (APA) 1, 3, 4, 8, 10, 176; ethical codes 369, 370
- American Psychology-Law Society (APLS) 13
- anacletic problems 211. *see also* attachment relationships; dependent personality styles
- anger management, and psychopathy 284
- anger scales: MMPI-A 134; MMPI-2 39, **51**; MMPI-2-RF **100**
- Antisocial Attitudes Scale, MMPI-2 39
- antisocial behavior, and psychopathy 233
- antisocial behavior scales: MMPI-A 124, 132; MMPI-2 23; MMPI-2-RF 80, 97; PAI 144; PCL-R 242-4; PCL:YV 273;
- Antisocial Features Scale, PAI 145, 152-3, 158, 159-60
- antisocial personality disorder 38; adolescents 121; personology model 191, **192**; and psychopathy 250
- antisocial personality disorder scales: ASEBA 334; MCMI 179, **182**, 182-3, 183-5, **184**; MMPI-2-RF 101-2; PAI 165-7, 169; PAI-A 163; PCL:YV **269**, 276; PCL-R 241, 246-7
- antisocial practices scales, **100**; MMPI-2 **51**; MMPI-2-RF; PCL-R **234**, 235
- Antisocial Process Screening Device (APSD) 270-1, 277
- anxiety: and divorce 366; and psychopathy 272-3; trait 232
- anxiety scales: MMPI-2 24, 28, 45; MMPI-2-RF 78, 98, 101-2; PAI 151, 157, 167; PCL:YV 272-3; RIM 214, 215
- APA. *see* American Psychological Association
- APLS (American Psychology-Law Society) 13
- APSD (Antisocial Process Screening Device) 270-1, 277
- ASEBA software 315, 316. *see also* Achenbach System of Empirically Based Assessment
- Asian Americans 35
- assessment measures 3, 5-7; *see also specific instruments*
- attachment relationships 357, 366
- Attention Deficit Hyperactivity Disorder (ADHD) 273
- Australia 31
- automatic scoring: ASEBA 315, 316; MCMI 176, 194-5; MMPI-2 22, 33, 36; MMPI-A 129, 130; MMPI-2-RF 64-5
- autosuggestion 2
- avoidant behavior 188, 211, 214, 215. *see also* guardedness
- back pain 83
- back random responding 147, 155
- Back-Infrequency Scale, MMPI-2 22
- BASC (Behavior Assessment System for Children) 330
- base rate (BR) scores, MCMI 179, 181, 190
- batterers' intervention program 92
- Beck Anxiety Inventory 10, 13, 330
- Beck Depression Inventory **8**, 10, 13, 330
- Beck Hopelessness Inventory 154
- Behavior Assessment System for Children (BASC) 330
- behavior-brain relationships 294, 302; *see also* neuropsychological assessment
- Behavior Problem Monitor (BPM) **315**, 315, **321**
- behavioral pediatrics 348
- behavioral scales: MMPI-2-RF 72, 77, 97, **98**; RIM 203
- Bender-Gestalt Visual Motor Test 7, **8**
- Benton Facial Recognition Test 304
- Best Interests of the Child standard (BIS) 366
- biases: eyewitness testimony 2; MMPI-2 35, 51; PAI 144-5, 145; racial 191; traditional clinical tests 15. *see also* response bias
- Bibliography of Published Studies Using the ASEBA* 329, 330, 333
- bidirectional variables, RIM 210
- bioevolutionary theory of personality 177, **178**

- bipolar disorder 75, 77–8. *see also* manic disorder scales
- bizarre mentation scales: MMPI–2 38, 45, **51**; MMPI–2–RF **100**
- Borderline Features Scale, PAI 145, 152, 158, 159, 167, 168
- borderline personality disorder scales: MCMI **186**; MMPI–2 31; PAI 152
- Boston Naming Test **8**, 303
- BPM (Behavior Problem Monitor) **315**, 315, **321**
- BR (base rate) scores, MCMI 179, 181, 190
- brain–behavior relationships 294, 302; *see also* neuropsychological assessment
- breadth of content coverage, PAI 143–4
- Bricklin Perceptual Scales 9
- Brief Problem Monitor **315**, **321**, 337
- brief-focused evaluation (BFE), PSI 360
- buffering factors, psychopathy 282, 354
- Buros *Mental Measurements Yearbook* 230, 368
- CAI. *see* competency to stand trial assessment instruments
- callousness 284
- Canada 247, **268**
- cancer patients, psychosocial adjustment 29
- Cannot Say (?) Scale: MMPI–A 109, 131; MMPI–2 22
- capital punishment hearings: neuropsychological assessment 303–6; PCL–R 245–6; surveys of test use 16
- Caregiver–Teacher Report Form (C–TRF) **315**
- case examples **339**; ASEBA 338–42 MCMI 192–5, **193**; MMPI–A 130–4, **131**, **133**; MMPI–2 42–51, **44–52**; MMPI–2–RF 92–102, **94–101**; neuropsychological assessment 303–6; PAI 163–9, **165**, **166**, **167**; PCL–R 251–4; PCL:YV 283–4; PSI 370–1; RIM 219–24, **221**, **222**, **223**
- Cashel Discriminant Function (CDF), PAI 149, 156, 164
- Category Test: surveys of test use 7, **8**
- Chaotic Representations, Grossman Facet Scale 181
- childhood abuse, neuropsychological assessment 305
- Child Behavior Checklist 124, 131, 335, 336, 337; surveys of test use 10, 13
- Child Behavior Checklist for Ages 6 to 18 (CBCL/6–18) 311–14, **312**, **314**, **321**, 324–5, 327–30, **328**, **329**, 332
- child custody evaluations 4, 10, 16–17; ASEBA 338–42; case example 370–1; expert testimony 1, 4; MCMI 187; MMPI–A 123; MMPI–2 39–40; MMPI–2–RF 92; PCL–R 245; PSI 359–70, **368**; RIM 216–17; surveys of test use 9, 13, 16. *see also* Achenbach System of Empirically Based Assessment; Parenting Stress Index
- Child Domain, PSI 346, **347**, 354, 355, 358, **368**, 371
- child molesters 237
- Child Psychopathy Scale (CPS) 271
- childhood trauma 273
- Children’s Apperception Test 9
- chronic pain. *see* pain
- civil law, expert testimony 3–4
- Civil Rights Act (1991) 71
- Classification of Violence Risk 296
- clinical assessment: case example 43, **44**, **45**, 51; child custody evaluations 40; criminal behavior 31; ethnic minorities 34, 35; and forensic assessment 4–5; K correction 33–4; post-traumatic stress disorder 41; risk assessments 38; surveys of test use 7–17, **8**, **11–12**, **14**, **15**; uses/limitations **15**, 16, 36; vs. specialized forensic tests 5–7. *for specific examples see clinical scales*
- clinical interviews, ASEBA 341
- Clinical Personality Patterns, MCMI 175
- clinical scales/applications: ASEBA 332–3; MCMI 187–9; MMPI–1 68; MMPI–A 113–15, 115, 120–2; MMPI–2 23, 27, 28–31, 37, 73; MMPI–2–RF 69, 70, 71, 73, 80–4; PAI **141–2**, 151–3, 156–8, 164–7; PCL–R 238; RIM 209–12
- CMR (Comprehension of Miranda Rights) 5–6
- coaching in test-taking: MMPI–2 33; RIM 218
- code types, MMPI–2 23, 25, 27, 28, 29, 30, 34
- coding criteria/guidelines, RIM 207
- cognitive-behavioral psychotherapy: indications 211; for psychopathy 284
- cognitive functioning: feigning 297, **298**; and forensic psychology 291; and parental divorce 365–6; psychotherapy indications 211; surveys of test use 9, 16. *see also* neuropsychological assessment
- cognitive functioning scales: MMPI–2 21; MMPI–2–RF 78, 91, **97**, 98; PAI 161–2; RIM 213
- collateral information 231–2, 267, 283. *see also* data collection
- common sense 2
- communication, neuropsychological assessment **293–4**
- Comparison Group data, MMPI–2–RF 76
- compensation-seeking evaluations, MMPI–2–RF 89
- competence, adaptive, ASEBA **329**
- competency to be executed 3
- competency to stand trial assessment instruments (CAI) 1, 3–4, 7, 9, 11, **12**; ASEBA 316, **317**; MMPI–A 122; MMPI–2 36–7; MMPI–2–RF 92–102, **94–101**; RIM 212–14; surveys of test use 9, 10, **11–12**, 13, 15
- Competitive Drive Scale, MMPI–2–RF **100**

- Comprehension of Miranda Rights (CMR) 5–6
 Comprehensive System (CS), RIM 202, 204, 205, 206, 213, 217
 compulsive personality disorder **193**
 compulsive/dependent personality styles 188
 computer administration. *see* automatic scoring
 conceptualization, RIM 202, 204, 207
 concussion 75, 78–80. *see also* traumatic brain injury
 conduct disorder: ASEBA 334; PCL:YV 272, 273; MMPI–A 115–16, 125, 126, 130, 132–4; PAI 166; PAI–A 163
 confidentiality 5
 confirmatory bias, traditional clinical tests 15
 Conners' Parent Rating Scale 10
 constricted Rorschach protocol 214, 215
 construct validity: forensic assessment instruments 7; MMPI–2–RF 76; PCL:YV 270–1; psychodiagnostic assessments 77–8. *see also* convergent validity
 content component scales: MMPI–A 113; MMPI–2 29, 39, 45–6, **51**
 content scales: MMPI–A **112**, 113, 114, 123, 125, 132, **133**; MMPI–2 24, 27, 34, 38–9, 45, **47**, 51, 68–9; MMPI–2–RF 73
 content validity, PCL:YV 270
 convergent validity: ABSD 270–1, 277; BASC 330; Beck inventories 154, 330; CBL 124; CPS 271; MCMI 183–5, **184**, **185**; MMPI 208; MMPI–A 111, 124; MMPI–2 28, 150, 330; MMPI–2–RF 76, 77, 80, 82, 89, 90, 91; PAI 149, 150, 151, 152–3, 154–5; PCL 183; PCL–R 241; PCL:YV 270–3, 276–7; PCL:SV 152–3; PSI 363; RIM 152, 208, 214
 cookbook approach to test interpretation 68, 76
 Cooperative Movement Scale, RIM 217
 Coping Deficit Index (CDI), RIM 206
 coronary heart disease, risk assessments 29
 Correction Scale, MMPI–2 23
 correctional interventions, MMPI–2 41–2
 Correctional Software Report, PAI 146
 Council of Representatives, APA 1, 4
 CPS (Child Psychopathy Scale) 271
 criminal behavior: and antisocial personality disorder 183; MMPI–2 31; motivations 274–5; PCL–R 242–4; RIM 212–14
 criminal law, expert testimony 3
 criminal responsibility: MMPI–2 37–8; RIM 212, 213; PCL–R **234**, 244, 247–8; surveys of test use 9, 13
 criminal risk assessment. *see* risk assessment; violence risk
 criterion keying 108
 critical item analysis, PSI 368
 Cronbach's alpha 74
 cross-cultural applicability: ASEBA 326–7, 336–7; PCL–R 239–40; PSI 352–3; RIM 217. *see also* ethnic minorities
 cross-informant comparisons, ASEBA 319, **322**, 330
 cross-informant syndromes, ASEBA 325
 cross-validation norm studies, PSI 352–3
 CS. *see* Comprehensive System
 CST (Competency Screening Test) **11**
 culpability, legal: *see* criminal responsibility
 cultural differences. *see* cross-cultural applicability; ethnic minorities
 cut scores: PCL–R 231, 249; PCL:YV 280
 cyclothymic disorder 220. *see also* bipolar disorder
 cynicism scales: MMPI–2 23, **51**; MMPI–2–RF **100**
 dangerous and severe personality disorder (DSPD) 246–7
 Dangerous Offenders Scale, PCL–R 247
 data collection: multimethod/multi-source approach 361–2; neuropsychological assessment **292–3**. *see also* collateral information
Daubert v. Merrell Dow Pharmaceuticals criteria 2, 3; child custody evaluations 10, 39; MCMI 185, 189, 190; MMPI–2 31–2, 32, 39; MMPI–2–RF 85; neuropsychological assessment 295; PSI 363; RIM 209
 Debasement Index, MCMI 176
 decision-making ability 291; legal culpability 247; RIM 216. *see also* neuropsychological assessment
 defensiveness: child custody evaluations 39–40; neuropsychological assessment 296
 defensiveness scales: MMPI–A 109, 117, 126, 132; PAI 148–9, 156, 163; PSI 367, 369–70; MMPI–2 32–3
 definitions: feigning 296; forensic psychology 1; malingering 32, 296; reliability 295; syndrome 324
 delinquent behavior 3; ASEBA 324; MMPI–A 123–4, 125, 132; MMPI–2 31, 42; PCL–R **234**; PCL:YV **268**, 272
 delirium 64
 delusional symptomatology: case example 48, 49; MMPI–2–RF 101–2
 demand characteristics, child custody evaluations 363–4
 demographic variables, MMPI–2 26
 demoralization scales: MMPI–2 23, 28, 41, 43; MMPI–2–RF 69–70, 71, 77, 87, 96–7; PAI 154
 dependent personality styles 188
 depression, and divorce 366
 depression scales: ASEBA 325; MCMI 183–5, **185**, 187; MMPI–A 115, 122; MMPI–2 23, **50**, **51**; MMPI–2–RF 77–8, 83, 87, **100**; PAI 145–6, 148, 151, 157; PCL:YV 272–3; PSI 347, 357; RIM 209, 210
 depth of content coverage, PAI 144

- Desirability Index, MCMI 176
- development, and psychopathy 282
- Diagnostic and Statistical Manual of Mental Disorders* (DSM–IV–TR) 32; ASEBA scales 319, **320**, 325, 326, 329, **329**, 330; RIM 223
- diagnostic criteria: feigning 296; personality disorders 175, 176, 179, 182, 250; psychopathy 250, 281
- diagnostic efficiency/power, MCMI 185, **186**, 186, 187, 194
- diagnostic predictions, MMPI–A 127
- diagnostic thresholds, cut scores 231, 249
- Digit Span subtest, Wechsler Adult Intelligence Scale 301
- diminished capacity 3
- diplomates, American Board of Forensic Psychology 4, 9–10, **11**
- Direct Observation Form (DOF), ASEBA 316, 324, 330, 332
- disability claims, MMPI–2–RF 92
- Disclosure Scale, MCMI 181
- Disconstraint Scales: MMPI–2 24, 39; MMPI–2–RF 72, 97, **99**, 100
- discriminant function, neuropsychological assessment 301
- discriminant validity: MMPI–A 124; MMPI–2 28; MMPI–2–RF 76; PAI 145
- disordered thinking scales: PAI 161–2; RIM 209, 212, 213
- dispositional planning, MMPI–A 122, 124
- Disruptive Behavior Rating Scale 124
- dissimulation 148–9. *see also* impression management
- distrust 125
- divergent validity, PCL:YV 272–3
- divorce 365, 365–6; . *see also* child custody evaluations
- domestic violence 188–9
- Domestic Violence Risk Appraisal Guide (DVRAG) 243
- Dominance Scale, PAI 154–5
- domination of/by others 183
- DRG (drug problems). *see* substance abuse
- driving offences, case example 283–4
- drug addiction. *see* substance abuse
- drug court completion 92
- drug trafficking, case example 92–3, **94–101**, 101–2, 283–4
- DSM. *see* *Diagnostic and Statistical Manual of Mental Disorders*
- DSPD (dangerous and severe personality disorder) 246–7
- Dusky v. United States* (1960) 36
- dynamic risk factors 38–9, 49
- Dysfunctional Negative Emotions Scale, MMPI–2 23
- Dysphoria Scale, MMPI–2–RF **100**
- early intervention, psychopathy 279, 280, 281. *see also* treatment
- EFA (exploratory factor analysis), ASEBA 325
- education level, MMPI–2 26, 39–40
- electroencephalography 297
- emotional damages, surveys of test use 9
- emotional functioning: psychotherapy indications 211; surveys of test use 16
- emotional functioning scales: MMPI–2 23, 28, 29; MMPI–2–RF 77, 91; PAI 167; RIM 202, 213, 216
- emotional poverty 230
- empathy 244, 246
- empirical keying 108
- empirically based syndromes, ASEBA 316, **318**, 319, 324, **329**
- employment screening, MMPI–2 30
- epileptic and non-epileptic seizures: MMPI–2–RF 75, 80–2; PAI 151; psychological assessment of medical patients 80–2
- Ethical Principles of Psychologists and Code of Conduct* (APA) 3
- ethical standards/codes 3, 295; APA 369, 370
- ethnic minorities: MCMI 191; MMPI–A 127–8; MMPI–2 26, 34–5, 42–51, **44–52**; MMPI–2–RF 72; PCL–R 239–40; PCL:YV 268, 279; RIM 217
- evolutionary psychology, and psychopathy 248
- executive functioning. *see* cognitive functioning
- expert testimony: child custody evaluations 4, 363; civil law 3–4; criminal law 3; historical perspectives 1, 2, 3; MMPI–2 31–2; neuropsychological assessment 295; psychopathy 250–1; PCL–R 248–9; responsibility of 16
- exploratory factor analysis (EFA), ASEBA 325
- Explosive Behavior Scales: MMPI–A 133; MMPI–2–RF **100**
- expressive anger, case example 48
- Extended Score Reports: MMPI–A 129; MMPI–2 33–4
- externalizing behaviors/disorders: ASEBA 324, **329**, 336; MMPI–2–RF 65, **66–7**, 72, 97, **98**; PCL:YV 273–4; and psychopathy 273–4. *see also* ADHD; aggression; conduct disorder; oppositional defiant disorder; substance abuse
- eyewitness testimony, historical perspectives 1, 2
- face validity 108; PSI 368, 369; RIM 217, 218
- facet subscales, MCMI 179, **180**, 181
- Factitious Disorder by Proxy 371
- factor analysis: ASEBA 324, 325; MMPI–A 117; MMPI–2–RF 69–70; PCL–R 233–5, **234**, **236**; PCL:YV 269, **269**; PSI 350
- fake bad. *see* feigning; inconsistency; malingering; symptom validity scales
- fake good: PSI 369; RIM 218. *see also* impression management

- false confessions 2
 false negatives, RIM 215
 Family Problems Scales: MMPI-2 46, **51**;
 MMPI-2-RF 99-100, **100**
 Fearless-Dominance Scale, MMPI-2-RF 80
 fearlessness trait 232
 Fears Subscales: MMPI-2 **51**; MMPI-2-RF **100**
 Federal Rules of Evidence 2
 feedback, therapeutic progress 120
 feigning: MMPI-2-RF 89; neuropsychological
 assessment 294, 296-302. *see also* malingering
 females, psychopathy 268. *see also* gender
 Finger Oscillation Test 7, **8**
 first-gate measures, PSI 356
 Five-Factor Model of Personality 241
 fixed/variable responding. *see* true response
 inconsistency; variable response inconsistency
 Flesch-Kincaid reading level index 64
 flexibility. *see* problem solving
 flooded Rorschach protocol 214, 215
 floor-effect detection of feigning 301, 302
 foreign language translations, MMPI-2 35
 forensic assessment instruments: ASEBA 333-8;
 MCMI 189-90; MMPI-A 122-30; MMPI-2
 7, 31-52; MMPI-2-RF 84-92; PAI 155-63,
 169; PCL-R 245-51; PCL:YV 279-83; PSI
 359-70, **368**; RIM 208-9, 212-19; specialized
 4-7, **6**
 forensic psychology: definition 1; developments in
 the field 1-5
 forensic scientist-practitioner approach 362-3
 forensically relevant instruments 6
 four-factor model, psychopathy **234**, 234-5, **236**,
 269, 275
 fraud, case example 251-4
Frye v. U.S. ruling 2, 3, 31
 Function of Rights in Interrogation (FRI) 5, 6

 Gastrointestinal Complaints Scale, MMPI-2-RF
 78, 81, **100**
 GCCT (Georgia Court Competency Test) **11**, **12**
 gender: MCMI 191; MMPI-A 125; MMPI-2 26;
 PCL-R 239, 242; PCL:YV 279; PSI 353-5; and
 psychopathy 268; and substance abuse 274
 gendered/non-gendered norms: MMPI-2-RF
 71-2; PAI 145
General Electric Company v. Joiner (1997) 3
 General Health Concerns Scale, MMPI-2-RF **100**
 generalizability: PCL-R 237-41; PCL:YV **268**; PSI
 350-5. *see also* convergent validity
 genetics, of personality traits 282-3
 Georgia Court Competency Test (GCCT) **11**, **12**
 Germany **268**
 grandiosity **234**, 284
 Grisso instruments **12**
 Grooved Pegboard Test **8**
 Grossman Facet Scales 179, **180**, 181, **193**

 guardedness 211, 214, 218. *see also* avoidant
 characteristics
 guilt **234**, 246
 gun crime 130-4

 Halstead-Reitan Neuropsychological Battery 10,
 11, **12**, **298**
 hand-scoring with templates, MMPI-2 36
 Hare Psychopathy Checklist (PCL) 183, 230, 232.
 see also Psychopathy Checklist
 harm reduction programs 244
 Harris-Lingoes Content subscales: MMPI-A
 112-13, 114, 129; MMPI-2 **50**
 harsh punishment 357, 358
 head injury. *see* traumatic brain injury
 Head Pain Complaints Scale, MMPI-2-RF 78, 81
 Health Concerns Scales: MMPI-2 29, **51**;
 MMPI-2-RF **100**
 heroin dependency 31. *see also* substance abuse
 hierarchical regression analyses, MMPI-2 34
 Higher-Order (H-O) Scales, MMPI-2-RF 65,
 65-6, 70, 71, 77, 91, 95, **96**
 hispanics 35. *see also* ethnic minorities
 historical perspectives, forensic psychology 1-3
 Historical-Clinical-Risk 20 (HCR-20) Scale,
 PCL-R 243, 244
 history of development: MCMI 176-8; MMPI
 67-8, 108-10; MMPI-A 110; MMPI-2 24-6,
 68-9; MMPI-2-RF 69-71; PAI 140, 143-5;
 PCL 232; PCL-R 232-3, 267; PCL:YV 267
 Hooper Visual Organization Test **8**
 hostility 46; case example 48, 49; and violence 250.
 see also aggression; violence
 Human Movement Responses Scale, RIM 217
 Humm-Wadsworth Temperament Scale 109
 hyperarousal 214, 215
 hypervigilance scales: PAI 167; RIM 206
 Hypochondriasis Scale, MMPI-2 23
 hypomania scales: MMPI-2 23, **50**; MMPI-2-RF
 80; RIM 220
 hypothesis generation, forensic psychology 366-7
 Hysteria Scale, MMPI-2 23, **50**

 Ideas of Persecution Scales: MMPI-2 23, 43;
 MMPI-2-RF 77, 101;
 Interdisciplinary Fitness Interview-Revised (IFI-R)
 11, **12**
 imagery, RIM 203, 204
 IMP (Interpersonal Measure of Psychopathy) 241
 Impatience Scale, MMPI-2-RF **100**
 impression management: PAI 148, 156, 159; PSI
 367, 369, 370; RIM 217, 218
 impulse control **234**; aggression 244; legal
 culpability 247; MMPI-A 116; MMPI-2 39
 Impulsive Antisociality Scale, MMPI-2-RF 80
 inconsistent responding. *see* variable response
 inconsistency

- Inefficacy Scale, MMPI-2-RF 98-9
- Infant-Toddler Social and Emotional Assessment (ITSEA) 330
- infrequency scales: MMPI 109; MMPI-2 22, 32, 35, 37; neuropsychological assessment 299; PAI 147, 155
- Infrequency Psychopathology Scales: MMPI-A 117; MMPI-2 22; MMPI-2-RF 87
- Infrequent Responses Scale, MMPI-2-RF 87
- insanity. *see* sanity evaluations
- insecure attachment 357
- insight, patient 43
- instrumental aggression 275
- intake measures, PSI 356
- intellectual functioning. *see* cognitive functioning
- intellectually disabled (ID) offenders, PCL-R 240-1
- intelligence tests: child custody evaluations 9; feigning **298**
- intercoder agreement, RIM 207
- Interest Scale, MMPI-2-RF 65, **67**, 71, 73, **98**
- internal consistency: MCMI 182-3; MMPI-A 114; MMPI-2 27; PAI 146-7
- internalizing scales: ASEBA 324, **329**, 336; MMPI-2-RF 65, **66**, 77, 91, **97**, 98; PCL:YV 272-3. *see also* anxiety; depression; self-harming; trauma
- international cross-validation norm studies, PSI 352-3. *see also* cross-cultural applicability; ethnic minorities
- Interpersonal Adjective Scales-Revised (IAS-R) 155
- interpersonal difficulties, case example 96
- Interpersonal Measure of Psychopathy (IMP) 241
- interpersonal scales: MMPI-A 123; MMPI-2-RF 65, **67**, **98**; PAI **142**, 154-5, 168; PCL-R **234**, 235; PCL:YV **269**, 284; RIM 211, 217
- interpretation of test results 370; ASEBA 337; MCMI 190, 191, 194-5; MMPI-A 128-30; MMPI-2 35-6; MMPI-2-RF 63-5; neuropsychological assessment **293**; PCL-R 249-50; PSI 355, 364-5, 368-70; RIM 203, 204, 210, 213, 217
- intersensory integration disorder, case example 370-1
- interventions. *see* treatment
- interviews, clinical, ASEBA 341
- interviews, semi-structured: PCL-R 231, 239; PCL:YV 267
- intraclass correlation coefficients (ICCs), RIM 207
- introjective problems, RIM 211
- introspective characteristics, RIM 211
- introversion 273; MMPI-2-RF 78, **99**, **100**
- invalid responding, MMPI-2-RF 90-2
- Inwald Personality Inventory (IPI) 30
- Iraq conflict, National Guard soldiers 75, 78-80
- Irritability Scale, MMPI-2-RF **100**
- Isolation Index, RIM 217
- item response theory (IRT) 239
- Jeffrey v. Dillard's Department Stores and Dillard's Bel Air Mall* 189
- Jenkins v. United States* 1, 2
- Journal of Clinical Child and Adolescent Psychology* 333
- journals, academic 4
- judgment, sound, RIM 216, 224
- juvenile delinquency. *see* delinquent behavior
- Juvenile Sex Offender Assessment Protocol-II (JSOAP-II) 276-7
- juveniles, competency to stand trial 9
- K-correction procedure: MMPI 116; MMPI-2 33-4, 40, 43
- kappa coefficients, RIM 207
- known groups designs 32
- Kumho Tire Co. v. Carmichael* 3
- labels, diagnostic: psychopathy 249, 279, 280-1, 283
- Lack of Drive Scale, MMPI-2-RF **100**
- language 5, 6; MMPI-2 25, 35, 68; PSI **353**
- Law and Neuroscience Project website 247
- learning disability, neuropsychological assessment 303-4, 306
- Least Detrimental Alternative (LDA) 366
- legal responsibility. *see* criminal responsibility
- Lie Scales: MMPI-A 109; MMPI-2 23
- life events, PSI 347, 348, **368**
- lifestyle scales: PCL-R **234**, 235; PCL:YV **269**
- literature, psychopathy 250-1
- Logical Memory Delayed Recognition (LMDR) subtest, Wechsler Memory Scale 302
- Long Term Offenders (LTO), PCL-R 247
- Low Positive Emotion Scales: MMPI-2 23; MMPI-2-RF 77
- Low Self-Esteem Subscales: MMPI-2 **51**; MMPI-2-RF **100**
- Luria-Nebraska Neuropsychological Battery, surveys of test use 10, **11**, **12**
- MacArthur Risk Study 242
- MacArthur Competency Assessment Tool-Criminal Adjudication (MacCAT-CA) **11**, 37
- MACI (Millon Adolescent Clinical Inventory) 10, 111
- major depression. *see* depression
- Malaise Scale, MMPI-2-RF 77, 81
- male juvenile delinquents. *see* delinquent behavior
- malingered 296; assessment 3, 4; neuropsychological tests 8; personal injury evaluations 40-1; surveys of test use **12**. *see also* feigning

- malinger scales: MMPI-2 14, 22-3, 27, 28, 32-3, 36, 37; MMPI-2-RF 88-9; PAI 150-1, 155, 156; RIM 218
- manic disorder scales: PAI 157-8, 159; RIM 220.
see also bipolar disorder
- Manual for Administration, MMPI-2-RF 74, 81
- marijuana abuse, neuropsychological assessment 304-5. *see also* substance abuse
- marital conflict, PSI 358
- Masculinity Scale, MMPI-2 23
- Massachusetts Youth Screening Instrument-Second Version (MAYSI-2) 5
- MCMI. *see* Millon Clinical Multiaxial Inventory
- Mechanical Physical Interests Scale, MMPI-2-RF 72
- Medical and Psychiatric Inventory 109
- medical. *see* clinical scales/applications
- Medico-Legal surveys of test use 13
- Megargee classification system 15, 42
- memory: feigning 297, **299**; and forensic psychology 291; malleability 2; MMPI-2-RF 90. *see also* neuropsychological assessment
- Mendota Juvenile Treatment Center (MJTC) 278
- mental health assessments. *see* neuropsychological assessment; psychological functioning
- Mental Measurements Yearbook* (Buros) 230, 368
- meta-analyses: MMPI-2 27, 32, 34, 40; PCL-R 242; RIM 208, 212
- mild traumatic brain injury, MMPI-2-RF 75, 78-80. *see also* traumatic brain injury
- Millon Adolescent Clinical Inventory (MACI) 10, 111
- Millon Clinical Multiaxial Inventory (MCMI) 175-6, 191; case example 192-5, **193**; clinical uses/limitations 187-9; feigning - neuropsychological assessment 300; forensic uses/limitations 7, 179, 189-90; history of development 176-8; personality spectrum prototypes **177**; personology model 177, 191, **192**; psychometric characterization 178-87, **182, 184**; surveys of test use 9, 10, **11, 12**, 10, 13
- Millon Personality Spectrometer 178
- Mini-Mental State Exam (MMSE) 330
- minimizers, RIM 218
- Minnesota Multiphasic Personality Inventory (MMPI): comparisons with other measures 208, 241; criminal responsibility evaluations 9; history of development 67-8; Megargee classification system 15; surveys of test use 7, **8, 9, 14**
- Minnesota Multiphasic Personality Inventory-Adolescent (MMPI-A) 21; administration/interpretation in forensic settings 128-30; case example 130-4, **131, 133**; clinical uses/limitations 120-2; description 110-11; ethnic minorities 127-8; forensic uses/limitations 122-30; history of development 110; psychometric characterization 114-20; revised version 121-2 MMPI-A-RF; scales 111-14, **112-13, 115**; surveys of test use 9, 10, 13, **14**
- Minnesota Multiphasic Personality Inventory-2 (MMPI-2) 9, 21; administration/interpretation in forensic settings 35-6; case example 42-51, **44, 45, 46, 47, 48, 49, 50-2**; childhood trauma 305; clinical uses/limitations 28-31; comparison with other instruments 150, 330; feigning assessment 299-300; forensic uses/limitations 7, 31-52; history of development 24-6, 68-9; Megargee classification system 15, 42; psychometric characterization 26-8; scales 4, 13, 15, 22-4, 32-3; short form 36; surveys of test use 7, **8, 9, 10, 11, 12, 13, 14**
- Minnesota Multiphasic Personality Inventory-2-Restructured Form (MMPI-2-RF) 63; administration/interpretation in forensic settings 63-5; case example 92-102, **94-101**; forensic uses/limitations 84-92; history of development 69-71; psychological assessment of medical patients 80-4; psychodiagnostic assessments 77-80; psychometric characterization 71-6; scales 65, **65-7**
- Minnesota Personality Schedule 109
- Miranda* rights: competency to waive 3, 5-6, 7, 10, **12, 16**; MMPI-A 122; neuropsychological assessment 291
- MJTC (Mendota Juvenile Treatment Center) 278
- MMPI. *see* Minnesota Multiphasic Personality Inventory
- moderating factors, psychopathy 282, 354
- modifier indices, MCMI 176. *see also* validity scales
- mood swings, PAI 167. *see also* emotional functioning
- moral decision making 247
- morale, PAI 154. *see also* demoralization scales
- motivation, crime/aggression, PCL:YV 274-5
- motor function, feigning 297, **299**
- multicultural assessment. *see* cross-cultural applicability; ethnic minorities
- multimethod/multi-source data gathering 361-2
- Multiple Specific Fears Scale, MMPI-2-RF 72, 98-9
- multivariate analysis, MMPI-2-RF 78
- Munchausen's Disorder by Proxy 371
- Münsterberg, Hugo 1
- murder: MMPI-A 125-6; MMPI-2 31
- narcissism, RIM 220
- National Guard soldiers, MMPI-2-RF 75, 78-80
- Native Americans, MMPI-2 34-5
- negative dissimulation, PAI 148-9, 156
- Negative Emotionality Scale, MMPI-2-RF **99**

- Negative Impression Management Scale, PAI 149–50, 156, 159, 160–1, 163
- Negative Predictive Power (NPP), PAI 148
- Negative Treatment Indicators Subscales, MMPI–2 **52**
- NEO–Personality Inventory (NEO–PI) 152, 154–5, 277
- neuroimaging 247–8, 297
- Neurological Complaints Scale, MMPI–2–RF 78, 81, **100**
- Neuropsychiatric Inventory (NPI) 330
- neuropsychological assessment 4, 17, 291, 302; case example 303–6; commonly used instruments **298–9**; feigning 296–302; forensic applications **292–4**; MMPI–2 31; principles 292, 294; surveys of test use 7–8, **8**, 13; test selection 294–6; violence risk 295–6. *see also* psychological functioning
- neuroscience applications 247–8, 297
- neurotic delinquents 272
- neuroticism scales: MMPI–2 31; MMPI–2–RF **99**; PAI 151
- non-compliance with medical treatment, PSI 357
- non-epileptic seizures (NES). *see* epileptic or non-epileptic seizures
- non-minimizers, RIM 218
- non-responsiveness measures: MMPI–2 22; MMPI–2–RF 75, 86–7
- Nonsupport Scale, PAI 154, 160, 167
- normative data/norms: ASEBA 324–5, 327–8, **328**; MCMI 179, 189; MMPI–A 110, 111, 120–1; MMPI–2 25, 26; MMPI–2–RF 71–3; PAI 145–6; PSI 350–5, 367; RIM 205–6
- not guilty by reason of insanity (NGRI): case example 42–51, **44**, **45**, **46**, **47**, **48**, **49**, **50–2**; MMPI–2 37. *see also* sanity evaluations
- NPP (Negative Predictive Power), PAI 148
- obsessive-compulsive personality, RIM 209, 210
- Offender Personality Disorder Pathway website 246–7
- Older Adult Behavior Checklist (OACBL) **315**, **321**, 326, 328, **328**, **329**
- Older Adult Self-Report (OASR) **315**, **321**, 326, **328**, 328, **329**
- On the Witness Stand* (Münsterberg) 2
- openness: psychotherapy indications 211; RIM 220
- Operation Enduring Freedom (OEF) veterans 79
- Oppositional Defiant Disorder (ODD) 273
- overpathologization: MCMI 190, 191; and psychopathy 282; RIM 206
- over-reporting measures: MMPI–A 126, 129–30; MMPI–2 32–3; MMPI–2–RF 65, 87–90, 93, 102; MMPI–2–RF 75
- PAI. *see* Personality Assessment Inventory
- pain: MMPI–2–RF 75, 83; PAI 151; treatment programs 29
- PAR (Psychological Assessment Resources) 130, 176
- paranoia scales: MMPI–2 23, 43, **50**; MMPI–2–RF 95–6; PAI 151, 158, 167, 168
- paranoid hypervigilance, case example 48, 49
- Parent Domain, PSI 346, **347**, 354–5, 357–8, **368**, 369, 371
- Parent Rating Scale 10
- parent reports **313**
- parental fitness/capacity: MMPI–2 40; MMPI–2–RF 92; surveys of test use 13, 15. *see also* child custody evaluations; Parenting Stress Index
- parental rights, termination 4
- Parenting Alliance Measure 368, 369
- Parenting Satisfaction Survey 368
- Parenting Stress Index (PSI) 17, 346–7, **347**; case example 370–1; clinical uses/limitations 355–8; forensic uses/limitations 359–70, **368**; history of development 348–50; language translations **353**; model 348–50, **349**; psychometrics 350–5, 367; surveys of test use 10
- Parenting Stress Index Short Form (PSI–SF) 350
- PAS (Personality Assessment Screener) 162
- passive/aggressive personality styles 188
- pathologization. *see* overpathologization
- PCL. *see* Psychopathy Checklist
- Pearson Assessments 21, 33, 43, 129, 130, 176
- People v. Stoll* 189
- perceptions: individual differences 2; RIM 202, 203, 219, 224
- Perceptual Thinking Index (PTI) 206
- persecutory delusions, case example 48, 49
- personal injury evaluations: expert testimony 1; MMPI–A 122–3; MMPI–2 40–1; MMPI–2–RF 92; RIM 214–16; surveys of test use 9, 16
- Personality Assessment Inventory (PAI) 140; case example 163–9, **165**, **166**, **167**; clinical uses/limitations 156–8, 164–7; comparison with MMPI–2–RF 82; feigning assessment 300–1; forensic uses/limitations 155–63, 169; history of development 140, 143–5; norms 145–6; psychometric characterization 146–51, 164; reading ability 64; scales **141–3**, 147–55; surveys of test use 10, **11**, **12**, 13; variants 111, 162–3
- Personality Assessment Inventory–Adolescent (PAI–A) 111, 162–3
- Personality Assessment Screener (PAS) 162
- personality disorder 175; adolescents 121; bioevolutionary theory 177, **178**; diagnosis 71; and psychopathy 274
- Personality Disorder Adjective Checklist 178
- personality disorder scales: MMPI–2–RF 65; PAI 152, 158. *see also* Millon Clinical Multiaxial Inventory

412 Subject Index

- personality measures 38, 76, 77, 125–6, 140, 208;
surveys of test use 9, 13, 16. *see also* Minnesota
Multiphasic Personality Inventory; Personality
Assessment Inventory; Rorschach Ink Blot Test
Personality Psychopathology Five Scales: MMPI–A
114; MMPI–2 24, 27, 28, 30; MMP1–2–RF 65,
67, 71, 73, 78, 97, **99**, 100; risk assessments 39,
49
personality spectrum prototypes, MCMI **177**
personality states, RIM 210
personality structure, and psychopathy 244
personality styles 178, 188
personality traits: genetic influence 282–3; RIM
210
personology model 177, 191, **192**
Physical Aggression Scale, PAI 154
plethysmograph **11**
positive dissimulation, PAI 148–9, 156
Positive Impression Management Scale, PAI 148,
156, 159, 163, 164
Positive Predictive Power (PPP), PAI 148, 150
post-traumatic stress disorder (PTSD) 4; MCMI
188; MMPI–2 41; MMPI–2–RF 75, 78–80, 87,
92; PAI 149–50, 151; RIM 214, 215
PPI (Psychopathic Personality Inventory) 80, 241
Predictive Indices, PAI **143**
Predictive Power, PAI 148, 150
predictive validity: MMPI–2 34; PCL–R 242, 246;
PCL:YV 275–7, 279–80; PSI 356–8, 368; RIM
210
preventive detention hearings, PCL–R 246–7
problem-solving skills: PSI 358; RIM 216
Profile Report, MMPI–A 129
Projective drawings, surveys of test use **11**, **12**
promiscuous sexual behavior 235, 252, 284
property crimes 125
prototypical distribution, MMPI–2 73
pseudoscience, RIM as 205
PSI. *see* Parenting Stress Index
PSY–5 scales. *see* Personality Psychopathology Five
Scales
Psychasthenia Scale, MMPI–2 23
Psychodiagnostics journal 204
Psychogenic Nonepileptic Seizures Scale,
MMPI–2–RF 81. *see also* epileptic or
non-epileptic seizures
Psychological Assessment Resources (PAR) 130,
176
psychological functioning 3, 4; ASEBA 334;
MMPI–A 126, 127; MMPI–2 29; PAI 144;
surveys of test use **11**. *see also*
neuropsychological assessment
psychological mindedness, RIM 211
Psychology–Law Listserv 13
psychometric characterization: ASEBA 324–5,
327–32, **329**, **331**; MCMI 178–87, **182**, **184**;
MMPI–A 114–20; MMPI–2 26–8;
MMPI–2–RF 71–6; neuropsychological
assessment 295; PCL–R 233–41; PCL:YV **268**;
PSI 350–5; RIM 205–9
Psychopathic Deviate Subscales, MMPI–2 23, **50**
Psychopathic Personality Inventory (PPI) 80, 241
psychopathology 17, 140; MMPI–2 21, 30–1, 29,
38; MMPI–2–RF 75, 76, 77, 80; surveys of test
use 13
psychopathy: and aggression 274–5; and antisocial
behavior 233; and antisocial personality disorder
250; and anxiety disorder 272–3; case example
251–4, 283–4; characterization 230;
externalizing behaviors/disorders 273–4;
four-factor model **234**; and personality disorder
274; RIM 220; treatment response 277–8
Psychopathy Checklist (PCL) 183, 230, 232
Psychopathy Checklist–Revised (PCL–R) 230–1;
case example 251–4; comparisons with other
measures 241; cut scores 231, 249; description
231–2; factor structure 233–5, **234**, **236**, **237**;
forensic uses/limitations 245–51; generalizability
237–41, 268; history of development 232–3,
267; misuses 248–9; qualifications for use
249–50; reliability 235–7; research contexts 251;
surveys of test use 10, **11**, 13; treatment response
244–5; validity 241–4
Psychopathology Checklist: Screening Version
(PCL–SV) 10, **11**, 124, 152–3, 238
Psychopathy Checklist: Youth Version (PCL:YV)
266; case example 283–4; crime/aggression
motivations 274–5; description 266–7; factor
analysis **269**, 269; forensic uses/limitations
279–83; generalizability **268**; history of
development 267; predictive validity 275–7;
psychopathic traits 266; reliability 269–70;
surveys of test use 13; treatment response 277–8;
validity 270–4
Psychopathy Content Scale, MCMI 271
psychosis: MMPI–2 31, 37, 45, 47; MMPI–2–RF
64, 88; **100**; PAI 151, 157–8; RIM 216
psychosocial adjustment: MMPI–2 29;
MMPI–2–RF 74; PSI 358
psychosocial environment 140. *see also* Personality
Assessment Inventory
psychotherapy, indications for 168–9; RIM
210–12. *see also* treatment
psychoticism scales. *see* psychosis
PTI (Perceptual Thinking Index) 206
PTSD. *see* post-traumatic stress disorder
punishment, harsh 357, 358

qualifications, forensic psychology 4
qualifications for use, PCL–R 249–50

race. *see* ethnic minorities
random responding. *see* true response inconsistency;
variable response inconsistency

- rapists 237
- Rarely Missed Index, Wechsler Memory Scale 302
- RC Scales. *see* Restructured Clinical Scales
- RDF (Rogers Discriminant Function), PAI 150, 163
- reactive aggression 275
- reading ability: ASEBA 311; MMPI-A 110, 111, 127, 129; MMPI-2 21; MMPI-2-RF 64; PAI 161
- reality testing, RIM 212–13, 223–4
- Reasoning and Understanding domains, MacArthur Competency Assessment Tool 37
- Receiver Operator Curve (ROC) analyses, MMPI-2-RF 78
- recidivism: MMPI-A 134; PCL-R 241, 243; PCL:YV 276, 278; and psychopathy 230, 279. *see also* violence risk
- reference data. *see* norms
- regression analyses, MMPI-2-RF 77, 91–2
- reinforcement seeking, scientific basis 177–8
- Reitan-Klove Sensory Perceptual Examination **8**
- reliability, assessment measures 2, 3; ASEBA 329; MCMI 181–3, **182**, 194; MMPI-2 26–7; MMPI-2-RF 73–4, 76; neuropsychological assessment 295; PAI 146–7; PCL-R 235–7; PCL:YV 269–70; RIM 207–8. *see also* test-retest reliability
- religion 68
- remorse **234**, 246, 284
- Repression Scale, MMPI-2 24
- research base. *see* scientific basis
- response bias: MCMI 190; MMPI-2 40, 51; MMPI-2-RF 90; neuropsychological assessment 300; PSI 364–5, 367
- response distortion, PAI 155–6
- response inhibition 247
- response style, MMPI-A 117, 123; PSI 369–70
- responses to interventions (RTIs). *see* treatment response
- responsibility for actions. *see* criminal responsibility
- Restandardization Committee, University of Minnesota Press 68, 69
- Restructured Clinical Scales: case example **46**; MMPI-2 13, 22, 23, 27, 28, 30, 38, 43, 51; MMPI-2-RF 65, **66**, 69–71, 73, 75, 77, 80, 83, 95, **96**
- retest reliability, RIM 207–8
- retrofitting traditional clinical tests 7
- Rey 15-Item Test **8**, **12**
- Richman Behavior Checklist (BCL) 330
- RIM. *see* Rorschach Inkblot Method
- risk assessments: coronary heart disease 29; MMPI-2 38–9, 49, **50**; PCL:YV 280; psychopathy 281; surveys of test use 13. *see also* violence risk
- risk management, PCL-R 244
- Rogers Discriminant Function (RDF) 150, 163
- Rorschach Comprehensive System Schizophrenia Index 152
- Rorschach Inkblot Method (RIM) 202, 224; case example 219–24, **221**, **222**, **223**; child custody evaluations 366; clinical uses/limitations 209–12; feigning assessment 301; forensic uses/limitations 208–9, 212–19; history of development 203–5; methodology 202–3; norms 205–6; psychometric characterization 205–9; surveys of test use 9, 10, **11**, **12**, 13, **14**
- RTIs (responses to interventions). *see* treatment response
- rule-breaking behavior 324, 325, 334. *see also* delinquent behavior
- rules of evidence 190
- sanity evaluations: malingering 32; MMPI-A 122; MMPI-2 37, 42–51, **44–52**; RIM 213, 214, 219–24, **221**, **222**, **223**; surveys of test use 10
- SAVRY (Structured Assessment of Violence Risk in Youth) 277
- S-B-R (Stanford-Binet Revised) test 10, **11**, **12**
- Schedule of Affective Disorders and Schizophrenia with School Age Children 124
- Schedule of Recent Events (SRE) 154
- schizoaffective disorder 223
- schizoid/avoidant personality style 188
- schizophrenia scales: MCMI 186–7; MMPI-2 23, 42–51, **44–52**; MMPI-2-RF 77–8, 87; PAI 151–2, 158, 162; PCL-R 242; RIM 209
- Schizophrenic Reaction, Childhood Type
- diagnostic category 324
- Schizotypal Characteristics Scale, MMPI-2-RF **100**
- SCICA (Semistructured Clinical Interview for Children and Adolescents) 316, 319, 332, 333
- SCID (Structured Clinical Interview for DSM-IV) 77
- scientific basis: ASEBA 335–6; child custody evaluations 361; expert testimony 3; MCMI 190; neuropsychological assessment 295; PCL-R 251; personality disorders diagnosis 176–7; PSI 355
- screening, PSI 356–8
- Seashore Rhythm Test **8**
- secure facilities 122
- seed scales, MMPI-2-RF 70
- seizure disorder. *see* epileptic and non-epileptic seizures
- self-concept, PAI 167–8
- self-control, RIM 213, 216
- self-deception scales: PAI 148; PSI 369–70
- Self-Depreciation Scale, MMPI-2-RF **100**
- Self-Doubt Scale, MMPI-2-RF **100**
- Self-Esteem Subscales: MMPI-2 **51**; MMPI-2-RF **100**
- self-fulfilling prophecies 270
- self-harming behavior, PCL:YV 272–3

- Self-Report Psychopathy Scale (SRPS) 241
- SEM. *see* standard error of measurement
- Semistructured Clinical Interview for Children and Adolescents (SCICA) 316, 319, 332, 333
- semi-structured interviews: PCL–R 231, 239; PCL:YV 267
- sensitivity to injury, MMPI–A 125
- sensory functioning, feigning 297, **299**
- Sentence Completion Test **8**, **11**, **12**
- sentencing, PCL:YV 270, 279, 281
- Severe Personality Pathology Scale, MCMI 175
- Severe Syndrome Scale, MCMI 175
- Sex Offender Registration and Notification Act (SORNA) 277
- Sex Offender Risk Appraisal Guide (SORAG) **11**, 243
- sex offenders: PCL–R 237, 243; risk assessments 3, 10, **11**, 39
- sexist language 25; MMPI–2 68
- sexual abuse, MMPI–A 123
- sexual promiscuity 235, 252, 284
- Sexual Violence Risk 20 (SVR–20) **11**, 243
- sexually violent predators (SVPs), PCL–R 246
- Shyness Scale, MMPI–2–RF 80, 99–100, **100**
- Si* Subscales, MMPI–A 112, **112–13**, 114, 129
- simulation study design, MMPI–2–RF 87–8
- SIPA (Stress Index for Parents of Adolescents) 347
- SIRS (Structured Interview of Reported Symptoms) 4, 8, **12**, 89, 149, 150, 151
- 16 Personality Factors (16FP) **12**
- social anxiety 273
- Social Avoidance Scales, MMPI–2–RF 78, 80
- Social Discomfort Subscales: MMPI–2 **51**; MMPI–2–RF **100**
- social environment, PAI 168
- Social Introversion Scale, MMPI–2 23, **50**
- social support, PSI 358
- social withdrawal, MMPI–A 123
- Society of Personality Assessment (SPA) 13
- socioeconomic status 39, 116
- soldiers, MMPI–2–RF 75, 78–80
- Somatic Complaints Scales: MMPI–2 23, 29; MMPI–2–RF 78, 80, 81, 82, 83, 87, **97**, 98; neuropsychological assessment 301; PAI 151, 157
- SORAG (Sex Offender Risk Appraisal Guide) **11**, 243
- SORNA (Sex Offender Registration and Notification Act) 277
- SPA (Society of Personality Assessment) 13
- Specialty Guidelines for Forensic Psychology* (APA) 1, 3, 4
- Specific Problem Scales: MMP1–2–RF 65, **66**, 71, 77, 78, 80
- specificity statistics, MCMI 190
- Speech Sounds Perception Test **8**
- spine surgery candidates, MMPI–2–RF 83
- Spouse Subscale, PSI 369
- SRE (Schedule of Recent Events) 154
- SRPS (Self-Report Psychopathy Scale) 241
- standard error of measurement (SEM): MMPI–2–RF 73–4; PAI 147; PCL–R 235
- standardization: MCMI 179; MMPI–2 26; MMPI–2–RF 71–3; PAI 145, 146
- Stanford Achievement Test 131
- Stanford–Binet Revised (S–B–R) test 10, **11**, **12**
- static factors, risk assessments 38
- stress, parental. *see* Parenting Stress Index
- stress scales: MMPI–2–RF 98–9; PAI 154; RIM 202, 213, 214, 215, 216
- Stress Index for Parents of Adolescents (SIPA) 347
- structural data, RIM 202–3, 218–19, 220, **222**, 224
- Structural Summary, MMPI–A 117, **118–19**, 120
- Structured Assessment of Violence Risk in Youth (SAVRY) 277
- Structured Clinical Interview for DSM–IV (SCID) 77
- Structured Interview of Reported Symptoms, Second Edition (SIRS–2) 4, 8, **12**, 89, 149, 150, 151
- structured interviews, personality disorders 176
- Submissiveness Scale, MMPI–2–RF **100**
- substance abuse: case example 163–9, **165**, **166**, **167**, 252, 284; neuropsychological assessment 304–5. *see also* alcohol abuse
- substance use scales: ASEBA **329**; MCMI 187–8; MMPI–A 125, 126; MMPI–2 29, 31, 39; PAI 153, 158–9, 162, 163, 164; PCL–R 238; PCL:YV **268**, 273, 274
- suicidal ideation scales: MMPI–2–RF **100**; PAI 144, 154, 160–1; RIM 210
- Superlative Scale, MMPI–2 23
- Supplementary Scales: MMPI–A **112**, 113, 114, 132, **133**; MMPI–2 24, 46, **48**, 51, 69; MMPI–2–RF 73
- supportive approaches, psychotherapy indications 211
- surveys of test use, traditional clinical tests 7–17, **11–12**
- suspiciousness, RIM 220
- SVR–20 (Sexual Violence Risk 20) **11**, 243
- SVPs (sexually violent predators), PCL–R 246
- Sweden **268**
- symptom validity scales: MMPI–2 33; MMPI–2–RF 22–3, 74, 87, 89; neuropsychological assessment 300; RIM 218
- Symptom Validity Test (SVT) 90, 91
- syndromes, ASEBA 316, **318**, 319, 324, **329**
- Tactual Performance Test **8**
- TAT (Thematic Apperception Test) 9, **11**, **12**
- TCI (Trauma Content Index) 214
- Teacher's Report Form (TRF) **315**, 324–5, 327–8, **328**, **329**, 330, 332, 333, 337

- Test Observation Form (TOF) 316, 332, 333
 Test of Memory Malinger (TOMM) 4, 8, **12**, 89
 test–retest reliability: MMPI–A 114; PAI 146–7
 Texture Scale, RIM 217
 theft, case example 251–4, 283–4
 Thematic Apperception Test (TAT) 9, **11**, **12**
 thematic data, RIM 203, 204, 219
 therapeutic assessment. *see* clinical assessment
 therapeutic change. *see* treatment response
 thought dysfunction scale, MMPI–2–RF 75, 95
 three-factor model of psychopathy 269
 TOF (Test Observation Form) 316, 332, 333
 TOMM (Test of Memory Malinger) 4, 8, **12**, 89
 TPI (Treatment Process Index) 161
 traditional clinical tests. *see* clinical assessment;
 clinical scales
 Trail Making Tests A and B 7, **8**, 304
 trauma, personal injury evaluations 40–1
 Trauma Content Index (TCI) 214
 Trauma Symptom Inventory 13
 traumatic brain injury: and aggression/violence risk
 295–6, 302; MMPI–2 27; MMPI–2–RF 75,
 78–80, 89; neuropsychological assessment 300,
 304
 traumatization, childhood 273
 treatment consideration scales: PAI **142**, 153–4,
 168–9
 treatment indications: PCL:YV 270; PSI 357; for
 psychotherapy 168–9; RIM 210–12
 treatment planning, PAI 161
 Treatment Process Index (TPI) 161
 Treatment Rejection Scale, PAI 154, 161
 treatment response, psychopathy 279, 280, 281
 treatment response measures: CBCL/6–18 315;
 MMPI–A 120; PCL–R 244–5; PCL:YV 277–8;
 RIM 211–12
 triage, PSI 356–8
 TRF (Teacher's Report Form) **315**, 324–5, 327–8,
 328, **329**, 330, 332, 333, 337
 true response inconsistency (TRIN) scales:
 MMPI–A 117, 126, 129, 131; MMPI–2 22, 27;
 MMPI–2–RF 73, 87; PAI 155
 trust 220
 T-scores: ASEBA 328, 330, 336; MMPI–A 120–1,
 131; MMPI–2 22, 23, 26, 43, 73; MMPI–2–RF
 72–3; PAI 145, 146, 147; PCL–R 237, 238,
 239, 240, 241; PSI 370
 two-factor model: psychopathy 233–4, 235, **237**
 Type A Subscales: MMPI–2 **51**; MMPI–2–RF
 100
 Uncommon Virtues Scale: MMPI–2–RF 74
 under-reporting measures: MMPI–A 129–30;
 MMPI–2 28; MMPI–2–RF 65, 75, 90, 93, 99;
 undersocialized delinquents 272
 unidirectional variables, RIM 210
 Uniform Marriage and Divorce Act (UMDA) 39,
 365
 United Kingdom 246
 United States 246, **268**, 351–2; Supreme Court 2,
 31
 University of Minnesota Hospital 25
 University of Minnesota Press 25;
 Restandardization Committee 68, 69
U.S. v. Huberty 32
 UT-scores, MMPI–2–RF 73
 'Validation of the Parenting Stress Index among
 Chinese Mothers in Hong Kong': (Tam, Chan
 and Wong) 352
 validity, assessment measures 3, 295; ASEBA
 329–30, **331**; MCMI 183–7, **184**, 194;
 MMPI–2–RF 74–5, 76; PAI 162, 164; PCL–R
 241–4; PCL:YV 270–4; PSI 356; RIM 208. *see*
 also construct validity; convergent validity;
 discriminant validity; divergent validity; face
 validity
 Validity Indicator Profile (VIP) 8, **12**
 validity indices: MCMI 176; PAI **143**
 validity scales: MCMI 176; MMPI 68; MMPI–A
 109, 116–17, 126, 131, 132; MMPI–2 4, 22,
 27–8, 33, 35–7, 39–40, **44**, **45**, 68;
 MMPI–2–RF **65**, 65, 70, 73–5, 84–7, 90–3,
 95; PAI **141**, 147–51
 variable response inconsistency (VRIN) scales:
 MMPI–A 117, 126, 129, 131; MMPI–2 22;
 MMPI–2–RF 73, 87, 93; PAI 147–8, 155
 Verbal Aggression Scale, PAI 154
 Victoria Symptom Validity Test 89
 Vietnam war, post-traumatic stress disorder 188
 violence/violent offenders: case example 251–4;
 and hostility 250. *see also* aggression; hostility
 Violence Potential Index (VPI) 160
 violence risk 3, 10, **11**, 38–9, 160; MMPI–A
 125–6, 126; neuropsychological assessment
 295–6; PAI 159–60; PCL–R 242–4; PCL:YV
 276, 277
 Violence Risk Appraisal Guide (VRAG) 10, **11**,
 242, 243, 244
 VIP (Validity Indicator Profile) 8, **12**
 vocabulary scales 5, 6. *see also* language
 volitional incapacity 213, 214
 VPI (Violence Potential Index) 160
 VRAG (Violence Risk Appraisal Guide) 10, **11**,
 242, 243, 244
 VRIN. *see* variable response inconsistency
 WAIS. *see* Wechsler Adult Intelligence Scale
 web-based surveys of test use 13
 websites: ASEBA 311; antisocial personality
 disorder 246–7; Law and Neuroscience Project
 247; PSI 347

416 Subject Index

- Wechsler Adult Intelligence Scales (various editions): neuropsychological assessment 31, 301–4; role in forensic psychology 7; surveys of test use 7, **8**, 9, 10, **11**, **12**, **14**
- Wechsler Memory Scale 7, 302
- Wisconsin Card Sorting Test (WCST) 7, **8**
- withdrawal measures: ASEBA 325; MMPI–A 123
- worry scale, MMPI–2–RF 98–9. *see also* anxiety; stress
- Youth Level of Service Case Management Inventory 277
- Youth Psychopathic Traits Inventory (YPI) 271
- Youth Self-Report (YSR) **315**, **321**, 324–5, 328, **329**, 330, 332, 336, 337; narrative reports/critical items 323–4, **323–4**; surveys of test use 10, 13